Grand Quiz Spring 2021

Subject Code MTH301 lecture 1 to 22

Solved By Riz Mughal

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I'm providing 100% correct quiz solution. You can visit my YouTube channel and get more information about all other subjects' quizzes and final year project (CS619).

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MTH301:Grand Quiz

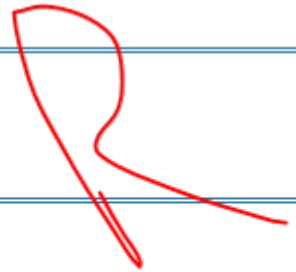
Quiz Start T

Question # 1 of 30 (Start time: 09:21:09 AM, 26 June 2021)

$Face\ of\ the\ parabola\ y = ax^2 + bx + c\ is\ opening\ upward\ when$

Select the correct option

- ○ b > 0
- ○ b < 0
- ○ a < 0
- ● a > 0



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Quiz Start Time: 09:21 AM,

Question # 2 of 30 (Start time: 09:21:29 AM, 26 June 2021)

If $w = x + y$ while $x = r + s$ and $y = r - s$ then which of the following is partial derivative of w w.r.t s?

Select the correct option

Reload Mat

●	0
○	-1
○	2
○	1

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Question # 3 of 30 (Start time: 09:22:23 AM, 26 June 2021)

The surface:z=f(x,y) falls (decreases) most rapidly at any point in the ---------- of Gradient(z).

Select the correct option

- ○ Direction
- ○ Direction perpendicular to
- ○ Arbitrary direction
- ● Opposite to the direction

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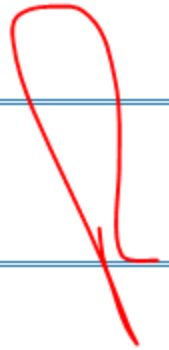
Question # 4 of 30 (Start time: 09:22:48 AM, 26 June 2021)

XY-Plane consist of all points where

Select the correct option

- xy=0
- xy=1
- z=0
- z=1



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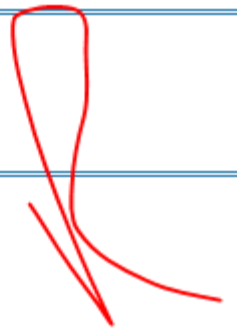
Question # 5 of 30 (Start time: 09:23:03 AM, 26 June 2021)

What is the condition of the extreme value theorem?

Select the correct option

- (●) A function must be continuous on a closed interval *
- ( ) A function must be continuous and differential
- ( ) No conditions are necessary
- ( ) A function must differential



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Quiz Start Time: 09:21

Question # 6 of 30 (Start time: 09:23:19 AM, 26 June 2021)

If $f(x, y, z) = x^2 + 2y - 3z^2$ then f_x, f_y and f_z are

Select the correct option

Reload

- $f_x = 2x, f_y = 2$and $f_z = 6z$
- $f_x = 2x, f_y = 2$and $f_z = -6z$
- $f_x = 2x, f_y = 2y$and $f_z = 6z$
- $f_x = 2x, f_y = 2y$and $f_z = -6z$

Click to Save Answer & Move to

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Question # 7 of 30 (Start time: 09:23:42 AM, 26 June 2021)

Can we evaluate the following integral in given order of integration

$$\int_{0}^{\frac{1}{2}} \int_{2x}^{1} e^{y^2} \, dy dx$$

Select the correct option

○	Yes
●	No

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Quiz Start Time

Question # 8 of 30 (Start time: 09:24:02 AM, 26 June 2021)

The $\int_0^1 2xe^{x^2}\,dx =$

Select the correct option

- $e+1$
- e^{x^2}
- $e-1$ (selected)
- e

Click to Save Answer & M

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Question # 9 of 30 (Start time: 09:24:18 AM, 26 June 2021)

If a vector of magnitude '4' is making angle 30-degree with Y-axis then its component vector along x-axis is --------

Select the correct option

- ○ 2*sqrt(3)
- ○ -2*sqrt(3)
- ● 2
- ○ -2

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Question # 10 of 30 (Start time: 09:24:34 AM, 26 June 2021)

The function

$$f(x,y,z)=\frac{(x^2+yz)}{(xy+z)}$$

is discontinuous when

Select the correct option

- $xy>z$
- $x+y+z=0$
- $xy+z=0$
- $x^2+yz=0$

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Quiz Start Time: (

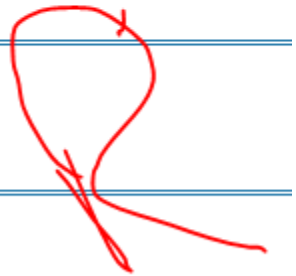
Question # 11 of 30 (Start time: 09:25:05 AM, 26 June 2021)

Value of the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at the point $(0, 3, 0)$ is _____

Select the correct option

Re

- ○ $2\sqrt{2}$
- ○ 9
- ● 3
- ○ $\sqrt{3}$



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Question # 12 of 30 (Start time: 09:25:26 AM, 26 June 2021)

$$\int_{1}^{3}\int_{0}^{1} xy\,dx\,dy = \text{- - - - - - - - - -}$$

Select the correct option

- $\frac{3}{4}$
- $\frac{7}{12}$
- $\frac{9}{4}$
- 3

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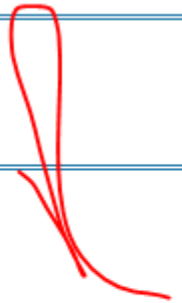
Question # 13 of 30 (Start time: 09:25:41 AM, 26 June 2021)

Range of the function is

$$f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$$

Select the correct option

- ○ $(0, +\infty)$
- ○ $(-\infty, 0)$
- ● $[0, +\infty)$
- ○ $(-\infty, +\infty)$



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Question # 14 of 30 (Start time: 09:25:55 AM, 26 June 2021)

Let $f(x, y)$ be a function with continuous second order partial derivatives in some circle centered at a critical point (x_0, y_0) and, let $D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - [f_{xy}(x_0, y_0)]^2$

Select the correct option

○	$f(x, y)$ has a relative minimum at (x_0, y_0).
○	No conclusion can be drawn.
●	$f(x, y)$ has a saddle point at (x_0, y_0).
○	$f(x, y)$ has a relative maximum at (x_0, y_0).

Click to Sa

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Quiz Start Time: 09:2

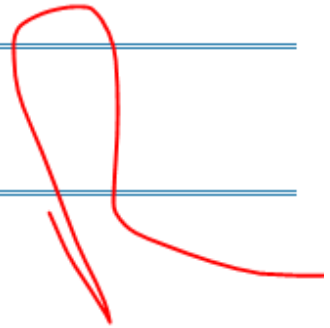
Question # 15 of 30 (Start time: 09:26:24 AM, 26 June 2021)

$x = 3\cos\theta,\ y = 3\sin\theta;\ 0 \leqslant \theta \leqslant \pi$, is a _____ form of a curve in the plane.

Select the correct option

Reloa

- parametric vector
- parametric
- explicit
- implicit



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Question # 16 of 30 (Start time: 09:26:46 AM, 26 June 2021)

If $w = f(x, y)$ where $x = g(r, s)$ then $\frac{\partial w}{\partial r} =$

Select the correct option

- $\frac{\partial w}{\partial r} = \frac{dw}{dx}\frac{\partial x}{\partial r} + \frac{dw}{dy}\frac{\partial y}{\partial r}$
- $\frac{\partial w}{\partial r} = \frac{dw}{dy}\frac{\partial y}{\partial r} + \frac{dw}{dx}\frac{\partial x}{\partial s}$
- $\frac{\partial w}{\partial r} = \frac{dw}{ds}\frac{\partial s}{\partial r} + \frac{dw}{dy}\frac{\partial y}{\partial r}$
- $\frac{\partial w}{\partial r} = \frac{dw}{dx}\frac{\partial x}{\partial w} - \frac{dw}{dy}\frac{\partial y}{\partial r}$

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Question # 17 of 30 (Start time: 09:27:03 AM, 26 June 2021)

$x - coordinate\ of\ vertex\ of\ parabola\ y = 3x^2 + 6x + 8\ is$

Select the correct option

- (●) -1
- ( ) 1
- ( ) 3
- ( ) 6

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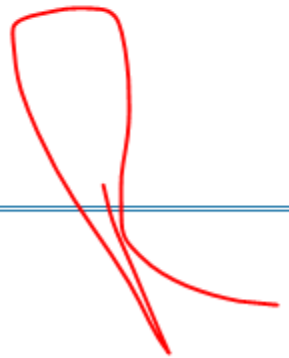
Question # 18 of 30 (Start time: 09:27:17 AM, 26 June 2021)

Length or magnitude of a unit vector is--------.

Select the correct option

- 0
- 1



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Question # 19 of 30 (Start time: 09:27:35 AM, 26 June 2021)

Which of the following is the representation of spherical coordinates

Select the correct option

- ○ (r, θ, z)
- ● (ρ, θ, φ)
- ○ (ρ, φ, z)
- ○ (x, y, z)

Click to Save An

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Quiz Start Time: 09:21 AM, 26

Question # 20 of 30 (Start time: 09:27:51 AM, 26 June 2021) Tot

Position of any point in the plane can be obtained by the two perpendicular lines known as x and y axes and together we call it as _____ coordinates system.

Select the correct option

- ○ Spherical
- ○ Rectangular
- ○ Cylindrical
- ● Cartesian

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Question # 21 of 30 (Start time: 09:28:07 AM, 26 June 2021)

Cylindrical coordinate system is used to determine the position of any object in

Select the correct option

- one dimension
- three dimension
- two dimension
- any dimension

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Question # 22 of 30 (Start time: 09:28:22 AM, 26 June 2021)

Which coordinate system uses two distances and one angle?

Select the correct option

- (●) cylindrical
- ( ) spherical
- ( ) rectangular
- ( ) none of these

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Question # 23 of 30 (Start time: 09:28:49 AM, 26 June 2021)

If $w = f(x)$ where $x = g(r, s)$ then $\frac{\partial w}{\partial s} =$

Select the correct option

- $\frac{\partial w}{\partial s} = \frac{dw}{ds}\frac{\partial s}{\partial s}$
- $\frac{\partial w}{\partial s} = \frac{dw}{dx}\frac{\partial x}{\partial s}$
- $\frac{\partial w}{\partial s} = \frac{dw}{dr}\frac{\partial r}{\partial s}$
- $\frac{\partial w}{\partial s} = \frac{dw}{ds}\frac{\partial s}{\partial r}$

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Question # 24 of 30 (Start time: 09:29:09 AM, 26 June 2021)

Partial derivative of

$$f(x,y) = x^2 + y^2 + xy$$

with respect to y is

Select the correct option

○	$x^2 + 2y + xy$
○	$x^2 + 2y + x$
●	$2y + x$
○	None of these

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Question # 25 of 30 (Start time: 09:29:26 AM, 26 June 2021)

In rectangular coordinate system, the point (1, 5) lies in

Select the correct option

- ○ fourth quadrant
- ● first quadrant
- ○ third quadrant
- ○ second quadrant

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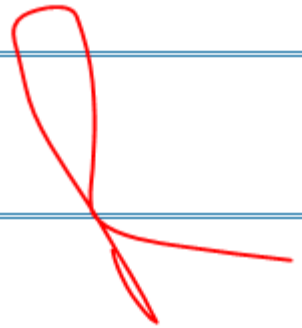
Question # 26 of 30 (Start time: 09:29:41 AM, 26 June 2021)

The differentiable functions are continuous ..... in its domain.

Select the correct option

- ○ nowhere
- ● everywhere
- ○ somewhere
- ○ only at (0,0)



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Question # 27 of 30 (Start time: 09:29:57 AM, 26 June 2021)

Double integral is used to find the ------ of solid.

Select the correct option

- ○ width
- ○ none of these
- ● volume
- ○ surface

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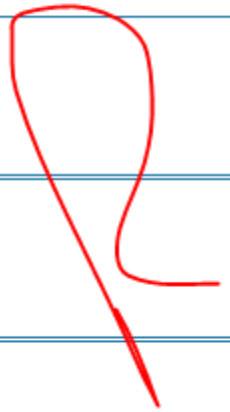
Question # 28 of 30 (Start time: 09:30:11 AM, 26 June 2021)

Value of the function f(x, y, z) = xy + z at the point (0, 1, 1) is _____.

Select the correct option

- ○ 2
- ● 1
- ○ 3
- ○ 0



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Quiz Start

Question # 29 of 30 (Start time: 09:30:26 AM, 26 June 2021)

Value of the function $f(x,y,z) = xy + z$ *at the* point $(1, x^3, y^2)$

Select the correct option

- $y + z$
- $x^3 + y^2$
- $xy + z^2$
- $x^3 + z$

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Quiz Start Time

Question # 30 of 30 (Start time: 09:30:44 AM, 26 June 2021)

The function of the form $f(x, y) = 3x^2x^5$ is _____ in the domain.

Select the correct option

- discontinuous
- may or may not be continuous
- continuous everywhere
- piecewise continuous

Click to Save Answer & M

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MTH301 MID FALL 2010

Suppose $f(x,\ y) = x^3 e^{xy}$. Which one of the following is correct?

$\frac{\partial f}{\partial x} = 3x^2 e^{xy} + x^3 y e^{xy}$

$\frac{\partial f}{\partial x} = 3x^2 y e^{xy}$

$\frac{\partial f}{\partial x} = 3x^2 e^{xy} + x^4 e^{xy}$

$\frac{\partial f}{\partial x} = 3x^2 e^{xy}$

Let R be a closed region in two dimensional space. What does the double integral over R calculates?
Area of R.
Radius of inscribed circle in R.
Distance between two endpoints of R.
None of these

What is the distance between points (3, 2, 4) and (6, 10, -1)?

$7\sqrt{2}$

$2\sqrt{6}$

$\sqrt{34}$

$7\sqrt{3}$

---------- planes intersect at right angle to form three dimensional space.

Three

4

8

There is one-to-one correspondence between the set of points on co-ordinate line and ----------

Set of real numbers

Set of integers

Set of natural numbers

Set of rational numbers

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}\ ,\ f_{yy}\ and\ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D = 0$ then ----------

f has relative maximum at (x_0, y_0)

f has relative minimum at (x_0, y_0)

f has saddle point at (x_0, y_0)

No conclusion can be drawn.

If $R = \{(x, y) / 0 \le x \le 2 \, and \, 0 \le y \le 3\}, then$

$\iint_R (1 - ye^{xy}) dA =$

$\int_0^2 \int_0^3 (1 - ye^{xy}) dy dx$

$\int_0^2 \int_0^3 (1 - ye^{xy}) dx dy$

$\int_2^3 \int_0^0 (1 - ye^{xy}) dx dy$

$\int_0^2 \int_2^3 (4xe^{2y}) dy dx$

Suppose $f(x, y) = 2xy \; where \; x = t^2 + 1 \; and \; y = 3 - t$. Which one of the following is true?

$\frac{df}{dt} = 6t - 4t^2 - 2$

$\frac{df}{dt} = 6t - 2$

$\frac{df}{dt} = 4t^3 + 6t - 6$

$\frac{df}{dt} = -6t^2 + 12t - 2$

Let i, j and k be unit vectors in the direction of x-axis, y-axis and z-axis respectively.

Suppose that $\vec{a} = 2i + 5j - k$. What is the magnitude of vector $\vec{a}$?

6
30
$\sqrt{30}$
$\sqrt{28}$

A straight line is ---------- geometric figure.
One-dimensional
Two-dimensional
Three-dimensional
Dimensionless

$If\ R = \{(x, y) / 0 \le x \le 2\ and\ 1 \le y \le 4\}, then$
$\iint_R (6x^2 + 4xy^3)dA =$

$\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$

$\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$

$\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$

$\int_2^4 \int_0^1 (6x^2 + 4xy^3)dxdy$

Which of the following formula can be used to find the Volume of a parallelepiped with adjacent edges formed by the vectors $\vec{a}$, $\vec{b}$ *and* $\vec{c}$?

$\left|\vec{a} \times \left(\vec{b} \times \vec{c}\right)\right|$

$\left|\vec{a} \bullet \left(\vec{b} \bullet \vec{c}\right)\right|$

$\left|\vec{a} \bullet \left(\vec{b} \times \vec{c}\right)\right|$

$\left|\vec{a} \times \left(\vec{b} \bullet \vec{c}\right)\right|$

The function $f(x,\ y) = \sqrt{y - x}$ is continuous in the region --------- and discontinuous elsewhere.

$x \neq y$

$x \leq y$

$x > y$

What is the relation between the direction of gradient at any point on the surface to the tangent plane at that point ?

parallel

perpendicular

opposite direction

No relation between them.

Suppose $f(x,\ y) = x^3 e^{xy}$. Which one of the statements is correct?

$\frac{\partial f}{\partial y} = 3x^3 e^{xy}$

$\frac{\partial f}{\partial y} = x^3 e^{xy}$

$\frac{\partial f}{\partial y} = x^4 e^{xy}$

$\frac{\partial f}{\partial y} = x^3 y e^{xy}$

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

perpendicular

parallel

in opposite direction

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}\ ,\ f_{yy}\ and\ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

Relative maximum at (x_0, y_0)

Relative minimum at (x_0, y_0)

Saddle point at (x_0, y_0)

No conclusion can be drawn.

If $R = \{(x, y) / 0 \le x \le 2 \, and \, -1 \le y \le 1\}, then$

$\iint_R (x + 2y^2) dA =$

$\int_{-1}^{1} \int_{0}^{2} (x + 2y^2) dy dx$

$\int_{0}^{2} \int_{1}^{-1} (x + 2y^2) dx dy$

$\int_{-1}^{1} \int_{0}^{2} (x + 2y^2) dx dy$

$\int_{1}^{2} \int_{-1}^{0} (x + 2y^2) dx dy$

$$f(x, y, z) = \frac{x^2 y}{z} + xyz$$

If

then what is the value of $f(1, 1, 1)$?

$f(1, 1, 1) = 1$

$f(1, 1, 1) = 2$

$f(1, 1, 1) = 3$

$f(1, 1, 1) = 4$

If $R = \{(x, y) / 0 \le x \le 4 \, and \, 0 \le y \le 9\}, then$

$\iint_R (3x - 4x\sqrt{xy})dA =$

$\int_0^9 \int_0^4 (3x - 4x\sqrt{xy})dydx$

$\int_0^4 \int_4^9 (3x - 4x\sqrt{xy})dxdy$

$\int_4^9 \int_0^0 (3x - 4x\sqrt{xy})dxdy$

$\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$

Let $f(x, y) = 2 + x^2 + \frac{y^2}{4}$

Q- *Find the gradient of f*

2MARKS

Q - Let the function $f(x, y)$ is continuous in the region R, where R is a rectangle as shown below.

complete the following equation

$$\iint_R f(x,\ y)\ dA = \int\ \int\ f(x,\ y)\ _____$$

2MARKS

Q.Find all critical points of the function

$f(x, y) = 4xy - x^3 - 2y^2$

$$\int_1^4 \int_0^2 (6x^2 + 4xy^3) dx\ dy$$

Evaluate

Q-Evaluate the following double integral.

$\iint \left(3 + 2x - 3y^2\right)\ dx\ dy$

3MARKS

$$y = \frac{1}{x^2}$$

Q- Let . If x changes from 3 to 3.3, find the approximate change in the value of y using differential dy.

3MARKS

MTH301-Calculus-II

Solved MCQS for Midterms papers

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3) Test phase + viva
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Every point in three dimensional space can be described by ------- coordinates

- **Three**

What are the direction cosines for the line joining the points (1, 3, 2) and (7, -2, 3)?

- $\frac{6}{\sqrt{62}}, \frac{-5}{\sqrt{62}} and \frac{1}{\sqrt{62}}$

The angles which a line makes with positive x ,y and z-axis are known as ----------

Direction angles

- **Plane parallel to xz-plane**

Domain of the function f (x, y, z)= $\sqrt{x^2, y^2, z^2}$ **is**

- **Entire 3D-Space**

Suppose f (x, y) = xy− 2y where x2 = +3t 1 and y = 2t. Which one of the following is true?

- $\frac{df}{dt} = -16t - t$

For a function f (x, y, z) , the equation ∂∂22xf +∂∂22yf +∂∂22zf =0 is known as ---------- --

- **Laplace's Equation (Ali)**

Magnitude of vector $\vec{a}$ **is 2, magnitude of vectorb is 3 and angle between them when placed tail to tail is 45 degrees. What is** $\vec{a}.\vec{b}$ **?**

- **4.2**

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Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- **Radius of inscribed circle in R**

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are ---------

- **Perpendicular**

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

- **In opposite direction**

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

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- **In opposite direction**

Let the function f x y(,) has continuous second-order partial derivatives) (fxx , fyy and fxy) in some circle centered at a critical point (x0, y0) and let D = fxx(x0, y0) fyy(x0, y0) − fxy2 (x0, y0) If D > 0 and fxx(x0, y0)

- **Relative maximum at**

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Let the function f x y(,) has continuous second-order partial derivatives (fxx , fyy and fxy) in some circle centered at a critical point (x0, y0) and let D = fxx(x0, y0) fyy(x0, y0) − fxy2(x0, y0) If D = 0 then ----------

- **No conclusion can be drawn.**

Every real number corresponds to _____ on the co-ordinate line.

- **A unique point**

There is one-to-one correspondence between the set of points on co-ordinate line and _____.

- **Set of real numbers**

Which of the following is associated to each point of three dimensional spaces?

- **A natural Number**

All axes are positive in _____octant.

- **First**

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- **Radius of inscribed circle in R**

Two surfaces are said to be orthogonal at appoint of their intersection if their normals at that point are_____.

- **Perpendicular**

By Extreme Value Theorem, if a function f (x y,) is continuous on a closed and bounded set R, then f (x, y) has both _____ on R.

- **Absolute maximum and absolute minimum value**

Let the function f (x y,) has continuous second-order partial derivatives (fxx , fyy and f xy) in some circle centered at a critical point (x0, y0) and let D = fxx (x0, y0) fyy (x0, y0) − fxy 2 (x0, y0) if D > 0 and fxx (x0, y0) < 0 then f has _____ . Relative maximum at

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- **Relative maximum at**

Let the function f (x y,) has continuous second-order partial derivatives ¬ ¬ ¬ ¬ (fxx , fyy and f xy) in some circle centered at a critical point (x0, y0) and let D = fxx (x0, y0) fyy (x0, y0) − fxy 2 (x0, y0) if

if D = 0 then _____.

- **No conclusion can be drawn.**

Every real number corresponds to ---------- on the co-ordinate line

- **Infinite number of points**

The is one –to – one correspondence between the set of the points on co-ordinate and ---------

- **Set of natural number**

Which of the following is associated to each point of three dimensional spaces?

- **A natural number**

All axes are positive in-------octant

- **First**

The spherical co-ordinates of a point are $\left(\sqrt{3},\frac{\pi}{3},\frac{\pi}{2}\right)$ when are its cylindrical co-ordinates?

- $\left(\sqrt{3}\sin\frac{\pi}{3},\frac{\pi}{2},\sqrt{3}\cos\frac{\pi}{3}\right)$

Suppose f(x,y)= xy -2y^2 Where x=3t +1 and y=2t which one of the following is true ?

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- $\frac{df}{dt} = -4t + 2$

Let w= f(x ,y ,z) and x =g(r, s) , y=h(r ,s), z=t(r , s) then by chain rule

$\frac{\partial w}{\partial r} =$

- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial x}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial x}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$

Is the function f(x,y) continuous at origin ? if not ,why?

$$f(x,y) = \begin{cases} 0 & if\ x \geq 0\ and\ y \geq 0 \\ 1 & otherwise \end{cases}$$

- **F(x ,y) is not defined**

Is the function f(x,y) continuous at origin ? if not ,why?

$$f(x,y) = \begin{cases} 0 & if\ (x,y) \neq 0 \\ 1 & if\ (x,y) = 0 \end{cases}$$

- **F(x, y) is not defined**

Let R be a closed region in two dimensional spaces. What does the double integral over R calculate?

- **Radius of inscribed circle in r**

Every real number corresponds to _____ on the co-ordinate line.

- **Infinite number of points**

There is one-to-one correspondence between the set of points on co-ordinate line and _____.

- **Set of natural numbers**

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Which of the following is associated to each point of three dimensional spaces?

- **A natural Number**

All axes are positive in _____octant

- **First**

By Extreme Value Theorem, if a function f (x y,) is continuous on a closed and bounded set R, then f (x y,) has both _____ on R.

- **Absolute maximum and absolute minimum value**

Let x be any point on co-ordinate line. What does the inequality -3 < x < 1 means?

- **The set of all real numbers between -3 and 1**

Which of the following number is associated to each point on a co-ordinate line?

- **A real number**

Which of the following set is the union of set of all rational and irrational numbers?

- **Set of real numbers**

---------- planes intersect at right angle to form three dimensional space.

- **Three**

each point of domain, the function ----------

- **Is defined**

Suppose f(x ,y)=xe^{3xy} Which one of the following is correct?

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- $\frac{\partial f}{\partial x} = 3x^2 e^{xy} + x^3 y e^{xy}$

Suppose f(x ,y)=xe^{3xy} Which one of the following is correct?

- $\frac{\partial f}{\partial y} = x^4 exy$

Suppose f (x, y) = 2xy where x = t 2 +1 and y = 3−t which one of the following is true?

- $\frac{\partial f}{\partial t} = -6t^2 + 12t - 2$

I, j and k are unit vectors in the direction of x-axis, y-axis and z-axis respectively. Suppose that → a = 2i +5 j –k

- $\sqrt{30}$

Which of the following are direction ratios for the line joining the points (1, 3, 5) and (2, −1, 4)?

- **1, -4 and -1**

Which of the following is geometrical representation of the equation y=x^2, in three dimensional space?

- **Parabola**

Which of the following is the interval notation of real line?

- **(-∞ , +∞)**

What is the general equation of parabola whose axis of symmetry is parallel to yaxis?

- **y = ax^2 +bx +c (a ≠ 0)**

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Which of the following is geometrical representation of the equation y=4, in three dimensional space?

- **A point on y-axis**

Plane is an example of ----------

- **Surface**

Function $f(x,y)=\sqrt{y-x}$ is continuous in the region --------- and discontinuous elsewhere.

- **X ≤ y**

At critical points function can assume

- **Zero value**

A plane can be perfectly determined by

- **One point and normal vector**

Intersection of two straight lines is ----------

- **Point**

Plane is a ---------- surface.

- **Two-dimensional**

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t)=\sin^2 t\,\hat{i}+(1-\cos 2t)\,\hat{j}$.

- **X=sin^{2}t ,y=1-cos2t ,z=0**

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What are the parametric equations that correspond to the following vector equation?

$$r(t) = (2t-1)\hat{i} - 3\sqrt{t}\,\hat{j} + \sin 3t\,\hat{k}$$

- **X=2t-1, y= $-3\sqrt{t}$, Z=sin3t**

What is the derivative of following vector-valued function?

$\vec{r}(t) = (\cos 5t\,, \tan t\,, 6\sin t)$

- $\vec{r}(t) = (-5\sin 5t\,, \sec^2 t\,, 6\cos t)$

The following differential is exact $dz = (x^2 y + y)dx - xdy$

- **False**

Which one of the following is correct Wallis Sine formula when n is even and n ≥ 2 ?

- $\int_0^{\frac{\pi}{2}} \sin^n x dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)} - - - - - - \frac{5}{6}\frac{3}{4}\frac{1}{2}$

Match the following equation in polar co-ordinates with its graph. r a where a is an arbitrary cons t

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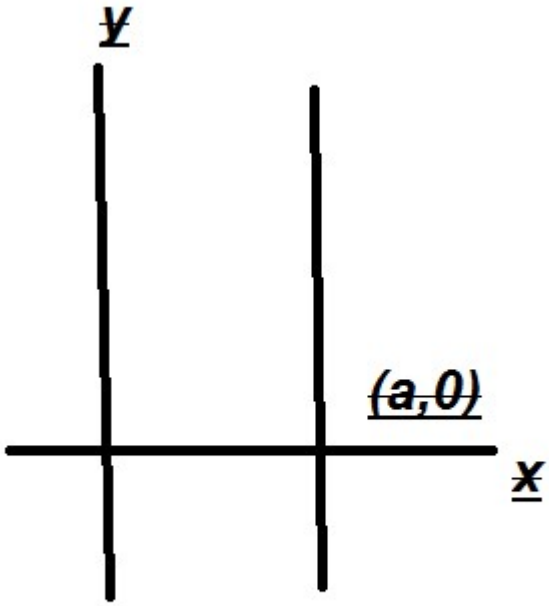

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by (r, π−θ) then the curve is said to be symmetric about which of the following?

- **Y-axis**

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by (-r, θ) then the curve is said to be symmetric about which of the following?

- **Pole**

What is the amplitude of a periodic function defined by f(x)= $\sin 3x$

- **0**

What is the period of a periodic function defined by $f(x) = 4\cos 3x$ **?**

- $\frac{2\pi}{3}$

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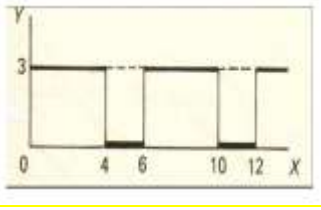
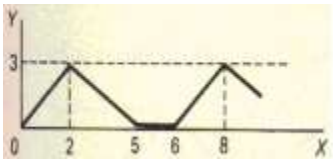
Match the following periodic function with its graph.

$$f(x)=\begin{cases}3 & 0<x<4\\ 0 & 4<x<6\end{cases}$$

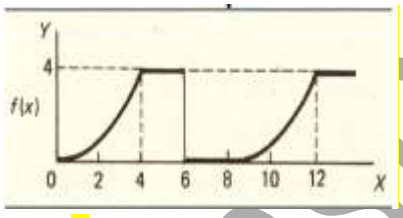
- 

What is the period of periodic function whose graph is as below?



6

What is the period of periodic function whose graph is as below?



- **8**

Let L denotes the Laplace Transform. If L { F(t)} = f (s) where s is a constant. And

$\lim_{x\to\infty}\left(\frac{f(t)}{t}\right)$ **Exists then which of the following equation holds?**

- $L\left(\frac{f(t)}{t}\right)=\int_{s}^{\infty} f(s)ds$

Which of the following is Laplace inverse transform of the function f(s) defined by $f(s)=\frac{3}{s-2}-\frac{2}{s}$ **?**

- $3e^{2t}-2$

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Let(x_1,y_1,z_1) and (x_2,y_2,z_2) be any two points in three dimensional space. What does the formula $\sqrt{(x_2-x_1)+(y_2-y_1)+(z_2-z_1)}$ calculates?

- **Midpoint of the line joining these two points**

Let the functions P (x, y) and Q (x, y) are finite and continuous inside and at the boundary of a closed curve C in the xy-plane. If (P dx +Q dy) is an exact differential then $\int_c^{\infty}(P\,dx+Q\,dy)=$

- **Zero**

What is Laplace transform of the function F (t) if F (t) = t?

- $L\{t\}=\frac{1}{s}$

What is the value of $L\{e^{5x}\}$ if L denotes Laplace transform?

- $L\{e^{5x}\}=\frac{1}{s-5}$

Evaluate the line integral $\int_c(3x+2y)dx+(2x-y)$ the line segment from (0, 0) to (0, 2).

- **2**

Evaluate the line integral $\int_c(2x+y)dx+(x^2-y)dy$ where C is the line segment from (0, 0) to (2, 0).

- **0**

Which of the following are direction ratios for the line joining the points (1, 3, 5) and (2, −1, 4)?

2, -3 and 20

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$if\ R = \{(X,Y)/0 \le X \le 2\ and\ 1 \le y \le 4\}, then \iint_R (6x^2 + 4xy^3)dx =$

- $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$

---------- Planes intersect at right angle to form three dimensional spaces.

- **Three**

If the positive direction of x, y axes are known then ---------- the positive direction of z-axis.

- **Left hand rule tells**

What is the distance between points (3, 2, 4) and (6, 10, -1)?

- $\frac{7}{\sqrt{2}}$

The equation ax +by+ cz + d = 0, where a, b, c, d are real numbers, is the general equation of which of the following?

- **Plane page**

Domain of the function $f(x,y) = \sqrt{y - x^2}$ *is*

- $y \ge x^2$

What is the period of a periodic function defined by f(x)=sinx/2?

- **4π**

The graph of an odd function is symmetrical about ----------

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- **Origin**

The path of integration of a line integral must be ----------

- **Continuous and single-valued**

Sign of line integral is reversed when ----------

- **Direction of path of integration is reversed.**

What is Laplace transform of the function f(t) if f(t) if f(t)=sin3t

- $l\{\sin 3t\} = \frac{3}{s^2+9}$

π is an example of ----------

- **Irrational numbers**

Straight line is a special kind of ----------

- **Curve**

An ordered triple corresponds to ---------- in three dimensional spaces.

- **A unique point**

The angles which a line makes with positive x ,y and z-axis are known as ----------

- **Direction angles**

Is the function f (x, y) continuous at origin? If not, why? $f(x,y) = 4xy + \sin 3x^2 y$

- **F (x, y) is continuous at origin**

Two vectors are opposite vectors if they have _____.

- **both (a) and (c)**

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If two dogs are pulling a bone with force=20 Newtons in opposite direction, the resultant force is _____

- **Zero Newton**

If the velocity (VI) of a car is 20m/s and velocity (V2) of bike is 40m/s, where both velocities are southward then which of the following would be the dot product of these velocities?

- **-800**

Limit of a function may exist at the point where it not defined

- **False**

Limit of a function may exist at the point where it not defined

- **False**

A_____of continuous functions is continuous

- **All (a), (b) and (c)**

A quotient of continuous function is continuous, expect where the denominator is _____

- **Zero**

Intersection of two surfaces is a _____ in three dimensional space Curve

- **Face the parabola (y=ax^2 + bx + c) is opening downward when A<0**

Which coordinate system uses two distances and one angle?

- **Cylindrical**

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Which of the following is a two dimensional coordinate system

- **Polar coordinate system**

Limit of a function exist may exist at the point where it not defined

- **False**

The number square root of 2 is

- **None of these /1.41**

A plane z=3 is _____ to XY plane.

- **Parallel**

Value of the function f(x,y,z)=xyz at point (2,2,2) is

- **8**

The number 3/11 is

In rectangular coordinate system the point (-3,2) is in

- **Second quadrant**

Which of the following function is not the function of one variable?

- **m=g(s) + content**

XZ- Plane consists of all points

- **where X=0**

Value of the function f(x,y,z)=x^2+ y^2 + z^ 2 at the point (0,1,0) is

- **1**

If w=f(x,y,z) where x=g(t), z=h(T) then

- **Dw/dt=@wdz/@xdt + @wdy/@ydt + @wdz/@zdt**

If f and its partial derivatives of the first, second and third orders are continuous on an open set, then at each point of the set_____

- **$@^2f/@y^2@x = @^2f/@y@x@y = @^2f/@x@y^2$**

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All axis are negative in ---------- octant

- **Eight**

In three dimensional space, intersection of two surfaces is a ----------

- **Plane**

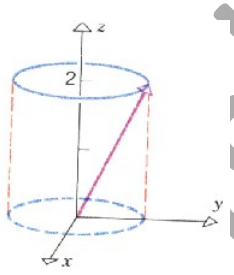
$$\frac{\partial^2 f}{\partial^2 x}+\frac{\partial^2 f}{\partial^2 y}+\frac{\partial^2 f}{\partial^2 z}=0$$

For a function $f(x,y,z)$, the equation is known as ----------

- **Laplace's Equation**

Match the following vector-valued function with its graph.

$r(t)=\cos t\,\hat{i}+\sin t\,\hat{j}+2\hat{k}$ **and** $0\le t\le 2\pi$

- 

What is the derivative of following vector-valued function?

- $\vec{r}'(t)=\left(4t^3,\ \frac{1}{2\sqrt{t+1}},\ \frac{-6}{t^3}\right)$

What is the derivative of following vector-valued function?

$\vec{r}(t)=\left(e^{t^2},\ t^2,\ \sec 2t\right)$

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- $\vec{r'}(t) = \left(2te^{t^2},\ 2t\ ,\ 2\sec 2t \tan 2t\right)$

The following differential is exact $dz = (x^2 + y^2)\ dx - 2xy\ dy$

- **False**

Which one of the following is correct Wallis Sine formula when n **is even and** $n \geq 2$ **?**

- $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)} \text{-------} \frac{5}{6}\frac{3}{4}\frac{1}{2}$

If $a > 0$**, then the equation, in polar co-ordinates, of the form** $r^2 = a^2 \cos 2\theta$ **represent which of the following family of curves?**

- **Leminscate**

Which of the following condition must be satisfied for a vector field $\vec{F}$ **to be a conservative vector field?**

- $\vec{F} = 0$

If $\lim_{(x,y,z)\to(x0,y0,z0)}$ **f(x,y,z) =L1 then**

- $\lim_{(x,y,z)\to(x0,y0,z0)}$ **cf(x,y,z) = cL1 If c is constant**

if $\lim_{(x,y)\to(x0,y0)}$ **F(x,y) = L1** $\lim_{(x,y)\to(x0,y0)}$ **g (x,y)= L2**

- $\lim_{(x,y)\to(x0,y0)}$ **f(x,y)/g(x,y) = L1/L2 ; L2 not equal to 0**

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The partial derivative with respect to x of $f(x,y) = 3x^2y+2y^2$ is _____

- **$6xy + 2y^2$**

Spherical coordinate @ is related to the cylindrical coordinate as_____

- **@=@**

How many real numbers exist between 1 and 5

- **Infinity**

Value of the function f(x,y,z)= square root $x^2+y^2+z^2$ at the point (0,3,0) is_____

- **3**

The number pi is.

- **Irrational**

Length of magnitude of a unit vector is _____

- **1**

The product of continuous

- **functions is continuous True**

F(x,y,z)= square root $x^2+y^2+z^2$

- **[0, +infinity**

The product of continuous

- **function is continuous True**

Range of the function $y=3x^2-2$ is

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- **All real numbers with y ≥ -2**

if the partial derivative (w,r,t 'x' or 'y') at any point on the surface "z=f(x,y) is zero, than tangent plane at that point will be _____ to XY- plane

- **parallel**

Two vectors A and B are _____ if and only if their scalar

- **is equal to zero. Perpendicular**

The partial derivative with respect to x of $f(x,y)= 3x^2y + 2y^2$ is _____

- **6xy**

The chain rule can be taken as the derivative of the order function and multiplying it items the derivative of the _____

- **The inner function**

Two vectors are opposite vectors if they have_____

- **Both (A) and (c)**

Magnitude of the force vector F = -30 Newton, is _____

- **ewton**

Two vectors A and B are_____ if and only if their social product is equal to zero

- **Perpendicular**

Which of the following should be the scalar multiple (A) such that force ; F_1=20N and F_2 = -60N satisfy the

- **A=-1/3**

A zero vector has magnitude equal to zero and _____

- **Direction Arbitrary**

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The chain rule states that the derivative of f(g(x)) is _____

- **f(g(x))-g(x)**

The order of differentiation in an nth order partial derivative can be change without affecting the final result whenever the function and all its partial derivatives of order less than or equal to 'n' are_____

- **Continuous**

For which of the following values of 'y' the partial derivative of z=2/y(x-1) EXIST

- **R-{ 1,0}**

To find the limit of $f(x,y) = x^2+y/x^2$ at (0,0) if we approach (0,0) through the line x-axis, we get the value

- **1**

Value of money is an example of _____

- **Quantity Scalar**

IF y = sin t and $t= e^{xu}$ then which of the following is partial derivative of w.r.t.x

- **ut cost**

The function z=F(x,y) represents a _____ in

- **space Line**

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If two dogs are pulling with force=20Newtons in opposite direction, then the resultant force is_____

- **Zero Newton**

Limit of a function may exist at the point where it not

- **defined False**

A_____ of continuous function is continuous

- **All options**

The function e^{xy} is continous

In whole region of xy plane

A quotient of continuous function is continuous, expect where the denominator is _____

- **Zero**

if the partial derivative (w,r,t 'x' or 'y') at any point on the surface "z=f(x,y) is zero, than tangent plane at that point will be parallel to _____

- **XY plane**

Two determine the position of any object in a room we need how many coordinates.

- **3(3 dimension)**

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Which of the following is a two dimensional coordinate system

- **Cylindrical coordinate system**

Which of the following is the representation of spherical coordinates

- **(p,θ, φ)**

In cartesian coordinate system all axes are at _____degree to each other

- **90**

intersection of two surfaces is a ____ in three dimensional space

- **Curve**

The function $f(x,y,z) = (x^2 +yz)/$

- **(xy+z) Xy+z=0**

For which of the following values of 'y' the partial derivative of z=2/y(x-1)

- **R-{1,0}**

YZ- plane of all points where

- **X=0**

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Domain of the following function y= x/ 3x-4

- **All real number**

Cylindrical coordinate system is used to the position of any

- **object is Any dimension**

To express the location of any object on earth which coordinate system is used

- **Cylindrica coordinate system**

Face of the parabola (y-ax^2 + bx+ c) is opening downward

- **when A<0**

Cylindrical coordinate "y" is related to the Cartesian coordinate as

- **R= cos0**

Which coordinate system uses two distances and one angle?

- **Rectangular**

Which of the following function is not the function of one variable y=

- **f{x,y}**

Which of the following is a two dimensional coordinate system

- **Cylindrical coordinate system**

The chain rule can be taken as the derivative of the outer function and multiplying it times the derivative of the _____

- **The inner function**

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The rule is called the chain rule because we use it to take derivatives of

____ by changing together their derivative

- **Composite function**

The partial derivative at any point on the surface represents the slope of tangent _____ at that point

- **Line**

Partial derivative f(x,y) = x^2+y^2+xy with respect to y is

- **None options**

Spherical coordinate 0 is related to the cylindrical coordinate as _____

- **@=@**

Which of the following is a two dimensional coordinate

- **system Polar coordinate system**

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Which coordinate system uses two distance and one angle?

- **Cylindrical**

A plane z=3 is _____ to XY-plane

- **Parallel**

Which of the following function is not the function of one variable

- **M=g(s)+ constant**

XZ-plane consists of all points where

- **x=0**

A zero vector has magnitude equal to zero and _____

- **direction No**

Which condition is necessary for a function to be continuous at any point (x,y)

- **All options**

If we = x + y while x=r + sandy = r-s then which of the following is partial derivative of w is w.r.tr?

- **1**

If g and h are continuous functions of one varialble, then f(x,y) = g(x) h(y) is a _____ function of x and y.

- **Continuous**

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The function of the form $f(x,y) = 3x^2x^5$ is _____ in the domain

- **continues everywhere**

If the velocity (v1) of a car is 20m/s eastward and velocity (v2) of bike is 40 m/s northward then which of the following would be the dot product of these velocities

- **-800**

Suppose that z=(x,y) is a function of two variable x and y, then partial derivative of function z with respect to x represents the _____ of the tangent of the curve

- **None OPTIONS**

The product of continuous functions is continuous

- **True**

The function e^{xy} is continuous In whole region of

- **xy plane**

The partial derivative of $f(x,y)=x^2y^{-2}$ with respect to x is _____

- **$x2_e$-y**

Let $f(x,y) = e^{2y}\cos x$ then $f_x(x,y)=$

- **e^{2y} sinx**

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If y=f(x) and x=g(t) then by chain rule dy/dt=

- **Dy/dx*dx/dt**

The partial derivative with respect to y of f(x,y) = x^2+2y^2+xy is _____

- **4y+x**

The mixed 2nd order partial derivatives of z=xy is equal to

xy

Scalar Triple Product of vectors A=[1,0,0], B=[0,1,0] and C=[0,0,1] is---------

- **1(not sure)**

he order of differentiation in an nth order partial derivative can be change without affecting the final result whenever the function and all its partial derivatives of order less than or equal to 'n' are ----------.

- **Differentiable (not sure)**

Let f(x,y)be a function with continuous second order partial derivatives in some circle centered at a critical point(x0,y0) and,

letD=fxx(x0,y0)fyy(x0,y0)−[fxy(x0,y0)]2IfD=0,then−−−−−−−−

- **f(x,y)has a saddle point at (x0,y0). (not sure)**

Geometrically the magnitude of Scalar Triple Product gives the----------.

- **Volume of Parallelepiped**

Value of money is an example of -------- quantity

- **Scalar**

Let f(x,y)be a function with continuous second order partial derivatives in some circle centered at a critical point(x0,y0) and,

LetD=fxx(x0,y0)fyy(x0,y0)−[fxy(x0,y0)]2. IfD>0andfxx(x0,y0)>0,then −−−−−

- f(x,y)has a saddle point at (x0,y0). (not sure)

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Two vectors of magnitudes [4,5] making an angle of 30degree, then the area of parallelogram associated with these two vectors is ---------

- Square units

For the function z=f(x, y, z),the total differentialdzis equal to

- dw = df = fx(x, y, z)dx +fy(x, y, z)dy +fz(x, y, z)dz

For the function z=f(x, y, z),the total differentialdzis equal to

- dz = df = fx(x, y)dx + fy(x, y)dy

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Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

An ordered pair corresponds to ---------- on the plane

Answer (Please select your correct option)

- A unique point
- A point in each quadrant
- Two points
- Infinite number of points

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Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

Plane is a special type of ----------.

Answer (Please select your correct option)

- Curve
- Surface
- Sphere
- Cone

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Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

There is one-to-one correspondence between the set of points on a co-ordinate line and ----------

Answer (Please select your correct option)

- Set of real numbers

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- Set of integers
- Set of natural numbers
- Set of rational numbers

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Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

What is the general equation of parabola whose axis of symmetry is parallel to y-axis?

Answer (Please select your correct option)

- $y = ax^2 + b \quad (a \neq 0)$
- $x = ay^2 + b \quad (a \neq 0)$
- $y = ax^2 + bx + c \quad (a \neq 0)$
- $x = ay^2 + by + c \quad (a \neq 0)$

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Question No : 5 of 26 Marks: 1 (Budgeted Time 1 Min)

If the spherical coordinates of a point are $\left(2, \frac{\pi}{4}, 0\right)$, then z-coordinate in rectangular coordinate system is . . . . . .

Answer (Please select your correct option)

- 0
- -2
- 2
- $\sqrt{2}$

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Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

The function $z = \frac{1}{\sqrt{x+y}}$ is discontinuous at origin because at the point (0,0), it . . . . . .

Answer (Please select your correct option)

- has a jump
- has a hole
- approaches towards infinity
- approaches towards zero

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Question No : 7 of 26 Marks: 1 (Budgeted Time 1 Min)

The function $f(x, y) = \sqrt{y - x}$ is continuous in the region --------- and discontinuous elsewhere.

Answer (Please select your correct option)

- $x \geq y$
- $x \neq y$
- $x \leq y$
- $x > y$

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Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

Let $w = f(x, y, z)$ and $x = g(r, s)$, $y = h(r, s)$, $z = t(r, s)$ then by chain rule $\frac{\partial w}{\partial r} = \ldots\ldots\ldots$

Answer (Please select your correct option)

- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

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Question No : 9 of 26 Marks: 1 (Budgeted Time 1 Min)

Every differentiable function is always . . . . . .

Answer (Please select your correct option)

- Piece wise continuous
- Discontinuous
- Continuous

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Question No : 10 of 26 Marks: 1 (Budgeted Time 1 Min)

The vector $\vec{a} \times \vec{b}$ is . . . . . to both $\vec{a}$ and $\vec{b}$.

Answer (Please select your correct option)

- Opposite
- Equal
- Parallel
- Orthogonal

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Question No : 11 of 26 Marks: 1 (Budgeted Time 1 Min)

Gradient of a scalar function always results in a . . . . . function.

Answer (Please select your correct option)

- Scalar
- Continuous
- Vector
- Constant

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Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

Tthe direction of gradient at any point on the surface is . . . . . to the tangent plane at that point.

Answer (Please select your correct option)

- parallel
- perpendicular
- opposite direction
- None of these.

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Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

$f_x(p_o)(x-x_o)+f_y(p_o)(y-y_o)+f_z(p_o)(z-z_o)=0$ is the equation of . . . . . .

Answer (Please select your correct option)

- Normal plane
- Point slope form of line
- Gradient of function $f(x,y,z)$
- Tangent plane

correct

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Question No : 14 of 26 Marks: 1 (Budgeted Time 1 Min)

For a function $z = f(x, y)$, the total differential is defined as . . . . . .

Answer (Please select your correct option)

- $dz = f_x(x,y)dx - f_y(x,y)dy$
- $dz = f_x(x,y)dx + f_y(x,y)dy$
- $dz = f_x(x,y)dy - f_y(x,y)dx$
- $dz = f_x(x,y)dy + f_y(x,y)dx$

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Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

For a function $f(x,y)$ to have both absolute maximum and minimum, it must be Continuous on . . . . . set R.

Answer (Please select your correct option)

- a closed and bounded

correct

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- an open and bounded
- a closed and unbounded
- an open and unbounded

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Question No : 16 of 26 Marks: 1 (Budgeted Time 1 Min)

The volume of parallelepiped with dimensions x, y, z is . . . . . .

Answer (Please select your correct option)

- $V = x^2y^2z^2$
- $V = x + y + z$
- $V = \sqrt{xyz}$
- $V = xyz$

correct

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Question No : 17 of 26 Marks: 1 (Budgeted Time 1 Min)

Let x be the length, width and height of a cube. The area of bottom will be . . . . . .

Answer (Please select your correct option)

- $A = x^2$
- $A = x^3$
- $A = xz$
- $A = x + x^2$

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Question No : 18 of 26 Marks: 1 (Budgeted Time 1 Min)

A double integral and iterated integral become identical provided that the integrand is . . . . . over the given rectangular region.

Answer (Please select your correct option)

- Bounded

correct

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- Discontinuous
- Continuous
- Defined

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Question No : 19 of 26 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x, y) : 2 \le x \le 4 \text{ and } 0 \le y \le 1\}$, then $\iint_R (4xe^{2y})dA = \ldots\ldots\ldots\ldots$

Answer (Please select your correct option)

- $\int_0^1 \int_2^4 (4xe^{2y})dydx$
- $\int_0^1 \int_2^4 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dydx$

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Question No : 20 of 26 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x,y) : 0 \le x \le 2 \text{ and } 0 \le y \le 3\}$, then $\iint_R (1 - ye^{xy})dA = \ldots\ldots\ldots$

Answer (Please select your correct option)

- $\int_0^2 \int_0^3 (1 - ye^{xy})dydx$
- $\int_0^2 \int_0^3 (1 - ye^{xy})dxdy$
- $\int_2^3 \int_0^0 (1 - ye^{xy})dxdy$
- $\int_0^2 \int_2^3 (4xe^{2y})dydx$

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Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

Which of the following is geometrical representation of set of real numbers?

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Answer (Please select your correct option)

- Co-ordinate line
- xy-plane
- Sphere
- Circular cylinder

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Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

There is one-to-one correspondence between the set of points on a co-ordinate line and ----------

Answer (Please select your correct option)

- Set of real numbers

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- Set of integers
- Set of natural numbers
- Set of rational numbers

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Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

An ordered triple corresponds to ---------- in a three dimensional space.

Answer (Please select your correct option)

- A unique point
- A point in each octant
- Three points
- Infinite number of points

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Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

If the positive directions of x and y axes are known then ---------- the positive direction of z-axis.

Answer (Please select your correct option)

- Horizontal rightward direction is
- Horizontal leftward direction is
- Left hand rule tells
- Right hand rule tells

correct not shoure

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Question No : 5 of 26 Marks: 1 (Budgeted Time 1 Min)

If a function is not defined at some point, then its limit ---------- exist at that point.

Answer (Please select your correct option)

- Always
- Never
- May

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Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

A composition of continuous functions . . . . . .

Answer (Please select your correct option)

- is always continuous

correct

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- may or may not be continuous
- is discontinuous
- is piece-wise continuous

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Question No : 7 of 26 Marks: 1 (Budgeted Time 1 Min)

According to the Euler's theorem, the order of partial differentiation can be changed, provided the function and all of its partial derivatives are. . . . . .

Answer (Please select your correct option)

- Piece-wise continuous
- Continuous
- Discontinuous

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Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

If $x = f(r,s)$ and $r = g(t)$, $s = h(t)$, then derivative of x with respect to t is written as. . . . . .

Answer (Please select your correct option)

- $\frac{\partial x}{\partial t}$
- (x,t)
- $\frac{\partial^2 x}{\partial t^2}$
- $\frac{dx}{dt}$

correct

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Question No : 9 of 26 Marks: 1 (Budgeted Time 1 Min)

A vector in a plane is always represented by a . . . . . .

Answer (Please select your correct option)

- Line
- Curve
- Line segment
- None of these.

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Question No : 10 of 26 Marks: 1 (Budgeted Time 1 Min)

If two vectors $\vec{a}$ and $\vec{b}$ have the relation $\vec{a} = \lambda\vec{b}$ where λ is a none-zero scalar then $\vec{a}$ and $\vec{b}$ are . . . . . to each other.

Answer (Please select your correct option)

- Orthogonal
- Non-parallel
- Parallel
- None of these

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Question No : 11 of 26 Marks: 1 (Budgeted Time 1 Min)

The function decreases most rapidly in the direction of . . . . . .

Answer (Please select your correct option)

- $-\nabla f$
- $-|\nabla f|$
- $\nabla f \times \hat{u}$
- $|\nabla f|$

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Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

Tthe direction of gradient at any point on the surface is . . . . . to the tangent plane at that point.

Answer (Please select your correct option)

- parallel
- perpendicular
- opposite direction
- None of these.

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Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

$2x^2 + y^2 = 4$ is the . . . . . form of equation of a curve.

Answer (Please select your correct option)

- Parametric
- Implicit
- Symmetric
- Explicit

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Question No : 14 of 26 Marks: 1 (Budgeted Time 1 Min)

If $f(x,y)$ has a relative extremum at a point (x_0,y_0) and both the first partial derivatives of f exist at this point, then . . . . . .

Answer (Please select your correct option)

- $f_x(x_0,y_0)=0$ and $f_y(x_0,y_0)\neq 0$
- $f_x(x_0,y_0)\leq 0$ and $f_y(x_0,y_0)\leq 0$
- $f_x(x_0,y_0)\geq 0$ and $f_y(x_0,y_0)\geq 0$
- $f_x(x_0,y_0)=0$ and $f_y(x_0,y_0)=0$

correct

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Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

For a function $f(x)$, if $f'(x)$ is equal to some non-zero constant, then $f(x)$ will have . . . . . .

Answer (Please select your correct option)

- At least one critical point

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- No critical point
- Non-zero critical points
- None of these

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Question No : 16 of 26 Marks: 1 (Budgeted Time 1 Min)

Let x, y, z be the length, width and height of an open rectangular box. The surface area of the box will be . . . . . .

Answer (Please select your correct option)

- $A = xy + 2yz + 2xz$

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- $A = yz + 4$
- $A = xz + yz + zx$
- $A = xyz$

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Question No : 17 of 26 Marks: 1 (Budgeted Time 1 Min)

Double integral of a function $f(x,y)$ represents . . . . . of the region between the surface defined by the function and the plane which contains its domain.

Answer (Please select your correct option)

- Perimeter
- Volume
- Area
- Circumference

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Question No : 18 of 26 Marks: 1 (Budgeted Time 1 Min)

For the double integral $\int_c^d \int_a^b f(x,y)dxdy$, order of integration does not matter provided that $f(x,y)$ is . . . . . .

Answer (Please select your correct option)

- Bounded
- Discontinuous
- Defined
- Continuous

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Question No : 19 of 26 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x,y) : 0 \le x \le 2 and -1 \le y \le 1\}$, then $\iint_R (x+2y^2)dA = \ldots\ldots\ldots\ldots\ldots$

Answer (Please select your correct option)

- $\int_{-1}^{1}\int_{0}^{2} (x+2y^2)dydx$
- $\int_{0}^{2}\int_{1}^{-1} (x+2y^2)dxdy$
- $\int_{-1}^{1}\int_{0}^{2} (x+2y^2)dxdy$
- $\int_{1}^{2}\int_{-1}^{0} (x+2y^2)dxdy$

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Question No : 20 of 26 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x,y) : 0 \le x \le 4 \text{ and } 0 \le y \le 9\}$, then $\iint_R (3x - 4x\sqrt{xy})dA = \ldots\ldots\ldots\ldots$

Answer (Please select your correct option)

- $\int_0^9 \int_0^4 (3x - 4x\sqrt{xy})dydx$
- $\int_0^4 \int_4^9 (3x - 4x\sqrt{xy})dxdy$
- $\int_4^9 \int_0^0 (3x - 4x\sqrt{xy})dxdy$
- $\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$

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MTH301 Short Notes Questions with Answer

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Question: What does it mean by the preservation of edge end point function in the definition of isomorphism of graphs?

Answer: Since you know that we are looking for two functions (Suppose one function is “f” and other function is “g”) which preserve the edge end point function and this preservation means that if we have vi as an end point of the edge ej then f(vi) must be an end point of the edge g(ej) and also the converse that is if f(vi) be an end point of the edge g(ej) then we must have vi as an end point of the edge ej. Note that vi and ej are the vertex and edge of one graph respectively where as f (vi) and g (ej) are the vertex and edge in the other graph respectively.

Question: Is there any method of identifying that the given graphs are isomorphic or not?(With out finding out two functions).

Answer: Unfortunately there is no such method which will identify whether the given graphs are isomorphic or not. In order to find out whether the two given graphs are isomorphic first we have to find out all the bijective mappings from the set vertices of one graph to the set of vertices of the other graph then find out all the bijective functions from the set of edges of one graph to the set of edges of the other graph. Then see which mappings preserve the edge end point function as defined in the definition of Isomorphism of graphs. But it is easy to identify that the two graphs are not isomorphic. First of all note that if there is any Isomorphic Invariant not satisfied by both the graphs, then we will say that the graphs are not Isomorphic. Note that if all the isomorphic Invariants are satisfied by two graphs then we can’t conclude that the graphs are isomorphic. In order to prove that the graphs are isomorphic we have to find out two functions which satisfied the condition as defined in the definition of Isomorphism of graphs.

Question: What is the application of isomorphism in real word?

Answer: There are many applications of the graph theory in computer Science as well as in the Practical life; some of them are given below. (1) Now you also go through the puzzles like that we have to go through these points without lifting the pencil and without repeating our path. These puzzles can be solved by the Euler and Hamiltonian circuits. (2) Graph theory as well as Trees has applications in “DATA STRUCTURE" in which you will use trees, especially binary trees in manipulating the data in your programs. Also there is a common application of the trees is "FAMILY TREE”. In which we represent a family using the trees. (3) Another example of the directed Graph is "The World Wide Web ". The files are the vertices. A link from one file to another is a directed edge (or arc). These are the few examples.

Question: Are Isomorphic graphs are reflexive, symmetric and transitive?

Answer: We always talk about " RELXIVITY"" SYMMETRIC" and TRANSIVITY of a relation. We never say that a graph is reflexive, symmetric or transitive. But also remember that we draw the graph of a relation which is reflexive and symmetric and the property of reflexivity and symmetric is evident from the graphs, we can’t draw the graph of a relation such that transitive property of the relation is evident. Now consider the set of all graphs say it G, this being a set ,so we can define a relation from the set G to itself. So we define the relation of Isomorphism on the set G x G.(By the definition of isomorphism) Our claim is that this relation is an " Equivalence Relation" which means that the relation of Isomorphism’s of two graphs is "REFLEXIVE" "SYMMETRIC" and "TRANSITIVE". Now if you want to draw the graph of this relation, then the vertices of this graph are the graphs from the set G.

Question: Why we can't use the same color in connected portions of planar

graph?

Answer: We define the coloring of graph in such a manner that we can't assign the same color to the adjacent vertices because if we give the same colors to the adjacent vertices then they are indistinguishable. Also note that we can give the same color to the adjacent vertices but such a coloring is called improper coloring and the way which we define the coloring is known as the proper coloring. We are interested in proper coloring that's why all the books consider the proper coloring

---

Question: What is meant by isomorphic invariant?

Answer: A property "P" of a graph is known as Isomorphic invariant. if the same property is found in all the graphs which are isomorphic to it. And all these properties are called isomorphic invariant (Also it clear from the words Isomorphic Invariant that the properties which remain invariant if the two graphs are isomorphic to each other).

Question: What is an infinite Face?

Answer: When you draw a Planar Graph on a plane it divides the plane into different regions, these regions are known as the faces and the face which is not bounded by the edges of the graph is known as the Infinite face. In other words the region which is unbounded is known as Infinite Face.

Question: What is "Bipartite Graph"?

Answer: A graph is said to be Bipartite if it's set of vertices can be divided into two disjoint sets such that no two vertices of the same set are adjacent by some edge of the graph. It means that the edges of one set will be adjacent with the vertices of the other set.

Question: What is chromatic number?

Answer: While coloring a graph you can color a vertex which is not adjacent with the vertices you already colored by choosing a new color for it or by the same color which you have used for the vertices which are not adjacent with this vertex. It means that while coloring a graph you may have different number of colors used for this purpose. But the least number of colors which are being used during the coloring of Graphs is known as the Chromatic number.

Question: What is the role of Discrete mathematics in our prectical life. what advantages will we get by learning it.

Answer: In many areas people have to faces many mathematical problems which can,t be solved in computer so discrete mathematics provide the facility to overcome these problems. Discrete math also covers the wide range of topics, starting with the foundations of Logic, Sets and Functions. It moves onto integer mathematics and matrices, number theory, mathematical reasoning, probability graphs, tree data structures and Boolean algebra.So that is why we need discrete math.

Question: What is the De Morgan's law .

Answer: De Morgan law states " Negation of the conjunction of two statements is logiacally equivalent to the disjunction of their negation and Negation of the disjunction of two statements is logically equivalent to the conjucnction of their negation". i.e. ~(p^q) = ~p v ~q and ~(p v q)= ~p ^ ~q For example: " The bus was late and jim is waiting "(this is an example of conjuction of two statements) Now apply neaggation on this statement you will get through De Morgan's law " The bus was not late or jim is not waiting" (this is the disjunction of negation of two statements). Now see both statement are logically equivalent.Thats what De Morgan want to say

---

Question: What is Tauology?

Answer: A tautology is a statement form that is always true regardless of the truth values of the statement variables. i.e. If you want to prove that (p v q) is tautology ,you have to show that all values of statement (p v q) are true regardless of the values of p and q.If all the values of the satement (p v q) is not true then this statement is not tautology.

Question: What is binary relations and reflexive,symmetric and transitive.

Answer: Dear student! First of all ,I will tell you about the basic meaning of relation i.e It is a logical or natural association between two or more things; relevance of one to another; the relation between smoking and heart disease. The connection of people by blood or marriage.

(b,a)=(3,1)also belongs to R.etc So clearly the above relation R is symmetric. And read the definition of transitive relation from the handouts and the book. You can easily understand it.

Question: What is the matrix relation .

Answer: Suppose that A and B are finite sets.Then we take a relation say R from A to B. From a rectangular array whose rows are labeled by the elements of A and whose columns are labeled by the elements of B. Put a 1 or 0 in each position of the array according as a belongs to A is or is not related to b belongs to B. This array is called the matrix of the relation. There are matrix relations of reflexive and symmetric relations. In reflexive relation, all the diagonal elements of relation should be equal to 1. For example if R = {(1,1), (1,3), (2,2), (3,2), (3,3)} defined on A = {1,2,3}. Then clearly R is reflexive. Simply in making matrix relation In the above example,as the defined set is A={1,2,3} so there are total three elements. Now we take 1, 2 and 3 horizontally and vertically.i.e we make a matrix from the relation R ,in the matrix you have now 3 columns and 3 rows. Now start to make the matrix ,as you have first order pair (1, 1) it means that 1 maps on itself and you write 1 in 1st row and in first column. 2nd order pair is (1, 3) it means that arrow goes from 1 to 3.Then you have to write 1 in 1st row and in 3rd column. (2, 2) means that arrow goes from 2 and ends itself. Here you have to write 1 in 2nd row and in 2nd column. (3,2) means arrow goes from 3 and ends at 2. Here you have to write 1 in 3rd row and in 2nd column. (3, 3) means that 3 maps on itself and you write 1 in 3rd row and in 3rd column. And where there is space empty or unfilled ,you have to write 0 there.

Question: what is binary relation.

Answer: Let A and B be any two sets. Then their cartesian product(or the product set) means a new set "A x B " which contains all the ordered pairs of the form (a,b) where a is in set A and b is in set B. Let we take any subset say 'R' of "A x B" ,then 'R' is called the binary

which contains in "A x B" and denote it by 'R'. Let R={(2,b),(3,a),(3,b)} Clearly R is a subset of "A x B" so 'R' is called the binary relation.

Question: Role of "Discrete Mathematics" in our prectical life. what advantages will we get by learning it.

Answer: Discrete mathematics concerns processes that consist of a sequence of individual steps. This distinguishes it from calculus, which studies continuously changing processes. While the ideas of calculus were fundamental to the science and technology of the industrial revolution, the ideas of discrete mathematics underline the science and technology specific to the computer age. Logic and proof: An important goal of discrete mathematics is to develop students' ability to think abstractly. This requires that students learn to use logically valid forms of argument, to avoid common logical errors, to understand what it means to reason from definition, and to know how to use both direct and indirect argument to derive new results from those already known to be true. Induction and Recursion: An exciting development of recent years has been increased appreciation for the power and beauty of "recursive thinking": using the assumption that a given problem has been solved for smaller cases, to solve it for a given case. Such thinking

knowledge of combinatorics, recurrence relations functions, and O-notation. Discrete Structures: Discrete mathematical structures are made of finite or count ably infinite collections of objects that satisfy certain properties. Those are sets, bolean of algebras, functions, finite start automata, relations, graphs and trees. The concept of isomorphism is used to describe the state of affairs when two distinct structures are the same intheir essentials and diffr only in the labeling of the underlying objects. Applications and modeling: Mathematics topic are best understood when they are seen ina variety of contexts and used to solve problems in a broad range of applied situations. One of the profound lessons of mathematics is that the same mathematical model can be used to solve problems in situations that appear superficially to be totally dissimilar. So in the end i want to say that discrete mathematics has many uses not only in computer science but also in the other fields too.

Question: what is the basic difference b/w sequences and series

Answer: A sequence is just a list of elements .In sequnce we write the terms of sequence as a list (seperated by comma's). e.g 2,3,4,5,6,7,8,9,... (in this we have terms 2,3,4,5,6,7,8,9 and so on).we write these in form of list seperated by comma's. And the sum of the terms of a sequence forms a series. e.g we have sequence 1,2,3,4,5,6,7 Now the series is sum of terms of sequence as 1+2+3+4+5+6+7.

Question: what is the purpose of permutations?

Answer: Permutation is an arrangement of objects in a order where repitition is not allowed. We need arrangments of objects in real life and also in mathematical problems.We need to know in how many ways we can arrange certain objects. There are four types of arrangments we have in which one is permutation.

Question: what is inclusion-exclusion principle

are disjoint finite sets, then n(AÈB) = n(A) + n(B) And if A and B are finite sets, then n(AÈB) = n(A) + n(B) - n(AÇB) For example If there are 15 girls students and 25 boys students in a class then how many students are in total. Now see if we take A ={ 15 girl students} and B={ 25 boys students} Here A and B are two disjoints sets then we can apply first rule n(AÈB) = n(A) + n(B) =15 + 25 =40 So in total there are 40 students in class. Take another Example for second rule. How many integers from 1 through 1000 are multiples of 3 or multiples of 5. Let A and B denotes the set of integers from 1 through 1000 that are multiples of 3 and 5 respectivly. n(A)= 333 n(B)=200 But these two sets are not disjoint because in A and B we have those elements which are multiple of both 3 and 5. so n(AÇB) =66 n(AÈB) = n(A) + n(B) - n(AÇB) =333 + 200 - 66 = 467

Question: How to use conditional probobility

Answer: Dear student In Conditional probability we put some condition on an event to be occur. e.g. A pair of dice is tossed. Find the probability that one of the dice is 2 if the sum is 6. If we have to find the probability that one of the dice is 2, then it is the case of simple probability. Here we put a condition that sum is six. Now A = { 2 appears in atleast one die} E = {sum is 6 } Here E = { (1,5), (2, 4), (3, 3), (4, 2), (5, 1) } Here two order pairs (2, 4) and (4, 2) satisfies the A. (i.e. belongs to A) Now A (intersection) B= { (2,4), (4,2)} Now by formula P(A/E) = P(A (intersection) E)/ P(E) = 2/5

Question: In which condition we use combination and in which condition permutation.

Answer: This depends on the statement of question. If in the statement of question you finds out that repetition of objects are not allowed and order matters then we use Permutation. e.g. Find the number of ways that a party of seven persons can arrange themselves in a row of seven chairs. See in this question repetition is no allowed because whenever a person is chosen for a particular seat r then he

exam. Find the number m of ways that the student can choose the eight questions See in this question repetition is not allowed that is when you choose one question then you cannot choose it again and also order does not matters(i.e either he solved Q1 first or Q2 first) so you use combination in this question.

Question: What is the differnce between edge and vertex

Answer: Vertices are nodes or points and edges are lines/arcs which are used to connect the vertices. e.g If you are making the graph to find the shortest path or for nay purpose of cites and roads between them which contain Lahore, Islamabad, Faisalabad , Karachi, and Multan. Then cities Lahore, Islamabad, Faisalabad , Karachi, and Multan are vertices and roads between them are edges.

Question: What is the differnce between yes and allowed in graphs.

Answer: Allowed mean that specific property can be occurs in that case but yes mean that specific property always occurs in that case. e.g. In Walk you may start and end at same point and may not be (allowed). But in Closed Walk you have to start and end at same point (yes).

Question: what is the meanging of induction? and also Mathematical Induction?

Answer: Basic meaning of induction is: a)The act or an instance of inducting. b) A ceremony or formal act by which a person is inducted, as into office or military service. In Mathematics. A two-part method of proving a theorem involving a positive integral variable. First the theorem is verified for the smallest admissible value of the integer. Then it is proven that if the theorem is true for any value of the integer, it is true for the next greater value. The final proof contains the two parts. As you have studied. It also means that presentation

voltage or current in a material by means of electric or magnetic fields, as in the secondary winding of a transformer when exposed to the changing magnetic field caused by an alternating current in the primary winding. In Biochemistry,it means that the process of initiating or increasing the production of an enzyme or other protein at the level of genetic transcription. In embryology,it means that the change in form or shape caused by the action of one tissue of an embryo on adjacent tissues or parts, as by the diffusion of hormones or chemicals.

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initiating or increasing the production of an enzyme or other protein at the level of genetic transcription. In embryology,it means that the change in form or shape caused by the action of one tissue of an embryo on adjacent tissues or parts, as by the diffusion of hormones or chemicals.

Question: What is "Hypothetical Syllogism".

Answer: Hypothetical syllogism is a law that if the argument is of the form p --> q q---> r Therefore p---> r Then it'll always be a tautology. i.e. if the p implies q and q implies r is true then its conclusion p implies r is always true.

Question: A set is define a well define collection of distinct objects so why an empty set is called a set although it has no element?

Answer: Some time we have collection of zero objects and we call them empty sets. e.g. Set of natural numbers greater than 5 and less than 5. A = { x belongs to N / 5< x < 5 } Now see this is a set which have collection of elements which are greater than 5 and less than 5 (from natural number).

Question: What is improper subset.

Answer: Let A and B be sets. A is proper subset of B, if, and only if, every element of A is in B but there is at least on element if B that is not in A. Now A is improper subset of B, if and only if, every element of A is in B and there is no element in B which is not in A. e.g. A= { 1, 2 , 3, 4} B= { 2, 1, 4, 3} Now A is improper subset of B. Because every element of A is in B and there is no element in B which is not in A

Question: How to check validity and unvalidity of argument through diagram.

those sets in the form of diagram. If diagram satisfies the conclusion then it is a valid argument otherwise invalid. e.g. If we have three premises S1: all my friends are musicians S2: John is my friend. S3: None of my neighbor are musicians. conclusion John is not my neighbor. Now we have three sets Friends, Musicians, neighbors. Now you see from premises 1 and 2 that friends are subset of musicians .From premises 3 see that neighbor is an individual set that is disjoint from set musicians. Now represent then in form of Venn diagram. Musicians neighbour Friends Now see that john lies in set friends which is disjoint from set neighbors. So their intersection is empty.Which shows that john is not his neighbor. In that way you can check the validity of arguments

Question: why we used venn digram?

Answer: Venn diagram is a pictorial representation of sets. Venn diagram can sometime be used to determine whether or not an argument is valid. Real life problems can easily be illustrate through Venn diagram if you first convert them into set form and then in Venn diagram form. Venn diagram enables students to organize similarities and differences visually or graphically. A Venn diagram is an illustration of the relationships between and among sets, groups of objects that share something in common.

Question: what is composite relation .

Answer: Let A, B, and C be sets, and let R be relation from A to B and let S be a relation from B to C. Now by combining these two relations we can form a relation from A to C. Now let a belongs to A, b belongs to B, and c belongs to C. We can write relations R as a R b and S as b

to x and our composite relation omits that common and relates directly 3 to x.) I do not understand your second question send it again. Now R O S ={(2,z), (3,x), (3,z)}

Question: What are the conditions to confirm functions .

Answer: The first condition for a relation from set X to a set Y to be a function is 1.For every element x in X, there is an element y in Y such that (x, y) belongs to F. Which means that every element in X should relate with distinct element of Y. e.g if X={ 1,2,3} and Y={x, y} Now if R={(1,x),(2,y),(1,y),(2,x)} Then R will not be a function because 3 belongs to X but is does not relates with any element of Y. so R={(1,x),(2,y),(3,y)} can be called a function because every element of X is relates with elements of Y. Second condition is : For all elements x in X and y and z in Y, if (x, y) belongs to F and (x, z) belongs to F, then y = z Which means that every element in X only relates with distinct element of Y. i.e. R={(1,x),(2,y),(2,x), (3,y)} cannot be called as function because 2 relates with x and y also.

Question: When a function is onto.

Answer: First you have to know about the concept of function. Function:It is a rule or a machine from a set X to a set Y in which each element of set X maps into the unique element of set Y. Onto Function: Means a function in which every element of set Y is the image of at least one element in set X. Or there should be no element left in set Y which is the image of no element in set X. If such case does not exist then the function is not called onto. For example:Let we define a function f : R R such that f(x)=x^2 (where ^ shows the symbol of power i.e. x raise to power 2). Clearly every element in the second set is the image of atleast one element in the first set. As for x=1 then f(x)=1^2=1 (1 is the image of 1 under the rule f) for x=2 then f(x)=2^2=4 (4 is the image of 2 under the rule f) for x=0 then f(x)=0^2=0 (0 is the image of 0 under the rule f) for x=-1 then f(x)=(-1)^2=1 (1 is the image of -1 under the rule f) So it is onto function.

Question: Is Pie an irrational number?

Answer: Pi π is an irrational number as its exact value has an infinite decimal expansion: Its decimal expansion never ends and does not repeat.

The numerical value of π truncated to 50 decimal places is:

3.14159 26535 89793 23846 26433 83279 50288 41971 69399 37510

Question: Difference between sentence and statement.

Answer: A sentence is a statement if it have a truth value otherwise this sentence is not a statement.By truth value i mean if i write a sentence "Lahore is capital of Punjab" Its truth value is "true".Because yes Lahore is a capital of Punjab. So the above sentence is a statement. Now if i write a sentence "How are you" Then you cannot answer in yes or no.So this sentence is not a statement. Every statement is a sentence but converse is not true.

Question: What is the truth table?

Answer: Truth table is a table which describe the truth values of a proposition. or we can say that Truth table display the complete behaviour of a proposition. There fore the purpose of truth table is to identify its truth values. A statement or a proposition in Discrete math can easily identify its truth value by the truth table. Truth tables are especially valuable in the determination of the truth values of propositions constructed from simpler propositions. The main steps while making a truth table are "first judge about the statement that how much symbols(or variables) it contain. If it has n symbols then total number of combinations=2 raise to power n. These all the combinations give the truth value of the statement from where we can judge that either the truthness of a statement or proposiotion is true or false. In all the combinations you have to put values either "F" or "T" against the variales.But note it that no row can be repeated. For example "Ali is happy and healthy" we denote "ali is happy" by p and "ali is healthy" by q so the above statement contain

two variables or symbols. The total no of combinations are =2 raise to power 2(as n=2) =4 which tell us the truthness of a statement.

Question: how empty set become a subset of every set.

Answer: If A & B are two sets, A is called a subset of B, if, and only if, every element of A is also an element of B. Now we prove that empty set is subset of any other set by a contra positive statement(of above statement) i.e. If there is any element in the the set A that is not in the set B then A is not a subset of B. Now if A={} and B={1,3,4,5} Then you cannot find an element which is in A but not in B. So A is subset of B.

Question: What is rational and irrational numbers.

Answer: A number that can be expressed as a fraction p/q where p and q are integers and q\not=0, is called a rational number with numerator p and denominator q. The numbers which cannot be expressed as rational are called irrational number. Irrational numbers have decimal expansions that neither terminate nor become periodic where in rational numbers the decimal expansion either terminate or become periodic after some numbers.

Question: what is the difference between graphs and spanning tree?

Answer: First of all, a graph is a "diagram that exhibits a relationship, often functional, between two sets of numbers as a set of points having coordinates determined by the relationship. Also called plot". Or A pictorial device, such as a pie chart or bar graph, used to illustrate quantitative relationships. Also called chart. And a tree is a

spanning tree for a graph G is a subgraph of G that contains every vertex of G and is a tree. It is not neccesary for a graph to always be a spanning tree. Graph becomes a spanning tree if it satisfies all the properties of a spanning tree.

Question: What is the probability ?

Answer: The definition of probability is : Let S be a finite sample space such that all the outcomes are equally likely to occur. The probability of an event E, which is a subset of S, is P(E) = (the number of outcomes in E)/ (the number of total outcomes in S) P(E) = n (E) / n (S) This definition is due to 'Laplace.' Thus probability is a concept which measures numerically the degree of certainty or uncertainty of the occurrence of an event. Explaination The basic steps of probability that u have to remember are as under 1. First list out all possible out comes. That is called the sample space S For example when we roll a die the all possible outcomes are the set S i.e. S = {1,2,3,4,5,6} 2. Secondly we have to find out all that possible outcomes, in which the probability is required . For example we are asked to find the probability of even numbers. First we decide any name of that event i.e E Now we check all the even numbers in S which are E = {2,4,6} Remember Event is always a sub-set of Sample space S. 3. Now we apply the definition of probability P(E) = (the number of outcomes in E)/ (the number of total outcomes in S) P(E) = n (E) / n (S) So from above two steps we have n (E) = 3 and n (S) = 6 then P(E) = 3 / 6 = 1/2 which is probability of an even number.

Question: what is permutation?

Answer: Permutation comes from the word permute which means " to change the order of." Basically permutation means a "complete change." Or the act of altering a given set of objects in a group. In Mathematics point of view it means that a ordered arrangement of the elements of a set (here the order of elements matters but repetition of the elements is not allowed).

Question: What is a function.

Answer: A function say 'f' is a rule or machine from a set A to the set B if for every element say a of A, there exist a unique element say b of set such that b=f(a) Where b is the image of a under f,and a is the pre-image. Note it that set A is called the domain of f and Y is called the codomain of f. As we know that function is a rule or machine in which we put an input,and we get an output.Like that a juicer machine.We take some apples(here apples are input) and we apply a rule or a function of juicer machine on it,then we get the output in the form of juice.

Question: What is p implies q.

Answer: p--- >q means to "go from hypothesis to a conclusion" where p is a hypothesis and q is a conclusion. And note it that this statement is conditioned because the "truth ness of statement p is conditioned on the truth ness of statement q". Now the truth value of p--->q is false only when p is true and q is false otherwise it will always true. E.g. consider an implication "if you do your work on Sunday ,I will give you ten rupees." Here p=you do your work on Sunday (is the hypothesis) , q=I will give you ten rupees (the conclusion or promise). Now the truth value of p---->q will false only when the promise is braked. i.e. You do your work on Sunday but you do not get ten rupees. In all other conditions the promise is not braked.

Question: What is valid and invalid arguments.

Answer: As "an argument is a list of statements called premises (or assumptions or hypotheses) which is followed by a statement called the conclusion. " A valid argument is one in which the premises entail(or imply) the conclusion. 1)It cannot have true premises and a false conclusion. 2)If its premises are true, its conclusion must be true. 3)If its conclusion is false, it must have at least one false premise. 4)All of the information in the conclusion is also in the premises. And an invalid agrument is one in which the premises do not entail(or imply) the conclusion. It can have true premises and a

false conclusion. Even if its premises are true, it may have a false conclusion. Even if its conclusion is false, it may have true premises. There is information in the conclusion that is not in the premises. To know them better,try to solve more and more examples and exercises.

Question: What is domain and co -domain.

Answer: Domain means "the set of all x-coordinates in a relation". It is very simple,Let we take a function say f from the set X to set Y. Then domain means a set which contain all the elements of the set X. And co domain means a set which contain all the elements of the set Y. For example: Let we define a function "f" from the set X={a,b,c,d} to Y={1,2,3,4}. such that f(a)=1, f(b)=2, f(c)=3, f(d)=1 Here the domain set is {a,b,c,d} And the co-domain set is {1,2,3,4} Where as the image set is {1,2,3}.Because f(a)=1 as 1 is the image of a under the rule 'f'. f(b)=2 as 2 is the image of b under the rule 'f'. f(c)=3 as 3 is the image of c under the rule 'f'. f(d)=1as 1 is the image of d under the rule 'f'. because "image set contains only those elements which are the images of elements found in set X". Note it that here f is one -one but not onto,because there is one element '4' left which is the image of nothing element under the rule 'f'.

Question: What is the difference between k-sample,k-selection, k-permutation and k-combination?

Answer: Actually, these all terms are related to the basic concept of choosing some elements from the given collection.

For it, two things are important:

1) Order of elements .i.e. which one is first, which one is

So we can get 4 kinds of selections:

1) The elements have both order and repetition. (It is called k-sample)

2) The elements have only order, but no repetition. (It is called k-permutation)

3) The elements have only repetition, but no order. (It is called k-selection)

4) The elements have no repetition and no order. (It is called k-combination)

Question: What is a combination?

Answer: A combination is an un-ordered collection of unique elements. Given S, the set of all possible unique elements, a combination is a subset of the elements of S. The order of the elements in a combination is not important (two lists with the same elements in different orders are considered to be the same combination). Also, the elements cannot be repeated in a combination (every element appears uniquely once

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Question: why is 0! equal to 1?

Answer: Since n! = n(n-1)!

Put n =1 in it.

1! = 1x(1 – 1)!

So 1 = 0!

0! = 1.

Question: What is the basic idea if Mathematical Induction?

Question: Define symmetric and anti-symmetric.

Answer:

Question: What is the main deffernce between Calculus and Discrete Maths?

Answer: Discrete mathematics is the study of mathematics which concerns to the study of discrete objects. Discrete math build students approach to think abstractly and how to handle mathematical models problems in computer While Calculus is a mathematical tool used to analyze changes in physical quantities. Or "Calculus is sometimes described as the mathematics of change." Also calculus played an important role in industrial area as well discrete math in computer.

Discrete mathematics concerns processes that consist of a sequence of individual steps. This distinguishes it from calculus, which studies continuously changing processes. the ideas of discrete mathematics underline the science and technology specific to the computer age. An important goal of discrete mathematics is to develop students' ability to think abstractly.

Question: Explain Valid Arguments.

A **valid argument** is one in which the premises imply the conclusion.

1) It cannot have true premises and a false conclusion.

2) If its premises are true, its conclusion must be true.

3) If its conclusion is false, it must have at least one false premise.

4) All of the information in the conclusion is also in the premises.

Question: What is the Difference between combinations and permutations?

Answer: When we talk of permutations and combinations in everyday talk we often use the two terms interchangeably. In mathematics, however, the two each have very specific meanings, and this distinction often causes problems

In brief, the permutation of a number of objects is the number of

simplistically) as

Permutations - position important (although choice may also be important)
Combinations - chosen important, which may help you to remember

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Question: What is the use of kruskal's algorithn in our daily life?

Answer: The Kruskal's algorithm is usually used to find minimum spanning tree i.e. the possible smallest tree that contains all the vertices. The standard application is to a problem like phone network design. Suppose, you have a business with several offices; you want to lease phone lines to connect them up with each other; and the phone company charges different amounts of money to connect different pairs of cities. You want a set of lines that connects all your offices with a minimum total cost. It should be a spanning tree, since if a network isn't a tree you can always remove some edges and save money. A less obvious application is that the minimum spanning tree can be used to approximately solve the traveling salesman problem. A convenient formal way of defining this problem is to find the shortest path that visits each point at least once.

Question: What is irrational number?

Answer: Irrational number An irrational number can not be expressed as a fraction. In decimal form, irrational numbers do not repeat in a pattern or terminate. They "go on forever" (infinity). Examples of irrational numbers are: pi= 3.141592654...

Question: Define membership table and truth table.

Answer: Membership table: A table displaying the membership of elements in sets. Set identities can also be proved using membership tables. An element is in a set, a 1 is used and an element is not in a set, a 0 is used. Truth table: A table displaying the truth values of propositions.

Question: Define function and example for finding domain and range of a function.

Answer:

Question: Why do we use konigsberg bridges problem?

Answer: Click on it.

Question: Explain the intersection of two sets?

Answer: Click on it.

Question: What is absurdity With example?

Answer:

Question: What is sequence and series?

Answer: Sequence A sequence of numbers is a function defined on the set of positive integer. The numbers in the sequence are called terms. Another way, the sequence is a set of quantities u1, u2, u3... stated in a definite order and each term formed according to a fixed pattern. U r =f(r) In example: 1,3,5,7,... 2,4,6,8,... 1 2 ,− 2 2 ,3 2 ,− 4 2 ,... Infinite sequence:- This kind of sequence is unending sequence like all natural numbers: 1, 2, 3, ... Finite sequence:- This kind of sequence contains only a finite number of terms. One of good examples are the page numbers. Series:- The sum of a finite or infinite sequence of expressions. 1+3+5+7+...

Question: Differntiate contigency and contradiction.

Answer:

Question: What is conditional statement, converse, inverse and contra-positive?

Answer:

Question: What is Euclidean algorithm?

Answer: In number theory, the **Euclidean algorithm** (also called **Euclid's algorithm**) is an algorithm to determine the greatest common divisor (GCD) of two integers.

Its major significance is that it does not require factoring the two integers, and it is also significant in that it is one of the oldest algorithms known, dating back to the ancient Greeks.

Question: what is the circle definition?

Answer: A circle is the locus of all points in a plane which are equidistant from a fixed point. The fixed point is called centre of that circle and the distance is called radius of that circle

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Question: What is bi-conditional statement?

Answer:

Question: Explain the difference between k-sample, k-selection, k-combination and k-permutation.

Answer:

Question: What is meant by Discrete?

Answer:

A type of data is discrete if there are only a finite number of values possible. Discrete data usually occurs in a case where there are only a certain number of values, or when we are counting something (using whole numbers). For example, 5 students, 10 trees etc.

Question: Explain D'Morgan Law.

Answer:

Question: What are digital circuits?

Answer: Digital circuits are electric circuits based on a number of discrete voltage levels.

In most cases there are two voltage levels: one near to zero volts and one at a higher level depending on the supply voltage in use. These two levels are often represented as L and H.

Question: What is absurdity or contradiction?

Answer: A statement which is always false is called an absurdity.

Question: What is contingency?

Answer: A statement which can be true or false depending upon the truth values of the variables is called a contingency.

Question: Is there any particular rule to solve Inductive Step in the mathematical Induction?

true.

In proving the result for n = k +1, the procedure changes, as it depends on the shape of the given statement.

Following steps are main:

1) You should simply replace n by k+1 in the left side of the statement.

2) Use the supposition of n = k in it.

3) Then you have to simplify it to get right side of the statement. This is the step,

where students usually feel difficulty.

Here sometimes, you have to open the brackets, or add or subtract some terms

or take some term common etc. This step of simplification to get right side of the given statement for n = n + 1 changes from question to question.

Now check this step in the examples of the Lessons 23 and 24.

Question: What is Inclusion Exclusion Principle?

Answer: Click on Inclusion Exclusion Principle.

Question: What is recusion?

Answer:

Question: Different notations of conditional implication.

Answer: If p than q. P implies q. If p , q. P only if q. P is sufficient for q.

Question: What is cartesion product?

Answer: Cartesian product of sets:- Let A and B be sets. The Cartesian product of A and B, denoted A x B (read “A cross B”) is the set of all ordered pairs (a, b), where a is in A and b is in B. For example: A = {1, 2, 3, 4, 5, 6} B = {a} A x B = {(1,a), (2,a), (3,a), (4,a), (5,a)}

Question: Define fraction and decimal expansion.

Answer: Fraction:- A number expressed in the form a/b where a is called the numerator and b is called the denominator. Decimal expansion:- The decimal expansion of a number is its representation in base 10 The number 3.22 3 is its integer part and 22 is its decimal part The number on the left of decimal point is integer part of the number and the number on the right of the decimal point is decimal part of the number.

Question: Explane venn diagram.

Answer: Venn diagram is a pictorial representation of sets. Venn diagram can sometime be used to determine whether or not an argument is valid. Real life problems can easily be illustrate through Venn diagram if you first convert them into set form and then in Venn diagram form. Venn diagram enables students to organize similarities and differences visually or graphically. A Venn diagram is an illustration of the relationships between and among sets, groups of objects that share something in common

Question: Write the types of functions.

Answer: Types of function:- Following are the types of function 1. One to one function 2. Onto function 3. Into function 4. Bijective function (one to one and onto function) One to one function:- A function f : A to B is said to be one to one if there is no repetition in the second element

A function f : A to B is said to be into function of Range of f is the subset of set B (co domain) Bijective function: Bijective function:- A function is said to be Bijective if it is both one to one and onto.

Question: Explain the pigeonhole principle.

Answer:

Question: What is conditional probability with example?.

Answer:

Question: Explain combinatorics.

Answer: Branch of mathematics concerned with the selection, arrangement, and combination of objects chosen from a finite set.

The number of possible bridge hands is a simple example; more complex problems include scheduling classes in classrooms at a large university and designing a routing system for telephone signals. No standard algebraic procedures apply to all combinatorial problems; a separate logical analysis may be required for each problem.

Question: How the tree diagram use in our real computer life?

Answer: Tree diagrams are used in data structure, compiler construction, in making algorithms, operating system etc.

Question: Write detail of cards.

Answer: Diamond Club Heart Spade A A A A 1 1 1 1 2 2 2 2 3 3 3 3 4 4 4 4 5 5 5 5 6 6 6 6 7 7 7 7 8 8 8 8 9 9 9 9 10 10 10 10 J J J J Q Q Q Q K K K K Where 26 cards are black & 26 are red. Also ‘A’ stands for ‘ace’ ‘J’ stands for ‘jack’ ‘Q’ stands for ‘queen’ ‘K’ stands for ‘king’

Question: what is the purpose of permutations?

Answer: Definition:- Possible arrangements of a set of objects in which the order of the arrangement makes a difference. For example, determining all the different ways five books can be arranged in order on a shelf. In mathematics, especially in abstract algebra and related areas, a permutation is a bijection, from a finite set X onto itself. Purpose of permutation is to establish significance without assumptions

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Question No: 1

Every point in three dimensional space can be described by --------coordinates.

- ► Two
- ► **Three**
- ► Four
- ► Eight

Question No: 2

What are the direction cosines for the line joining the points (1, 3, 2) and (7, -2, 3)?

- ► $\frac{1}{7}, \frac{-3}{2}$ *and* $\frac{2}{3}$
- ► $\frac{7}{11}, \frac{-6}{11}$ *and* $\frac{6}{11}$
- ► $\frac{8}{3\sqrt{10}}, \frac{1}{3\sqrt{10}}$ *and* $\frac{5}{3\sqrt{10}}$
- ► $\frac{6}{\sqrt{62}}, \frac{5}{\sqrt{62}}$ *and* $\frac{1}{\sqrt{62}}$

Question No: 3

The angles which a line makes with positive x ,y and z-axis are known as ----------

- ► Direction cosines
- ► Direction ratios
- ► **Direction angles**

Question No: 4

Which of the following is geometrical representation of the equation $y = 4$, in three dimensional space?

- ► A point on y-axis (Farhan)
- ► Plane parallel to xy-plane
- ► Plane parallel to yz-axis
- ► **Plane parallel to xz-plane (Ali)**

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Question No: 5

Domain of the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ is

- ► **Entire 3D-Space**
- ► Entire 3D-Space except origin
- ► $(0, \infty)$
- ► $(-\infty, \infty)$

Question No: 6

If $f(x, y) = x^2 y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2} =$

- ► $2xy + \frac{1}{x^2}$
- ► $2y + \frac{1}{x^2}$
- ► $2xy - \frac{1}{x^2}$
- ► $2y - \frac{1}{x^2}$

Question No: 7

Suppose $f(x, y) = xy - 2y^2$ *where* $x = 3t + 1$ *and* $y = 2t$. Which one of the following is true?

- ► $\frac{df}{dt} = -4t + 2$
- ► $df = -16t - t$ ~~dt~~
- ► $\frac{df}{dt} = 18t + 2$

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► $\frac{df}{dt} = -10t^2 + 8t + 1$

Question No: 8

For a function $f(x, y, z)$, the equation $\frac{\partial^2 f}{\partial^2 x} + \frac{\partial^2 f}{\partial^2 y} + \frac{\partial^2 f}{\partial^2 z} = 0$ is known as ----------

► Gauss Equation

► Euler's equation

► **Laplace's Equation**

► Stoke's Equation

Question No: 9

Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3 and angle between them when placed tail to tail is 45 degrees. What is $\vec{a} \cdot \vec{b}$?

► 4.5

► 6.2

► 5.1

► **4.2**

Question No: 10

Is the function $f(x, y)$ continuous at origin? If not, why?

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$f(x, y) = \frac{xy}{x^2 + y^2}$

- $f(x, y)$ is continuous at origin
- $\lim_{(x, y)\to(0, 0)} f(x, y)$ does not exist

- $f(0, 0)$ is defined and $\lim_{(x, y)\to(0, 0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 11

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- Area of R.??(ali)
- **Radius of inscribed circle in R**
- Distance between two endpoints of R.
- None of these

Question No: 12

Which of the following formula can be used to find the Volume of a parallelepiped with adjacent edges formed by the vectors a , b *and* c ?

- $\left|a \times \left(b \times c\right)\right|$
- $\left|a \text{ g}\left(b \text{ g}c\right)\right|$
- $\left|a \text{ g } b \times c\right|$
- $\left|a \times \left(b \text{ g}c\right)\right|$

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Question No: 13

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are ---------

- ► Parallel
- ► **Perpendicular**
- ► In opposite direction

Question No: 14

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

- ► perpendicular
- ► parallel
- ► **In opposite direction**

Question No: 15

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}, f_{yy} \text{ and } f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

- ► **Relative maximum at** (x_0, y_0)
- ► Relative minimum at (x_0, y_0)

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► Saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 16

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy}$ and $f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

If $D = 0$ then ----------

► f has relative maximum at (x_0, y_0)

► f has relative minimum at (x_0, y_0)

► f has saddle point at (x_0, y_0)

► **No conclusion can be drawn.**

Question No: 17

If $R = \{(x, y) / 0 \le x \le 2 \text{ and } 1 \le y \le 4\}$, *then*

$$\iint_R (6x^2 + 4xy^3)dA =$$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$

► $\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$

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- $\int_{2}^{4}\int_{0}^{1}(6x^2 + 4xy^3)dxdy$

Question No: 18

If $R = \{(x, y) / 0 \le x \le 2 \text{ and } -1 \le y \le 1\}$, *then*
$\iint_R (x + 2y^2)dA =$

- $\int_{-1}^{1}\int_{0}^{2}(x + 2y^2)dydx$
- $\int_{0}^{2}\int_{1}^{-1}(x + 2y^2)dxdy$
- $\int_{-1}^{1}\int_{0}^{2}(x + 2y^2)dxdy$
- $\int_{1}^{2}\int_{-1}^{0}(x + 2y^2)dxdy$

Question No: 19

If $R = \{(x, y) / 2 \le x \le 4 \text{ and } 0 \le y \le 1\}$, *then*
$\iint_R (4xe^{2y})dA =$

- $\int_{0}^{1}\int_{2}^{4}(4xe^{2y})dydx$
- $\int_{0}^{1}\int_{2}^{4}(4xe^{2y})dxdy$

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- $\int_{1}^{4}\int_{0}^{2}(4xe^{2y})dxdy$

- $\int_{1}^{4}\int_{0}^{2}(4xe^{2y})dydx$

Question No: 20

$If\ R=\{(x,y)/0\le x\le 4\ and\ 0\le y\le 9\},then$
$\iint_{R}(3x-4x\sqrt{xy})dA=$

- $\int_{0}^{9}\int_{0}^{4}(3x-4x\sqrt{xy})dydx$

- $\int_{0}^{4}\int_{4}^{9}(3x-4x\sqrt{xy})dxdy$

- $\int_{4}^{9}\int_{0}^{0}(3x-4x\sqrt{xy})dxdy$

- $\int_{0}^{4}\int_{0}^{9}(3x-4x\sqrt{xy})dydx$

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1. Every real number corresponds to_____on the co-ordinate line.

- Infinite number of points
- Two points (one positive and one negative)
- A unique point??f
- None of these

2. There is one-to-one correspondence between the set of points on co-ordinate line and _____.

- Set of real numbers
- Set of integers
- Set of natural numbers
- Set of rational numbers

3. Which of the following is associated to each point of three dimensional spaces?

- A real number
- An ordered pair
- An ordered triple
- **A natural Number**

4. All axes are positive in_____octant.

- **First**
- Second
- Fourth
- Eighth

5. The spherical co-ordinates of a point are $\left(\sqrt{3}, -, -\right)$.What are its cylindrical co-ordinates?

- $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}, -, 0\right)$
- $\left(\sqrt{3}\cos\frac{\pi}{3}, \sqrt{3}\sin\frac{\pi}{3}, 0\right)$

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- $\left(\sqrt{3}\sin\frac{\pi}{3},\frac{\pi}{2},\sqrt{3}\cos\frac{\pi}{3}\right)$
- $\left(\sqrt{3},\frac{\pi}{3},0\right)$

6. Suppose $f(x,y)=xy-2y^2$ *where* $x=3t+1$ *and* $y=2t$. Which one of the following is true?

- $\frac{df}{dt}=4t+2$
- $\frac{df}{dt}=-16t-t$
- $\frac{df}{dt}=18t+2$
- $\frac{df}{dt}=-10t^2+8t+1$
- $\frac{df}{dt}$
- $\frac{df}{dt}$

7. Let $w=f(x,y,z)$ *and* $x=g(r,s)$, $y=h(r,s)$, $z=t(r,s)$ then by chain rule $\frac{\partial w}{\partial r}=$

- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$

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- $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

8. Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3and angle between them when placed tail to tail is 45 degrees. What is$\vec{a}.\vec{b}$?

- 4.5
- **6.2**
- 5.1
- **4.2**

9. Is the function $f(x,y)$ continuous at origin? If not, why?

$$f(x,y)=\begin{cases}0 & if\ x\geq 0\ and\ y\geq 0\\ 1 & otherwise\end{cases}$$

- $f(x,y)$ is continuous at origin
- $f(0,0)$ is not defined
- $f(0,0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist
- $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exist but these two numbers are not equal.

10. Is the function $f(x,y)$ continuous at origin? If not, Why?

$$f(x,y)=\begin{cases}\frac{3x^2y}{x^2+y^2} & if(x,y)\neq 0\\ 0 & if\ (x,y)=0\end{cases}$$

- $f(x,y)$ is continuous at origin
- $f(0,0)$ **is not defined**
- $f(0,0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist

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- $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exist but these two numbers are not equal.

11. Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- Area of R
- **Radius of inscribed circle in R.**
- Distance between two endpoints of R.
- None of these

12. Which of the following formula can be used to find the volume of a parallelepiped with adjacent edges formed by the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$?

- $|\vec{a} \times (\vec{b} \times \vec{c})|$
- $|\vec{a} . (\vec{b} . \vec{c})|$
- $|\vec{a} . (\vec{b} \times \vec{c})|$
- $|\vec{a} \times (\vec{b} . \vec{c})|$

13. Two surfaces are said to be orthogonal at appoint of their intersection if their normals at that point are_____.

- Parallel
- **Perpendicular**
- In opposite direction
- Same direction

14. By Extreme Value Theorem, if a function $f(x,y)$ is continuous on a closed and bounded set R, then $f(x,y)$ has both _____on R.

- **Absolute maximum and absolute minimum value**
- Relative maximum and relative minimum value
- Absolute maximum and relative minimum value
- Relative maximum and absolute minimum value

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15. Let the function $f(x,y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy}$ *and* $f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f^2_{xy}(x_0, y_0)$ if $D > 0$ *and* $f_{xx}(x_0, y_0) < 0$ *then* f *has*_____.

- **Relative maximum at (x_0, y_0)**
- Relative minimum at (x_0, y_0)
- Saddle point at (x_0, y_0)
- No conclusion can be drawn.

16. Let the function $f(x,y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy}$ *and* $f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f^2_{xy}(x_0, y_0)$ if

if $D = 0$ then_____.

- f has relative maximum at (x_0, y_0)
- f has relative minimum at (x_0, y_0)
- f has saddle point at (x_0, y_0)
- **No conclusion can be drawn.**

17. 1 7 *If* $R = R_1 \cup R_2$ *,where* R_1 *and* R_2 are no over lapping regions then .

$$\iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA =$$

- $\iint_R f(x,y)dA$
- **$\iint_{R_1} f(x,y)dA \cup \iint_{R_2} f(x,y)dA$**
- $\iint_R f(x,y)dV$
- $\iint_R f(x,y)dV \cap \iint_{R_2} f(x,y)dA$

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18. 1 8 . *If* $R = \{(x,y)/0 \le x \le 2 \text{ and } 1 \le y \le 4\}$ *,then* $\iint_R (6x^2 + 4xy^3)dA =$

- $\int_1^0 \int (6x^2 + 4xy^3)\,dydx$
- $\int_0^2 \int_1^4 (6x^2 + 4xy^3)\,dydx$
- $\int_1^0 \int_4^2 (6x^2 + 4xy^3)\,dydx$
- $\int_2^0 \int_1^4 (6x^2 + 4xy^3)\,dydx$

19. 1 9 . *If* $R = \{(x,y)/2 \le x \le 4 \text{ and } 0 \le y \le 1\}$ *, then* $\iint_R (4xe^{2y})dA =$

- $\int_0^2 \int (4xe^{2y})\,dydx$
- $\int_0^2 \int_1^4 (4xe^{2y})\,dxdy$
- $\int_1^0 \int_4^2 (4xe^{2y})\,dxdy$
- $\int_1^0 \int_4^2 (4xe^{2y})\,dydx$

20. 2 0 . *If* $R = \{(x,y)/0 \le x \le 4 \text{ and } 9 \le y \le 4\}$ *,then* $\iint_R (3x - 4x\sqrt{xy})dA =$

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- $\int_0^0 \int_0^0 \left(3x - 4x\sqrt{xy}\right) dydx$
- $\int_0^4 \int_0^0 \left(3x - 4x\sqrt{xy}\right) dxdy$
- $\int_4^0 \int_0^0 \left(3x - 4x\sqrt{xy}\right) dxdy$
- $\int_0^0 \int_0^0 \left(3x - 4x \quad xy\right) dydx$

Question No: 1

Which of the following number is associated to each point on a co-ordinate line?

- An integer
- **A real number**
- A rational number
- A natural number

Question No: 2

If $a > 0$, then the parabola $y = ax^2 + bx + c$ opens in which of the following direction?

- **Positive x - direction**
- Negative x - direction
- Positive y - direction
- Negative y - direction

Question No: 3

Rectangular co-ordinate of a point is $(1, \sqrt{3}, -2)$. What is its spherical co-ordinate?

□□□□□□□▶□ $\left(2, 2, \frac{\pi}{3}, \frac{3\pi}{2}\right)$

□□□□□□□▶□ $\left(2, 2, \frac{\pi}{2}, \frac{3\pi}{4}\right)$

□□□□□□□▶□ $\left(2, 2, \frac{\pi}{3}, \frac{3\pi}{4}\right)$

□□□□□□□▶□ $\left(2, \frac{\pi}{3}, \frac{3\pi}{4}\right)$

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$$\begin{pmatrix} \sqrt{} & 3 & 4 \end{pmatrix}$$

Question No: 4

If a function is not defined at some point, then its limit---------- exist at that point.

□□□□□□□►□Always

□□□□□□□►□Never

□□□□□□□►□**May**

Question No: 5

Suppose $f(x, y) = x^3 e^{xy}$. Which one of the statements is correct?

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$\frac{\partial f}{\partial y} = 3x^3e^{xy}$

$\frac{\partial f}{\partial y} = x^3e^{xy}$

$\frac{\partial f}{\partial y} = x^4e^{xy}$

$\frac{\partial f}{\partial y} = x^3ye^{xy}$

Question No: 6

If $f(x, y) = x^2y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2}$ =

$2xy + \frac{1}{x^2}$

$2y + \frac{1}{x^2}$

$2xy - \frac{1}{x^2}$

$2y - \frac{1}{x^2}$

Question No: 7

Suppose $f(x, y) = xy - 2y^2$ *where* $x = 3t + 1$ *and* $y = 2t$. Which one of the following is true?

$\frac{df}{dt} = -4t + 2$

$\frac{df}{dt} = -16t - t$

$\frac{df}{dt} = 18t + 2$

$\frac{df}{dt} = -10t^2 + 8t + 1$

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Question No: 8

Is the function $f(x, y)$ continuous at origin? If not, why?

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$$f(x, y) = \begin{cases} 0 & If\ x \geq 0\ and\ y \geq 0 \\ 1 & Otherwise \end{cases}$$

□□□□□□□►□ $f(x, y)$ **is continuous at origin**
□□□□□□□►□ $f(0, 0)$ is not defined

□□□□□□□►□ $f(0, 0)$ is defined but $\lim_{(x, y)\to(0, 0)} f(x, y)$ does not exist
□□□□□□□►□ $f(0, 0)$ is defined and $\lim_{(x, y)\to(0, 0)}$ exists but these two numbers are not equal.

Question No: 9
What is the relation between the direction of gradient at any point on the surface to the tangent plane at that point ?

□□□□□□□►□parallel
□□□□□□□►□perpendicular
□□□□□□□►□opposite direction
□□□□□□□►□No relation between them.

Question No: 10
Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------
► perpendicular
► parallel
► in opposite direction

Question No: 11
By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R, then $f(x, y)$ has both---------- on R.

► Absolute maximum and absolute minimum value
► Relative maximum and relative minimum value

Question No: 12
Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy}\ and\ f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

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If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

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► **Relative maximum at** (x_0, y_0)

► Relative minimum at (x_0, y_0)

► Saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 13

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

If $D = 0$ then ----------

► f has relative maximum at (x_0, y_0)

► f has relative minimum at (x_0, y_0)

► f has saddle point at (x_0, y_0)

► **No conclusion can be drawn.**

Question No: 14

The function $f(x, y) = \frac{}{\sqrt{y-x}}$ is continuous in the region ---------- and discontinuous elsewhere.

► $x \neq y$

► $x \leq y$

► $x > y$

Question No: 15

Plane is an example of ----------

► Curve

► **Surface**

► Sphere

► Cone

Question No: 16

If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then

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$\int_{R_1}\int f(x, y)\, d+A \qquad f(x\int_{R_2},\int$

- $\iint_R f(x, y)\, dA$
- $\int_{R_1}\int f(x, y)\, d\cup A \qquad f(x\int_{R_2}$
- $\iint_R f(x, y)\, dV$
- $\int_{R_1}\int f(x, y)\, d\cap A \qquad f(x$

Question No: 17

If $R = \{(x, y) / 0 \le x \le 2 \text{ and } 1 \le y \le 4\}$, *then*

$\iint_R (6x^2 + 4xy^3)dA =$

- $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$
- $(6x^2 + 4xy^3)dxdy$
- $\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$
- $(6x^2 + 4xy^3)dxdy$
- $\int_1^4 \int_0^2$
- $\int_2^4 \int_0^1$

Question No: 18

If $R = \{(x, y) / 0 \le x \le 2 \text{ and } -1 \le y \le 1\}$, *then*

$\iint_R (x + 2y^2)dA =$

- $\int_{-1}^1 \int_0^2 (x + 2y^2)dydx$
- $\int_0^2 \int_1^{-1} (x + 2y^2)dxdy$

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- $\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dxdy$
- $\int_{1}^{2}\int_{-1}^{0}(x+2y^2)dxdy$

Question No: 19

If $R=\{(x,y)/0\le x\le 2 \text{ and } 0\le y\le 3\}$, *then*

$\iint_R (1-ye^{xy})dA =$

- $\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dydx$
- $\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dxdy$
- $\int_{0}^{2}\int_{0}^{3}(4xe^{2y})dydx$
- $\int_{2}^{3}\int_{0}^{0}$
- $\int_{0}^{3}\int_{2}^{0}$

Question No: 20

If $R=\{(x,y)/0\le x\le 4 \text{ and } 0\le y\le 9\}$, *then*

$\iint_R (3x-4x\sqrt{xy})dA =$

- $\int_{0}^{9}\int_{0}^{4}(3x-4x\sqrt{xy})dydx$
- $\int_{0}^{4}\int_{0}^{9}(3x-4x\,xy)dxdy$
- $\int_{0}^{4}\int_{0}^{9}(3x-4x\sqrt{xy})dxdy$
- $\int_{0}^{4}\int_{0}^{9}(3x-4x\sqrt{xy})dydx$
- $\int_{4}^{9}\int_{0}^{0}$
- $\int_{0}^{9}\int_{0}^{4}$

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Question # 01:

Rectangular co-ordinates of a point are $\left(3,3,-3\sqrt{3}\right)$.

a. Convert Rectangular co-ordinates to Spherical co-ordinates
b. Convert Rectangular co-ordinates to Cylindrical co-ordinates.
c. Verify your answer by converting back Rectangular co-ordinates from any one of these, that is, either from Spherical co-ordinates or Cylindrical co-ordinates.

Solution:

Rectangular co-ordinates (x, y, z) to Spherical co-ordinates (ρ, θ, ϕ)

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{45}$$

$$\tan\theta = \frac{y}{x} = 1 \Rightarrow \theta = \frac{\pi}{4}$$

$$\cos\phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}} \Rightarrow \phi = Cos^{-1}\left(\frac{-3\sqrt{3}}{\sqrt{45}}\right) = Cos^{-1}\left(\frac{-\sqrt{3}}{\sqrt{5}}\right)$$

$$\left(\sqrt{45}, \frac{\pi}{4}, Cos^{-1}\left(\frac{-\sqrt{3}}{\sqrt{5}}\right)\right)$$

Rectangular co-ordinates (x, y, z) to Cylindrical co-ordinates (r, θ, z)

$$r = \sqrt{x^2 + y^2} = \sqrt{18}$$

$$\tan\theta = \frac{y}{x} = 1 \Rightarrow \theta = \frac{\pi}{4}$$

$$z = -3\sqrt{3}$$

$$\left(\sqrt{18}, \frac{\pi}{4}, -3\sqrt{3}\right)$$

Spherical co-ordinates (ρ, θ, ϕ) to Rectangular co-ordinates (x, y, z)

$$x = \rho\, Sin\phi\, Cos\,\theta = \sqrt{45}.\frac{\sqrt{2}}{\sqrt{5}}.\frac{1}{\sqrt{2}} = 3$$

$$y = \rho\ Sin\phi\, Sin\,\theta = \sqrt{45}.\frac{\sqrt{2}}{\sqrt{5}}.\frac{1}{\sqrt{2}} = 3$$

$$z = \rho\, Cos\,\phi = \sqrt{45}.\frac{-\sqrt{3}}{\sqrt{5}} = -3\sqrt{3}$$

$$\left(3, 3, -3\sqrt{3}\right)$$

Cylindrical co-ordinates (r, θ, z) to Rectangular co-ordinates (x, y, z)

$$x = r\cos\theta = \sqrt{18}.Cos\frac{\pi}{4} = \sqrt{18}.\frac{1}{\sqrt{2}} = \sqrt{9} = 3$$

$$y = r\sin\theta = \sqrt{18}.\frac{1}{\sqrt{2}} = 3$$

$$z = z = -3\sqrt{3}$$

Rectangular Co-ordinates	Spherical Co-ordinates	Cylindrical Co-ordinates
$(3,3,-3\sqrt{3})$	$\left(\sqrt{45}, Cos^{-1}(\frac{-\sqrt{3}}{\sqrt{5}}), \frac{\pi}{4}\right)$	$\left(\sqrt{18}, \frac{\pi}{4}, -3\sqrt{3}\right)$

Question # 02:

Describe the set of all points in xyz-coordinate system at which *f* is continuous.

$$f(x,y,z) = \sqrt{10-x}\ \ y\ln z$$

Solution:

$$f(x,y,z) = \sqrt{10-x}\ \ y\ln z$$

$$D = \{(x,y,z) : x \le 10, z > 0\}$$

Question # 03:

By considering different path approach, find whether

$$\lim_{(x,y)\to(0,0)} \frac{x^2}{3x^2 + 2y^2}$$ exist or not.

(Note: In order to get full marks, do all necessary steps)

Solution:

$$\lim_{(x,y)\to(0,0)} \frac{x^2}{3x^2+2y^2}$$

$$put\ y = mx$$

$$= \lim_{(x,mx)\to(0,0)} \frac{x^2}{3x^2+2m^2x^2}$$

$$= \lim_{(x,mx)\to(0,0)} \frac{x^2}{x^2(3+2m^2)}$$

$$= \frac{1}{3+2m^2}$$

$$\lim_{(x,y)\to(0,0)} \frac{x^2}{3x^2+2y^2}$$

$$put\ y = x$$

$$= \lim_{x\to 0} \frac{x^2}{5x^2} = \frac{1}{5}$$

Since along different paths we have different limits, hence the limit does not exist.

Question # 01 Marks = 10

Use chain rule to find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial \phi}$ if $w = x^2 + y^2 + z^2$ where

$$x = r\cos\theta\sin\phi$$
$$y = r\cos\theta\cos\phi$$
$$z = r\sin\theta$$

Solution:

To find $\frac{\partial w}{\partial r}$, using chain rule, we have

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$$

Substituting the values, we get

$$\frac{\partial w}{\partial r} = (2x)(\cos\theta\sin\phi) + (2y)(\cos\theta\cos\phi) + (2z)(\sin\theta)$$

Putting the values of x, y and z, it yields

$$\begin{aligned}\frac{\partial w}{\partial r} &= 2r\cos^2\theta\sin^2\phi + 2r\cos^2\theta\cos^2\phi + 2r\sin^2\theta \\ &= 2r\left[\cos^2\theta(\sin^2\phi + \cos^2\phi) + \sin^2\theta\right] \\ &= 2r\left(\cos^2\theta + \sin^2\theta\right) \\ &= 2r\end{aligned}$$

Similarly, for $\frac{\partial w}{\partial \phi}$, using chain rule, we get

$$\frac{\partial w}{\partial \phi} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial \phi} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial \phi} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial \phi}$$

Substituting the values, it becomes

$$\frac{\partial w}{\partial \phi} = (2x)(r\cos\theta\cos\phi) - (2y)(r\cos\theta\sin\phi) + (2z)(0)$$

Putting the values of x and y, it yields

$$\frac{\partial w}{\partial \phi} = 2r^2\cos^2\theta\sin\phi\cos\phi - 2r^2\cos^2\theta\cos\phi\sin\phi$$
$$= 0$$

Question # 02 Marks = 10

Find the equation of tangent plane to the surface $z = \ln\sqrt{x^2 + y^2}$ at the point ***P***(-1,0,0) .

Solution:

The given surface is $z = \ln\sqrt{x^2 + y^2}$ which can be written as

$$f(x, y, z) = \ln\sqrt{x^2 + y^2} - z = 0 \text{ -----(1)}$$

Taking partial derivatives of eq.(1) with respect to x, y and z, we get

$$f_x = \frac{x}{x^2 + y^2}$$
$$f_y = \frac{y}{x^2 + y^2}$$
$$f_z = -1$$

At P(-1,0,0), It becomes

$$f_x(-1,0,0) = -1$$
$$f_y(-1,0,0) = 0$$
$$f_z(-1,0,0) = -1$$

The general form of equation of tangent plane is

$$f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0) = 0$$

which becomes

$$-(x+1) + 0(y-0) - 1(z-0) = 0$$
$$x + z + 1 = 0$$

which is the required equation of tangent plane.

Question # 03 Marks = 05

Find the critical points of the function given as

$f(x, y) = x^3 - 3xy - y^3$

Solution:

The given function is $f(x, y) = x^3 - 3xy - y^3$

Taking partial derivatives of the given function with respect to x and y, we get

$$f_x = 3x^2 - 3y$$
$$f_y = -3x - 3y^2$$

For the critical points, put $f_x = f_y = 0$ which yields

$$x^2 - y = 0 \Rightarrow x^2 = y \quad ---- (1)$$
$$-x - y^2 = 0 \Rightarrow x = -y^2 \quad ----- (2)$$

Substituting eq. (2) in eq. (1), we get

$$y^4 - y = 0 \Rightarrow y(y^3 - 1) = 0 \Rightarrow y = 0,\ y = 1$$

Substituting these values of y in eq. (2), we get

$$x = 0,\ x = -1.$$

Thus, the required critical points are $(0,0)\ and\ (-1,1)$

(Note: In order to get full marks, do all necessary steps)

Question No: 1

If $f(x,y)=x^2+2y^2-x^2y$, find the local extrema and saddle points of f .

Solution:

$f(x,y)=x^2+2y^2-x^2y$

$f_x=2x-2xy$

$f_y=4y-x^2$

For critical points

$f_x=0=2x-2xy \Rightarrow 2x(1-y)=0 \Rightarrow x=0 \text{ or } y=1$

$f_y=0=4y-x^2 \Rightarrow x^2=4y$ ----------(1)

Using x=0 in (1) we have y=0

(0,0) is critical point.

Using y=1 in (1) we have

$x^2=4 \Rightarrow x=\pm 2$

(2,1) and (-2,1) are critical points.

Now we check on these three critical points for extremum.

$f_{xx}=2-2y$

$f_{yy}=4$

$f_{xy}=-2x$

$D=f_{xx}.f_{yy}-[f_{xy}]^2=(2-2y)(4)-[-2x]^2$

For (0,0) $D=f_{xx}.f_{yy}-[f_{xy}]^2=2(4)-0=8>0$

$f_{xx}=2>0$

So f is minimum at (0,0).

For (2,1) $D=f_{xx}.f_{yy}-[f_{xy}]^2=(0)(4)-[-4]^2=-16<0$

So (2,1) is a saddle point.

For (-2,1) $D=f_{xx}.f_{yy}-[f_{xy}]^2=(0)(4)-[4]^2=-16<0$

So (-2,1) is a saddle point.

Question No: 2

Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} (x+y)dydx$;a>0

Solution:

$$\int_0^a \int_0^{\sqrt{a^2-x^2}} (x+y)dydx$$

$$=\int_0^a (xy+\frac{y^2}{2})\,|_0^{\sqrt{a^2-x^2}}\ \mathrm{dx}$$

$$=\int_0^a (x\sqrt{a^2-x^2}+\frac{a^2-x^2}{2})\ \mathrm{dx}$$

$$=\frac{1}{-2}\int_0^a -2x\sqrt{a^2-x^2}\,dx\ +\frac{1}{2}\int_0^a (a^2-x^2)\mathrm{dx}$$

$$=\frac{1}{-2}\frac{(a^2-x^2)^{\frac{1}{2}+1}}{\frac{1}{2}+1}|_0^a\ +\frac{1}{2}\ (a^2x-\frac{x^3}{3})\,|_0^a$$

$$=\frac{1}{-2}\frac{(a^2-a^2)^{\frac{3}{2}}}{\frac{3}{2}}-\frac{1}{-2}\frac{(a^2-0)^{\frac{3}{2}}}{\frac{3}{2}}+\frac{1}{2}\ (a^3-\frac{a^3}{3})$$

$$=0+\frac{a^3}{3}\ +\frac{a^3}{3}=\frac{2a^3}{3}$$

Question No: 3

Evaluate $\int_{\sqrt{\pi}}^{\sqrt{2\pi}} \int_0^{x^2} \sin\frac{y}{x}\,dydx$

Solution:

$$\int_{\sqrt{\pi}}^{\sqrt{2\pi}} \int_{0}^{x^2} \sin\frac{y}{x}\,dy\,dx$$

$$= \int_{\sqrt{\pi}}^{\sqrt{2\pi}} -\frac{(\cos\frac{y}{x})}{\frac{1}{x}}\Big|_0^{x^2}\ \mathrm{dx}$$

$$= \int_{\sqrt{\pi}}^{\sqrt{2\pi}} (x - x\cos x)\ \mathrm{dx}$$

$$= \int_{\sqrt{\pi}}^{\sqrt{2\pi}} x\,dx - \int_{\sqrt{\pi}}^{\sqrt{2\pi}} (x\cos x)\,dx$$

$$= \frac{x^2}{2}\Big|_{\sqrt{\pi}}^{\sqrt{2\pi}} - [x\sin x|_{\sqrt{\pi}}^{\sqrt{2\pi}} - \int_{\sqrt{\pi}}^{\sqrt{2\pi}} \sin x\ \mathrm{dx}]$$

$$= \frac{2\pi}{2} - \frac{\pi}{2} - [x\sin x|_{\sqrt{\pi}}^{\sqrt{2\pi}} - (-\cos x)|_{\sqrt{\pi}}^{\sqrt{2\pi}}]$$

$$= \frac{2\pi}{2} - \frac{\pi}{2} - [\sqrt{2\pi}\sin\sqrt{2\pi} - \sqrt{\pi}\sin\sqrt{\pi} + \cos\sqrt{2\pi} - \cos\sqrt{\pi}]$$

$$= \frac{\pi}{2} - \sqrt{2\pi}\sin\sqrt{2\pi} + \sqrt{\pi}\sin\sqrt{\pi} - \cos\sqrt{2\pi} + \cos\sqrt{\pi}$$

(Note: In order to get full marks, do all necessary steps)

Question No: 1 Marks =10

Evaluate the following integral by converting it to polar coordinates.

$$\iint_R \frac{1}{1+x^2+y^2} dA$$

where R is the sector in the first quadrant bounded by $y = 0$ and $y = x$ and $x^2 + y^2 = 4$.

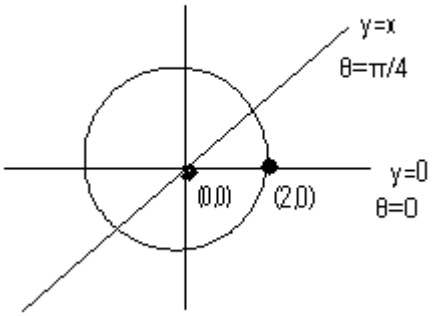
Solution:

Given integral is

$$\iint_R \frac{1}{1+x^2+y^2} dA$$

For conversion to polar coordinates, we have

$$x^2 + y^2 = r^2 \Rightarrow \frac{1}{1+x^2+y^2} = \frac{1}{1+r^2}$$



r ranges from 0 to 2

θ ranges from 0 to $\frac{\pi}{4}$

$dA = rdrd\theta$

Thus, the integral in polar coordinates becomes

$$\int_0^{\pi/4} \int_0^2 \frac{1}{1+r^2} rdrd\theta$$

Multiplying and dividing by 2

$$\frac{1}{2}\int_0^{\pi/4}\int_0^{2}\frac{2r}{1+r^2}drd\theta$$

$$=\frac{1}{2}\int_0^{\pi/4}\left[\ln(1+r^2)\right]_0^2 d\theta \qquad \text{[By the rule of integration } \int\frac{f'(x)}{f(x)}dx=\ln f(x)\text{]}$$

$$=\frac{1}{2}\int_0^{\pi/4}(\ln 5-\ln 1)d\theta$$

$$=\frac{\ln 5}{2}\int_0^{\pi/4}d\theta$$

$$=\frac{\ln 5}{2}[\theta]_0^{\pi/4}$$

$$=\frac{\ln 5}{2}[\frac{\pi}{4}-0]$$

$$=\frac{\pi\ln 5}{8}$$

which is the required answer.

Question No: 2 Marks =10

Find the arc length of the curve

$$\vec{r}(t)=(\cos^3 t)\hat{i}+(\sin^3 t)\hat{j}+2\hat{k} \qquad 0\le t\le \pi/2$$

Solution:

The given position vector is

$$\vec{r}(t)=(\cos^3 t)\hat{i}+(\sin^3 t)\hat{j}+2\hat{k} \text{ where } 0\le t\le \pi/2$$

In parametric form,

$$x=\cos^3 t,\ \ y=\sin^3 t,\ \ z=2$$
$$\Rightarrow \frac{dx}{dt}=-3\cos^2 t\sin t,\ \ \frac{dy}{dt}=3\sin^2 t\cos t,\ \ \frac{dz}{dt}=0$$

The arc length of the curve is

$$s=\int_0^{\pi/2}\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2}\,dt$$
$$=\int_0^{\pi/2}\sqrt{9\cos^4 t\sin^2 t+9\sin^4 t\cos^2 t+0}\,dt$$
$$=\int_0^{\pi/2}\sqrt{9\cos^2\sin^2 t(\cos^2 t+\sin^2 t)}\,dt$$
$$=3\int_0^{\pi/2}\cos t\sin t dt$$
$$=\left[\frac{3\sin^2 t}{2}\right]_0^{\pi/2}$$
$$=\frac{3}{2}\left[\sin^2\frac{\pi}{2}-\sin^2 0\right]$$
$$=\frac{3}{2}$$

which is the required answer.

(Note: In order to get full marks, do all necessary steps)

Question # 1:

Evaluate the line integral

$\int_c (ydx-x^2dy)$, where C is the curve x=t, y=$\frac{1}{2}t^2$ *where* $0\le t\le 2$.

Solution:

$$\int_c (ydx - x^2 dy)$$

$$= \int_0^2 ydx - x^2 dy$$

$$= \int_0^2 [\frac{1}{2}t^2 - t^2.t]dt$$

$$= \frac{1}{2}\int_0^2 t^2 dt - \int_0^2 t^3 dt$$

$$= \frac{1}{2}.\frac{t^3}{3}|_0^2 - \frac{t^4}{4}|_0^2$$

$$= \frac{4}{3} - 4$$

$$= -\frac{8}{3}$$

Question # 2:

Use Green's theorem to evaluate the integral $\oint_c ((x^2 - y)dx + xdy)$, where C is the circle

$x^2 + y^2 = 4$ and c is oriented counterclockwise.

Solution:

$$\oint_c ((x^2 - y)dx + xdy)$$

Using Green's Theorem.

$$\oint_c \text{Pdx+Qdy} = -\iint_R (\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x})dxdy$$

$$\frac{\partial P}{\partial y} = -1, \frac{\partial Q}{\partial x} = 1$$

$$-\iint_R (-1-1)dA$$

$$= 2\iint_R dA$$

$$= 2\int_0^{2\pi}\int_0^2 rdrd\theta$$

$$= 4\int_0^{2\pi} d\theta$$

$$= 8\pi$$

Question # 3:

If $A = (x^4 - y^2z^2)i + (x^2 + y^2)j - x^3y^3z^3k$, determine curl of A at the point (1,4,-3).

Solution:

$$CurlA = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^4 - y^2z^2 & x^2 + y^2 & -x^3y^3z^3 \end{vmatrix}$$

$$CurlA = -3x^3y^2z^3\ \text{i} + (3x^2y^3z^3 - 2y^2z)\ \text{j} + (2\text{x}+2\text{yz}^2)\ \text{k}$$

$$CurlA = 1296i - 5088j + 74k \quad \text{at } (1, 4, -3)$$

Question No: 1 Marks =15

Determine a half range cosine series to represent the function given by

$$f(t) = 2t + 3 \quad 0 < t < 2$$
$$f(t) = f(t+4)$$

Solution:

To obtain a cosine series, i.e., a series with no sine term involved, we need an even function. Therefore, we assume the wave form to be symmetric about y-axis.

Now, to find the expressions for the Fourier coefficient, we have

$$a_\circ = \frac{4}{T}\int_0^{T/2} f(t)dt \quad \text{where T=4 is period of the given function.}$$
$$= \frac{4}{4}\int_0^{4/2} (2t+3)dt$$
$$= \left|\frac{2t^2}{2} + 3t\right|_0^2$$
$$= \left\{(2)^2 + 3(2)\right\} - \left\{(0)^2 + 3(0)\right\}$$
$$= 4 + 6 + 0$$
$$= 10$$

$$a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos n\omega t dt$$
$$= \frac{4}{4}\int_0^{4/2} (2t+3)\cos n\omega t dt$$

Integrating by parts, we get

$$a_n = \left|(2t+3)\frac{\sin n\omega t}{n\omega}\right|_0^2 - \int_0^2 2\frac{\sin n\omega t}{n\omega}dt$$

$$= \left|(2t+3)\frac{\sin n\omega t}{n\omega}\right|_0^2 - \frac{2}{n\omega}\left|\frac{(-\cos n\omega t)}{n\omega}\right|_0^2$$

$$= \frac{7\sin n\omega 2}{n\omega} + \frac{2}{n^2\omega^2}[\cos n\omega 2 - 1]$$

Since, $\omega = \frac{2\pi}{T} = \frac{\pi}{2}$

Therefore,

$$a_n = \frac{7\sin n\pi}{n\omega} + \frac{2}{n^2\omega^2}[\cos n\pi - 1]$$

$\sin n\pi = 0,$

$$\cos n\pi = \begin{cases} 1 & \text{(n is even)} \\ -1 & \text{(n is odd)} \end{cases}$$

So,

$$a_n = \begin{cases} 0 & \text{(n is even)} \\ \frac{-4}{n^2\omega^2} & \text{(n is odd)} \end{cases}$$

Thus,

$$f(t) = \frac{a_\circ}{2} + \sum_{n=1}^{\infty} a_n \cos n\omega t$$

$$= 5 - \frac{4}{\omega^2}\left\{\cos \omega t + \frac{1}{9}\cos 3\omega t + \frac{1}{25}\ldots\right\}$$

which is the required fourier series.

Question No: 2 **Marks =15**

Verify Stokes' theorem for a hemisphere S defined as $x^2 + y^2 + z^2 = 9 \quad (z \geq 0)$ where a vector field $F = z^2\hat{i} + 2x\hat{j} - y\hat{k}$ exists over the surface and around its boundary c.

Solution:

Stokes' theorem states that

$$\int_S CurlF.ds = \oint_c F.dr$$

(a)

$$\oint_c F.dr = \oint_c (z^2\hat{i} + 2x\hat{j} - y\hat{k}).(\hat{i}dx + \hat{j}dy + \hat{k}dz)$$

$$= \oint_c z^2dx + 2xdy - ydz$$

For converting it to polar coordinates,

$x = r\cos\theta, \quad y = r\sin\theta, \quad z = 0$

where r=3 is the radius of the sphere.

$x = 3\cos\theta, \quad y = 3\sin\theta, \quad z = 0$

$dx = -3\sin\theta d\theta, \quad dy = 3\cos\theta d\theta, \quad dz = 0$

$0 \leq \theta \leq 2\pi$

Thus, the above integral becomes,

$$\int_0^{2\pi} 2(3\cos\theta)(3\cos\theta d\theta)$$

$$= 18\int_0^{2\pi} \cos^2\theta d\theta$$

$$= 18\int_0^{2\pi} \frac{1+\cos 2\theta}{2} d\theta$$

$$= 9\left[\int_0^{2\pi} d\theta + \int_0^{2\pi} \cos 2\theta d\theta\right]$$

$$= 9\left[\theta\right]_0^{2\pi} + 9\left[\frac{\sin 2\theta}{2}\right]_0^{2\pi}$$

$$= 9(2\pi - 0) + \frac{9}{2}(\sin 4\pi - \sin 0)$$

$$\oint_c F.dr = 18\pi \quad \text{Eq.(A)}$$

(*b*)

$$curlF = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 & 2x & -y \end{vmatrix}$$

$$= -\hat{i} + 2z\hat{j} + 2\hat{k}$$

$$\int_s curlF.d\vec{s} = \int_s curlF.\hat{n}ds$$

where

$$\hat{n} = \frac{\nabla s}{|\nabla s|} = \frac{2x\hat{i} + 2y\hat{j} + 2z\hat{k}}{\sqrt{4x^2 + 4y^2 + 4z^2}}$$

$$= \frac{x\hat{i} + y\hat{j} + z\hat{k}}{3}$$

So,

$$\int_s curlF.d\vec{s} = \int_s (-\hat{i} + 2z\hat{j} + 2\hat{k}).(\frac{x\hat{i} + y\hat{j} + z\hat{k}}{3})ds$$

$$= \frac{1}{3}\int_s (-x + 2yz + 2z)ds$$

For converting it to spherical coordinates, we have

$$x = 3\sin\phi\cos\theta$$
$$y = 3\sin\phi\sin\theta$$
$$z = 3\cos\phi$$
$$ds = 9\sin\phi d\phi d\theta$$

So, the above integral becomes,

$$\int_s curlF.d\vec{s}$$
$$= \frac{1}{3}\int_s \{-3\sin\cos\theta + 2(3\sin\phi\sin\theta)(3\cos\phi) + 6\cos\phi\}9\sin\phi d\phi d\theta$$
$$= 3\int_s \{-3\sin^2\phi\cos\theta + 18\sin^2\phi\sin\theta\cos\phi + 6\cos\phi\sin\phi\} d\phi d\theta$$
$$= 3\int_0^{2\pi}\int_0^{\pi/2} -3\sin^2\phi\cos\theta d\phi d\theta + 3\int_0^{2\pi}\int_0^{\pi/2} 18\sin^2\phi\sin\theta\cos\phi d\phi d\theta + 3\int_0^{2\pi}\int_0^{\pi/2} 6\cos\phi\sin\phi d\phi d\theta$$
$$= I_1 + I_2 + I_3 \quad \text{Eq.(1)}$$

$$I_1 = -9\int_0^{2\pi}\int_0^{\pi/2} \sin^2\phi\cos\theta d\phi d\theta$$
$$= -9\int_0^{2\pi}\int_0^{\pi/2} \frac{1-\cos 2\phi}{2}\cos\theta d\phi d\theta$$
$$= -9\int_0^{2\pi}\left[\frac{\phi}{2} - \frac{\sin 2\phi}{4}\right]_0^{\pi/2}\cos\theta d\theta$$
$$= -9\int_0^{2\pi}\left[\frac{\pi}{4} - \frac{\sin\pi}{4}\right]\cos\theta d\theta$$
$$= \frac{-9\pi}{4}\int_0^{2\pi}\cos\theta d\theta$$
$$= \frac{-9\pi}{4}\left[\sin\theta\right]_0^{2\pi}$$
$$= \frac{-9\pi}{4}(0)$$
$$I_1 = 0 \qquad \text{Eq.(2)}$$

$$I_2 = 54\int_0^{2\pi}\int_0^{\pi/2} \sin^2\phi \sin\theta\cos\phi d\phi d\theta$$

$$= 54\int_0^{2\pi}\left[\frac{\sin^3\phi}{3}\right]_0^{\pi/2} \sin\theta d\theta$$

$$= \frac{54}{3}\int_0^{2\pi}\left[\sin^3\frac{\pi}{2} - \sin 0\right]\sin\theta d\theta$$

$$= 18\int_0^{2\pi} \sin\theta d\theta$$

$$= -18\left[\cos\theta\right]_0^{2\pi}$$

$$= -18\left[\cos 2\pi - \cos 0\right]$$

$$= -18(1-1) = 0 \quad \text{Eq.(3)}$$

$$I_3 = 18\int_0^{2\pi}\int_0^{\pi/2} \cos\phi \sin\phi d\phi d\theta$$

$$= 18\int_0^{2\pi}\left[\frac{\sin^2\phi}{2}\right]_0^{\pi/2} d\theta$$

$$= 9\int_0^{2\pi}(1-0)d\theta$$

$$= 9\left[\theta\right]_0^{2\pi}$$

$$= 18\pi \quad \text{Eq.(4)}$$

Substituting the values of eqs.(2), (3), (4) in eq. (1), we get

$$\int_s curlF.d\vec{s} = 18\pi \quad \text{Eq.(B)}$$

Hence, from the equality of Eq. (A) and (B), Stokes' theorem is verified.

(Note: In order to get full marks, do all necessary steps)

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Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

Write down the equation of a plane passing through the point $(2, -3, 1)$ and perpendicular to the vector $\vec{n} = 16\,\hat{i} + 6\,\hat{j} + 6\,\hat{k}$.

Answer (Please click here to Add Answer)

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Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

State the condition when $\iint_R f(x,y)dA = \int_c^d \int_a^b f(x,y)dxdy = \int_a^b \int_c^d f(x,y)dydx$ where R is the region of integration.

Answer (Please click here to Add Answer)

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Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

Find the critical points for the function. $f(x,y)$ where as $f_x = 2x + y$ and $f_y = 2y + x$.

Answer (Please click here to Add Answer)

Made By: Waqar Siddhu

Question No : 24 of 26 Marks: 3 (Budgeted Time 6 Min)

Evaluate the following double integral.

$$\iint \left(3 + 2x - 3y^2\right) dx\, dy$$

Answer (Please click here to Add Answer)

Made By: Waqar Siddhu

Question No : 25 of 26 Marks: 5 (Budgeted Time 10 Min)

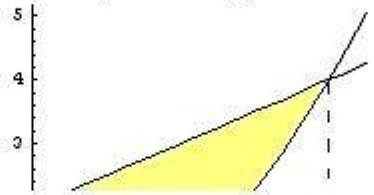
Consider a function $f(x,y) = x^3 - 3xy + y^3$. One of its critical point is $(1, 1)$. Find, whether $f(x,y)$ has relative maxima, relative minima or saddle point at $(1, 1)$.

Answer (Please click here to Add Answer)

Made By: Waqar Siddhu

Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)

Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ and $y = x + 2$.



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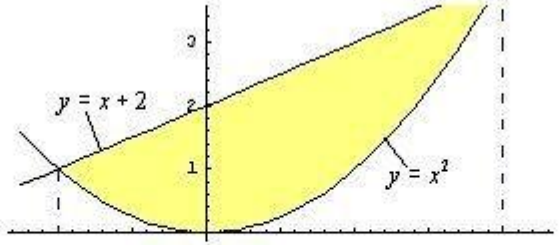
Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)



Answer (Please click here to Add Answer)

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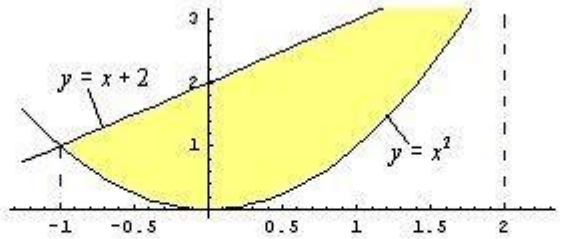
Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)



Answer (Please click here to Add Answer)

Made By: Waqar Siddhu

Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

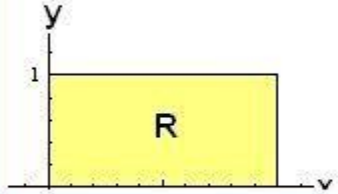
Given that $f_{xx} = -6x$, $f_{yy} = -6y$, $f_{xy} = 3$ for some $f(x,y)$. Check whether $f(x,y)$ has relative maximum, relative minimum or a saddle point at P(1,1).

Answer (Please click here to Add Answer)

Made By: Waqar Siddhu

Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

Let the function $f(x,y)$ is continuous in the region R, where R is a rectangle as shown below.



Answer (Please click here to Add Answer)

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Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

Let $y = 3x^2 - 5$. If x changes from 2 to 2.1, find the approximate change in the value of y using differential dy.

Answer (Please click here to Add Answer)

Made By: Waqar Siddhu

Question No : 24 of 26 Marks: 3 (Budgeted Time 6 Min)

If the order of integration for the integral $\int_0^1 \int_y^1 e^{x^2} dx dy$ is changed. Find the change in the limits of new integral.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

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Question No : 25 of 26 Marks: 5 (Budgeted Time 10 Min)

Find equation of normal line (in parametric form) to the surface $f(x, y, z) = xy + 2yz - xz^2 + 10$ at the point $(-5, 5, 1)$.

Answer (Please click here to Add Answer)

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Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)

Evaluate $\int_1^4 \int_0^2 (6x^2 + 4xy^3)\,dx\,dy$

Answer (Please click here to Add Answer)

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FINALTERM EXAMINATION

Spring 2010
MTH301- Calculus II

Time: 90 min
Marks: 60

Student Info	
Student ID:	
Center:	
Exam Date:	

For Teacher's Use Only

Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

Question No: 1 (Marks: 1) - Please choose one

Intersection of two straight lines is ----------

- Surface
- Curve
- Plane
- ***Point***

Question No: 2 (Marks: 1) - Please choose one

Plane is a ---------- surface.

- One-dimensional
- ***Two-dimensional***
- Three-dimensional
- Dimensionless

Question No: 3 (Marks: 1) - Please choose one

Let w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$\frac{\partial w}{\partial r} =$

- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

Question No: 4 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t)=\sin^2 t\,\hat{i}+(1-\cos 2t)\,\hat{j}$

- $x=\sin^2 t \quad , \quad y=1-\cos 2t \quad , \quad z=0$
- $y=\sin^2 t \quad , \quad x=1-\cos 2t \quad , \quad z=0$

▶ $x = \sin^2 t \quad , \quad y = 1 - \cos 2t \quad , \quad z = 1$

▶ $x = \sin^2 t \quad , \quad y = \cos 2t \quad , \quad z = 1$

Question No: 5 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$r(t) = (2t - 1)\hat{i} - 3\sqrt{t}\,\hat{j} + \sin 3t\,\hat{k}$

▶ $z = 2t - 1 \quad , \quad x = -3\sqrt{t} \quad , \quad y = \sin 3t$

▶ $y = 2t - 1 \quad , \quad x = -3\sqrt{t} \quad , \quad z = \sin 3t$

▶ $x = 2t - 1 \quad , \quad z = -3\sqrt{t} \quad , \quad y = \sin 3t$

▶ $x = 2t - 1 \quad , \quad y = -3\sqrt{t} \quad , \quad z = \sin 3t$

Question No: 6 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = (\cos 5t,\ \tan t\ ,\ 6\sin t)$

▶ $\vec{r}'(t) = \left(\frac{\sin 5t}{5},\ \sec t\ ,\ 6\cos t \right)$

- $\vec{r}'(t) = (\frac{-\sin 5t}{5}, \sec t\ , 6\cos t)$

- $\vec{r}'(t) = (-5\sin 5t, \sec^2 t\ , 6\cos t)$

- $\vec{r}'(t) = (\sin 5t, \sec^2 t\ , -6\cos t)$

Question No: 7 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(t^4, \sqrt{t+1}\ , \frac{3}{t^2} \right)$

- $\vec{r}'(t) = \left(4t^3, \frac{1}{\sqrt{t+1}}\ , \frac{-6}{t^3} \right)$

- $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}\ , \frac{6}{t^3} \right)$

- $\vec{r}'(t) = \left(4t^4, \frac{1}{2\sqrt{t+1}}\ , \frac{-6}{t^3} \right)$

- $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}\ , \frac{-6}{t^3} \right)$

Question No: 8 (Marks: 1) - Please choose one

The following differential is exact

$dz = (x^2 y + y)\ dx - x\ dy$

▶ True

▶ **False**

Question No: 9 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots\cdots\cdots \frac{5}{6} \frac{3}{4} \frac{1}{2}$ page #182

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots\cdots\cdots \frac{6}{7} \frac{4}{5} \frac{2}{3}$

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \cdots\cdots\cdots \frac{6}{5} \frac{4}{3} \frac{2}{1}$

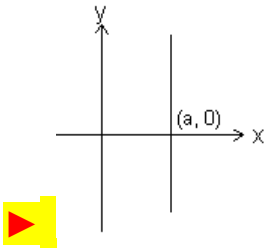
▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \cdots\cdots\cdots \frac{6}{5} \frac{4}{3} \frac{2}{1}$

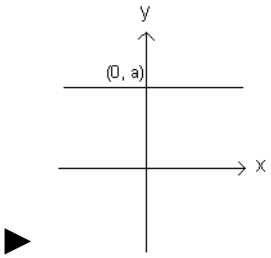
Question No: 10 (Marks: 1) - Please choose one

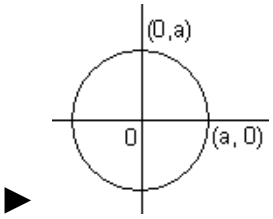
Match the following equation in polar co-ordinates with its graph.

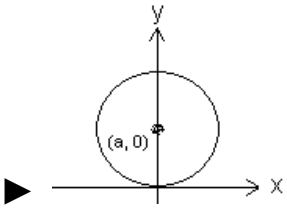
$r\cos\theta = a$

where a is an arbitrary cons tan*t*









Question No: 11 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

► Initial line

► **Y-axis**

► Pole

Question No: 12 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(-r, \theta)$ then the curve is said to be symmetric about which of the following?

► Initial line

► y-axis

► ***Pole***

Question No: 13 (Marks: 1) - Please choose one

What is the amplitude of a periodic function defined by $f(x) = \sin\frac{x}{3}$?

► **0**

► 1

- $\frac{1}{3}$
- Does not exist

Question No: 14 (Marks: 1) - Please choose one

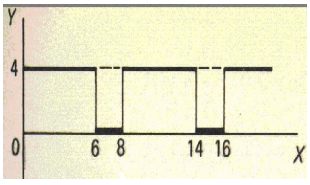
What is the period of a periodic function defined by $f(x) = 4\cos 3x$?

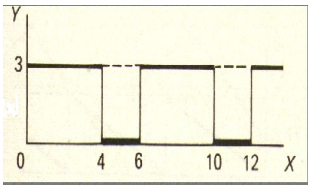
- $\frac{\pi}{4}$
- $\frac{\pi}{3}$
- $\frac{2\pi}{3}$
- π

Question No: 15 (Marks: 1) - Please choose one

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 3 & 0 < x < 4 \\ 0 & 4 < x < 6 \end{cases}$$

- 

► 

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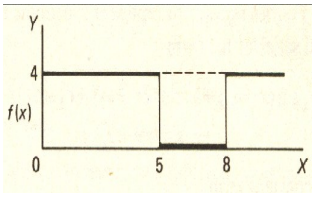
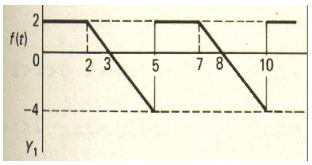
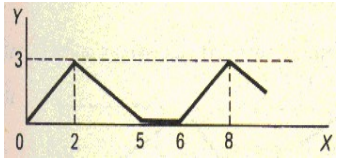
► 

► 

Question No: 16 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 2

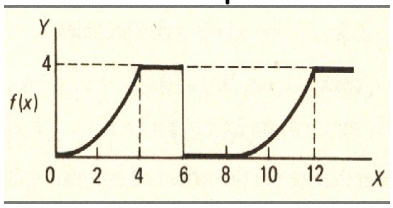
► 5

► 6

► 8

Question No: 17 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



- ▶ 0
- ▶ 4
- ▶ 6
- ▶ **8**

Question No: 18 (Marks: 1) - Please choose one

Let L denotes the Laplace Transform.

If $L\{F(t)\} = f(s)$ where s is a constant.and $\lim_{t\to 0}\left(\frac{F(t)}{t}\right)$ exists then which of the following equation holds?

- ▶ $L\left(\frac{F(t)}{t}\right) = f(s+a)$
- ▶ $L\left(\frac{F(t)}{t}\right) = f(s-a)$

- $L\left(\frac{F(t)}{t}\right)=\int_{s}^{\infty} f(s)\,ds$

- $L\left(\frac{F(t)}{t}\right)=-\frac{d}{ds}\{f(s)\}$

Question No: 19 (Marks: 1) - Please choose one

Which of the following is Laplace inverse transform of the function $f(s)$ defined by $f(s)=\frac{3}{s-2}-\frac{2}{s}$?

- $3te^{2t}-2$
- $3e^{2t}-2t$
- $3e^{2t}-2$
- None of these.

Question No: 20 (Marks: 1) - Please choose one

Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be any two points in three dimensional space. What does the formula $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$ calculates?

- Distance between these two points

► **Midpoint of the line joining these two points**

► Ratio between these two points

Question No: 21 (Marks: 1) - Please choose one

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane. If $(P\,dx + Q\,dy)$ is an exact differential then

$\oint_C (P\,dx + Q\,dy) =$

► **Zero**

► One

► Infinite

Question No: 22 (Marks: 1) - Please choose one

What is Laplace transform of the function $F(t)$ if $F(t) = t$?

► $L\{t\} = \frac{1}{s}$

► $L\{t\} = \frac{1}{s^2}$

- $L\{t\} = e^{-s}$
- $L\{t\} = s$

Question No: 23 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

- $L\{e^{5t}\} = \frac{1}{s-5}$
- $L\{e^{5t}\} = \frac{s}{s^2+25}$
- $L\{e^{5t}\} = \frac{5}{s^2+25}$
- $L\{e^{5t}\} = \frac{5!}{s^6}$

Question No: 24 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx+(2x-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

- 1
- 0

► 2

► -2

Question No: 25 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx+(x^2-y)\,dy$ where C is the line segment from (0, 0) to (2, 0).

► 0

► -4

► 4

► Do not exist

Question No: 26 (Marks: 1) - Please choose one

Which of the following are direction ratios for the line joining the points $(1, 3, 5)$ and $(2, -1, 4)$?

► 3, 2 and 9

► 1, -4 and -1

► 2, -3 and 20

► 0.5, -3 and 5/4

Question No: 27 (Marks: 1) - Please choose one

$If\ R = \{(x,y)/0 \le x \le 2\ and\ 1 \le y \le 4\},\ then$
$\iint_R (6x^2 + 4xy^3)dA =$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$

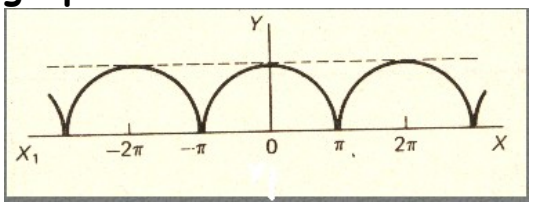
► $\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$

► $\int_2^4 \int_0^1 (6x^2 + 4xy^3)dxdy$

Question No: 28 (Marks: 1) - Please choose one

Which of the following is true for a periodic function whose graph is as below?

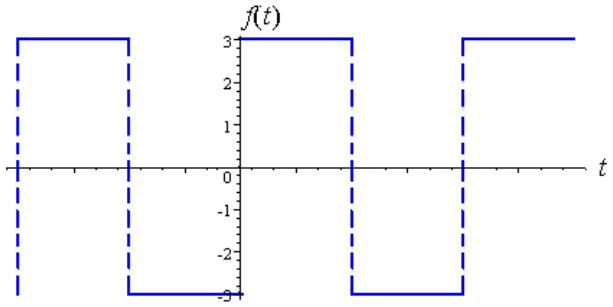


► **Even function**

► Odd function

► Neither even nor odd function

Question No: 29 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above.

► **An odd function**

► An even function

► Neither even nor odd

Question No: 30 (Marks: 1) - Please choose one

At each point of domain, the function ----------

► **Is defined**

► Is continuous

► Is infinite

► Has a limit

Question No: 31 (Marks: 2)

Determine whether the following differential is exact or not.

$dz = 4x^3y^3\ dx + 3x^4y^2\ dy$

Solution:

$$dz = 4x^3y^3\ dx + 3x^4y^2\ dy$$

$$\frac{\partial p}{\partial y} = 12x^3y^2$$

$$\frac{\partial Q}{\partial X} = 12x^3y^2$$

$$\frac{\partial p}{\partial y} = \frac{\partial Q}{\partial X}$$

yes

Question No: 32 (Marks: 2)

Evaluate

$$\int_{-\pi}^{\pi} \sin\ nx\ dx$$

where n is an integer other than zero.

Solution:

$$\int_{-\pi}^{\pi} \sin\ nx\ dx$$

$$= \left[\frac{-\cos nx}{n}\right]_{-\pi}^{\pi}$$

$$= \left[\frac{-\cos n\pi}{n} + \frac{\cos n\pi}{n}\right]$$

$$= \frac{1}{n}(-\cos n\pi + \cos n\pi)$$

$$= 0$$

Question No: 33 (Marks: 2)

Find Laplce transform of the function $F(t)$ if $F(t) = e^{3t}$

Solution:

$$L(e^{3t}) = \int_0^{\infty} e^{3t} - e^{-st}$$

$$= \int_0^{\infty} e^{-(s-3)t}.dt$$

$$= \{\frac{e^{-(s-3)t}}{-(s-3)}\} \lim 0 - \infty$$

$$= \frac{-1}{s-3}(\frac{1}{e^{(s-3)t}})$$

$$= \frac{-1}{s-3}(0-1)$$

$$= \frac{1}{s-3} \text{.....Ans}$$

Question No: 34 (Marks: 3)

Determine the Fourier co-efficient a_0 of the periodic function defined below:

$f(x) = 2x+1 \qquad 0 < x < 2$

Solution:

$a_\circ = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx$

$f(x) = (2x+1)$

$(0,2)$

$= \int_0^2 (2x+1)dx$

$= \left[x^2 + x\right]_0^2$

$= 6$

Question No: 35 (Marks: 3)

Determine whether the following differential is exact or not.

$dz = (3x^2e^{2y} - 2y^2e^{3x})\,dx + (2x^3e^{2y} - 2ye^{3x})\,dy$

Solution:

dz = Pdx + Qdy

Therefore,

For dz to be an exact differential it must satisfy $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

But this test fails becuase $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$

Not Exact

Question No: 36 (Marks: 3)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$

Solution:

$$\int_0^{\frac{\pi}{2}} \sin^3 x dx$$

$$= \frac{n-1}{n}.$$

$$= \frac{3-1}{3}$$

$$= \frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \sin^5 x dx$$

$$= \frac{n-1}{n}.\frac{n-3}{n-2}$$

$$= \frac{5-1}{5}.\frac{5-3}{5-2}$$

$$= \frac{4}{5}.\frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$$

$$= \frac{2}{3} + \frac{4}{5}.\frac{2}{3}$$

Question No: 37 (Marks: 5)

Evaluate the following line integral which is independent of path.

$$\int_{(0,0)}^{(3,2)} (2xe^y)\, dx + (x^2e^y)\, dy$$

Solution:

$$p = \frac{\partial z}{\partial x} = 2e^y \qquad \int 2e^y dx$$

$$Q = \frac{\partial z}{\partial y} = x^2 e^y \qquad \int x^2 e^y dy$$

$$z = \int_{(0,0)}^{(3,2)} 2xe^y + x^2 ye^y$$

$$z = 6e^2 + 18e^2$$

$$z = 24e^2$$

Question No: 38 (Marks: 5)

Determine the Fourier coefficients b_n for a periodic function $f(t)$ of period 2 defined by

$$f(t) = \begin{cases} 4\ (1 + t) & -1 < t < 0 \\ 0 & 0 < t < 1 \end{cases}$$

Solution:

$$bn = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx dx$$

$$= \frac{1}{\pi}\int_{-1}^{1} 4(1+t)\sin nx dx$$

$$= \frac{1}{\pi}\left[\frac{-4(1+t)\cos nx}{n}\right]_{-1}^{1}$$

$$= \frac{-4(1+t)}{\pi n}\left[\cos n(1) - \cos n(-1)\right]$$

$$= \frac{-4(1+t)}{\pi n}(\cos n + \cos n)$$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$$\vec{F}(x,\ y,\ z) = (4x - z)\hat{i} + (3y + z)\hat{j} + (y - x)\hat{k}$$

……

ASSALAM O ALAIKUM
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FINALTERM EXAMINATION
Spring 2010
MTH301- Calculus II (Session - 2)

Time: 90 min
Marks: 60

Student Info	
StudentID:	$$
Center:	OPKST
ExamDate:	19 Aug 2010

For Teacher's Use Only									
Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

Question No: 1 (Marks: 1) - Please choose one

---------- planes intersect at right angle to form three dimensional space.

- **Three**
- Four
- Eight
- Twelve

Question No: 2 (Marks: 1) - Please choose one

If the positive direction of x, y axes are known then ---------- the positive direction of z-axis.

- Horizontal rightward direction is
- Vertical upward direction is
- **Left hand rule tells**
- Right hand rule tells

Question No: 3 (Marks: 1) - Please choose one

What is the distance between points (3, 2, 4) and (6, 10, -1)?

- $7\sqrt{2}$
- $2\sqrt{6}$
- $\sqrt{34}$
- $7\sqrt{3}$

Question No: 4 (Marks: 1) - Please choose one

The equation $ax+by+cz+d=0$, where a,b,c,d are real numbers, is the general equation of which of the following?

- ► **Plane**
- ► Line
- ► Curve
- ► Circle

Question No: 5 (Marks: 1) - Please choose one

The spherical co-ordinates of a point are $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$. What are its cylinderical co-ordinates?

- ► $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}, 0\right)$
- ► $\left(\sqrt{3}\cos\frac{\pi}{3}, \sqrt{3}\sin\frac{\pi}{3}, 0\right)$
- ► $\left(\sqrt{3}\sin\frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3}\cos\frac{\pi}{3}\right)$
- ► $\left(\sqrt{3}, \frac{\pi}{3}, 0\right)$

Question No: 6 (Marks: 1) - Please choose one

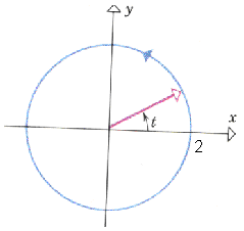
Domain of the function $f(x, y)=\sqrt{y-x^2}$ is

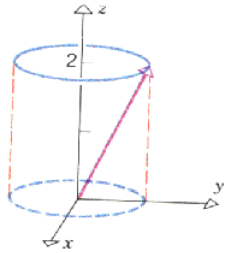
- ► $y < x^2$
- ► $y \geq x^2$
- ► $y \neq x^2$
- ► Entire space

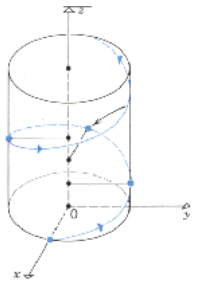
Question No: 7 (Marks: 1) - Please choose one

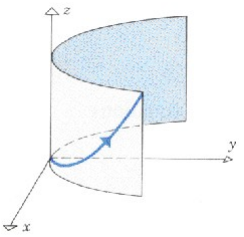
Match the following vector-valued function with its graph.

$r(t) = \cos t\,\hat{i} + \sin t\,\hat{j} + 2\hat{k}$ and $0 \le t \le 2\pi$

▶ 

▶ 

▶ 

▶ 

Question No: 8 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1 - \cos 2t)\,\hat{j}$

▶ $x = \sin^2 t \quad , \quad y = 1 - \cos 2t \quad , \quad z = 0$

▶ $y = \sin^2 t \quad , \quad x = 1 - \cos 2t \quad , \quad z = 0$

- $x = \sin^2 t \quad , \quad y = 1 - \cos 2t \quad , \quad z = 1$
- $x = \sin^2 t \quad , \quad y = \cos 2t \quad , \quad z = 1$

Question No: 9 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$r(t) = (2t-1)\hat{i} - 3\sqrt{t}\,\hat{j} + \sin 3t\,\hat{k}$

- $z = 2t-1 \quad , \quad x = -3\sqrt{t} \quad , \quad y = \sin 3t$
- $y = 2t-1 \quad , \quad x = -3\sqrt{t} \quad , \quad z = \sin 3t$
- $x = 2t-1 \quad , \quad z = -3\sqrt{t} \quad , \quad y = \sin 3t$
- $x = 2t-1 \quad , \quad y = -3\sqrt{t} \quad , \quad z = \sin 3t$

Question No: 10 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t = 1$? If not, why?

$\vec{r}(t) = \left(\frac{t+1}{t-1}, t^2, 2t \right)$

- $\vec{r}(t)$ is continuous at $t = 1$
- $\vec{r}(1)$ is not defined
- $\vec{r}(1)$ is defined but $\lim_{t \to 1} \vec{r}(t)$ does not exist
- **$\vec{r}(1)$ is defined and $\lim_{t \to 1} \vec{r}(t)$ exists but these two numbers are not equal.**

Question No: 11 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

- $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}\cdots\cdots\cdots\frac{5}{6}\frac{3}{4}\frac{1}{2}$
- $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}\cdots\cdots\cdots\frac{6}{7}\frac{4}{5}\frac{2}{3}$
- $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}\cdots\cdots\cdots\frac{6}{5}\frac{4}{3}\frac{2}{1}$
- $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}\cdots\cdots\cdots\frac{6}{5}\frac{4}{3}\frac{2}{1}$

Question No: 12 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Cosine formula when n is odd and $n \geq 3$?

- $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}\cdots\cdots\cdots\frac{5}{6}\frac{3}{4}\frac{1}{2}$
- $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}\cdots\cdots\cdots\frac{6}{5}\frac{4}{3}\frac{2}{1}$
- $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}\cdots\cdots\cdots\frac{6}{5}\frac{4}{3}\frac{2}{1}$
- $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}\cdots\cdots\cdots\frac{6}{7}\frac{4}{5}\frac{2}{3}$

Question No: 13 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

- Initial line
- **y-axis**
- Pole

Question No: 14 (Marks: 1) - Please choose one

If $a > 0$, then the equation, in polar co-ordinates, of the form $r^2 = a^2 \cos 2\theta$ represent which of the following family of curves?

► **Leminscate**

► Cardiods

► Rose curves

► Spiral

Question No: 15 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x) = \sin \frac{x}{2}$?

► $\frac{\pi}{2}$

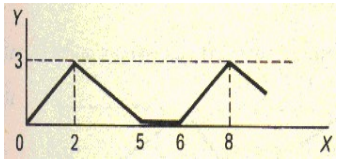
► π

► $\frac{3\pi}{2}$

► 4π

Question No: 16 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 2

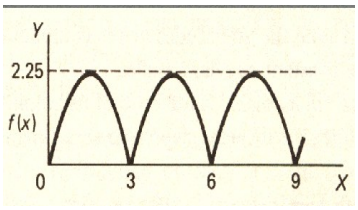
► 5

► **6**

► 8

Question No: 17 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 0

► 2.25

► 3

► 6

Question No: 18 (Marks: 1) - Please choose one

Let L denotes the Laplace Transform.

If $L\{F(t)\} = f(s)$ where s is a constant, then which of the following equation holds?

► $L\{t\ F(t)\} = -\frac{d}{ds}\{f(s)\}$

► $L\{t\ F(t)\} = f(s+t)$

► $L\{t\ F(t)\} = f(s)$

► $L\{t F(t)\} = \int_{s}^{\infty} f(s)\ ds$

Question No: 19 (Marks: 1) - Please choose one

The graph of an odd function is symmetrical about ----------

► x-axis

► y-axis

► **origin**

Question No: 20 (Marks: 1) - Please choose one

Consider the function $f(x, y, z) = \sqrt{1-x^2-y^2-z^2}$. What is the value of $f\left(0, \frac{1}{2}, \frac{1}{2}\right)$?

► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \sqrt{\frac{1}{2}}$

► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 2$

► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}$

► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 0$

Question No: 21 (Marks: 1) - Please choose one

The path of integration of a line integral must be ----------

► straight and single-valued

► **continuous and single-valued**

► straight and multiple-valued

► continuous and multiple-valued

Question No: 22 (Marks: 1) - Please choose one

Sign of line integral is reversed when ----------

► path of integration is divided into parts.

► path of integration is parallel to y-axis.

► **direction of path of integration is reversed.**

► path of integration is parallel to x-axis.

Question No: 23 (Marks: 1) - Please choose one

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane.

If $(P\,dx + Q\,dy)$ is an exact differential then

$\oint_C (P\,dx + Q\,dy) =$

► Zero

► **One**

► Infinite

Question No: 24 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

- $L\{e^{5t}\} = \frac{1}{s-5}$
- $L\{e^{5t}\} = \frac{s}{s^2+25}$
- $L\{e^{5t}\} = \frac{5}{s^2+25}$
- $L\{e^{5t}\} = \frac{5!}{s^6}$

Question No: 25 (Marks: 1) - Please choose one

What is laplace transform of the function $F(t)$ if $F(t) = \sin 3t$?

- $L\{\sin 3t\} = \frac{3}{s^2+9}$
- $L\{\sin 3t\} = \frac{s}{s^2+9}$
- $L\{\sin 3t\} = \frac{1}{s-3}$
- $L\{\sin 3t\} = \frac{3!}{s^4}$

Question No: 26 (Marks: 1) - Please choose one

If L denotes laplace transform then

$L\{te^{5t}\} =$

- $L\{te^{5t}\} = \frac{1}{s^2-5}$
- $L\{te^{5t}\} = \frac{1}{s^2+5}$
- $L\{te^{5t}\} = \frac{1}{(s+5)^2}$
- $L\{te^{5t}\} = \frac{1}{(s-5)^2}$

Question No: 27 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx+(2x-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

- ► 1
- ► 0
- ► 2
- ► **-2**

Question No: 28 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx+(x^2-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

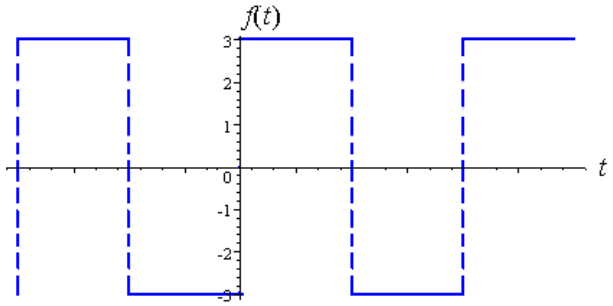
- ► -4
- ► **-2**
- ► 0
- ► 2

Question No: 29 (Marks: 1) - Please choose one

Divergence of a vector function is always a ----------

- ► Scalar
- ► **Vector**

Question No: 30 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above

► **An odd function**

► An even function

► Neither even nor odd

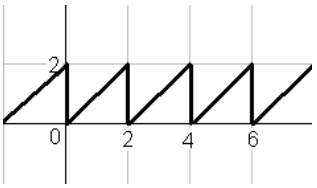
Question No: 31 (Marks: 2)

Does the following limit exist? If yes find its value, if no give reason

$$\lim_{t\to 0}\left[(e^{2t}+5)\hat{i}+(t^2+2t-3)\hat{j}+\left(\frac{1}{t}\right)\hat{k}\right]$$

Question No: 32 (Marks: 2)

Define the periodic function whose graph is shown below.



Question No: 33 (Marks: 2)

Find Laplace Transform of the function $F(t)$ if $F(t)=t^4$

Solution:

The Laplace transform of the given function will be:

$f(t) = t^4$

$L\{t^4\} = \frac{4!}{s^5}$

Question No: 34 (Marks: 3)

Determine whether the following differential is exact or not.

$dz = (4x^3y + 2xy^3)\ dx + (x^4 + 3x^2y^2)\ dy$

Question No: 35 (Marks: 3)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$

Solution:

$\int_0^{\frac{\pi}{2}} \sin^8 x dx = \frac{7}{8}.\frac{5}{6}.\frac{3}{4}.\frac{1}{2}---------\frac{\pi}{2}$

Question No: 36 (Marks: 3)

Find Laplace transform of the function $F(t)$ if

$F(t) = e^{2t} \sin 3t$

Solution:

Laplace transform will be

$L(t) = e^{2t}$.....1

$= \frac{1}{s-2}$

$L(t) = \sin 3t$.....2

$L(t) = \frac{a}{s^2 + 3^2}$

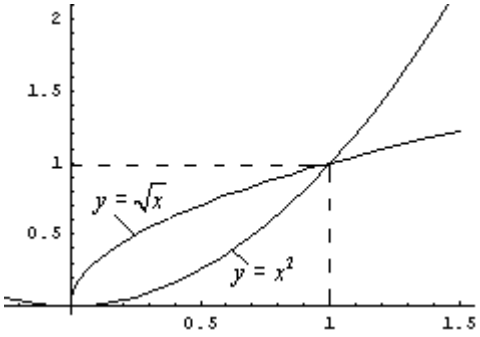
$L(t) = \frac{a}{s^2 + 9}$

Combining,

$L(t) = (\frac{1}{s-2})(\frac{a}{s^2+9})$

Question No: 37 (Marks: 5)

Using definite integral, find area of the region that is enclosed between the curves $y = x^2$ *and* $y = \sqrt{x}$



Question No: 38 (Marks: 5)

Determine the fourier co-efficient b_n of the following function.

$f(x) = x^2$ $\quad 0 < x < 2\pi$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (4x - z)\hat{i} + (3y + z)\hat{j} + (y - x)\hat{k}$

ASSALAM O ALAIKUM
All Dearz fellows
ALL IN ONE MTH301
Final term PAPERS & MCQz
Created BY Farhan & Ali
BS (cs) 2nd sem
Hackers Group
From Mandi Bahauddin
Remember us in your prayers

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FINALTERM EXAMINATION
Fall 2009

MTH301- Calculus II

Time: 120 min
Marks: 80

Question No: 1 (Marks: 1) - Please choose one

π is an example of ----------

- ► Irrational numbers
- ► Rational numbers
- ► Integers
- ► Natural numbers

Question No: 2 (Marks: 1) - Please choose one

Straight line is a special kind of ----------

- ► Surface
- ► Curve
- ► Plane

► Parabola

Question No: 3 (Marks: 1) - Please choose one

An ordered triple corresponds to ---------- in three dimensional space.

► A unique point

► A point in each octant

► Three points

► Infinite number of points

Question No: 4 (Marks: 1) - Please choose one

The angles which a line makes with positive x ,y and z-axis are known as ----------

► Direction cosines

► Direction ratios

► Direction angles

Question No: 5 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

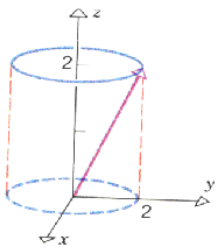
$f(x, y) = 4xy + \sin 3x^2 y$

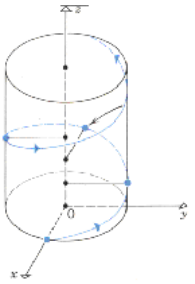
- ▶ $f(x, y)$ is continuous at origin
- ▶ $f(0, 0)$ is not defined
- ▶ $f(0, 0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- ▶ $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

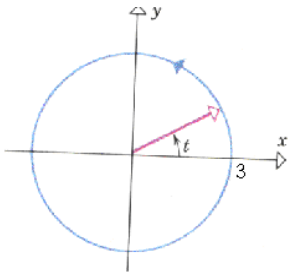
Question No: 6 (Marks: 1) - Please choose one

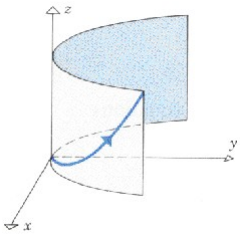
Match the following vector-valued function with its graph.

$r(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ and $0 \le t \le 2\pi$

▶ 

▶ 

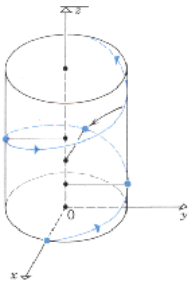
▶ 

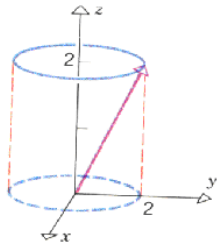
▶ 

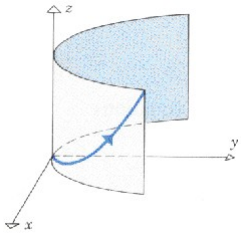
Question No: 7 (Marks: 1) - Please choose one

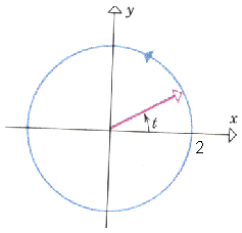
Match the following vector-valued function with its graph.

$r(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ and $t \geq 0$

▶ 

▶ 

▶ 

▶ 

Question No: 8 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$

▶ $x = \sin^2 t \quad , \quad y = 1 - \cos 2t \quad , \quad z = 0$

▶ $y = \sin^2 t \quad , \quad x = 1 - \cos 2t \quad , \quad z = 0$

▶ $x = \sin^2 t \quad , \quad y = 1 - \cos 2t \quad , \quad z = 1$

▶ $x = \sin^2 t \quad , \quad y = \cos 2t \quad , \quad z = 1$

Question No: 9 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t = 0$? If not, why?

$\vec{r}(t) = (4\cos t, \sqrt{t}\ , 4\sin t)$

▶ $\vec{r}(0)$ is not defined

▶ $\vec{r}(0)$ is defined but $\lim_{t \to 0} \vec{r}(t)$ does not exist

▶ $\vec{r}(0)$ is defined and $\lim_{t \to 0} \vec{r}(t)$ exists but these two numbers are not equal.

▶ $\vec{r}(t)$ is continuous at $t = 0$

Question No: 10 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = (\cos 5t,\ \tan t\ ,\ 6\sin t)$

- $\vec{r}'(t) = \left(\frac{\sin 5t}{5},\ \sec t\ ,\ 6\cos t\right)$
- $\vec{r}'(t) = (\frac{-\sin 5t}{5},\ \sec t\ ,\ 6\cos t)$
- **$\vec{r}'(t) = (-5\sin 5t,\ \sec^2 t\ ,\ 6\cos t)$**
- $\vec{r}'(t) = (\sin 5t,\ \sec^2 t\ ,\ -6\cos t)$

Question No: 11 (Marks: 1) - Please choose one

The following differential is exact

$dz = (3x^2 y + 2)\ dx + (x^3 + y)\ dy$

- **True**
- False

Question No: 12 (Marks: 1) - Please choose one

The following differential is exact

$dz = (3x^2 + 4xy)\ dx + (2x^2 + 2y)\ dy$

► True

► False

Question No: 13 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is odd and $n \geq 3$?

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \text{--------} \frac{5}{6} \frac{3}{4} \frac{1}{2}$

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \text{--------} \frac{6}{5} \frac{4}{3} \frac{2}{1}$

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \text{--------} \frac{6}{7} \frac{4}{5} \frac{2}{3}$

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \text{--------} \frac{6}{5} \frac{4}{3} \frac{2}{1}$

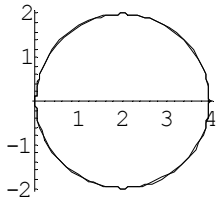
Question No: 14 (Marks: 1) - Please choose one

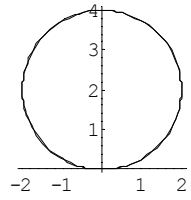
Which of the following is correct?

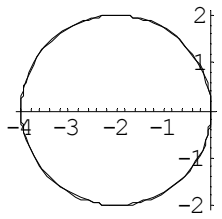
- $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{16}$
- $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3\pi}{16}$
- $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{8}$
- $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{2\pi}{3}$

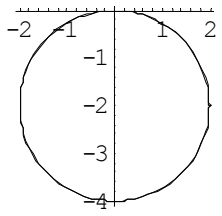
Question No: 15 (Marks: 1) - Please choose one

Match the following equation in polar co-ordinates with its graph.
$r = 4\sin\theta$

- 

- 

▶ 

▶ 

Question No: 16 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

▶ Initial line

▶ y-axis

▶ Pole

Question No: 17 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

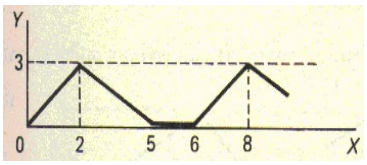
▶ $\frac{\pi}{2}$

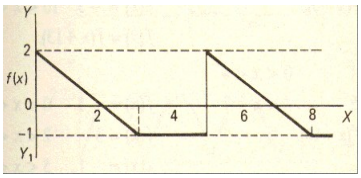
- π
- $\frac{3\pi}{2}$
- 4π

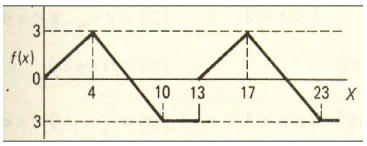
Question No: 18 (Marks: 1) - Please choose one

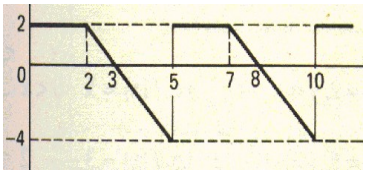
Match the following periodic function with its graph.

$$f(x)=\begin{cases}\frac{3}{4}x & 0<x<4\\ 7-x & 4<x<10\\ -3 & 10<x<13\end{cases}$$

- 

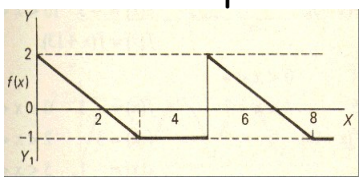
- 

- 

- 

Question No: 19 (Marks: 1) - Please choose one

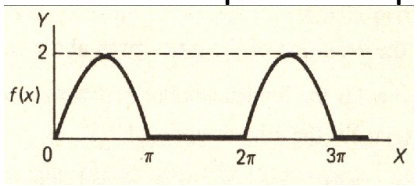
What is the period of periodic function whose graph is as below?



- ► 2
- ► 3
- ► 4
- ► 5

Question No: 20 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



- ► 0
- ► 2
- ► π

► 2π

Question No: 21 (Marks: 1) - Please choose one

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

► $\left(2, \frac{-\pi}{4}\right)$

► $\left(2, \frac{-\pi}{2}\right)$

► $\left(2, \frac{-\pi}{3}\right)$

► $\left(2, \frac{3\pi}{4}\right)$

Question No: 22 (Marks: 1) - Please choose one

The function $f(x) = x^3 e^x$ is ----------

► Even function

► Odd function

- Neither even nor odd

Question No: 23 (Marks: 1) - Please choose one

The graph of an even function is symmetrical about ----------

- x-axis
- y-axis
- origin

Question No: 24 (Marks: 1) - Please choose one

At which point the vertex of parabola, represented by the equation $y = x^2 - 4x + 3$, occurs?

- $(0, 3)$
- $(2, -1)$
- $(-2, 15)$
- $(1, 0)$

Question No: 25 (Marks: 1) - Please choose one

The equation $y = x^2 - 4x + 2$ represents a parabola. Find a point at which the vertex of given parabola occurs?

- $(2, -2)$
- $(-4, 34)$
- $(0, 0)$
- $(-2, 14)$

Question No: 26 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$f(x, y) = \dfrac{xy}{x^2 + y^2}$

- $f(x, y)$ is continuous at origin
- $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 27 (Marks: 1) - Please choose one

Sign of line integral is reversed when ----------

- path of integration is divided into parts.

► path of integration is parallel to y-axis.

► direction of path of integration is reversed.

► path of integration is parallel to x-axis.

Question No: 28 (Marks: 1) - Please choose one

What is Laplace transform of a function F(t)?

(s is a constant)

► $\int_0^s e^{-st} F(t)\, dt$

► $\int_0^\infty e^{st} F(t)\, dt$

► $\int_{-\infty}^\infty e^{-st} F(t)\, dt$

► $\int_0^\infty e^{-st} F(t)\, dt$

Question No: 29 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

► $L\{e^{5t}\} = \frac{1}{s-5}$

► $L\{e^{5t}\} = \frac{s}{s^2+25}$

► $L\{e^{5t}\} = \frac{5}{s^2+25}$

► $L\{e^{5t}\} = \frac{5!}{s^6}$

Question No: 30 (Marks: 1) - Please choose one

What is the Laplace Inverse Transform of $\frac{1}{s+1}$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = t+1$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}+e^{t}$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{t}$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}$

Question No: 31 (Marks: 1) - Please choose one

What is Laplace Inverse Transform of $\frac{5}{s^2+25}$

- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 5t$
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 5t$
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 25t$
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 25t$

Question No: 32 (Marks: 1) - Please choose one

What is $L\{-6\}$ if L denotes Laplace Transform?

- $L\{-6\} = \frac{1}{s+6}$
- $L\{-6\} = \frac{-6}{s}$
- $L\{-6\} = \frac{s}{s^2+36}$
- $L\{-6\} = \frac{-6}{s^2+36}$

Question No: 33 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx+(2x-y)\,dy$ where C is the line segment from (0, 0) to (2, 0).

- 6
- -6
- 0
- Do not exist

Question No: 34 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx+(x^2-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

- -4
- -2
- 0
- 2

Question No: 35 (Marks: 1) - Please choose one

Plane is an example of ----------

- Curve
- Surface
- Sphere
- Cone

Question No: 36 (Marks: 1) - Please choose one

$If\ R = \{(x,y)/0 \le x \le 2\ and\ -1 \le y \le 1\}, then$

$\iint_R (x+2y^2)dA =$

- $\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dydx$
- $\int_{0}^{2}\int_{1}^{-1}(x+2y^2)dxdy$

▶ $\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dxdy$

▶ $\int_{1}^{2}\int_{-1}^{0}(x+2y^2)dxdy$

Question No: 37 (Marks: 1) - Please choose one

To evaluate the line integral, the integrand is expressed in terms of x, y, z with

▶ $dr = dx\,\hat{i} + dy\,\hat{j}$

▶ $dr = dx\,\hat{i} + dy\,\hat{j} + dy\,\hat{k}$

▶ $dr = dx + dy + dz$

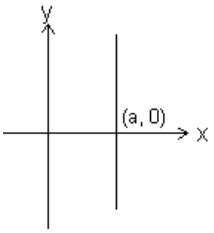
▶ $dr = dx + dy$

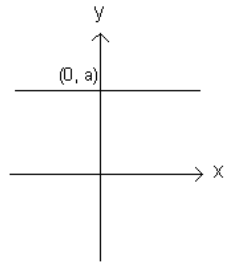
Question No: 38 (Marks: 1) - Please choose one

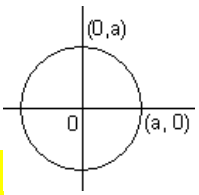
Match the following equation in polar co-ordinates with its graph.

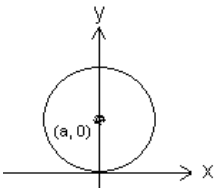
$r = a$

where a is an arbitrary constant.



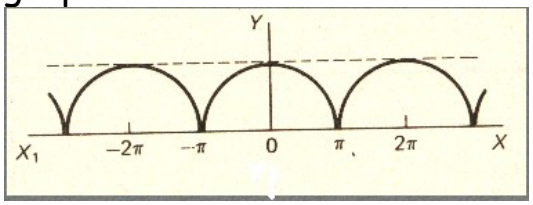






Question No: 39 (Marks: 1) - Please choose one

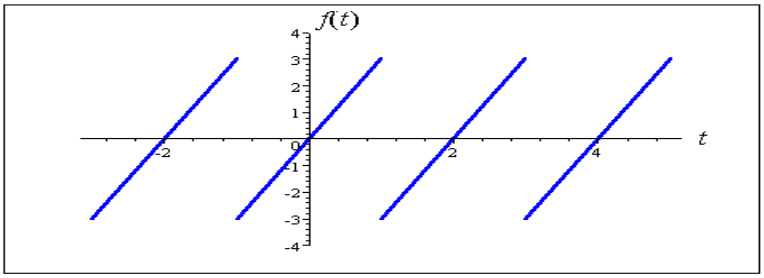
Which of the following is true for a periodic function whose graph is as below?



- Even function

- Odd function
- Neither even nor odd function

Question No: 40 (Marks: 1) - Please choose one



The graph of "saw tooth wave" given above is ----------

- An odd function
- An even function
- Neither even nor odd

Question No: 1 (Marks: 2) - Please choose one

Laplace transform of 't ' is

- $\frac{1}{s}$
- $\frac{1}{s^2}$
- e^{-s}
- s

Question No: 2 (Marks: 2) - Please choose one

Symmetric equation for the line through (1,3,5) and (2,-2,3) is

- $x-2=-\frac{y+2}{3}=-\frac{z-3}{5}$
- $x+2=-\frac{y+3}{5}=-\frac{z+5}{2}$
- $x-1=-\frac{y-3}{5}=-\frac{z-5}{2}$
- $x+1=\frac{y+3}{5}=\frac{z-5}{5}$

Question No: 3 (Marks: 1) - Please choose one

The level curves of f(x, y) = y Cscx are parabolas.

- True.
- False.

Question No: 4 (Marks: 1) - Please choose one

The equation $z = r$ is written in

- Rectangular coordinates
- Cylindrical coordinates
- Spherical coordinates
- None of the above

FINALTERM EXAMINATION
Spring 2010
MTH301- Calculus II (Session - 4)
Ref No: Time: 90 min
Marks: 60

Student Info	
StudentID:	
Center:	OPKST
ExamDate:	07 Aug 2010

For Teacher's Use Only									
Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

Question No: 1 (Marks: 1) - Please choose one

There is one-to-one correspondence between the set of points on co-ordinate line and ----------

- ► Set of real numbers
- ► Set of integers
- ► Set of natural numbers
- ► Set of rational numbers

Question No: 2 (Marks: 1) - Please choose one

Straight line is a special kind of ----------

- ► Surface
- ► Curve
- ► Plane
- ► Parabola

Question No: 3 (Marks: 1) - Please choose one

$\lim_{(x,\, y)\to(0,\, 0)} \frac{xy^2}{x^2+y^2} =$

- ► ∞
- ► 0
- ► 1
- ► 0.5

Question No: 4 (Marks: 1) - Please choose one

If $f(x,\, y) = x^2y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2}$ =

- ▶ $2xy+\frac{1}{x^2}$
- ▶ $2y+\frac{1}{x^2}$
- ▶ $2xy-\frac{1}{x^2}$
- ▶ $2y-\frac{1}{x^2}$

Question No: 5 (Marks: 1) - Please choose one

Suppose $f(x,\ y)=xy-2y^2$ *where* $x=3t+1$ *and* $y=2t$. Which one of the following is true?

□□□□□□□▶□ $\frac{df}{dt}=-4t+2$

□□□□□□□▶□ $\frac{df}{dt}=-16t-t$

□□□□□□□▶□ $\frac{df}{dt}=18t+2$

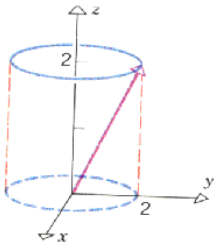
□□□□□□□▶□ $\frac{df}{dt}=-10t^2+8t+1$

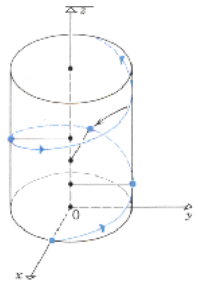
Question No: 6 (Marks: 1) - Please choose one

Match the following vector-valued function with its graph.

$r(t)=3\cos t\,\hat{i}+3\sin t\,\hat{j}$ and $0\le t\le 2\pi$

□□□□□□□▶□





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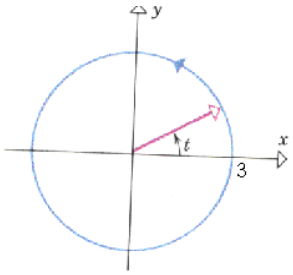

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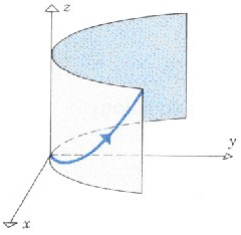

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Question No: 7 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$

- ▶ $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z = 0$
- ▶ $y = \sin^2 t \quad , \quad x = 1-\cos 2t \quad , \quad z = 0$
- ▶ $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z = 1$
- ▶ $x = \sin^2 t \quad , \quad y = \cos 2t \quad , \quad z = 1$

Question No: 8 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t = \frac{\pi}{2}$? If not, why?

$\vec{r}(t) = (\tan t,\ \sin t^2\ ,\ \cos t)$

□□□□□□□►□ $\vec{r}(t)$ is continuous at $t = \frac{\pi}{2}$

□□□□□□□►□ $\vec{r}\left(\frac{\pi}{2}\right)$ is not defined

□□□□□□□►□ $\vec{r}\left(\frac{\pi}{2}\right)$ is defined but $\lim_{t \to \frac{\pi}{2}} \vec{r}(t)$ does not exist

□□□□□□□►□ $\vec{r}\left(\frac{\pi}{2}\right)$ is defined and $\lim_{t \to \frac{\pi}{2}} \vec{r}(t)$ exists but these two numbers are not equal.

Question No: 9 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(t^4,\ \sqrt{t+1}\ ,\ \frac{3}{t^2}\right)$

□□□□□□□►□ $\vec{r}'(t) = \left(4t^3,\ \frac{1}{\sqrt{t+1}}\ ,\ \frac{-6}{t^3}\right)$

□□□□□□□►□ $\vec{r}'(t) = \left(4t^3,\ \frac{1}{2\sqrt{t+1}}\ ,\ \frac{6}{t^3}\right)$

□□□□□□□►□ $\vec{r}'(t) = \left(4t^4,\ \frac{1}{2\sqrt{t+1}}\ ,\ \frac{-6}{t^3}\right)$

□□□□□□□►□ $\vec{r}'(t) = \left(4t^3,\ \frac{1}{2\sqrt{t+1}}\ ,\ \frac{-6}{t^3}\right)$

Question No: 10 (Marks: 1) - Please choose one

The following differential is exact

$dz = (6xy + 2y^2 - 5)\ dx + (3x^2 + 4xy - 6)\ dy$

□□□□□□□►True

□□□□□□□►False

Question No: 11 (Marks: 1) - Please choose one

The following differential is exact

$dz = (x^2 y + y)\ dx - x\ dy$

□□□□□□□►True

□□□□□□□►False

Question No: 12 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

□□□□□□□► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots\cdots\cdots \frac{5}{6} \frac{3}{4} \frac{1}{2}$

□□□□□□□► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots\cdots\cdots \frac{6}{7} \frac{4}{5} \frac{2}{3}$

□□□□□□□► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \cdots\cdots\cdots \frac{6}{5} \frac{4}{3} \frac{2}{1}$

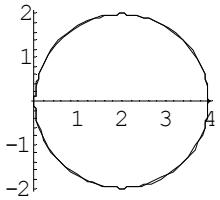
□□□□□□□► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \cdots\cdots\cdots \frac{6}{5} \frac{4}{3} \frac{2}{1}$

Question No: 13 (Marks: 1) - Please choose one

Match the following equation in polar co-ordinates with its graph.

$r = 4 \sin \theta$

□□□□□□□► □



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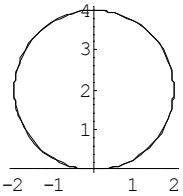

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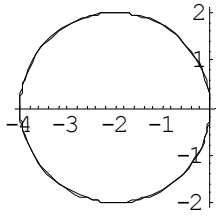

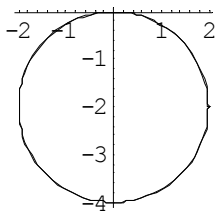

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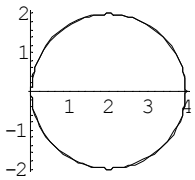
Question No: 14 (Marks: 1) - Please choose one

Match the following equation in polar co-ordinates with its graph.

$r = -4\sin\theta$



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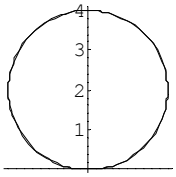

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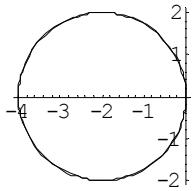

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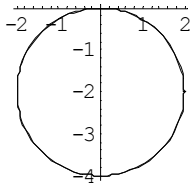

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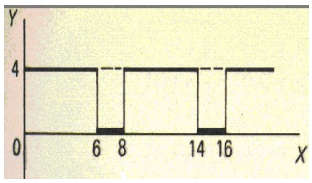
Question No: 15 (Marks: 1) - Please choose one

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 3 & 0 < x < 4 \\ 0 & 4 < x < 6 \end{cases}$$



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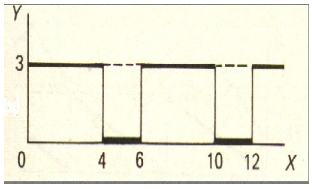
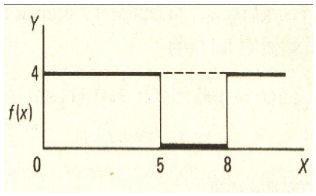
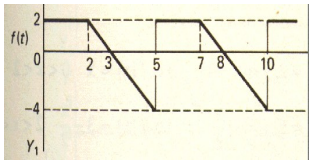
□□□□□□□▶□ 

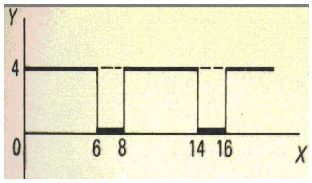
□□□□□□□▶□ 

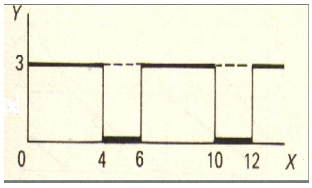
□□□□□□□▶□ 

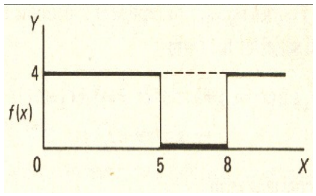
Question No: 16 (Marks: 1) - Please choose one

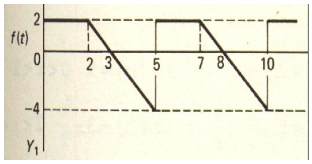
Match the following periodic function with its graph.

$$f(x)=\begin{cases}4 & 0<x<5\\ 0 & 5<x<8\end{cases}$$

□□□□□□□▶□ 

□□□□□□□▶□ 

□□□□□□□▶□ 

□□□□□□□▶□ 

Question No: 17 (Marks: 1) - Please choose one

Which of the following condition must be satisfied for a vector field $\vec{F}$ to be a conservative vector field?

□□□□□□□▶ □Line integral of $\vec{F}$ along a curve, depends only on the endpoints of that curve, not on the particular route taken.

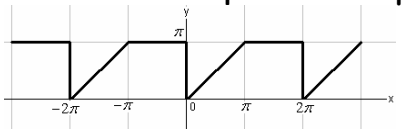
□□□□□□□▶ □Divergence of $\vec{F}$ should be zero

□□□□□□□▶ □Gradient of $\vec{F}$ should be zero.

□□□□□□□▶ □ $\vec{F} = 0$

Question No: 18 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



□□□□□□□▶ □ π

□□□□□□□▶ □ $-\pi$

□□□□□□□▶ □ 2π

□□□□□□□▶ □ -2π

Question No: 19 (Marks: 1) - Please choose one

Let L denotes the Laplace Transform.

According to First Shift Theorem, if $L\{F(t)\} = f(s)$ then which of the following equation holds?

s *and* a are constants.

□□□□□□□▶ □ $L\{e^{-at}F(t)\} = f(s-a)$

□□□□□□□▶ □ $L\{e^{-at}F(t)\} = f(s+a)$

□□□□□□□▶ □ $L\{e^{-at}F(t)\} = f(s)$

□□□□□□□▶ □ $L\{e^{-at}F(t)\} = f(a)$

Question No: 20 (Marks: 1) - Please choose one

Polar co-ordinates of a point are $\left(7, \frac{-\pi}{4}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

□□□□□□□▶ □ $\left(7, \frac{3\pi}{4}\right)$

□□□□□□□►□$\left(-7, \frac{3\pi}{4}\right)$

□□□□□□□►□$\left(-7, \frac{-\pi}{4}\right)$

□□□□□□□►□$\left(7, \frac{-3\pi}{4}\right)$

Question No: 21 (Marks: 1) - Please choose one

The graph of an even function is symmetrical about ----------

□□□□□□□►□x-axis

□□□□□□□►□y-axis

□□□□□□□►□origin

Question No: 22 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$f(x, y) = \frac{xy}{x^2 + y^2}$

□□□□□□□►□ $f(x, y)$ is continuous at origin

□□□□□□□►□ $\lim_{(x,y)\to(0, 0)} f(x, y)$ does not exist

□□□□□□□►□ $f(0, 0)$ is defined and $\lim_{(x,y)\to(0, 0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 23 (Marks: 1) - Please choose one

Consider the function $f(x, y, z) = \sqrt{1 - x^2 - y^2 - z^2}$. What is the value of $f\left(0, \frac{1}{2}, \frac{1}{2}\right)$?

□□□□□□□►□ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \sqrt{\frac{1}{2}}$

□□□□□□□►□ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 2$

□□□□□□□►□ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}$

□□□□□□□►□ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 0$

Question No: 24 (Marks: 1) - Please choose one

Sign of line integral is reversed when ----------

□□□□□□□► □path of integration is divided into parts.

□□□□□□□► □path of integration is parallel to y-axis.

□□□□□□□► □direction of path of integration is reversed.

□□□□□□□► □path of integration is parallel to x-axis.

Question No: 25 (Marks: 1) - Please choose one

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane.

If $(P\,dx + Q\,dy)$ is an exact differential then

$\oint_C (P\,dx + Q\,dy) =$

► Zero

► One

► Infinite

Question No: 26 (Marks: 1) - Please choose one

What is laplace transform of the function $F(t)$ if $F(t) = \cos 2t$?

► $L\{\cos 2t\} = \dfrac{2}{s^2+4}$

► $L\{\cos 2t\} = \dfrac{1}{s-2}$

► $L\{\cos 2t\} = \dfrac{s}{s^2+4}$

► $L\{\cos 2t\} = \dfrac{2!}{s^3}$

Question No: 27 (Marks: 1) - Please choose one

What is $L\{-6\}$ if L denotes Laplace Transform?

- $L\{-6\} = \frac{1}{s+6}$
- $L\{-6\} = \frac{-6}{s}$
- $L\{-6\} = \frac{s}{s^2+36}$
- $L\{-6\} = \frac{-6}{s^2+36}$

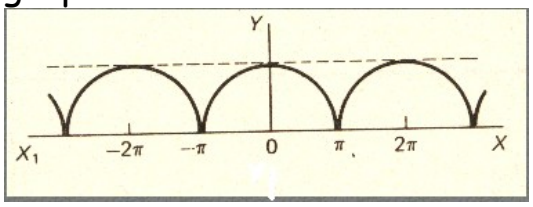
Question No: 28 (Marks: 1) - Please choose one

Curl of vector function is always a ----------

- Scalar
- Vector

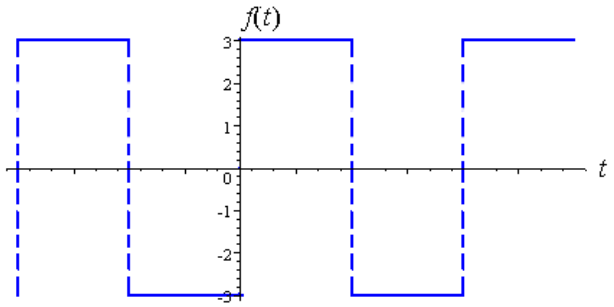
Question No: 29 (Marks: 1) - Please choose one

Which of the following is true for a periodic function whose graph is as below?



- Even function
- Odd function
- Neither even nor odd function

Question No: 30 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above

- ► An odd function
- ► An even function
- ► Neither even nor odd

Question No: 31 (Marks: 2)

Evaluate the line integral $\int_C 2x\ dx$ where C is the line segment from $(0, 2)$ to $(2, 6)$

Question No: 32 (Marks: 2)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x\ dx$

Question No: 33 (Marks: 2)

Find Laplace Transform of the function $F(t)$ if $F(t) = \sin 2t$.

Question No: 34 (Marks: 3)

Find $div\ \vec{F}$, if $\vec{F} = (3x + y)\hat{i} + xy^2z\hat{j} + (xz^2)\hat{k}$

Question No: 35 (Marks: 3)

Determine whether the following differential is exact or not.

$dz = (4x^3y + 2xy^3)\ dx + (x^4 + 3x^2y^2)\ dy$

Question No: 36 (Marks: 3)

Prove whether the following function is even, odd or neither.

$f(x) = x^3 e^x$

Question No: 37 (Marks: 5)

Evaluate the following line integral which is independent of path.

$$\int_{(2,-2)}^{(-1,0)} (2xy^3)\,dx + (3y^2x^2)\,dy$$

Question No: 38 (Marks: 5)

Determine the fourier co-efficient b_n of the following function.

$f(x) = x^2 \qquad 0 < x < 2\pi$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (3x + y)\hat{i} + xy^2z\,\hat{j} + xz^2\,\hat{k}$

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Mindhacker124@gmail.com

Hearthacker124@gmail.com

- At critical points function can assume
  - maximum value.
  - minimum value.
  - both maximum and minimum value.
  - **zero value**

- A plane can be perfectly determined by
  - one point.
  - **one point and normal vector.**
  - one point and parallel vector.
  - one normal vector.

- The length and width of a rectangle are measured with the errors of at most 4 % and 3%, respectively. Use differentials to approximate the maximum percentage error in the calculated area.

Solution:

Let
x = length

y = width

A = area

Then $A = xy$ So

$$\frac{\partial A}{\partial x} = y$$

$$\frac{\partial A}{\partial y} = x$$

Therefore,

$$\partial A \approx \frac{\partial A}{\partial x} + \frac{\partial A}{\partial y}\partial y$$

$$= y\partial x + x\partial y$$

We desire percentage change in A, which is relative change multiplied by 100 so let's work out relative change first. This is given by

$$\frac{\partial A}{A} \approx \frac{y\partial x}{A} + \frac{x\partial y}{A}$$

$$= \frac{\partial x}{x} + \frac{\partial y}{y}$$

Since

$$A = xy$$

$$-0.04 \le \frac{\partial x}{x} \le 0.04 \text{ and } -0.03 \le \frac{\partial y}{y} \le 0.04$$

The maximum possible value for $\frac{\partial A}{A}$ given the above constraints. This happens when $\frac{\partial x}{x} = 0.04$ and $\frac{\partial y}{y} = 0.03$, giving $\frac{\partial A}{A} = 0.07$. This is relative error, so the percentage error is 7%.

Convert the Cartesian coordinates $(1, \sqrt{2}, -2)$ into cylindrical coordinates.

Solution:

We are to convert (x,y,z) int (r, θ,z)

Given:

$$x = 1$$

$$y = \sqrt{2}$$

$$z = -2$$

We know that:

$$r = \sqrt{x^2 + y^2}$$

Put value of x and y in the above equation:

$r = \sqrt{(1)^2 + (\sqrt{2})^2}$

$r = \sqrt{1+2}$

$r = \sqrt{3}$

$\tan\theta = \frac{y}{x}$

$\theta = \tan^{-1}(\frac{y}{x})$

$\theta = \tan^{-1}(\frac{\sqrt{2}}{1})$

$\theta = \tan^{-1}\sqrt{2}$

$\theta = 54.73$

$z = -2$

So the cylindrical coordinates are $(\sqrt{3}, 54.73, -2)$

==========

$\vec{a} = \hat{i} + \hat{j} - \hat{k}$

- A unit vector in the direction of $\vec{a}$ is
  - $\frac{\left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}} \right\rangle}{\left\langle \frac{1}{3}, \frac{1}{3}, \frac{-1}{3} \right\rangle}$
  - $\left\langle \frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right\rangle$
  - $\left\langle \frac{1}{2}, \frac{1}{2}, \frac{-1}{2} \right\rangle$

Find equation of the tangent plane to the surface $z = x^3 + y^3$ at the point $(2,1,3)$.

Solution:

$f_x = 3x - y$

$f_y = 3y - x$

$f_z = -1$

$f_x(P) = 6 - 1 = 5$

$f_y(P) = 3 - 2 = 1$

$f_z(P) = -1$

Equation of tangent line to the surface through P is:

$5(x-2)+1(y-1)-1(z-3)=0$

$5x-10+y-1-z+3=0$

Hence:

$5x+y-z-8=0$

==========

Dot product of $\vec{a} = \hat{i}+2\hat{j}+\hat{k}$ *and* $\vec{b} = 2\hat{i}+3\hat{j}+2\hat{k}$ is

- 8
- 5
- **10**
- 13

- Evaluate the double integral $\iint_R x^2 dA$; R is the region bounded by $y = \frac{16}{x}$, $y = x$, *and* $x = 8$.

Solution:

Area of R = $\int \int_R x^2 dA$

$= \int_2^8 \int_{\frac{16}{x}}^{x} x^2 dy dx$

$= \int_2^8 \int_{\frac{16}{x}}^{x} x^2 \Big|_{\frac{16}{x}}^{x} dx$

$= \int_2^8 x^3 - 16x dx$

$$= \int_2^8 \frac{x^4}{4} - \frac{16x^2}{2}$$

$$= \int_2^8 \frac{x^4}{4} - 8x^2$$

$$= \left(\frac{8^4}{4} - 8\times 8\right) - \left(\frac{2^4}{4} - 8\times 2\right)$$

$$= \mathbf{972}$$

- Domain of $\sqrt{x} + \sqrt{y}$ is
  - $x + y > 0$
  - $x \geq 0\ , y \geq 0$
  - $x > 0, y > 0$
  - $x + y \geq 0$

- Use double integrals to compute the volume of the solid bounded by the *cylinder* $x^2 + y^2 = 9$ *and the planes* $y + z = 9$ *and* $z = 0$

Solution:

$$\iint_R (9 - y)da$$

R as a type 1 region we find that:

$$V = \int_{-2}^{2} \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (9 - y)\, dydx$$

$$= \int_{-2}^{2} \left[9y - \frac{1}{2}y^2 \right]_{y=-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dx$$

$$= \int_{-2}^{2} 18\sqrt{9 - x^2}\, dx$$

$$= 18\left(\frac{1}{2}\pi 9^2\right) = 729\pi$$

Created 2010/9 mid term ….
ASSALAM O ALIKUM all fellows
ALL IN ONE MTH301 MIDTERM PAPERS
solved (4)
SOLVED BY Farhan & Ali
BS (cs) 2nd sem
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Remember us in your prayers

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MIDTERM EXAMINATION

Spring 2010
MTH301- Calculus II

Ref No: 1499814
Time: 60 min
Marks: 40

Student Info	
Student ID:	
Center:	
Exam Date:	

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Q No.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**Total**
Marks									
Q No.	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	
Marks									
Q No.	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	
Marks									
Q No.	**25**	**26**							
Marks									

Question No: 1 (Marks: 1) - Please choose one

Every point in three dimensional space can be described by ------- coordinates.

► Two

► **Three**

► Four

► Eight

Question No: 2 (Marks: 1) - Please choose one

What are the direction cosines for the line joining the points (1, 3, 2) and (7, -2, 3)?

► $\frac{1}{7}, \frac{-3}{2}$ *and* $\frac{2}{3}$

► $\frac{7}{11}, \frac{-6}{11}$ *and* $\frac{6}{11}$

► $\frac{8}{3\sqrt{10}}, \frac{1}{3\sqrt{10}}$ *and* $\frac{5}{3\sqrt{10}}$

► $\frac{6}{\sqrt{62}}, \frac{-5}{\sqrt{62}}$ *and* $\frac{1}{\sqrt{62}}$

Question No: 3 (Marks: 1) - Please choose one

The angles which a line makes with positive x ,y and z-axis are known as ----------

► Direction cosines

► Direction ratios

► **Direction angles**

Question No: 4 (Marks: 1) - Please choose one

Which of the following is geometrical representation of the equation $y = 4$, in three dimensional space?

► A point on y-axis (Farhan)

► Plane parallel to xy-plane

► Plane parallel to yz-axis

► **Plane parallel to xz-plane (Ali)**

Question No: 5 (Marks: 1) - Please choose one

Domain of the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ is

► **Entire 3D-Space**

- Entire 3D-Space except origin
- $(0, \infty)$
- $(-\infty, \infty)$

Question No: 6 (Marks: 1) - Please choose one

If $f(x, y) = x^2 y - y^3 + \ln x$
then $\frac{\partial^2 f}{\partial x^2} =$

- $2xy + \frac{1}{x^2}$
- $2y + \frac{1}{x^2}$
- $2xy - \frac{1}{x^2}$
- $2y - \frac{1}{x^2}$

Question No: 7 (Marks: 1) - Please choose one

Suppose $f(x, y) = xy - 2y^2$ *where* $x = 3t + 1$ *and* $y = 2t$. Which one of the following is true?

- $\frac{df}{dt} = -4t + 2$
- $\frac{df}{dt} = -16t - t$
- $\frac{df}{dt} = 18t + 2$

► $\frac{df}{dt} = -10t^2 + 8t + 1$

Question No: 8 (Marks: 1) - Please choose one

For a function $f(x, y, z)$, the equation $\frac{\partial^2 f}{\partial^2 x} + \frac{\partial^2 f}{\partial^2 y} + \frac{\partial^2 f}{\partial^2 z} = 0$ is known as ----------

► Gauss Equation

► Euler's equation

► **Laplace's Equation (Ali)**

► Stoke's Equation

Question No: 9 (Marks: 1) - Please choose one

Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3 and angle between them when placed tail to tail is 45 degrees. What is $\vec{a} \cdot \vec{b}$?

► 4.5

► 6.2

► 5.1

► **4.2**

Question No: 10 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$f(x, y) = \frac{xy}{x^2 + y^2}$

- ▶ $f(x, y)$ is continuous at origin
- ▶ $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- ▶ $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 11 (Marks: 1) - Please choose one

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- ▶ Area of R.??(ali)
- ▶ **Radius of inscribed circle in R**
- ▶ Distance between two endpoints of R.
- ▶ None of these

Question No: 12 (Marks: 1) - Please choose one

Which of the following formula can be used to find the Volume of a parallelepiped with adjacent edges formed by the vectors a , b *and* c ?

- ▶ $\left|a \times \left(b \times c\right)\right|$
- ▶ $\left|a \, g\left(b \, gc\right)\right|$
- ▶ $\left|a \, g\left(b \times c\right)\right|$
- ▶ $\left|a \times \left(b \, gc\right)\right|$

Question No: 13 (Marks: 1) - Please choose one

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are ---------

► Parallel

► **Perpendicular**

► In opposite direction

Question No: 14 (Marks: 1) - Please choose one

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

► perpendicular

► parallel

► **In opposite direction (Farhan)**

Question No: 15 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}\ ,\ f_{yy}\ and\ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}^{\ 2}(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

► **Relative maximum at** (x_0, y_0)

► Relative minimum at (x_0, y_0)

► Saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 16 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}\ ,\ f_{yy}\ and\ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D = 0$ then ----------

► f has relative maximum at (x_0, y_0)

► f has relative minimum at (x_0, y_0)

► f has saddle point at (x_0, y_0)

► **No conclusion can be drawn.**

Question No: 17 (Marks: 1) - Please choose one

$If\ R = \{(x, y) / 0 \le x \le 2\ and\ 1 \le y \le 4\}, then$

$\iint_R (6x^2 + 4xy^3)dA =$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$

► $\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$

- $\int_{2}^{4}\int_{0}^{1}(6x^2+4xy^3)dxdy$

Question No: 18 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 2\ and\ -1\le y\le 1\},then$

$\iint_R (x+2y^2)dA=$

- $\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dydx$
- $\int_{0}^{2}\int_{1}^{-1}(x+2y^2)dxdy$
- $\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dxdy$
- $\int_{1}^{2}\int_{-1}^{0}(x+2y^2)dxdy$

Question No: 19 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/2\le x\le 4\ and\ 0\le y\le 1\},then$

$\iint_R (4xe^{2y})dA=$

- $\int_{0}^{1}\int_{2}^{4}(4xe^{2y})dydx$
- $\int_{0}^{1}\int_{2}^{4}(4xe^{2y})dxdy$

- $\int_{1}^{4}\int_{0}^{2}(4xe^{2y})dxdy$
- $\int_{1}^{4}\int_{0}^{2}(4xe^{2y})dydx$

Question No: 20 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 4\ and\ 0\le y\le 9\},then$
$\iint_{R}(3x-4x\sqrt{xy})dA=$

- $\int_{0}^{9}\int_{0}^{4}(3x-4x\sqrt{xy})dydx$
- $\int_{0}^{4}\int_{4}^{9}(3x-4x\sqrt{xy})dxdy$
- $\int_{4}^{9}\int_{0}^{0}(3x-4x\sqrt{xy})dxdy$
- $\int_{0}^{4}\int_{0}^{9}(3x-4x\sqrt{xy})dydx$

Question No: 21 (Marks: 2)

Suppose that the surface $f(x, y, z)$ has continuous partial derivatives at the point (a, b, c). Write down the equation of tangent plane at this point.

Question No: 22 (Marks: 2)

Evaluate the following double integral.

$\iint (3x - y)\ dy\ dx$

$\iint \left(3x^2 y/2 - xy^2/2\right)\ \ dx \int\iint (3x - y)\ dy\ dx$

$\int \frac{3x^1 y}{1} - \frac{y^2}{2} dx$

$\frac{3x^2 y}{2} - \frac{xy^2}{2} + c$

Answer

$\int \frac{3x^1 y}{1} - \frac{y^2}{2} dx$

$\frac{3x^2 y}{2} - \frac{xy^2}{2} + c$

Question No: 23 (Marks: 3)

Evaluate the following double integral.

$\iint \left(3 + 2x - 3y^2\right)\ dx\ dy$

$= \int \left(3x + x^2 + 3xy^2\right) dy$

$= 3xy + x^2 y + 3xy^3$

$\iint \left(3 + 2x - 3y^2\right)\ dx\ dy$

$= \int \left(3x + x^2 + 3xy^2\right) dy$

$= 3xy + x^2 y + 3xy^3$

Question No: 24 (Marks: 3)

Let $f(x, y, z) = yz^3 - 2x^2$

Find the gradient of f.

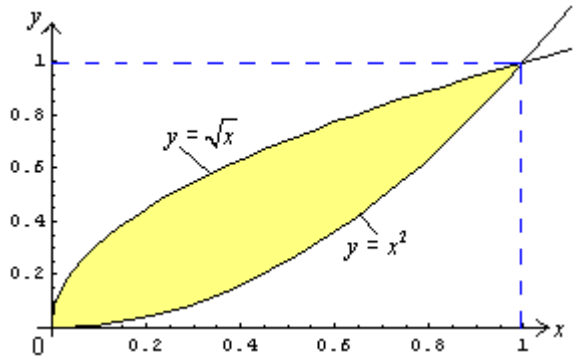
Question No: 25 (Marks: 5)

Find all critical points of the function

$f(x, y) = y^2 + xy + 3y + 2x + 3$

Question No: 26 (Marks: 5)

Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ and $y = \sqrt{x}$.



PAPER NO.2

1. Every real number corresponds to_____on the co-ordinate line.

- Infinite number of points
- Two points (one positive and one negative)
- A unique point??f
- None of these

2. There is one-to-one correspondence between the set of points on co-ordinate line and _____.

- Set of real numbers
- Set of integers
- Set of natural numbers
- Set of rational numbers

3. Which of the following is associated to each point of three dimensional spaces?

- A real number
- An ordered pair
- An ordered triple

- **A natural Number**

4. All axes are positive in _____octant.

- **First**
- Second
- Fourth
- Eighth

5. The spherical co-ordinates of a point are $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$.What are its cylindrical co-ordinates?

- $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}, 0\right)$
- $\left(\sqrt{3}\cos\frac{\pi}{3}, \sqrt{3}\sin\frac{\pi}{3}, 0\right)$
- $\left(\sqrt{3}\sin\frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3}\cos\frac{\pi}{3}\right)$
- $\left(\sqrt{3}, \frac{\pi}{3}, 0\right)$

6. Suppose $f(x, y) = xy - 2y^2$ *where* $x = 3t + 1$ *and* $y = 2t$. Which one of the following is true?

- $\frac{df}{dt} = -4t + 2$
- $\frac{df}{dt} = -16t - t$
- $\frac{df}{dt} = 18t + 2$
- $\frac{df}{dt} = -10t^2 + 8t + 1$

7. Let $w = f(x, y, z)$ *and* $x = g(r, s)$, $y = h(r, s)$, $z = t(r, s)$ then by chain rule $\frac{\partial w}{\partial r} =$

- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$

- $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

8. Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3and angle between them when placed tail to tail is 45 degrees. What is$\vec{a}.\vec{b}$?

- 4.5
- **6.2**
- 5.1
- **4.2**

9. Is the function $f(x,y)$ continuous at origin? If not, why?

$$f(x,y)=\begin{cases}0 \ if \ x\geq 0 \ and \ y\geq 0\\ 1 \qquad\quad otherwise\end{cases}$$

- $f(x,y)$ is continuous at origin
- $f(0,0)$ is not defined
- $f(0,0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist
- $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exist but these two numbers are not equal.

10. Is the function $f(x,y)$ continuous at origin? If not, Why?

$$f(x,y)=\begin{cases}\frac{3x^2y}{x^2+y^2} & if(x,y)\neq 0\\ 0 & if\ (x,y)=0\end{cases}$$

- $f(x,y)$ is continuous at origin
- $f(0,0)$ **is not defined**
- $f(0,0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist

- $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exist but these two numbers are not equal.

11. Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- Area of R
- **Radius of inscribed circle in R.**
- Distance between two endpoints of R.
- None of these

12. Which of the following formula can be used to find the volume of a parallelepiped with adjacent edges formed by the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$?

- $|\vec{a} \times (\vec{b} \times \vec{c})|$
- $|\vec{a} . (\vec{b} . \vec{c})|$
- $|\vec{a} . (\vec{b} \times \vec{c})|$
- $|\vec{a} \times (\vec{b} . \vec{c})|$

13. Two surfaces are said to be orthogonal at appoint of their intersection if their normals at that point are_____.

- Parallel
- **Perpendicular**
- In opposite direction
- Same direction

14. By Extreme Value Theorem, if a function $f(x,y)$ is continuous on a closed and bounded set R, then $f(x,y)$ has both _____ on R.

- **Absolute maximum and absolute minimum value**
- Relative maximum and relative minimum value
- Absolute maximum and relative minimum value
- Relative maximum and absolute minimum value

15. Let the function $f(x,y)$ has continuous second-order partial derivatives $(f_{xx}\ ,\ f_{yy}\ and\ f_{xy})$ in some circle centered at a critical point (x_0,y_0) and let $D=f_{xx}(x_0,y_0)\ \ f_{yy}(x_0,y_0)-f_{xy}^2(x_0,y_0)$ if
$D>0\ \ and\ \ f_{xx}(x_0,y_0)<0\ \ then\ \ f\ has$_____ .

- **Relative maximum at (x_0,y_0)**
- Relative minimum at (x_0,y_0)
- Saddle point at (x_0,y_0)
- No conclusion can be drawn.

16. Let the function $f(x,y)$ has continuous second-order partial derivatives $(f_{xx}\ ,\ f_{yy}\ and\ f_{xy})$ in some circle centered at a critical point (x_0,y_0) and let $D=f_{xx}(x_0,y_0)\ \ f_{yy}(x_0,y_0)-f_{xy}^2(x_0,y_0)$ if
if $D=0$ then _____.

- f has relative maximum at (x_0,y_0)
- f has relative minimum at (x_0,y_0)
- f has saddle point at (x_0,y_0)
- **No conclusion can be drawn.**

17. $If\ R=R_1\cup R_2\ ,where\ R_1\ and\ R_2$ are no over lapping regions then $\iint_{R_1} f(x,y)dA+\iint_{R_2} f(x,y)dA=$

- $\iint_R f(x,y)dA$
- $\iint_{R_1} f(x,y)dA\cup\iint_{R_2} f(x,y)dA$
- $\iint_R f(x,y)dV$
- $\iint_R f(x,y)dV\cap\iint_{R_2} f(x,y)dA$

18. *If* $R=\{(x,y)/0\le x\le 2 \text{ and } 1\le y\le 4\}$ *,then* $\iint_R (6x^2+4xy^3)dA=$

- $\int_1^4\int_0^2 (6x^2+4xy^3)\,dydx$
- $\int_0^2\int_1^4 (6x^2+4xy^3)\,dydx$
- $\int_1^4\int_0^2 (6x^2+4xy^3)\,dydx$
- $\int_2^4\int_0^1 (6x^2+4xy^3)\,dydx$

19. *If* $R=\{(x,y)/2\le x\le 4 \text{ and } 0\le y\le 1\}$ *, then* $\iint_R (4xe^{2y})dA=$

- $\int_0^1\int_2^4 (4xe^{2y})\,dydx$
- $\int_0^1\int_2^4 (4xe^{2y})\,dxdy$
- $\int_1^4\int_0^2 (4xe^{2y})\,dxdy$
- $\int_1^4\int_0^2 (4xe^{2y})\,dydx$

20. *If* $R=\{(x,y)/0\le x\le 4 \text{ and } 0\le y\le 9\}$ *,then* $\iint_R (3x-4x\sqrt{xy})dA=$

- $\int_0^9\int_0^4 (3x-4x\sqrt{xy})dydx$
- $\int_0^9\int_0^4 (3x-4x\sqrt{xy})dxdy$
- $\int_4^9\int_0^0 (3x-4x\sqrt{xy})dxdy$

- $\int_0^4 \int_0^9 \left(3x - 4x\sqrt{xy}\right) dydx$

21. Suppose that the surface $f(x,y,z)$ has continuous partial derivatives at the point (a,b,c) Write down the equation of tangent plane at this point.

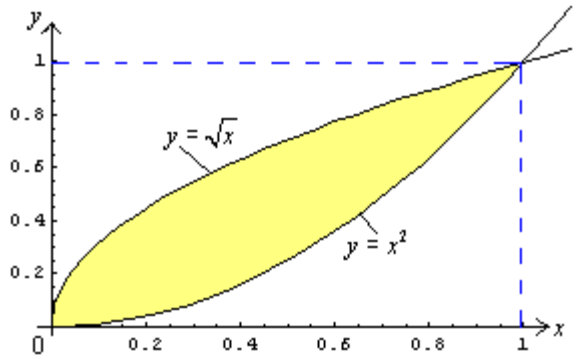
22. Evaluate the following double integral $\int\int \left(12xy^2 - 8x^3\right) dydx$.

23. Evaluate the following double integral $\int\int \left(3 + 2x - 3y^2\right) dxdy$

24. Let $f(x,y,z) = xy^2e^z$ Find the gradient of f .

25. Find, Equation of Tangent plane to the surface $f(x,y,z) = x^2 + y^2 + z - 9$ at the point $(1,2,4)$.

26. Use the double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ *and* $y = \sqrt{x}$.



MIDTERM EXAMINATION
Spring 2010
MTH301- Calculus II (Session - 3)

Time: 60 min
Marks: 40

Student Info	
StudentID:	
Center:	**OPKST**
ExamDate:	**6/3/2010 12:00:00 AM**

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Q No.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**Total**

Marks									
Q No.	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	
Marks									
Q No.	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	
Marks									
Q No.	**25**	**26**							
Marks									

Question No: 1 (Marks: 1) - Please choose one

Which of the following number is associated to each point on a co-ordinate line?

► An integer

► **A real number**

► A rational number

► A natural number

Question No: 2 (Marks: 1) - Please choose one

If $a > 0$, then the parabola $y = ax^2 + bx + c$ opens in which of the following direction?

► **Positive x - direction**

► Negative x - direction

► Positive y - direction

► Negative y - direction

Question No: 3 (Marks: 1) - Please choose one

Rectangular co-ordinate of a point is $(1, \sqrt{3}, -2)$. What is its spherical co-ordinate?

► $\left(2\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{2}\right)$

► $\left(2\sqrt{2}, \frac{\pi}{2}, \frac{3\pi}{4}\right)$

► $\left(2\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4}\right)$

► $\left(\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4}\right)$

Question No: 4 (Marks: 1) - Please choose one

If a function is not defined at some point, then its limit ---------- exist at that point.

► Always

► Never

► **May**

Question No: 5 (Marks: 1) - Please choose one

Suppose $f(x, y) = x^3 e^{xy}$. Which one of the statements is correct?

□□□□□□□▶□ $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$

□□□□□□□▶□ $\frac{\partial f}{\partial y} = x^3 e^{xy}$

□□□□□□□▶□ $\frac{\partial f}{\partial y} = x^4 e^{xy}$

□□□□□□□▶□ $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

Question No: 6 (Marks: 1) - Please choose one

If $f(x, y) = x^2 y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2}$ =

□□□□□□□▶□ $2xy + \frac{1}{x^2}$

□□□□□□□▶□ $2y + \frac{1}{x^2}$

□□□□□□□▶□ $2xy - \frac{1}{x^2}$

□□□□□□□▶□ $2y - \frac{1}{x^2}$

Question No: 7 (Marks: 1) - Please choose one

Suppose $f(x, y) = xy - 2y^2$ *where* $x = 3t + 1$ *and* $y = 2t$. Which one of the following is true?

□□□□□□□▶□ $\frac{df}{dt} = -4t + 2$

□□□□□□□

▶□ $\frac{df}{dt} = -16t - t$

□□□□□□□▶□ $\frac{df}{dt} = 18t + 2$

□□□□□□□▶□ $\frac{df}{dt} = -10t^2 + 8t + 1$

Question No: 8 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} 0 & If\ x \geq 0\ and\ y \geq 0 \\ 1 & Otherwise \end{cases}$$

□□□□□□□►□ $f(x, y)$ **is continuous at origin**

□□□□□□□►□ $f(0, 0)$ is not defined

□□□□□□□►□ $f(0, 0)$ is defined but $\lim_{(x,y)\to(0, 0)} f(x, y)$ does not exist

□□□□□□□►□ $f(0, 0)$ is defined and $\lim_{(x,y)\to(0, 0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 9 (Marks: 1) - Please choose one

What is the relation between the direction of gradient at any point on the surface to the tangent plane at that point ?

□□□□□□□►□parallel

□□□□□□□►□perpendicular

□□□□□□□►□opposite direction

□□□□□□□►□No relation between them.

Question No: 10 (Marks: 1) - Please choose one

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

► perpendicular

► parallel

► in opposite direction

Question No: 11 (Marks: 1) - Please choose one

By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R, then $f(x, y)$ has both ---------- on R.

► Absolute maximum and absolute minimum value

► Relative maximum and relative minimum value

Question No: 12 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}\ ,\ f_{yy}\ and\ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

► **Relative maximum at** (x_0, y_0)

► Relative minimum at (x_0, y_0)

► Saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 13 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}\ ,\ f_{yy}\ and\ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}^{\ 2}(x_0, y_0)$

If $D = 0$ then ----------

► f has relative maximum at (x_0, y_0)

► f has relative minimum at (x_0, y_0)

► f has saddle point at (x_0, y_0)

► **No conclusion can be drawn.**

Question No: 14 (Marks: 1) - Please choose one

The function $f(x, y) = \sqrt{y - x}$ is continuous in the region --------- and discontinuous elsewhere.

► $x \neq y$

► $x \leq y$

► $x > y$

Question No: 15 (Marks: 1) - Please choose one

Plane is an example of ----------

► Curve

► **Surface**

► Sphere

► Cone

Question No: 16 (Marks: 1) - Please choose one

If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then

$\iint_{R_1} f(x, y)\, dA \qquad \iint_{R_2} f(x,$

- ▶ $\iint_R f(x, y)\, dA$
- ▶ $\iint_{R_1} f(x, y)\, dA \qquad \iint_{R_2} f(x$
- ▶ $\iint_R f(x, y)\, dV$
- ▶ $\iint_{R_1} f(x, y)\, dA \qquad \iint_{R_2} f(x$

Question No: 17 (Marks: 1) - Please choose one

If $R = \{(x, y) / 0 \le x \le 2 \text{ and } 1 \le y \le 4\}$, *then*

$\iint_R (6x^2 + 4xy^3)dA =$

- ▶ $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$
- ▶ $\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$
- ▶ $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$
- ▶ $\int_2^4 \int_0^1 (6x^2 + 4xy^3)dxdy$

Question No: 18 (Marks: 1) - Please choose one

If $R = \{(x, y) / 0 \le x \le 2 \text{ and } -1 \le y \le 1\}$, *then*

$\iint_R (x + 2y^2)dA =$

- ▶ $\int_{-1}^1 \int_0^2 (x + 2y^2)dydx$
- ▶ $\int_0^2 \int_1^{-1} (x + 2y^2)dxdy$

- $\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dxdy$
- $\int_{1}^{2}\int_{-1}^{0}(x+2y^2)dxdy$

Question No: 19 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 2\ and\ 0\le y\le 3\},then$

$\iint_R (1-ye^{xy})dA=$

- $\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dydx$
- $\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dxdy$
- $\int_{2}^{3}\int_{0}^{0}(1-ye^{xy})dxdy$
- $\int_{0}^{2}\int_{2}^{3}(4xe^{2y})dydx$

Question No: 20 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 4\ and\ 0\le y\le 9\},then$

$\iint_R (3x-4x\sqrt{xy})dA=$

- $\int_{0}^{9}\int_{0}^{4}(3x-4x\sqrt{xy})dydx$
- $\int_{0}^{4}\int_{4}^{9}(3x-4x\sqrt{xy})dxdy$
- $\int_{4}^{9}\int_{0}^{0}(3x-4x\sqrt{xy})dxdy$
- $\int_{0}^{4}\int_{0}^{9}(3x-4x\sqrt{xy})dydx$

Question No: 21 (Marks: 2)

Evaluate the following double integral.

$\iint\left(2xy+y^3\right)\,dx\,dy$

Question No: 22 (Marks: 2)

$Let\ f(x,\ y) = 2 + x^2 + \frac{y^2}{4}$

$Find\ the\ gradient\ of\ f$

Question No: 23 (Marks: 3)

Evaluate the following double integral.

$\iint \left(3 + 2x - 3y^2\right)\ dx\ dy$

Question No: 24 (Marks: 3)

$Let\ f(x,\ y,\ z) = yz^3 - 2x^2$

$Find\ the\ gradient\ of\ f.$

Question No: 25 (Marks: 5)

Find Equation of a Tangent plane to the surface $f(x,\ y,\ z) = x^2 + 3y + z^3 - 9$ at the point $(2,\ -1,\ 2)$

Question No: 26 (Marks: 5)

Evaluate the iterated integral

$\int_2^4 \int_{\frac{x}{2}}^{\sqrt{x}} \left(xy\right)\ dy\,dx$

MTH301 Calculus II

Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

Every real number corresponds to ---------- on the co-ordinate line.

Answer (Please select your correct option)

- Infinite number of points
- Two points (one positive and one negative)
- A unique point
- None of these

MTH301 Calculus II

Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

There is one-to-one correspondence between the set of points on co-ordinate line and ----------

Answer (Please select your correct option)

- Set of real numbers
- Set of integers
- Set of natural numbers
- Set of rational numbers

MTH301 Calculus II

Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

Which of the following is associated to each point of three dimensional space?

Answer (Please select your correct option)

- A real number
- An ordered pair
- An ordered triple
- A natural number

MTH301 Calculus II

Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

All axes are positive in ---------- octant.

Answer (Please select your correct option)

- First
- Second
- Fourth
- Eighth

MTH301 Calculus II

Question No : 5 of 26 Marks: 1 (Budgeted Time 1 Min)

The spherical co-ordinates of a point are $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$. What are its cylinderical co-ordinates?

Answer (Please select your correct option)

- $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}, 0\right)$
- $\left(\sqrt{3}\cos\frac{\pi}{3}, \sqrt{3}\sin\frac{\pi}{3}, 0\right)$
- $\left(\sqrt{3}\sin\frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3}\cos\frac{\pi}{3}\right)$
- $\left(\sqrt{3}, \frac{\pi}{3}, 0\right)$

MTH301 Calculus II

Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

Suppose $f(x, y) = xy - 2y^2$ *where* $x = 3t + 1$ *and* $y = 2t$. Which one of the following is true?

Answer (Please select your correct option)

- $\frac{df}{dt} = -4t + 2$
- $\frac{df}{dt} = -16t - t$
- $\frac{df}{dt} = 18t + 2$
- $\frac{df}{dt} = -10t^2 + 8t + 1$

MTH301 Calculus II

Question No : 7 of 26 Marks: 1 (Budgeted Time 1 Min)

Let w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$\frac{\partial w}{\partial r}$ =

Answer (Please select your correct option)

- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

MTH301 Calculus II

Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3 and angle between them when placed tail to tail is 45 degrees. What is $\vec{a} \cdot \vec{b}$?

Answer (Please select your correct option)

- 4.5
- 6.2
- 5.1
- 4.2

MTH301 Calculus II

Question No : 9 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} 0 & \text{If } x \geq 0 \text{ and } y \geq 0 \\ 1 & \text{Otherwise} \end{cases}$$

Answer (Please select your correct option)

- $f(x, y)$ is continuous at origin
- $f(0, 0)$ is not defined
- $f(0, 0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

MTH301 Calculus II

Question No : 10 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} \dfrac{3x^2y}{x^2 + y^2} & \text{if } (x, y) \neq 0 \\ 0 & \text{if } (x, y) = 0 \end{cases}$$

Answer (Please select your correct option)

- $f(x, y)$ is continuous at origin
- $f(0, 0)$ is not defined
- $f(0, 0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

MTH301 Calculus II

Question No : 11 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

Answer (Please select your correct option)

- Area of R.
- Radius of inscribed circle in R.
- Distance between two endpoints of R.
- None of these

MTH301 Calculus II

Question No : 12 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following formula can be used to find the Volume of a parallelepiped with adjacent edges formed by the vectors $\vec{a}$, $\vec{b}$ *and* $\vec{c}$?

Answer (Please select your correct option)

- $\left|\vec{a} \times \left(\vec{b} \times \vec{c}\right)\right|$
- $\left|\vec{a} \bullet \left(\vec{b} \bullet \vec{c}\right)\right|$
- $\left|\vec{a} \bullet \left(\vec{b} \times \vec{c}\right)\right|$
- $\left|\vec{a} \times \left(\vec{b} \bullet \vec{c}\right)\right|$

MTH301 Calculus II

Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are ---------

Answer (Please select your correct option)

- Parallel
- Perpendicular
- In opposite direction

MTH301 Calculus II

Question No : 14 of 26 Marks: 1 (Budgeted Time 1 Min)

By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R, then $f(x, y)$ has both ---------- on R.

Answer (Please select your correct option)

- Absolute maximum and absolute minimum value
- Relative maximum and relative minimum value

MTH301 Calculus II

Question No : 15 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

Answer (Please select your correct option)

- (●) Relative maximum at (x_0, y_0)
- ( ) Relative minimum at (x_0, y_0)
- ( ) Saddle point at (x_0, y_0)
- ( ) No conclusion can be drawn.

MTH301 Calculus II

Question No : 16 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D = 0$ then ----------

Answer (Please select your correct option)

- (●) f has relative maximum at (x_0, y_0)
- ( ) f has relative minimum at (x_0, y_0)
- ( ) f has saddle point at (x_0, y_0)
- ( ) No conclusion can be drawn.

MTH301 Calculus II

Question No : 17 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then

$$\iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA =$$

Answer (Please select your correct option)

- $\iint_{R} f(x,y)dA$
- $\iint_{R_1} f(x,y)dA \cup \iint_{R_2} f(x,y)dA$
- $\iiint_{R} f(x,y)dV$
- $\iint_{R_1} f(x,y)dA \cap \iint_{R_2} f(x,y)dA$

MTH301 Calculus II

Question No : 18 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $R = \{(x,y) / 0 \le x \le 2 \text{ and } 1 \le y \le 4\}$ *,then*

$$\iint_{R} (6x^2 + 4xy^3)dA =$$

Answer (Please select your correct option)

- $\int_{1}^{4}\int_{0}^{2} (6x^2 + 4xy^3)dydx$
- $\int_{0}^{2}\int_{1}^{4} (6x^2 + 4xy^3)dxdy$
- $\int_{1}^{4}\int_{0}^{2} (6x^2 + 4xy^3)dxdy$
- $\int_{2}^{4}\int_{0}^{1} (6x^2 + 4xy^3)dxdy$

MTH301 Calculus II

Question No : 19 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $R = \{(x,y) / 2 \le x \le 4 \text{ and } 0 \le y \le 1\}$, then

$\iint_R (4xe^{2y})dA =$

Answer (Please select your correct option)

- $\int_0^1 \int_2^4 (4xe^{2y})dydx$
- $\int_0^1 \int_2^4 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dydx$

MTH301 Calculus II

Question No : 20 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $R = \{(x,y) / 0 \le x \le 4 \text{ and } 0 \le y \le 9\}$, then

$\iint_R (3x - 4x\sqrt{xy})dA =$

Answer (Please select your correct option)

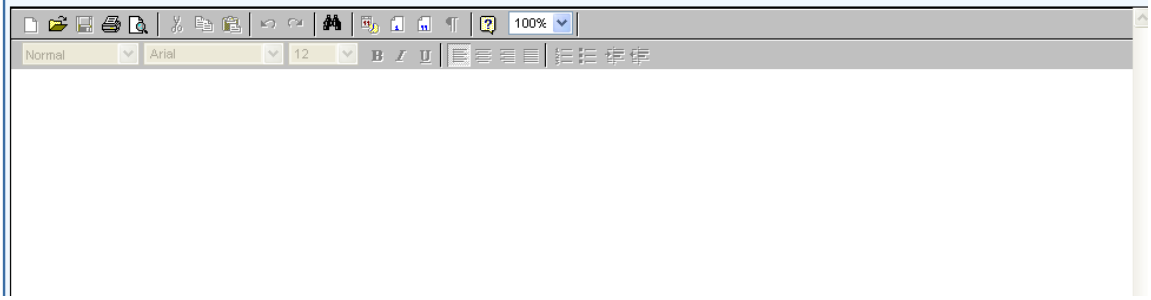
- $\int_0^9 \int_0^4 (3x - 4x\sqrt{xy})dydx$
- $\int_0^4 \int_4^9 (3x - 4x\sqrt{xy})dxdy$
- $\int_4^9 \int_0^0 (3x - 4x\sqrt{xy})dxdy$
- $\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$

MTH301 Calculus II

Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

Suppose that the surface $f(x, y, z)$ has continuous partial derivatives at the point (a, b, c). Write down the equation of tangent plane at this point.

Answer (Please click here to Add Answer)



MTH301 Calculus II

Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

Evaluate the following double integral.

$$\iint (12xy^2 - 8x^3)\, dy\, dx$$

MTH301 Calculus II

Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

Evaluate the following double integral.

$$\iint (3 + 2x - 3y^2)\, dx\, dy$$

Answer (Please click here to Edit Answer)

MTH301 Calculus II

Question No : 24 of 26 Marks: 3 (Budgeted Time 6 Min)

Let $f(x, y, z) = xy^2e^z$

Find the gradient of f.

Answer (Please click here to Add Answer)

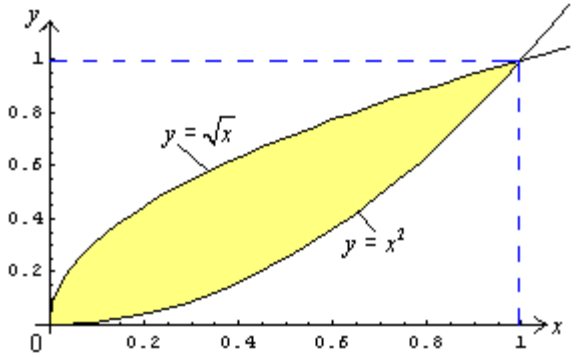
MTH301 Calculus II

Question No : 25 of 26 Marks: 5 (Budgeted Time 10 Min)

Find, Equation of Tangent plane to the surface $f(x, y, z) = x^2 + y^2 + z - 9$ at the point $(1, 2, 4)$

Answer (Please click here to Add Answer)

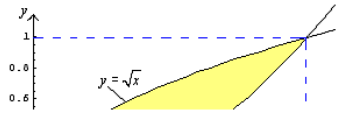
Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ and $y = \sqrt{x}$.



MTH301 Calculus II

Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)

Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ and $y = \sqrt{x}$.



Answer (Please click here to Add Answer)

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ASSALAM O ALAIKUM

All Dear fellows

ALL IN ONE MTH301 Calculus II

Midterm solved papers

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Remember me in your prayers

MIDTERM EXAMINATION

Spring 2010

MTH301- Calculus II (Session - 3)

1. Every real number corresponds to_____on the co-ordinate line.

- **Infinite number of points**
- Two points (one positive and one negative)
- A unique point
- None of these

2. There is one-to-one correspondence between the set of points on co-ordinate line and _____.

- Set of real numbers
- Set of integers
- **Set of natural numbers**
- Set of rational numbers

3. Which of the following is associated to each point of three dimensional space?

- A real number
- An ordered pair
- An ordered triple
- **A natural Number**

4. All axes are positive in _____octant.

- **First**
- Second
- Fourth
- Eighth

5. The spherical co-ordinates of a point are $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$.What are its cylindrical co-ordinates?

- $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}, 0\right)$
- $\left(\sqrt{3}\cos\frac{\pi}{3}, \sqrt{3}\sin\frac{\pi}{3}, 0\right)$
- $\left(\sqrt{3}\sin\frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3}\cos\frac{\pi}{3}\right)$
- $\left(\sqrt{3}, \frac{\pi}{3}, 0\right)$

6. Suppose $f(x,y) = xy - 2y^2$ *where* $x = t^3 + 1$ *and* $y = t$. Which one of the following is true?

- $\frac{df}{dt} = -4t + 2$
- $\frac{df}{dt} = -16t - t$
- $\frac{df}{dt} = 18t + 2$
- $\frac{df}{dt} = -10t^2 + 8t + 1$

7. Let $w = f(x,y,z)$ *and* $x = g(r\ s)$, $y = h(r\ s)$, $z = t(r\ s)$ then by chain rule $\frac{\partial w}{\partial r} =$

- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$

- $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

8. Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3and angle between them when placed tail to tail is 45 degrees. What is$\vec{a}.\vec{b}$?

- 4.5
- **6.2**
- 5.1
- 4.2

9. Is the function $f(x,y)$ continuous at origin? If not, why?$f(x,y)=\begin{cases}0 & if\ x\geq 0\ and\ y\geq 0\\ 1 & otherwise\end{cases}$

- $f(x,y)$ is continuous at origin
- $f(0,0)$ is not defined
- $f(0,0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist
- $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exist but these two numbers are not equal.

10. Is the function $f(x,y)$ continuous at origin? If not, Why?$f(x,y)=\begin{cases}\frac{3x^2y}{x^2+y^2} & if(x,y)\neq 0\\ 0 & if\ (x,y)=0\end{cases}$

- $f(x,y)$ is continuous at origin
- $f(0,0)$ is not defined
- $f(0,0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist
- $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exist but these two numbers are not equal.

11. Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- Area of R
- Radius of inscribed circle in R.
- Distance between two endpoints of R.
- None of these

12. Which of the following formula can be used to find the volume of a parallelepiped with adjacent edges formed by the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$?

- $|\vec{a} \times (\vec{b} \times \vec{c})|$
- $|\vec{a} . (\vec{b} . \vec{c})|$
- $|\vec{a} . (\vec{b} \times \vec{c})|$
- $|\vec{a} \times (\vec{b} . \vec{c})|$

13. Two surfaces are said to be orthogonal at appoint of their intersection if their normals at that point are_____.

- Parallel
- Perpendicular
- In opposite direction
- Same direction

14. By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R, then $f(x, y)$ has both _____ on R.

- Absolute maximum and absolute minimum value
- Relative maximum and relative minimum value
- Absolute maximum and relative minimum value
- Relative maximum and absolute minimum value

15. Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}$, f_{yy} *and* $f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ \ f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$ if

$D > 0$ *and* $f_{xx}(x_0, y_0) < 0$ *then* f *has* _____ .

- Relative maximum at (x_0, y_0)
- Relative minimum at (x_0, y_0)
- Saddle point at (x_0, y_0)
- No conclusion can be drawn.

16. Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}$, f_{yy} *and* $f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ \ f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$ if

if $D = 0$ then _____.

- f has relative maximum at (x_0, y_0)
- f has relative minimum at (x_0, y_0)
- f has saddle point at (x_0, y_0)
- No conclusion can be drawn.

17. *If* $R = R_1 \cup R_2$ *where* R_1 *and* R_2 are no over lapping regions then

$$\iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA =$$

- $\iint_R f(x,y)dA$
- $\iint_{R_1} f(x,y)dA \cup \iint_{R_2} f(x,y)dA$
- $\iint_R f(x,y)dV$
- $\iint_R f(x,y)dV \cap \iint_{R_2} f(x,y)dA$

18. *If* $R = \{(x, y) / 1 \le x \le 2 \text{ and } \le y \le 4\}$ *then* $\iint_R \left(6x^2 + 4xy^3\right)dA =$

- $\int_1^4 \int_0^2 \left(6x^2 + 4xy^3\right) dydx$
- $\int_0^2 \int_1^4 \left(6x^2 + 4xy^3\right) dydx$
- $\int_1^4 \int_0^2 \left(6x^2 + 4xy^3\right) dydx$
- $\int_2^4 \int_0^1 \left(6x^2 + 4xy^3\right) dydx$

19. *If* $R = \{(x, y) / 2 \le x \le 4 \text{ and } \le y \le 1\}$ *then,* $\iint_R \left(4xe^{2y}\right)dA =$

- $\int_0^1 \int_2^4 \left(4xe^{2y}\right) dydx$

- $\int_{0}^{1}\int_{2}^{4}\left(4xe^{2y}\right)dxdy$
- $\int_{1}^{4}\int_{0}^{2}\left(4xe^{2y}\right)dxdy$
- $\int_{1}^{4}\int_{0}^{2}\left(4xe^{2y}\right)dydx$

20. *If* $R=\{(x,y) / 1\le x\le 4 \text{ and } 0\le y\le 9\}$ *then* $\iint_{R}\left(3x-4x\sqrt{xy}\right)dA=$

- $\int_{0}^{9}\int_{0}^{4}\left(3x-4x\sqrt{xy}\right)dydx$
- $\int_{0}^{9}\int_{0}^{4}\left(3x-4x\sqrt{xy}\right)dxdy$
- $\int_{4}^{9}\int_{0}^{0}\left(3x-4x\sqrt{xy}\right)dxdy$
- $\int_{0}^{4}\int_{0}^{9}\left(3x-4x\sqrt{xy}\right)dydx$

21. Suppose that the surface $f(x,y,z)$ has continuous partial derivatives at the point (a,b,c) Write down the equation of tangent plane at this point.

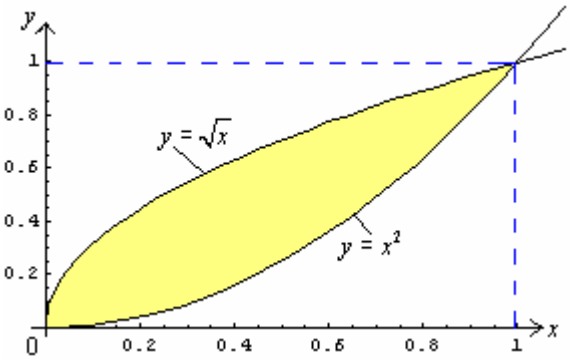
22. Evaluate the following double integral $\int\int\left(12xy^2-8x^3\right)dydx$.

23. Evaluate the following double integral $\int\int\left(3+2x-3y^2\right)dxdy$

24. Let $f(x,y,z)=xy^2e^z$ Find the gradient of f .

25. Find, Equation of Tangent plane to the surface $f(x,y,z)=x^2+y^2+z-9$ at the point $(1,2,4)$.

26. Use the double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y=x^2$ *and* $y=\sqrt{x}$.



MIDTERM EXAMINATION

Fall 2009

MTH301- Calculus II

Question No: 1 (Marks: 1) - Please choose one

i. _____ Let x be any point on co-ordinate line. What does the inequality -3 < x < 1 means?

a. ► The set of all integers between -3 and 1

b. ► The set of all natural numbers between -3 and 1.

c. ► The set of all rational numbers between -3 and 1

d. ► The set of all real numbers between -3 and 1

Question No: 2 (Marks: 1) - Please choose one

_____

Which of the following number is associated to each point on a co-ordinate line?

► An integer

► A real number

► A rational number

- ▶ A natural number

Question No: 3 (Marks: 1) - Please choose one

Which of the following set is the union of set of all rational and irrational numbers?

- ▶ Set of rational numbers
- ▶ Set of integers
- **▶ Set of real numbers**
- ▶ Empty set.

Question No: 4 (Marks: 1) - Please choose one

π is an example of ----------

- **▶ Irrational numbers**
- ▶ Rational numbers
- ▶ Integers
- ▶ Natural numbers

Question No: 5 (Marks: 1) - Please choose one

Which of the following is associated to each point on a plane?

- ▶ A real number
- ▶ A natural number

► An ordered pair

► An ordered triple

Question No: 6 (Marks: 1) - Please choose one

_____ ------

---------- planes intersect at right angle to form three dimensional space.

► **Three**

► Four

► Eight

► Twelve

Question No: 7 (Marks: 1) - Please choose one

_____ At

each point of domain, the function ----------

► **Is defined**

► Is continuous

► Is infinite

► Has a limit

Question No: 8 (Marks: 1) - Please choose one

_____

What is the general equation of parabola whose axis of symmetry is parallel to y-axis?

► $y = ax^2 + b \qquad (a \neq 0)$

- $x = ay^2 + b \qquad (a \neq 0)$
- $y = ax^2 + bx + c \qquad (a \neq 0)$
- $x = ay^2 + by + c \qquad (a \neq 0)$

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Question No: 9 (Marks: 1) - Please choose one

The spherical co-ordinates (ρ, θ, ϕ), of a point are $\left(2, \frac{\pi}{4}, 0\right)$. What are the rectangular co-ordinates of this point?

- $(0,\ 0,\ \sqrt{2})$
- $(2, 0, 0)$
- $(0, 0, 2)$
- $(\sqrt{2}, 0, \sqrt{2})$

Question No: 10 (Marks: 1) - Please choose one

Let $f(x,\ y) = y^2 x^4 e^x + 2$.

$$\frac{\partial^5 f}{\partial y^3 \partial x^2}$$

Which method is best suited for evaluation of ?

- Normal method of finding the higher order mixed partial derivatives
- Chain Rule

► Laplacian Method

► **Euler's method for mixed partial derivative**

Question No: 11 (Marks: 1) - Please choose one

Suppose $f(x, y) = x^3 e^{xy}$. Which one of the following is correct?

► $\frac{\partial f}{\partial x} = 3x^2 e^{xy} + x^3 y e^{xy}$

► $\frac{\partial f}{\partial x} = 3x^2 e^{xy} + x^4 e^{xy}$

► $\frac{\partial f}{\partial x} = 3x^2 e^{xy}$

► $\frac{\partial f}{\partial x} = 3x^2 y e^{xy}$

Question No: 12 (Marks: 1) - Please choose one

Suppose $f(x, y) = x^3 e^{xy}$. Which one of the statements is correct?

► $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$

► $\frac{\partial f}{\partial y} = x^3 e^{xy}$

► $\frac{\partial f}{\partial y} = x^4 e^{xy}$

- ► $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

Question No: 13 (Marks: 1) - Please choose one

Suppose $f(x, y) = 2xy \text{ where } x = t^2 + 1 \text{ and } y = 3 - t$. Which one of the following is true?

- ► $\frac{df}{dt} = 6t - 4t^2 - 2$
- ► $\frac{df}{dt} = 6t - 2$
- ► $\frac{df}{dt} = 4t^3 + 6t - 6$
- ► $\frac{df}{dt} = -6t^2 + 12t - 2$

Question No: 14 (Marks: 1) - Please choose one

Let i, j and k be unit vectors in the direction of x-axis, y-axis and z-axis respectively. Suppose that $\vec{a} = 2i + 5j - k$. What is the magnitude of vector $\vec{a}$?http://vustudents.ning.com

- ► 6
- ► 30

► $\sqrt{30}$

► $\sqrt{28}$

Question No: 15 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \frac{xy}{x^2 + y^2}$$

► $f(x, y)$ is continuous at origin

► $\lim_{(x,y)\to(0,0)} f(x, y)$ **does not exist**

► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 16 (Marks: 1) - Please choose one

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

► Area of R.

► Radius of inscribed circle in R.

► Distance between two endpoints of R.

► None of these

Question No: 17 (Marks: 1) - Please choose one

What is the relation between the direction of gradient at any point on the surface to the tangent plane at that point ?

- ► parallel
- ► perpendicular
- ► opposite direction
- ► No relation between them.

Question No: 18 (Marks: 1) - Please choose one

_____ Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

- ► perpendicular
- ► parallel
- ► in opposite direction

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Question No: 19 (Marks: 1) - Please choose one

_____ Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}, \ f_{yy} \ and \ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

- ► **Relative maximum at (x_0, y_0)**
- ► Relative minimum at (x_0, y_0)
- ► Saddle point at (x_0, y_0)
- ► No conclusion can be drawn.

Question No: 20 (Marks: 1) - Please choose one

Which of the following are direction ratios for the line joining the points $(1, 3, 5)$ and $(2, -1, 4)$?

► 3, 2 and 9

► 1, -4 and -1

► 2, -3 and 20

► 0.5, -3 and 5/4

Question No: 21 (Marks: 1) - Please choose one

If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then

$$\iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA =$$

$\iint_R f(x,y)dA$

►

$\iint_{R_1} f(x,y)dA \cup \iint_{R_2} f(x,y)dA$

►

$\iint_R f(x,y)dV$

►

$$\iint_{R_1} f(x,y)dA \cap \iint_{R_2} f(x,y)dA$$

►

Question No: 22 (Marks: 1) - Please choose one

If $R = \{(x,y)/2 \le x \le 4 \, and \, 0 \le y \le 1\}, then$

$\iint_R (4xe^{2y})dA =$

$\int_0^1 \int_2^4 (4xe^{2y})dydx$

►

$\int_0^1 \int_2^4 (4xe^{2y})dxdy$

►

$\int_1^4 \int_0^2 (4xe^{2y})dxdy$

►

$\int_1^4 \int_0^2 (4xe^{2y})dydx$

►

Question No: 23 (Marks: 1) - Please choose one http://vustudents.ning.com

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$If\ R = \{(x, y) / 0 \le x \le 2\ and\ 0 \le y \le 3\}, then$

$\iint_R (1 - ye^{xy})dA =$

$\int_0^2 \int_0^3 (1 - ye^{xy})dydx$

►

$\int_0^2 \int_0^3 (1 - ye^{xy})dxdy$

►

$\int_2^3 \int_0^0 (1 - ye^{xy})dxdy$

►

$\int_0^2 \int_2^3 (4xe^{2y})dydx$

►

Question No: 24 (Marks: 1) - Please choose one

Which of the following is geometrical representation of the equation $y = x^2$, in three dimensional space?

► Parabola

► Straight line

► Half cylinder

► Cone

Question No: 25 (Marks: 3)

Consider the point $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$ in spherical coordinate system. Find the rectangular coordinatesof this point.

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Question No: 26 (Marks: 5)

Consider a function $f(x, y) = 4xy - x^4 - y^4$ .One of its critical point is $(1, 1)$. Find whether $(1, 1)$ is relative maxima, relative minima or saddle point of $f(x, y)$.

Question No: 27 (Marks: 10)

Find Directional derivative of the function, $f(x, y) = x^2y - 4y^3$, at the point $(2, 1)$ in the direction of vector, $\vec{u} = \frac{\sqrt{3}}{2}i + \frac{1}{2}j$

MIDTERM EXAMINATION

Spring 2010

MTH301- Calculus II

Question No: 1 (Marks: 1) - Please choose one

Which of the following is the interval notation of real line?

► **(-∞ , +∞)**

► (-∞ , 0)

► (0 , +∞)

Question No: 2 (Marks: 1) - Please choose one

What is the general equation of parabola whose axis of symmetry is parallel to y-axis?

► $y = ax^2 + b \qquad (a \neq 0)$

► $x = ay^2 + b \qquad (a \neq 0)$

► $y = ax^2 + bx + c \qquad (a \neq 0)$

► $x = ay^2 + by + c \qquad (a \neq 0)$

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Question No: 3 (Marks: 1) - Please choose one

Which of the following is geometrical representation of the equation $y = 4$, in three dimensional space?

► **A point on y-axis**

► Plane parallel to xy-plane

► Plane parallel to yz-axis

► Plane parallel to xz-plane

Question No: 4 (Marks: 1) - Please choose one

Suppose $f(x,\ y) = x^3 e^{xy}$. Which one of the statements is correct?

- ▶ $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$
- ▶ $\frac{\partial f}{\partial y} = x^3 e^{xy}$
- ▶ $\frac{\partial f}{\partial y} = x^4 e^{xy}$
- ▶ $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

Question No: 5 (Marks: 1) - Please choose one

If $f(x,\ y) = x^2 y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2}$ =

- ▶ $2xy + \frac{1}{x^2}$
- ▶ $2y + \frac{1}{x^2}$
- ▶ $2xy - \frac{1}{x^2}$

$2y - \frac{1}{x^2}$

▶

Question No: 6 (Marks: 1) - Please choose one

Let

w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$\frac{\partial w}{\partial r} =$

$\frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$

▶

$\frac{\partial w}{\partial r}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$

▶

$\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$

▶

$\frac{\partial w}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

▶

Question No: 7 (Marks: 1) - Please choose one

Is

the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} \frac{3x^2 y}{x^2 + y^2} & if \ (x, y) \neq 0 \\ 0 & if \ (x, y) = 0 \end{cases}$$

► $f(x, y)$ is continuous at origin

► $f(0, 0)$ **is not defined**

► $f(0, 0)$ **is defined but** $\lim_{(x,y)\to(0,0)} f(x, y)$ **does not exist**

► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 8 (Marks: 1) - Please choose one

_____ Let R be a closed region in two dimensional space. What does the double integral over R calculates?

► Area of R.

► Radius of inscribed circle in R.

► Distance between two endpoints of R.

► **None of these (not sure)**

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Question No: 9 (Marks: 1) - Please choose one

_____ Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are ---------

► Parallel

► **Perpendicular**

► In opposite direction

Question No: 10 (Marks: 1) - Please choose one

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

► perpendicular

► parallel

► in opposite direction

Question No: 11 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}, \ f_{yy} \ and \ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) > 0$ then f has ----------

► Relative maximum at (x_0, y_0)

► Relative minimum at (x_0, y_0)

► Saddle point at (x_0, y_0)

► No conclusion can be drawn. http://vustudents.ning.com

Question No: 12 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}, \ f_{yy} \ and \ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

► Relative maximum at (x_0, y_0)

► Relative minimum at (x_0, y_0)

► Saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 13 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}\ ,\ f_{yy}\ and\ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D < 0$ then f has ----------

► Relative maximum at (x_0, y_0)

► Relative minimum at (x_0, y_0)

► Saddle point at (x_0, y_0)

► No conclusion can be drawn

Question No: 14 (Marks: 1) - Please choose one

Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be any two points in three dimensional space. What does the formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ calculates?

► **Distance between these two points**

► Midpoint of the line joining these two points

► Ratio between these two points

Question No: 15 (Marks: 1) - Please choose one

_____ The function $f(x, y) = \sqrt{y - x}$ is continuous in the region --------- and discontinuous elsewhere.

► $x \neq y$

► $x \leq y$

► $x > y$ http://vustudents.ning.com

Question No: 16 (Marks: 1) - Please choose one

_____

Plane is an example of ----------

► Curve

► Surface

► Sphere

► Cone

Question No: 17 (Marks: 1) - Please choose one

_____ If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then

$$\iint_{R_1} f(x,y)dA+\iint_{R_2} f(x,y)dA=$$

$\iint_{R} f(x,y)dA$

►

$\iint_{R_1} f(x,y)dA\cup\iint_{R_2} f(x,y)dA$

►

$\iint_{R} f(x,y)dV$

►

$\iint_{R_1} f(x,y)dA\cap\iint_{R_2} f(x,y)dA$

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►

Question No: 18 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 2\ and\ 1\le y\le 4\},then$

$\iint_{R} (6x^2+4xy^3)dA=$

$\int_{1}^{4}\int_{0}^{2} (6x^2+4xy^3)dydx$

►

$\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$

▶

$\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$

▶

$\int_2^4 \int_0^1 (6x^2 + 4xy^3)dxdy$

▶

Question No: 19 (Marks: 1) - Please choose one

If $R = \{(x, y) / 0 \le x \le 2 \, and \, 0 \le y \le 3\}$, *then*

$\iint_R (1 - ye^{xy})dA =$

$\int_0^2 \int_0^3 (1 - ye^{xy})dydx$

▶

$\int_0^2 \int_0^3 (1 - ye^{xy})dxdy$

▶

$\int_2^3 \int_0^0 (1 - ye^{xy})dxdy$

▶

$\int_0^2 \int_2^3 (4xe^{2y})dydx$

▶

Question No: 20 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 4 and\ 0\le y\le 9\}, then$

$\iint_R (3x-4x\sqrt{xy})dA=$

http://vustudents.ning.com

- $\int_0^9 \int_0^4 (3x-4x\sqrt{xy})dydx$

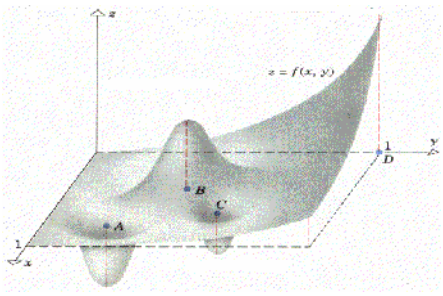
- $\int_0^4 \int_4^9 (3x-4x\sqrt{xy})dxdy$

- $\int_4^9 \int_0^0 (3x-4x\sqrt{xy})dxdy$

- $\int_0^4 \int_0^9 (3x-4x\sqrt{xy})dydx$

Question No: 21 (Marks: 2)

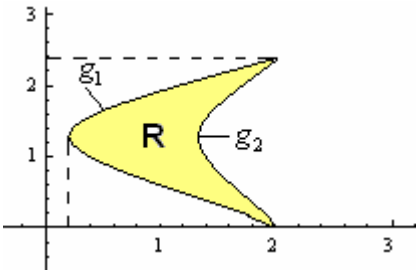
Following is the graph of a function of two variables



In its whole domain, state whether the function has relative maximum value or absolute maximum value at point B. Also, justify your answer

Question No: 22 (Marks: 2)

Let the function $f(x, y)$ is continuous in the region R, where R is bounded by graph of functions g_1 and g_2 (as shown below). http://vustudents.ning.com



In the following equation, replace question mark (?) with the correct value.

$$\iint_R f(x, y)\, dA = \int_{?}^{?} \int_{?}^{?} f(x, y) _____?____$$

Question No: 23 (Marks: 3)

Evaluate the following double integral.

$$\iint \left(3 + 2x - 3y^2\right) dx\, dy$$

Question No: 24 (Marks: 3)

Let $f(x, y, z) = yz^3 - 2x^2$

Find the gradient of f.

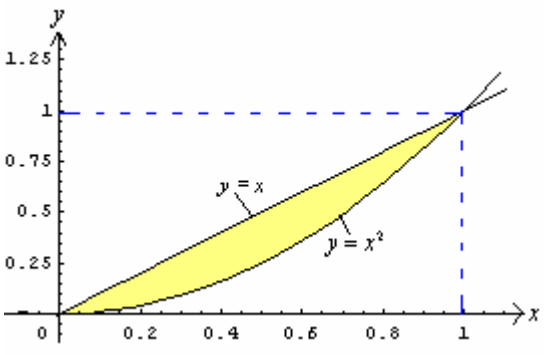
Question No: 25 (Marks: 5)

Find, Equation of Normal line (in parametric form) to the surface

$f(x, y, z) = xy + 2yz - xz^2 + 10$ at the point $(-5, 5, 1)$

Question No: 26 (Marks: 5)

Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x \text{ and } y = x^2$, in the first quadrant.



MIDTERM EXAMINATION

Spring 2010

MTH301- Calculus II (Session - 3)

Question No: 1 (Marks: 1) - Please choose one

Which of the following number is associated to each point on a co-ordinate line?

► An integer

► A real number

► A rational number

► A natural number

Question No: 2 (Marks: 1) - Please choose one

If $a > 0$, then the parabola $y = ax^2 + bx + c$ opens in which of the following direction?

► Positive x - direction

► Negative x - direction

► Positive y - direction

► Negative y - direction

Question No: 3 (Marks: 1) - Please choose one

Rectangular co-ordinate of a point is $(1, \sqrt{3}, -2)$. What is its spherical co-ordinate?

► $\left(2\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{2}\right)$

$\left(2\sqrt{2}, \frac{\pi}{2}, \frac{3\pi}{4}\right)$

$\left(2\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4}\right)$

$\left(\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4}\right)$

Question No: 4 (Marks: 1) - Please choose one

If a function is not defined at some point, then its limit ---------- exist at that point.

Always

Never

May

Question No: 5 (Marks: 1) - Please choose one

Suppose $f(x, y) = x^3 e^{xy}$. Which one of the statements is correct?

$\frac{\partial f}{\partial y} = 3x^3 e^{xy}$

$\frac{\partial f}{\partial y} = x^3 e^{xy}$

$\frac{\partial f}{\partial y} = x^4 e^{xy}$

$\frac{\partial f}{\partial y} = x^3 y e^{xy}$

Question No: 6 (Marks: 1) - Please choose one

If $f(x, y) = x^2 y - y^3 + \ln x$

$\frac{\partial^2 f}{\partial x^2}$

then =

. $2xy + \frac{1}{x^2}$

. $2y + \frac{1}{x^2}$

. $2xy - \frac{1}{x^2}$

. $2y - \frac{1}{x^2}$

Question No: 7 (Marks: 1) - Please choose one

Suppose $f(x, y) = xy - 2y^2 \ \ where \ \ x = 3t + 1 \ \ and \ \ y = 2t$. Which one of the following is true?

. $\frac{df}{dt} = -4t + 2$

. $\frac{df}{dt} = -16t - t$

. $\frac{df}{dt} = 18t + 2$

. $\frac{df}{dt} = -10t^2 + 8t + 1$

Question No: 8 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} 0 & If \ \ x \geq 0 \ and \ \ y \geq 0 \\ 1 & Otherwise \end{cases}$$

- ► $f(x, y)$ is continuous at origin
- ► $f(0, 0)$ is not defined
- ► $f(0, 0)$ is defined but $\lim_{(x,y)\to(0, 0)} f(x, y)$ does not exist
- ► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0, 0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 9 (Marks: 1) - Please choose one

What is the relation between the direction of gradient at any point on the surface to the tangent plane at that point ?

- ► parallel
- ► **perpendicular**
- ► opposite direction
- ► No relation between them.

Question No: 10 (Marks: 1) - Please choose one

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

- ► perpendicular
- ► parallel
- ► **in opposite direction**

Question No: 11 (Marks: 1) - Please choose one

By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R, then $f(x, y)$ has both ---------- on R.

► **Absolute maximum and absolute minimum value**

► Relative maximum and relative minimum value

Question No: 12 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}, \ f_{yy} \ and \ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

► **Relative maximum at (x_0, y_0)**

► Relative minimum at (x_0, y_0)

► Saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 13 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}, \ f_{yy} \ and \ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D = 0$ then ----------

► f has relative maximum at (x_0, y_0)

► f has relative minimum at (x_0, y_0)

► f has saddle point at (x_0, y_0)

► **No conclusion can be drawn.**

Question No: 14 (Marks: 1) - Please choose one

The function $f(x, y) = \sqrt{y - x}$ is continuous in the region --------- and discontinuous elsewhere.

- ► $x \neq y$
- ► $x \leq y$
- ► $x > y$

Question No: 15 (Marks: 1) - Please choose one

Plane is an example of ----------

- ► Curve
- ► **Surface**
- ► Sphere
- ► Cone

Question No: 16 (Marks: 1) - Please choose one

If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then

$$\iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA =$$

- ► $\iint_{R} f(x,y)dA$
- ► $\iint_{R_1} f(x,y)dA \cup \iint_{R_2} f(x,y)dA$

$$\iint_R f(x,y)dV$$

►

$$\iint_{R_1} f(x,y)dA \cap \iint_{R_2} f(x,y)dA$$

►

Question No: 17 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 2\ and\ 1\le y\le 4\},\ then$

$\iint_R (6x^2+4xy^3)dA=$

$$\int_1^4 \int_0^2 (6x^2+4xy^3)dydx$$

►

$$\int_0^2 \int_1^4 (6x^2+4xy^3)dxdy$$

►

$$\int_1^4 \int_0^2 (6x^2+4xy^3)dxdy$$

►

$$\int_2^4 \int_0^1 (6x^2+4xy^3)dxdy$$

►

Question No: 18 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 2\ and\ -1\le y\le 1\},\ then$

$\iint_R (x+2y^2)dA=$

$\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dydx$

►

$\int_{0}^{2}\int_{1}^{-1}(x+2y^2)dxdy$

►

$\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dxdy$

►

$\int_{1}^{2}\int_{-1}^{0}(x+2y^2)dxdy$

►

Question No: 19 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 2\ and\ 0\le y\le 3\},then$

$\iint_R (1-ye^{xy})dA=$

$\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dydx$

►

$\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dxdy$

►

$\int_{2}^{3}\int_{0}^{0}(1-ye^{xy})dxdy$

►

$\int_{0}^{2}\int_{2}^{3}(4xe^{2y})dydx$

►

Question No: 20 (Marks: 1) - Please choose one

$If\ R = \{(x, y) / 0 \le x \le 4\ and\ 0 \le y \le 9\}, then$

$\iint_R (3x - 4x\sqrt{xy})dA =$

$\int_0^9 \int_0^4 (3x - 4x\sqrt{xy})dydx$

►

$\int_0^4 \int_4^9 (3x - 4x\sqrt{xy})dxdy$

►

$\int_4^9 \int_0^0 (3x - 4x\sqrt{xy})dxdy$

►

$\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$

►

Question No: 21 (Marks: 2)

Evaluate the following double integral.

$\iint \left(2xy + y^3\right)\ dx\ dy$

Question No: 22 (Marks: 2)

$Let\ \ f(x,\ y) = 2 + x^2 + \frac{y^2}{4}$

$Find\ the\ gradient\ of\ f$

Question No: 23 (Marks: 3)

Evaluate the following double integral.

$$\iint \left(3+2x-3y^2\right)\, dx\, dy$$

Question No: 24 (Marks: 3)

Let $f(x,\ y,\ z) = yz^3 - 2x^2$
Find the gradient of f.

Question No: 25 (Marks: 5)

Find Equation of a Tangent plane to the surface $f(x,\ y,\ z) = x^2 + 3y + z^3 - 9$ at the point $(2,\ -1,\ 2)$

Question No: 26 (Marks: 5)

Evaluate the iterated integral

$$\int_2^4 \int_{\frac{x}{2}}^{\sqrt{x}} \left(xy\right)\, dy\, dx$$

MIDTERM EXAMINATION

MTH301

Suppose $f(x, y) = x^3 e^{xy}$. Which one of the following is correct?

$\frac{\partial f}{\partial x} = 3x^2 e^{xy} + x^3 y e^{xy}$

$\frac{\partial f}{\partial x} = 3x^2 y e^{xy}$

$\frac{\partial f}{\partial x} = 3x^2 e^{xy} + x^4 e^{xy}$

$\frac{\partial f}{\partial x} = 3x^2 e^{xy}$

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

Area of R.

Radius of inscribed circle in R.

Distance between two endpoints of R.

None of these

What is the distance between points (3, 2, 4) and (6, 10, -1)?

$7\sqrt{2}$

$2\sqrt{6}$

$\sqrt{34}$

$7\sqrt{3}$

---------- planes intersect at right angle to form three dimensional space.

Three

4

8

12

There is one-to-one correspondence between the set of points on co-ordinate line and ----------

Set of real numbers

Set of integers

Set of natural numbers

Set of rational numbers

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}\ ,\ \ f_{yy}\ and\ \ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D = 0$ then ----------

f has relative maximum at (x_0, y_0)

f has relative minimum at (x_0, y_0)

f has saddle point at (x_0, y_0)

No conclusion can be drawn.

$If\ R = \{(x, y) / 0 \le x \le 2\ and\ 0 \le y \le 3\}, then$

$\iint_R (1 - ye^{xy})dA =$

$\int_0^2 \int_0^3 (1 - ye^{xy})dydx$

$\int_0^2 \int_0^3 (1 - ye^{xy})dxdy$

$\int_2^3 \int_0^0 (1 - ye^{xy})dxdy$

$\int_0^2 \int_2^3 (4xe^{2y})dydx$

Suppose $f(x,\ y) = 2xy\ \ where\ x = t^2 + 1\ \ and\ \ y = 3 - t$. Which one of the following is true?

$\frac{df}{dt} = 6t - 4t^2 - 2$

$\frac{df}{dt} = 6t - 2$

$\frac{df}{dt} = 4t^3 + 6t - 6$

$\frac{df}{dt} = -6t^2 + 12t - 2$

Let i, j and k be unit vectors in the direction of x-axis, y-axis and z-axis respectively. Suppose that $\vec{a} = 2i + 5j - k$. What is the magnitude of vector $\vec{a}$?

6

30

$\sqrt{30}$

$\sqrt{28}$

A straight line is ---------- geometric figure.

One-dimensional

Two-dimensional

Three-dimensional

Dimensionless

$If\ R = \{(x, y) / 0 \le x \le 2\ and\ 1 \le y \le 4\}, then$

$\iint_R (6x^2 + 4xy^3) dA =$

$\int_1^4 \int_0^2 (6x^2 + 4xy^3) dy dx$

$\int_0^2 \int_1^4 (6x^2 + 4xy^3) dx dy$

$\int_1^4 \int_0^2 (6x^2 + 4xy^3) dx dy$

$\int_2^4 \int_0^1 (6x^2 + 4xy^3) dx dy$

Which of the following formula can be used to find the Volume of a parallelepiped with adjacent edges formed by the vectors $\vec{a}$, $\vec{b}$ *and* $\vec{c}$?

$\left|\vec{a}\times\left(\vec{b}\times\vec{c}\right)\right|$

$\left|\vec{a}\bullet\left(\vec{b}\bullet\vec{c}\right)\right|$

$\left|\vec{a}\bullet\left(\vec{b}\times\vec{c}\right)\right|$

$\left|\vec{a}\times\left(\vec{b}\bullet\vec{c}\right)\right|$

The function $f(x,\ y)=\sqrt{y-x}$ is continuous in the region --------- and discontinuous elsewhere.

$x\neq y$

$x\leq y$

$x>y$

What is the relation between the direction of gradient at any point on the surface to the tangent plane at that point ?

parallel

perpendicular

opposite direction

No relation between them.

Suppose $f(x,\ y)=x^3e^{xy}$. Which one of the statements is correct?

$\frac{\partial f}{\partial y}=3x^3e^{xy}$

$\frac{\partial f}{\partial y}=x^3e^{xy}$

$\frac{\partial f}{\partial y}=x^4e^{xy}$

$\frac{\partial f}{\partial y}=x^3ye^{xy}$

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

perpendicular

parallel

in opposite direction

Let the function $f(x, y)$ has continuous second-order partial derivatives $\left(f_{xx}\ ,\ \ f_{yy}\ and\ \ f_{xy}\right)$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\ f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

Relative maximum at (x_0, y_0)

Relative minimum at (x_0, y_0)

Saddle point at (x_0, y_0)

No conclusion can be drawn.

$If\ R = \{(x, y)/0 \le x \le 2\ and\ -1 \le y \le 1\}, then$

$\iint_R (x + 2y^2)dA =$

$\int_{-1}^{1}\int_{0}^{2} (x + 2y^2)dydx$

$\int_{0}^{2}\int_{1}^{-1} (x + 2y^2)dxdy$

$\int_{-1}^{1}\int_{0}^{2} (x + 2y^2)dxdy$

$\int_{1}^{2}\int_{-1}^{0} (x + 2y^2)dxdy$

$f(x, y, z) = \frac{x^2 y}{z} + xyz$

If

then what is the value of $f(1, 1, 1)$?

$f(1, 1, 1) = 1$

$f(1, 1, 1) = 2$

$f(1, 1, 1) = 3$
$f(1, 1, 1) = 4$

If $R = \{(x, y) / 0 \le x \le 4 \, and \, 0 \le y \le 9\}$, *then*
$\iint_R (3x - 4x\sqrt{xy})dA =$

$\int_0^9 \int_0^4 (3x - 4x\sqrt{xy})dydx$

$\int_0^4 \int_4^9 (3x - 4x\sqrt{xy})dxdy$

$\int_4^9 \int_0^0 (3x - 4x\sqrt{xy})dxdy$

$\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$

Let $f(x, y) = 2 + x^2 + \frac{y^2}{4}$

Q- *Find the gradient of* f

2MARKS

Q - Let the function $f(x, y)$ is continuous in the region R, where R is a rectangle as shown below.complete the following equation

$\iint_R f(x, y) \, dA = \int \int f(x, y)$ _____

2MARKS

Q.Find all critical points of the function
$f(x, y) = 4xy - x^3 - 2y^2$

$\int_1^4 \int_0^2 (6x^2 + 4xy^3)dx \, dy$

Evaluate

Q-Evaluate the following double integral.

$\iint \left(3+2x-3y^2\right) dx\ dy$

3MARKS

$$y=\frac{1}{x^2}$$

Q- Let . If x changes from 3 to 3.3, find the approximate change in the value of y using differential dy.

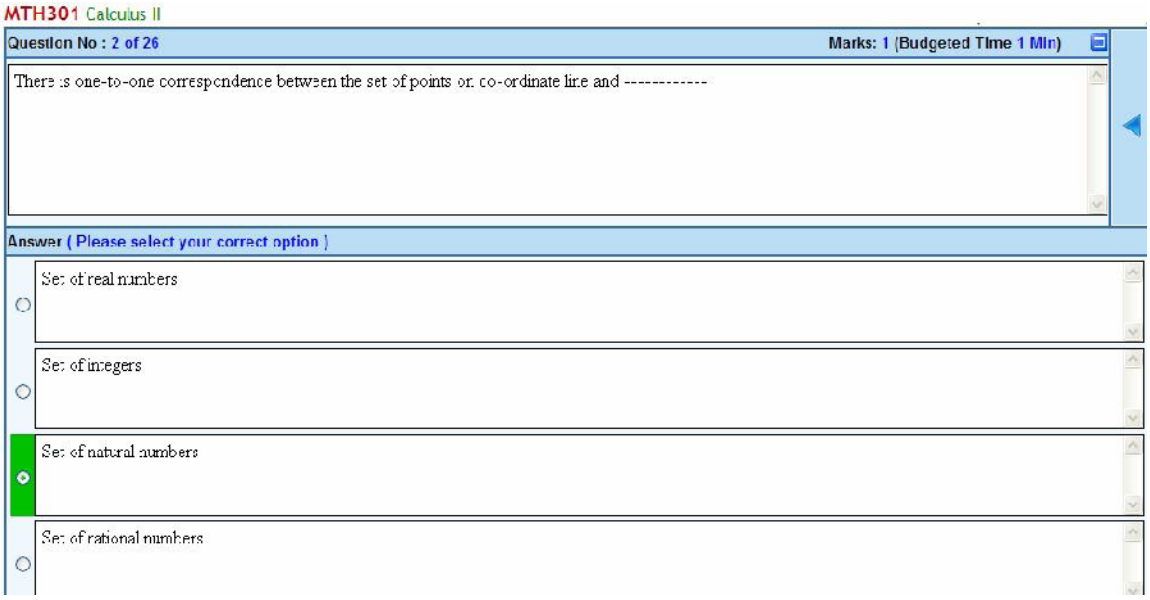
3MARKS

MIDTERM EXAMINATION

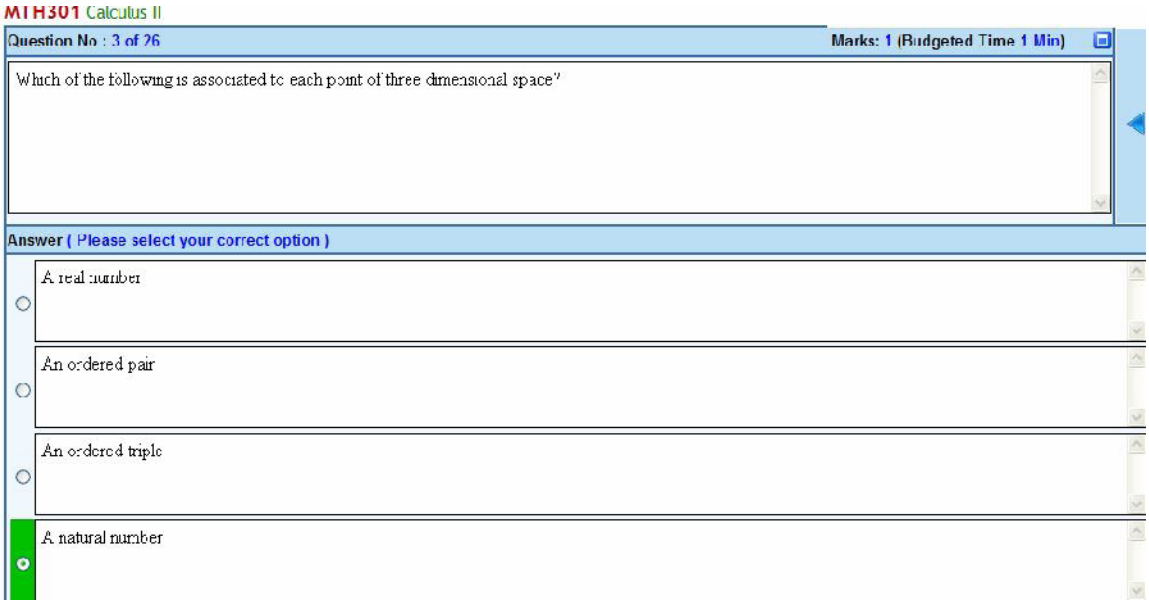
MTH301

That all is collected by removing the mistakes of the one paper file, hope it helpful…!

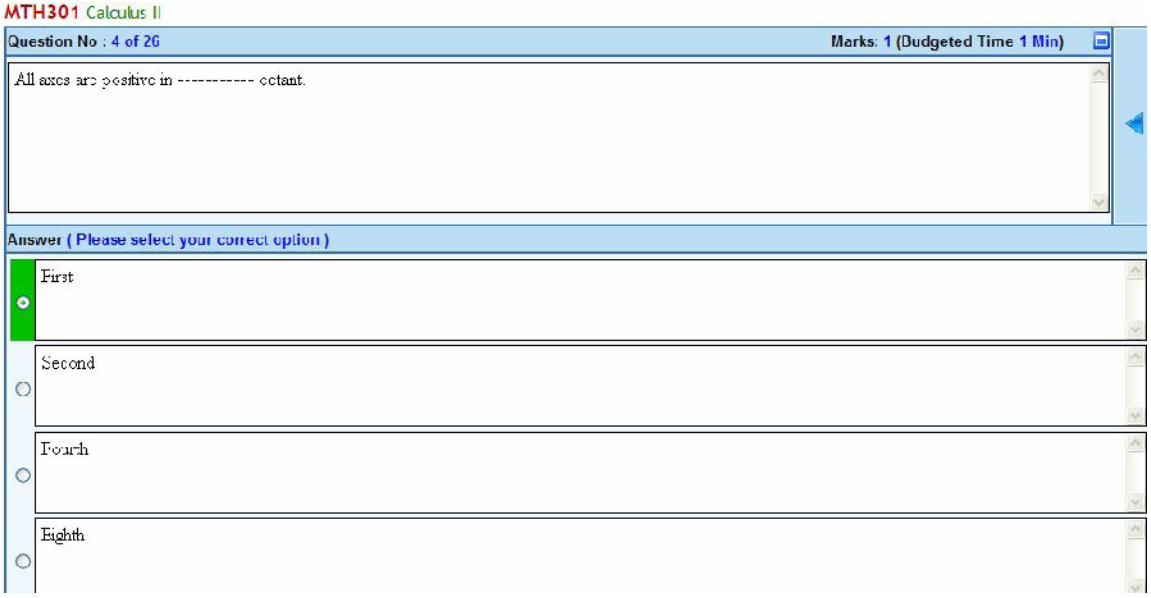
In that the answer is "unique point"



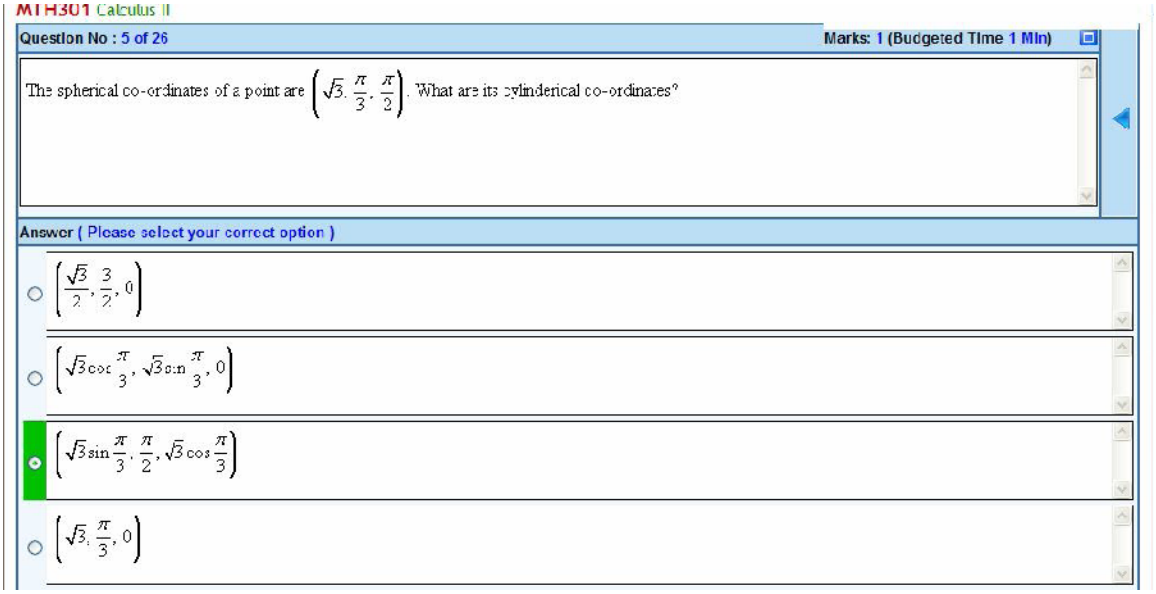
In that the correct answer is the "Set of Real Numbers".



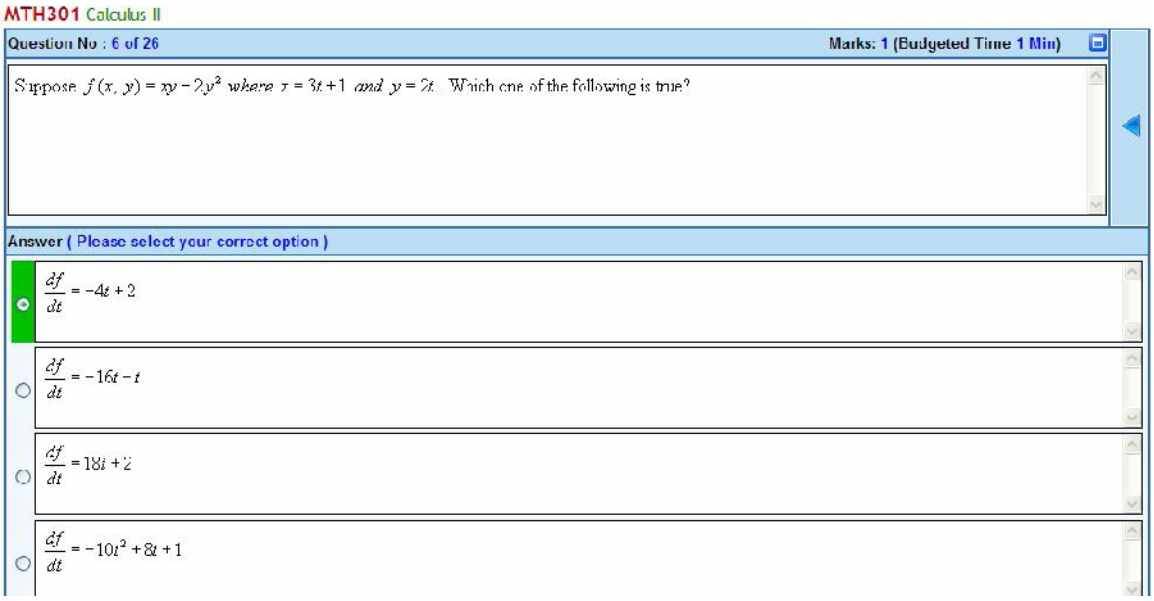
In this the exact answer is "an ordered triple".



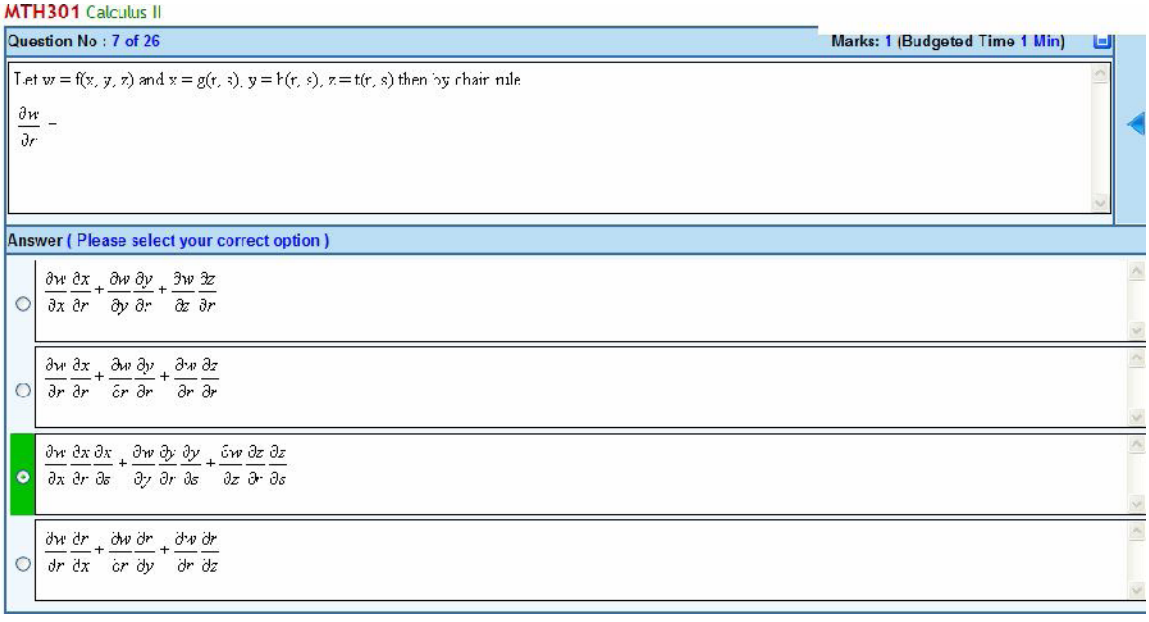
It's right



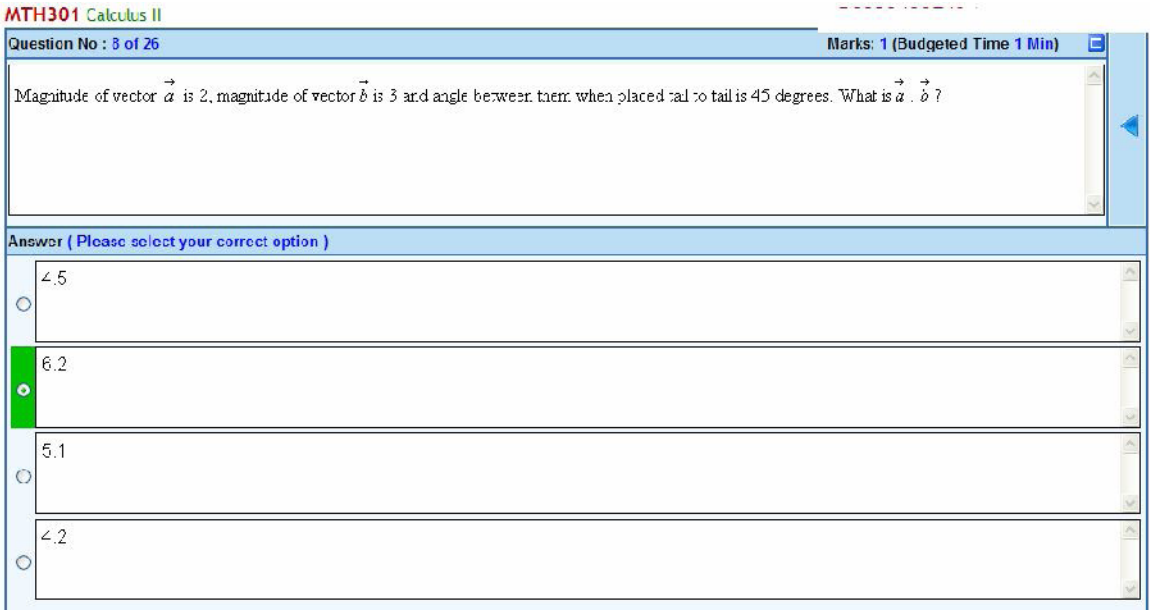
This one is also wrong as the answer is "4th one".because the central point of spherical is pie/3 that should be same in cylindrical so, that's only occur in last one…



That's right



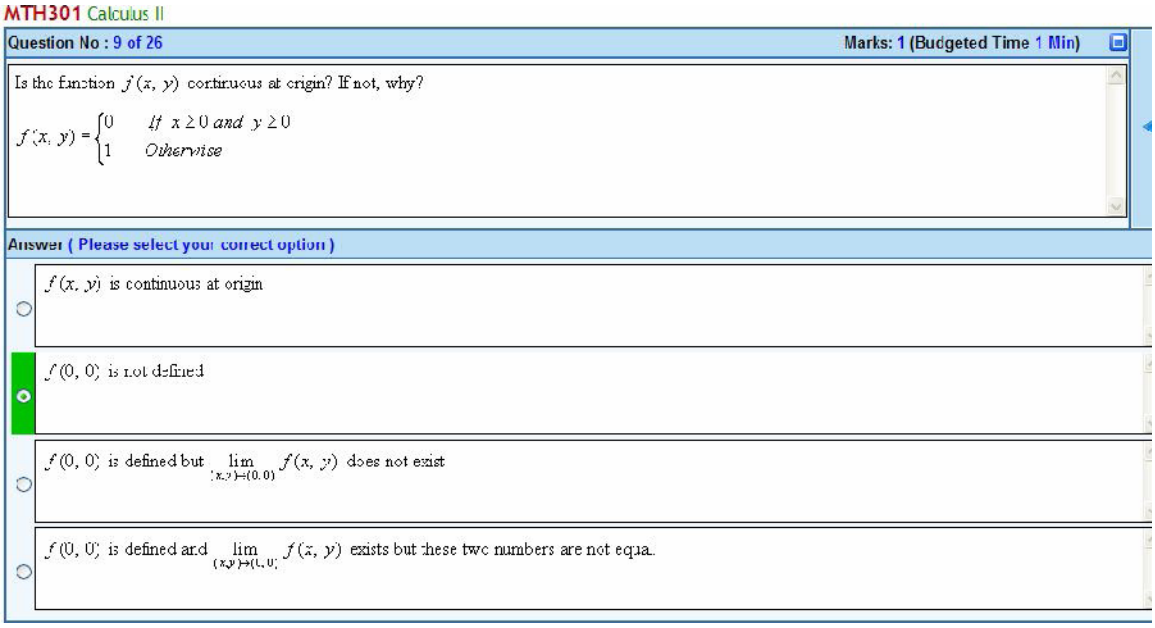
Here's answer is the "2nd" one…!



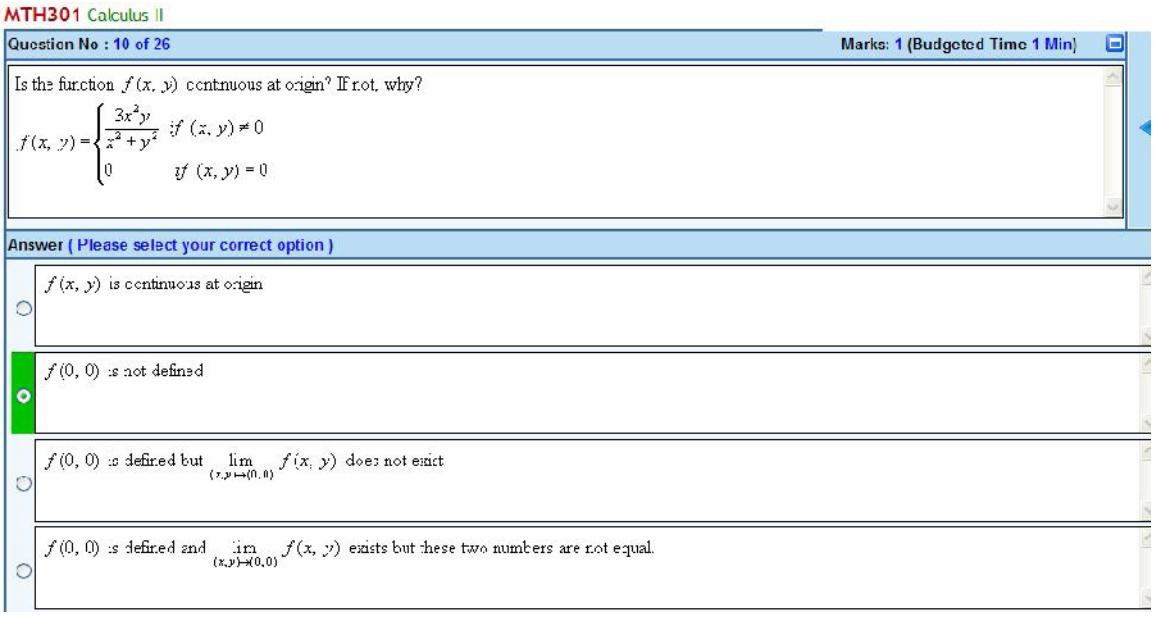
That can't say exactly guys, its formula is following…!

a.b = |a||b| cosØ

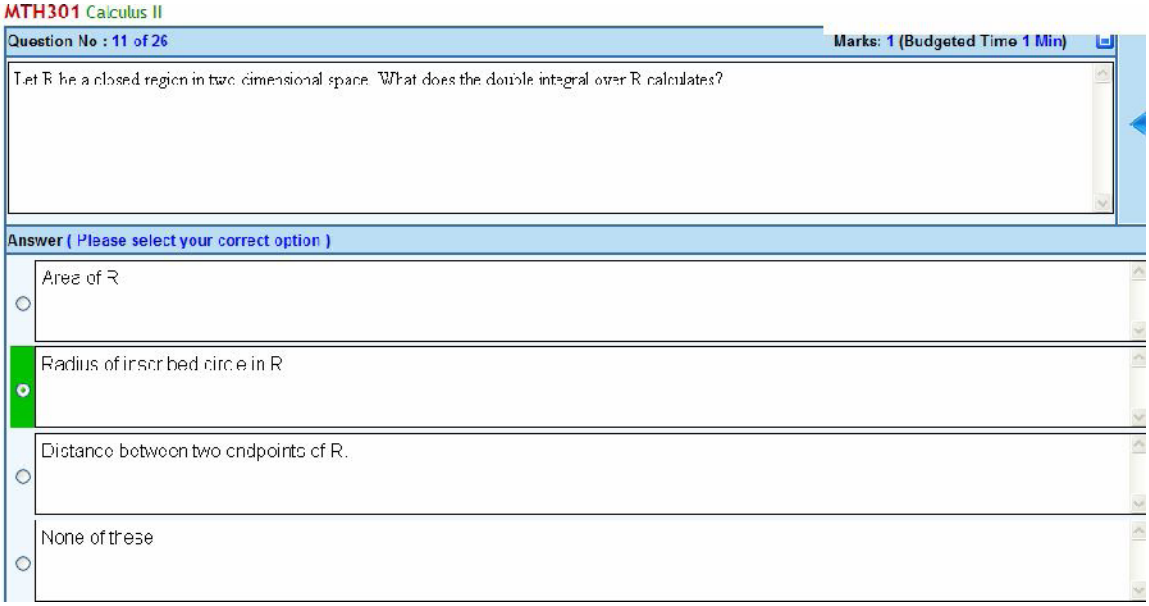
Ø=angle between the vectors



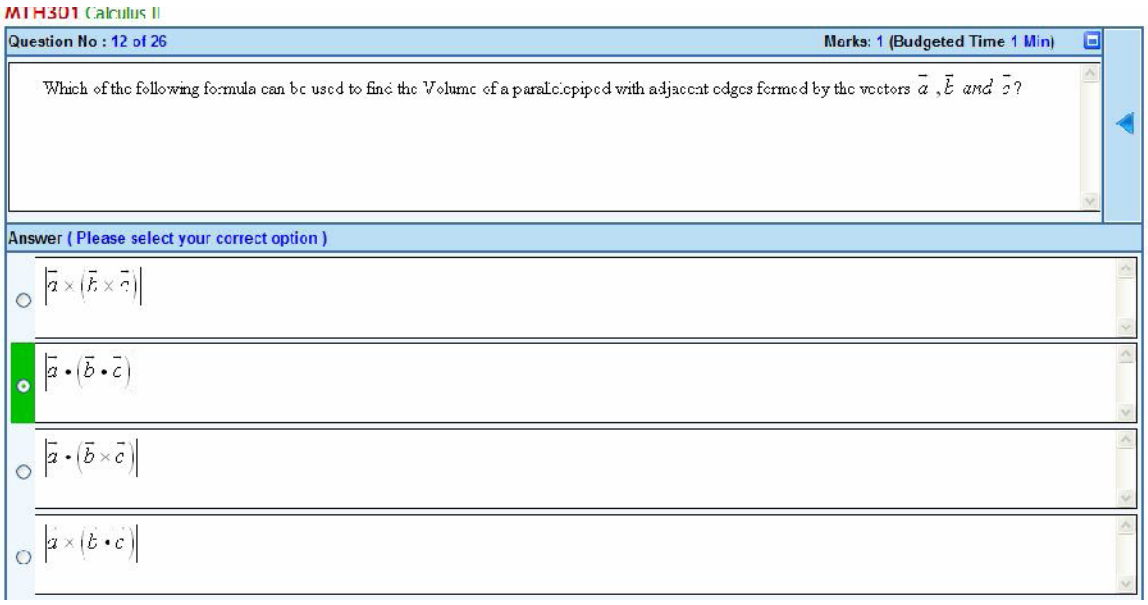
That's correct answer…!



But that's wrong the correc one is the 1st..

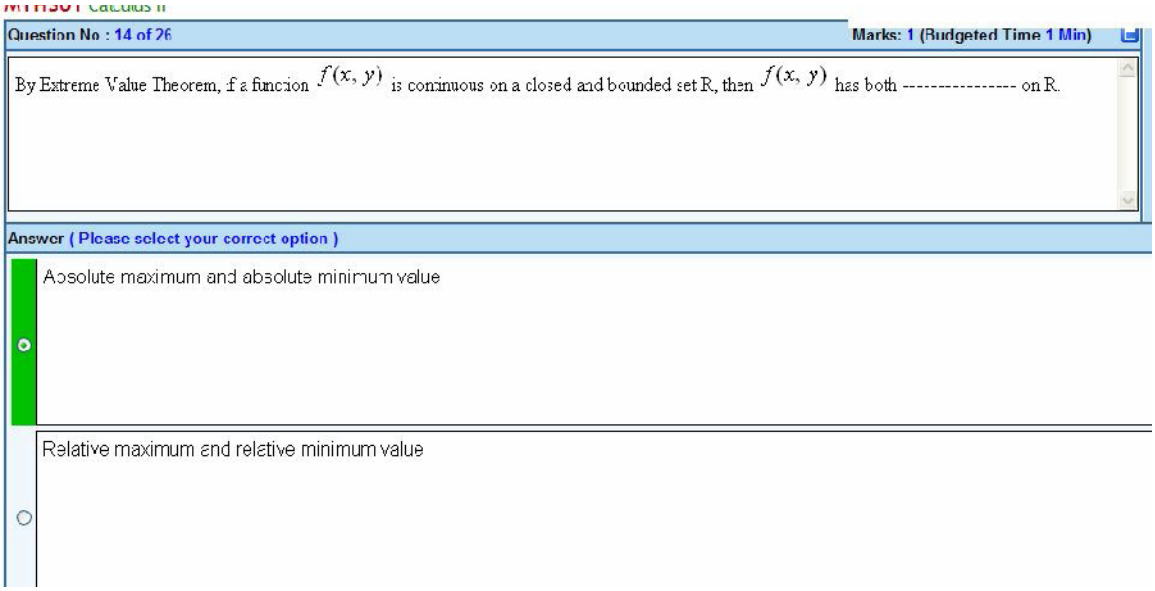


That's the first answer, as that was been define in the lecture 22…!

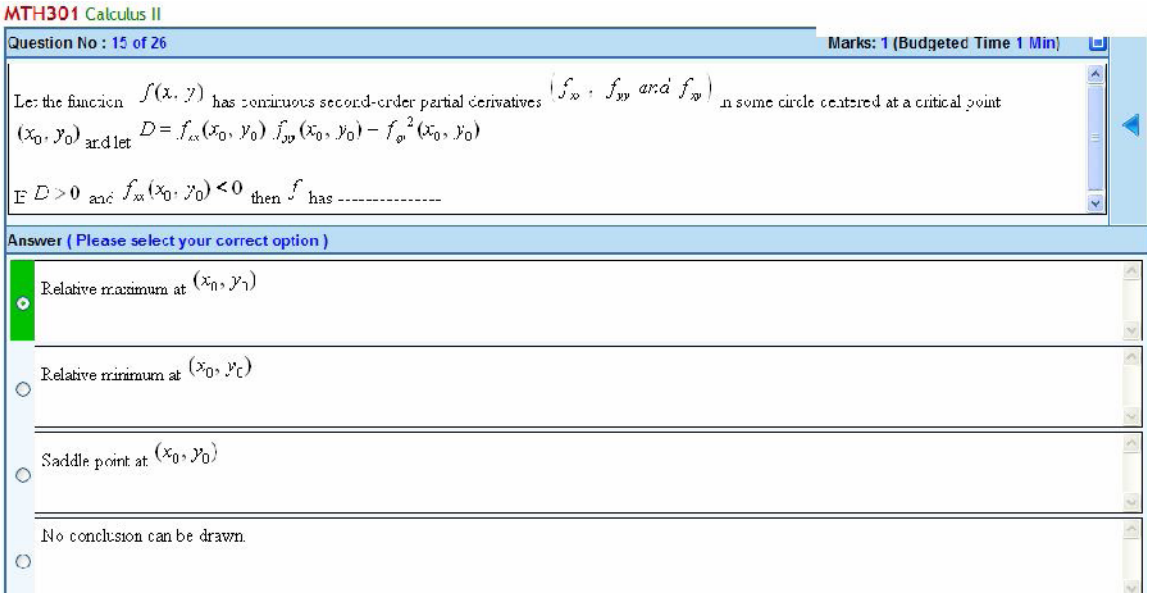


Here's the answer is the last one, you can check that on the start of lecture 11th, handout…!

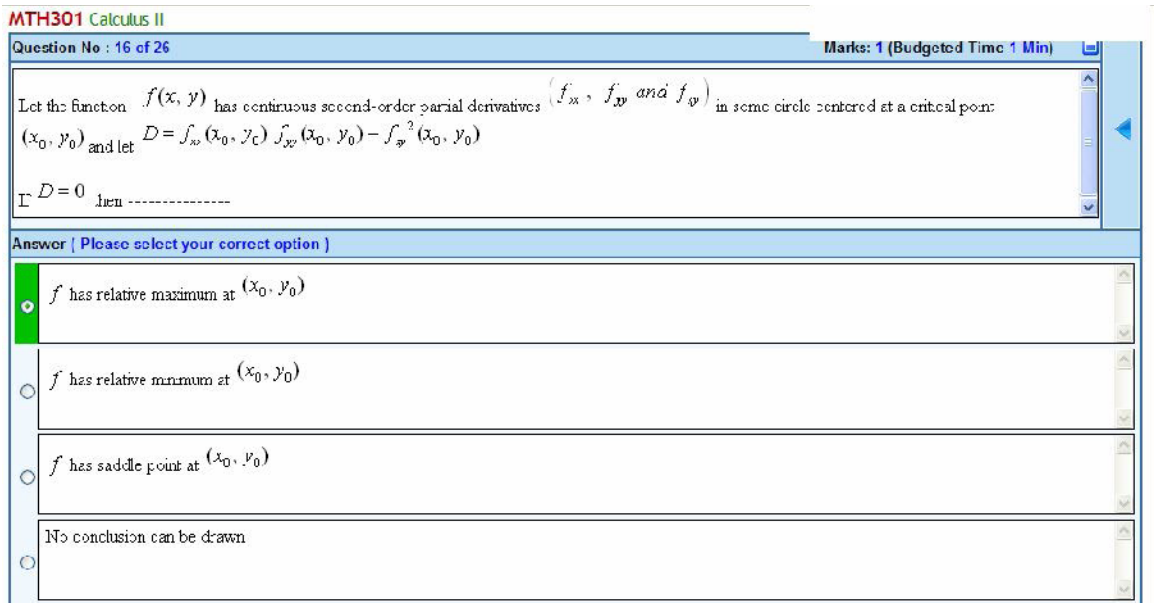
Its answer is the "perpendicular"…!



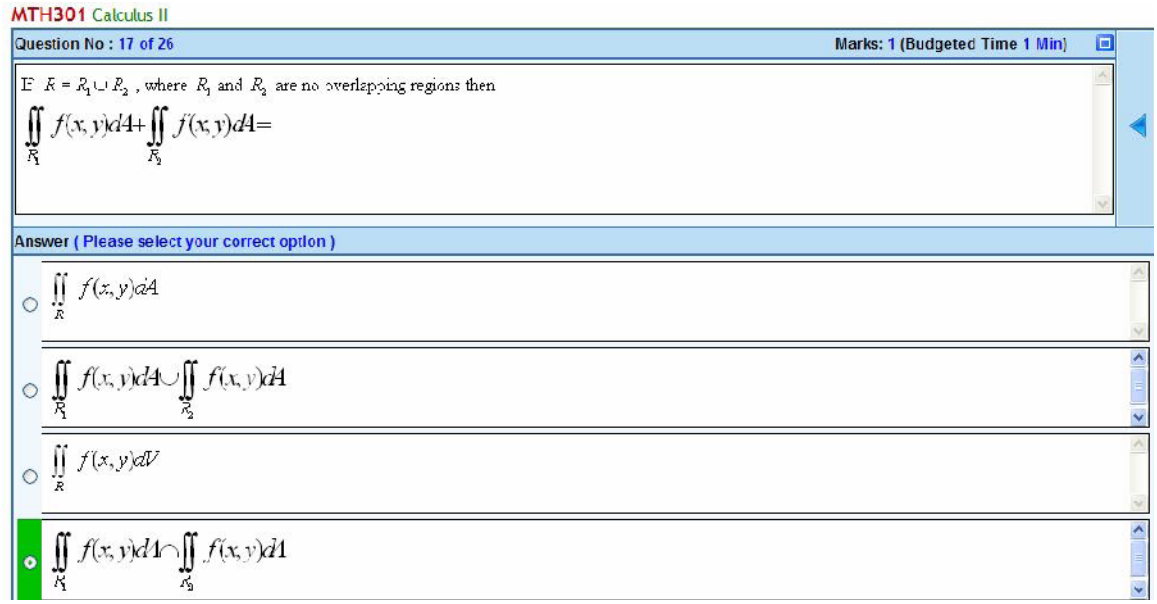
That's right answer…!



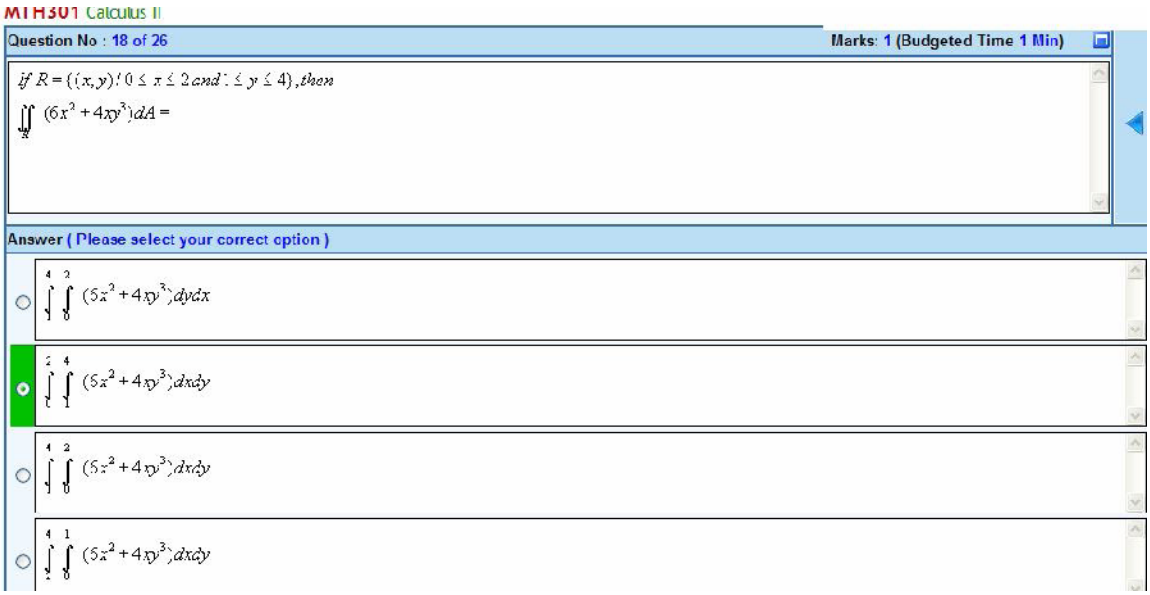
This is also right one..!



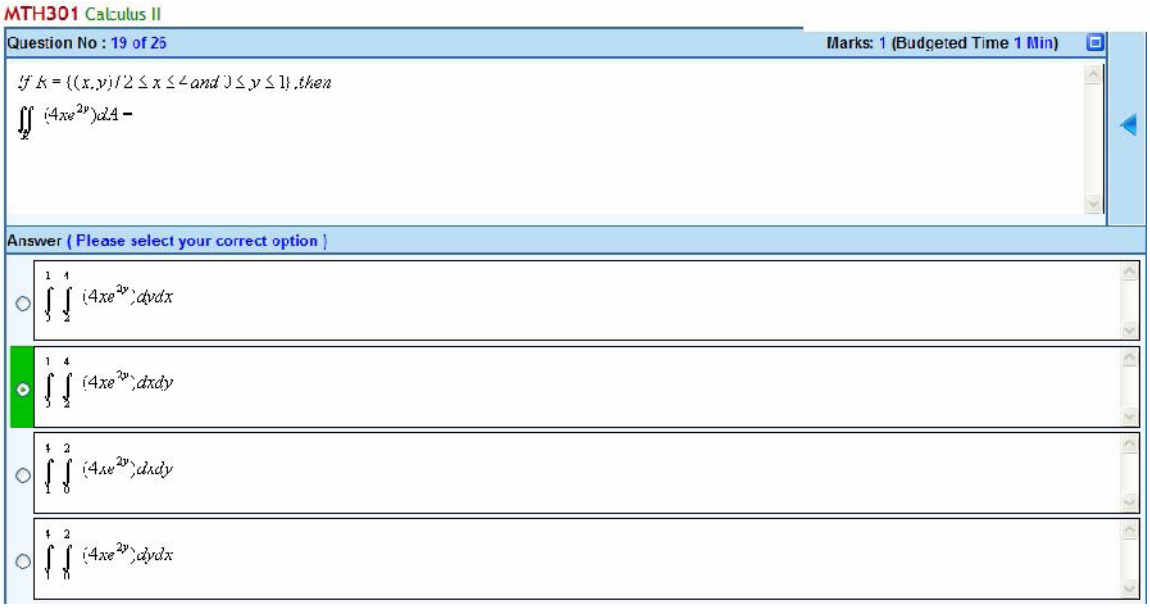
"No conclusion can be drawn" is the answere…!



Its answer is the first one option…!

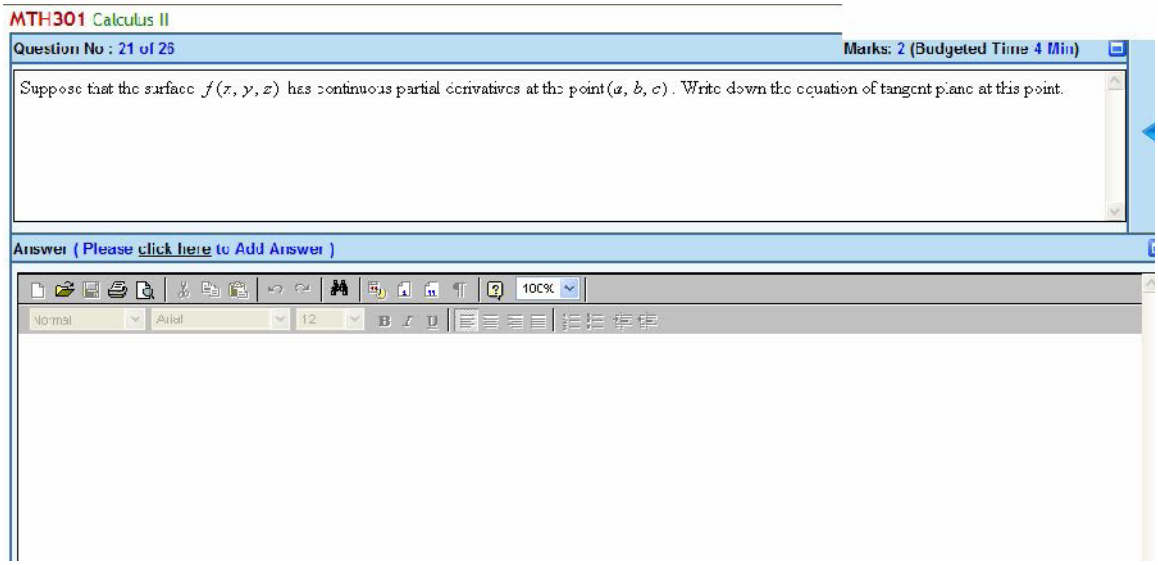


Its answer is the 3rd option



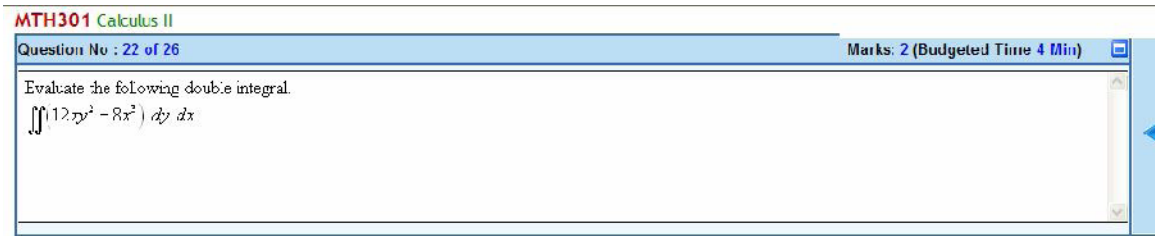
It's right one…!

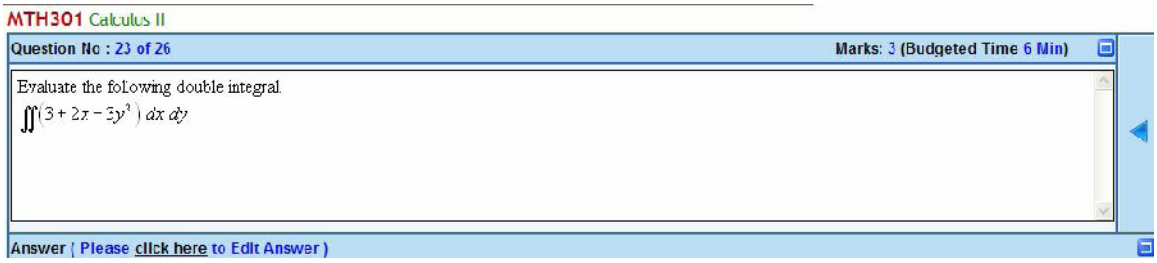
This one's also right…



Its equation is

$$f_x(a)(x-x_0)+f_y(b)(y-y_0)+f_z(c)(z-z_0)=0$$





These both can be solve by the following example the me gonna paste…!

Evaluate the following double integral.

$$\iint\left(3+2x-3y^2\right)\,dx\,dy$$

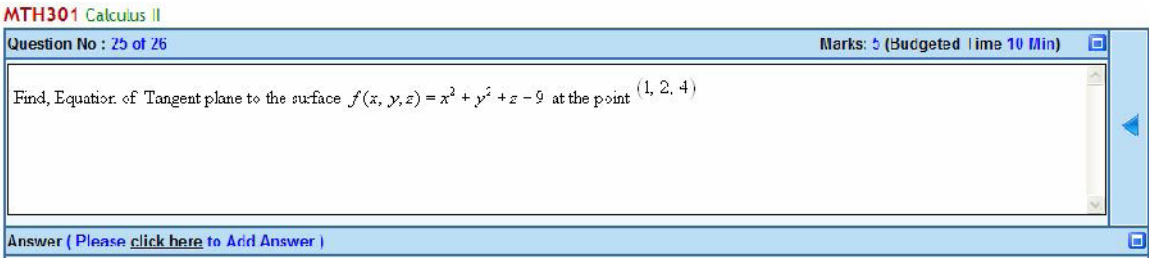
Solution:

$$\iint\left(3+2x-3y^2\right)\,dx\,dy$$

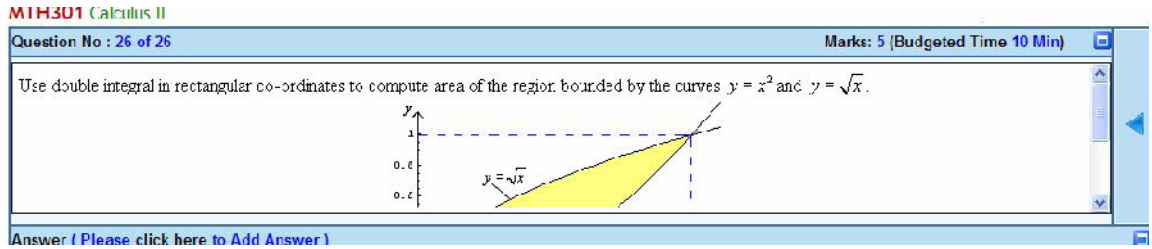
$$\int(3x+2x^2-3xy^2)dy$$

$$3xy+2x^2y-xy^3$$

didn't got this question dear's…!



You can use the above tangent equation to sove this…!



Its answer is the under mentioned…!

as our require is area of the following region

$V = \iint_R 1 dA = \iint_R dA$ we can compute area by keeping f(x,y) const

$$= \int_0^1 \int_{x^2}^{x} dy dx$$

$$= \int_0^1 [x]_{x^2}^{x}$$

$$= \int_0^1 (x - x^2)$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1$$

$$= \left[\frac{1}{2} - \frac{1}{3}\right]$$

$$= \left[\frac{3-2}{6}\right]$$

$= \frac{1}{6}$ that's exactly what we require...!

MIDTERM EXAMINATION
Spring 2010
MTH301- Calculus II (Session - 2)

Ref No: 1499740
Time: 60 min
Marks: 40

Student Info	
StudentID:	**BC080402259**
Center:	**OPKST**
ExamDate:	**6/4/2010 12:00:00 AM**

For Teacher's Use Only

Q No.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**Total**
Marks									
Q No.	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	
Marks									
Q No.	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	
Marks									
Q No.	**25**	**26**							
Marks									

Question No: 1 (Marks: 1) - Please choose one

Which of the following is the interval notation of real line?

- (-∞ , +∞)
- (-∞ , 0)
- (0 , +∞)

Question No: 2 (Marks: 1) - Please choose one

What is the general equation of parabola whose axis of symmetry is parallel to y-axis?

- $y = ax^2 + b \quad (a \neq 0)$
- $x = ay^2 + b \quad (a \neq 0)$
- $y = ax^2 + bx + c \quad (a \neq 0)$
- $x = ay^2 + by + c \quad (a \neq 0)$

Question No: 3 (Marks: 1) - Please choose one

Which of the following is geometrical representation of the equation $y = 4$, in three dimensional space?

- A point on y-axis
- Plane parallel to xy-plane
- Plane parallel to yz-axis
- Plane parallel to xz-plane

Question No: 4 (Marks: 1) - Please choose one

Suppose $f(x, y) = x^3 e^{xy}$. Which one of the statements is correct?

- $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$
- $\frac{\partial f}{\partial y} = x^3 e^{xy}$
- $\frac{\partial f}{\partial y} = x^4 e^{xy}$

- ▶ $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

Question No: 5 (Marks: 1) - Please choose one

If $f(x, y) = x^2 y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2}$ =

- ▶ $2xy + \frac{1}{x^2}$
- ▶ $2y + \frac{1}{x^2}$
- ▶ $2xy - \frac{1}{x^2}$
- ▶ $2y - \frac{1}{x^2}$

Question No: 6 (Marks: 1) - Please choose one

Let w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$\frac{\partial w}{\partial r} =$

- ▶ $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$
- ▶ $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- ▶ $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- ▶ $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

Question No: 7 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} \frac{3x^2 y}{x^2 + y^2} & if\ (x, y) \neq 0 \\ 0 & if\ (x, y) = 0 \end{cases}$$

- ▶ $f(x, y)$ is continuous at origin

- ► $f(0, 0)$ is not defined
- ► $f(0, 0)$ is defined but $\lim_{(x,y)\to(0, 0)} f(x, y)$ does not exist
- ► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0, 0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 8 (Marks: 1) - Please choose one

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- ► Area of R.
- ► Radius of inscribed circle in R.
- ► Distance between two endpoints of R.
- ► None of these

Question No: 9 (Marks: 1) - Please choose one

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are ---------

- ► Parallel
- ► Perpendicular
- ► In opposite direction

Question No: 10 (Marks: 1) - Please choose one

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

- ► perpendicular
- ► parallel
- ► in opposite direction

Question No: 11 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) > 0$ then f has ----------

- ► Relative maximum at (x_0, y_0)
- ► Relative minimum at (x_0, y_0)
- ► Saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 12 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

► Relative maximum at (x_0, y_0)

► Relative minimum at (x_0, y_0)

► Saddle point at (x_0, y_0)

► No conclusion can be drawn.

Question No: 13 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D < 0$ then f has ----------

► Relative maximum at (x_0, y_0)

► Relative minimum at (x_0, y_0)

► Saddle point at (x_0, y_0)

► No conclusion can be drawn

Question No: 14 (Marks: 1) - Please choose one

Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be any two points in three dimensional space. What does the formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ calculates?

► Distance between these two points

► Midpoint of the line joining these two points

► Ratio between these two points

Question No: 15 (Marks: 1) - Please choose one

The function $f(x, y) = \sqrt{y - x}$ is continuous in the region --------- and discontinuous elsewhere.

- $x \neq y$
- $x \leq y$
- $x > y$

Question No: 16 (Marks: 1) - Please choose one

Plane is an example of ----------

- Curve
- Surface
- Sphere
- Cone

Question No: 17 (Marks: 1) - Please choose one

If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then

$$\iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA =$$

- $\iint_{R} f(x,y)dA$
- $\iint_{R_1} f(x,y)dA \cup \iint_{R_2} f(x,y)dA$
- $\iint_{R} f(x,y)dV$
- $\iint_{R_1} f(x,y)dA \cap \iint_{R_2} f(x,y)dA$

Question No: 18 (Marks: 1) - Please choose one

$If\ R = \{(x,y) / 0 \leq x \leq 2\ and\ 1 \leq y \leq 4\},\ then$

$$\iint_{R} (6x^2 + 4xy^3)dA =$$

- $\int_{1}^{4}\int_{0}^{2}(6x^2+4xy^3)dydx$
- $\int_{0}^{2}\int_{1}^{4}(6x^2+4xy^3)dxdy$
- $\int_{1}^{4}\int_{0}^{2}(6x^2+4xy^3)dxdy$
- $\int_{2}^{4}\int_{0}^{1}(6x^2+4xy^3)dxdy$

Question No: 19 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 2\ and\ 0\le y\le 3\},then$
$\iint_R (1-ye^{xy})dA=$

- $\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dydx$
- $\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dxdy$
- $\int_{2}^{3}\int_{0}^{0}(1-ye^{xy})dxdy$
- $\int_{0}^{2}\int_{2}^{3}(4xe^{2y})dydx$

Question No: 20 (Marks: 1) - Please choose one

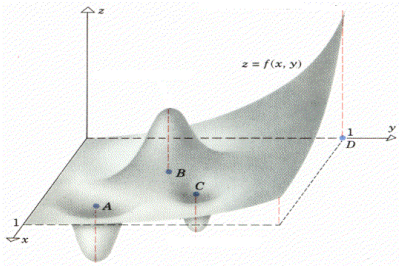
$If\ R=\{(x,y)/0\le x\le 4\ and\ 0\le y\le 9\},then$
$\iint_R (3x-4x\sqrt{xy})dA=$

- $\int_{0}^{9}\int_{0}^{4}(3x-4x\sqrt{xy})dydx$
- $\int_{0}^{4}\int_{4}^{9}(3x-4x\sqrt{xy})dxdy$
- $\int_{4}^{9}\int_{0}^{0}(3x-4x\sqrt{xy})dxdy$

▶ $\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$

Question No: 21 (Marks: 2)

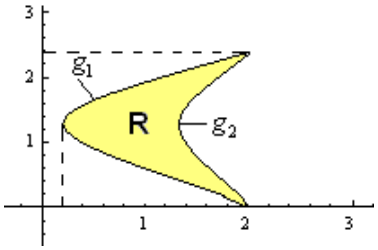
Following is the graph of a function of two variables



In its whole domain, state whether the function has relative maximum value or absolute maximum value at point B. Also, justify your answer

Question No: 22 (Marks: 2)

Let the function $f(x, y)$ is continuous in the region R, where R is bounded by graph of functions g_1 and g_2 (as shown below).



In the following equation, replace question mark (?) with the correct value.

$$\iint_R f(x, y)\, dA = \int_?^? \int_?^? f(x, y)\ _____?____$$

Question No: 23 (Marks: 3)

Evaluate the following double integral.

$\iint (3 + 2x - 3y^2)\, dx\, dy$

Question No: 24 (Marks: 3)

Let $f(x, y, z) = yz^3 - 2x^2$

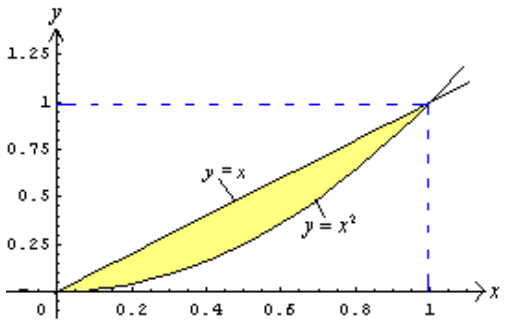
Find the gradient of f.

Question No: 25 (Marks: 5)

Find, Equation of Normal line (in parametric form) to the surface $f(x, y, z) = xy + 2yz - xz^2 + 10$ at the point $(-5, 5, 1)$

Question No: 26 (Marks: 5)

Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x$ and $y = x^2$, in the first quadrant.



MIDTERM EXAMINATION
Spring 2010
MTH301- Calculus II (Session - 3)

Time: 60 min
Marks: 40

Student Info	
StudentID:	
Center:	OPKST
ExamDate:	6/3/2010 12:00:00 AM

For Teacher's Use Only									
Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26							
Marks									

Question No: 1 (Marks: 1) - Please choose one

Which of the following number is associated to each point on a co-ordinate line?

► An integer

► **A real number**

► A rational number

► A natural number

Question No: 2 (Marks: 1) - Please choose one

If $a>0$, then the parabola $y=ax^2+bx+c$ opens in which of the following direction?

► **Positive x - direction**

► Negative x - direction

► Positive y - direction

► Negative y - direction

Question No: 3 (Marks: 1) - Please choose one

Rectangular co-ordinate of a point is $(1, \sqrt{3}, -2)$. What is its spherical co-ordinate?

$\left(2\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{2}\right)$

. ►

$\left(2\sqrt{2}, \frac{\pi}{2}, \frac{3\pi}{4}\right)$

. ►

$\left(2\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4}\right)$

. ►

$\left(\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4}\right)$

. ►

Question No: 4 (Marks: 1) - Please choose one

If a function is not defined at some point, then its limit ---------- exist at that point.

. ► Always

. ► Never

. ▶ May

Question No: 5 (Marks: 1) - Please choose one

Suppose $f(x,\ y) = x^3 e^{xy}$. Which one of the statements is correct?

. ▶ $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$

. ▶ $\frac{\partial f}{\partial y} = x^3 e^{xy}$

. ▶ $\frac{\partial f}{\partial y} = x^4 e^{xy}$

. ▶ $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

Question No: 6 (Marks: 1) - Please choose one

If $f(x,\ y) = x^2 y - y^3 + \ln x$

$\frac{\partial^2 f}{\partial x^2}$

then =

. ▶ $2xy + \frac{1}{x^2}$

. ▶ $2y + \frac{1}{x^2}$

. ▶ $2xy - \frac{1}{x^2}$

. ▶ $2y - \frac{1}{x^2}$

Question No: 7 (Marks: 1) - Please choose one

Suppose $f(x,\ y) = xy - 2y^2 \ \ where\ \ x = 3t + 1 \ \ and \ \ y = 2t$. Which one of the following is true?

$\frac{df}{dt} = -4t + 2$

$\frac{df}{dt} = -16t - t$

$\frac{df}{dt} = 18t + 2$

$\frac{df}{dt} = -10t^2 + 8t + 1$

Question No: 8 (Marks: 1) - Please choose one

Is the function $f(x,\ y)$ continuous at origin? If not, why?

$$f(x,\ y) = \begin{cases} 0 & If\ \ x \geq 0\ and\ \ y \geq 0 \\ 1 & Otherwise \end{cases}$$

- $f(x,\ y)$ is continuous at origin
- $f(0,\ 0)$ is not defined
- $f(0,\ 0)$ is defined but $\lim_{(x,y)\to(0,\ 0)} f(x,\ y)$ does not exist
- $f(0,\ 0)$ is defined and $\lim_{(x,y)\to(0,\ 0)} f(x,\ y)$ exists but these two numbers are not equal.

Question No: 9 (Marks: 1) - Please choose one

What is the relation between the direction of gradient at any point on the surface to the tangent plane at that point ?

- parallel
- **perpendicular**
- opposite direction
- No relation between them.

Question No: 10 (Marks: 1) - Please choose one

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

- perpendicular
- parallel
- **in opposite direction**

Question No: 11 (Marks: 1) - Please choose one

By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R, then $f(x, y)$ has both ---------- on R.

- ► **Absolute maximum and absolute minimum value**
- ► Relative maximum and relative minimum value

Question No: 12 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

- ► **Relative maximum at** (x_0, y_0)
- ► Relative minimum at (x_0, y_0)
- ► Saddle point at (x_0, y_0)
- ► No conclusion can be drawn.

Question No: 13 (Marks: 1) - Please choose one

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}{}^2(x_0, y_0)$

If $D = 0$ then ----------

- ► f has relative maximum at (x_0, y_0)
- ► f has relative minimum at (x_0, y_0)
- ► f has saddle point at (x_0, y_0)
- ► **No conclusion can be drawn.**

Question No: 14 (Marks: 1) - Please choose one

The function $f(x, y) = \sqrt{y - x}$ is continuous in the region --------- and discontinuous elsewhere.

- $x \neq y$
- $x \leq y$
- $x > y$

Question No: 15 (Marks: 1) - Please choose one

Plane is an example of ----------

- Curve
- **Surface**
- Sphere
- Cone

Question No: 16 (Marks: 1) - Please choose one

If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then $\iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA =$

- $\iint_R f(x,y)dA$
- $\iint_{R_1} f(x,y)dA \cup \iint_{R_2} f(x,y)dA$
- $\iint_R f(x,y)dV$
- $\iint_{R_1} f(x,y)dA \cap \iint_{R_2} f(x,y)dA$

Question No: 17 (Marks: 1) - Please choose one

If $R = \{(x, y) / 0 \le x \le 2 \, and \, 1 \le y \le 4\}, then$

$\iint_R (6x^2 + 4xy^3)dA =$

$\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$

▶

$\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$

▶

$\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$

▶

$\int_2^4 \int_0^1 (6x^2 + 4xy^3)dxdy$

▶

Question No: 18 (Marks: 1) - Please choose one

If $R = \{(x, y) / 0 \le x \le 2 \, and \, -1 \le y \le 1\}, then$

$\iint_R (x + 2y^2)dA =$

$\int_{-1}^1 \int_0^2 (x + 2y^2)dydx$

▶

$\int_0^2 \int_1^{-1} (x + 2y^2)dxdy$

▶

$\int_{-1}^1 \int_0^2 (x + 2y^2)dxdy$

▶

$\int_1^2 \int_{-1}^0 (x + 2y^2)dxdy$

▶

Question No: 19 (Marks: 1) - Please choose one

If $R=\{(x,y)/0\le x\le 2 \text{ and } 0\le y\le 3\}$, *then*

$\iint_R (1-ye^{xy})dA=$

$\int_0^2 \int_0^3 (1-ye^{xy})dydx$

▶

$\int_0^2 \int_0^3 (1-ye^{xy})dxdy$

▶

$\int_2^3 \int_0^0 (1-ye^{xy})dxdy$

▶

$\int_0^2 \int_2^3 (4xe^{2y})dydx$

▶

Question No: 20 (Marks: 1) - Please choose one

If $R=\{(x,y)/0\le x\le 4 \text{ and } 0\le y\le 9\}$, *then*

$\iint_R (3x-4x\sqrt{xy})dA=$

$\int_0^9 \int_0^4 (3x-4x\sqrt{xy})dydx$

▶

$\int_0^4 \int_4^9 (3x-4x\sqrt{xy})dxdy$

▶

$\int_4^9 \int_0^0 (3x-4x\sqrt{xy})dxdy$

▶

$$\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$$

▶

Question No: 21 (Marks: 2)

Evaluate the following double integral.

$\iint \left(2xy + y^3\right) dx\ dy$

Question No: 22 (Marks: 2)

$Let\ f(x,\ y) = 2 + x^2 + \frac{y^2}{4}$

$Find\ the\ gradient\ of\ f$

Question No: 23 (Marks: 3)

Evaluate the following double integral.

$\iint \left(3 + 2x - 3y^2\right) dx\ dy$

Question No: 24 (Marks: 3)

$Let\ f(x,\ y,\ z) = yz^3 - 2x^2$

$Find\ the\ gradient\ of\ f.$

Question No: 25 (Marks: 5)

Find Equation of a Tangent plane to the surface $f(x,\ y,\ z) = x^2 + 3y + z^3 - 9$ at the point $(2,\ -1,\ 2)$

Question No: 26 (Marks: 5)

Evaluate the iterated integral

$\int_2^4 \int_{\frac{x}{2}}^{\sqrt{x}} (xy)\, dy\, dx$

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11-27-2011, 12:52 PM #1

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Total 26 Questions:
20 MCQS. (Which is mostly from past papers)
6 Long questions:

Long Questions are here:

Q: 21: Write down the equation of a plane passing through the point (3,-4, 2) and perpendicular to the vector n=4i+3j+6k

Q: 22: Lets xyz is three positive integers, whose sum is 18 and the product of two integers with square of third integers will be maximum. Wright the function to be maximized in the function x and y.

Q: 23: Find the critical point for the function f(x, y) =x^2+y^2along the line y+2-2xat which the absolute extrema of the function can exist.

Q: 24: Find the equation of normal line (symmetric form) to the surface f(x, y) =$x^2+y^2+z^2$-6 at the point (-1, 2, 1).

Q: 25: Using second partial derivative test show that the function f(x, y) =xy (24-x-y) is maximum at x=8 and y=8.

Q: 26: Find the gradient of the f(x, y, z) =yz^3- $2x^2$.

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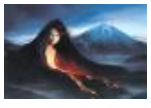
12-01-2011, 03:27 PM #3



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Best of luck!

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Question No : 1 of 26 Marks: 1 (Budgeted Time 1 Min)

Which of the following is geometrical representation of set of real numbers?

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Answer (Please select your correct option)

- Co-ordinate line
- xy-plane
- Sphere
- Circular cylinder

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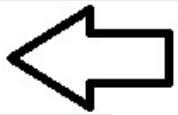
Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

There is one-to-one correspondence between the set of points on a co-ordinate line and ----------

Answer (Please select your correct option)

- Set of real numbers
- Set of integers
- Set of natural numbers
- Set of rational numbers



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Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

An ordered triple corresponds to ---------- in a three dimensional space.

Answer (Please select your correct option)

- A unique point
- A point in each octant
- Three points
- Infinite number of points

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Question No : 4 of 26 Marks: 1 (Budgeted Time 1 Min)

If the positive directions of x and y axes are known then ---------- the positive direction of z-axis.

Answer (Please select your correct option)

- Horizontal rightward direction is
- Horizontal leftward direction is
- Left hand rule tells
- Right hand rule tells

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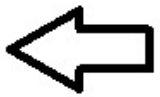
Question No : 5 of 26 Marks: 1 (Budgeted Time 1 Min)

If a function is not defined at some point, then its limit ---------- exist at that point.

Answer (Please select your correct option)

- Always
- Never
- May



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Question No : 6 of 26 Marks: 1 (Budgeted Time 1 Min)

A composition of continuous functions . . . . . .

Answer (Please select your correct option)

- is always continuous

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- may or may not be continuous
- is discontinuous
- is piece-wise continuous

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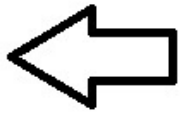
Question No : 7 of 26 Marks: 1 (Budgeted Time 1 Min)

According to the Euler's theorem, the order of partial differentiation can be changed, provided the function and all of its partial derivatives are. . . . . .

Answer (Please select your correct option)

- Piece-wise continuous
- Continuous
- Discontinuous



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Question No : 8 of 26 Marks: 1 (Budgeted Time 1 Min)

If $x = f(r,s)$ and $r = g(t)$, $s = h(t)$, then derivative of x with respect to t is written as. . . . . .

Answer (Please select your correct option)

- $\frac{\partial x}{\partial t}$
- (x,t)
- $\frac{\partial^2 x}{\partial t^2}$
- $\frac{dx}{dt}$

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Question No : 9 of 26 Marks: 1 (Budgeted Time 1 Min)

A vector in a plane is always represented by a . . . . . .

Answer (Please select your correct option)

- Line
- Curve
- Line segment
- None of these.

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Question No : 10 of 26 Marks: 1 (Budgeted Time 1 Min)

If two vectors $\vec{a}$ and $\vec{b}$ have the relation $\vec{a} = \lambda\vec{b}$ where λ is a none-zero scalar then $\vec{a}$ and $\vec{b}$ are . . . . . to each other.

Answer (Please select your correct option)

- Orthogonal
- Non-parallel
- Parallel
- None of these

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Question No : 11 of 26 Marks: 1 (Budgeted Time 1 Min)

The function decreases most rapidly in the direction of . . . . . .

Answer (Please select your correct option)

- $-\nabla f$
- $-|\nabla f|$
- $\nabla f \times \hat{u}$
- $|\nabla f|$

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Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

Tthe direction of gradient at any point on the surface is . . . . . to the tangent plane at that point.

Answer (Please select your correct option)

- parallel
- perpendicular
- opposite direction
- None of these.

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Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

$2x^2 + y^2 = 4$ is the . . . . . form of equation of a curve.

Answer (Please select your correct option)

- Parametric
- Implicit
- Symmetric
- Explicit

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Question No : 14 of 26 Marks: 1 (Budgeted Time 1 Min)

If $f(x,y)$ has a relative extremum at a point (x_0,y_0) and both the first partial derivatives of f exist at this point, then . . . . . .

Answer (Please select your correct option)

- $f_x(x_0,y_0) = 0$ and $f_y(x_0,y_0) \neq 0$
- $f_x(x_0,y_0) \leq 0$ and $f_y(x_0,y_0) \leq 0$
- $f_x(x_0,y_0) \geq 0$ and $f_y(x_0,y_0) \geq 0$
- $f_x(x_0,y_0) = 0$ and $f_y(x_0,y_0) = 0$

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Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

For a function $f(x)$, if $f'(x)$ is equal to some non-zero constant, then $f(x)$ will have . . . . . .

Answer (Please select your correct option)

- At least one critical point
- No critical point
- Non-zero critical points
- None of these

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Question No : 16 of 26 Marks: 1 (Budgeted Time 1 Min)

Let x, y, z be the length, width and height of an open rectangular box. The surface area of the box will be . . . . . .

Answer (Please select your correct option)

- $A = xy + 2yz + 2xz$
- $A = yz + 4$
- $A = xz + yz + zx$
- $A = xyz$

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Question No : 17 of 26 Marks: 1 (Budgeted Time 1 Min)

Double integral of a function $f(x,y)$ represents . . . . . of the region between the surface defined by the function and the plane which contains its domain.

Answer (Please select your correct option)

- Perimeter
- Volume
- Area
- Circumference

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Question No : 18 of 26 Marks: 1 (Budgeted Time 1 Min)

For the double integral $\int_c^d \int_a^b f(x,y)dxdy$, order of integration does not matter provided that $f(x,y)$ is . . . . . .

Answer (Please select your correct option)

- Bounded
- Discontinuous
- Defined
- Continuous

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Question No : 19 of 26 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x, y) : 0 \le x \le 2 and -1 \le y \le 1\}$, then $\iint_R (x + 2y^2) dA =$ . . . . . .

Answer (Please select your correct option)

- $\int_{-1}^{1} \int_{0}^{2} (x + 2y^2) dy dx$
- $\int_{0}^{2} \int_{1}^{-1} (x + 2y^2) dx dy$
- $\int_{-1}^{1} \int_{0}^{2} (x + 2y^2) dx dy$ ⇦

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- $\int_{1}^{2} \int_{-1}^{0} (x + 2y^2) dx dy$

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Question No : 20 of 26 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x, y) : 0 \le x \le 4 \text{ and } 0 \le y \le 9\}$, then $\iint_R (3x - 4x\sqrt{xy})dA = \ldots\ldots\ldots\ldots$

Answer (Please select your correct option)

- $\int_0^9 \int_0^4 (3x - 4x\sqrt{xy})dydx$
- $\int_0^4 \int_4^9 (3x - 4x\sqrt{xy})dxdy$
- $\int_4^9 \int_0^0 (3x - 4x\sqrt{xy})dxdy$
- $\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$ ⇦

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Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

Given that $f_{xx} = -6x$, $f_{yy} = -6y$, $f_{xy} = 3$ for some $f(x,y)$. Check whether $f(x,y)$ has relative maximum, relative minimum or a saddle point at P(1,1).

Answer (Please click here to Add Answer)

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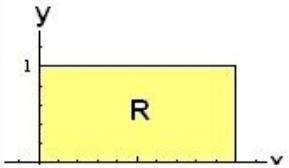
Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

Let the function $f(x,y)$ is continuous in the region R, where R is a rectangle as shown below.



Answer (Please click here to Add Answer)

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Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

Let $y = 3x^2 - 5$. If x changes from 2 to 2.1, find the approximate change in the value of y using differential dy.

Answer (Please click here to Add Answer)

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Question No : 24 of 26 Marks: 3 (Budgeted Time 6 Min)

If the order of integration for the integral $\int_0^1 \int_y^1 e^{x^2} dxdy$ is changed. Find the change in the limits of new integral.

Answer (Please click here to Add Answer)

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Question No : 25 of 26 Marks: 5 (Budgeted Time 10 Min)

Find equation of normal line (in parametric form) to the surface $f(x, y, z) = xy + 2yz - xz^2 + 10$ at the point $(-5, 5, 1)$.

Answer (Please click here to Add Answer)

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Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)

Evaluate $\int_1^4 \int_0^2 (6x^2 + 4xy^3)\,dx\,dy$

Answer (Please click here to Add Answer)

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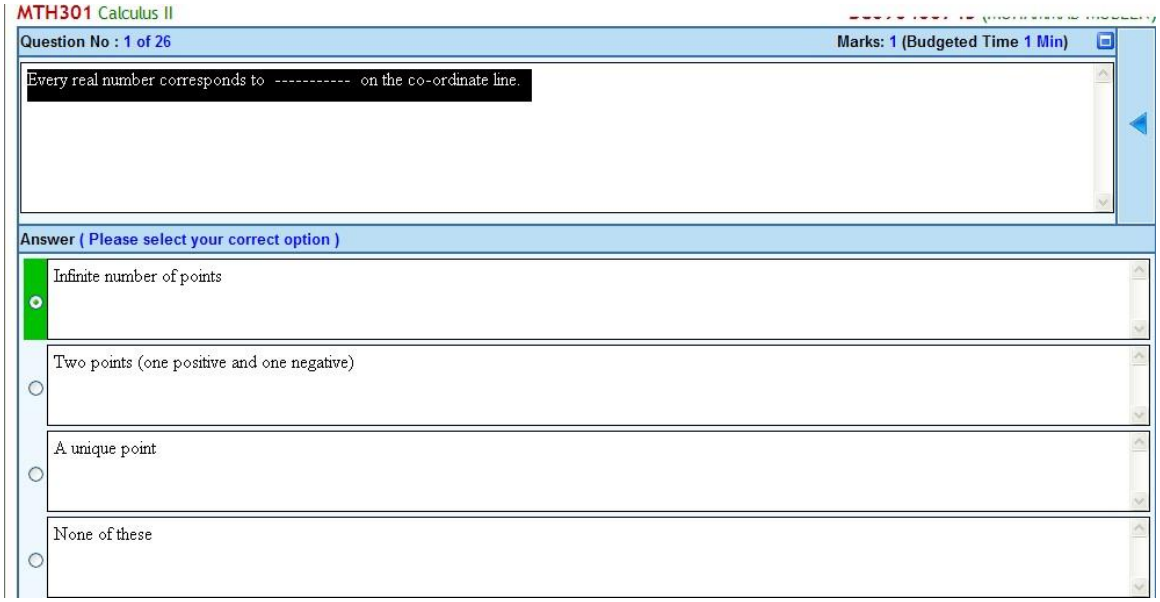


MTH301 Calculus II

Question No : 2 of 26 Marks: 1 (Budgeted Time 1 Min)

There is one-to-one correspondence between the set of points on co-ordinate line and ----------

Answer (Please select your correct option)

- Set of real numbers
- Set of integers
- Set of natural numbers
- Set of rational numbers

MTH301 Calculus II

Question No : 3 of 26 Marks: 1 (Budgeted Time 1 Min)

Which of the following is associated to each point of three dimensional space?

Answer (Please select your correct option)

- A real number
- An ordered pair
- An ordered triple
- A natural number

MTH301 Calculus II

Question No : 4 of 26 **Marks: 1 (Budgeted Time 1 Min)**

All axes are positive in ---------- octant.

Answer (Please select your correct option)

- First
- Second
- Fourth
- Eighth

MTH301 Calculus II

Question No : 5 of 26 **Marks: 1 (Budgeted Time 1 Min)**

The spherical co-ordinates of a point are $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$. What are its cylinderical co-ordinates?

Answer (Please select your correct option)

- $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}, 0\right)$
- $\left(\sqrt{3}\cos\frac{\pi}{3}, \sqrt{3}\sin\frac{\pi}{3}, 0\right)$
- $\left(\sqrt{3}\sin\frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3}\cos\frac{\pi}{3}\right)$
- $\left(\sqrt{3}, \frac{\pi}{3}, 0\right)$

MTH301 Calculus II

Question No : 6 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Suppose $f(x, y) = xy - 2y^2$ *where* $x = 3t + 1$ *and* $y = 2t$. Which one of the following is true?

Answer (Please select your correct option)

- $\frac{df}{dt} = -4t + 2$
- $\frac{df}{dt} = -16t - t$
- $\frac{df}{dt} = 18t + 2$
- $\frac{df}{dt} = -10t^2 + 8t + 1$

MTH301 Calculus II

Question No : 7 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Let w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$\frac{\partial w}{\partial r} =$

Answer (Please select your correct option)

- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

MTH301 Calculus II

Question No : 8 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3 and angle between them when placed tail to tail is 45 degrees. What is $\vec{a} \cdot \vec{b}$?

Answer (Please select your correct option)

- 4.5
- 6.2
- 5.1
- 4.2

MTH301 Calculus II

Question No : 9 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} 0 & \text{If } x \geq 0 \text{ and } y \geq 0 \\ 1 & \text{Otherwise} \end{cases}$$

Answer (Please select your correct option)

- $f(x, y)$ is continuous at origin
- $f(0, 0)$ is not defined
- $f(0, 0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

MTH301 Calculus II

Question No : 10 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} \frac{3x^2y}{x^2+y^2} & if\ (x, y) \neq 0 \\ 0 & if\ (x, y) = 0 \end{cases}$$

Answer (Please select your correct option)

- $f(x, y)$ is continuous at origin
- $f(0, 0)$ is not defined
- $f(0, 0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

MTH301 Calculus II

Question No : 11 of 26 **Marks: 1 (Budgeted Time 1 Min)**

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

Answer (Please select your correct option)

- Area of R.
- Radius of inscribed circle in R.
- Distance between two endpoints of R.
- None of these

MTH301 Calculus II

Question No : 12 of 26 Marks: 1 (Budgeted Time 1 Min)

Which of the following formula can be used to find the Volume of a parallelepiped with adjacent edges formed by the vectors $\bar{a}$, $\bar{b}$ *and* $\bar{c}$?

Answer (Please select your correct option)

- $\left|\bar{a} \times (\bar{b} \times \bar{c})\right|$
- $\left|\bar{a} \bullet (\bar{b} \bullet \bar{c})\right|$
- $\left|\bar{a} \bullet (\bar{b} \times \bar{c})\right|$
- $\left|\bar{a} \times (\bar{b} \bullet \bar{c})\right|$

MTH301 Calculus II

Question No : 13 of 26 Marks: 1 (Budgeted Time 1 Min)

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are ---------

Answer (Please select your correct option)

- Parallel
- Perpendicular
- In opposite direction

MTH301 Calculus II

Question No : 14 of 26 Marks: 1 (Budgeted Time 1 Min)

By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R, then $f(x, y)$ has both ---------- on R.

Answer (Please select your correct option)

- Absolute maximum and absolute minimum value
- Relative maximum and relative minimum value

MTH301 Calculus II

Question No : 15 of 26 Marks: 1 (Budgeted Time 1 Min)

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has ----------

Answer (Please select your correct option)

- (●) Relative maximum at (x_0, y_0)
- ( ) Relative minimum at (x_0, y_0)
- ( ) Saddle point at (x_0, y_0)
- ( ) No conclusion can be drawn.

MTH301 Calculus II

Question No : 16 of 26 Marks: 1 (Budgeted Time 1 Min)

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D = 0$ then ----------

Answer (Please select your correct option)

- (●) f has relative maximum at (x_0, y_0)
- ( ) f has relative minimum at (x_0, y_0)
- ( ) f has saddle point at (x_0, y_0)
- ( ) No conclusion can be drawn.

MTH301 Calculus II

Question No : 17 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then

$$\iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA =$$

Answer (Please select your correct option)

- $\iint_R f(x,y)dA$
- $\iint_{R_1} f(x,y)dA \cup \iint_{R_2} f(x,y)dA$
- $\iiint_R f(x,y)dV$
- $\iint_{R_1} f(x,y)dA \cap \iint_{R_2} f(x,y)dA$

MTH301 Calculus II

Question No : 18 of 26 **Marks: 1 (Budgeted Time 1 Min)**

If $R = \{(x,y) / 0 \le x \le 2 \text{ and } 1 \le y \le 4\}$, then

$$\iint_R (6x^2 + 4xy^3)dA =$$

Answer (Please select your correct option)

- $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$
- $\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$
- $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$
- $\int_2^4 \int_0^1 (6x^2 + 4xy^3)dxdy$

MTH301 Calculus II

Question No : 19 of 26 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x,y) / 2 \le x \le 4 \text{ and } 0 \le y \le 1\}$, then

$\iint_R (4xe^{2y})dA =$

Answer (Please select your correct option)

- $\int_0^1 \int_2^4 (4xe^{2y})dydx$
- $\int_0^1 \int_2^4 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dydx$

MTH301 Calculus II

Question No : 20 of 26 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x,y) / 0 \le x \le 4 \text{ and } 0 \le y \le 9\}$, then

$\iint_R (3x - 4x\sqrt{xy})dA =$

Answer (Please select your correct option)

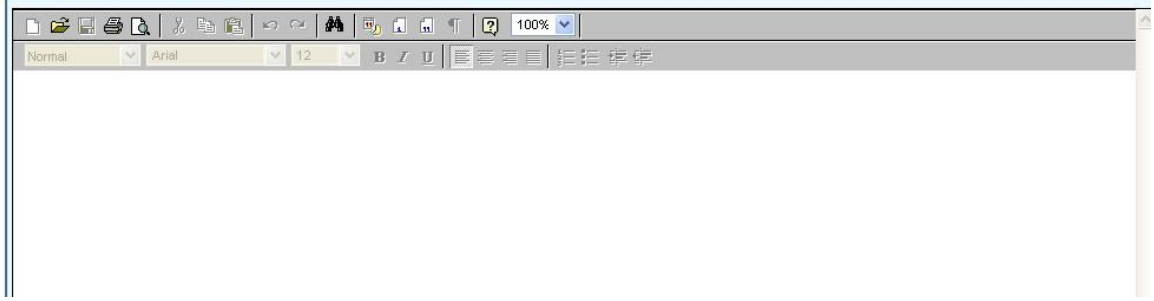
- $\int_0^9 \int_0^4 (3x - 4x\sqrt{xy})dydx$
- $\int_0^4 \int_4^9 (3x - 4x\sqrt{xy})dxdy$
- $\int_4^9 \int_0^0 (3x - 4x\sqrt{xy})dxdy$
- $\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$

MTH301 Calculus II

Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

Suppose that the surface $f(x, y, z)$ has continuous partial derivatives at the point (a, b, c). Write down the equation of tangent plane at this point.

Answer (Please click here to Add Answer)



MTH301 Calculus II

Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

Evaluate the following double integral.

$$\iint \left(12xy^2 - 8x^3\right) dy\, dx$$

MTH301 Calculus II

Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

Evaluate the following double integral.

$$\iint \left(3 + 2x - 3y^2\right) dx\, dy$$

Answer (Please click here to Edit Answer)

MTH301 Calculus II

Question No : 24 of 26 Marks: 3 (Budgeted Time 6 Min)

Let $f(x, y, z) = xy^2e^z$

Find the gradient of f.

Answer (Please click here to Add Answer)

MTH301 Calculus II

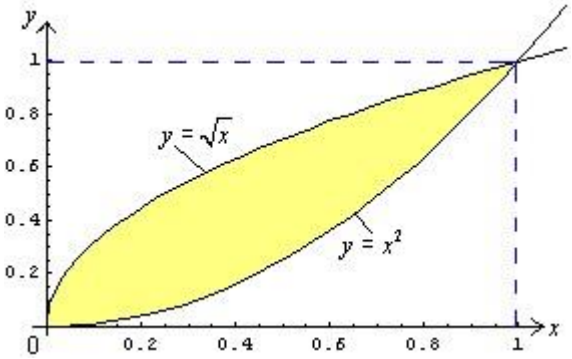
Question No : 25 of 26 Marks: 5 (Budgeted Time 10 Min)

Find, Equation of Tangent plane to the surface $f(x, y, z) = x^2 + y^2 + z - 9$ at the point $(1, 2, 4)$

Answer (Please click here to Add Answer)

Use double integral in rectangular co-ordinates to compute area of the region bounded

by the curves $\sqrt{\ }$ and .



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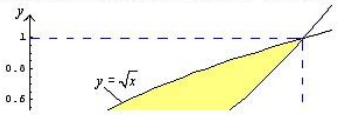
MTH301 Calculus II

Question No : 26 of 26 Marks: 5 (Budgeted Time 10 Min)

Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ and $y = \sqrt{x}$.



Answer (Please click here to Add Answer)

Calculus II

MTH301

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Knowledge beyond the boundaries

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Lecture No-1 Introduction

– Calculus is the mathematical tool used to analyze changes in physical quantities.
– Calculus is also Mathematics of Motion and Change.
– Where there is motion or growth, where variable forces are at work producing acceleration, Calculus is right mathematics to apply.

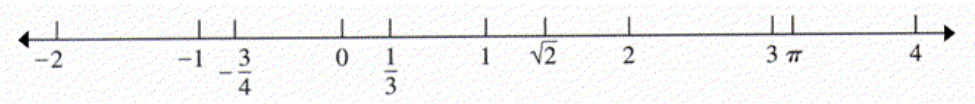
Differential Calculus **Deals with the Problem *of* Finding**

(1)Rate of change.
(2)Slope of curve.

Velocities and acceleration of moving bodies. Firing angles that give cannons their maximum range. The times when planets would be closest together or farthest apart.

Integral Calculus
Deals with the Problem of determining a *Function* from information about its *rates of Change.*
Integral Calculus Enables Us
(1) To calculate lengths of curves.
(2) To find areas of irregular regions in plane.
(3) To find the volumes and masses of arbitrary solids
(4) To calculate the future location of a body from its present position and knowledge of the forces acting on it.

Reference Axis System
Before giving the concept of Reference Axis System we recall you the concept of real line and locate some points on the real line as shown in the figure below, also remember that the real number system consist of both Rational and Irrational numbers that is we can write set of real numbers as union of rational and irrational numbers.



Here in the above figure we have locate some of the rational as well as irrational numbers and also note that there are infinite real numbers between every two real numbers.
Now if you are working in two dimensions then you know that we take the two mutually perpendicular lines and call the horizontal line as x-axis and vertical line as y-axis and where these lines cut we take that point as origin.
Now any point on the x-axis will be denoted by an order pair whose first element which is also known as abscissa is a real number and other element of the order pair which is also known as ordinate will has 0 values.
Similarly any point on the y-axis can be representing by an order pair. Some points are shown in the figure below. Also note that these lines divide the plane into four regions, First ,Second ,Third and Fourth quadrants respectively. We take the positive real numbers at the right side of the origin and negative to the left side, in the case of x-axis. Similarly for y-axis and also shown in the figure.

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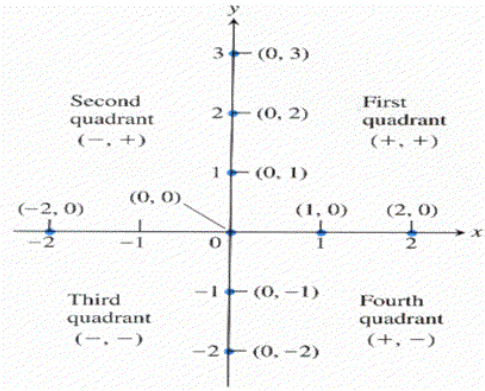
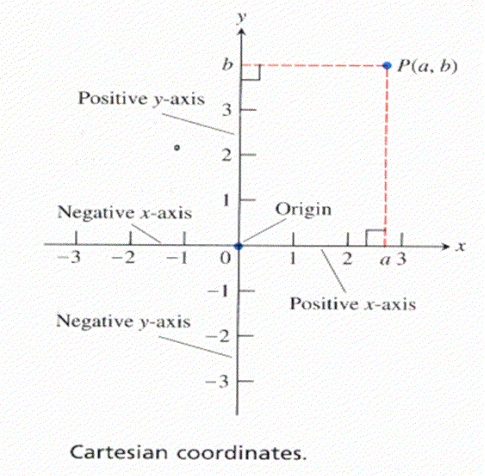


Location of a point

Now we will illustrate how to locate the point in the plane using x and y axis. Draw two perpendicular lines from the point whose position is to be determined. These lines will intersect at some point on the x-axis and y-axis and we can find out these points. Now the distance of the point of intersection of x-axis and perpendicular line from the origin is the X-C ordinate of the point P and similarly the distance from the origin to the point of intersection of y-axis and perpendicular line is the Y-coordinate of the point P as shown in the figure below.



Cartesian coordinates.

In space we have three mutually perpendicular lines as reference axis namely x ,y and z axis. Now you can see from the figure below that the planes x= 0 ,y=0 and z=0 divide the space into eight octants. Also note that in this case we have (0,0,0) as origin and any point in the space will have three coordinates.

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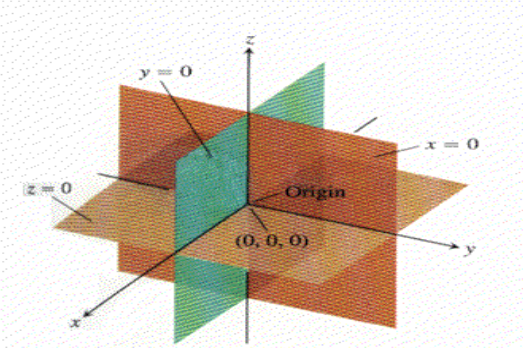

The planes $x = 0$, $y = 0$, and $z = 0$ divide space into eight octants.

Sign of co-ordinates in different octants

First of all note that the equation x=0 represents a plane in the 3d space and in this plane every point has its x-coordinate as 0, also that plane passes through the origin as shown in the figure above. Similarly y=0 and z=0 are also define a plane in 3d space and have properties similar to that of x=0.Such that these planes also pass through the origin and any point in the plane y=0 will have y-coordinate as 0 and any point in the plane z=0 has z-coordinate as 0. Also remember that when two planes intersect we get the equation of a line and when two lines intersect then we get a plane containing these two lines. Now note that by the intersection of the planes x=0 and z=0 we get the line which is our y-axis. Also by the intersection of x=0 and y=0 we get the line which is z-axis, similarly you can easily see that by the intersection of z=0 and y=0 we get line which is x-axis.

Now these three planes divide the 3d space into eight octants depending on the positive and negative direction of axis.

The octant in which every coordinate of any point has positive sign is known as first octant formed by the positive x, y and z –axis. Similarly in second octant every points has x-coordinate as negative and other two coordinates as positive correspond to negative x-axis and positive y and z axis.

Now one octant is that in which every point has x and y coordinate negative and z-coordinate positive, which is known as the third octant. Similarly we have eight octants depending on the sign of coordinates of a point. These are summarized below.

First octant	(+, +, +)	Formed by positive sides of the three axis.
Second octant	(-, +, +)	Formed by –ve x-axis and positive y and z-axis.
Third octant	(-, -, +)	Formed by –ve x and y axis with positive z-axis.
Fourth octant	(+, -, +)	Formed by +ve x and z axis and –ve y-axis.
Fifth octant	(+, +, -)	Formed by +ve x and y axis with -ve z-axis.
Sixth octant	(-, +, -)	Formed by –ve x and z axis with positive y-axis.
Seventh octant	(-, -, -)	Formed by –ve sides of three axis.
Eighth octant	(+, -, -)	Formed by -ve y and z-axis with +ve x-axis.

(Remember that we have two sides of any axis one of positive values and the other is of negative values)

Now as we told you that in space we have three mutually perpendicular lines as reference axis. So far you are familiar with the reference axis for 2d which consist of two perpendicular lines namely x-axis and y-axis. For the reference axis of 3d space we need another perpendicular axis which can be obtained by the cross product of the two vectors, now the direction of that vector can be obtained by Right hand rule. This is illustartaed below with diagram.

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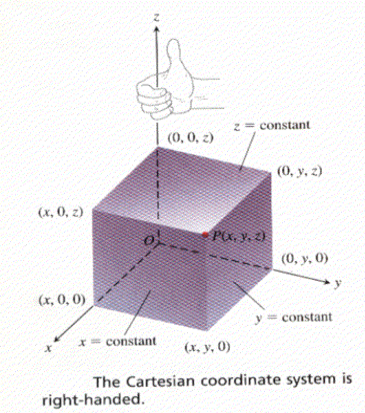

The Cartesian coordinate system is right-handed.

Concept of a Function

Historically, the term, function, denotes the dependence of one quantity on other quantity. The quantity x is called the independent variable and the quantity y is called the dependent variable. We write y = *f* (x) and we read y is a function of x.
The equation y = 2x defines y as a function of x because each value assigned to x determines unique value of y.

Examples of function

– The area of a circle depends on its radius r by the equation $A= \pi r^2$ so, we say that A is a function of r.
– The volume of a cube depends on the length of its side x by the equation $V= x^3$ so, we say that V is a function of x.
– The velocity *V* of a ball falling freely in the earth's gravitational field increases with time t until it hits the ground, so we say that V is function of t.
– In a bacteria culture, the number n of present after one day of growth depends on the number N of bacteria present initially, so we say that N is function of n.

Function of Several Variables

Many functions depend on more than one independent variable.

Examples

The area of a rectangle depends on its length *l* and width *w* by the equation
A = l w , so we say that *A* is a function of *l* and *w.*
The volume of a rectangular box depends on the length *l*, width *w* and height *h* by the equation
V = l w h so, we say that *V* is a function of *l* , *w* and *h.*
The area of a triangle depends on its base length l and height h by the equation
A= ½ l h, so we say that A is a function of l and h.
The volume V of a right circular cylinder depends on its radius r and height h by the equation V= πr2h so, we say that V is a function of r and h.

Home Assignments:

In the first Lecture we recall some basic terminologies which are essential and prerequisite for this course. You can find the Home Assignments on the last page of Lecture # 1 at LMS.

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Lecture No-2 Values of functions:

Consider the function $f(x) = 2x^2 - 1$, then $f(1) = 2(1)^2 - 1 = 1$, $f(4) = 2(4)^2 - 1 = 31$, $f(-2) = 2(-2)^2 - 1 = 7$

$f(t-4) = 2(t-4)^2 - 1 = 2t^2 - 16t + 31$

These are the values of the function at some points.

Example

Now we will consider a function of two variables, so consider the function $f(x,y) = x^2y+1$ then $f(2,1) = (2^2)1+1=5$, $f(1,2) = (1^2)2+1=3$, $f(0,0) = (0^2)0+1=1$, $f(1,-3) = (1^2)(-3)+1=-2$, $f(3a,a) = (3a)^2a+1=9a^3+1$, $f(ab,a-b) = (ab)^2(a-b)+1=a^3b^2-a^2b^3+1$

These are values of the function at some points.

Example:

Now consider the function $f(x,y) = x + \sqrt[3]{xy}$ then

(a) $f(2,4) = 2 + \sqrt[3]{(2)(4)} = 2 + \sqrt[3]{8} = 2 + 2 = 4$

(b) $f(t,t^2) = t + \sqrt[3]{(t)(t^2)} = t + \sqrt[3]{t^3} = t + t = 2t$

(c) $f(x,x^2) = x + \sqrt[3]{(x)(x^2)} = x + \sqrt[3]{x^3} = x + x = 2x$

(d) $f(2y^2,4y) = 2y^2 + \sqrt[3]{(2y^2)(4y)} = 2y^2 + \sqrt[3]{8y^3} = 2y^2 + 2y$

Example:

Now again we take another function of three variables

$$f(x,y,z) = \sqrt{1-x^2-y^2-z^2} \text{ Then}$$

$$f(0,\frac{1}{2},\frac{1}{2}) = \sqrt{1-0-(\frac{1}{2})^2-(\frac{1}{2})^2} = \sqrt{\frac{1}{2}}$$

Example:

Consider the function $f(x,y,z) = xy^2z^3+3$ then at certain points we have

$f(2,1,2) = (2)(1)^2(2)^3+3=19$, $f(0,0,0) = (0)(0)^2(0)^3+3=3$, $f(a,a,a) = (a)(a)^2(a)^3+3=a^6+3$
$f(t,t^2,-t) = (t)(t^2)^2(-t)^3+3=-t^8+3$, $f(-3,1,1) = (-3)(1)^2(1)^3+3=0$

Example:

Consider the function $f(x,y,z) = x^2y^2z^4$ where $x(t) = t^3$ $y(t) = t^2$ and $z(t)=t$

(a) $f(x(t),y(t),z(t)) = [x(t)]^2[y(t)]^2[z(t)]^4 = [t^3]^2[t^2]^2[t]^4 = t^{14}$
(b) $f(x(0),y(0),z(0)) = [x(0)]^2[y(0)]^2[z(0)]^4 = [0^3]^2[0^2]^2[0]^4 = 0$

Example:

Let us consider the function $f(x,y,z) = xyz + x$ then

$f(xy,y/x,xz) = (xy)(y/x)(xz) + xy = xy^2z+xy$.

Example:

Let us consider $g(x,y,z) = z \sin(xy)$, $u(x,y,z) = x^2z^3$, $v(x,y,z) = Pxyz$, $w(x,y,z) = \frac{xy}{z}$ Then.

$g(u(x,y,z), v(x,y,z), w(x,y,z)) = w(x,y,z) \sin(u(x,y,z)\, v(x,y,z))$

Now by putting the values of these functions from the above equations we get

$g(u(x,y,z), v(x,y,z), w(x,y,z)) = \frac{xy}{z} \sin[(x^2z^3)(Pxyz)] = \frac{xy}{z} \sin[(Px^3z^4)]$.

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Example:

Consider the function $g(x,y) = y \sin(x^2y)$ and $u(x,y) = x^2y^3$ $v(x,y) = \pi xy$ Then

$g(u(x,y), v(x,y)) = v(x,y) \sin([u(x,y)]^2 v(x,y))$

By putting the values of these functions we get

$g(u(x,y), v(x,y)) = \pi xy \sin([x^2y^3]^2 \pi xy) = \pi xy \sin(x^5y^7)$.

Function of One Variable

A function f of one real variable x is a rule that assigns a unique real number f(x) to each point x in some set D of the real line.

Function of two Variables

A function f in two real variables x and y, is a rule that assigns unique real number f (x,y) to each point (x,y) in some set D of the xy-plane.

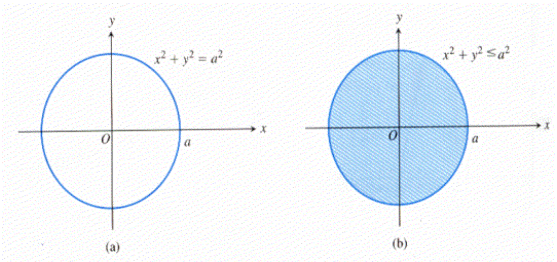
Function of three variables:

A function f in three real variables x, y and z, is a rule that assigns a unique real number f (x,y,z) to each point (x,y,z) in some set D of three dimensional space.

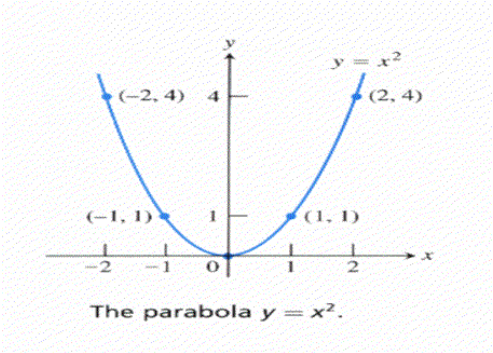
Function of n variables:

A function f in n variable real variables $x_1, x_2, x_3, \ldots\ldots, x_n$, is a rule that assigns a unique real number $w = f(x_1, x_2, x_3, \ldots\ldots, x_n)$ to each point $(x_1, x_2, x_3, \ldots\ldots, x_n)$ I n some set D of n dimensional space.

Circles and Disks:



PARABOLA



Parabola $y = -x^2$

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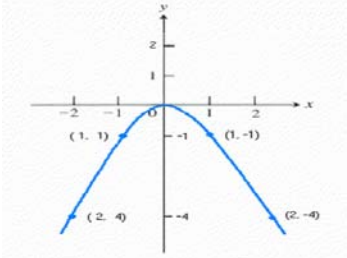
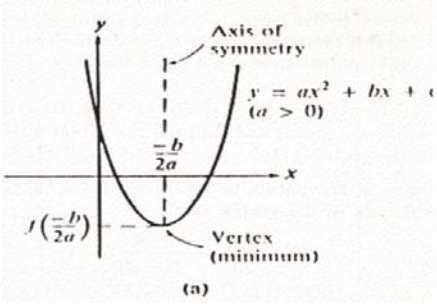
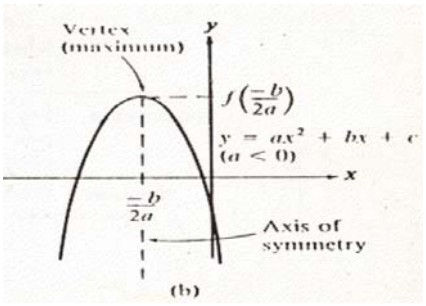


General equation of the Parabola opening upward or downward is of the form $y = f(x) = ax^2+bx + c$. Opening upward if $a > 0$. Opening downward if $a < 0$.
x co-ordinate of the vertex is given by $x_0 = -b/2a$. So the y co-ordinate of the vertex is $y_0= f(x_0)$ axis of symmetry is $x = x_0$. As you can see from the figure below

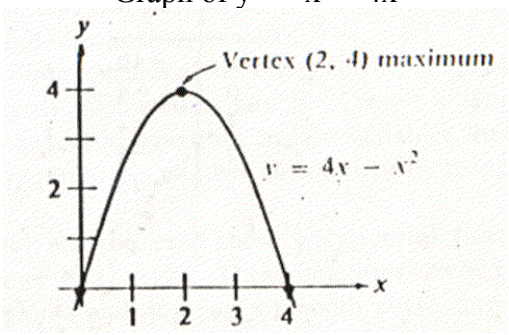




Sketching of the graph of parabola $y = ax^2+bx + c$
Finding vertex: x – co-ordinate of the vertex is given by $x_0= - b/2a$
So, y – co-ordinate of the vertex is $y_0= a x_0^2+b x_0 + c$. Hence vertex is $V(x_0 , y_0)$.

Example: Sketch the parabola $y = - x^2 + 4x$
Solution: Since $a = -1 < 0$, parabola is opening downward. Vertex occurs at $x = - b/2a = (-4)/2(-1) =2$. Axis of symmetry is the vertical line $x = 2$. The y-co-ordinate of the vertex is $y = -(2)^2 + 4(2) = 4$. Hence vertex is $V(2 , 4)$. The zeros of the parabola (i.e. the point where the parabola meets x-axis) are the solutions to $-x^2 +4x = 0$ so $x = 0$ and $x = 4$. Therefore (0,0)and (4,0) lie on the parabola. Also (1,3) and (3,3) lie on the parabola.

Graph of $y = - x^2 + 4x$



Example $y = x^2 - 4x+3$
Solution: Since $a = 1 > 0$, parabola is opening upward.Vertex occurs at $x = - b/2a = (4)/2 =2$.Axis of symmetry is the vertical line $x = 2$. The y co-ordinate of the vertex is $y = (2)^2 - 4(2) + 3 = -1$.Hence vertex is $V(2 , -1)$The zeros of the parabola (i.e. the point where the parabola meets x-axis) are the solutions to $x^2 - 4x + 3 = 0$, so $x = 1$ and $x = 3$.Therefore (1,0)and (3,0) lie on the parabola. Also (0 ,3) and (4, 3) lie on the parabola.

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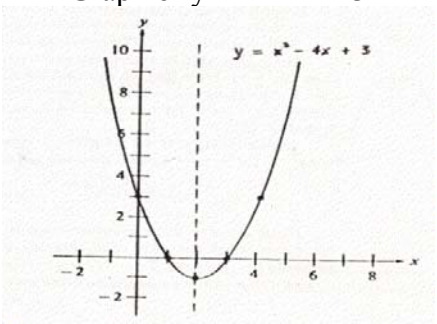
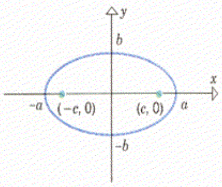
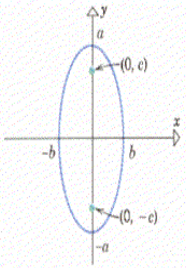
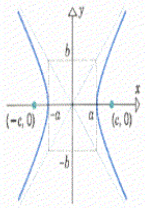
Graph of $y = x^2 - 4x+3$



Ellipse

ORIENTATION	DESCRIPTION	STANDARD EQUATION
	• Foci and major axis on the x-axis. • Minor axis on the y-axis. • Center at the origin. • x-intercepts: $\pm a$. • y-intercepts: $\pm b$. • $a \geq b$	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
	• Foci and major axis on the y-axis. • Minor axis on the x-axis. • Center at the origin. • x-intercepts: $\pm b$. • y-intercepts: $\pm a$. • $a \geq b$	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$

Hyperbola

ORIENTATION	DESCRIPTION	STANDARD EQUATION	ASYMPTOTE EQUATIONS
	• Foci on the x-axis. • Conjugate axis on the y-axis. • Center at the origin.	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$y = \frac{b}{a}x$ $y = -\frac{b}{a}x$

Home Assignments:

In this lecture we recall some basic geometrical concepts which are prerequisite for this course and you can find all these concepts in the chapter # 12 of your book Calculus By Howard Anton.

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Lecture No-3 Elements of three dimensional geometry

Distance formula in three dimension

Let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be two points such that PQ is not parallel to one of the coordinate axis Then $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ Which is known as Distance fromula between the points P and Q.

Example of distance formula

Let us considerthe points A (3, 2, 4), B (6, 10, –1),
and C (9, 4, 1)
Then

$$|AB| = \sqrt{(6-3)^2 + (10-2)^2 + (-1-4)^2} = \sqrt{98} = 7\sqrt{2}$$

$$|AC| = \sqrt{(9-3)^2 + (4-2)^2 + (1-4)^2} = \sqrt{49} = 7$$

$$|BC| = \sqrt{(9-6)^2 + (4-10)^2 + (1+1)^2} = \sqrt{49} = 7$$

Mid point of two points

If R is the middle point of the line segment PQ, then the co-ordinates of the middle points are
x= (x1+x2)/2 ,
y= (y1+y2)/2 ,
z= (z1+z2)/2

Let us consider tow points A(3,2,4) and B(6,10,-1)
Then the co-ordinates of mid point of AB is

[(3+6)/2,(2+10)/2,(4-1)/2]
= (9/2,6,3/2)

Direction Angles

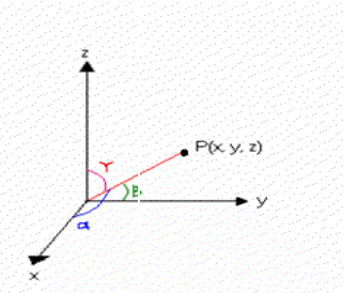
The **direction angles** α β, γ of a lineare defined as

α = Angle between lineand the positive x-axis

β = Angle between line and the positive y-axis

γ = Angle between lineand the positive z-axis.

By definition, each of these angles lies between 0 and π



Direction Ratios

Cosines of direction angles are called direction cosines
Any multiple of direction cosines are called direction numbers or direction ratios of the line L.

Given a point, finding its Direction cosines

y-axis

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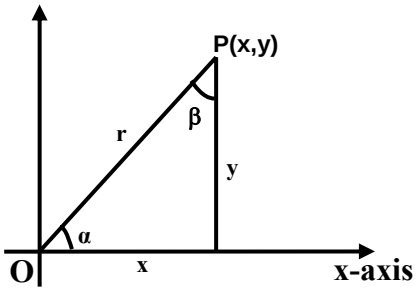
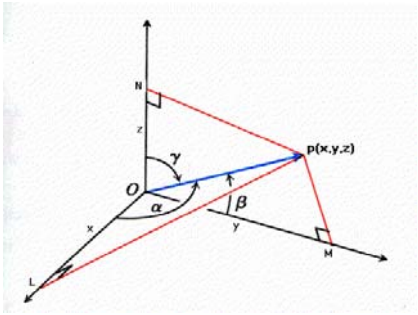


From triangle we can write
cos α = **x/r**
cos β = **y/r**

Direction angles of a Line



The angles which a line makes with positive x,y and z-axis are known as Direction Angles. In the above figure the blue line has direction angles as α, and which are the angles which blue line makes with x,y and z-axis respectively.

Direction cosines:

Now if we take the cosine of the Direction Angles of a line then we get the Direction cosines of that line. So the Direction Cosines of the above line are given by

$$\cos\alpha = \frac{x}{OP} = \frac{x}{\sqrt{x^2+y^2+z^2}}$$

$$\cos\beta = \frac{y}{OP} = \frac{y}{\sqrt{x^2+y^2+z^2}}$$

Similarly,

$$\cos\gamma = \frac{z}{OP} = \frac{z}{\sqrt{x^2+y^2+z^2}}$$

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1.$$

Direction cosines and direction ratios of a line joining two points

•For a line joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, x_2)$ the direction ratios are

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$x_2 - x_1, y_2 - y_1, z_2 - z_1$ and the directions cosines are $\frac{x_2 - x_1}{|PQ|}, \frac{y_2 - y_1}{|PQ|}$ *and* $\frac{z_2 - z_1}{|PQ|}$.

Example For a line joining two points P(1,3,2) and Q(7,-2,3) the direction ratios are

$$7 - 1, -2 - 3, 3 - 2$$
$$6, \quad -5, \quad 1$$

and the directions cosines are

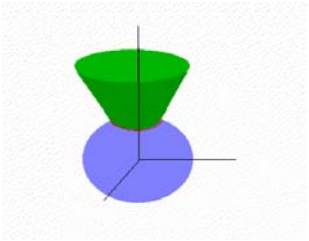
$$6/\sqrt{62}, \quad -5/\sqrt{62}, \quad 1/\sqrt{62}$$

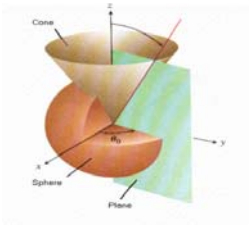
In two dimensional space the graph of an equation relating the variables x and y is the set of all point (x, y) whose co-ordinates satisfy the equation. Usually, such graphs are curves. In three dimensional space the graph of an equation relating the variables x, y and z is the set of all point (x, y, z) whose co-ordinates satisfy the equation. Usually, such graphs are surfaces.

Intersection of two surfaces

- Intersection of two surfaces is a curve in three dimensional space.
- It is the reason that a curve in three dimensional space is represented by two equations representing the intersecting surfaces.

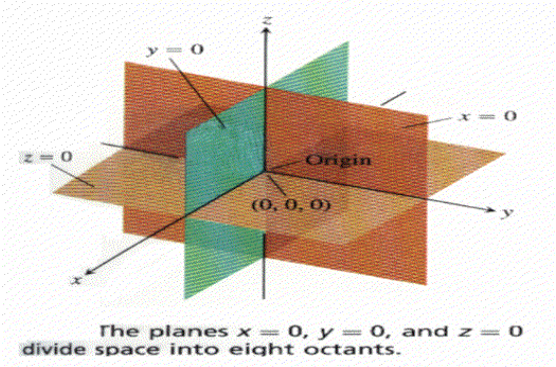
Intersection of Cone and Sphere





Intersection of Two Planes

If the two planes are not parallel, then they intersect and their intersection is a straight line. Thus, two non-parallel planes represent a straight line given by two simultaneous linear equations in x, y and z and are known as non-symmetric form of equations of a straight line.



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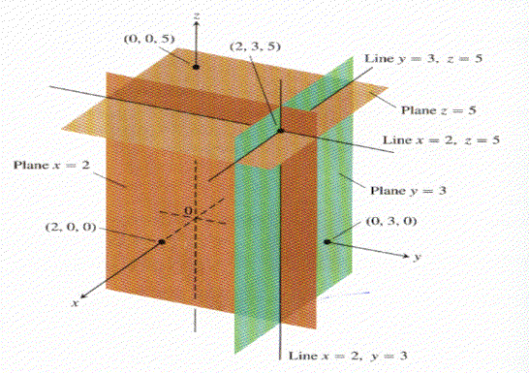
REGION	DESCRIPTION	EQUATION
xy-plane	Consists of all points of the form (x, y, 0)	z = 0
xz-plane	Consists of all points of the form (x, 0, z)	y = 0
yz-plane	Consists of all points of the form (0, y, z)	x = 0
x-axis	Consists of all points of the form (x, 0, 0)	y = 0, z = 0
y-axis	Consists of all points of the form (0, y, 0)	z = 0, x = 0
z-axis	Consists of all points of the form (0, 0, z)	x = 0, y = 0

Planes parallel to Co-ordinate Planes



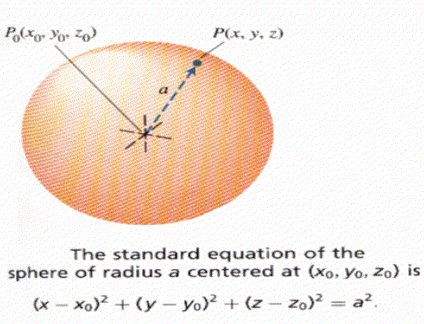
General Equation of Plane

Any equation of the form

$$ax + by + cz + d = 0$$

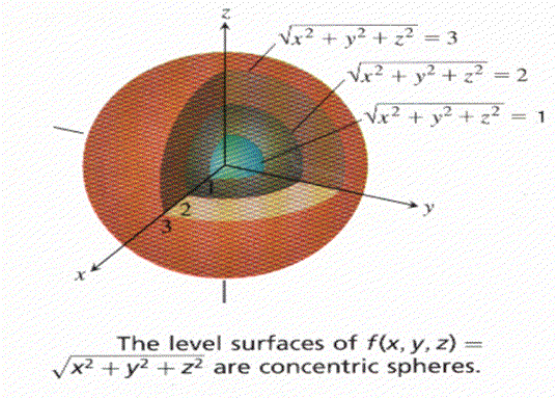
where a, b, c, d are real numbers,represent a plane.

Sphere



The standard equation of the sphere of radius a centered at (x_0, y_0, z_0) is

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2.$$



The level surfaces of $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ are concentric spheres.

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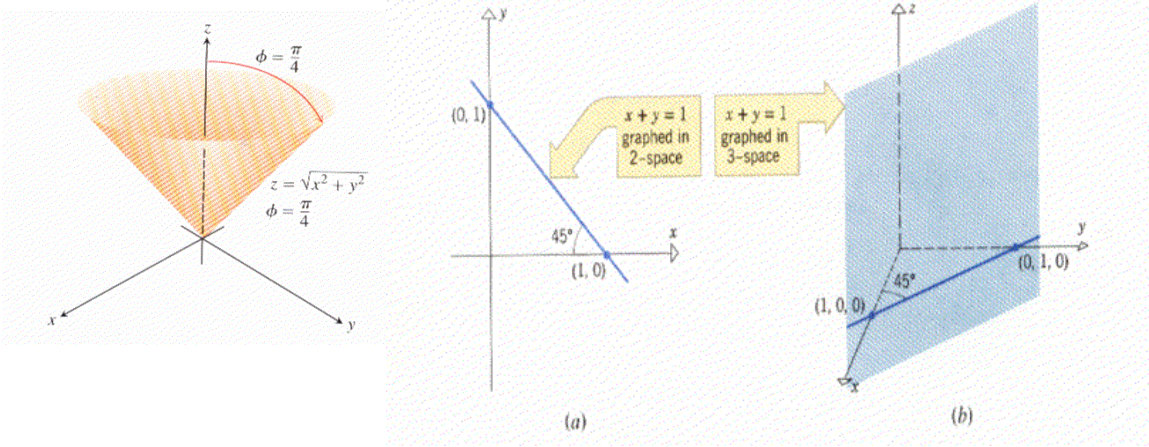
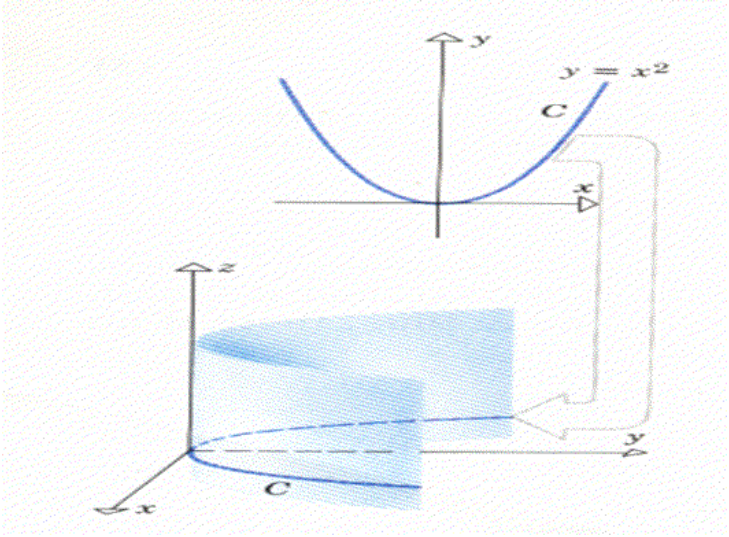
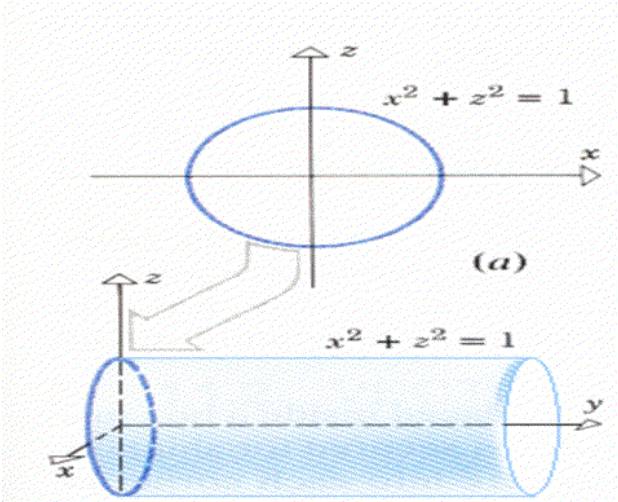
Right Circular Cone





Horizontal Circular Cylinder



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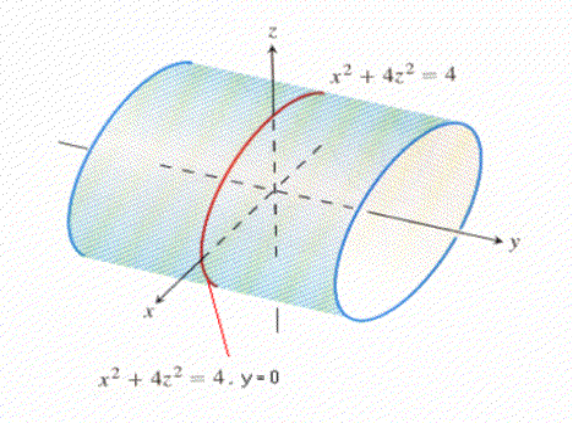
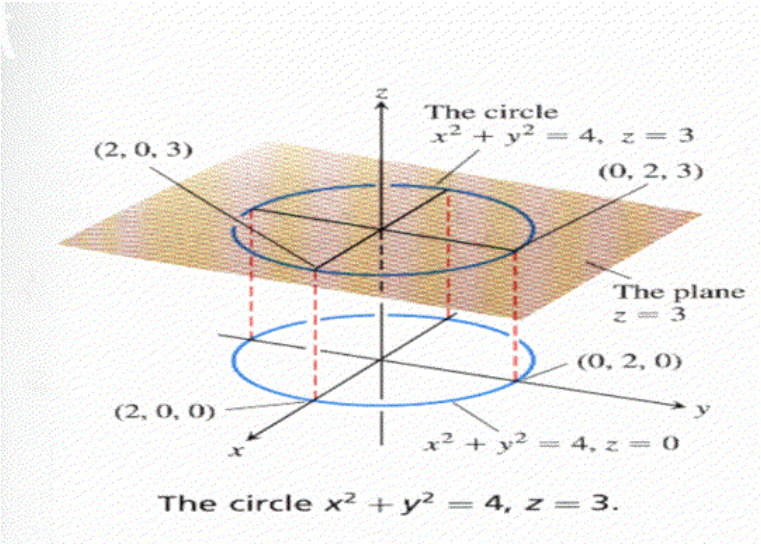
Horizontal Elliptic Cylinder





Overview of Lecture # 3

Chapter # 14
Three Diamentional Space
Page # 657

Book **CALCULUS by HOWARD ANTON**

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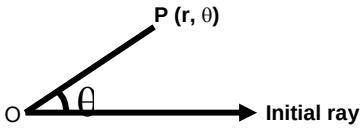
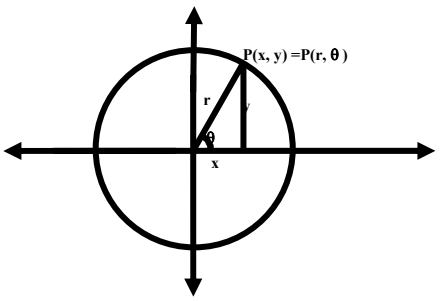
Lecture -4 Polar co-ordinates

You know that position of any point in the plane can be obtained by the two perpendicular lines known as x and y axis and together we call it as Cartesian coordinates for plane. Beside this coordinate system we have another coordinate system which can also use for obtaining the position of any point in the plane. In that coordinate system we represent position of each particle in the plane by "r" and "θ "where "r" is the distance from a fixed point known as pole and θ is the measure of the angle.



"O" is known as pole.

Conversion formula from polar to Cartesian coordinates and vice versa



From above diagram and remembering the trigonometric ratios we can write $x = r\cos\theta$, $y = r\sin\theta$. Now squaring these two equations and adding we get,

$$x^2 + y^2 = r^2$$

Dividing these equations we get

$$y/x = \tan\theta$$

These two equations gives the relation between the Plane polar and Plane Cartesian coordinates.

Rectangular co-ordinates for 3d

Since you know that the position of any point in the 3d can be obtained by the three mutually perpendicular lines known as x ,y and z – axis and also shown in figure below, these coordinate axis are known as Rectangular coordinate system.

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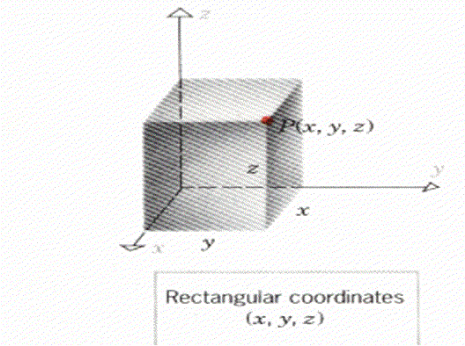
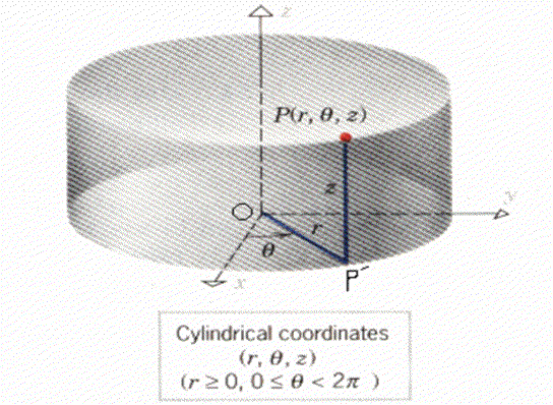


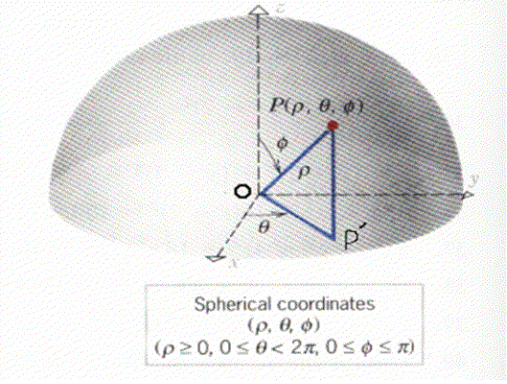
Cylindrical co-ordinates

Beside the Rectangular coordinate system we have another coordinate system which is used for getting the position of the any particle is in space known as the cylindrical coordinate system as shown in the figure below.



Spherical co-ordinates

Beside the Rectangular and Cylindrical coordinate systems we have another coordinate system which is used for getting the position of the any particle is in space known as the spherical coordinate system as shown in the figure below.



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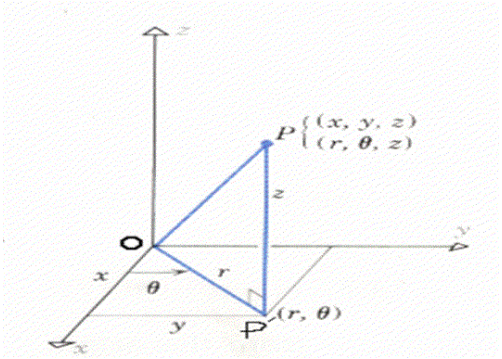
Conversion formulas between rectangular and cylindrical co-ordinates

Now we will find out the relation between the Rectangular coordinate system and Cylindrical coordinates. For this consider any point in the space and consider the position of this point in both the axis as shown in the figure below.



In the figure we have the projection of the point P in the xy-Plane and write its position in plane polar coordinates and also represent the angle θ now from that projection we draw perpendicular to both of the axis and using the trigonometric ratios find out the following relations.

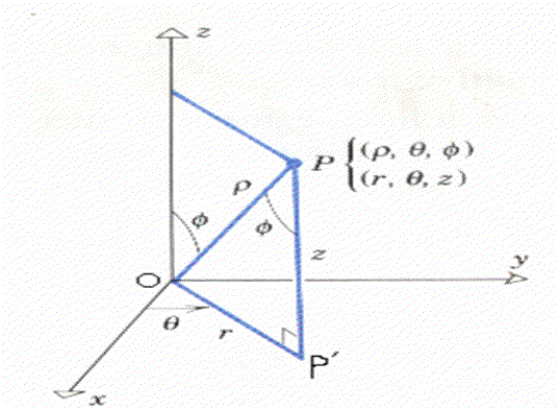
$(r, \theta, z) \rightarrow (x, y, z)$

$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$

$r = \sqrt{x^2+y^2}, \quad \tan\theta = \frac{y}{x}, \quad z = z$

Conversion formulas between cylindrical and spherical co-ordinates

Now we will find out the relation between spherical coordinate system and Cylindrical coordinate system. For this consider any point in the space and consider the position of this point in both the axis as shown in the figure below.



First we will find the relation between Planes polar to spherical, from the above figure you can easily see that from the two right angled triangles we have the following relations.

$(\rho, \theta, \phi) \rightarrow (r, \theta, z)$

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$r = \rho \sin\phi\,,\ \ \theta = \theta,\ \ z = \rho \cos\phi$

Now from these equations we will solve the first and second equation for ρ and ϕ. Thus we have

$(r, \theta, z) \rightarrow (\rho, \theta, \phi)$

$\rho = \sqrt{r^2 + z^2}\ \ \theta = \theta,\ \ \tan\phi = \frac{r}{z}$

Conversion formulas between rectangular and spherical co-ordinates

(ρ, θ, Φ) → (x, y, z)

Since we know that the relation between Cartesian coordinates and Polar coordinates are

x = r cos θ, y = r sin θ and z = z .We also know the relation between Spherical and cylindrical coordinates are,

$r = \rho \sin\phi\,,\ \ \theta = \theta,\ \ z = \rho \cos\phi$

Now putting this value of "r" and "z" in the above formulas we get the relation between spherical coordinate system and Cartesian coordinate system. Now we will find

$$(x, y, z) \rightarrow (\rho, \theta, \Phi)$$

$$\begin{aligned} x^2 + y^2 + z^2 &= (\rho \sin\Phi \cos\theta)^2 + (\rho \sin\Phi \sin\theta)^2 + (\rho \cos\Phi)^2 \\ &= \rho^2\{\sin^2\Phi(\cos^2\theta + \sin^2\theta) + \cos^2\Phi)\} \\ &= \rho^2(\sin^2\Phi + \cos^2\Phi)^2 = \rho^2 \end{aligned}$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$\text{Tan}\theta = y/x$ and $\text{Cos}\,\Phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$

Constant surfaces in rectangular co-ordinates

The surfaces represented by equations of the form

$$\mathbf{x = x_0,\ y = y_0,\ z = z_0}$$

where x_o, y_o, z_o are constants, are planes parallel to the xy-plane, xz-plane and xy-plane, respectively. Also shown in the figure

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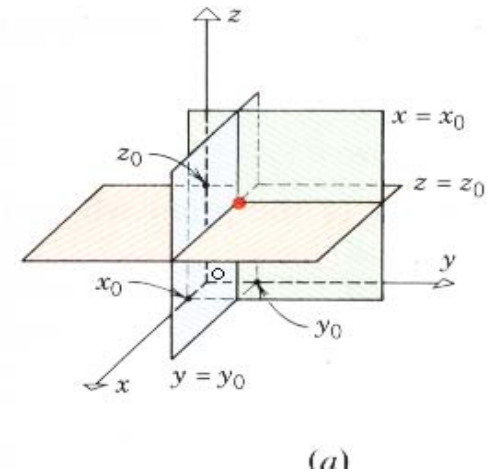


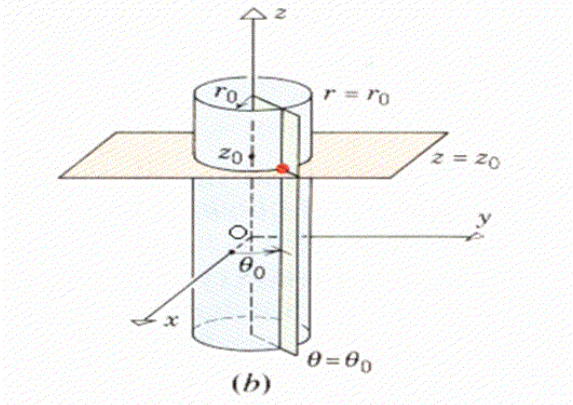
(a)

Constant surfaces in cylindrical co-ordinates

The surface **$r = r_0$ is a right cylinder** of radius r_0 centered on the z-axis. At each point (r, θ, z) this surface on this cylinder, r has the value r_0 , z is unrestricted and

$$0 \leq \theta < 2\pi.$$

The surface **$\theta = \theta_0$ is a half plane** attached along the z-axis and making angle θ_0 with the positive x-axis. At each point (r, θ, z) on the surface ,θ has the value θ_0, z is unrestricted and r ≥0. The surfaces **$z = z_0$ is a horizontal plane.** At each point (r, θ, z) this surface z has the value z_0 , but r and θ are unrestricted as shown in the figure below.



(b)

Constant surfaces in spherical co-ordinates

The surface **$\rho = \rho_0$ consists of all points whose distance ρ from origin is ρ_0.** Assuming that **ρ_0** to be nonnegative, this is a sphere of radius **ρ_0** centered at the origin. The surface **$\theta = \theta_0$ is a half plane** attached along the z-axis and making angle θ_0 with the positive x-axis. The surface $\Phi = \Phi_0$ consists of all points from which a line segment to the origin makes an angle of Φ_0 with the positive z-axis. Depending on whether $0 < \Phi_0 < \pi/2$ or $\pi/2 < \Phi_0 < \pi$, this will be a cone opening up or opening down. If $\Phi_0 = \pi/2$, then the cone is flat and the surface is the xy-plane.

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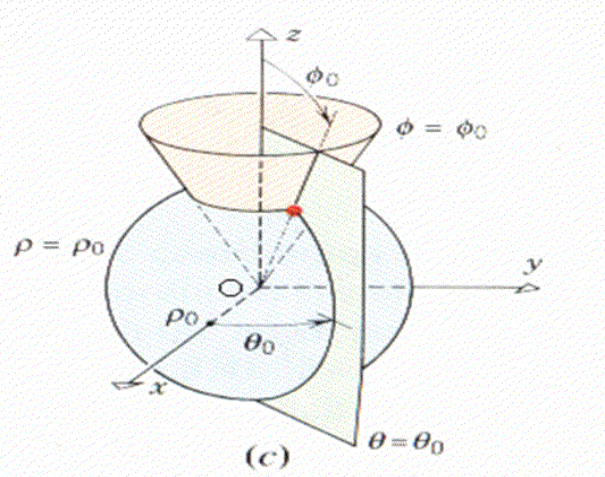

Spherical Co-ordinates in Navigation

Spherical co-ordinates are related to longitude and latitude coordinates used in navigation. Let us consider a right handed rectangular coordinate system with origin at earth's center, positive z-axis passing through the north pole,and x-axis passing through the prime meridian. Considering earth to be a perfect sphere of radius ρ = 4000 miles, then each point has spherical coordinates of the form (4000,θ,Φ) where Φ and θ determine the latitude and longitude of the point. Longitude is specified in degree east or west of the prime meridian and latitudes is specified in degree north or south of the equator.

Domain of the Function

- In the above definitions the set D is the domain of the function.
- The Set of all values which the function assigns for every element of the domain is called the Range of the function.
- When the range consist of real numbers the functions are called the real valued function.

NATURAL DOMAIN

Natural domain consists of all points at which the formula has no divisions by zero and produces only real numbers.

Examples

Consider the Function $\varpi = \sqrt{y - x^2}$. Then the domain of the function is $y \geq x^2$ Which can be shown in the plane as

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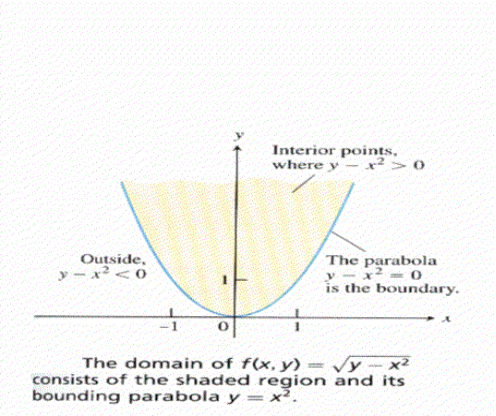

The domain of $f(x, y) = \sqrt{y - x^2}$ consists of the shaded region and its bounding parabola $y = x^2$.

and Range of the function is $[0, \infty)$.

Domain of function w = 1/xy is the whole xy- plane Excluding x-axis and y-axis, because at x and y axis all the points has x and y coordinates as 0 and thus the defining formula for the function gives us 1/0. So we exclude them.

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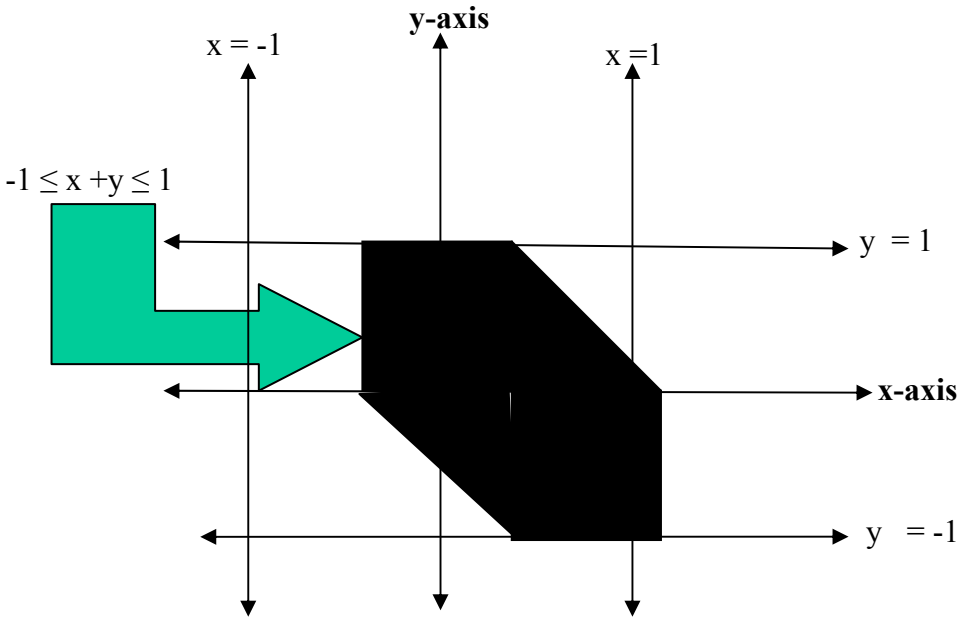
Lecture No-5 Limit of Multivariable Function

$f(x,y)= \sin^{-1}(x+y)$

Domain of f is the region in which $-1 \le x+y \le 1$



Domains and Ranges

Functions	Domain	Range
$\omega = \sqrt{x^2 + y^2 + z^2}$	Entire space	$[0, \infty)$
$\omega = \frac{1}{x^2 + y^2 + z^2}$	$(x,y,z) \neq (0,0,0)$	$(0, \infty)$
$\omega = xy \ln z$	Half space $z > 0$	$(-\infty, \infty)$

Examples of domain of a function

$f(x, y) = xy\sqrt{y - 1}$ Domain of f consists of region in xy plane where $y \ge 1$

$f(x, y) = \sqrt{x^2 + y^2 - 4}$

Domain of f consists of region

in xy plane where $x^2 + y^2 \ge 4$

As shown in the figure

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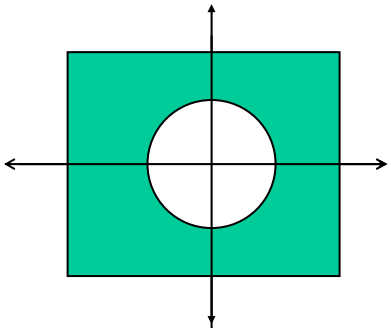
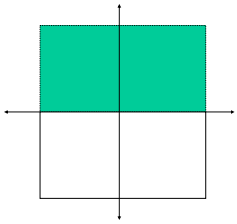


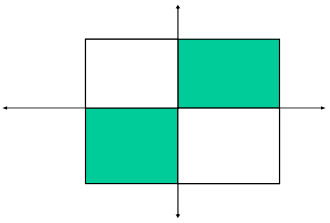


f(x, y) = lnxy

Domain of *f* consists of region lying in first and third quadrants in xy plane as shown in above figure right side.

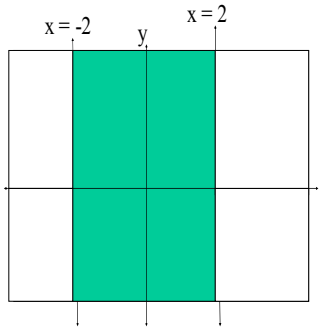
$f(x, y,z) = e^{xyz}$

Domain of f consists of region of three dimensional space



$$f(x, y) = \frac{\sqrt{4 - x^2}}{y^2 + 3}$$

Domain of f consists of region in xy plane $x^2 \leq 4, -2 \leq x \leq 2$



$$f(x, y, z) = \sqrt{25 - x^2 - y^2 - z^2}$$

Domain of f consists of region in three dimensional space occupied by sphere centre at (0, 0, 0) and radius 5.

$$f(x, y) = \frac{x^3 + 2x^2y - xy - 2y^2}{x + 2y}$$

f(0,0) is not defined but we see that limit exits.

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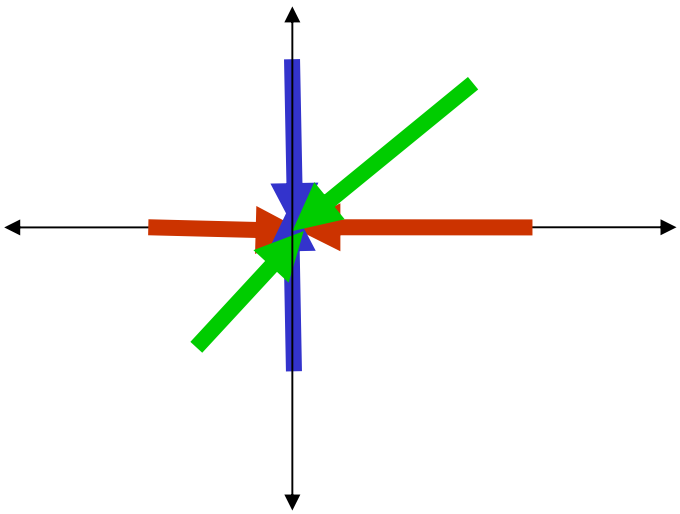
Approaching to (0,0) through x-axis	f (x,y)	Approaching to (0,0) through y-axis	f (x,y)
(0.5,0)	0.25	(0,0.1)	-0.1
(0.25,0)	0.0625	(0,0.001)	-0.001
(0.1,0)	0.01	(0,0.00001)	0.00001
(-0.25,0)	0.0625	(0,-0.001)	0.001
(-0.1,0)	0.01	(0,-0.00001)	0.00001

Approaching to (0,0) through y = x	f (x,y)
(0.5,0.5)	-0.25
(0.1,0.1)	-0.09
(0.01,0.01)	-0.0099
(-0.5,-0.5)	0.75
(-0.1,-0.1)	0.11
(-0.01,-0.01)	0.0101



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Example

$$f(x,y) = \frac{xy}{x^2+y^2}$$

f(0,0) is not defined and we see that limit also does not exit.

Approaching to (0,0) through x-axis (y = 0)	f (x,y)	Approaching to (0,0) through y = x	f (x,y)
(0.5,0)	0	(0.5,0.5)	0.5
(0.1,0)	0	(0.25,0.25)	0.5
(0.01,0)	0	(0.1,0.1)	0.5
(0.001,0)	0	(0.05,0.05)	0.5
(0.0001,0)	0	(0.001,0.001)	0.5
(-0.5,0)	0	(-0.5,-0.5)	0.5
(-0.1,0)	0	(-0.25,-0.25)	0.5
(-0.01,0)	0	(-0.1,-0.1)	0.5
(-0.001,0)	0	(-0.05,-0.05)	0.5
(-0.0001,0)	0	(-0.001,-0.001)	0.5

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = 0 \text{ (along } y = 0)$$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = 0.5 \text{ (along } y = x)$$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} \text{ does not exist.}$$

Example

$$\lim_{(x,\,y)\to(0,\,0)} \frac{xy}{x^2+y^2}$$

Let (x, y) approach $(0, 0)$ along the line $y = x$.

$$f(x,y) = \frac{xy}{x^2+y^2} = \frac{x \cdot x}{x^2+x^2} = \frac{1}{1+1} \quad x \neq 0.$$

$$= \frac{1}{2}$$

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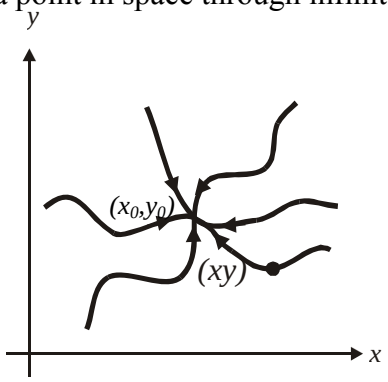
Let (x, y) approach $(0, 0)$ along
the line $y = 0$.

$f(x, y) = \frac{x.(0)}{x^2 + (0)^2} = 0$, $x \neq 0$.

Thus $f(x, y)$ assumes two different
values as (x,y) approaches (0,0)
along two different paths.

$\lim_{(x, y) \to (0, 0)} f(x, y)$ does not exist.

We can approach a point in space through infinite paths some of them are shown in the figure below.



Rule for Non-Existence of a Limit

If in

$$\lim_{(x, y) \to (a, b)} f(x, y)$$

We get *two or more different values as we approach (a, b) along different paths*, then

$$\lim_{(x, y) \to (a, b)} f(x, y)$$

does not exist.

The paths along which (a, b) is approached may be *straight lines or plane curves through (a, b).*

Example

$$\lim_{(x, y) \to (2, 1)} \frac{x^3 + x^2 y - x - y^2}{x + 2y}$$

$$= \frac{\lim_{(x, y) \to (2, 1)} (x^3 + 2x^2 y - xy - 2y^2)}{\lim_{(x, y) \to (2, 1)} (x + 2y)}$$

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$$= \frac{\lim_{(x,y)\to(2,1)} (x^3 + 2x^2y - x - y^2)}{\lim_{(x,y)\to(2,1)} (x + 2y)}$$

Example

$$\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2 + y^2}}$$

We set

$$x = r\cos\theta,\ y = r\sin\theta$$

$$\text{then } \frac{x}{\sqrt{x^2 + y^2}} = \frac{r\cos\theta \cdot r\sin\theta}{\sqrt{r^2\cos^2\theta + r^2\sin^2\theta}}$$

$$= r\cos\theta\sin\theta \quad \text{, for } r > 0$$

Since $r = \sqrt{x^2 + y^2}$, $r \to 0$ as $(x, y) \to (0, 0)$,

$$\lim_{(x,y)\to(0,0)} \frac{x}{\sqrt{x^2+y^2}} = \lim_{r\to 0} r\cos\theta\sin\theta = 0,$$

since $|\cos\theta \sin\theta| \le 1$ for all value of θ.

RULES FOR LIMIT

If $\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L_1$ *and* $\lim_{(x,y)\to(x_0,y_0)} g(x,y) = L_2$ Then

(a) $\lim_{(x,y)\to(x_0,y_0)} cf(x,y) = cL_1$ (if c is constant)

(b) $\lim_{(x,y)\to(x_0,y_0)} \{f(x,y) + g(x,y)\} = L_1 + L_2$

$\lim_{(x,y)\to(x_0,y_0)} \{f(x,y) - g(x,y)\} = L_1 - L_2$

(d) $\lim_{(x,y)\to(x_0,y_0)} \{f(x,y)g(x,y)\} = L_1L_2$

(e) $\lim_{(x,y)\to(x_0,y_0)} \frac{f(x,y)}{g(x,y)} = \frac{L_1}{L_2}$ (if $L_2 = 0$)

$\lim_{(x,y)\to(x_0,y_0)} c = c$ (c a constant), $\lim_{(x,y)\to(x_0,y_0)} x_0 = x_0$, $\lim_{(x,y)\to(x_0,y_0)} y_0 = y_0$

Similarly for the function of three variables.

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Overview of lecture# 5

In this lecture we recall you all the limit concept which are prerequisite for this course and you can find all these concepts in the chapter # 16 (topic # 16.2)of your Calculus By Howard Anton.

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Lecture No -6 Geometry of continuous functions

Geometry of continuous functions in one variable or Informal definition of continuity of function of one variable.

A function is continuous if we draw its graph by a pen then the pen is not raised so that there is no gap in the graph of the function

Geometry of continuous functions in two variables or Informal definition of continuity of function of two variables.

The graph of a continuous function of two variables to be constructed from a thin sheet of clay that has been hollowed and pinched into peaks and valleys without creating tears or pinholes.

Continuity of functions of two variables

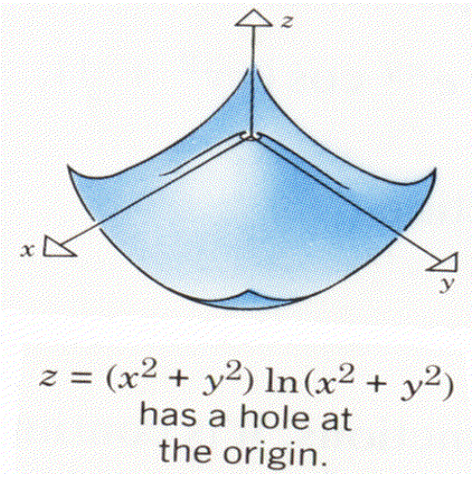
A function f of two variables is called continuous at the point (x_0,y_0) if

1. $f(x_0,y_0)$ if defined.
2. $\lim_{(x,y)\to(x_0,y_0)} f(x,y)$ exists.
3. $\lim_{(x,y)\to(x_0,y_0)} f(x,y) = f(x_0,y_0)$.

The requirement that $f(x_0,y_0)$ must be defined at the point (x_0,y_0) eliminates the possibility of a hole in the surface $z = f(x_0,y_0)$ above the point (x_0,y_0).

Justification of three points involving in the definition of continuity.

(1) Consider the function of two variables $x^2 + y^2 \ln(x^2 + y^2)$ now as we know that the Log function is not defined at 0, it means that when x = 0 and y = 0 our function $x^2 + y^2 \ln(x^2 + y^2)$ is not defined. Consequently the surface $z = x^2 + y^2 \ln(x^2 + y^2)$ will have a hole just above the point (0,0)as shown in the graph of $x^2 + y^2 \ln(x^2 + y^2)$



(2) The requirement that $\lim_{(x,y)\to(x_0,y_0)} f(x,y)$ exists ensures us that the surface z = f(x,y) of the function f(x,y) doesn't become infinite at (x_0,y_0) or doesn't oscillate widely.

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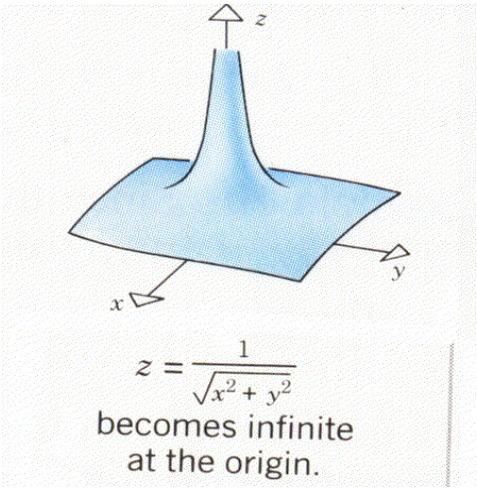
Consider the function of two variables $\frac{1}{\sqrt{x^2+y^2}}$ now as we know that the Natural domain of the function is whole the plane except origin. Because at origin we have x = 0 and y =0 in the defining formula of the function we will have at that point 1/0 which is infinity. Thus the limit of the function $\frac{1}{\sqrt{x^2+y^2}}$ does not exists at origin. Consequently the surface $z = \frac{1}{\sqrt{x^2+y^2}}$ will approaches towards infinity when we approaches towards origin as shown in the figure above.



(3) The requirement that

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = f(x_0,y_0)$$

ensures us that the surface z = f(x,y) of the function f(x,y) doesn't have a vertical jump or step above the point (x_0,y_0).

Consider the function of two variables

$$f(x,y) = \begin{cases} 0 & \text{if } x \geq 0 \text{ and } y \geq 0 \\ 1 & \text{otherwise} \end{cases}$$

now as we know that the Natural domain of the function is whole the plane. But you should note that the function has one value "0" for all the points in the plane for which both x and y have nonnegative values. And value "1" for all other points in the plane. Consequently the surface

$z = f(x,y) = \begin{cases} 0 & \text{if } x \geq 0 \text{ and } y \geq 0 \\ 1 & \text{otherwise} \end{cases}$ has a jump as shown in the figure

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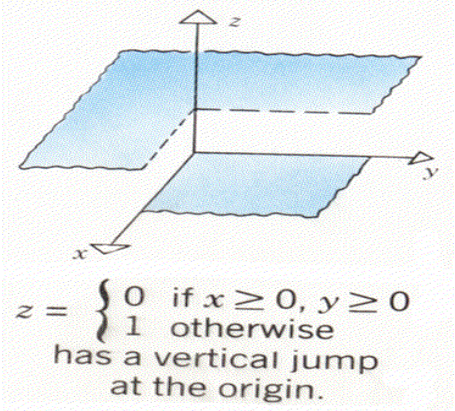

Example
Check whether the limit exists or not for the function

$$\lim_{(x,y)\to(0,0)} f(x,y) = \frac{x^2}{x^2+y^2}$$

Solution:

First we will calculate the Limit of the function along x-axis and we get

$$\lim_{(x,y)\to(0,0)} f(x,y) = \frac{x^2}{x^2+0} = 1 \textbf{ (Along x-axis)}$$

Now we will find out the limit of the function along y-axis and we note that the limit is $\lim_{(x,y)\to(0,0)} f(x,y) = \frac{y^2}{y^2+0} = 1$ **(Along y-axis).** Now we will find out the limit of the function along the line y = x and we note that $\lim_{(x,y)\to(0,0)} f(x,y) = \frac{x^2}{x^2+x^2} = \frac{1}{2}$ **(Along y = x)**
It means that limit of the function at (0,0) doesn't exists because it has different values along different paths. Thus the function cannot be continuous at (0,0). And also note that the function id not defined at (0,0) and hence it doesn't satisfy two conditions of the continuity.

Example
Check the continuity of the function at (0,0)

$$f(x,y) = \begin{cases} \dfrac{\sin(x^2+y^2)}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 1 & \text{if } (x,y) = (0,0) \end{cases}$$

Solution:

First we will note that the function is defined on the point where we have to check the Continuity that is the function has value at (0,0). Next we will find out the Limit of the function at (0,0) and in evaluating this limit we use the result $\lim_{x\to 0} \frac{\sin x}{x} = 1$ and note that

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$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} \frac{\text{Sin}(x^2+y^2)}{x^2+y^2}$$

$$=1 = f(0,0)$$

This shows that f is continuous at (0,0)

CONTINUITY OF FUNCTION OF THREE VARIABLES

A function f of three variables is called *continuous at a point* (x_0,y_0,z_0) if

1. $f(x_0,y_0,z_0)$ if defined.
2. $\lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x,y,z)$ exists;
3. $\lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x,y,z) = f(x_0,y_0,z_0)$.

EXAMPLE

Check the continuity of the function

$$f(x,y,z) = \frac{y+1}{x^2+y^2-1}$$

Solution:

First of all note that the given function is not defined on the cylinder $x^2+y^2-1=0$ .Thus the function is not continuous on the cylinder $x^2+y^2-1=0$ And continuous at all other points of its domain.

RULES FOR CONTINOUS FUNCTIONS

(a) If g and h are continuous functions of one variable, then f(x, y) = g(x)h(y) is a continuous function of x and y

(b) If g is a continuous function of one variable and h is a continuous function of two variables, then their composition f(x, y) = g(h(x,y)) is a continuous function of x and y.

A composition of continuous functions is continuous.

A sum, difference, or product of continuous functions is continuous.

A quotient of continuous function is continuous, expect where the denominator is zero.

EXAMPLE OF PRODUCT OF FUNCTIONS TO BE CONTINUED

In general, any function of the form $f(x, y) = Ax^my^n$ (m and n non negative integers) is continuous because it is the product of the continuous functions Ax^m and y^n.

The function $f(x, y) = 3x^2y^5$ is continuous because it is the product of the continuous functions $g(x) = 3x^2$ and $h(y) = y^5$.

CONTINUOUS EVERYWHERE

A function f that is continuous at each point of a region R in 2-dimensional space or 3-dimensional space is said to be *continuous on R.* A function that is continuous at every point in 2-dimensional space or 3-dimensional space is called continuous *everywhere* or simply *continuous.*

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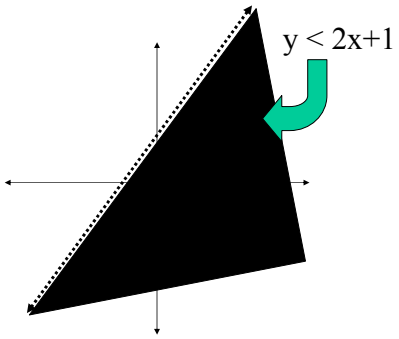
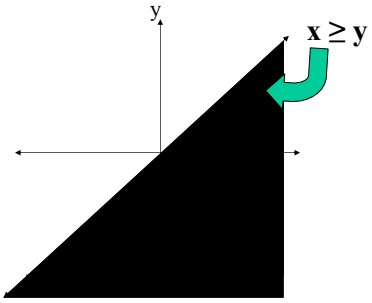
Example:

(1)f (x,y) = ln(2x – y +1)
The function f is continuous in the whole region where 2x > y-1, y < 2x+1.And its region is shown in figure below.



(2) $f(x,y) = e^{1-xy}$
The function f is continuous in the whole region of xy-plane.
(3) $f(x,y) = \tan^{-1}(y-x)$
The function f is continuous in the whole region of xy- plane.
(4) $f(x,y) = \sqrt{y-x}$
The function is continuous where $x \geq y$



Partial derivative
Let f be a function of x and y. If we hold y constant, say $y = y_0$ and view x as a variable, then $f(x, y_0)$ is a function of x-alone. If this function is differentiable at $x = x_o$, then the value of this derivative is denoted by $f_x(x_o , y_0)$ and is called the Partial derivative of f with respect of x at the point (x_o , y_0) .
Similarly, if we hold x constant, say $x = x_0$ and view y as a variable, then $f(x_0 , y)$ is a function of y alone. If this function is differentiable at $y = y_0$, then the value of this derivative is denoted by $f_y(x_0 , y_0)$ and is called the Partial derivative of f with respect of y at the point (x_0 , y_0)

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Example

$$f(x,y) = 2x^3y^2 + 2y + 4x$$

Treating y as a constant and differentiating with respect to x, we obtain

$$f_x(x,y) = 6x^2y^2 + 4$$

Treating x as a constant and differentiating with respect to y, we obtain

$$f_y(x,y) = 4x^3y + 2$$

Substituting $x = 1$ and $y = 2$ in these partial-derivative formulas yields.

$$f_x(1,2) = 6(1)^2(2)^2 + 4 = 28$$
$$f_y(1,2) = 4(1)^3(2) + 2 = 10$$

Example

$$Z = 4x^2 - 2y + 7x^4y^5$$
$$\frac{\partial z}{\partial x} = 8x + 28x^3y^5$$
$$\frac{\partial z}{\partial y} = -2 + 35x^4y^4$$

Example

$$z = f(x,y) = x^2 \sin^2 y$$

Then to find the derivative of f with respect to x we treat y as a constant therefore

$$\frac{\partial z}{\partial x} = f_x = 2x\sin^2 y$$

Then to find the derivative of f with respect to y we treat x as a constant therefore

$$\frac{\partial z}{\partial y} = f_y = x^2 2\sin y\cos y$$
$$= x^2 \sin 2y$$

Example

$$z = \ln\left(\frac{x^2+y^2}{x+y}\right)$$

By using the properties of the ln we can write it as

$$z = \ln(x^2 + y^2) - \ln(x+y)$$
$$\frac{\partial z}{\partial x} = \frac{1}{x^2+y^2} \cdot 2x - \frac{1}{x+y}$$
$$= \frac{2x^2 + 2xy - x^2 - y^2}{(x^2+y^2)(x+y)}$$
$$= \frac{x^2 + 2xy - y^2}{(x^2+y^2)(x+y)}$$

Similarly, (or by symmetry)

$$\frac{\partial z}{\partial y} = \frac{y^2 + 2xy - x^2}{(x^2+y^2)(x+y)}$$

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Example

$$z = x^4 \sin(xy^3)$$

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}\left[x^4 \sin(xy^3)\right]$$

$$= x^4 \frac{\partial}{\partial x}\left[\sin(xy^3)\right] + \sin(xy^3)\frac{\partial}{\partial x}(x^4)$$

$$= x^4 \cos(xy^3)y^3 + \sin(xy^3)4x^3$$

$$\frac{\partial z}{\partial x} = x^4 y^3 \cos(xy^3) + \sin(xy^3)$$

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}\left[x^4 \sin(xy^3)\right]$$

$$= x^4 \frac{\partial}{\partial y}\left[\sin(xy^3)\right] + \sin(xy^3)\frac{\partial}{\partial y}(x^4)$$

$$= x^4 \cos(xy^3).3xy^2 + \sin(xy^3).0$$

$$= 3x^5 y^2 \cos(xy^3)$$

Example

$$z = \cos(x^5 y^4)$$

$$\frac{\partial z}{\partial x} = -\sin(x^5 y^4)\frac{\partial}{\partial x}(x^5 y^4)$$

$$= -5x^4 y^4 \sin(x^5 y^4)$$

$$\frac{\partial z}{\partial y} = -\sin(x^5 y^4)\frac{\partial}{\partial y}(x^5 y^4)$$

$$= -4x^5 y^3 \sin(x^5 y^4)$$

Example

$$w = x^2 + 3y^2 + 4z^2 - xyz$$

$$\frac{\partial w}{\partial x} = 2x - yz$$

$$\frac{\partial w}{\partial y} = 6y - xz$$

$$\frac{dw}{dz} = 8z - xy$$

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Lecture No -7 Geometric meaning of partial derivative

Geometric meaning of partial derivative

$z = f(x, y)$

Partial derivative of f with respect of x is denoted by

$$\frac{\partial z}{\partial x} \text{ or } f_x \text{ or } \frac{\partial f}{\partial x}$$

Partial derivative of f with respect of y is denoted by

$$\frac{\partial z}{\partial y} \text{ or } f_y \text{ or } \frac{\partial f}{\partial y}$$

Partial Derivatives

Let z = f(x, y) be a function of two variable defined on a certain domain D.
For a given change Δx in x, keeping y as it is, the change Δz in z, is given by

$$\Delta z = f(x + \Delta x, y) - f(x, y)$$

If the ratio

$$\frac{\Delta z}{\Delta x} = \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

approaches to a finite limit as Δx →0, then this limit is called Partial derivative of f with respect of x.
Similarly for a given change Δy in y, keeping x as it is, the change Δz in z, is given by

$$\Delta z = f(x, y + \Delta y) - f(x, y)$$

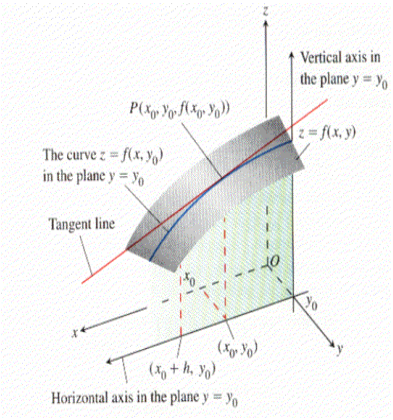
If the ratio

$$\frac{\Delta z}{\Delta y} = \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

approaches to a finite limit as Δy →0, then this limit is called Partial derivative of f with respect of y.

Geometric Meaning of Partial Derivatives

Suppose $z = f(x, y)$ is a function of two variables. The graph of f is a surface. Let P be a point on the graph with coordinates $(x_o, y_o, f(x_o, y_o))$. If a point starting from P, changes its position on the surface such that y remains constant, then the locus of this point is the curve of intersection of $z = f(x, y)$ and $y =$ constant. On this curve, $\frac{\partial z}{\partial x}$ is derivative of $z = f(x, y)$ with respect to x with y constant.



Thus $\frac{\partial z}{\partial x}$ = slope of the tangent to this curve at P

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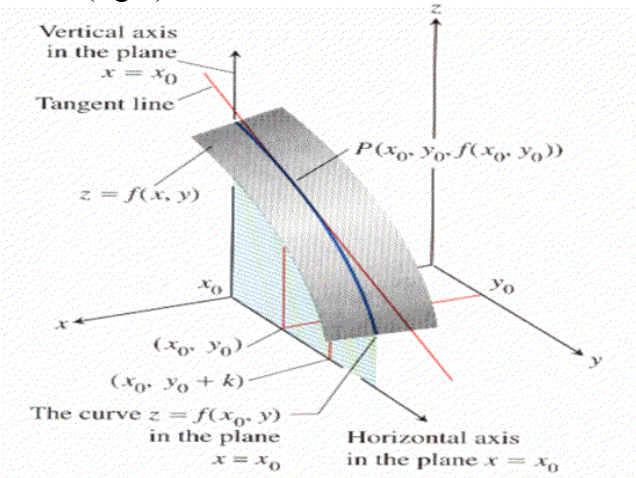
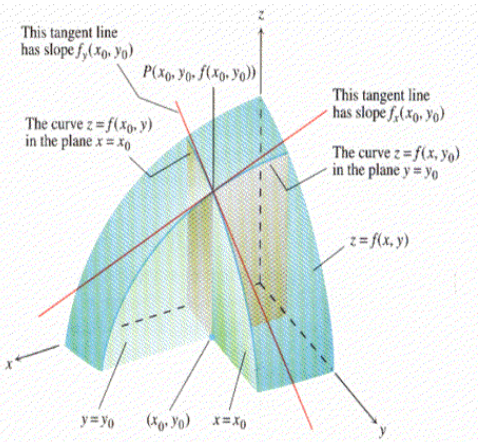
Similarly, $\frac{\partial z}{\partial y}$ is the gradient of the tangent at P to the curve of intersection of $z = f(x, y)$ and x = constant.

As shown in the figure below (left) Also together these tangent lines are shown in figure below (right).





Partial Derivatives of Higher Orders

The partial derivatives f_x and f_y of a function f of two variables x and y, being functions of x and y, may possess derivati ves. In such cases, the **second order** partial derivatives are defined as below.

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}(f_x) = (f_x)_x = f_{xx} = f_x^2$$

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}(f_x) = (f_x)_y = f_{xy}$$

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}(f_y) = (f_y)_x = f_{yx}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}(f_y) = (f_y)_y = f_{yy} = f_y^2$$

Thus, there are four second order partial derivatives for a function $z = f(x, y)$. The partial derivatives f_{xy} and f_{yx} are called **mixed second partials** and **are not equal** in general. Partial derivatives of order more than two can be defined in a similar manner.

Example

$$\frac{\partial z}{\partial x} = \frac{1}{\sqrt{1 - \frac{x^2}{y^2}}} \cdot \left(\frac{1}{y}\right) = \frac{1}{\sqrt{y^2 - x^2}}$$

$$\frac{\partial z}{\partial y} = \frac{1}{\sqrt{1 - \frac{x^2}{y^2}}} \cdot \frac{-x}{y^2} = \frac{-x}{y\sqrt{y^2 - x^2}}$$

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{-1}{2}(y^2 - x^2)^{-3/2} 2y = \frac{-y}{(y^2 - x^2)^{3/2}}$$

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$$\frac{\partial^2 z}{\partial x\,\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)$$

$$= \frac{-1}{y\sqrt{y^2 - x^2}} - \frac{x}{y}\left[\frac{x}{(y^2 - x^2)^{3/2}}\right]$$

$$= \frac{-y^2 + x^2 - x^2}{y(y^2 - x^2)^{3/2}} = \frac{-y}{(y^2 - x^2)^{3/2}}$$

Hence

$$\frac{\partial^2 z}{\partial x\,\partial y} = \frac{\partial^2 z}{\partial y\,\partial x}$$

Example

$$f(x, y) = x\cos y + ye^x$$

$$\frac{\partial f}{\partial x} = \cos y + ye^x$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = -\sin y + e^x$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = ye^x$$

$$f(x, y) = x\cos y + y\,e^x$$

$$\frac{\partial f}{\partial y} = -x\sin y + e^x$$

$$\frac{\partial^2 f}{\partial x \partial y} = -\sin y + e^x$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = -x\cos y$$

Laplace's Equation

For a function ***w = f(x,y,z)***

The equation

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

is called **Laplace's equation** .

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Example

$$f(x, y) = e^x \sin y + e^y \cos x,$$

$$\frac{\partial f}{\partial x} = e^x \sin y - e^y \sin x$$

$$\frac{\partial^2 f}{\partial x^2} = e^x \sin y - e^y \cos x$$

$$\frac{\partial f}{\partial y} = e^x \cos y + e^y \cos x$$

$$\frac{\partial^2 f}{\partial y^2} = -e^x \sin y + e^y \cos x$$

Adding both partial second order derivative, we have

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = e^x \sin y - e^y \cos x$$

$$- e^x \sin y + e^y \cos x \quad = 0$$

Euler's theorem
The mixed derivative theorem

If f(x,y) and its partial derivatives f_x, f_y, f_{xy} and f_{yx} are defined throughout an open region containing a point (a, b) and are all continuous at (a, b), then

$$f_{xy}(a , b) = f_{yx}(a , b)$$

Advantage of Euler's theorem

$$w = xy + \frac{e^y}{y^2 + 1}$$

The symbol $\partial^2 w / \partial x \, \partial y$ tell us to differentiate first with respect to y and then with respect to x. However, if we postpone the differentiation with respect to y and differentiate first with respect to x, we get the answer more quickly.

$$\frac{\partial w}{\partial x} = y \qquad \frac{\partial^2 w}{\partial y \partial x} = 1$$

Overview of lecture# 7
Chapter # 16 Partial derivatives
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Lecture No- 8 More About Euler Theorem Chain Rule

In general, the order of differentiation in an nth order partial derivatice can be change without affecting the final result whenever the function and all its partial derivatives of order $\leq$ n are continuous. For example, if f and its partial order derivatives of the first, second, and third orders are continuous on an open set, then at each point of the set,

$$f_{xyy} = f_{yxy} = f_{yyx}$$

or in another notation.

$$\frac{\partial^3 f}{\partial y^2 \partial x} = \frac{\partial^3 f}{\partial y \partial x \partial y} = \frac{\partial^3 f}{\partial x \partial y^2}$$

Order of differentiation

For a function

$$f(x,y) = y^2 x^4 e^x + 2$$

If we are interested to find $\frac{\partial^5 f}{\partial y^3 \partial x^2}$, that is, differentiating in the order firstly w.r.t. x and then w.r.t. y, calculation will involve many steps making the job difficult. But if we differentiate this function with respect to y, firstly and then with respect to x secondly then the value of this fifth order derivative can be calculated in a few steps.

$$\frac{\partial^5 f}{\partial x^2 \partial y^3} = 0$$

EXAMPLE

$$f(x,y) = \frac{x+y}{x-y}$$

$$f_x(x,y) = \frac{(x-y)\frac{\partial}{\partial x}(x+y) - (x+y)\frac{\partial}{\partial x}(x-y)}{(x-y)^2}$$

$$= \frac{(x-y)(1) - (x+y)(1)}{(x-y)^2}$$

$$= \frac{-2y}{(x-y)^2}$$

$$f_y(x,y) = \frac{(x-y)\frac{\partial}{\partial y}(x+y) - (x+y)\frac{\partial}{\partial y}(x-y)}{(x-y)^2}$$

$$= \frac{(x-y)(1) - (x+y)(-1)}{(x-y)^2}$$

$$= \frac{2x}{(x-y)^2}$$

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EXAMPLE

$$f(x,y) = x^3 e^{-y} + y^3 \sec\sqrt{x}$$

$$f_x(x,y) = 3x^2 e^{-y} + y^3 \sec\sqrt{x}\tan\sqrt{x}\frac{1}{2\sqrt{x}}$$

$$f_y(x,y) = -x^3 e^{-y} + 3y^2 \sec\sqrt{x}$$

EXAMPLE

$$f(x,y) = x^2 y e^{xy}$$

$$\begin{aligned} f_x(x,y) &= 2xye^{xy} + x^2y^2e^{xy} \\ &= xye^{xy}(2+xy) \\ f_x(1,1) &= (1)(1)e^{(1)(1)}[2+(1)(1)] \\ &= 3e \end{aligned}$$

$$f(x,y) = x^2 y e^{xy}$$

$$\begin{aligned} f_y(x,y) &= x^2e^{xy} + x^3ye^{xy} \\ &= x^2e^{xy}(1+xy) \\ f_y(1,1) &= (1)(1)e^{(1)(1)}[1+(1)(1)] \\ &= 2e \end{aligned}$$

Example

$$f(x,y) = x^2\cos(xy)$$

$$\begin{aligned} f_x(x,y) &= 2x\cos(xy) - x^2y\sin(xy) \\ f_x(1/2,\pi) &= 2(1/2)\cos(\pi/2) - (1/2)^2(\pi)\sin(\pi/2) \\ &= -\pi/4 \end{aligned}$$

$$\begin{aligned} f_y(x,y) &= -x^3\sin(xy) \\ f_y(1/2,\pi) &= -(1/2)^3\sin(\pi/2) \\ &= -1/8 \end{aligned}$$

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EXAMPLE

$$w = (4x - 3y + 2z)^5$$

$$\frac{\partial w}{\partial x} = 20(4x - 3y + 2z)^4$$

$$\frac{\partial^2 w}{\partial y \partial x} = -24\ (4x - 3y + 2z)^3$$

$$\frac{\partial^3 w}{\partial z \partial y \partial x} = -1440(4x - 3y + 2z)^2$$

$$\frac{\partial^4 w}{\partial z^2 \partial y \partial x} = -576\ (4x - 3y + 2z)$$

Chain Rule in function of One variable

Given that w= f(x) and x = g(t), we find $\frac{dw}{dt}$ *as follows:*

From w = f(x), we get $\frac{dw}{dx}$

From x = g(t), we get $\frac{dx}{dt}$

Then

$$\frac{dw}{dt} = \frac{dw}{dx}\ \frac{dx}{dt}$$

Example

w = x + 4, x = Sint

By Substitution

w = Sint + 4

$\frac{dw}{dt}$ = Cost

w = x + 4 $\Rightarrow \frac{dw}{dx} = 1$

x = Sint $\Rightarrow \frac{dx}{dt}$ = Cost

By Chain Rule

$\frac{dw}{dt} = \frac{dw}{dt} \times \frac{dx}{dt} = 1 . \times \text{Cost} = \text{Cost}$

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Chain rule in function of one variable

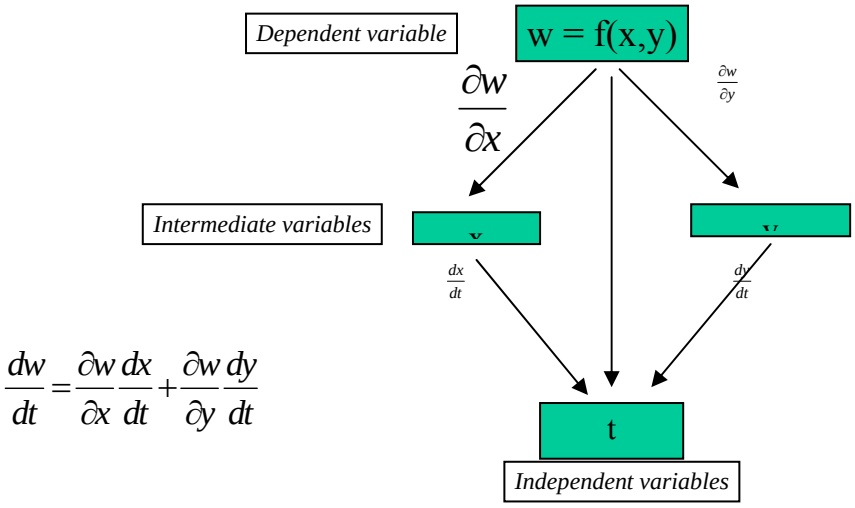
y is a function of u, u is a function of v
v is a function of w, w is a function of z
z is a function of x. Ultimately y is function of x

so we can talk about $\frac{dy}{dx}$

and by chain rule it is given by

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dv}\frac{dv}{dw}\frac{dw}{dz}\frac{dz}{dx}$$

w = f(x,y), x = g(t), y = f(t)



$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt}$$

EXAMPLE BY SUBSTITUTION

$$w = xy$$
$$x = \cos t,\ y = \sin t$$
$$w = \cos t \sin t$$
$$= \frac{1}{2}\ 2 \sin t \cos t$$
$$= \frac{1}{2}\ \sin 2t$$
$$\frac{dw}{dt} = \frac{1}{2}\ \cos 2t.2$$
$$= \cos 2t$$

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EXAMPLE

$$w = xy, x = \cos t, \text{and } y = \sin t$$

$$\frac{\partial w}{\partial y} = x \qquad \frac{\partial w}{\partial x} = y$$

$$\frac{dx}{dt} = -\sin t, \quad \frac{dy}{dt} = \cos t,$$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt}$$

$$= (\sin t)(-\sin t) + (\cos t)(\cos t)$$

$$= -\sin^2 t + \cos^2 t = \cos 2t$$

EXAMPLE

$$z = 3x^2 y^3$$

$$x = t^4, \quad y = t^2$$

$$\frac{\partial z}{\partial x} = 6xy^3, \qquad \frac{\partial z}{\partial y} = 9x^2y^2$$

$$\frac{dx}{dt} = 4t^3, \qquad \frac{dy}{dt} = 2t$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

$$= (6xy^3)(4t^3) + 9x^2y^2(2t)$$

$$= 6(t^4)(t^6)(4t^3) + 9(t^8)(t^4)(2t)$$

$$= 24t^{13} + 18t^{13} = 42t^{13}$$

EXAMPLE

$$z = \sqrt{1 + x - 2xy^4}$$

$$x = \ln$$

$$y = t$$

$$\frac{\partial z}{\partial x} = \frac{1}{2\sqrt{1 + x - 2xy^4}}$$

$$\frac{\partial z}{\partial y} = \frac{1}{2\sqrt{1 + x - 2xy^4}}.(-8xy^3) = \frac{-4xy^3}{\sqrt{1 + x - 2xy^4}}$$

$$\frac{dx}{dt} = \frac{1}{t}, \quad \frac{d}{dt} =$$

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$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\cdot\frac{dy}{dt}$$

$$= \frac{1 - 2y^4}{2\sqrt{1 + x - 2xy^4\, t}}\,\frac{1}{t} - \frac{4xy^3}{\sqrt{1 + x - 2xy^4}}\cdot 1$$

$$= \frac{1}{\sqrt{1 + x - 2xy^4}}\left[\frac{1-}{2t} - 4xy^3\right]$$

$$= \frac{1}{\sqrt{1 + lint - 2\,(lnt)\,t^4}}\left[\frac{1-2t^4}{2t} - 4\,(lnt)\,t^3\right]$$

$$= \frac{1}{\sqrt{1 + lnt - 2t^4\, lnt}}\left[\frac{1}{2t} - t^3 - 4t^3\, lnt\right]$$

EXAMPLE

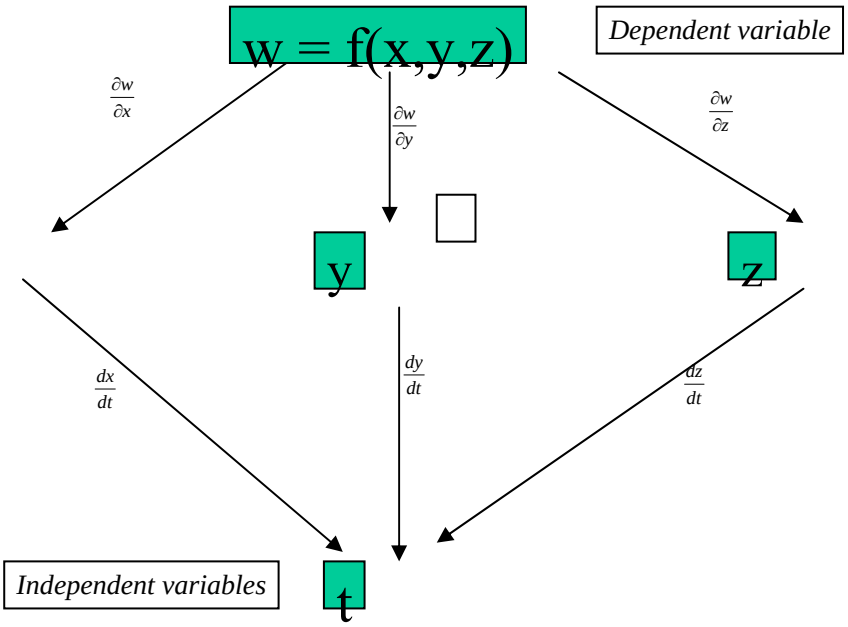
$$z = \ln(2x^2 + y)$$

$$x = \sqrt{t}, \qquad y = t^{2/3}$$

$$\frac{\partial z}{\partial x} = \frac{1}{2x^2 + y}\cdot 4x = \frac{4}{2x^2 + y}$$

$$\frac{\partial z}{\partial y} = \frac{1}{2x^2 + y}, \quad \frac{dx}{dt} = \frac{1}{2}\frac{1}{\sqrt{t}}, \quad \frac{dy}{dt} = \frac{2}{3}t^{-1/3}$$

w = f(x,y,z), x = g(t) ,y = f(t), z = h(t)



Overview of Lecture#8

Chapter # 16
Topic # 16.4
Page # 799 Book Calculus By Haward Anton

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Lecture No - 9 Examples

First of all we revise the example which we did in our 8th lecture.

Consider w = f(x,y,z) Where

$x = g(t), y = f(t), z = h(t)$

Then

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt}$$

Example:

$\mathbf{w = x^2 + y + z + 4}$

$\mathbf{x = e^t, \quad y = cost, \quad z = t + 4}$

$\frac{\partial w}{\partial x} = 2x, \quad \frac{\partial w}{\partial y} = 1, \quad \frac{\partial w}{\partial z} = 1$

$\frac{dx}{dt} = e^t, \quad \frac{dy}{dt} = -Sint, \quad \frac{dz}{dt} = 1$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\cdot\frac{dz}{dt}$$

$= (2x)(e^t) + (1) \cdot (-Sint) + (1)(1)$

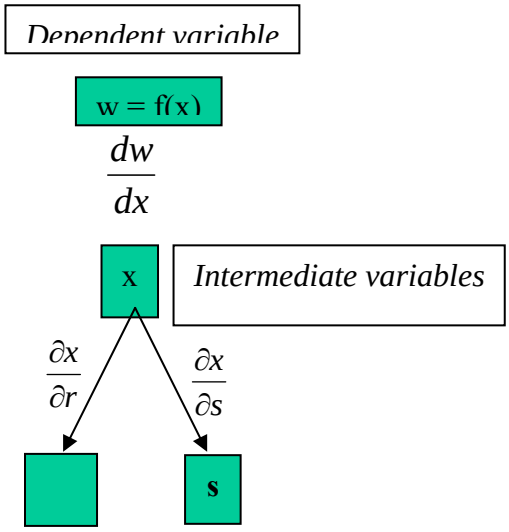
$= 2(e^t)(e^t) - Sint + 1$

$= 2e^{2t} - Sint + 1$

Consider

$w = f(x)$, *where* $x = g(r, s)$. Now it is clear from the figure that "x" is intermediate variable and we can write.

$$\frac{\partial w}{\partial r} = \frac{dw}{dx}\frac{\partial x}{\partial r} \text{ and } \frac{\partial w}{\partial s} = \frac{dw}{dx}\frac{\partial x}{\partial s}$$



Example:

$\mathbf{w = Sin\ x + x^2, \quad x = 3r + 4s}$

$\frac{dw}{dx} = Cosx + 2x$

$\frac{\partial x}{\partial r} = 3 \qquad \frac{\partial x}{\partial s} = 4$

$$\frac{\partial w}{\partial r} = \frac{dw}{dx}\cdot\frac{\partial x}{\partial r}$$

$= (Cosx + 2x) \cdot 3$

$= 3\,Cos(3r+4s) + 6(3r + 4s)$

$= 3\,Cos(3r + 4s) + 18r + 24s$

$$\frac{\partial w}{\partial s} = \frac{dw}{dx}\cdot\frac{\partial x}{\partial s}$$

$= (Cosx + 2x) \cdot 4$

$= 4\,Cosx + 8x$

$= 4\,Cos(3r + 4s) + 8(3r + 4s)$

$= 4\,Cos(3r + 4s) + 24r + 32s$

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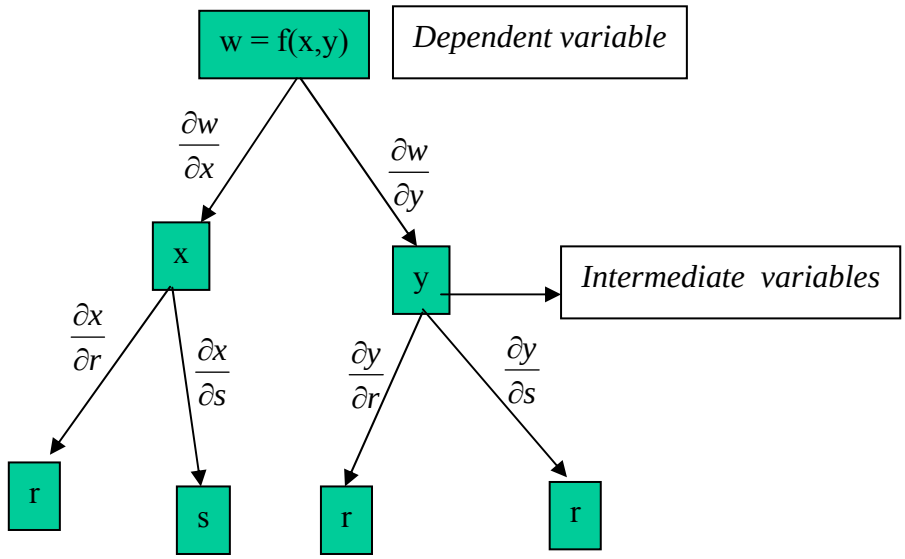
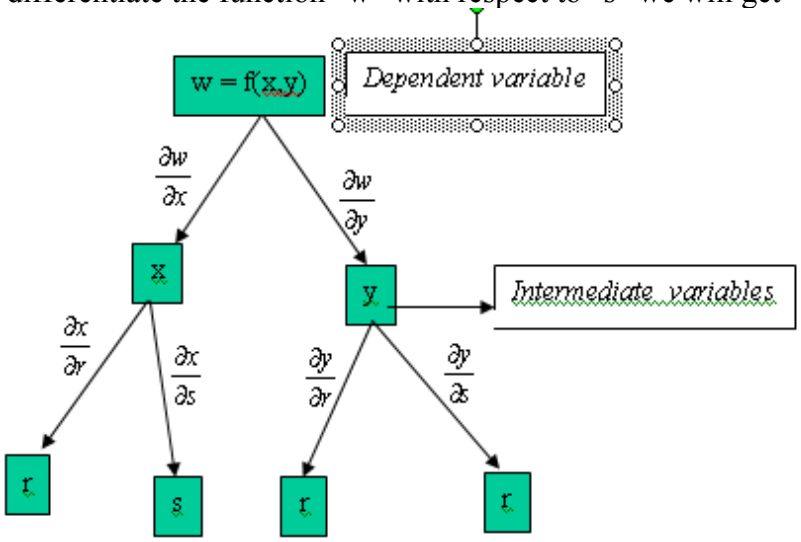
Consider the function $w = f(x,y)$, *Where* $x = g(r, s)$, $y = h(r, s)$



$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r}$$

Similarly if you differentiate the function "w" with respect to "s" we will get



And we have

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s}$$

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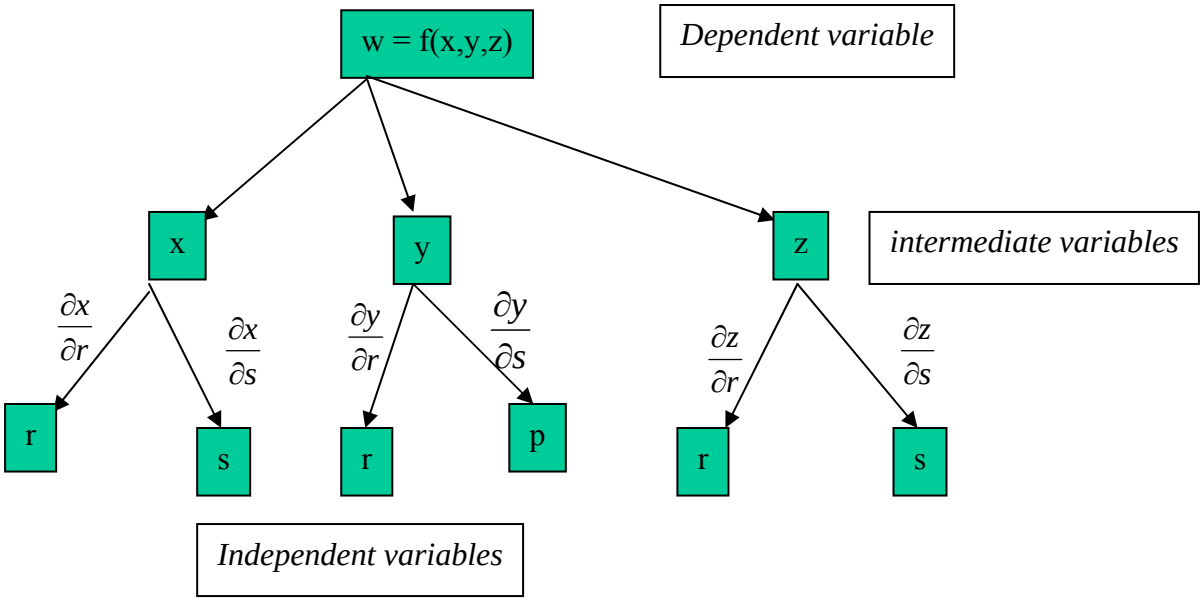
Consider the function $w = f(x,y,z)$, *Where* $x = g(r, s)$, $y = h(r,s)$, $z = k(r, s)$



Thus we have

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$$

Similarly if we differentiate with respect to "s" then we have,

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial s}$$

Example:

Consider the function $w = x + 2y + z^2$, $x = \frac{r}{s}, y = r^2 + \ln s,\ z = 2r$

First we will calculate

$$\frac{\partial w}{\partial x} = 1 \quad \frac{\partial w}{\partial y} = 2 \quad \frac{\partial w}{\partial z} = 2z \quad \frac{\partial x}{\partial r} = \frac{1}{s} \quad \frac{\partial y}{\partial r} = 2r \quad \frac{\partial z}{\partial r} = 2$$

Now as we know that $\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$ By putting the values from above we get

$$\frac{\delta w}{\delta r} = (1)\left(\frac{1}{s}\right) + (2)(2r) + (2z)(2)$$

$$= \frac{1}{s} + 4r + (4r)(2) = \frac{1}{s} + 12r$$

Now

$$\frac{\partial x}{\partial s} = -\frac{r}{s^2} \quad \frac{\partial y}{\partial s} = \frac{1}{s} \quad \frac{\partial z}{\partial s} = 0$$

So we can calculate

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$$

$$= (1)\left(-\frac{r}{s^2}\right) + (2)\left(\frac{1}{s}\right) + (2z)(0)$$

$$= \frac{2}{s} - \frac{r}{s^2}$$

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Remembering the different Forms of the chain rule:
The best thing to do is to draw appropriate tree diagram by placing the dependent variable on top, the intermediate variables in the middle, and the selected independent variable at the bottom. To find the derivative of dependent variable with respect to the selected independent variable, start at the dependent variable and read down each branch of the tree to the independent variable, calculating and multiplying the derivatives along the branch. Then add the products you found for the different branches.

The Chain Rule for Functions of Many Variables

Suppose $\omega = f(x, y, \ldots, \upsilon)$ is a differentiable function of the variables x, y, ….., υ (a finite set) and the x, y, …, υ are differentiable functions of p, q, , t (another finite set). Then ω is a differentiable function of the variables p through t and the partial derivatives of ω with respect to these variables are given by equations of the form

$$\frac{\partial \omega}{\partial p} = \frac{\partial \omega}{\partial x}\frac{\partial x}{\partial p} + \frac{\partial \omega}{\partial y}\frac{\partial y}{\partial p} + \ldots\ldots + \frac{\partial \omega}{\partial \upsilon}\frac{\partial \upsilon}{\partial p}.$$

The other equations are obtained by replacing p by q, …, t, one at a time. One way to remember last equation is to think of the right-hand side as the dot product of two vectors with components.

$\left(\frac{\partial \omega}{\partial x}, \frac{\partial \omega}{\partial y} \ldots\ldots \frac{\partial \omega}{\partial \upsilon}\right)$ and $\left(\frac{\partial x}{\partial p}, \frac{\partial y}{\partial p} \ldots\ldots \frac{\partial \upsilon}{\partial p}\right)$

Derivatives of ω with respect to the intermedaite variables

Derivatives of the intermedaite variables with respect to the selected independent variable

Example:

$$w = \ln(e^r + e^s + e^t + e^u)$$

Taking "ln" of both sides of the given equation we get

$$e^w = e^r + e^s + e^t + e^u$$

Now Taking partial derivative with respect to "r, s , u , and t" we get

$$e^w w_r = e^r \Rightarrow w_r = e^{r-w} \quad , \quad e^w w_s = e^s \Rightarrow w_s = e^{s-w} \quad , \quad e^w w_u = e^u \Rightarrow w_u = e^{u-w} \text{ and}$$

$$e^w w_t = e^t \Rightarrow w_t = e^{t-w}$$

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Now since we have $w_r = e^{r-w}$ Now Differentiate it partially w.r.t. "s"

$$w_{rs} = e^{r-w}(-w_s)$$
$$= -e^{r-w}e^{s-w} \qquad \text{(Here we use the value of } w_s \text{)}$$
$$w_{rs} = -e^{r+s-2w}$$

Now differentiate it partially w.r.t. "t" and using the value of w_t we get,

$$w_{rst} = -e^{r+s-2w}(-2w_t)$$
$$= 2e^{r+s-2w}e^{t-w}$$
$$w_{rst} = 2e^{r+s+t-3w}$$

Now differentiate it partially w.r.t. "u" we get,

$w_{rstu} = 2e^{r+s-3w}(-3w_u)$ and by putting the value of w_u, we get,

$$w_{rstu} = -6e^{r+s+t-3w}(e^{u-w})$$
$$w_{rstu} = -6e^{r+s+t+u-4w}$$

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Lecture No -10 Introduction to vectors

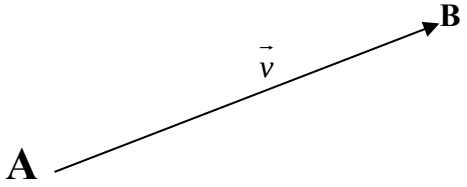
Some of things we measure are determined by their magnitude. But some times we need magnitude as well as direction to describe the quantities.

For Example, To describe a force, We need direction in which that force is acting (Direction) as well as how large it is (Magnitude).

Other Example is the body's Velocity; we have to know where the body is headed as well as how fast it is.

Quantities that have direction as well as magnitude are usually represented by arrows that point the direction of the action and whose lengths give magnitude of the action in term of a suitably chosen unit.

A vector in the plane is a directed line segment.



$$\vec{v} = \overrightarrow{AB}$$

Vectors are usually described by the single bold face roman letters or letter with an arrow. The vector defined by the directed line segment from point A to point B is written as $\overrightarrow{AB}$.

Magnitude or Length Of a Vector :

Magnitude of the vector $\vec{v}$ is denoted by

$|\vec{v}| = |\overrightarrow{\mathbf{AB}}|$ is the length of the line segment AB

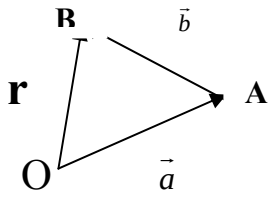
Unit vector

Any Vector whose Magnitude or length is 1 is a unit vector.

Unit vector in the direction of vector $\vec{v}$ is denoted by $\hat{v}$.
and is given by

$$\hat{v} = \frac{\vec{v}}{|v|}$$

Addition Of Vectors



This diagram shows three vectors, in two vectors one vector $\overrightarrow{OA}$ is connected with tail of vector $\overrightarrow{AB}$. The tail of third vector $\overrightarrow{OB}$ is connected with the tail of OA and head is connected with the head of vector $\overrightarrow{AB}$.This third vector is called Resultant vector.

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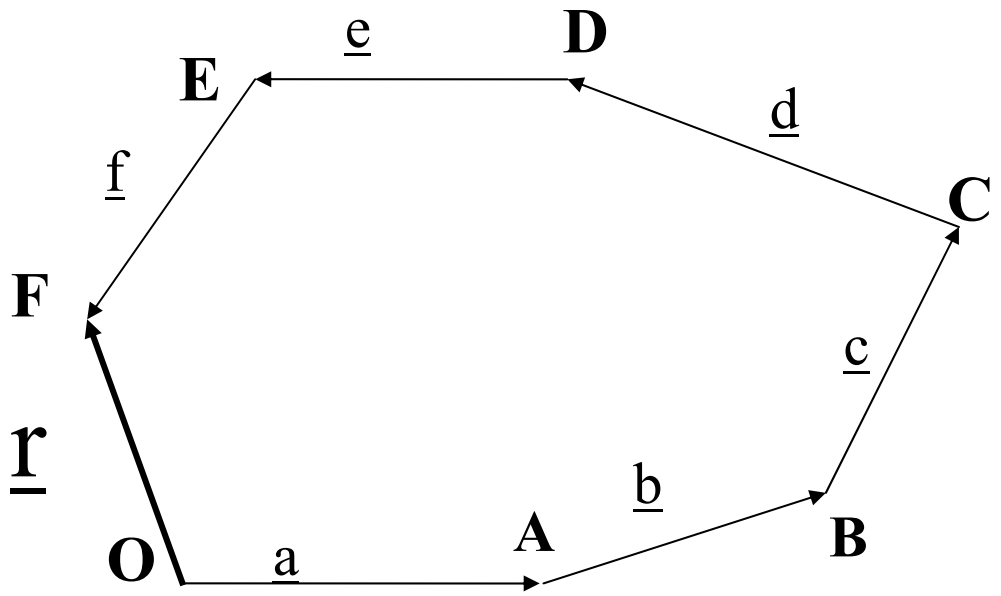
The resultant vector can be written as

$$\underline{r} = \underline{a} + \underline{b}$$

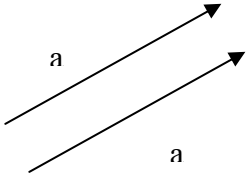
Similarly

$$\underline{r} = \underline{a} + \underline{b} + \underline{c} + \underline{d} + \underline{e} + \underline{f}$$



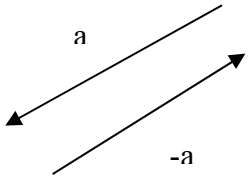
Equal vectors

Two vectors are equal or same vectors if they have same magnitude and direction.



Opposite vectors

Two vectors are opposite vectors if they have same magnitude and opposite direction.



Parallel vectors

Two vector are parallel if one vector is scalar multiple of the other.

$$\underline{b} = \lambda \underline{a}$$

where λ is a non zero scalar.

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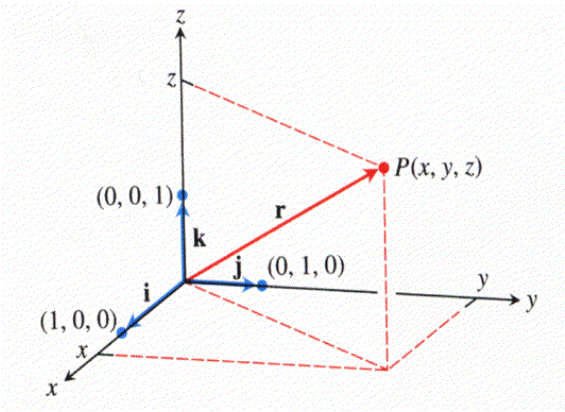


$\underline{r} = x\,\boldsymbol{i} + y\,\boldsymbol{j} + z\,\boldsymbol{k}$

Addition and subtraction of two vectors in rectangular component:

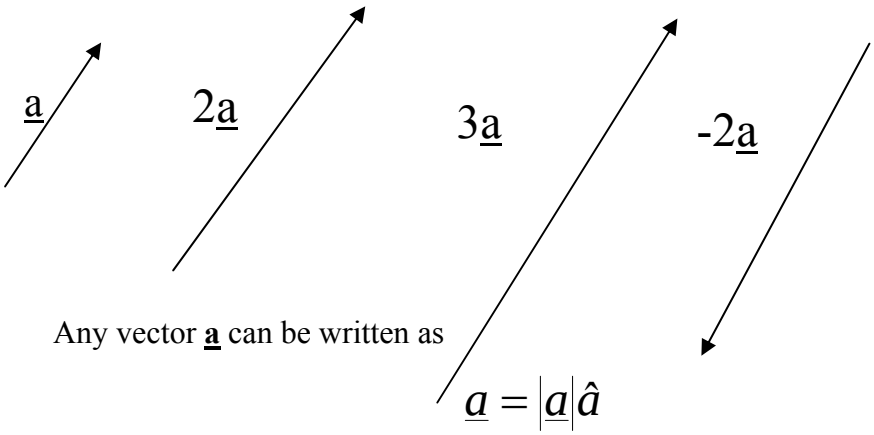
Let $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$

$$\mathbf{a} + \mathbf{b} = (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) + (b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k})$$
$$= (a_1 + b_1)\mathbf{i} + (a_2 + b_2)\mathbf{j} + (a_3 + b_3)\mathbf{k}$$
$$\mathbf{a} - \mathbf{b} = (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) - (b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k})$$
$$= (a_1 - b_1)\mathbf{i} + (a_2 - b_2)\mathbf{j} + (a_3 - b_3)\mathbf{k}$$

I th component of first vector is added (subtracted) to the ith component of second vector, jth component of first vector is added (subtracted) to the jth component of second vector, similarly kth component of first vector is added (subtracted) to the kth component of second vector,

Multiplication of a vector by a scalar



Any vector **a** can be written as

$$\underline{a} = |\underline{a}|\hat{a}$$

Scalar product

Scalar product (dot product) ("a dot b") of vector a and b is the number

$$a.b = |a|\,|b| \cos\theta.$$

where θ is the angle between a and b.

In word, a.b is the length of a times the length of b times the cosine of the angle between a and b.

Remark:-

$$\mathbf{a.b = b.a}$$

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This is known as commutative law.

Some Results of Scalar Product

$$a.b = |a|\,|b| \cos\theta.$$

1. $a \perp b$ If
   This means that if a is perpendicular to b.
   Then $a.b=0$
   Also
   $i.j= 0 = j.i$
   $j.k= 0 =k.j$
   $k.i= 0 =i.k$

2. If $a \parallel b$
   That means a is parallel to b.
   Then
   $$a.b = |a|\,|b|$$
   If we replace b by a then
   $$\mathbf{a.a} = |\mathbf{a}|\,|\mathbf{a}|$$
   $$\mathbf{a.a} = |\mathbf{a}|^2$$
   $$\sqrt{|\mathbf{a}|} = \mathbf{a.a}$$
   so i.i=j.j=k.k=1

Example

If $\mathbf{a} = 3\mathbf{k}$ and $\mathbf{b} = \sqrt{2}\mathbf{i} + \sqrt{2}\,\mathbf{k}$, then

$$\mathbf{a}.\ = |\mathbf{a}|\,|\mathbf{b}| \cos\theta$$
$$= (3)\,(2) \cos\frac{\pi}{4}$$
$$= 6 \cdot \frac{\sqrt{2}}{2} = 3\sqrt{2}.$$

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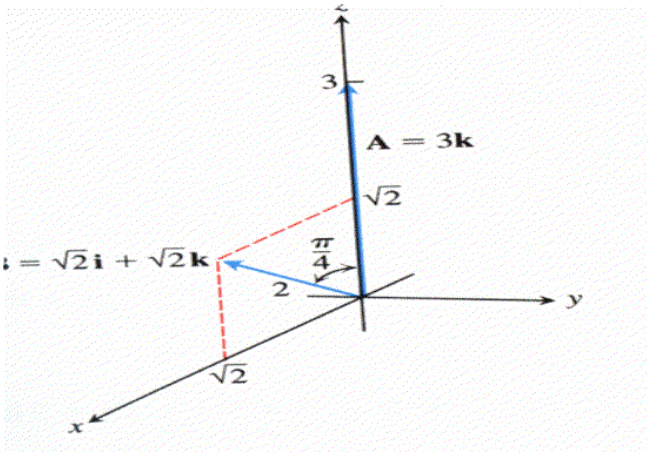

EXPRESSION FOR a.b IN COMPONENT FORM

$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and
$\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$

$$\begin{aligned}\mathbf{a}.\mathbf{b} &= (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) \cdot (b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}) \\ &= a_1\mathbf{i} \cdot (b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}) + a_2\mathbf{j} \cdot (b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}) \\ &\quad + a_3\mathbf{k} \cdot (b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}) \\ &= a_1b_1\mathbf{i}\cdot\mathbf{i} + a_1b_2\mathbf{i}.\mathbf{j} + a_1b_3\mathbf{i}\cdot\mathbf{k} + a_2b\mathbf{j}\cdot\mathbf{i} + a_2b_2\mathbf{j}\cdot\mathbf{j} \\ &\quad + a_2b_3\mathbf{j}\cdot\mathbf{k} + a_3b_1\mathbf{k}\cdot\mathbf{i} + + a_3b_2\mathbf{k}\cdot\mathbf{j} + a_3b_3\mathbf{k}\cdot\mathbf{k} \\ &= a_1b_1(1) + a_1b_2(0) + a_1b_3(0) + a_2b_1(0) + a_2b_2(1) \\ &\quad + a_2b_3(0) + a_3b_1(0) + + a_3b_2(0) + a_3b_3(1) \\ &= a_1b_1 + a_2b_2 + a_3b_3\end{aligned}$$

In dot product ith component of vector a will multiply with ith component of vector b , jth component of vector a will multiply with jth component of vector b and kth component of vector a will multiply with kth component of vector b

Angle Between two vectors

The angle between two nonzero vectors **a** and **b** is

$$\theta = \cos^{-1}\left(\frac{\mathbf{a}.\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}\right)$$

Since the values of the arc lie in $[0, \pi]$, above equation automatically gives the angle made by **a** and **b**.

Example

a = i − 2j − 2k and b = 6i + 3j + 2k.

$\mathbf{a}.\mathbf{b} = (1)(6) + (-2)(3) + (-2)(2)$
$= 6 - 6 - 4 = -4$

$|\mathbf{a}| = \sqrt{(1)^2 + (-2)^2 + (-2)^2} = \sqrt{9} = 3$

$|\mathbf{b}| = \sqrt{(6)^2 + (3)^2 + (2)^2} = \sqrt{49} = 7$

$$\theta = \cos^{-1}\left(\frac{\mathbf{a}.\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}\right)$$

$$= \cos^{-1}\left(\frac{-4}{(3)(7)}\right) = \cos^{-1}\left(-\frac{4}{21}\right) \approx 1.76 \text{ rad}$$

Perpendicular (Orthogonal) Vectors

a and b are perpendicular if and only if $\mathbf{a}.\mathbf{b} = 0$.

This has two parts If "a" and "b" are per perpendicular then a.b=0. And if a.b=0 "a" and "b" will be perpendicular.

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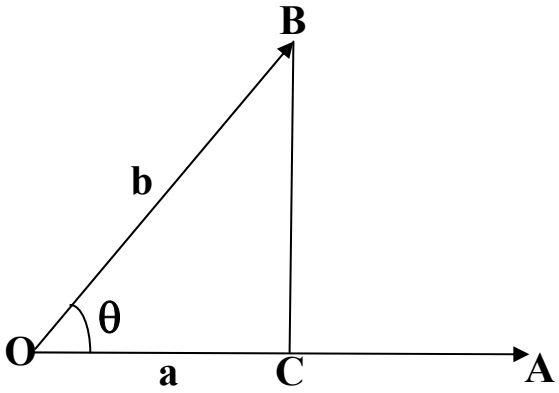
Vector Projectio

Consider the Projection of a vector b on a vector a making an angle θ with each other



From right angle triangle OCB
COS θ =Base / hypotenuse

$$\text{COS}\,\theta = \frac{|\vec{OC}|}{|\mathbf{b}|}$$

$$|\vec{OC}| = |\mathbf{b}|\,\text{Cos}\theta$$

$$= \frac{|\mathbf{b}|\,|\mathbf{a}|\,\text{Cos}\theta}{|\mathbf{a}|}$$

$$= \frac{\mathbf{b.a}}{|\mathbf{a}|} = \mathbf{b} \cdot \frac{\mathbf{a}}{|\mathbf{a}|}$$

$$= \mathbf{b} \cdot \frac{\mathbf{a}}{|\mathbf{a}|}$$

The number $|\mathbf{b}|\cos\theta$ is called the **scalar component of B in the direction of a** .

since $|\mathbf{b}|\cos\theta = B.\ \frac{\mathbf{a}}{|\mathbf{a}|}$,

we can find the scalar component by "dotting" **b** with the direction of **a**

Example

Vector Projection of $\mathbf{b} = 6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$
onto $\mathbf{a} = \mathbf{i} - 2\,\mathbf{j} - 2\mathbf{k}$ is

$$\text{proj}_{\mathbf{a}}\, b = \frac{\mathbf{b} \cdot \mathbf{a}}{\mathbf{a} \cdot \mathbf{a}}\,\mathbf{a}$$

$$= \frac{6 - 6 - 4}{1 + 4 + 4}\,(\mathbf{i} - 2\mathbf{j} - 2\mathbf{k})$$

$$= -\frac{4}{9}(\mathbf{i} - 2\mathbf{j} - 2\mathbf{k})$$

$$= \frac{4}{9}\mathbf{i} + \frac{8}{9}\mathbf{j} + \frac{8}{9}\mathbf{k}.$$

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he scalar component of **b** in the direction of **a**

$$|\mathbf{b}|\cos\theta = \mathbf{b} \cdot \frac{\mathbf{a}}{|\mathbf{a}|} = (6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}) \cdot \left(\frac{1}{3}\mathbf{i} - \frac{2}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}\right)$$

$$= 2 - 2 - \frac{4}{3} = -\frac{4}{3}.$$

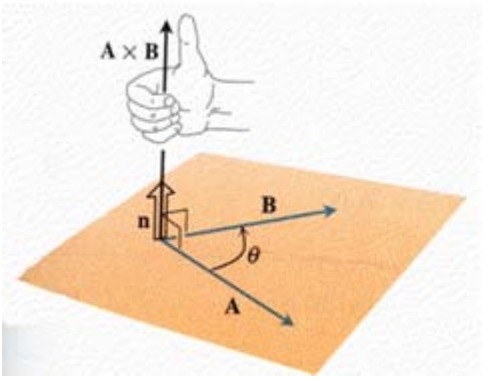
The Cross Product of Two Vectors in Space

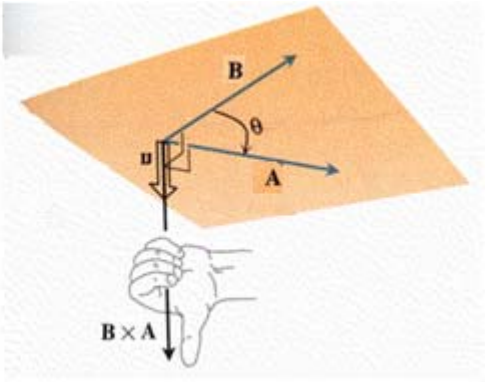
Consider two nonzero vectors **a** and **b** in space. The **vector product** **a** × **b** ("**a** cross **b**") to be the vector

a × **b** = (|**a**||**b**| sin θ) **n**

where **n** is a vector determined by right hand rule.

Right-hand rule





We start with two nonzero nonparallel vectors **A** and **B** .We select a unit vector **n** Perpendicular to the plane by the **right handed rule.** This means we choose n to be the unit vector that points the way your right thumb points when your fingers curl through the angle 0 from **A** to **B** .

The vector A*B is orthogonal to both **A** and **B.**

Some Results of Cross Product

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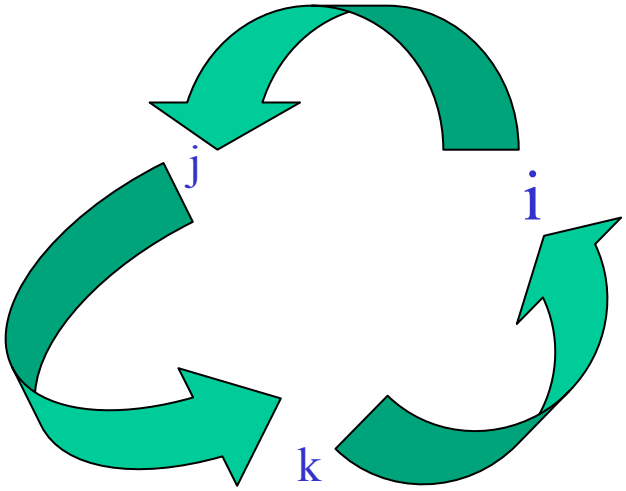
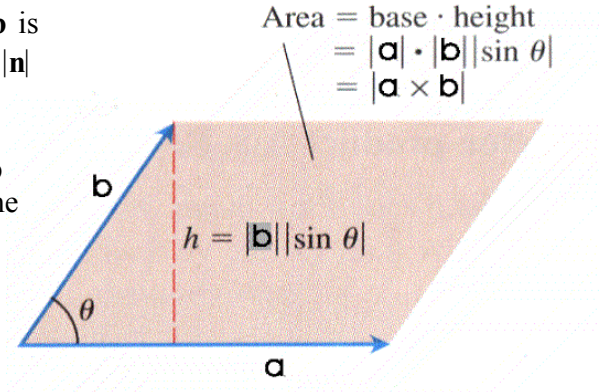
$\mathbf{a} \times \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}|\,\text{Sin}\,\theta\,\hat{\mathbf{n}}$
If $\mathbf{a} \| \mathbf{b}$
then $\mathbf{a} \times \mathbf{b} = 0$
So $\mathbf{a} \times \mathbf{a} = 0$
$\mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = 0$
If $\mathbf{a} \perp \mathbf{b}$
then $\mathbf{a} \times \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}|\,\hat{\mathbf{n}}$
$\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{i} = -\mathbf{k}$
$\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{j} = -\mathbf{i}$
$\mathbf{k} \times \mathbf{i} = \mathbf{j}$ $\mathbf{i} \times \mathbf{k} = -\mathbf{j}$
Note that this product is not commutative.



The Area of a Parallelogram

Because n is a unit
the magnitude of $\mathbf{a} \times \mathbf{b}$ is
$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}|\,|\sin\theta|\,|\mathbf{n}|$
$= |A|\,|\mathbf{b}|\,\sin\theta$
This is the area of the
determined by $\mathbf{a}$ and $\mathbf{b}$
$|\mathbf{a}|$ being the base of the
and $|\mathbf{b}|\,|\sin\theta|$ the



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$\mathbf{a} \times \mathbf{b}$ from the components of a and b

Suppose

$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$,

$\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$.

$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) \times (b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}) \\ &= a_1b_1\mathbf{i} \times \mathbf{i} + a_1b_2\mathbf{i} \times \mathbf{j} + a_1b_3\mathbf{i} \times \mathbf{k} \\ &\quad + a_2b_1\mathbf{j} \times \mathbf{i} + a_2b_2\mathbf{j} \times \mathbf{j} + a_2b_3\mathbf{j} \times \mathbf{k} \\ &\quad + a_3b_1\mathbf{k} \times \mathbf{i} + a_3b_2\mathbf{k} \times \mathbf{j} + a_3b_3\mathbf{k} \times \mathbf{k}\end{aligned}$$

$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

$\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$.

$$= (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}.$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Example:

Let

$\mathbf{a} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = -4\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ **. Then**

$$a \times b = \begin{vmatrix} i & j & k \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix}$$

$$a \times b = i(1-3) - j(2+4) + k(6+4)$$

$a \times b = -2i - 6j + 10k$ is the required cross product of **a** and **b.**

Over view of Lecture # 10

Chapter# 14

Article # 14.3, 14.4

Page # 679

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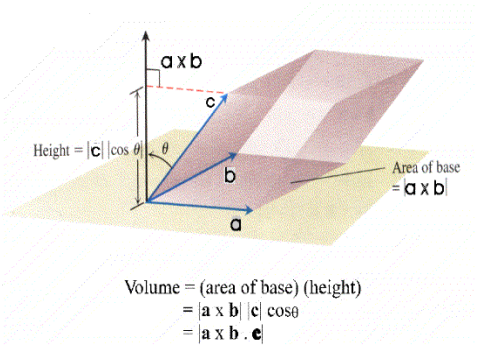
Lecture No -11 The Triple Scalar or Box Product

The product $(\mathbf{a}\times\mathbf{b})\,.\,\mathbf{c}$ is called the **triple scalar product** of **a**, **b**, and **c** (in that order).
As $|\,(\mathbf{a}\times\mathbf{b}).\mathbf{c}\,| = |a\times\mathbf{b}|\,|c|\,|\cos\theta|$
the absolute value of the product is the volume of the parallelepiped (parallelogram-sided box) determin-ed by a,b and c



By treating the planes of b and c and of c and a as he base planes of the parallelepiped determined by a, b and c
we see that

$$(a * b).c = (b\times c).a=(c\times a).b$$

Since the dot product is commutative, $(a\times b).c = a.(b\times c)$

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k},$$
$$\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}.$$
$$\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}.$$

$$\mathbf{a}\,.\,(\mathbf{b}\times\mathbf{c}) = \mathbf{a}.\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \mathbf{a}\,.\left[\begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}\mathbf{i} - \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}\mathbf{j} + \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}\mathbf{k}\right]$$

$$= a_1\begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2\begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3\begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

$$= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Example

$$\mathbf{a} = \mathbf{i} + 2\mathbf{j} - \mathbf{k},\quad \mathbf{b} = -2\mathbf{i} + 3\mathbf{k},\quad \mathbf{c} = 7\mathbf{j} - 4\mathbf{k}.$$

$$\mathbf{a}\,.(\mathbf{b}\times\mathbf{c}) = \begin{vmatrix} 1 & 2 & -1 \\ -2 & 0 & 3 \\ 0 & 7 & -4 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 3 \\ 7 & -4 \end{vmatrix} - 2\begin{vmatrix} -2 & 3 \\ 0 & -4 \end{vmatrix} - \begin{vmatrix} -2 & 0 \\ 0 & 7 \end{vmatrix}$$

$$= -21 - 16 + 14$$

$$= -23$$

The volume is

$$|\mathbf{a}\,.(\mathbf{b}\times\mathbf{c})| = 23.$$

When we solve **a.(b×c)** then answer is -23 . if we get negative value then Absolute value make it positive and also volume is always positive.

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Gradient of a Scalar Function

$$\nabla \equiv \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z},$$

∇ is called "del operator"

Gradient ϕ is a vector operator

gradient ϕ is a vector operator

defined as ϕ is a vector operator defined as

$$\text{grad}\,\phi = \left[\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right]\phi$$

$$\text{grand}\,\phi = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right)\phi$$

$$= \nabla\phi,$$

$$= \nabla\,\phi,$$

$$\nabla \equiv \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z},$$

∇ is called "del operator"

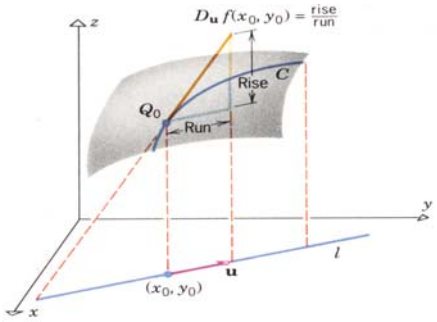
∇ "del operator" is a vector quantity. Grad means gradient. Gradient is also vector quantity. $\nabla\phi$ is vector and ϕ is scalar quantity, Every component of $\nabla\phi$ will operate with the .

Directional Derivative

If f(x,y) is differentiable at (x_0,y_0), and if $\mathbf{u} = (u_1, u_2)$ is a unit vector, then the ***directional derivative*** of f at (x_0, y_0) in the direction of $\mathbf{u}$ is defined by

$D_{\mathbf{u}} f(x_0,y_0) = f_x(x_0,y_0)u_1 + f_y(x_0,y_0)u_2$

It should be kept in mind that there are infinitely many directional derivatives of z = f(x,y) at a point (x_0,y_0), one for each possible choice of the direction vector $\mathbf{u}$



Remarks (Geometrical interpretation)

The directional derivative $D_{\mathbf{u}} f(x_0,y_0)$ can be interpreted algebraically as the instantaneous rate of change in the direction of $\mathbf{u}$ at (x_0,y_0) of z=f(x,y) with respect to the distance parameter s described above, or geometrically as the rise over the run of the tangent line to the curve C at the point Q_0

Example

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The directional derivative of
$f(x,y) = 3x^2y$ at the point (1, 2) in the direction of the vector $a = 3i + 4j$.

$$f(x,y) = 3x^2y$$

$f_x(x,y) = 6xy$, $f_y(x,y) = 3x^2$

so that

$f_x(1,2) = 12$, $f_y(1,2) = 3$

$\mathbf{a} = 3\mathbf{i}+4\mathbf{j}$

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{\|\mathbf{a}\|} = \frac{1}{\sqrt{25}}(3\mathbf{i}+4\mathbf{j})$$

$$= \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$$

$$D_{\mathbf{u}}f(1,2) = 12\left(\frac{3}{5}\right) + 3\left(\frac{4}{5}\right)$$

$$= \frac{48}{5}$$

Note:
Formula for the directional derivative can be written in the following compact form using the gradient notation

$$D_{\mathbf{u}}f(x,y) = \nabla f(x,y) \cdot \hat{\mathbf{u}}$$

The dot product of the gradient of f with a unit vector $\hat{\mathbf{u}}$ produces the

f_x means that function f(x,y) is differentiating partially with respect to x and
f_y means that function f(x,y) is differentiating partially with respect to y.

Example

$f(x,y) = 2x^2 + y^2$, $P_0(-1,1)$

$\mathbf{u} = 3\mathbf{i} - 4\mathbf{j}$

$|\mathbf{u}| = \sqrt{3^2 + (-4)^2} = 5$

$\hat{u} = \frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}$

$f_x = 4x$ $\quad f_x(-1,1) = -4$

$f_y = 2y$ $\quad f_y(-1,1) = 2$

$D_{\mathbf{u}}f(-1,1) = f_x(-1,1)u_1 + f_y(-1,1)u_2$

$$= -\frac{12}{5} - \frac{8}{5} = -4$$

Another example, In this example we have to find directional derivative of the function $f(x,y) = 2x^2 + y^2$ at the point $P_0(-1,1)$ in the direction of **u = 3i – 4j. To find the directional derivative we again use the above formula**

Remarks

If $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is a unit vector making an angle θ with the positive x - axis, then

$u_1 = \cos\theta$ and $u_2 = \sin\theta$

$D_{\mathbf{u}}f(x_0,y_0) = f_x(x_0,y_0)u_1 + f_y(x_0,y_0)u_2$

can be written in the form

$D_{\mathbf{u}}f(x_0,y_0) = f_x(x_0,y_0)\cos\theta + f_y(x_0,y_0)\sin\theta$

Example

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The directional derivative of e^{xy} at $(-2,0)$ in the direction of the unit vector **u** that makes an angle of $\pi/3$ with the positive x-axis.

$$f(x,y) = e^{xy}$$

$f_x(x.y) = ye^{xy}, \quad f_y(x, y) = x e^{xy}$

$fx(-2,0) = 0, \quad fy(2,) = -2$

$$D_{\mathbf{u}} f(-2, 0) = f_x(-2, 0)\cos\frac{\pi}{3} + f_y(-2, 0)\sin\frac{\pi}{3}$$

$$= 0\left(\frac{1}{2}\right) + (-2)\left(\frac{\sqrt{3}}{2}\right)$$

$$= -\sqrt{3}$$

Gradient of function

If *f* is a function of x and
then the gradient of is defined

$$\nabla f(x,y) = f_x(x,y)\mathbf{i} + f_y(x,y)\mathbf{j}$$

Directional Derivative

Formula for the directional derivative can be written in the following compact form using the gradient

$$D_{\mathbf{u}} f(x, y) = \nabla f(x, y) \cdot \hat{\mathbf{u}}$$

The dot product of the gradient *f* with a unit $\hat{\mathbf{u}}$ produces directional derivative of f in direction **u**.

EXAMPLE

$f(x,y) = 2xy - 3y^2, \quad P_0\ (5,5)$

$\mathbf{u} = 4\mathbf{i} + 3\mathbf{j} \quad |\mathbf{u}| = \sqrt{4^2 + 3^2} = 5$

$$\hat{u} = \frac{\mathbf{u}}{|\mathbf{u}|} = \frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j}$$

$f_x = 2y, \qquad fy = 2x - by$

$f_x\ (5, 5) = 10, \qquad fy\ (5,5) = -20$

$\nabla f = 10\mathbf{i} - 20\mathbf{j}$

$D_{\mathbf{u}} f(5,5) = \nabla f \,.\, \hat{u}$

$$= 10\left(\frac{4}{5}\right) - 20\left(\frac{3}{5}\right)$$

$$= -4$$

,In this example we have to find directional derivative of the function $f(x, y) = 2x\ y - 3y^2$ at the point $P_0(5,5)$ in the direction of **u = 4i + 3j. To find the directional derivative we again use the above formula**

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EXAMPLE

Directional derivative of the function
$f(x, y) = xe^y + \cos(xy)$ at the point
$(2,0)$ in the direction of $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j}$.

$\hat{\mathbf{u}} = \dfrac{\mathbf{a}}{|\mathbf{a}|} = \dfrac{3}{5}\mathbf{i} - \dfrac{4}{5}\mathbf{j}.$

$f_x(x,y) = e^y - y\sin(xy)$

$f_y(x,y) = xe^y - x\sin(xy)$

The partial derivatives of f
at (2, 0) are

$f_x(2,0) = e^0 - 0 = 1$

$f_y(2,0) = 2e^0 - 2.0 = 2$

The gradient of f at (2, 0)

$\nabla f_{(2,0} = {}_x(2, 0)\,\mathbf{i} + {}_y(2, 0)\,\mathbf{j}$

$= \mathbf{i} + \mathbf{j}$

The derivative of f at 0) in the
direction of a is

$({}_{\mathbf{u}}f)_{(2,0} = \nabla f_{(2,0} \cdot$

$= \mathbf{j}) \left(\dfrac{3}{5}\mathbf{i} - \dfrac{4}{5}\mathbf{j}\right)$

$\dfrac{3}{5} - \dfrac{8}{5} = -$

Properties of Directional Derivatives

$D_{\mathbf{u}}f = \nabla f \cdot \hat{\mathbf{u}} = |\nabla f| \cos\theta$

1. The function f increases most rapidly when $\cos\theta = 1$, or when $\mathbf{u}$ is the direction of ∇f. That is, at each point P in its domain, **f increases most rapidly in the direction of the gradient vector ∇ f at P.** The derivative in this direction is

$$D_{\mathbf{u}}f = |\nabla f| \cos(0) = |\nabla f|.$$

2. Similarly, ***f* decreases most rapidly in the direction of $-\nabla$f.** The derivative in this direction is

$D_{\mathbf{u}}f = |\nabla f| \cos(\pi) = -|\nabla f|.$

3. **Any direction $\hat{\mathbf{u}}$ orthogonal of the gradient is a direction of zero change in f** because θ then equals $\pi/2$ and

$D_{\mathbf{u}}f = |\nabla f| \cos(\pi/2) = |\nabla f|.0 = 0$

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$$f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$$

a) The direction of rapid change .

The function increases most rapidly in the direction of ∇f at (1,1). The gradient is

$(\nabla f)_{(1,1)} = (x\mathbf{i} + y\mathbf{j})_{(1,1)} = \mathbf{i} + \mathbf{j}$.

its direction is

$$\hat{\mathbf{u}} = \frac{\mathbf{i} + \mathbf{j}}{|\mathbf{i} + \mathbf{j}|}$$

$$= \frac{\mathbf{i} + \mathbf{j}}{\sqrt{(1)^2 + (1)^2}}$$

$$= \frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}.$$

(1,1) are the directions orthogonal to ∇f.

b) **The directions of zero change**

The directions of zero change at

$$\hat{\mathbf{n}} = -\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$$

and $-\hat{\mathbf{n}} = \frac{1}{\sqrt{2}}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j}$.

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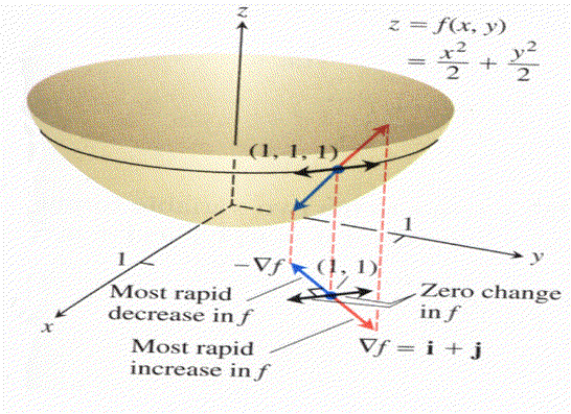

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Lecture No -12 Tangent planes to the surfaces

Normal line to the surfaces

If **C** is a smooth parametric curve on three dimension, then tangent line to **C** at the point P_0 is the line through P_0 along the unit tangent vector to the C at the P_0.The concept of a tangent plane builds on this definition.

If $P_0(x_0,y_0,z_0)$ is a point on the Surface **S,** and if the tangent lines at P_0 to all the smooth curves that pass through P_0 and lies on the surface S all lie in a common plane, then we shall regards that plane to be the ***tangent plane*** to the surface S at P_0.

Its normal (the straight line through P_0 and perpendicular to the tangent) is called the ***surface normal*** of S at P_0.

Different forms of equation of straight line in two dimensional space

1. **Slope intercept form of the Equation of a line.**

$$y = mx + c$$

Where **m** is the slope and **c** is y intercept.

2.Point_Slope Form

Let m be the slope and $P_0(x_0, y_0)$ be the point of required line, then

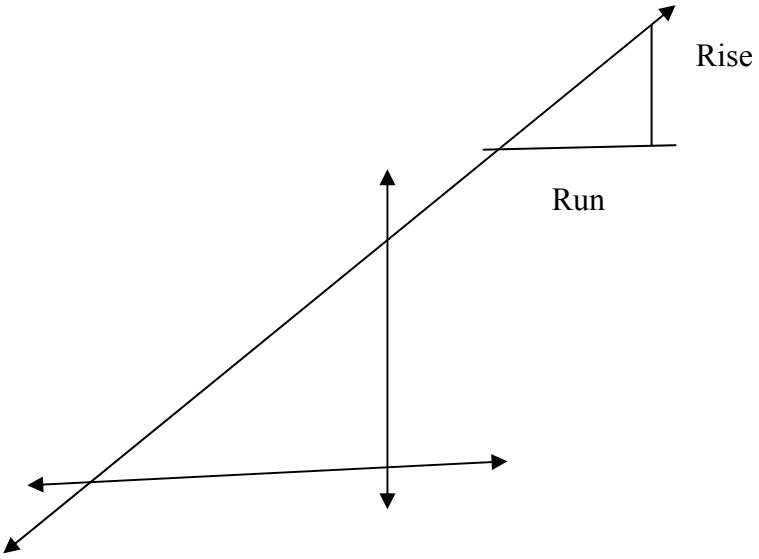
$$\mathbf{y - y_0 = m\,(x - x_0)}$$

3. **General Equation of straight line**

$$Ax + By + C = 0$$

$$m = \text{slope of line} = \frac{\text{Rise}}{\text{Run}} = \frac{b}{a}$$

$$y - y_0 = \frac{b}{a}(x - x_0)$$



Parametric equation of a line

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Parametric equation of a line in two dimensional space passing through the point (x_0, y_0) and parallel to the vector a**i** + b**j** is given by

$$\mathbf{x = x_0 + at}, \qquad \mathbf{y = y_0 + bt}$$

Eliminating t from both equation we get

$$\frac{x - x_0}{a} = \frac{y - y_0}{b}$$

$$y - y_0 = \frac{b}{a}(x - x_0)$$

Parametric vector form:

$$\mathbf{r}(t) = (x_0 + at)\,\mathbf{i} + (y_0 + bt)\,\mathbf{j},$$

Equation of line in three dimensional

Parametric equation of a line in three dimensional space passing through the point (x_0, y_0, z_0) and parallel to the vector ai + bj + ck is given by

$$\mathbf{x = x_0 + at}, \qquad \mathbf{y = y_0 + bt}, \qquad \mathbf{z = z_0 + ct}$$

Eliminating t from these equations we get

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

EXAMPLE

Parametric equations for the straight line through the point A (2, 4, 3) and parallel to the vector $\mathbf{v} = 4\mathbf{i} + 0\mathbf{j} - 7\mathbf{k}$.

$x_0 = 2, y_0 = 4, z_0 = 3$

and a = 4, b = 0 , c = - 7.

The required parametric equations of the straight line are

$$\mathbf{x = 2 + 4t},$$
$$\mathbf{y = 4 + 0t},$$
$$\mathbf{z = 3 - 7t}$$

Different form of equation of curve

Curves in the plane are defined in different ways

Explicit form:

$$y = f(x)$$

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Example

$$y = \sqrt{9-x^2}, \qquad -3 \le x \le 3.$$

Implicit form:

$$F(x, y) = 0$$

Example

$$x^2 + y^2 = 9, \qquad -3 \le x \le 3, \qquad 0 \le y \le 3$$

Parametric form:

$$x = f(t) \text{ and } y = g(t)$$

Example

$$x = 3\cos\theta, \qquad y = 3\sin\theta, \qquad 0 \le \theta \le \pi$$

$$x = 3\cos\theta, \quad y = 3\sin\theta$$

$$x^2 + y^2 = 9\cos^2\theta + 9\sin^2\theta$$

$$= 9(\mathrm{Cos}^2\theta + \mathrm{Sin}^2\theta)$$

$$x^2 + y^2 = 9$$

Parametric vector form:

$$\mathbf{r}(t) = f(t)\,\mathbf{i} + g(t)\,\mathbf{j}, \qquad a \le t \le b.$$

$$\mathbf{r}(t) = 3\cos\theta\mathbf{i} + 3\sin\theta\mathbf{j}, \; 0 \le \theta \le \pi.$$

Equation of a plane

A plane can be completely determined if we know its one point and direction of perpendicular (normal) to it.

Let a plane passing through the point $P_0\,(x_0, y_0, z_0)$ and the direction of normal to it is along the vector

$$\mathbf{n} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$$

Let P (x, y, z) be any point on the plane then the line lies on it so that **n** $\perp$ $\overrightarrow{P_0P}$ ($\perp$ means perpendicular to)

$\overrightarrow{P_0P} = (x - x_0)\,\mathbf{i} + (y - y_0)\,\mathbf{j} + (z - z_0)\,\mathbf{k}$

$\mathbf{n} \,.\, \overrightarrow{P_0P} = 0$

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$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$

is the required equation of the plane

Here we use the theorem ,let a and b be two vectors, if a and b are perpendicular then a.b=0 so **n** and P_0P **a**re perpendicular vector so n.P0P=0

REMARKS

Point normal form of equation of plane is

$$\mathbf{a(x - x_0) + b(y - y_0) + c(z - z_0) = 0}$$

We can write this equation as

$$\mathbf{ax + by + cz - ax_0 - by_0 - cz_0 = 0}$$

$$\mathbf{ax + by + cz + d = 0}$$

where $\mathbf{d = - ax_0 - by_0 - cz_0}$

Which is the equation of plane

EXAMPLE

An equation of the plane passing through the point (3, - 1, 7) and perpendicular to the vector **n** = 4**i** + 2**j** - 5**k**.

A point-normal form of the equation is

$$\mathbf{4(x - 3) + 2(y + 1) - 5(z - 7) = 0}$$

$$\mathbf{4x + 2y - 5z + 25 = 0}$$

Which is the same form of the equation of plane **ax + by + cz + d = 0**

The general equation of straight line is ax + by + c = 0

Let (x_1, y_1) and (x_2, y_2) be two points on this line then

$ax_1 + by_1 + c = 0$

$ax_2 + by_2 + c = 0$

Subtracting above equation

$a(x_2 - x_1) + b(y_2 - y_1) = 0$

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$\mathbf{v} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j}$
is a vector in the direction of line
$\phi(x, y) = ax + by$
$\phi_x = a, \quad \phi_y = b$
$\nabla\phi = a\mathbf{i} + b\mathbf{j} = \mathbf{n}$
$\nabla\phi \cdot \mathbf{r} = 0$
Then n and **v** are perpendicular
The general equation of plane is
$ax + by + cz + d = 0$
For any two points (x_1, y_1, z_1) and (x_2, y_2, z_2) lying on this plane we have
$ax_1 + by_1 + cz_1 + d = 0 \qquad (1)$
$a x_2 + by_2 + cz_2 + d = 0 \qquad (2)$

Subtracting equation (1) from (2)
have
$a (x_2 - x_1) + b (y_2 - y_1) + c (z_2 - z_1) = 0$

$(a\mathbf{i} + b\mathbf{j} + c\mathbf{k}).\left[(X_2\text{-}X_1)\, i + (Y_2\text{-}Y_1)\, j + (Z_2\text{-}Z_1)\, k\right]$

Here we use the definition of dot product of two vectors.

$\phi = ax + by + cz$
$\phi_x = a, \quad \phi_y = b, \quad \phi_z = c$
$\nabla\phi = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$
Where $\mathbf{v} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$

$\nabla\phi$ is always normal to the plane.

Gradients and Tangents to Surfaces

$f(x, y) = c$
$z = f(x,y), \; z = c$

If a differentiable function f(x,y) has a constant value c along a smooth curve having parametric equation

$x = g(t), \quad y = h(t), \; r = g(t)i + h(t)j$
differentiating both sides of this equation with respect to t leads to the equation

$$\frac{d}{dt} f(g(t), h(t)) = \frac{d}{dt}(c)$$

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$$\frac{\partial f}{\partial x}\frac{dg}{dt}+\frac{\partial f}{\partial y}\frac{dh}{dt}=0 \qquad \text{Chain Rule}$$

$$\left(\frac{\partial f}{\partial x}\boldsymbol{i}+\frac{\partial f}{\partial y}\boldsymbol{j}\right).\left(\frac{dg}{dt}i+\frac{dh}{dt}\boldsymbol{j}\right)=0$$

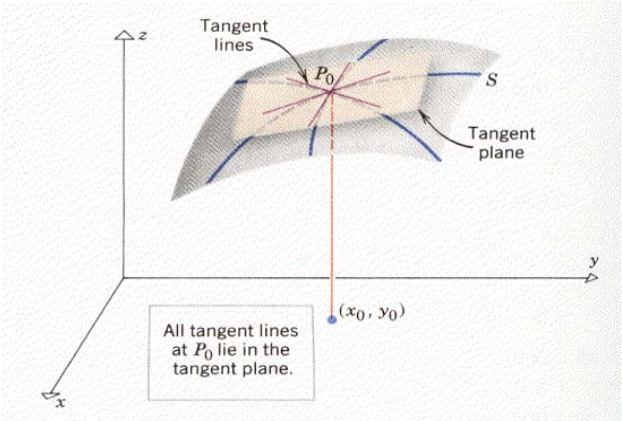
$$\nabla f.\frac{d\boldsymbol{r}}{dt}=0$$

∇f is normal to the tangent vector d **r**/dt,
so it is normal to the curve through (x_0, y_0).

Tangent Plane and Normal Line

Consider all the curves through the point **P0(x0, y0, z0)** on a surface **f(x, y, z) = 0.** The plane containing all the tangents to these curves at the point P0(x0, y0, z0) is called the ***tangent plane to the surface at the point P0***.

The straight lines perpendicular to all these tangent lines at P_0 is called the ***normal line to the surface at P0*** if fx, fy, fz are all continuous at P_0 and not all of them are zero, then gradient *f* (i.e fx**i** + fy**j** + fz**k**) at P0 gives the direction of this normal vector to the surface at P0.



Tangent plane

Let P_0 (x_0, y_0, z_0) be any point on the Surface f(x,y,z) = 0. If f(x,y,z) is differentiable at $p_0(x_0,y_0,z_0)$ then the tangents plane at the point P_0 (x_0,y_0,z_0) has the equation

EXAMPLE

$$9x^2 + 4y^2 - z^2 = 36 \qquad P\ (2,3,6).$$

$$\boldsymbol{f(x,y,z) = 9x^2 + 4y^2 - z^2 - 36}$$

$$f_x = 18x, \qquad f_y = 8y, \qquad f_z = -2z$$

$$f_x(P) = 36, \qquad f_y(P) = 24, \qquad f_z(P) = -12$$

Equations of Tangent Plane to the surface

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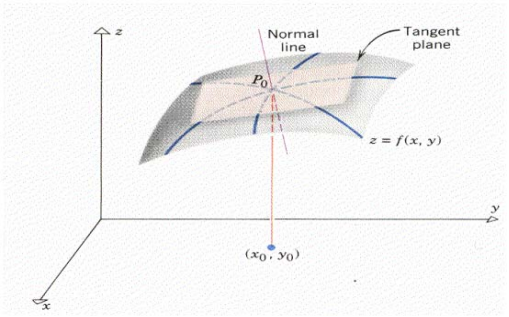
through P is

$36(x - 2) + 24(y - 3) - 12(z - 6) = 0$
$3x + 2y - z - 6 = 0$

EXAMPLE

$z = x \cos y - ye^x$ (0,0,0).



$\cos y - ye^x - z = 0$

$f(x,y,z) = \cos y - ye^x - z$

$f_x(0,0,0) = (\cos y - ye^x)_{(0,0)} = 1 - 0.1 = 1$

$f_y(0,0,0) = (-x \sin y - e^x)_{(0,0)} = 0 - 1 = -1.$

$f_z(0,0,0) = -1$

The tangent plane is

$f_x(0,0,0)(x - 0) + f_y(0,0,0)(y - 0) + f_z(0,0,0)(z - 0) = 0$

$1(x - 0) - 1(y - 0) - 1(z - 0) = 0,$

$x - y - z = 0.$

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Lecture No -13 Orthogonal Surface

In this Lecture we will study the following topics

- **Normal line**
- **Orthogonal Surface**
- **Total differential for function of one variable**
- **Total differential for function of two variables**

Normal line

Let $P_0(x_0,y_0,z_0)$ be any point on the surface f(x,y,z)=0 If f(x,y,z) is differentiable at $P_0(x_0,y_0,z_0)$ then the normal line at the point $P_o(x_0,y_0,z_0)$ has the equation

$$x = x_0+f_x(P_0)t, \quad y = y_0+f_y(P_0)t, \quad z = z_0+f_z(P_0)t$$

Here $\mathbf{f_x}$ means that the function f(x,y,z) is partially differentiable with respect to x And $\mathbf{f_x(P_0)}$ means that the function **f(x,y,z)** is partially differentiable with respect to x at the point $\mathbf{P_0(x_0,y_0,z_0)}$

$\mathbf{f_y}$ means that the function f(x,y,z) is partially differentiable with respect to y And $\mathbf{f_y(P_0)}$ means that the function **f(x,y,z)** is partially differentiable with respect to y at the point $\mathbf{P_0(x_0,y_0,z_0)}$

Similarly

fz means that the function f(x,y,z) is partially differentiable with respect to z And $\mathbf{f_z(P_0)}$ means that the function **f(x,y,z)** is partially differentiable with respect to z at the point $\mathbf{P_0(x_0,y_0,z_0)}$

EXAMPLE

Find the Equation of the tangent plane and normal of the surface f(x,y,z)= x2+y2+z2-4 at the point P(1,-2,3)

$$\mathbf{f(x,y,z) = x^2+y^2+z^2 - 14}$$

$$\mathbf{P\ (1, -2, 3).}$$

$$f_x = 2x, \quad f_y = 2y, \quad f_z = 2z$$

$$f_x(p_0) = 2, \quad f_y\ (P_0) = -4, \quad f_z\ (P_0) = 6$$

Equation of the tangent p lane to the surface at P is

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$$2(x-1)-4(y+2)+6(z-3)=0$$

$$x-2y+3z-14=0$$

Equations of the normal line of the surface through P are

$$\frac{x-1}{2}=\frac{y+2}{-4}=\frac{z-3}{6}$$

$$\frac{x-1}{1}=\frac{y+2}{-2}=\frac{z-3}{3}$$

EXAMPLE

Find the equation of the tangent plane and normal plane

$$4x^2-y^2+3z^2=10 \quad P(2,-3,1)$$

$$f(x,y,z)=4x^2-y^2+3z^2-10$$

$$f_x=8x, \qquad f_y=-2y, \qquad f_z=6z$$

$$f_x(P)=16, \qquad f_y(P)=6, \qquad f_z(P)=6$$

Equations of Tangent Plane to the surface through P is

$$16(x-2)+6(y+3)+6(z-1)=0$$

$$8x+3y+3z=10$$

Equations of the normal line to the surface through P are

$$\frac{x-2}{16}=\frac{y+3}{6}=\frac{z-1}{6}$$

$$\frac{x-2}{8}=\frac{y+3}{3}=\frac{z-1}{3}$$

Example

$$z=\frac{1}{2}x^7y^2$$

$$f(x,y,z)=\frac{1}{2}x^7y^2-z$$

$$f_x=\frac{7}{2}x^6.y^2, \qquad f_y=-x^7.y^3, \qquad f_z=-1$$

$$f_x(2,4,4)=\frac{7}{2}(2)^6(4)^2=14$$

$$f_y(2,4,4)=-(2)^7(4)^3=-2$$

$$f_z(2,4,4)=-1$$

Equation of Tangent at (2, 4, 4) is given by

$$f_x(2,4,4)(x-2)+f_y(2,4,4)(y-4)+f_z(2,4,4)(z-4)=0$$

$$14(x-2)+(-2)(y-4)-(z-4)=0$$

$$14x-2y-z-16=0$$

The normal line has equations

$$x=2+f_x(2,4,4)t, \quad y=4+f_y(2,4,4)t, \quad z=4+f_z(2,4,4)t$$

$$x=2+14t, \qquad y=4-2t, \qquad z=4-t$$

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ORTHOGONAL SURFACES

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are orthogonal. They are Said to intersect orthogonally if they are orthogonal at every point common to them.

CONDITION FOR ORTHOGONAL SURFACES

Let (x, y, z) be any point of intersection of

$$f(x, y, z) = 0 \text{----} \quad (1)$$

and $g(x, y, z) = 0$ -----(2)

Direction ratios of a line normal to (1) are f_x, f_y, f_z

Similarly, direction rations of a line normal to (2) are g_x, g_y, g_z

The two normal lines are orthogonal if and only if

$$f_x g_x + f_y g_y + f_z g_z = 0$$

EXAMPLE

Show that given two surfaces are orthogonal or not

$f(x,y,z) = x^2 + y^2 + z - 16$ (1)

$g(x,y,z) = x^2 + y^2 - 638$ (2)

Adding (1) and (2)

$$x^2 + y^2 = \frac{63}{4}, \quad z = \frac{1}{4} \qquad (3)$$

$f_x = 2x, \quad f_y = 2y \quad f_z = 1$

$g_x = 2x, \quad g_y = 2y \quad g_z = -63$

$f_x g_x + f_y g_y + f_z g_z$

$= 4(x^2 + y^2) - 63$ using (3)

$f_x g_x + f_y g_y + f_z g_z$

$$= 4\left(\frac{63}{4}\right) - 63 = 0$$

Since they satisfied the condition of orthogonality so they are orthogonal.

Differentials of a functions

For a function $y = f(x)$

$dy = f'(x)\,dx$

is called the differential of functions $f(x)$

dx the differential of x is the same as the actual change in x

i.e. $dx = \nabla x$ where as dy the differential of y is the approximate change in the value of the functions which is different from the actual change ∇y in the value of the functions.

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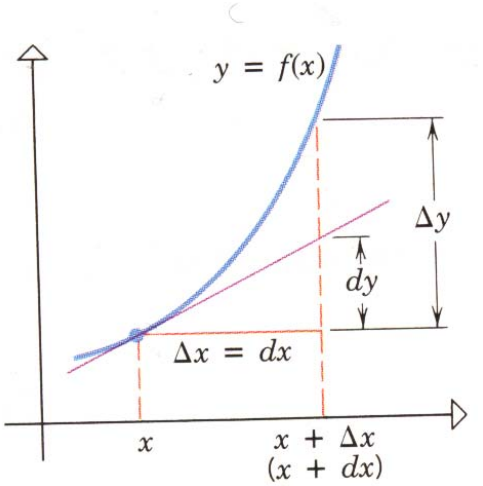
Distinction between the increment Δy and the differential dy



Approximation to the curve

If *f* is differentiable at x , then the tangent line to the curve y = f(x) at x_0 is a reasonably good approximation to the curve y = f(x) for value of x near x_0 Since the tangent line passes through the point $(x_0, f(x_0))$ and has slope $f'(x_0)$, the point-slope form of its equation is

$$y - f'(x_0) = f(x_0)(x - x_0) \quad \text{or}$$
$$y = f(x_0) + f'(x_0)(x - x_0)$$

EXAMPLE

$f(x) = \sqrt{x}$

$x = 4$ and $dx = \Delta x = 3$

$$\Delta y = \sqrt{x + \Delta x} - \sqrt{x}$$
$$= \sqrt{7} - \sqrt{4} \approx .65$$

If $y = \sqrt{x}$, then

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}} \quad \text{so} \quad dy = \frac{1}{2\sqrt{x}}\, dx$$
$$= \frac{1}{2\sqrt{x}}\,(3) = \frac{3}{4} = .75$$

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EXAMPLE

Using differentials approximation for the value of cos 61 °.

Let $y = \cos x$ and $x = 60°$

then $dx = 61° - 60° = 1°$

$$\Delta y \approx dy = -\sin x\, dx = -\sin 60° (1°)$$

$$= \frac{\sqrt{3}}{2}\left(\frac{1}{180}\pi\right)$$

Now $y = \cos x$

$$y+\Delta y = \cos(x+\Delta x) = \cos(x+dx)$$

$$= \cos(60°+1°) = \cos 61°$$

$$\cos 61° = y+\Delta y = \cos x + \Delta y$$

$$\approx \cos 60° - \frac{\sqrt{3}}{2}\left(\frac{1}{180}\pi\right)$$

$$\cos 61° \approx \frac{1}{2} - \frac{\sqrt{3}}{2}\left(\frac{1}{180}\pi\right)$$

$$= 0.5 - 0.01511 = 0.48489$$

$$\cos 61° \approx 0.48489$$

EXAMPLE

A box with a square base has its height twice is width. If the width of the box is 8.5 inches with a possible error of ± 0.3 inches

Let x and h be the width and the height of the box respectively, then its volume V is given by

$$V = x^2 h$$

Since $h = 2x$, so (1) take the form

$$V = 2x^3$$

$$dV = 6x^2\, dx$$

Since $x = 8.5$, $dx = \pm 0.3$, so putting these values in (2), we have

$$dV = 6(8.5)^2(\pm 0.3) = \pm 130.05$$

This shows that the possible error in the volume of the box is ±130.05.

TOTAL DIFFERENTIAL

If we move from (x_0, y_0) to a point $(x_0 + dx, y_0 + dy)$ nearby, the resulting differential in f is

$$df = f_x(x_0, y_0)\, dx + f_y(x_0, y_0)\, dy$$

This change in the linearization of

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f is called the total differential of *f*.

EXACT CHANGE

Area = xy

x = 10, y = 8 Area = 80
x = 10.03 y = 8.02 Area = 80.4406
Exact Change in area = 80.4406 – 80
= 0.4406

EXAMPLE

A rectangular plate expands in such a way that its length changes from 10 to 10.03 and its breadth changes from 8 to 8.02.

Let x and y the length and breadth of the rectangle respectively, then its area is

$$A = xy$$

$$dA = A_x\,dx + A_y dy = ydx + xdy$$

By the given conditions

$$x = 10,\ dx = 0.03,\ y = 8,\ dy = 0.02.$$

$$dA = 8(0.03) + 10(0.02) = 0.44$$

Which is exact Change

EXAMPLE

The volume of a rectangular parallelepiped is given by the formula V = xyz. If this solid is compressed from above so that z is decreased by 2% while x and y each is increased by 0.75% approximately

$$V = xyz$$

$$dV = V_x d\,x + V_y dy + V_z dx$$

$$dV = yzdx + xzdy + zydz \quad (1)$$

$$dx = \frac{0.75}{100}x,\ dy = \frac{0.75}{100}y,\ dz = -\frac{2}{100}z$$

Putting these values in (1), we have

$$dV = \frac{0.75}{100}xyz + \frac{0.75}{100}xyz - \frac{2}{100}xyz$$

$$= -\frac{0.5}{100}xyz = -\frac{0.5}{100}V$$

This shows that there is 0.5 % decrease in the volume.

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EXAMPLE

A formula for the area Δ of a triangle is $\Delta = \frac{1}{2}\, ab \sin C$. Approximately what error is made in computing Δ if a is taken to be 9.1 instead of 9, b is taken to be 4.08 instead of 4 and C is taken to be $30°3'$ instead of $30°$.

By the given conditions

$$a = 9,\ b = 4,\ C = 30°,$$

$$da = 9.1 - 9 = 0.1,$$

$$db = 4.08 - 4 = 0.08$$

$$dC = 30°3' - 3' = \left(\frac{3}{60}\right)°$$

$$= \frac{3}{60} \times \frac{\pi}{180} \text{ radians}$$

Putting these values in (1), we have

$$\Delta = \frac{1}{2}\, ab \sin C$$

$$d\Delta = \frac{\partial}{\partial a}\left(\frac{1}{2} ab \sin C\right) da + \frac{\partial}{\partial b}\left(\frac{1}{2} ab \sin C\right) db + \frac{\partial}{\partial C}\left(\frac{1}{2} ab \sin C\right) dC$$

$$d\Delta = \frac{1}{2}\, b \sin C da + \frac{1}{2}\, a \sin C\, db + \frac{1}{2}\, ab \cos C dC$$

$$d\Delta = \frac{1}{2} 4 \sin 30°(0.1) + \frac{1}{2} 9 \sin 30°(0.08) + \frac{1}{2}\, 36 \cos 30° \left(\frac{\pi}{3600}\right)$$

$$d\Delta = 2\left(\frac{1}{2}\right)(0.1) + \frac{1}{2}\left(\frac{1}{2}\right)(0.08) + 18\left(\frac{\sqrt{3}}{2}\right)\left(\frac{3.14}{3600}\right) = 0.293$$

$$\% \text{ change in area} = \frac{0.293}{\Delta} \times 100$$

$$= \frac{0.293}{9} \times 10 = 3.25\%$$

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Lecture No -14 Extrema of Functions of Two Variables

In this lecture we shall fined the techniques for finding the highest and lowest points on the graph of a function or , equivalently, the largest and smallest values of the function.

The graph of many functions form hills and valleys. The tops of the hills are relative maxima and the bottom of the valleys are called relative minima. Just as the top of a hill on the earth's terrain need not be the highest point on the earth , so a relative maximum need not be the highest point on the entire graph .

Absolute maximum

A function f of two variables on a subset of $\mathbf{R}^2$ is said to have an D **absolute (global) maximum** value on D if there is some point (x_0, y_0) of D such that value of f on D

$$f(x_0, y_0) \geq f(x, y) \text{ for all } (x, y) \in D$$

In such a case $f(x_0, y_0)$ is the **absolute maximum**

Relative extremum and absolute extremum

If f has a relative maximum or a relative minimum at (x_0, y_0), then we say that f has a relative extremum at (x_0, y_0), and if f has an absolute maximum or absolute minimum at $(x_0 y_0)$, then we say that f has an absolute extremum at (x_0, y_0).

Absolute minimum

A function f of two variables on a subset D of $\mathbf{R}^2$ is said to have an **absolute (global) minimum** value on D if there is some point (x_0, y_0) of D such that

$$f(x_0, y_0) \leq f(x, y) \text{ for all } (x, y) \in D.$$

In such a case $f(x_0, y_0)$ is the **absolute minimum** value of f on D.

Relative (local) maximum

The function f is said to have a relative (local) maximum at some point (x0,y0) of its domain D if there exists an open disc *K centered* at (x0,y0) and of radius r

$$K=\{(x, y) \in \mathbf{R}^2 : (x - x_0)^2 + (y - y_0)^2 < r^2\}$$

With $K \in D$ such that

$$f(x_0, y_0) \geq f(x, y) \text{ for all } (x, y)$$

Relative (local) minimum

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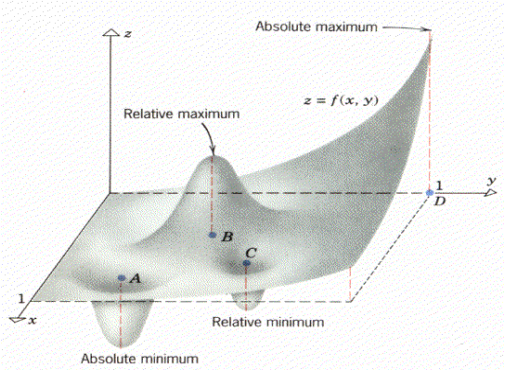
The function f is said to have a **relative(local) minimum** at some point (x_0, y_0) of **D** if there exists an open disc K centred at (x_0, y_0) and of radius r with $K \subset D$ such that

$$f(x_0, y_0) \leq f(x, y) \text{ for all } (x, y) \in K.$$



Extreme Value Theorem

If f (x, y) is continuous on a closed and bounded set R, then f has both an absolute maximum and on absolute minimum on R .

Remarks

If any of the conditions the Extreme Value Theorem fail to hold, then there is no guarantee that an absolute maximum or absolute minimum exists on the region R. Thus, a discontinuous function on a closed and bounded set need not have any absolute extrema, and a continuous function on a set that is not closed and bounded also need not have any absolute extrema.

Extreme values or extrema of *f*

The maximum and minimum values of *f are* referred to as extreme values of extrema of f .Let a function f of two variables be defined on an open disc

$$K = \{(x, y): (x - x_0)^2 + (y - y_0)^2 < r^2\}.$$

Suppose $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ both exist on K

If f has relative extrema at (x_0, y_0),then

$$f_x(x_0, y_0) = 0 = f_y(x_0, y_0).$$

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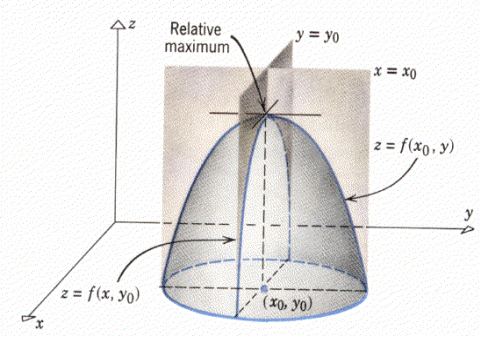

Saddle Point

A differentiable function f(x, y) has a saddle has a saddle point (a, b) if in every open disk centered at (a, b) there are domain points (x, y) where f (x, y) > f (a, b) and domain points (x, y) where f (x, y) < f (a, b). The corresponding point (a, b, f (a, b)) on the surface z = f (x, y) is called a saddle point of the surface

Remarks

Thus, the only points where a function f(x,y) can assume extreme values are critical points and boundary points. As with differentiable functions if a single variable, not every critical point gives rise to o a local extremum. A differentiable function of a single variable might have a point of inflection. A differentiable function of two variable might have a saddle point.

EXAMPLE

Fine the critical points of the given function

$$f(x, y) = x^3 + y^3 - 3axy, a > 0.$$

f_x, f_y exist at all points of the domain of f.

$f_x = 3x_2 - 3ay$, $f_y = 3y^2 - 3ax$

For critical points $f_x = f_y = 0$.

Therefore, $x^2 - ay = 0$ (1)

and $ax - y^2 = 0$ (2)

Substituting the value of x from (2) into (1), we have

$$\frac{y^4}{a^2} - ay = 0$$

$$y(y^3 - a^3) = 0$$

$y = 0$, $y = a$

and so

$x = 0$, $x = a$.

The critical points are (0, 0) and (a, a).

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Overview of lecture # 14

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Extreme Value Theorem	16.9.3	834
Exercise set	Q#1,3,5,7,9,11,13,15,17	841

Book

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Lecture No - 15 Example

EXAMPLE

$$f(x, y) = \sqrt{x^2 + y^2}$$

$$f_x(x, y) = \frac{x}{\sqrt{x^2 + y^2}}$$

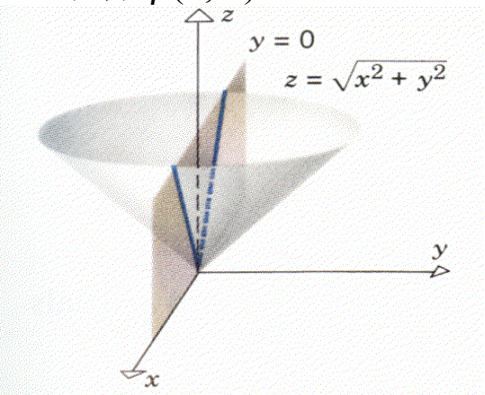
$$f_y(x, y) = \frac{y}{\sqrt{x^2 + y^2}}$$

The partial derivatives exist at all points of the domain of f except at the origin which is in the domain of f. Thus (0, 0) is a critical point of f

Now $f_x(x, y) = 0$ only if $x = 0$ and
$f_y(x, y) = 0$ only if $y = 0$

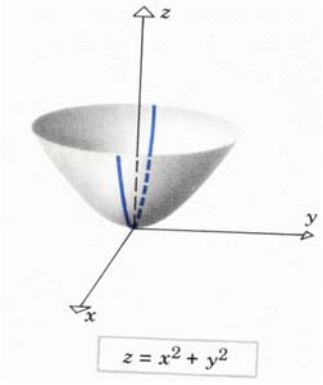
The only critical point is (0,0) and f(0,0)=0

Since $f(x, y) \geq 0$ for all (x, y), $f(0, 0) = 0$ is the absolute minimum value of f.



Example

$z = f(x, y) = x^2 + y^2$ (Paraboloid)
$f_x(x, y) = 2x$, $f_y(x, y) = 2y$
when $f_x(x, y) = 0$, $f_y(x, y) = 0$
we have (0, 0) as critical point.



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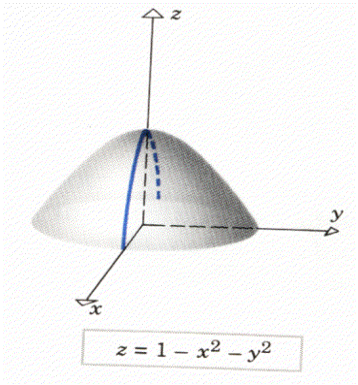
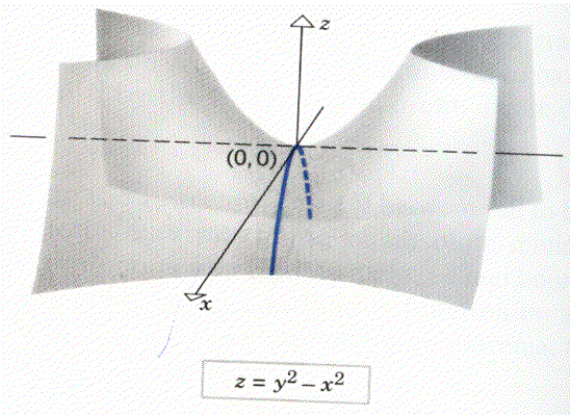
EXAMPLE

$z = g(x, y) = 1 - x^2 - y^2$ (Paraboloid)
$g_x(x, y) = -2x, \quad g_y(x, y) = -2y$
when $g_x(x, y) = 0, \quad g_y(x, y) = 0$
we have $(0, 0)$ as critical point.



EXAMPLE

$z = h(x,y) = y^2 - x^2$ (Hyperbolic paraboloid)
$h_x(x, y) = -2x, \quad h_y(x, y) = 2y$
when $h_x(x, y) = 0, \quad h_y(x, y) = 0$
we have $(0, 0)$ as critical point.



EXAMPLE

$$\mathbf{f(x, y) = \sqrt{x^2 + y^2}}$$

$$f_x = \frac{x}{\sqrt{x^2 + y^2}} \quad f_y = \frac{y}{\sqrt{x^2 + y^2}}$$

The point (0,0) is critical point of f because the partial derivatives do not both exist. It is evident geometrically that $f_x(0,.0)$ does not exist because the trace of the cone in the plane y=0 has a corner at the origin.

The fact that $f_x(0,0)$ does not exist can also be seen algebraically by noting that $f_x(0,0)$ can be interpreted as the derivative with respect to x of the function

$$f(x, 0) = \sqrt{x^2 + 0} = |x| \quad \text{at } x = 0.$$

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But $|x|$ is not differentiable at x = 0, so $f_x(0,0)$ does not exist. Similarly, $f_y(0,0)$ does not exist. The function f has a relative minimum at the critical point (0,0).

The Second Partial Derivative Test

Let *f* be a function of two variables with continuous second order partial derivatives in some circle centered at a critical point (x_0, y_0), and let

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f^2_{xy}(x_0, y_0)$$

(a) If **D > 0** and $f_{xx}(x_0,y_0) > 0$, then f has a **relative minimum** at (x_0,y_0).

(b) If **D > 0** and $f_{xx}(x_0,y_0) < 0$, then f has a **relative maximum** at (x_0,y_0).

(c) If **D < 0**, then f has a **saddle point** at (x_0,y_0).

(d) If **D = 0**, then no conclusion can be drawn.

REMARKS

If a function *f* of two variables has an absolute extremum (either an absolute maximum or an absolute minimum) at an interior point of its domain, then this extremum occurs at a critical point.

EXAMPLE

$$f(x,y) = 2x^2 - 4x + xy^2 - 1$$

$f_x(x, y) = 4x - 4 + y^2$, $f_{xx}(x, y) = 4$

$f_y(x, y) = 2xy$, $f_{yy}(x, y) = 2x$

$f_{xy}(x, y) = f_{yx}(x, y) = 2y$

For critical points, we set the first partial derivatives equal to zero. Then

$4x - 4 + y^2 = 0$ (1)

and $2xy = 0$ (2)

we have $x = 0$ or $y = 0$

$x = 0$, then from (1), $y = \pm 2$.

$y = 0$, then from (1), $x = 1$.

Thus the critical points are **(1,0), (0, 2), (0, − 2).**

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We check the nature of each point.

$$f_{xx}(1,0) = 4,$$
$$f_{yy}(1, 0) = 2,$$
$$f_{xy}(1, 0) = 0$$

$$\mathbf{D= f_{xx}(1, 0).f_{yy}(1, 0) - [f_{xy}(1, 0)]^2}$$
$$= 8 > 0$$

and $f_{xx}(1, 0)$ is positive. Thus f has a relative minimum at (1, 0).

$$f_{xx}(0,-2) = 4,$$
$$f_{yy}(0,-2) = 0,$$
$$f_{xy}(0,-2) = -4$$

$$\mathbf{D= f_{xx}(0, -2).f_{yy}(0, -2) - [f_{xy}(0, -2)]^2}$$
$$= -16 < 0.$$

$$f_{xx}(0, 2) = 4,$$
$$f_{yy}(0, 2) = 0,$$
$$f_{xy}(0, 2) = 4$$

$$\mathbf{D= f_{xx}(0, 2).f_{yy}(0, 2) - [f_{xy}(0, 2)]^2}$$
$$= -16 < 0.$$

Therefore, f has a saddle point at (0,2).

Therefore, f has a saddle point at (0,– 2).

EXAMPLE

$$f(x,y) = e^{-(x^2+y^2+2x)}$$

$$\mathbf{f_x(x, y)=-2 (x+1)e^{-(x^2+y^2+2x)}},$$
$$\mathbf{f_y(x, y) = -2ye^{-(x^2+y^2+2x)}}$$

For critical points

$f_x(x,y) = 0, \quad x + 1 = 0, \quad x = -1$ and

$f_y(x, y) = 0, \quad y = 0$

Hence critical point is (-1,0).

$$f_{xx}(x,y) = [(-2x - 2)^2 - 2]e^{-(x^2+y^2+2x)}$$
$$f_{xx}(-1, 0) = -$$
$$f_{yy}(x,y) = [4y^2 - 2]e^{-(x^2+y^2+2x)}$$
$$f_{yy}(-1, 0) = -$$
$$f_{xy}(x,y) = -2y(-2x - 2)e^{-(x^2+y^2+2x)}$$
$$f_{xy}(-1, 0) = 0$$
$$\mathbf{D = f_{xx}(-1,0)\, f_{yy}(-1, 0) - f^2_{xy}(-1, 0)}$$
$$= (-2e)(-2e) > 0$$

This shows that f is maximum at (-1, 0).

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EXAMPLE

$$f(x,y) = 2x^4 + y^2 - x^2 - 2y$$

$f_x(x, y) = 8x^3 - 2x,$ $f_y(x, y) = 2y - 2$

$f_{xx}(x, y) = 24x^2 - 2,$ $f_{yy}(x,y) = 2,$

$f_{xy}(x, y) = 0$

For critical points

$f_x(x, y) = 0,$

$2x(4x^2 - 1) = 0,$ $x = 0, 1/2, -1/2$

$f_y(x, y) = 0,$

$2y - 2 = 0,$ $y = 1$

Solving above equation we have the critical points $(0,1), \left(-\frac{1}{2}, 1\right) \left(\frac{1}{2}, 1\right)$.

$f_{xx}(0,1) = -2,$ $f_{yy}(0, 1) = 2,$

$f_{xy}(0, 1) = 0$

$D = f_x(0, 1)\, f_{yy}(0, 1) - f^2_{xy}(0, 1)$

$= (-2)(2) - 0 = -4 < 0$

This shows that (0, 1) is a saddle point.

$f_{xx}\left(\frac{1}{2}, 1\right) = 4,$ $f_{yy} = \left(\frac{1}{2}, 1\right) = 2$

$f_{xy}\left(\frac{1}{2}, 1\right) = 0$

$D = f_{xx}\left(\frac{1}{2},1\right) f_{yy}\left(\frac{1}{2},1\right) - f^2_{xy}\left(-\frac{1}{2},1\right)$

$= (4)(2) - 0 = 8 > 0$

$f_{xx}\left(\frac{1}{2},1\right) = 4 > 0$, so f is minimum at $\left(\frac{1}{2},1\right)$.

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Example

Locate all relative extrema and saddle points of

$$f(x, y) = 4xy - x^4 - y^4.$$

$f_x(x, y) = 4y - 4x^3$, $f_y(x, y) = 4x - 4y^3$

For critical points

$f_x(x, y) = 0$

$4y - 4x^3 = 0$ (1)

$y = x^3$

$f_y(x, y) = 0$

$4x - 4y^3 = 0$ (2)

$x = y^3$

Solving (1) and (2), we have the critical points (0,0), (1, 1),(−1, −1).

Now $f_{xx}(x, y) = -12x^2$, $f_{xx}(0, 0) = 0$

$f_{yy}(x, y) = -12y^2$, $f_{yy}(0,0) = 0$

$f_{xy}(x, y) = 4$, $f_{xy}(0, 0) = 4$

$D = f_{xx}(0,0) f_y(0,0) - f^2_{xy}(0,0)$

$= (0)(0) - (4)^2 = -16 < 0$

This shows that (0,0) is the saddle point.

$f_{xx}(x, y) = -12x^2$, $f_{xx}(1,1) = -12 < 0$

$f_{yy}(x,y) = -12y^2$, $f_y(1,1) = -12$

$f_{xy}(x, y) = 4$, $f_{xy}(1,1) = 4$

$D = f_{xx}(1,1) f_{yy}(1,1) - f^2_{xy}(1, 1)$

$= (-12)(-12) - (4)^2 = 128 > 0$

This shows that f has relative maximum at (1,1).

$f_{xx}(x,y) = -12x^2$, $f_{xx}(-1, -1) = -12 < 0$

$f_y(x, y) = -2y^2$, $f_y(-1, -1) = -12$

$f_{xy}(x, y) = 4$, $f_{xy}(-1, -1) = 4$

$D = f_{xx}(-1,-1) f_{yy}(-1,-1) - f^2_{xy}(-1,-1)$

$= (-12)(-12) - (4)^2 = 128 > 0$

This shows that f has relative maximum (−1, − 1).

Over view of lecture # 15 Book

Calculus by HOWARD ANTON

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Lecture No -16 Extreme Valued Theorem

EXTREME VALUED THEOREM

If the function f is continuous on the closed interval [a, b], then f has an absolute maximum value and an absolute minimum value on [a, b]

Remarks

An absolute extremum of a function on a closed interval must be either a relative extremum or a function value at an end point of the interval. Since a necessary condition for a function to have a relative extremum at a point C is that C be a critical point, we may determine the absolute maximum value and the absolute minimum value of a continuous function f on a closed interval [a, b] by the following procedure.

1. Find the critical points of f on [a, b] and the function values at these critical.
2. Find the values of f (a) and f (b).
3. The largest and the smallest of the above calculated values are the absolute maximum value and the absolute minimum value respectively

Example

Find the absolute extrema of

$$f(x)= x^3+ x^2-x+1 \quad \text{on} \quad [-2,1/2]$$

Since f is continuous on **[-2,1/2],** the extreme value theorem is applicable. For this

$$f'(x) =3x^2+2x-1$$

This shows that f(x) exists for all real numbers, and so the only critical numbers of **f** will be the values of x for which f (x)=0.

Setting $f'(x) = 0$, we have

$(3x - 1)(x + 1) = 0$

from which we obtain

$x = -1$ and $x = \frac{1}{3}$

The critical points of f are –1 and $\frac{1}{3}$, and each of these points is in the given closed interval $(-2, \frac{1}{2})$ We find the function values at the critical points and at the end points of the interval, which are given below.

$f(-2) = -1,\ f(-1) = 2,$

$f\left(\frac{1}{3}\right) = \frac{22}{27},\ f\left(\frac{1}{2}\right) = \frac{7}{8}$

The absolute maximum value of f on $(-2, \frac{1}{2})$ is therefore 2, which occurs at -1, and the absolute min. value of f on $(-2, \frac{1}{2})$ is -1, which occurs at the left end point -2.

Find the absolute extrema of

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$f(x) = (x \quad 2)^{2/3}$ on [1, 5].

Since f is continuous on [1. 5], the extreme-value ttheorem is applicable.

Differentiating f with respect to x, we get

$$f'(x) = \frac{2}{3(x-2)^{1/3}}$$

There is no value of x for which $f'(x) = 0$. However, since $f'(x)$ does not exist at 2, we conclude that 2 is a critical point of f,

so that the absolute extrema occur either at 2 or at one of the end points of the interval. The function values at these points are given below.

$f(1) = 1$, $f(2) = 0$, $f(5) = \sqrt[3]{9}$

From these values we conclude that the absolute minimum value of f on [1,5] is 0, occurring at 2, and the absolute maximum value of f on [1, 5] is $\sqrt[3]{9}$,occurring at 5.

Find the absolute extrema of

$h(x) = x^{2/3}$ on $[-2, 3]$.

$$h'(x) = \frac{2}{3}x^{-1/3} = \frac{2}{3x^{1/3}}$$

$h'(x)$ has no zeros but is undefined at x = 0.

The values of h at this one critical point

and at the endpoints x = − 2 and x = 3 are

$h(0) = 0$

$h(-2) = (-2)^{2/3} = 4^{1/3}$

$h(3) = (3)^{2/3} = 9^{1/3}$.

The absolute maximum value is $9^{1/3}$ assumed at x = 3; the absolute minimum is 0, assumed at x = 0.

How to Find the Absolute Extrema of a Continuous Function f of Two Variables on a Closed and Bounded Region R.

Step 1.

Find the critical points of f that lie in the interior of R.

Setp 2.

Find all boundary points at which the absolute extrema can occur,

Step 3.

Evaluate f(x,y) at the points obtained in the previous steps. The largest of these values is the absolute maximum and the smallest the absolute minimum.

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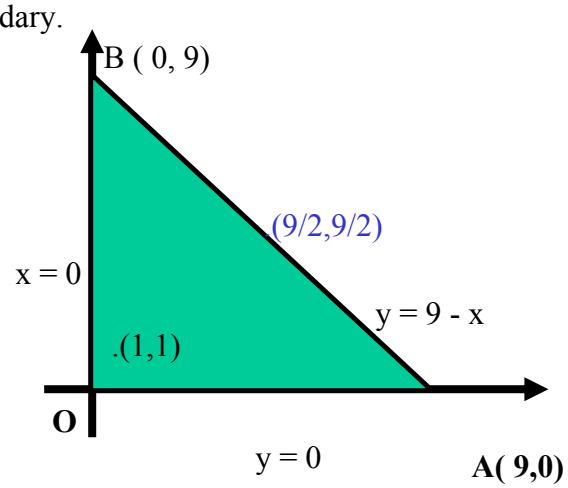
Find the absolute maximum and minimum value of

$$f(x,y) = 2 + 2x + 2y - x^2 - y^2$$

On the triangular plate in the first quadrant bounded by the lines x=0,y=0,y=9-x
Since f is a differentiable, the only places where f can assume these values are points inside the triangle having vertices at O(0,0), A(9,0)and B(0,9) where $f_x = f_y = 0$ and points of boundary.



For interior points:

We have $f_x = 2-2x = 0$ and $f_y = 2-2y = 0$
yielding the single point (1,1)

For boundary points we take take the triangle one side at time :

1. On the segment OA, y=0
   $U(x) = f(x, 0) = 2+2x-x^2$

may be regarded as function of x defined on the closed interval $0 \le x \le 9$ Its extreme values may occur at the endpoints x=0 and x=9 which corresponds to points (0, 0) and (9, 0) and U(x) has critical point where

$U'(x) = 2-2x = 0$ Then x=1

On the segment OB, x=0 and

$V(y) = f(0,y) = 2+2y-y^2$

Using symmetry of function f, possible points are (0,0),(0,9) and (0,1)

3. The interior points of AB.

With y = 9 - x, we have

$f(x, y) = 2+2x+2(9-x)-x^2-(9- x)^2$

$W(x) = f(x, 9-x) = - 61 +18x - 2x^2$

Setting $w'(x) = 18 - 4x = 0$, x = 9/2.

At this value of x, y = 9 – 9/2

Therefore we have $(\frac{9}{2}, \frac{9}{2})$ as a critical point.

(x, y)	(0,0)	(9,0)	(1, 0)	$\left(\frac{9}{2}, \frac{9}{2}\right)$
f(x,y)	2	–61	3	$-\frac{41}{2}$

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(x, y)	(0, 9)	(0,1)	(1,1)
f(x,y)	– 61	3	4

The absolute maximum is 4 which f assumes at the point (1,1) The absolute minimum is -61 which f assumes at the points (0, 9) and (9,0)

EXAMPLE

Find the absolute maximum and the absolute minimum values of

$f(x,y)=3xy-6x-3y+7$

on the closed triangular region **r** with the vertices (0,0), (3,0) and (0,5) .

$f(x, y) = 3xy - 6x - 3y + 7$

$f_x(x, y) = 3y - 6, \quad f_y(x, y) = 3x - 3$

For critical points

$f_x(x, y) = 0$

$3y - 6 = 0$

$y = 2$

$f_y(x, y) = 0$

$3x - 3 = 0$

$x = 1$

Thus, (1, 2) is the only critical point in the interior of R. Next, we want to determine the location of the points on the boundary of R at which the absolute extrema might occur. The boundary extrema might occur. The boundary each of which we shall treat separately.

i)The line segment between (0, 0) and (3, 0):

On this line segment we have y=0 so (1) simplifies to a function of the single variable x,

$$u(x)=f(x, 0) = -6x + 7, \ 0 \leq x \leq 3$$

This function has no critical points because $u'(6)=-6$ is non zero for all x . Thus, the extreme values of u(x) occur at the endpoints x = 0 and x=3 , which corresponds to the points (0, 0) and (3,0) on R

ii) The line segment between the (0,0) and (0,5)

On this line segment we have x=0 ,so single variable y,

$$v(y) = f(0, y) = -3y + 7, \ 0 \leq y \leq 5$$

This function has no critical points because $v'(y)=-3$ is non zero for all y.Thus ,the extreme values of v(y) occur at the endpoints y = 0 and y=5 which correspond to the point (0,0) and (0,5) or R

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iii) The line segment between (3,0) and (0,5)

In the XY- plan , an equation for the line segment

$$y = -\frac{5}{3}x + 5,\ 0 \le x \le 3$$

so (1) simplifies to a function of the single variable x,

$$w(x) = f\left(x, -\frac{5}{3}x + 5\right)$$

$$= -5x^2 + 14x - 8,\quad 0 \le x \le 3$$

$$w'(x) = -10x + 14$$

$$w'(x) = 0$$

$$10x + 14 = 0$$

$$x = \frac{7}{5}$$

This shows that x =7/5 is the only critical point of w. Thus, the extreme values of w occur either at the critical point x=7/5 or at the endpoints x=0 and x=3.The endpoints correspond to the points (0, 5) and (3, 0) of R, and from (6) the critical point corresponds to [7/5,8/3]

(x, y)	(0, 0)	(3, 0)	(0, 5)	$\left(\frac{7}{5}, \frac{8}{3}\right)$	(1, 2)
f(x,y)	7	– 11	– 8	$-\frac{9}{5}$	1

Finally, table list the values of f(x,y) at the interior critical point and at the points on the boundary where an absolute extremum can occur. From the table we conclude that the absolute maximum value of f is f(0,0)=7 and the absolute minimum values is f(3,0)=-11.

OVER VIEW:

Maxima and Minima of functions of two variables. Page # 833

Exercise: 16.9 Q #26,27,28,29.

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Lecture No -17 Examples

EXAMPLE

Find the absolute maximum and minimum values of f(x,y)=xy-x-3y on the closed triangular region R with vertices (0, 0), (0, 4), and (5, 0).

$$f(x,y) = xy - x - 3y \qquad (1)$$

$f_x(x, y) = y - 1, \quad f_y(x, y) = x - 3$

For critical points

$f_x(x, y) = 0, \; y - 1 = 0$

$$y = 1 \qquad (2)$$

$f_y(x, y) = 0, \; 3x - 3 = 0$

$$x = 3 \qquad (3)$$

Thus, (3, 1) is the only critical point in the interior of R. Next, we want to determine the location of the points on the boundary of R at which the absolute extrema might occur. The boundary of R consists of three line segments, each of which we shall treat separately.

(i) The line segment between (0, 0) and (5,0)

On this line segment we have y = 0, so (1) simplifies to a function of the single variable x,

$$u(x) = f(x, 0) = -x, \; 0 \le x \le 5 \qquad (4)$$

The function has no critical points because the $u'(x)=-1$ is non zero for all x.Thus,the exteme values of u(x) occure at the endpoints x=0 and x=5 , which corresponds to the points (0,0) and (5,0) of R.

ii) The line segment between (0,0) and (0,4)

On this line segment we have x = 0, so (1) simplifies to a function of the single variable y,

$$v(y)=f(0, y) = -3y, \; 0 \le y \le 4. \qquad (5)$$

This function has no critical points because $v'(y)= -3$ is nonzero for all y. Thus, the extreme values of v(y) occur at the endpoints y =0 and y=4 ,which correspond to the point (0,0) and (0,4) or R.

iii) The line segment between (5,0) and (0,4)

In the xy-plan, an equation is

$$y = -\frac{4}{5}x + 4, 0 \le x \le 5 \qquad (6)$$

so (1) simplifies to a function of the single variable x,

$$w(x) = f\left(x, -\frac{4}{5}x + 4\right)$$

$$= x\left(-\frac{4}{5}x + 4\right) - x - 3\left(-\frac{4}{5}x + 4\right)$$

$$= -\frac{4}{5}x^2 + \frac{27}{5}x - 12$$

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$$w'(x) = -\frac{8}{5}x + \frac{27}{5}$$

$$by\ w'(x) = 0,\ we\ get\ x = \frac{27}{8}$$

This shows that $x = \frac{27}{8}$ is the only critical point of w. Thus, the extreme values of w occur either at the critical point $x = \frac{27}{8}$ or at the endpoints x = 0 and x = 5. The endpoints correspond to the points (0, 4) and (5, 0) of R, and from (6) the critical point corresponds to $\left[\frac{27}{8}, \frac{13}{10}\right]$

(x, y)	(0, 0)	(5, 0)	(0, 4)	(27/8, 13/10)	(3, 1)
$f(x, y)$	0	-5	-12	-231/80	-3

Finally, from the table below, we conclude that the absolute maximum value of f is f (0,0) = 0 and the absolute minimum value is $f(0, 4) = -12$

Example

Find three positive numbers whose sum is 48 and such that their product is as large as possible

Let x,y and z be the required numbers, then we have to maximize the product

f(x,y)=xy(48-x-y)

Since

$f_x = 48y - 2xy - y^2$, $f_y = 48x - 2xy - x^2$

solving

$f_x = 0$, $f_y = 0$

we get x=16, y=16, z=16

Since x+y+z=48

$f_{xx}(x,y) = -2y$, $f_{xx}(16, 16) = -32 < 0$

$f_{xy}(x, y) = 48-2x-2y$, $f_{xy}(16, 16) = -16$

$f_{yy}(x, y) = -2x$, $f_{yy}(16, 16) = -32$

$D = f_{xx}(16,16) f_{yy}(16,16) - f^2_{xy}(16,16)$

$= (-32)(-32) - (16)^2 = 768 > 0$

For x = 16, y = 16 we have z = 16 since x + y + z = 48

Thus, the required numbers are 16, 16, 16.

Example

Find three positive numbers whose sum is 27 and such that the sum of their squares is as small as possible

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Let x, y, z be the required numbers, then
we have to

$$f(x,y) = x^2 + y^2 + z^2$$
$$= x^2 + y^2 + (27 - x - y)^2$$

Since x+y+z = 27

$f_x = 4x+2y-54$, $f_y = 2x+4y-54$,
$f_{xx} = 4$, $f_{yy} = 4$, $f_{xy} = 2$

Solving $f_x = 0$, $f_y = 0$
We get x = 9, y = 9, z = 9
Since x + y + z = 27

$$D = f_{xx}(9,9)\, f_{yy}(9,9) - [f_{xy}(9,9)]^2$$
$$= (4)(4) - 2^2 = 12 > 0$$

This shows that f is minimum
x = 9, y = 9, z = 9, so the required numbers are 9, 9, 9.

Example

Find the dimensions of the rectangular box of maximum volume that can be inscribed in a sphere of radius 4.

Solution:

The volume of the parallelepiped with dimensions x, y, z is

$$V = xyz$$

Since the box is inscribed in the sphere of radius 4, so equation of sphere is $x^2 + y^2 + z^2 = 4^2$ from this equation we can write $z = \sqrt{16 - x^2 - y^2}$ and putting this value of "z" in above equation we get $V = xy\sqrt{16 - x^2 - y^2}$ .Now we want to find out the maximum value of this volume, for this we will calculate the extreme values of the function "V". For extreme values we will find out the critical points and for critical points we will solve the equations $V_x = 0$ and $V_y = 0$ .Now we have

$$V_x = y\sqrt{16 - x^2 - y^2} + \frac{xy(-2x)}{2\sqrt{16 - x^2 - y^2}}$$

$$\Rightarrow V_x = y\left\{\frac{-2x^2 - y^2 + 16}{\sqrt{16 - x^2 - y^2}}\right\} \quad Now \;\; V_x = 0 \Rightarrow y\left\{\frac{-2x^2 - y^2 + 16}{\sqrt{16 - x^2 - y^2}}\right\} = 0$$

$$\Rightarrow -2x^2 - y^2 + 16 = 0 \Rightarrow 2x^2 + y^2 = 16 \ldots\ldots\ldots\ldots\ldots\ldots (a)$$

Similarly we have

$$V_y = x\sqrt{16 - x^2 - y^2} + \frac{xy(-2y)}{2\sqrt{16 - x^2 - y^2}}$$

$$\Rightarrow V_y = x\left\{\frac{-x^2 - 2y^2 + 16}{\sqrt{16 - x^2 - y^2}}\right\} \quad Now \;\; V_y = 0 \Rightarrow x\left\{\frac{-x^2 - 2y^2 + 16}{\sqrt{16 - x^2 - y^2}}\right\} = 0$$

$$\Rightarrow -x^2 - 2y^2 + 16 = 0 \Rightarrow x^2 + 2y^2 = 16 \ldots\ldots\ldots\ldots\ldots\ldots (b)$$

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Solving equations (a) and (b) we get the $x = \frac{4}{\sqrt{3}}$ *and* $y = \frac{4}{\sqrt{3}}$

Now $V_{xx} = \frac{xy(2x^2 + 3y^2 - 48)}{(16 - x^2 - y^2)^{\frac{3}{2}}}$ (We obtain this by using quotient rule of differentiation)

$V_{xx}(\frac{4}{\sqrt{3}}, \frac{4}{\sqrt{3}}) = -\frac{16}{\sqrt{3}} \prec 0$

Also we have to calculate $V_{yy} = \frac{xy(3x^2 + 2y^2 - 48)}{(16 - x^2 - y^2)^{\frac{3}{2}}}$ and $V_{yy}(\frac{4}{\sqrt{3}}, \frac{4}{\sqrt{3}}) = -\frac{16}{\sqrt{3}} \prec 0$ Also note that $V_{xy}(\frac{4}{\sqrt{3}}, \frac{4}{\sqrt{3}}) = -\frac{8}{\sqrt{3}}$ Now as we have the formula for the second order partial derivative is $f_{xx}.f_{yy} - (f_{xy})^2$ and putting the values which we calculated above we note that $f_{xx}(\frac{4}{\sqrt{3}}, \frac{4}{\sqrt{3}}).f_{yy}(\frac{4}{\sqrt{3}}, \frac{4}{\sqrt{3}}) - (f_{xy}(\frac{4}{\sqrt{3}}, \frac{4}{\sqrt{3}}))^2 = +\frac{320}{3} \succ 0$ Which shows that the function V has maximum value when $x = \frac{4}{\sqrt{3}}$ *and* $y = \frac{4}{\sqrt{3}}$. So the dimension of the rectangular box are $x = \frac{4}{\sqrt{3}}, y = \frac{4}{\sqrt{3}}$ *and* $z = \frac{4}{\sqrt{3}}$.

Example

A closed rectangular box with volume of 16 ft^3 is made from two kinds of materials. The top and bottom are made of material costing Rs. 10 per square foot and the sides from material costing Rs.5 per square foot. Find the dimensions of the box so that the cost of materials is minimized

Let x, y, z, and C be the length, width, height, and cost of the box respectively. Then it is clear form that

C=10(xy+xy)+5(xz+xz)+5(yz+yz)----------(1)

C=20xy+10(x+y)z

The volume of the box is given by

xyz=16----------(2)

Putting the value of z from (2) in (1), we have

$C = 20xy + 10\ (x + y)\frac{16}{xy}$

$C = 20\ xy + \frac{160}{y} + \frac{160}{x}$

$C_x = 20y - \frac{160}{x^2}$, $C_y = 20x - \frac{160}{y^2}$

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For critical points

$C_x = 0$

$20y - \frac{160}{x^2} = 0$ and $C_y = 0$

$20x - \frac{160}{y^2} = 0$

Solving these equations, we have x = 2, y = 2. Thus the critical point is (2, 2).

$C_{xx}(x, y) = \frac{320}{x^3}$

$C_{xx}(2, 2) = \frac{320}{8} = 40 > 0$

$C_{yy}(x, y) = \frac{320}{y^3}$

$C_{yy}(2, 2) = \frac{320}{8} = 40$

$C_{xy}(x, y) = 20$

$C_{xy}(2, 2) = 20$

$C_{xx}(2,2)\, C_{yy}(2,2) - C^2_{xy}(2,2) = (40)(40)-(20)^2 = 1200 > 0$

This shows that S has relative minimum at x = 2 and y = 2. Putting these values in (2), we have z = 4, so when its dimensions are $2 \times 2 \times 4$.

Example

Find the dimensions of the rectangular box of maximum volume that can be inscribed in a sphere of radius a.

Solution:

The volume of the parallelepiped with dimensions x, y, z is

$V = xyz$

Since the box is inscribed in the sphere of radius 4, so equation of sphere is $x^2 + y^2 + z^2 = 4^2$ from this equation we can write $z = \sqrt{a^2 - x^2 - y^2}$ and putting this value of "z" in above equation we get $V = xy\sqrt{a^2 - x^2 - y^2}$ .Now we want to find out the maximum value of this volume, for this we will calculate the extreme values of the function "V". For extreme values we will find out the critical points and for critical points we will solve the equations $V_x = 0$ and $V_y = 0$ .Now we have

$$V_x = y\sqrt{a^2 - x^2 - y^2} + \frac{xy(-2x)}{2\sqrt{a^2 - x^2 - y^2}}$$

$$\Rightarrow V_x = y\left\{\frac{-2x^2 - y^2 + a^2}{\sqrt{a^2 - x^2 - y^2}}\right\} \; Now \; V_x = 0 \Rightarrow y\left\{\frac{-2x^2 - y^2 + a^2}{\sqrt{a^2 - x^2 - y^2}}\right\} = 0$$

$$\Rightarrow -2x^2 - y^2 + 16 = 0 \Rightarrow 2x^2 + y^2 = a^2 \ldots\ldots\ldots\ldots\ldots\ldots (a)$$

Similarly we have

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$$V_y = x\sqrt{a^2 - x^2 - y^2} + \frac{xy(-2y)}{2\sqrt{a^2 - x^2 - y^2}}$$

$$\Rightarrow V_y = x\left\{\frac{-x^2 - 2y^2 + a^2}{\sqrt{a^2 - x^2 - y^2}}\right\} \text{ Now } V_y = 0 \Rightarrow x\left\{\frac{-x^2 - 2y^2 + a^2}{\sqrt{a^2 - x^2 - y^2}}\right\} = 0$$

$$\Rightarrow -x^2 - 2y^2 + a^2 = 0 \Rightarrow x^2 + 2y^2 = a^2 \ldots\ldots\ldots\ldots\ldots\ldots(b)$$

Solving equations (a) and (b) we get the $x = \frac{a}{\sqrt{3}}$ *and* $y = \frac{a}{\sqrt{3}}$

Now $V_{xx} = \frac{xy(2x^2 + 3y^2 - 3a^2)}{(a^2 - x^2 - y^2)^{\frac{3}{2}}}$ (We obtain this by using quotient rule of differentiation)

$$V_{xx}(\frac{a}{\sqrt{3}}, \frac{a}{\sqrt{3}}) = -\frac{a^2}{\sqrt{3}} \prec 0$$

Also we have to calculate $V_{yy} = \frac{xy(3x^2 + 2y^2 - 3a^2)}{(a^2 - x^2 - y^2)^{\frac{3}{2}}}$ and $V_{yy}(\frac{a}{\sqrt{3}}, \frac{a}{\sqrt{3}}) = -\frac{a^2}{\sqrt{3}}$ Also note that $V_{xy}(\frac{a}{\sqrt{3}}, \frac{a}{\sqrt{3}}) = -\frac{2a}{\sqrt{3}}$ Now as we have the formula for the second order partial derivative is $f_{xx}.f_{yy} - (f_{xy})^2$ and putting the values which we calculated above we note that $f_{xx}(\frac{a}{\sqrt{3}}, \frac{a}{\sqrt{3}}).f_{yy}(\frac{a}{\sqrt{3}}, \frac{a}{\sqrt{3}}) - (f_{xy}(\frac{a}{\sqrt{3}}, \frac{a}{\sqrt{3}}))^2 = +\frac{20a^2}{3} \succ 0$ Which shows that the function V has maximum value when $x = \frac{a}{\sqrt{3}}$ *and* $y = \frac{a}{\sqrt{3}}$. So the dimension of the rectangular box are $x = \frac{a}{\sqrt{3}}, y = \frac{a}{\sqrt{3}}$ *and* $z = \frac{a}{\sqrt{3}}$.

Example:

Find the points o the plane x + y + z = 5 in the first octant at which $f(x,y,z) = xy^2z^2$ has maximum value.

Solution:

Since we have $f(x,y,z) = xy^2z^2$ and we are given the plane x + y + z = 5 from this equation we can write x = 5 – y – z . Thus our function "f' becomes $f((5 - y - z),y,z) = (5 - y - z)y^2z^2$ Say this function u(y,z) That is $u(y,z) = (5 - y - z)y^2z^2$ Now we have to find out extrema of this function. On simplification we get

$u(y,z) = 5\,y^2z^2 - y^3z^2 - y^2z^3$

$u_y = 10yz^2 - 3y^2z^2 - 2yz^3$

$= yz^2(10 - 3y - 2z)$

$u_z = 10y^2z - 2y^3z - 3y^2z^2$

$= y^2z\,(10 - 2y - 3z)$

$u_y = 0, \quad u_z = 0$

$y = 0, \quad z = 0$

$10 - 3y - 2z = 0$

$10 - 2y - 3z = 0$

On solving above equations we get $-10 + 5z = 0 \Rightarrow z = 2$ and $10 - 3y - 4 = 0 \Rightarrow y = 2$

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$u_{yy} = 10z^2 - 6yz^2 - 2z^3$
$u_{zz} = 10y^2 - 2y^3 - 6y^2z$
$u_{yz} = 20yz - 6y^2z - 6yz^2$
at
$y = 2, \quad z = 2$
$u_{yy}(2,2) = 40 - 48 - 16 = -24 < 0$
$y_{zz}(2,2) = 40 - 16 - 48 = -24$
$u_{yz}(2,2) = 80 - 48 - 48 = -16$
$D = u_{yy}\, u_{zz} - (uyz)^2$
$= (-24)(-24) - (-16)^2$
$= 576 - 256$
$= 320 > 0$
For $y = 2$ and $z = 2$
We have $x = 5 - 2 - 2 = 1$

Example:

Find all points of the plane x+y+z=5 in the first octant at which $f(x,y,z)=xy^2z^2$ has a maximum value.

$f(x,y,z) = xy^2z^2 = xy^2(5 - x - y)^2$
Since $x+y+z = 5$
$f_x = y^2(5-3x-y)(5-x-y)$,
$f_y = 2xy(5-x-2y)(5-x-y)$
Solving $f_x = 0$, $f_y = 0$, we get
$x=1, y=2, z=2 \ \therefore \ x+y+z = 5$

$f_{xx}=-y^2(5-3x-y) - 3y^2(5-x-5)$
$f_{xy}=2y(5-x-y)(5-3x-y)-y^2(5-3x-y)$
$\quad - y^2(5 - x - y)$
$f_{yy}=2x(5-x-y)(5-x-2y)-2xy(5-x-2y)$
$\quad - 4xy(5 - x - y)$
$f_{xx}(1, 2, 2) = -24 < 0$
$f_{yy}(1, 2, 2) = -16$
$f_{xy}(1, 2, 2) = -8$
$f_{xx}\, f_{yy} - (f_{xy})^2 = (-24)(-16)-(-8)^2$
$= 320 > 0$

Hence "f" has maximum value when x = 1 and y = 2. Thus the points where the function has maximum value is x = 1,y = 2 and z = 2.

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Lecture No -18 Revision of Integration

Example:

Consider the following integral $\int_0^1 (xy + y^2)dx$ Integrating we get

$$\int_0^1 (xy + y^2)dy = x\int_0^1 ydx + \int_0^1 y^2 dx$$

$$= x\left|\frac{y^2}{2}\right|_0^1 + \left|\frac{y^3}{3}\right|_0^1 = y(\frac{1}{2}) + y^2$$

$$\Rightarrow \int_0^1 (xy + y^2)dx = y(\frac{1}{2}) + y^2$$

Example:

Consider the following integral $\int_0^1 (xy + y^2)dy$ Integrating we get

$$\int_0^1 (xy + y^2)dy = x\int_0^1 ydx + \int_0^1 y^2 dx$$

$$= x\left|\frac{y^2}{2}\right|_0^1 + \left|\frac{y^3}{3}\right|_0^1 = x(\frac{1}{2}) + \frac{1}{3}$$

$$\Rightarrow \int_0^1 (xy + y^2)dx = \frac{x}{2} + \frac{1}{3}$$

Double Integral

Symbolically, the double integral of two variables "x" and "y" over the certain region R of the plane is denoted by $\iint_R f(x, y)dxdy$.

Example:

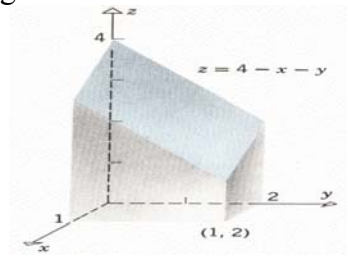
Use a double integral to find out the solid bounded above by the plane Z = 4 – x – y and below by the rectangle $R = \{(x, y) : 0 \le x \le 1, 0 \le y \le 2\}$

Solution:

We have to find the region "R"out the volume "V" over that is,

$$V = \iint_R (4 - x - y)dA$$

And the solid is shown in the figure below.



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$V = \iint_R (4 - x - y)dA = \int_0^2 \int_0^1 (4 - x - y)dxdy$

$\int_0^2 \int_0^1 (4 - x - y)dxdy = \int_0^2 \left| 4x - \frac{x^2}{2} - xy \right|_{x=0}^{x=1} dy$ After putting the upper and lower limits we get

$\int_0^2 \left(\frac{7}{2} - y \right) dy = \left| \frac{7}{2} y - \frac{y^2}{2} \right|_0^2$ again after putting the limits we get the required volume of the solid $V = \iint_R (4 - x - y)dA = 5$.

Example:

Evaluate the double integral $\int_0^1 \int_0^1 (xy + y^2)dxdy$

Solution:

First we will integrate the given function with respect to "x' and our integral becomes $\int_0^1 \int_0^1 (xy + y^2)dydx = \int_0^1 \left(x \left| \frac{y^2}{2} \right|_0^1 + \left| \frac{y^3}{3} \right|_0^1 \right) dy$

and after applying the limits we have,

$\int_0^1 \int_0^1 (xy + y^2)dydx = \int_0^1 \left(\frac{x}{2} + \frac{1}{3} \right) dy$ *integrating we get*

$\int_0^1 \int_0^1 (xy + y^2)dydx = \left| \frac{x^2}{4} + \frac{x}{3} \right|_0^1 = \frac{1}{4} + \frac{1}{3} = \frac{7}{12}$

Iterated or Repeated Integral

The expression $\int_c^d \left[\int_a^b f(x, y)dx \right] dy$ is called iterated or repeated integral. Often the brackets are omitted and this expression is written as

$\int_c^d \int_a^b f(x, y)dxdy = \int_c^d \left[\int_a^b f(x, y)dx \right] dy$ Where we have $\int_a^b f(x, y)dx$ yields a function of "y", which is then integrated over the interval $c \le y \le d$.

Similarly $\int_a^b \int_c^d f(x, y)dydx = \int_a^b \left[\int_c^d f(x, y)dy \right] dx$ Where we have $\int_c^d f(x, y)dy$ yields a function of "x"which is then integrated the interval $a \le x \le b$.

Example:

Evaluate the integral $\int_0^1 \int_0^2 (x + 3)dydx$.

Solution:

Here we will first integrate with respect to "y" and get a function of "x" then we will integrate that function with respect to "x" to get the required answer. So

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$\int_0^1\int_0^2 (x+3)dydx = \int_0^1 (x+3)\left|y\right|_0^2 dx$ and after putting the limits we get, $\int_0^1 (x+3)\left|y\right|_0^2 dx = \int_0^1 2(x+3)dx = 2\left|\frac{x^2}{2}+3x\right|_0^1$ and the required value of the double integral is $\int_0^1\int_0^2 (x+3)dydx = 2(\frac{1}{2}+3) = 7$.

Now if we change the order of integration so we get $\int_0^2\int_0^1 (x+3)dxdy$ Then we have $\int_0^2\int_0^1 (x+3)dxdy = \int_0^2 \left|(\frac{x^2}{2}+3)\right|_0^1 dy = \int_0^2 \frac{7}{2}dy = \frac{7}{2}\left|y\right|_0^2 = 7$ .Now you note that the value of the integral remain same if we change the order of integration. Actually we have a stronger result which we sate as a theorem.

Theorem:

Let R be the rectangle defined by the inequalities **a < x < b** and **c < y < d.** If f(x, y) is continuous on this rectangle, then $\iint_R f(x,y)dA = \int_c^d\int_a^b f(x,y)dxdy = \int_a^b\int_c^d f(x,y)dxdy$.

Remark:

This powerful theorem enables us to evaluate a double integral over a rectangle by calculating an iterated integral. Moreover the theorem tells us that the **"order of integration in the iterated integral does not matter".**

Example:

Evaluate the integral $\int_0^{\ln 2}\int_0^{\ln 3} e^{x+y}dxdy$

Solution:

First we will integrate the function with respect to "x". Note that we can write e^{x+y} as $e^x.e^y$ So we have, $\int_0^{\ln 2} e^y \left|e^x\right|_0^{\ln 3} dy = \int_0^{\ln 2} e^y(3-1)dy$ Here we use the fact that "e" and "ln" are inverse function of each other. So we have $e^{\ln 3} = 3$ .Thus we get,

$\int_0^{\ln 2} e^y \left|e^x\right|_0^{\ln 3} dy = 2\int_0^{\ln 2} e^y dy = 2\left|e^y\right|_0^{\ln 2} = 2(2-1) = 2$ *is the required answer.*

Example:

Evaluate the integral $\int_0^{\ln 3}\int_0^{\ln 2} e^{x+y}dydx$ **(Note that in this example we change the order of integration)**

Solution:

First we will integrate the function with respect to "y". Note that we can write e^{x+y} as $e^x.e^y$ So we have, $\int_0^{\ln 3} e^x \left|e^y\right|_0^{\ln 2} dy = \int_0^{\ln 3} e^x(2-1)dy$ Here we use the fact that "e" and "ln" are inverse function of each other. So we have $e^{\ln 2} = 2$ .Thus we get,

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$$\int_0^{\ln 3} e^x \left| e^y \right|_0^{\ln 2} dx = \int_0^{\ln 3} e^x dy = \left| e^x \right|_0^{\ln 3} = (3-1) = 2 \text{ is the required answer.}$$

Note that in both cases our integral has the same value.

Over view:

Double integrals Page # 854-857

Exercise Set 17.1 (page 857): 1,3,5,7,9,11,13,15,17,19

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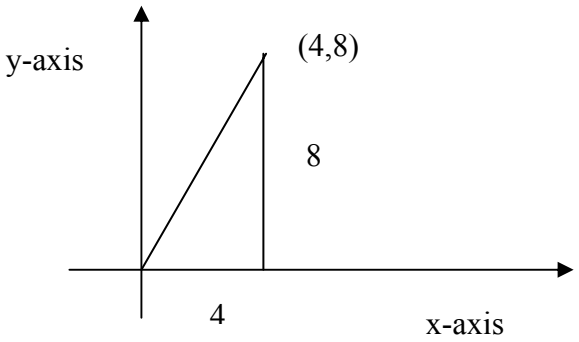
Lecture No -19 Use Of Integrals

Area as anti-derivatives

$$\int_0^4 2x\,dx = |x^2|_0^4$$
$$= (4)^2 = 16$$



Area of triangle =1/2 base x altitude
$= \frac{1}{2}(4)(8) = 16$

volume as anti-derivative

$$\text{Volume} = \int_0^2 \int_0^3 5\,dydx$$
$$= \int_0^2 |5y|_0^3 dx = \int_0^2 15\,dx$$
$$= |15x|_0^2 = 30$$

$0 \le x \le 2,\quad 0 \le y \le 3,\quad 0 \le z \le 5$
Volume = 2 x 3 x 5 = 30
The following results are analogous to the result of the definite integrals of a function of single variable.

$$\iint_R cf(x,y)dxdy$$
$$= c\iint_R f(x,y)dxdy \text{(c a constant)}$$

$$\iint_R [f(x,y) + g(x,y)]\,dxdy$$
$$= \iint_R f(x,y)\,dxdy + \iint_R g(x,y)\,dxdy$$
$$\iint_R [f(x,y) - g(x,y)]\,dxdy$$
$$= \iint_R f(x,y)\,dxdy - \iint_R g(x,y)\,dxdy$$

Use double integral to find the volume under the surface $z = 3x^3 + 3x^2y$ and the rectangle $\{(x,y): 1 \le x \le 3,\ 0 \le y \le 2\}$.

$$\text{Volume} = \int_0^2 \int_1^3 (3x^3 + 3x^2y)\,dx\,dy$$
$$= \int_0^2 \left| \frac{3x^4}{4} + x^3y \right|_1^3 dy$$
$$= \int_0^2 \left[\frac{3(3)^4}{4} - \frac{3}{4} + (3)^3y - y \right] dy$$
$$= \int_0^2 [60 + 26y]\,dy$$
$$= |60y + 13y^2|_0^2 = 172$$

Use double integral to find the volume of solid in the first octant enclosed by the surface z = x2 and the planes x=2, y =0, y = 3 and z = 0

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$$\text{Volume} = \int_0^2 \int_0^3 x^2 \, dy dx$$

$$= \int_0^2 \left| x^2 y \right|_0^3 dx = \int_0^2 [3x^2] dx$$

$$= \left| x^3 \right|_0^2 = 8$$

$\iint_R f(x,y)dA \geq 0$ if $f(x,y) \geq 0$ on R

$\iint_R f(x,y)dA \geq \iint_R g(x,y)dA$

if $f(x,y) \geq g(x,y)$

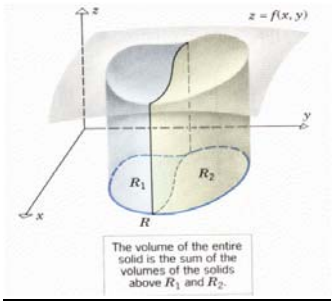
If $f(x,y)$ is nonnegative on a region R, then subdividing R into two regions R_1 and R_2 has the effect to subdividing the solid between R and $z = f(x,y)$ into two solids, the sum of whose volumes is the volume of the entire solid

$$\iint_R f(x, y)\, dA = \iint_{R1} f(x, y)\, dA + \iint_{R2} f(x, y)\, dA$$

The volume of the solid S can also be obtained using cross sections perpendicular to the y -axis.

$$\text{Vol}(S) = \int_c^d A(y)\, dy \qquad (1)$$

Where A(y) represents the area of the cross section perpendicular to the y-axis taken at the point y

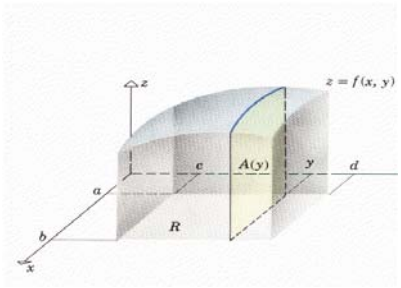


How to compute cross sectional area

For each fixed y in the interval $c \leq y \leq d$, the function f(x,y) is a function of x alone ,and A(y)may be viewed as the area under the graph of this function along the interval a<x<b, Thus

$$A(y) = \int_a^b f(x, y)\, dx$$

Substiuting this expression in (1) yields.



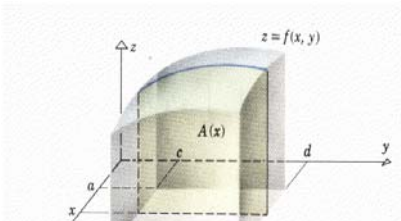
$$\text{Vol}(S) = \int_c^d \left[\int_a^b f(x, y)\, dx \right] dy$$

$$= \int_0^d \int_a^b f(x, y)\, dx\, dy$$

Similarly the volume of the side S can also be obtained using sections perpendicular to the x-axis

$$\text{Vol}(S) = \int_a^b A(x) \qquad (3)$$

Where A(x) is the area of the cross section perpendicular to the x-axis taken at the point x.



For each fixed x in the interval $a \le x \le b$ the function f(x,y) is a function of alone, so that the area A(x) is given by

$$A(x) = \int_{}^{d} f(x,y)dy$$

Substituting this expression in (3) yields

$$\text{Vol (S)} = \int_a^b \left[\int_c^d f(x, y)\, dy \right] dx$$

$$= \int_a^b \int_c^d f(x,y)\, dy\, dx \qquad (4)$$

From eq (2) and eq (4) we have

$$\iint_R f(x,y)\, dA = \int_c^d \int_a^b f(x,y)\, dx\, dy$$

$$= \int_a^b \int_c^d f(x,y)\, dy\, dx$$

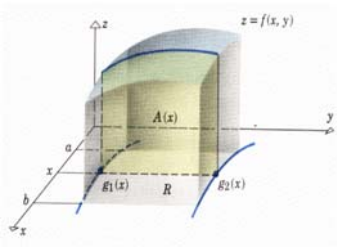
Double integral for non-rectangular region

Type I region is bounded the left and right by the vertical lines x=a and x=b and is bounded below and above by continuous curves $y=g_{1(x)}$ and $y=g_{2(x)}$ where

$$g_{1(x)} \le g_{2(x)} \text{ for } a \le x \le b$$

If R is a type I region on which f(x, y) is continuous, then

$$\iint_R f(x,y)dA = \int_a^b \int_{g_{1(x)}}^{g_{2(x)}} f(x,y)dydx \quad (1)$$



By the method of cross section, the volume of S is also given by

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$$\text{Vol (S)} = \int_a^b A(x)\,dx \qquad (2)$$

where A(x) is the area of the cross section at the fixed point x this cross section area extends from $g_{1(x)}$ to $g_{2(x)}$ in the y-direction,

$$\text{so , } A(x) = \int_{g_1(x)}^{g_2(x)} f(x, y)\,dy$$

Substituting this in (2) we obtain

$$\text{Vol (S)} = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\,dy\,dx$$

Since the volume of S is also given by

$$\iint_R f(x, y)\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\,dy\,dx$$

Type II region is bounded below and above by horizontal lines y=c and y=d and is bounded in the left and right by continuous curves $x=h_1(y)$ and $x=h_2(y)$ satisfying $h_1(y) \le h_2(y)$ for for $c \le y \le d$.

. If R is a type II region on which f (x, y) is continuous, then

$$\iint_R f(x,y)dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)dx\,dy$$

Similarly, the partial definite integral with respect to $\int_c^d$ f(x,y) is evaluated by holding x fixed and integrating with respect to y.

An integral of the form $\int_c^d f(x, y)\,dy$ produces a function of x.

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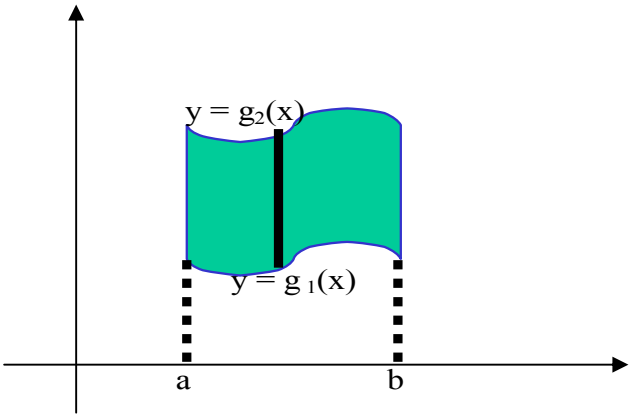
Lecture No -20 Double integral for non-rectangular region

Double integral for non-rectangular region

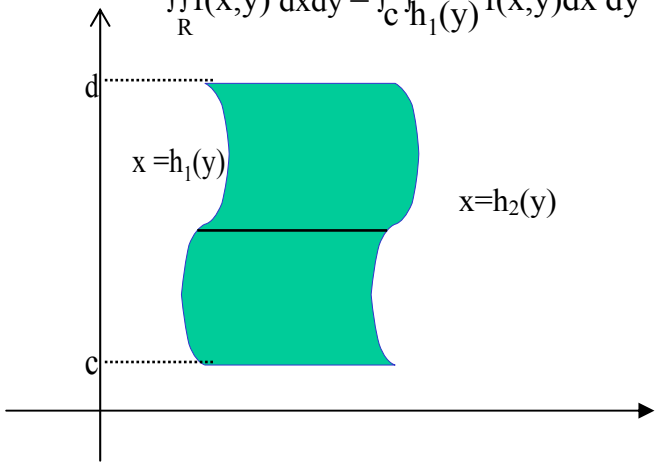
Type I region is bounded the left and right by vertical lines x=a and x=b and is bounded below and above by curves $y=g_1(x)$ and $y=g_2(x)$ where $g_1(x) \leq g_2(x)$ for $a \leq x \leq b$

$$\iint_R f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx$$



Type II region is bounded below and above by the horizontal lines y=c and y=d and is bounded on the left and right by the continuous curves $x= h_1(y)$ and $x=h_2(y)$ satisfying $h_1(y) \leq h_2(y)$ for $c \leq y \leq d$

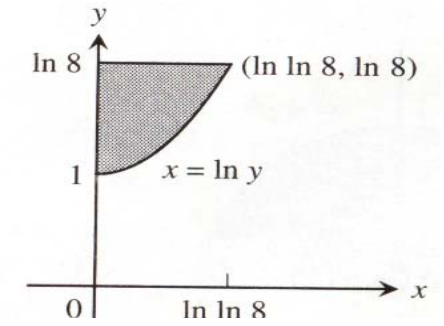
$$\iint_R f(x,y)\, dxdy = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)dx\, dy$$



Write double integral of the function *f(x,y)*on the region whose sketch is given

$$\int_1^{\ln 8} \int_0^{\ln y} f(x,y)\, dx$$

$$\int_0^{\ln(\ln 8)} \int_{e^x}^{\ln 8} f(x,y)\, y$$



Write double integral of the function *f(x,y)*on the region whose sketch is given

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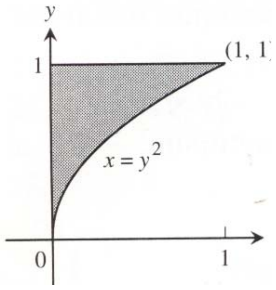
$$\int_0^1 \int_0^{y^2} f(x,y)\, dx\, dy$$

$$\int_0^1 \int_{\sqrt{x}}^1 f(x, y)\, dy\, dx$$



EXAMPLE

Draw the region and evaluate an equivalent integral with the order of integration reversed

$$\int_0^2 \int_x^{2x} (4x + 2)\, dy\, dx$$

The region of integration is given by the inequalities $x^2 \leq y \leq 2x$ and $0 \leq x \leq 2$.

$$\int_0^4 \int_{y/2}^{\sqrt{y}} (4x + 2)\, dx\, dy.$$

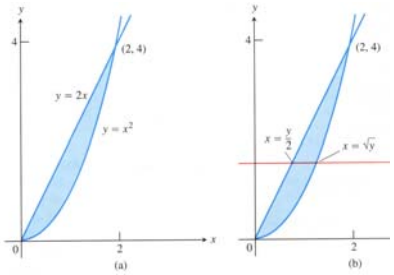
$$= \int_0^4 \left| 2x^2 + 2x \right|_{y/2}^{\sqrt{y}} dy$$

$$= \int_0^4 \left[2y + 2\sqrt{y} - \frac{y}{2} - y \right] dy$$

$$= \left| y^2 + \frac{4}{3} y^{3/2} - \frac{y^3}{6} - \frac{y^2}{2} \right|_0^4$$

$$= \left[16 + \frac{4}{3}(4)^{3/2} - \frac{(4)^3}{6} - 8 \right]$$

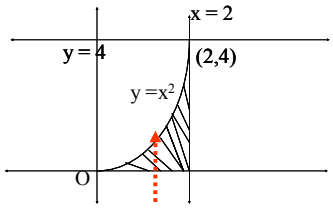
$$= 16 + \frac{4}{3}(8) - \frac{64}{6} - 8 = 8$$

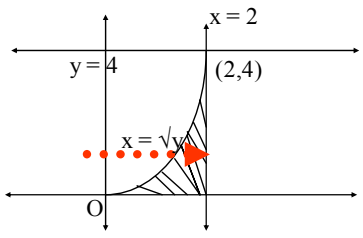


EXAMPLE

Evaluate $I = \int_0^4 \int_{\sqrt{y}}^2 y \cos x^5\, dx\, dy$ The integral is over the region $0 \leq y \leq 4$, $x = \sqrt{y}$ and $x = 2$

$$I = \int_0^2 \int_0^{x^2} y \cos x^5\, dy\, dx$$





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$$= \int_0^2 \left[\frac{y^2}{2}\cos x^5\right]_0^{x2} dx$$

$$= \int_0^2 \frac{x^4}{2}\cos x^5\, dx$$

$$= \frac{1}{10}\int_0^2 \cos x^5 \,.\, (5x^4)\, dx$$

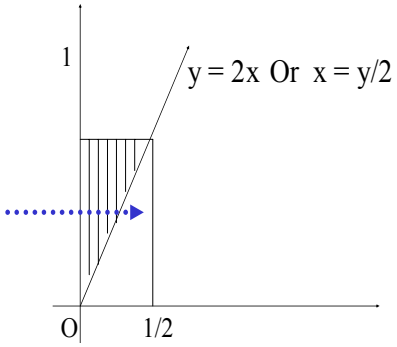
$$= \left[\frac{1}{10}\sin x^5\right]_0^2 = \frac{1}{10}\sin 32$$

Evaluate $I = \int_0^{1/2}\int_{2x}^{1} e^{y^2}\, dy\, dx$

The integral cannot be evaluated I the given order since e^{y^2} has no antiderivative. We shall change the order of integration. The region R which integration is performed is given by $0 \le x \le \frac{1}{2}$, $y = 2x$ and $y = 1$

This region is also enclosed by

$x = 0$, $x = \frac{y}{2}$ and $0 \le y \le 1$



Thus

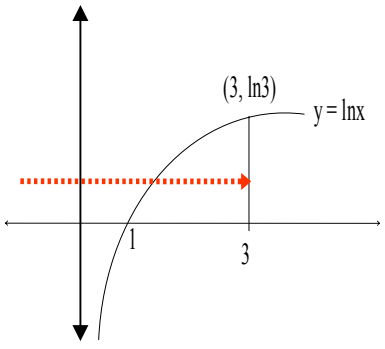
$$I = \int_0^1\int_0^{y/2} e^{y^2}\, dx\, dy$$

$$= \int_0^1 \frac{y}{2}\, e^{y^2}\, dy$$

$$= \left[\frac{1}{4}\, e^{y^2}\right]_0^1 = \frac{1}{4}(e-1)$$

$$\int_1^3\int_0^{\ln x} x\, dy\, dx$$

Reversing the order of integration



$$= \int_0^{\ln 3}\int_{e^y}^{3} x\, dx\, dy$$

$$= \int_0^{\ln 3}\left|\frac{x^2}{2}\right|_{e^y}^{3} dy = \frac{1}{2}\int_0^{\ln 3}[9 - e^{2y}]\, dy$$

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$$= \frac{1}{2}\left|9y - \frac{e^{2y}}{2}\right|_0^{\ln 3}$$

$$= \frac{1}{2}\left[9\ln 3 - \frac{e^{2\ln 3}}{2} + \frac{e^0}{2}\right]$$

$$= \frac{1}{2}\left[9\ln 3 - \frac{9}{2} + \frac{1}{2}\right]$$

$$= \frac{1}{2}\left[9\ln 3 - 4\right]$$

$$= \frac{9}{2}\ln 3 - 2$$

Over view of Lecture # 20

Book Calsulus By Howard Anton
Chapter # !7 Article # 17.2
Page (858-863) Exercise set 17.2
21,22,23,25,27,35,37,38

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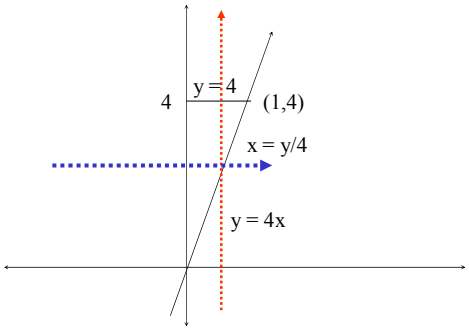
Lecture No -21 **Examples**

$$\int_0^1 \int_{4x}^4 e^{-y^2}\, dy\, dx$$



Reversing the order of integration

$$\int_0^4 \int_0^{y/4} e^{-y^2}\, dx\, dy$$

$$= \int_0^4 \left| x\, e^{-y^2} \right|_0^{y/4} dy \quad = \int_0^4 \frac{y}{4}\, e^{-y^2}\, dy$$

$$= \frac{-1}{8} \int_0^4 e^{-y^2} (-2y)\, dy$$

$$= -\frac{1}{8} \left| e^{-y^2} \right|_0^4 = -\frac{1}{8} \left| e^{-16} - c^0 \right|$$

$$= \frac{1}{8} \left(1 - \frac{1}{e^{16}} \right)$$

Example

Calculate

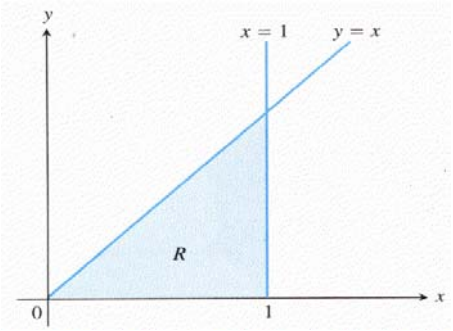
$$\iint_R \frac{\sin x}{x}\, dA.$$

where R is the triangle in the xy- plane bounded by the x-axis ,the line y=x and the line x=1

We integrate first with respect to y and then with respect to x, we find

$$\int_0^1 \left(\int_0^x \frac{\sin x}{x}\, dy \right) dx$$

$$= \int_0^1 \left(y \frac{\sin x}{x} \Big]_{y=0}^{y=x} \right) dx$$



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$$= \int_0^1 \sin x\,dx = -\cos(0) + 1 \approx 0.46$$

$$= \int_0^1 \sin x\,dx = -\cos(0) + 1 \approx 0.46$$

EXAMPLE

$$\int_0^2 \int_{y/2}^1 e^{x^2}\,dxdy$$

Since there is no elementary antiderivative of e^{x^2}, the integral

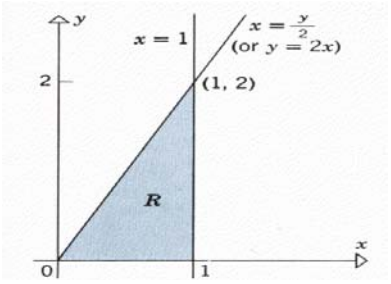
$$\int_0^2 \int_{y/2}^1 e^{x^2}\,dxdy$$

cannot be evaluated by performing the x-integration first.

To evaualte this integral , we express is as an equivalent iterated integral with the order if integration reversed . For the inside integration, y is fixed and x varies from he line x = y/2 to the line x = 1. For the outside integration, y varies from 0 to 2, so the given iterated integral is equal to a double integral over the triangular region R.

To reverse the order of integration, we treat R as a type I region, which enables us to write the given integral as

$$\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2}\,dxdy$$



By changing the order of integration we get,

$$\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2}\,dxdy = \int_0^1 \int_0^{2x} e^{x^2}\,dydx$$

$$= \int_0^1 [e^{x^2} y]_{y=0}^{2x}\,dx$$

$$= \int_0^1 2xe^{x^2}\,dx$$

$$= e^{x^2}\Big|_0^1 = e - 1$$

EXAMPLE

Use a double integral to find the volume of the solid that is bounded above by the palne Z=4-x-y and below by the rectangle $R = \{(x,y): 0 \le x \le 1,\ 0 \le y \le 2\}$

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$$V=\iint_R (4-x-y)\,dA$$

$$=\int_0^2\int_0^1 (4-x-y)\,dx\,dy$$

$$=\int_0^2 \left[4x-\frac{x^2}{2}-xy\right]_{x=0}^{1} dy$$

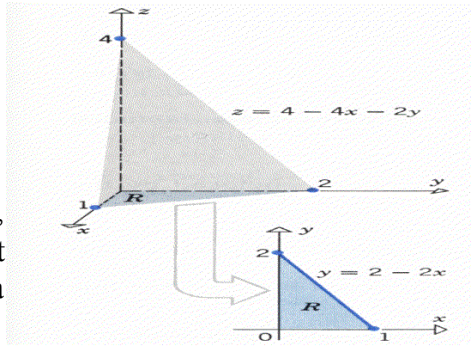
$$=\int_0^2 \left(\frac{7}{2}-y\right) dy=\left[\frac{7}{2}y-\frac{y^2}{2}\right]_0^2=5$$

EXAMPLE

Use a double integral to find the volume of the tetrahedron bounded by the coordinate planes and the plane z=4-4x-2y The tetrahedron is bounded above by the plane.

$$z=4-4x-2y \quad \text{----------(1)}$$

and below by the triangular region R



Thus, the volume is given by

$$V = \iint_R (4 - 4x - 2y)\,dA$$

The region R is bounded by the x-axis, the y-axis, and the line $y = 2 - 2x$ [set $z = 0$ in (1)], so that treating R as a type I region yields.

$$V=\iint_R (4-4x-2y)\,dA$$

$$=\int_0^1\int_0^{2-2x} (4-4x-2y)\,dy\,dx$$

$$=\int_0^1 [4y-4xy-y^2]_{y=0}^{2-2x} dx$$

$$=\int_0^1 (4-8x+4x^2)dx$$

$$=\frac{4}{3}$$

Find the volume of the solid bounded by the cylinder x2+y2 = 4 and the planes y + z = 4 and z = 0.

The solid is bounded above by the plane $z = 4 - y$ and below by the region R within the circle $x^2 + y^2 = 4$.

The volume is given by

$$V = \iint_R (4 - y)\,dA$$

Treating R as a type I region we obtain

$$V=\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4-y)\,dy\,dx$$

$$=\int_{-2}^{2}\left[4y-\frac{1}{2}y^2\right]_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx$$

$$=\int_{-2}^{2} 8\sqrt{4-x^2}\,dx$$

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$$= 8\left| \frac{x\sqrt{4-x^2}}{2} + \frac{4}{2}\sin^{-1}\frac{x}{2} \right|_{-2}^{2}$$

$$= 8\left| 2\sin^{-1}(1) - 2\sin^{-1}(-1) \right|$$

$$= 8[2(\frac{\pi}{2}) + 2(\frac{\pi}{2})]$$

$$= 8(2\pi) = 16\pi$$

EXAMPLE

Use double integral to find the volume of the solid that is bounded above by the paraboiled $z=9x^2 + y^2$,below by the plane $z=0$ and laterally by the planes

$$x = 0, \quad y = 0, \quad x = 3, \quad y = 2$$

$$\text{Volume} = \int_0^3 \int_0^2 (9x^2 + y^2)\, dy\, dx$$

$$= \int_0^3 \left[9x^2 y + \frac{y^3}{3} \right]_0^2 dx$$

$$= \int_0^3 \left(18x^2 + \frac{8}{3} \right) dx$$

$$= \left| 6x^3 + \frac{8}{3}x \right|_0^3$$

$$= 6\,(27) + 8$$

$$= 170$$

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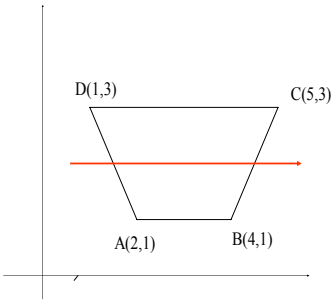
Lecture No -22 Examples

EXAMPLE

$$\iint_R xy\, dA$$

R is the region bounded by the Trapezium with the vertices (1, 3) (5, 3) (2, 1) (4, 1)

Slope of AD $= \frac{3-1}{1-2} = -2$



Equation of line AD

$y - 1 = -2(x - 2)$

$y - 1 = -2x + 4$

$-2x = y - 5$

$x = -\frac{y-5}{2}$

Slope of BC $= \frac{3-1}{5-4} = 2$

Equation of line BC

$y - 1 = 2(x - 4) \Rightarrow y - 1 = 2x - 8 \Rightarrow 2x = y + 7 \Rightarrow x = \frac{y+7}{2}$

$$\int_1^3 \int_{-(y-5)/2}^{(y+7)/2} xy\, dx\, dy = \int_1^3 y \left| \frac{x^2}{2} \right|_{-(y-5)/2}^{(y+7)/2} dy = \int_1^3 (3y^2 + 3y)\, dy$$

$$= \left| y^3 + \frac{3y^2}{2} \right|_1^3 = (3)^3 + \frac{3(3)^2}{2} - 1 - \frac{3}{2} = 38$$

EXAMPLE

Use double integral to find the volume of the wedge cut from the cylinder $4x^2+y^2=9$ by the plane z=0 and z=y+3

Solution:

Since we can write $4x^2 + y^2 = 9$ as $\frac{x^2}{\left(\frac{3}{2}\right)^2} + \frac{y^2}{9} = 1$ this is equation of ellipse.

Now the Lower and upper limits for "x" are $x = \frac{-\sqrt{9-y^2}}{2}$ *and* $x = \frac{\sqrt{9-y^2}}{2}$

And upper and lower limits for "x" are -3 and 3 respectively. So the required volume is given by $\int_{-3}^{3} \int_{\frac{-\sqrt{9-y^2}}{2}}^{\frac{\sqrt{9-y^2}}{2}} (y+3)\, dx\, dy$

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$$= \int_{-3}^{3} \left| xy + 3x \right|_{\frac{-\sqrt{9-y^2}}{2}}^{\frac{\sqrt{9-y^2}}{2}} dy$$

$$= \int_{-3}^{3} \left[y\left(\frac{\sqrt{9-y^2}}{2}\right) + 3\left(\frac{\sqrt{9-y^2}}{2}\right) + y\left(\frac{\sqrt{9-y^2}}{2}\right) + 3\left(\frac{\sqrt{9-y^2}}{2}\right) \right] dy$$

$$= \int_{-3}^{3} \left[y\sqrt{9-y^2} + 3\sqrt{9-y^2} \right] dy$$

$$= -\frac{1}{2}\int_{-3}^{3} \sqrt{9-y^2}\,(-2y)dy + 3\int_{-3}^{3} \sqrt{9-y^2}\,dy$$

$$= -\frac{1}{2}.\frac{2}{3}\left| (9-y^2)^{\frac{3}{2}} \right|_{-3}^{3} + 3\int_{-3}^{3} \sqrt{9-y^2}\,dy$$

$$= -\frac{1}{3}.[0] + 3\int_{-3}^{3} \sqrt{9-y^2}\,dy = 3\left| \frac{y}{2}\sqrt{9-y^2} + \frac{9}{2}\sin^{-1}\left(\frac{y}{2}\right) \right|_{-3}^{3} = \frac{27\pi}{2}$$

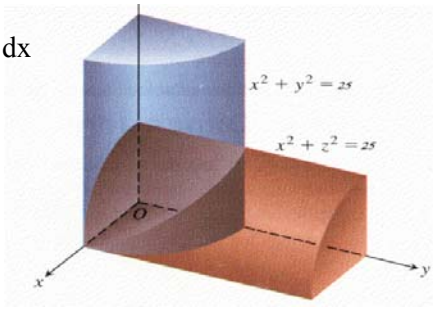
EXAMPLE

Use double integral to find the volume of solid common to the cylinders $x^2+ y^2=25$ and $x^2+ z^2=25$

Volume $= 8\int_0^5 \int_0^{\sqrt{25-x^2}} \sqrt{25-x^2}\ dy\ dx$

$= 8\int_0^5 \sqrt{25-x^2}\ \left| y \right|_0^{\sqrt{25-x^2}}\ dx$

$= 8\int_0^5 (25-x^2)\ dx$



$$= 8\left| 25x - \frac{x^3}{3} \right|_0^5 = 8\left(125 - \frac{125}{3}\right) = 8\left(\frac{375-125}{3}\right) = 8\left(\frac{250}{3}\right) = \frac{2000}{3}$$

AREA CALCUALTED AS A DOUBLE INTEGRAL

$$V = \iint_R 1\ dA = \iint_R dA \qquad (1)$$

However, the solid has congruent cross sections taken parallel to the xy-plan so that

V = area of base x height=area of R.1 = area of R

Combining this with (1) yields the area formula

$$\text{area of R} = \iint_R dA \qquad (2)$$

EXAMPLE

Use a double integral to find the area of the region R enclosed between the parabola $y = ½\ x^2$ and the line y=2x

$$\text{area of R} = \iint_R dA$$

$$= \int_0^4 \int_{x^2/2}^{2x} dy\ dx$$

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$$= \int_0^4 \left[y\right]_{y=x^2/2}^{2x} dx = \int_0^4 \left(2x - \frac{1}{2}x^2\right) dx = \left[x^2 - \frac{x^3}{6}\right]_0^4 = \frac{16}{3}$$

Treating R as type IIyields.

area of R$= \iint_R dA = \int_0^8 \int_{y/2}^{\sqrt{2y}} dx\, dy$

$$= \int_0^8 \left[x\right]_{y=x^2/2}^{\sqrt{2x}} dx = \int_0^8 \left(\sqrt{2y} - \frac{1}{2}y\right) dy = \left[\frac{2\sqrt{2}}{3}y^{3/2} - \frac{y^2}{4}\right]_0^8 = \frac{16}{3}$$

EXAMPLE

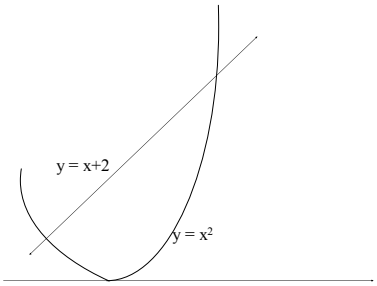
Find the area of the region R enclosed by the parabola $y=x^2$ and the line $y=x+2$

$$\int_{-1}^{2} \int_{y=x^2}^{y=x+2} dy\, dx$$

$$= \int_{-1}^{2} \left[y\right]_{x^2}^{x+2} dx$$

$$= \int_{-1}^{2} [x + 2 - x^2] dx$$

$$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2} = \frac{9}{2}$$



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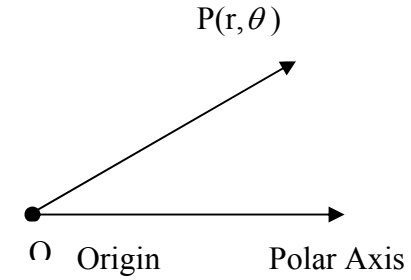
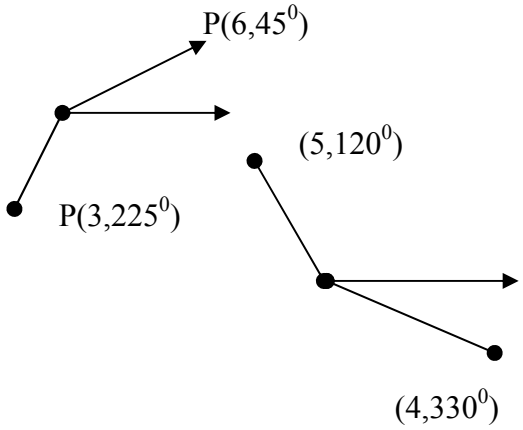
Lecture No - 23 Polar Coordinate Systems

POLAR COORDIANTE SYSTEMS

To form a polar coordinate system in a plane, we pick a fixed point O, called the origin or pole, and using the origin as an endpoint we construct a ray, called the polar axis. After selecting a unit of measurement, we may associate with any point P in the plane a pair of polar coordinates (r, θ), where r is the distance from P to the origin and θ measures the angle from the polar axis to the line segment OP.

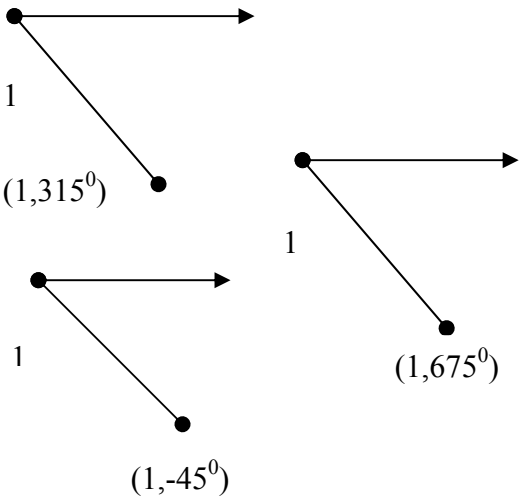


The number r is called the radial distance of P and θ is called a polar angle of P. In the points (6, 45°), (3, 225°), (5, 120°), and (4, 330°) are plotted in polar coordinate systems.



THE POLAR COORDINATES OF A POINT ARE NOT UNIQUE.

For example, the polar coordinates
(1, 315°), (1, – 45°), and (1, 675°)
all represent the same point
In general, if a point P has polar co-ordinate
(r, θ), then for any integer n=0,1,2,3,…….
(r, θ +n.360⁰) and (r, θ +n.360⁰)
are also polar co-ordinates of p
In the case where P is the origin, the line line seg
Because there is no clearly defined polar angle ir
Polar angle θ may be used. Thus, for every θ ma
(0, θ) is the origin.



NEGATIVE VALUES OF R

When we start graphing curves in polar coordinates, it will be desirable to allow negative values for r. This will require a special definition. For motivation, consider the point P with polar coordinates (3, 225⁰) We can reach this point by rotating the polar axis 225⁰ and then moving forward from the origin 3 units along the terminal side of the angle. On the other hand, we can also reach the point P by rotating the polar axis 45⁰ and then moving backward 3 units from the origin along the extension of the terminal side of the angle

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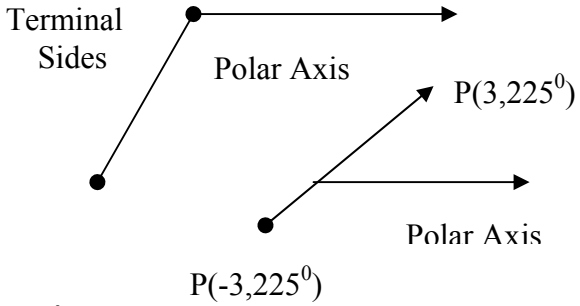
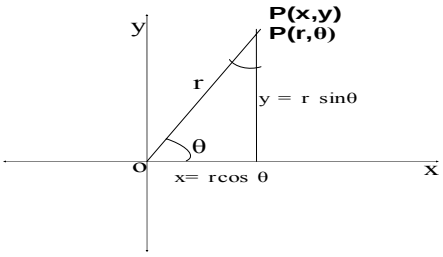
This suggests that the point (3, 225 might also be denoted by (− 3, 45 with the minus sign serving to indicate that the point is on the extension of the angle's terminal side rather than on the terminal side itself.

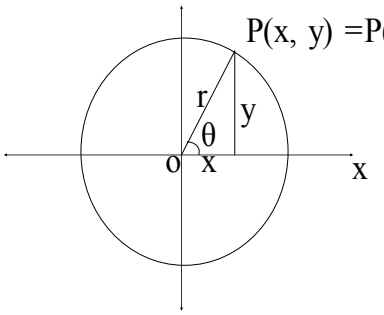


Since the terminal side of the angle $\theta + 180^0$ is the extension of the terminal side other angle θ, We shall define.
(-r, θ) and (r, $\theta + 180^0$) to be polar coordinates for the same point .
With r=3 and $\theta = 45^0$ in (2) if follows that (-3, 45^0) and (-3, 225^0) represent the same point.

RELATION BETWEEN POLAR AND RECTANGULAR COORDINATES



CONVERSION FORMULA FROM POLAR TO CARTESIAN COORDINATES AND VICE VERSA



$x = r \cos\theta$

$y = r \sin\theta$

$x^2 + y^2 = r^2$

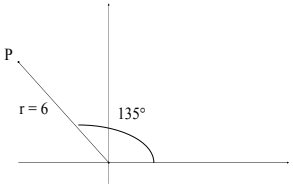
$y/x = \tan\theta$

Example

Find the rectangular coordinates of the point P whose polar co-ordinates are $(6,135^0)$

Solution:

Substituting the polar coordinates
r = 6 and $\theta = 135^0$ in $x = \cos\theta$ and $y = \sin\theta$ yeilds



$x = 6 \cos 135° = 6(-\sqrt{2}/2) = -3\sqrt{2}$

$y = 6 \sin 135° = 6(\sqrt{2}/2) = 3\sqrt{2}$

Thus, the rectangular coordinates of the point P are $(-3\sqrt{2}, 3\sqrt{2})$

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Example:

Find polar coordinates of the point P whose rectangular coordinates are $(-2, 2\sqrt{3})$

Solution:

We will find polar coordinates (r, θ) of P such that r >0 and $0 \le \theta \le 2\pi$.

$$r = \sqrt{x^2 + y^2} = \sqrt{(-2)^2 + (2\sqrt{3})^2} = \sqrt{4+12} = \sqrt{16} = 4$$

$$\tan\theta = \frac{y}{x} = \frac{2\sqrt{3}}{-2} = -\sqrt{3} \Rightarrow \theta = \tan^{-1}(-\sqrt{3}) = \frac{2\pi}{3}$$

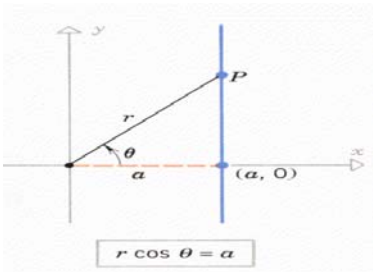
From this we have (-2,2$\sqrt{3}$) lies in the second quadrant of P. All other Polar co-ordinates of P have the form

$$(4, \frac{2\pi}{3} + 2n\pi) \quad or \quad (-4, \frac{5\pi}{3} + 2n\pi) \quad , \quad Where\ n\ is\ \text{int}eger$$

LINES IN POLAR COORDIANTES

A line perpendicular to the x-axis and passing through the point with xy co-ordinates with (a,0) has the equation x=a . To express this equation in polar co-ordinates we substitute x = r cos θ $\Rightarrow$ a = rcos θ --------(1)

This result makes sense geometrically since each point P (r, θ) on this line will yield the value a for r cos θ.

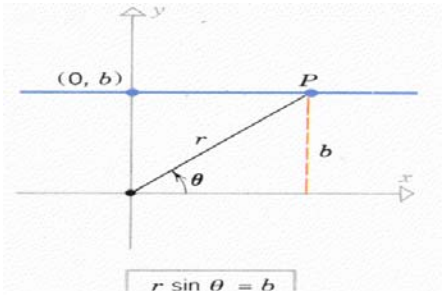


A line parallel to the x-axis that meets the y-axis it the point with xy-coordinates (0, b) has the equation y = b.

Substituting y = r sin θ yields.

r sin θ = b (2)

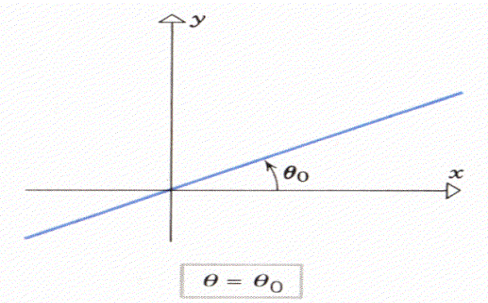
as the polar equation of this line. This makes sense geometrically since each point P (r, θ) on this line will yield the value b for r sin θ



For Any constant θ_0, the equation

$\theta = \theta_0$ (3)

is satisfied by the coordinates of all points of the form P (r, θ_0), regardless of the value of r. Thus, the equation represents the line through the origin making an angle of θ_0 (radians) with the polar axis.



By substitution x = rcosθ and y = rsinθ in the equation Ax +By +C = 0 . We obtain the general polar form of the line,

$$r\,(A\cos\theta + B\sin\theta) + C = 0$$

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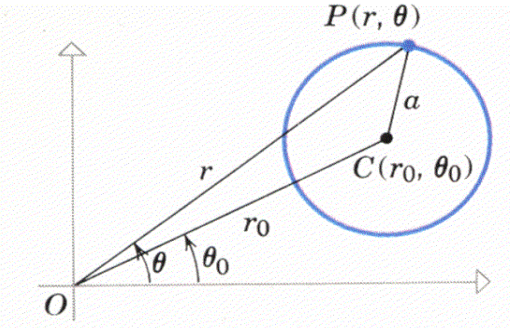
CIRCLES IN POLAR COORDINATES

Let us try to find the polar equation of a circle whose radius is a and whose center has polar coordinates (r_0, θ_0). If we let P(r, θ) be an arbitrary point on the circle, and if we apply the law of cosines to the triangle OCP we obtain

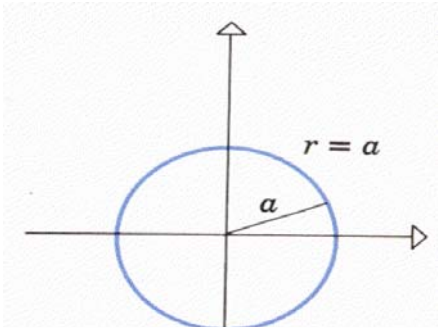


$$r^2 - 2rr_0 \cos(\theta - \theta_0) + r_0^2 = a^2 \qquad (1)$$

SOME SPECIAL CASES OF EQUATION OF CIRCLE IN POLAR COORDINATES

A circle of radius a, centered at the origin, has an especially simple polar equation. If we let $r_0 = 0$ in (1), we obtain $r^2 = a^2$ or, since $a \geq 0$, $r = a$

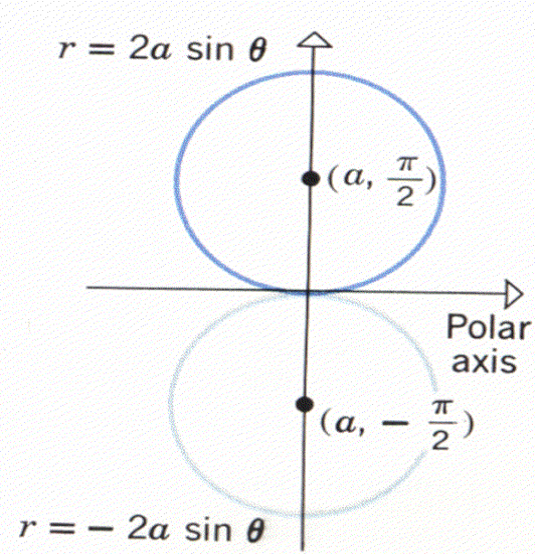
This equation makes sense geometrically since the circle of radius a, centered at the origin, consists of all points P (r, θ) for which r = a, regardless of the value of θ



If a circle of radius a has its center on the x-axis and passes through the origin, then the polar coordinates of the center are either

(a, 0) or (a, π)

depending on whether the center is to the right or left of the origin



Lecture No -24 Sketching

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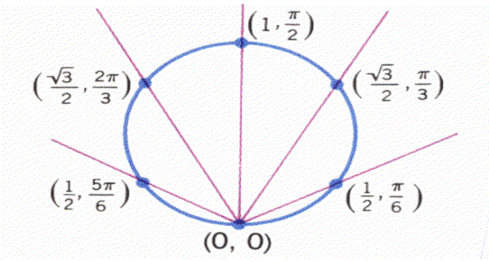
Draw graph of the curve having the equation r = sin θ

By substituting values for θ at increments of $\frac{\pi}{6}(30^0)$ and calculating r , we can construct The following table:

θ (radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$
r = sin θ	0	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$

θ (radians)	π	$\frac{7\pi}{6}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$	2π
r = sin θ	0	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	0

Note that there are 13 pairs listed in Table, but only 6 points plotted in This is because the pairs from θ = π on yield duplicates of the preceding points. For example, (– ½, 7π/6) and (1/2, π/6) represent the same point. The points appear to lie on a circle.



hat this is indeed the case may be seen by expressing the given equation in terms of x and y. We first multiply the given equation through by r to obtain $r^2 = r \sin\theta$ which can be rewritten as

$$x^2 + y^2 = y \quad \text{or} \quad x^2 + y^2 - y = 0$$

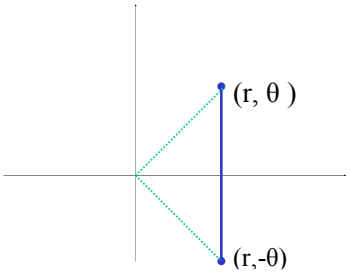
or on completing the square $x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4}$ .This is a circle of radius 1\2 centered at the point (0,1/2) in the xy-plane.

Sketching of Curves in Polar Coordinates

1.SYMMETRY

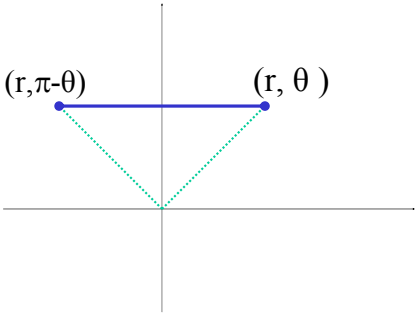
(i) Symmetry about the Initial Line

If the equation of a curve remains unchanged when (r,θ) is replaced by either $(r,-\theta)$ in its equation ,then the curve is symmetric about initial line.



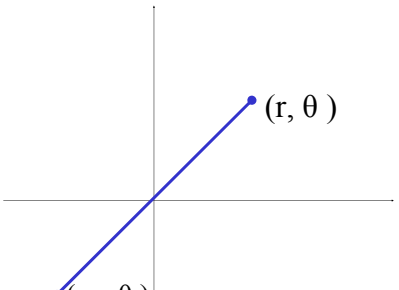
(ii) Symmetry about the y-axis

If when (r, θ) is replaced by either $(r,\pi-\theta)$ in The equation of a curve and an equivalent equation is obtained ,then the curve is symmetric about the line perpendicular to the initial i.e, the y-axis



(ii) Symmetry about the Pole

If the equation of a curve remains unchanged



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when either (-r, θ) or is substituted for (r, θ) in its equation ,then the curve is symmetric about the pole. In such a case ,the center of the curve.

2. Position Of The Pole Relative To The Curve

See whether the pole on the curve by putting r=0 in the equation of the curve and solving for θ.

3. Table Of Values

Construct a sufficiently complete table of values. This can be of great help in sketching the graph of a curve.

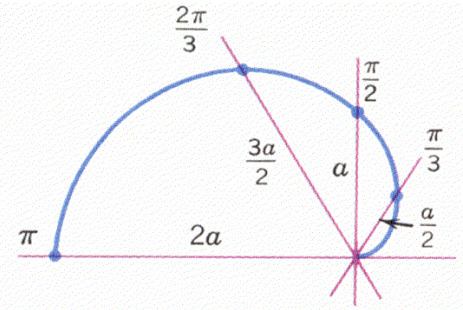
II Position Of The Pole Relative To The Curve.

When r = 0 , θ = 0 .Hence the curve passes through the pole.

III. Table of Values

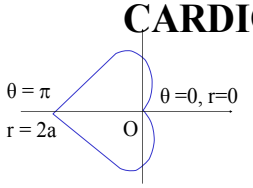
θ	0	π/3	π/2	2π/3	π
r=a (1−cosθ)	0	a/2	a	3a/2	2a

As θ varies from 0 to π, cos θ decreases steadily from 1 to − 1, and 1 − cos θ increases steadily from 0 to 2. Thus, as θ varies from 0 to π, the value of r = a (1 − cos θ) will increase steadily from an initial value of r = 0 to a final value of r = 2a.

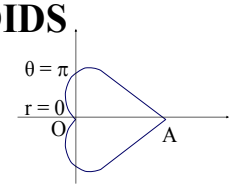


On reflecting the curve in about the x-axis, we obtain the curve.

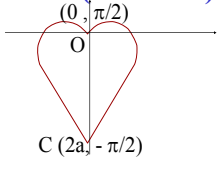
CARDIOIDS



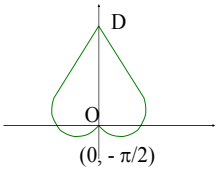
r = a (1 – cos θ)



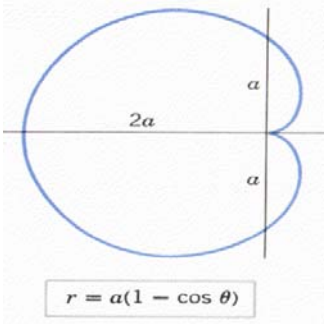
r = a (1 + cos θ)



r = a (1 – sin θ)



r = a (1 + sin θ)



CARDIDOIDS AND LIMACONS

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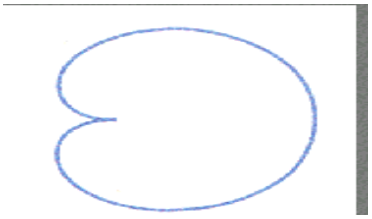
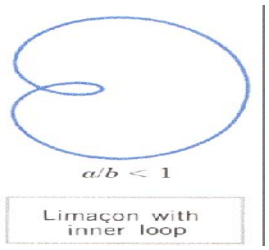
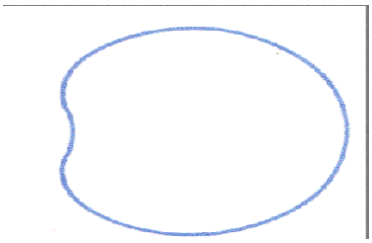
$r=a+b\sin\theta$, $r=a-b\sin\theta$

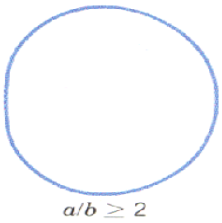
$r=a+b\cos\theta$, $r=a-b\cos\theta$

The equations of above form produce polar curves called limacons. Because of the heart-shaped appearance of the curve in the case a = b, limacons of this type are called cardioids. The position of the limacon relative to the polar axis depends on whether sin θ or cos θ appears in the equation and whether the + or – occurs.









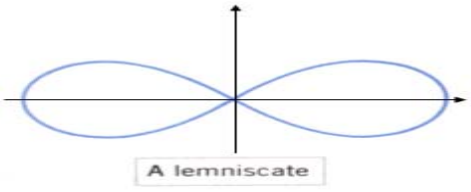
LEMINSCATE

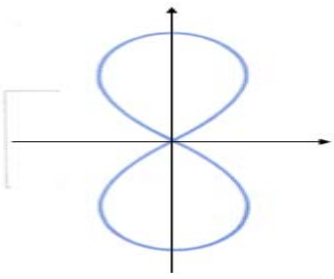
If a > 0, then equation of the form

$r^2 = a^2\cos 2\theta$, $r^2 = -a^2\cos 2\theta$

$r^2 = a^2\sin 2\theta$, $r^2 = -a^2\sin 2\theta$

represent propeller-shaped curves, called lemiscates (from the Greek word "lemnicos" for a looped ribbon resembling the Fig 8. The lemniscates are centered at the origin, but the position relative to the polar axis depends on the sign preceding the a^2 and whether sin 2θ or cos 2θ appears in the equation.





Example

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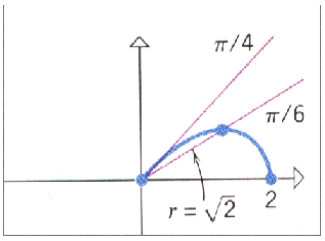
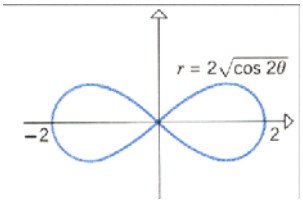
$$r^2 = 4 \cos 2\theta$$

The equation represents a lemniscate. The graph is **symmetric** about the **x-axis** and the **y-axis**. Therefore, we can obtain each graph by first sketching the portion of the graph in the range $0 \leq \theta < \pi/2$ and then reflecting that portion about the x- and y-axes.The curve passes through the origin when $\theta = \pi/4$, so the line $\theta = \pi/4$ is tangent to the curve at the origin.As θ varies from 0 to $\pi/4$, the value of $\cos 2\theta$ decreases steadily from 1 to 0, so that rdecreases steadily from 2 to 0.For θ in the range $\pi/4 < \theta < \pi/2$, the quantity $\cos 2\theta$ is negative, so there are no real values of r satisfying first equation. Thus, there are no points on the graph for such θ. The entire graph is obtained by reflecting the curve about the x-axis and then reflecting the resulting curve about the y-axis.



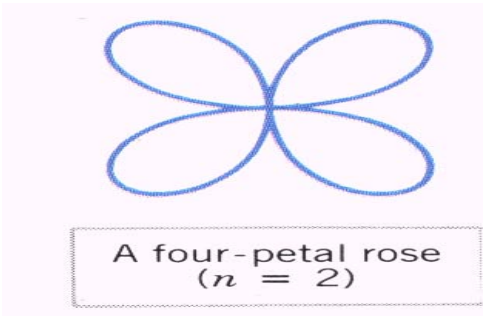


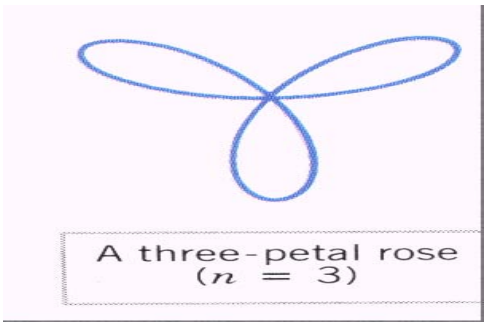
ROSE CURVES

Equations of the form

$r = a \sin n\theta$ and $r = a \cos n\theta$

represent flower-shaped curves called roses. The rose has **n equally spaced petals or loops if n is odd** and **2n equally spaced petals if n is even**





The orientation of the rose relative to the polar axis depends on the sign of the constant a and whether $\sin\theta$ or $\cos\theta$ appears in the equation.

SPIRAL

A curve that "winds around the origin" infinitely many times in such a way that r increases (or decreases) steadily as θ increases is called a spiral. The most common example is the spiral of Archimedes, which has an equation of the form.

$r = a\theta \quad (\theta \geq 0)$ or $r = a\theta \quad (\theta \leq 0)$

In these equations, θ is in radians and a is positive.

EXAMPLE

Sketch the curve **$r = \theta \ (\theta \geq 0)$** in polar coordinates.

This is an equation of spiral with a = 1; thus, it represents an Archimedean spiral.

Since r = 0 when $\theta = 0$, the origin is on the curve and the polar axis is tangent to the spiral.

A reasonably accurate sketch may be obtained by plotting the intersections of the spiral with the x and y axes and noting that r increases steadily as θ increases. The intersections with the x-axis occur when

$\theta = 0, \ \pi, \ 2\pi, \ 3\pi, \ \ldots\ldots$

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at which points r has the values

$r = 0,\ \pi,\ 2\pi,\ 3\pi,\ldots..$

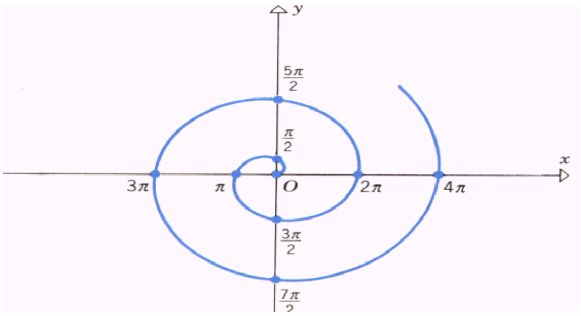
and the intersections with the y-axis occur when

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots\ldots$$

at which points r has the values

$$r = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots\ldots$$

Starting from the origin, the Archimedean spirals $r = \theta$ ($\theta \geq 0$) loops counterclockwise around the origin.



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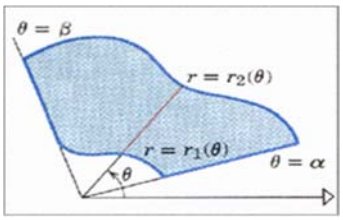
Lecture No -25 Double integrals in polar co-ordinates

Double integrals in which the integrand and the region of integration are expressed in polar coordinates are important for two reasons: First, they arise naturally in many applications, and second, many double integrals in rectangular coordinates are more easily evaluated if they are converted to polar coordinates. The function $z = f(r,\theta)$ to be integrated over the region R As shown in the Fig.



INTEGRALS IN POLAR COORDIATES

When we defined the double integral of a function over a region R in the xy-plane, we began by cutting R into rectangles whose sides were parallel to the coordinate axes. These were the natural shapes to use because their sides have either constant x-values or constant y-values. In polar coordintes, the natural shape is a "polar rectangle" whose sides have constant r and θ- values.

Suppose that a function f (r, θ) is defined over a region R that is bounded by the ray $\theta = \alpha$ and $\theta = \beta$ and by the continuous curves $r = r_1(\theta)$ and $r = r_2(\theta)$. Suppose also that $0 \leq r_1(\theta) \leq r_2(\theta) \leq a$ for every value of θ between α and β. Then R lies in a fan-shaped region Q defined by the inequalities $0 \leq r \leq a$ and $\alpha \leq \theta \leq \beta$.

Then the double integral in polar coordinates is given as

$$\iint_R f(r,\theta)\, dA = \int_{\theta=\alpha}^{\theta=\beta} \int_{r=r_1(\theta)}^{r=r_2(\theta)} f(r,\theta)\, dr\, d\theta$$

How to find limits of integration from sketch

Step 1. Since θ is held fixed for the first integration, draw a radial line from the origin through the region R at a fixed angle θ. This line crosses the boundary of R at most twice. The innermost point of intersection is one the curve $r = r_1(\theta)$ and the outermost point is on the curve $r = r_2(\theta)$. These intersections determine the r-limits of integration.

Step 2. Imagine rotating a ray along the positive x-axis one revolution counterclockwise about the origin. The smallest angle at which this ray intersects the region R is $\theta = \alpha$ and the largest angle is $\theta = \beta$. This yields the θ-limits of the integration.

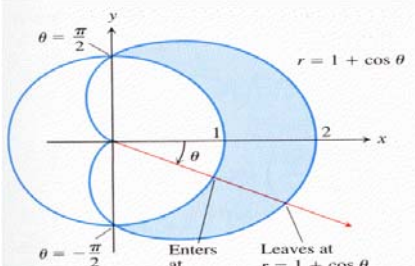
EXAMPLE

Find the limits of integration for integrating f (r, θ) over the region R that lies inside the cardioid $r = 1 + \cos\theta$ and outside the circle $r = 1$.

Solution:

Step 1. We sketch the region and label the bounding curves.

Step 2. The r-limits of integration. A typical ray from



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The origin enters R where r =1 and leaves where

r =1+cos θ.

Step 3. The θ-limits of integration. The rays from the

origin that intersect R run from $\theta = -\frac{\pi}{2}$ to $\theta = \frac{\pi}{2}$.

The integral is $\int_{-\pi/2}^{\pi/2}\int_{1}^{1+\cos\theta} f(r,\theta)\, r\, dr\, d\theta$

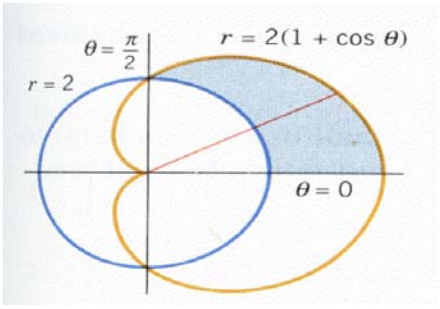
$= 2\int_{0}^{\pi/2}\int_{1}^{1+\cos\theta} f(r,\theta)\, r\, dr\, d\theta$

EXAMPLE

Evalaute $\iint_R \sin\theta\, dA$

Where R is the region ion the first quadrant that is outside the circle r = 2 and inside the cardioid r = 2(1+cosθ).

Solution:



$\iint_R \sin\theta\, dA = \int_0^{\pi/2}\int_2^{2(1+\cos\theta)} (\sin\theta)\, r\, dr\, d\theta$

$= \int_0^{\pi/2} \frac{1}{2} r^2 \sin\theta \Big]_{r=2}^{2(1+\cos\theta)} d\theta$

$= 2\int_0^{\pi/2} [(1+\cos\theta)^2 \sin\theta - \sin\theta]\, d\theta$

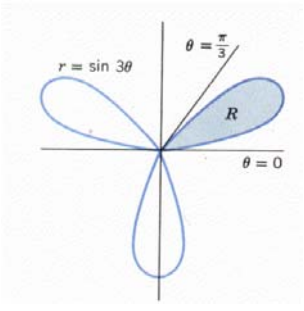
$= 2\left[-\frac{1}{3}(1+\cos\theta)^3 + \cos\theta\right]_0^{\pi/2} = \left[-\frac{1}{3} - \left(-\frac{5}{3}\right)\right] = \frac{8}{3}$

EXAMPLE

Use a double polar integral to find the area enclosed by the three-petaled rose r = sin 3θ.

We calculate the area of the petal R in the first quadrant and multiply by three.

Solution:



$A = 3\iint_R dA = 3\int_0^{\pi/3}\int_0^{\sin 3\theta} r\, dr\, d\theta = \int_0^{\pi/3} \frac{r^2}{2}\Big|_0^{\sin 3\theta} dr\, d\theta$

$= \frac{3}{2}\int_0^{\pi/3} \sin^2 3\theta\, d\theta = \frac{3}{4}\int_0^{\pi/3} (1-\cos 6\theta)\, d\theta$

$= \frac{3}{4}\left[\theta - \frac{\sin 6\theta}{6}\right]_0^{\pi/3} = \left[\frac{3}{4}\theta - \frac{3}{24}\sin 6\theta\right]_0^{\pi/3} = \frac{1}{4}\pi$

EXAMPLE

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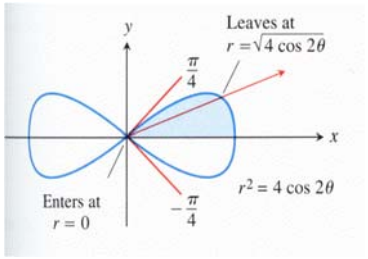
Find the area enclosed by the lemniscate $r^2 = 4\cos 2\theta$. The total area is four times the first-quadrant portion.

Solution:



$$A = 4\int_0^{\pi/4}\int_0^{\sqrt{4\cos 2\theta}} r\,dr\,d\theta = A\int_0^{\pi/4}\left[\frac{r2}{2}\right]_{r=0}^{r=\sqrt{4\cos 2\theta}} d\theta$$

$$= 4\int_0^{\pi/4} 2\cos 2\theta\, d\theta = 4\sin 2\theta\Big]_0^{\pi/4} = 4.$$

CHANGING CARTESIAN INTEGRALS INTO POLAR INTEGRALS

The procedure for changing a Cartesian integral $\iint_R f(x, y)\,dx\,dy$ into a polar integral has two steps.

Step 1. Substiute $x = r\cos\theta$ and $y = r\sin\theta$, and replace $dx\,dy$ by $r\,dr\,d\theta$ in the Cartesian integral.

Step 2. Supply polar limits of integration for the boundary of R. The Cartesian integral then becomes

$$\iint_R f(x,y)\,dx\,dy = \iint_G f(r\cos\theta, r\sin\theta) r\,dr\,d\theta$$

where G denotes the region of integration in polar coordinates.

Notice that dx dy is not replaced by dr dθ but by r dr dθ.

EXAMPLE

Evalaute the double integral $\int_0^1\int_0^{\sqrt{1-x^2}} (x^2 + y^2)\,dy\,dx$ by changing to polar coordinates.

The region of integration is bounded by

$0 \le y \le \sqrt{1 - x^2}$ and $0 \le x \le 1$

$y = \sqrt{1 - x^2}$ is the circle

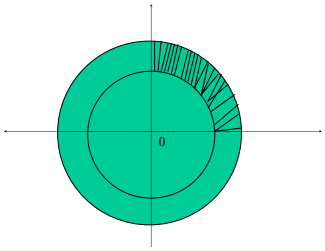
$x^2 + y^2 = 1, \quad r = 1$

On changing into the polar coordinates, the given integral is

$$\int_0^{\pi/2}\int_0^1 r^3\,dr\,d\theta = \int_0^{\pi/2}\left|\frac{r^4}{4}\right|_0^1 d\theta = \int_0^{\pi/2}\frac{1}{4}\,d\theta = \left|\frac{\theta}{4}\right|_0^{\pi/2} = \frac{\pi}{8}$$

EXAMPLE

Evaluate $I = \int_D\int \frac{dx\,dy}{x^2 + y^2}$ by changing to polar coordinates,



where D is the region in the first quadrant between the circles $x^2+y^2=a^2$ and $x^2+y^2=b^2$, $0<a<b$

$$I = \int_0^{\pi/2}\int_a^b \frac{r\,dr\,d\theta}{r^2} = \int_0^{\pi/2}[\ln r]_a^b\,d\theta$$

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$$= \int_0^{\pi/2} \ln\left(\frac{b}{a}\right) d\theta = \left[\theta \ln\left(\frac{b}{a}\right)\right]_0^{\pi/2} = \frac{\pi}{2} \ln\left(\frac{b}{a}\right).$$

EXAMPLE

Evaluate the double integral $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy\, dx$ by changing to polar coordinates.

The region of integration is bounded by $0 < y < \sqrt{1-x^2}$ and $0 \leq x \leq 1$

$y = \sqrt{1-x^2}$ is the circle $x^2 + y^2 = 1$, $r = 1$

On changing into the polar coordinates, the given integral is

$$\int_0^{\pi/2} \int_0^1 r^3 dr d\theta = \int_0^{\pi/2} \left|\frac{r^4}{4}\right|_0^1 d\theta = \int_0^{\pi/2} \frac{1}{4} d\theta = \frac{1}{4} |\theta|_0^{\pi/2} = \frac{1}{4}(\pi/2) = \pi/8$$

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Lecture No - 26 EXAMPLES

EXAMPLE

Evalaute $I=\int_0^4 \int_0^{\sqrt{4y-y^2}} (x^2+y^2)dx\,dy$ by changing into polar coordinates.

The region of integration is bounded by $0 \le x \le \sqrt{4y - y^2}$ and $0 \le y \le 4$

Now $x = \sqrt{4y - y^2}$ is the circle $x^2+y^2-4y=0 \Rightarrow x^2 + y^2 = 4y$.In polar coordinates this takes the form $r^2 = 4r \sin\theta$, $r = 4 \sin\theta$

On changing the integral into polar coordinates, we have

$$I = \int_0^{\pi/2} \int_0^{4\sin\theta} r^2 . r\,dr\,d\theta = \int_0^{\pi/2} 64 \sin^4\theta\,d\theta = 64.\frac{3.1}{4.2} . \frac{\pi}{2} = 12\pi$$ (using Walli's formula)

EXAMPLE

Evalaute $\iint_R e^{x^2+y^2}\,dy\,dx$.,where R is the semicircular region bounded by the x-axis and the curve $y = \sqrt{1 - x^2}$

In Cartesian coordinates, the integral in question is a nonelementary integral and there is no direct way to integrate $e^{x^2+y^2}$ with respect to either x or y.

Substituting $x = r\cos\theta$, $y = r\sin\theta$, and replacing dy dx by r dr dθ enables us to evaluate the integral as

$$\iint_R e^{x^2+y^2}dy\,dx = \int_0^{\pi}\int_0^{1} e^{r^2} r\,dr\,d\theta = \int_0^{\pi} \left[\frac{1}{2} e^{r^2}\right]_0^1 d\theta = \int_0^{\pi} \frac{1}{2}(e-1)d\theta = \frac{\pi}{2}(e - 1).$$

EXAMPLE

Let R_a be the region bounded by the circle $x^2 + y^2 = a^2$. Define

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}dx\,dy = \lim_{a\to\infty} \iint_R e^{-(x^2+y^2)}dx\,dy$$

To evaluate this improper integral.

$$I=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \exp\{-(x^2 + y^2)\}\,dx\,dy = \lim_{a\to\infty} \iint_{Da} \exp\{-(x^2+ y^2)\}\,dx\,dy$$

$$= \lim_{a\to\infty} \int_0^{2\pi}\int_0^{a} \exp\{-r^2\} r\,dr\,d\theta = \lim_{a\to\infty} \int_0^{2\pi} \frac{1}{2}(1 - \exp\{-a^2\}d\theta = \lim_{a\to\infty} \frac{1}{2}(1 - \exp\{-a^2\}\,\theta\Big|_0^{2\pi}$$

$$= \pi - \lim_{a\to\infty} \frac{\pi}{\exp\{-a^2\}} = \pi$$

EXAMPLE

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Prove that $\int_0^{\infty} e^{-t^2}\,dt = \frac{1}{2}\sqrt{\pi}$.

$$I = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \exp\{-(x^2+y^2)\}\,dx\,dy = \int_{-\infty}^{\infty} \exp\{-y^2\}\left[\int_{-\infty}^{\infty} \exp\{-x^2\}dx\right] dy = \left[\int_{-\infty}^{\infty} \exp\{-y^2\}dy\right]\left[\int_{-\infty}^{\infty} \exp\{-x^2\}dx\right]$$

$$= \left[\int_{-\infty}^{\infty} \exp\{-t^2\}dt\right]\left[\int_{-\infty}^{\infty} \exp\{-t^2\}\,dt\right] = \lim_{a\to\infty}\left[\int_{-a}^{a} \exp\{-t^2\}\,dt\right]^2 = 4\lim_{a\to\infty}\left[\int_0^a \exp\{-t^2\}\,dt\right]^2$$

Hence we have $4\lim_{a\to\infty}\left[\int_0^a \exp\{-t^2\}\,dt\right]^2 = \lim_{a\to\infty}\int_{-a}^{a}\int_{-a}^{a} \exp\{-(x^2+y^2)\}dx\,dy$ $4\lim_{a\to\infty}\left[\int_0^{\infty} \exp\{-t^2\}dt\right]^2 = \pi$

$$\lim_{a\to\infty}\left[\int_0^{\infty} \exp\{-t^2\}dt\right]^2 = \pi/4 \Rightarrow \int_0^{\infty} \exp\{-t^2\}\,dt = \frac{\sqrt{\pi}}{2}$$

THEOREM

Let G be the rectangular box defined by the inequalities
$a \le x \le b, \quad c \le y \le d, \qquad k \le z \le \ell$

If f is continuous on the region G, then $\iiint_G f(x, y, z)\,dV = \int_a^b \int_c^d \int_k^{\ell} f(x, y, z)\,dz\,dy\,dx$

Moreover, the iterated integral on the right can be replaced with any of the five other iterated integrals that result by altering the order of integration.

$$= \int_c^d \int_k^{\ell} \int_a^b f(x, y, z)\,dy\,dz\,dx = \int_c^d \int_a^b \int_k^{\ell} f(x, y, z)\,dy\,dx\,dz = \int_a^b \int_c^d \int_k^{\ell} f(x, y, z)\,dx\,dy\,dz$$

$$= \int_a^b \int_k^{\ell} \int_c^d f(x, y, z)\,dx\,dz\,dy = \int_k^{\ell} \int_a^b \int_c^d f(x, y, z)\,dz\,dx\,dy$$

EXAMPLE

Evalaute the triple integral $\iiint_G 12xy^2z^3\,dV$ over the rectangular box G defined by the inequalities $-1 \le x \le 2, 0 \le y \le 3, 0 \le z \le 2$.

We first integrate with respect to z, holding x and y fixed, then with respect to y holding x fixed, and finally with respect to x.

$$\iiint_G 12xy^2z^3\,dV = \int_{-1}^{2}\int_0^3\int_0^2 12xy^2z^3\,dz\,dy\,dx = \int_{-1}^{2}\int_0^3 \left[3xy^2z^4\right]_{z=0}^{2} dy\,dx = \int_{-1}^{2}\int_0^3 48xy^2\,dy\,dx$$

$$= \int_{-1}^{2} \left[16xy^3\right]_{y=0}^{3} dx = \int_{-1}^{2} 432x\,dx = 216x^2\Big]_{-1}^{2} = 648$$

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EXAMPLE

Evalaute $\iiint_R$ (x – 2y + z) dx dy dz Region R : $0 \le x , 1,\ 0 \le y \le x^2,\ 0 \le z \le x + y$

$$= \int_0^1 \int_0^{x^2} \int_0^{x+y} (x - 2y + z)\, dz\, dy\, dx = \int_0^1 \int_0^{x^2} \left| \frac{(x - 2y + z)^2}{2} \right|_0^{x+y} dy\, dx$$

$$= \int_0^1 \int_0^{x^2} \left[\frac{(x-2y+x+y)^2}{2} - \frac{(x-2y)^2}{2} \right] dy\, dx = \frac{1}{2} \int_0^1 \int_0^{x^2} (3x2 - 3y2)\, dy\, dx = \frac{3}{2} \int_0^1 \left| x^2y - \frac{y^3}{3} \right|_0^{x^2} dx$$

$$= \frac{3}{2} \int_0^1 \left(x4 - \frac{x^6}{z} \right) dx = \frac{3}{2} \left| \frac{x^5}{5} - \frac{x^7}{21} \right|_0^1 = \frac{3}{2} \left[\frac{1}{5} - \frac{1}{21} \right] = \frac{8}{35}$$

Example:

Evalaute $\iiint_S$ xyzd x dy dz Where S = $\{(x,y,z): x^2+y^2+z^2 \le 1,\ x \ge 0,\ y \ge 0,\ z \ge 0\}$

S is the sphere $x^2 + y^2 + z^2 = 1$ .Since x, y, z are al +ve so we have to consider only the +ve octant of the sphere.

Now $x^2 + y^2 + z^2 = 1$. So that $z = \sqrt{1 - x^2 - y^2}$

The Projection of the sphere on xy plan is the circle $x^2 + y^2 = 1$.

This circle is covered as y-varies from 0 to $\sqrt{1 - x^2}$ and x varies from 0 to 1.

$$\iiint_R xyz\, dx\, dy\, dz = \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\, dz\, dy\, dx = \int_0^1 \int_0^{\sqrt{1-x^2}} xy \left| \frac{z^5}{2} \right|_0^{\sqrt{1-x^2-y^2}} dy\, dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} xy \left(\frac{1 - x^2 - y^2}{2} \right) dy\, dx = \frac{1}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} x\,(y - x^2y - y^3)\, dy\, dx$$

$$= \frac{1}{2} \int_0^1 x \left(\frac{y^2}{2} - \frac{x^2y^2}{2} - \frac{y^4}{4} \right) \Big|_0^{\sqrt{1-x^2}} dx = \frac{1}{4} \int_0^1 x \left[1 - x^2 - x^2\,(1-x^2) - \frac{(1-x^2)^2}{2} \right] dx$$

$$= \frac{1}{8} \int_0^1 (x - 2x^3 + x^5)\, dx = \frac{1}{8} \left| \frac{x^2}{2} - \frac{x^4}{2} + \frac{x^6}{6} \right|_0^1 = \frac{1}{8} \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{6} \right) = \frac{1}{48}$$

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Lecture No -27 Vector Valued Functions

Recall that a function is a rule that assigns to each element in its domain one and only one element in its range. Thus far, we have considered only functions for which the domain and range are sets of real numbers; such functions are called real-valued functions of a real variable or sometimes simply real-valued functions. In this section we shall consider functions for which the domain consists of real numbers and the range consists of vectors in 2-space or 3-space; such functions are called vector-valued functions of a real variable or more simply vector-valued functions. In 2-space such functions can be expressed in the form.

r (t) = (x (t), y (t)) = x (t) i + y (t) j

and in 3-space in the form

r (t) = (x (t), y (t), z (t))i+y(t)j+z (t)k

where x(t), y(t), and z(t) are real-valued functions of the real variable t. These real-valued functions are called the component functions or components of **r**. As a matter of notation, we shall denote vector-valued functions with boldface type [**f(t), g(t)**, and **r** (t) and real-valued functions, as usual, with lightface italic type [*f(t), g(t), and r(t)*].

EXAMPLE

r (t) = (ln t) **i** + $\sqrt{t^2 + 2}$ **j** + (cos tπ)**k**

then the component functions are x(t)=ln*t*,y(t)=$\sqrt{t^2 + 2}$,and z (t) = costπ

The vector that r(t) associates with t = 1 is r(1)=(ln 1) **i**+$\sqrt{3}$**j** +(cos π) **k**=$\sqrt{3}$**j**– **k**

The function **r** is undefined if $t \leq 0$ because ln *t* is undefined for such *t*.

If the domain of a vector-valued function is not stated explicitly, then it is understood to consist of all real numbers for which every component is defined and yields a real value. This is called the natural domain of the function. Thus the natural domain of a vector-valued function is the intersection of the natural domains of its components.

PARAMETRIC EQUATIONS IN VECTOR FORM

Vector-valued functions can be used to express parametric equations in 2-space or 3-space in a compact form.

For example, consider the parametric equations x = x(t), y = y (t)

Because two vectors are equivalent if and only if their corresponding components are equal, this pair of equations can be replaced by the single vector equation.

x = x(t), y = y (t)

x **i** + y **j** = x (t)**i** + y (t) **j**

Similarly, in 3-space the three parametric equations

x = x(t), y = y (t), z = z (t)

can be replaced by the single vector equation

x**i** + y**j** + z**k** = x(t)**i** + y(t)**j** + z(t)**k**

if we let r = x **i** +y **j** and **r**(t) = x(t) **i** + y(t) **j** in 2-sapce

and let **r** = x**i** + y**j** + z**k** and **r**(t) = x(t) **i** + y(t)**j** + z(t)**k**

in 3-space, then both (2) and (4) can be written as **r** = **r**(t)

which is the vector form of the parametric equations in (1) and (3). Conversely, every vector equation of form (5) can be rewritten as parametric equations by equating components on the two sides.

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EXAMPLE

Express the given parametric equations as a single vector equation.

(a) $x = t^2$, $y = 3t$

(b) $x = \cos t$, $y = \sin t$, $z = t$

(a) Using the two sides of the equations as components of a vector yields.

$$x\,\mathbf{i} + y\,\mathbf{j} = t^2\,\mathbf{i} + 3t\,\mathbf{j}$$

(b) Proceeding as in part (a) yields

$$x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$$

EXAMPLE

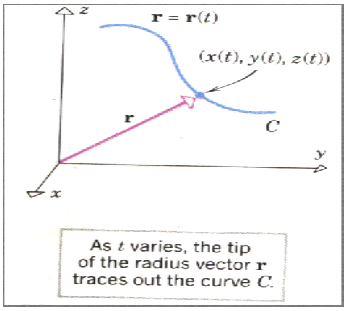
Find parametric equations that correspond to the vector equation

$$x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k} = (t^3 + 1)\,\mathbf{i} + 3\,\mathbf{j} + e^t\,\mathbf{k}$$

Equating corresponding components yields.

$x = t^3 + 1$, $y = 3$, $z = e^t$

GRAPHS OF VECOR-VALUED FUNCTOINS



One method for interpreting a vector-valued function r(t) in 2-space or 3-space geometrically is to position the vector r = r (t) with its initial point at the origin, and let C be the curve generated by the tip of the vector r as the parameter t varies

The vector r, when positioned in this way, is called the radius vector or position vector of C, and C is called the graph of the function r (t) or, equivalently, the graph of the equation r = r (t). The vector equation r = r (t) is equivalent to a set of parametric equations, so C is also called the graph of these parametric equations.

EXAMPLE

Sketch the graph of the vector-valued function $\mathbf{r(t) = (\cos t)i + (\sin t)j}$, $\mathbf{0 \le t \le 2\pi}$

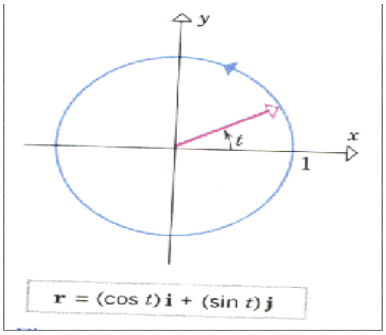
The graph of r (t) is the graph of the vector equation

$x\mathbf{i}+y\mathbf{j} = (\cos t)\mathbf{i} + (\sin t)\mathbf{j}$, $0 \le t \le 2\pi$

or equivalently, it is the graph of the parametric equations

$x = \cos t$, $y = \sin t$ $(0 \le t \le 2\pi)$

This is a circle of radius 1 that is centered at the origin with the direction of increasing t counterclockwise. The graph and a radius vector are shown in Fig.



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EXAMPLE

Sketch the graph of the vector-valued function r(t) = (cos t)i+(sin t)j+2k, $0 \leq t \leq 2\pi$

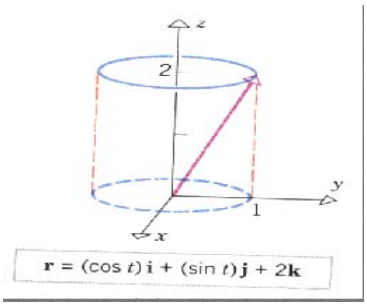
The graph of **r**(t) is the graph of the vector equation

x**i**+y**j**+z**k**=(cost)**i**+(sint)**j**+2k, $0 \leq t \leq 2\pi$

or, equivalently, it is the graph o the parametric equations

x=cos t, y = sin t, z=2 $(0 \leq t \leq 2\pi)$

From the last equation, the tip o the radius vecor traces a curve in the plane z = 2, and from the first two equations and the preceding example, the curve is a circle of radius 1 centered on the z-axis and traced counterclockwise looking down the z-axis. The graph and a radius vector are shown in Fig.



EXAMPLE

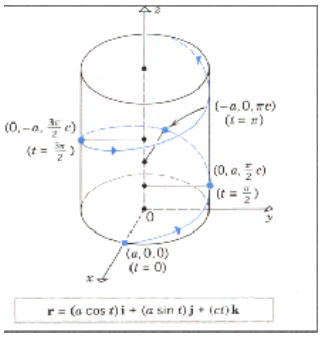
Sketch the graph of the vector-valued function r(t) = (a cos t)i + (a sin t)j + (ct)k where a and c are positive constant.

The graph of **r**(t) is the graph of the parametric equations.

x = a cos t, y = a sin t, z = ct

As the parameter t increases, the value of z = ct also increases, so the point (x, y, z) moves upward. However, as t increases, the point (x, y, z) also moves in a path directly over the circle. x = a cos t, y = a sin t in the xy-plane. The combination of these upward and circular motions produces a corkscrew-shaped curve that wraps around a right-circular cylinder of radius a centered on the z-axis.

This curve is called a circular helix.



EXAMPLE

Describe the graph of the vector equation r = (– 2 + t)i + 3tj + (5 – 4t)k

The corresponding parametric equations are x = – 2 + t, y = 3t, z = 5 – 4t

The graph is the line in 3-space that passes through the point (– 2, 0, 5) and is parallel to the vector **i** + 3**j** – 4**k**.

EXAMPLE

The graph of the vector-valued function r (t) = t **i** + t^2 **j** + t^3 **k** is called a twisted cubic. Show that this curve lies o the parabolic cylinder y = x^2, and sketch the graph for $t \geq 0$

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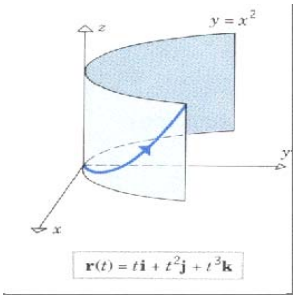
The corresponding parametric equations are $x = t$, $y = t^2$, $z = t^3$
Eliminating the parameter t in the equations for x and y yields $y = x^2$, so the curve lies on the parabolic cylinder with this equation. The curve starts at the origin for t = 0; as t increases, so do x, y, and z, so the curve is traced in the upward direction, moving away from the origin along the cylinder.



GRAPHS OF CONSTANT VECOR-VALUED FUNCTIONS

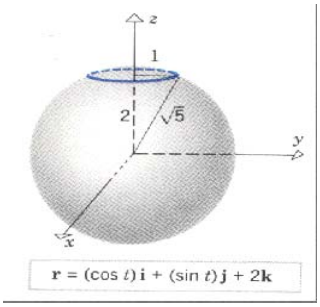
If c is a constant vector in the sense that it does not depend on a parameter, then the graph of $\mathbf{r} = \mathbf{c}$ is a single point since the radius vector remains fixed with its tip at **c**.
If $\mathbf{c} = x_0\mathbf{i} + y_0\mathbf{j}$ (in 2-space), then the graph is the point (x_0, y_0), and if $c = x_0\mathbf{i} + y_0\mathbf{j} + z_0\mathbf{k}$ (in 3-space), then the graph is the point (x_0, y_0, z_0).

EXAMPLE

The graph of the equation $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$ is the point (2, 3, – 1) in 3-space.
If r(t) is a vector-valued function, then for each value of the parameter t, the expression ||r(t)|| is a real-valued function of t because the norm (or length of r(t) is a real number.
For example,
If $\mathbf{r}(t) = t\,\mathbf{i} + (t - 1)\,\mathbf{j}$
Then $\|r(t)\| = \sqrt{t^2 + (t-1)^2}$ which is a real-valued function of t.

EXAMPLE

The graph of $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + 2\mathbf{k}$, $0 \le t \le 2\pi$
is a circle of radius 1 centered on the z-axis and lying in the plane $z = 2$. This circle lies on the surface of a sphere of radius $\sqrt{5}$ because for each value of t
$\|r(t)\| = \sqrt{\cos^2 t + \sin^2 t + t} = \sqrt{1 + 4} = \sqrt{5}$
which shows that each point on the circle is a distance of $\sqrt{5}$ units from the origin.



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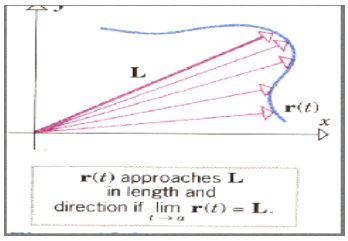
Lecture No -28 Limits of Vector Valued Functions

The limit of a vector-valued functions is defined to be the vector that results by taking the limit of each component. Thus, for a function $r(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$ in 2-space we define.

$$\lim_{t\to\alpha} r(t) = (\lim_{t\to\alpha} x(t))\mathbf{i} + (\lim_{t\to\alpha} y(t))\mathbf{j}$$

and for a function $r(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ in 3-space we define.

$$\lim_{t\to\alpha} r(t) = (\lim_{t\to\alpha} x(t))\mathbf{i} + (\lim_{t\to\alpha} y(t))\mathbf{j} + (\lim_{t\to\alpha} z(t))\mathbf{k}$$



If the limit of any component does not exist, then we shall agree that the limit of r (t) does not exist. These definitions are also applicable to the one-sided and infinite limits $\lim_{t\to\alpha^+}$, $\lim_{t\to\alpha^-}$, $\lim_{t\to+\infty}$, and $\lim_{t\to-\infty}$. It follows from (1) and (2) that

$$\lim_{t\to\alpha} r(t) = L$$

if and only if the components of r(t) approach the components of L as $t \to a$. Geometrically, this is equivalent to stating that the length and direction of r (t) approach the length and direction of L as $t \to \alpha$

CONTINUITY OF VECTOR-VALUED FUNCTIONS

The definition of continuity for vector-valued functions is similar to that for real-valued functions. We shall say that r is continuous at t_0 if

1. $r(t_0)$ is defined;
2. $\lim_{t\to t_0} r(t)$ exists;
3. $\lim_{t\to t_0} r(t) = r(t_0)$.

It can be shown that r is continuous at t_0 if and only if each component of r is continuous. As with real-valued functions, we shall call r continuous everywhere or simply continuous if r is continuous at al real values of t. geometrically, the graph of a continuous vector-valued function is an unbroken curve.

DERIVATIVES OF VECOR-VALUED FUNCTIONS

The definition of a derivative for vector-valued functions is analogous to the definition for real-valued functions.

DEFINITION

The derivative $r'(t)$ of a vector-valued function r(t) is defined by

$$\mathbf{r'(t) = \lim_{h\to 0} \frac{r(t+h) - r(t)}{h}}$$

Provided this limit exists.

For computational purposes the following theorem is extremely useful; it states that the derivative of a vector-valued function can be computed by differentiating each components.

THEOREM

(a) If $r(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$ is a vector-valued function in 2-space, and if x(t) and y(t) are differentiable, then

$$r'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j}$$

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(b) If $r(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ is a vector-valued function in 3-space, and if x(t), y(t), and z(t) are differentiable, then

$$r'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k}$$

We shall prove part (a). The proof of (b) is similar.

Proof (a):

$$\mathbf{r}'(t) = \lim_{h\to 0}\frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} = \lim_{h\to 0}\frac{[x(t+h)-x(t)]}{h}\mathbf{i} + \lim_{h\to 0}\frac{[y(t+h) - y(t)]}{h}\mathbf{j}$$

$$= x'(t)\mathbf{i} + y'(t)\mathbf{j}$$

As with real-valued functions, there are various notations for the derivative of a vector-valued function. If r = r (t), then some possibilities are

$\frac{d}{dt}[r(t)]$, $\frac{dr}{dt}$, $\mathbf{r}'(t)$, and $\mathbf{r}'$

EXAMPLE

Let $\mathbf{r(t) = t^2 i + t^3 j}$. Find $\mathbf{r'(t)}$ and $\mathbf{r'(1)}$

$$\mathbf{r}'(t) = \frac{d}{dt}[t^2]\,\mathbf{i} + \frac{d}{dt}[t^3]\,\mathbf{j}$$

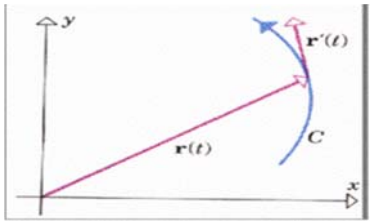
$$= 2t\,\mathbf{i}\;\; 3t^2\,\mathbf{j}$$

Substituting t=1 yields

$\mathbf{r}'(1) = 2\mathbf{i}+3\mathbf{j}$.

TAGENT VECTORS AND TANGENT LINES
GEOMETRIC INTERPRETATIONS OF THE DERIVATIVE.

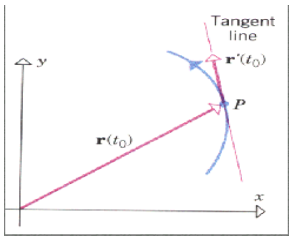
Suppose that C is the graph of a vector-valued function r(t) and that $\mathbf{r}'(t)$ exists and is nonzero for a given value of t. If the vector $\mathbf{r}'(t)$ is positioned with its initial point at the terminal point of the radius vector



DEFINITION

Let P be a point on the graph of a vector-valued function $\mathbf{r}(t)$, and let $\mathbf{r}(t_0)$ be the radius vector from the origin to P

If $\mathbf{r}'(t_0)$ exists and $\mathbf{r}'(t_0) \neq \mathbf{0}$, then we call $\mathbf{r}'(t_0)$ the tangent vector to the graph of $\mathbf{r}$ at $\mathbf{r}(t_0)$



REMARKS

Observe that the graph of a vector-valued function can fail to have a tangent vector at a point either because the derivative in (4) does not exist or because the derivative is zero at the point.If a vector-valued function $\mathbf{r}(t)$ has a tangent vector $\mathbf{r}'(t_0)$ at a point onits graph, then the line that is parallel to $\mathbf{r}'(t_0)$ and passes through the tip of the radius vector $\mathbf{r}(t_0)$ is called the tangent line of the graph of $\mathbf{r}(t)$ at $\mathbf{r}(t_0)$

Vector equation of the tangent line is

$$\mathbf{r = r\,(t_0) + t\,r'(t_0)}$$

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EXAMPLE

Find parametric equation of the tangent line to the circular helix

x = cost, y = sint, z = 1 at the point where t = π/6

To find a vector equation of the tangent line, then we shall equate components to obtain the parametric equations. A vector equation r=r(t) of the helix is

$x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$

Thus, $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$

$\Rightarrow \mathbf{r}'(t) = (-\sin t)\mathbf{i} + (\cos t)\mathbf{j} + k$

At the point where $t = \pi/6$, these vectors are

$$\mathbf{r}\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}\mathbf{i} + \frac{1}{2}\mathbf{j} + \frac{\pi}{6}\mathbf{k} \quad \text{and}$$

$$\mathbf{r}'\left(\frac{\pi}{6}\right) = -\frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j} + \mathbf{k}$$

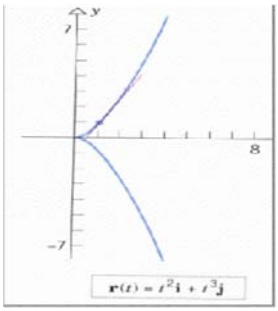
so from (5) with $t_0 = \pi/6$ a vector equation of the tangent line is

$$\mathbf{r}\left(\frac{\pi}{6}\right) + t\,\mathbf{r}'\left(\frac{\pi}{6}\right) = \left(\frac{\sqrt{3}}{2}\mathbf{i} + \frac{1}{2}\mathbf{j} + \frac{\pi}{6}\mathbf{k}\right) + t\left(-\frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j} + \mathbf{k}\right)$$

Simplifying, then equating the resulting components with the corresponding components of $r = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ yields the parametric equation.

$$x = \frac{\sqrt{3}}{2} - \frac{1}{2}t, \quad y = \frac{1}{2} + \frac{\sqrt{3}}{2}t, \quad z = \frac{\pi}{6} + t$$

EXAMPLE



The graph of $\mathbf{r}(t) = t^2\mathbf{i} + t^3\mathbf{j}$ is called a semicubical parabola

Find a vector equation of the tangent line to the graph of **r**(t) at

(a) the point (0,0) (b) the point (1,1)

The derivative of **r**(t) is

$\mathbf{r}'(t) = 2t\mathbf{i} + 3t^2\mathbf{j}$

(a) The point (0, 0) on the graph of **r** corresponds to t = 0. As this point we have $\mathbf{r}'(0) = 0$, so there is no tangent vector at the point and consequently a tangent line does not exist at this point.

(b) The point (1, 1) on the graph of r corresponds to t = 1, so from (5) a vector equation of the tangent line at this point is $\mathbf{r} = \mathbf{r}(1) + t\,\mathbf{r}'(1)$

From the formulas for r (t) and $\mathbf{r}'(t)$ with t = 1, this equation becomes

$$\mathbf{r} = (\mathbf{i} + \mathbf{j}) + t(2\mathbf{i} + 3\mathbf{j})$$

If **r** is a vector-valued function in 2-space or 3-space, then we say that **r**(t) is smoothly parameterized or that **r** is a smooth function of t if the components of r have continuous derivatives with respect to t and $\mathbf{r}'(t) \neq \mathbf{0}$ for any value of t. Thus, in 3-space $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$

is a smooth function of t if $x'(t)$, $y'(t)$, and $z'(t)$ are continuous and there is no value of t at which al three derivatives are zero. A parametric curve C in 2-space or 3-space will be called smooth if it is the graph of some smooth vector-valued function.

It can be shown that a smooth vector-valued function has a tangent line at every point on its graph.

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PROPERTIES OF DERIVATIVES

(Rules of Differentiation).

In either 2-space or 3-space let $\mathbf{r}(t)$, $\mathbf{r}_1(t)$, and $\mathbf{r}_2(t)$ be vector-valued functions, f(t) a real-valued function, k a scalar, and c a fixed (constant) vector. Then the following rules of differentiation hold:

$$\frac{d}{dt}[c] = 0$$

$$\frac{d}{dt}[kr(t)] = k\frac{d}{dt}[r(t)]$$

$$\frac{d}{dt}[\mathbf{r}_1(t) + \mathbf{r}_2(t)] = \frac{d}{dt}[\mathbf{r}_1(t)] + \frac{d}{dt}[\mathbf{r}_2(t)]$$

$$\frac{d}{dt}[\mathbf{r}_1(t) - \mathbf{r}_2(t)] = \frac{d}{dt}[\mathbf{r}_1(t)] - \frac{d}{dt}[\mathbf{r}_2(t)]$$

$$\frac{d}{dt}[f(t)\mathbf{r}(t)] = f(t)\frac{d}{dt}[\mathbf{r}(t)] + \mathbf{r}(t)\frac{d}{dt}[f(t)]$$

In addition to the rules listed in the foregoing theorem, we have the following rules for differentiating dot products in 2-space or 3-space and cross products in 3-space:

$$\frac{d}{dt}[\mathbf{r}_1(t).\mathbf{r}_2(t)] = \mathbf{r}_1.\frac{dr_2}{dt} + \frac{dr_1}{dt}.\mathbf{r}_2 \quad (6)$$

$$\frac{d}{dt}[\mathbf{r}_1(t)\times\mathbf{r}_2(t)] = \mathbf{r}_1\times\frac{dr_2}{dt} + \frac{dr_1}{dt}\times\mathbf{r}_2 \quad (7)$$

REMARK:

In (6), the order of the factors in each term on the right does not matter, but in (7) it does.

In plane geometry one learns that a tangent line to a circle is perpendicular to the radius at the point of tangency. Consequently, if a point moves along a circular arc in 2-space, one would expect the radius vector and the tangent vector at any point on the arc to be perpendicular. This is the motivation for the following useful theorem, which is applicable in both 2-space and 3-space.

THEOREM:

If r (t) is a vector-valued function in 2-space or 3-space and ||r(t)|| is constant for all t, then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$

that is, $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are orthogonal vectors for all t.It follows from (6) with $\mathbf{r}_1(t)=\mathbf{r}_2(t)=\mathbf{r}(t)$ that

$$\frac{d}{dt}[\mathbf{r}(t).\mathbf{r}(t)] = \mathbf{r}(t).\frac{d\mathbf{r}}{dt} + \frac{d\mathbf{r}}{dt}.\mathbf{r}(t)$$

or, equivalently, $\frac{d}{dt}[||\mathbf{r}(t)||]^2 = 2\mathbf{r}(t).\frac{d\mathbf{r}}{dt}$

But $||\mathbf{r}(t)||^2$ is constant, so its derivative is zero. Thus $2r(t).\frac{d\mathbf{r}}{dt} = 0$ that is $\mathbf{r}(t).\frac{d\mathbf{r}}{dt} = 0$

That is the $\mathbf{r}(t)$ is perpendicular $\frac{d\mathbf{r}}{dt}$

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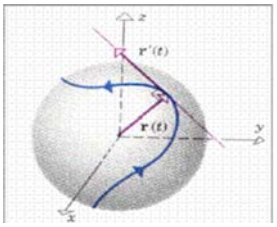
EXAMPLE

Just as a tangent line to a circular arc in 2-space is perpendicular to the radius at the point of tangency, so a tangent line to a curve on the surface of a sphere in 3-space is perpendicular to the radius at the point of tangency.



To see that this is so, suppose that the graph of **r**(t) lies on the surface of the sphere of radius k > 0 centered at the origin.For each value of *t* we have ||**r**(t)||=k,

$$\mathbf{r}(t).\ \mathbf{r}'(t) = 0$$

which implies that the radius vector **r**(t) and the tangent vector $\mathbf{r}'(t)$ are perpendicular. This completes the argument because the tangent line, where it exists, is parallel to the tangent vector.

INTEGRALS OF VECTOR VALUED FUNCTION

(a) If r(t)=x(t)I +y(t) j is a vector-valued function in 2-space ,the we define.

$$\int r(t)dt = (\int x(t)dt)i + (\int y(t)dt)\,j \qquad (1a)$$

$$\int_a^b r(t)dt = (\int_a^b x(t)dt)i + (\int_a^b y(t)dt)\,j \qquad (1b)$$

(b) If r(t) = x (t) **i** + y(t) **j** + z(t)k is a vector-valued function in 3-space, then we define.

$$\int \mathbf{r}(t)dt = \left(\int x(t)dt\right)\mathbf{i} + \left(\int y(t)\,dt\right)\mathbf{j} + \left(\int z(t)dt\right)\mathbf{k} \qquad \textbf{(2a)}$$

$$\int_a^b r(t)dt = \left(\int_a^b x(t)dt\right)\mathbf{i} + \left(\int_a^b y(t)dt\right)\mathbf{j} + \left(\int_a^b z(t)dt\right)\mathbf{k} \qquad \textbf{(2b)}$$

$Let\ r(t) = 2ti + 3t^2 j$

$(a)\int r(t)dt \qquad (b)\int_0^2 r(t)dt$

$$\int r(t)dt = \int (2ti + 3t^2 j)dt = (\int 2tdt)i + (\int 3t^2 dt)\,j$$

$$(t^2 + C_1)i + (t^3 + C_2)\,j = t^2 i + C_1 i + t^3 j + C_2 j = t^2 i + t^3 j + C_1 i + C_2 j = t^2 i + t^3 j + C$$

$Where\ C = C_1 i + C_2 j\ is\ an\ arbitrary\ vector\ constant\ of\ integration$

$$(b)\int_0^2 r(t)dt = \int_0^2 (2ti + 3t^2 j)dt = (\int_0^2 2tdt)i + (\int_0^2 3t^2 dt)\,j = \left[t^2\right]_0^2 i + \left[t^3\right]_0^2 j = (2^2 - 0)i + (2^3 - 0)\,j = 4i + 8j$$

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PROPERTEIS OF INTEGRALS

$$\int cr(t)\,dt = c\int r(t)\,dt \qquad (3)$$

$$\int [r_1(t)+r_2(t)]\,dt = \int r1(t)dt + \int r_2(t)\,dt \qquad (4)$$

$$\int [r_1(t)-r_2(t)]dt = \int r_1(t)\,dt - \int r_2(t)\,dt \qquad (5)$$

These properties also hold for definite integrals of vector-valued functions. In addition, we leave it for the reader to show that if r is a vector-valued function in 2-space or 3-space, then

$$\frac{d}{dt}\left[\int r(t)\,dt\right] = r(t) \qquad (6)$$

This shows that an indefinite integrals of **r**(t) is, in fact, the set of antiderivatives of r(t), just as for real-valued functions.

If **r**(t) is any antiderivative or **r**(t) in the sense that $\mathbf{R}'(t) = \mathbf{r}(t)$, then

$$\int r(t)\,dt = \mathbf{R}(t) + \mathbf{C} \qquad (7)$$

where C is an arbitrary vector constant of integration. Moreover,

$$\int_a^b \mathbf{r}(t)\,dt = \mathbf{R}(t)\Big]_a^b = \mathbf{R}(b) - \mathbf{R}(a).$$

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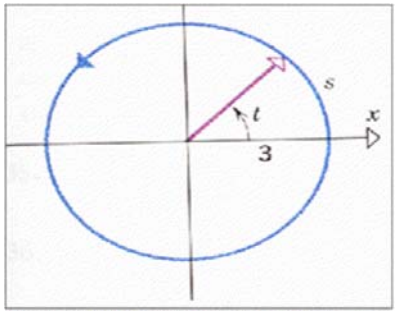
Lecture No -29 Change of parameter

It is possible for different vector-valued functions to have the same graph.
For example, the graph of the function
$r = (3 \cos t)\mathbf{i} + (3 \sin t)\mathbf{j}$, $0 \leq t \leq 2\pi$
is the circular of radius 3 centered at the origin with counterclockwise orientation. The parameter t can be interpreted geometrically as the positive angle in radians from the x-axis to the radius vector. For each value of t, let s be the length of the arc subtended by this angle on the circle



The parameters s and t are related by
$t = s/3$, $0 < s < 6\pi$
if we substitute this in (10), we obtain a vector-valued function of the parameter s, namely
$r = 3 \cos (s/3)\mathbf{i} + 3 \sin (s/3)\mathbf{j}$, $0 \leq s \leq 6\pi$
whose graph is also the circle of radius 3 centered at the origin with counterclockwise orientation .In various problems it is helpful to change the parameter in a vector-valued function by making an appropriate substitution. For example, we changed the parameter above from t to s by substituting t=s/3 in (10).
In general, if g is a real-valued function, then substituting $t = g(u)$ in r(t) changes the parameter from t to u.

When making such a change of parameter, it is important to ensure that the new vector-valued function of u is smooth if the original vector-valued function of t is smooth. It can be proved that this will be so if g satisfies the following conditions:

1. g is differentiable.
2. g' is continuous.
3. $g'(u) \neq 0$ for any u in the domain of g.
4. The range of g is the domain of **r**.

If g satisfies these conditions, then we call $t = g(u)$ a smooth change of parameter. Henceforth, we shall assume that all changes of parameter are smooth, even if it is not stated explicitly.

ARC LENGTH

Because derivatives of vector-valued functions are calculated by differentiating components, it is natural do define integrals of vector-functions in terms of components.

EXAMPLE

If $x'(t)$ and $y'(t)$ are continuous for $a \leq t \leq b$, then the curve given by the parametric equations
$x = x(t)$, $y = y(t)$ $(a \leq t \leq b)$ (9)
has arc length

$$\mathbf{L} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \qquad (10)$$

This result generalizes to curves in 3-spaces exactly as one would expect:
If $x'(t)$, $y'(t)$, and $z'(t)$ are continuous for $a \leq t \leq b$, then the curve given by the parametric equations
$x = x(t)$, $y = y(t)$, $z = z(t)$ $(a \leq t \leq b)$

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has arc length

$$\mathbf{L} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt \qquad (12)$$

EXAMPLE

Find the arc length of that portion of the circular helix

x = cos t, y = sin t, z = t

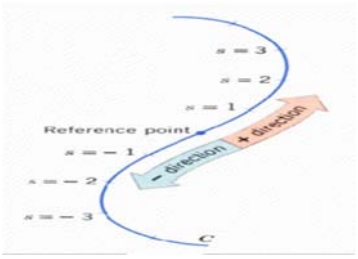
From t = 0 to t = π

The arc length is

$$\mathbf{L} = \int_0^{\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt = \int_0^{\pi} \sqrt{(-\sin t)^2 + (\cos t)^2 + 1}\, dt$$

$$= \int_0^{\pi} \sqrt{2}\ dt = \sqrt{2}\ \pi$$

ARC LENTH AS A PARAMETER



For many purposes the best parameter to use for representing a curve in 2-space or 3-space parametrically is the length of arc measured along the curve from some fixed reference point. This can be done as follows:

Step 1: Select an arbitrary point on the curve C to serve as a reference point.

Step 2: Starting from the reference point, choose one direction along the curve to be the positive direction and the other to be the negative direction.

Step 3: If P is a point on the curve, let s be the "signed" arc length along C from the reference point to P, where s is positive if P is in the positive direction from the reference point, and s is negative if P is in the negative direction.

By this procedure, a unique point P on the curve is determined when a value for s is given. For example, s = 2 determines the point that is 2 units along the curve in the positive direction from the reference point, and $s = -\frac{3}{2}$ determines the point that is $\frac{3}{2}$ units along the curve in the negative direction from the reference point.

Let us now treat s as a variable. As the value of s changes, the corresponding point P moves along C and the coordinates of P become functions of s. Thus, in 2-space the coordinates of P are (x(x,), y(s)), and in 3-space they are (x(s), y(s), z(s)). Therefore, in 2-space the curve C is given by the parametric equations x = x(s), y = y (s)

and in 3-space by x = x(s), y = y(s), z = z (s)

REMARKS

When defining the parameter s, the choice of positive and negative directions is arbitrary. However, it may be that the curve C is already specified in terms of some other parameter t, in which case we shall agree always to take the direction of increasing t as the positive direction for the parameter s. By so doing, s will increase as t increases and vice versa.

The following theorem gives a formula for computing an arc-length parameter s when the curve C is expressed in terms of some other parameter t. This result will be used when we want to change the parameterization for C from t to s.

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THEOREM

(a) Let C be a curve in 2-space given parametrically by

$x = x(t)$, $y = y(t)$

where $x'(t)$ and $y'(t)$ are continuous functions. If an arc-length parameter s is introduced with its reference point at $(x(t_0), y(t_0))$, then the parameters s and t are related by

$$s = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du \qquad (13a)$$

(b) Let C be a curve in 3-space given parametrically by

$x = x(t)$, $y = y(t)$, $z = z(t)$

where $x'(t)$, $y'(t)$, and $z'(t)$ are continuous functions. If an arc-length parameter s is introduced with its reference point at $(x(t_0), y(t_0), z(t_0))$, then the parameters s and t are related by

$$s = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2 + \left(\frac{dz}{du}\right)^2}\, du \qquad (13b)$$

Proof

If $t > t_0$, then from (10) (with *u* as the variable of integration rather than *t*) it follows that

$$\int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du \qquad (14)$$

represents the arc length of that portion of the curve C that lies between $(x(t_0), y(t_0))$ and $(x(t), y(t))$. If $t < t_0$, then (14) is the negative of this arc length. In either case, integral (14) represents the "signed" arc length *s* between these points, which proves (13a).

It follows from Formulas (13a) and (13b) and the Second Fundamental Theorem of Calculus (Theorem 5.9.3) that in 2-space.

$$\frac{ds}{dt} = \frac{d}{dt}\left[\int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du\right] = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

and in 3-space

$$\frac{ds}{dt} = \frac{d}{dt}\left[\int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2 + \left(\frac{dz}{du}\right)^2}\, dt\right] = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$$

Thus, in 2-space and 3-space, respectively,

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \qquad (15a)$$

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \qquad (15b)$$

REMARKS:

Formulas (15a) and (15b) reveal two facts worth noting. First, ds/dt does not depend on t_0; that is, the value of ds/dt is independent of where the reference point for the parameter s is located. This is to be expected since changing the position of the reference point shifts each value of s by a constant (the arc length between the reference points), and this constant drops out when we differentiate. The second fact to be noted from (15a) and (15b) is that $ds/dt \geq 0$ for all t. This is also to be expected since s increases with t by the remark preceding Theorem 15.3.2. If the curve C is smooth, then it follows from (15a) and (15b) that $ds/dt \geq 0$ for all t .

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EXAMPLE

$x = 2t + 1,\ y = 3t - 2$ (16)

using arc length s as a parameter, where the reference point for s is the point (1, – 2).

In formula (13a) we used u as the variable of integration because t was needed as a limit of integration. To apply (13a), we first rewrite the given parametric equations with u in place of t; this gives

from which we obtain

$x = 2u + 1, \quad y = 3u - 2$

$\frac{dx}{du} = 2, \quad \frac{dy}{du} = 3$

we see that the reference point (1,–2) corresponds to $t = t_0 = 0$

$$s = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du = \int_{t_0}^{t} \sqrt{13}\, du = \sqrt{13u}\,\Big]_{u=0}^{u=t} = \sqrt{13t}$$

Therefore, $t = \frac{1}{\sqrt{13}}\, s$

Substituting this expression in the given parametric equations yields.

$$x = 2\left(\frac{1}{\sqrt{13}}s\right) + 1 = \frac{2}{\sqrt{13}}\, s + 1$$

$$y = 3\left(\frac{1}{\sqrt{13}}s\right) - 2 = \frac{3}{\sqrt{13}}\, s - 2$$

EXAMPLE

Find parametric equations for the circle $x = a \cos t,\ y = a \sin t\ \ (0 \le t \le 2\pi)$ using arc length s as a parameter, with the reference point for s being (a, 0), where $a > 0$.

We first replace t by u in the given equations so that $x = a \cos u, \quad y = a \sin u$

And $\frac{dx}{du} = -a \sin u, \quad \frac{dy}{du} = a \cos u$

Since the reference point (a, 0) corresponds to t = 0, we obtain

$$s = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du = \int_{t_0}^{t} \sqrt{(-a \sin u)^2 + (a \cos u)^2}\, du = \int_{0}^{t} a\, du = au\,\Big]_{u=0}^{u=t} = at$$

Solving for t in terms of s yields $t = s/a$

Substituting this in the given parametric equations and using the fact that $s = at$ ranges from 0 to $2\pi a$ as t ranges from 0 to 2π, we obtain

$x = a\cos(s/a),\ y = a \sin(s/a)\ (0 \le s \le 2\pi a)$

Example

Find Arc length of the curve $\mathbf{r}(t) = t^3\mathbf{i} + t\mathbf{j} + \frac{1}{2}\sqrt{6}\, t^2\, \mathbf{k},\ 1 \le t \le 3$

Here $x = t^3,\ y = t,\ z = \frac{1}{2}\sqrt{6}\, t^2$

$\frac{dx}{dt} = 3t^2, \quad \frac{dy}{dt} = 1, \quad \frac{dz}{dt} = \sqrt{6}\, t$

$$\text{Arc length} = \int_1^3 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt = \int_1^3 \sqrt{9t^4 + 1 + 6t^2}\, dt = \int_1^3 \sqrt{(3t^2 + 1)^2}\, dt$$

$$= \left|t^3 + t\right|_1^3 = (3)^3 + 3 - (1)^3 - 1 = 27 + 3 - 1 - 1 = 28$$

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EXAMPLE

Calculate $\frac{\mathbf{dr}}{\mathbf{du}}$ by chain Rule.

$\mathbf{r} = e^{t}\mathbf{i} + 4e^{-t}\mathbf{j}$

$\frac{dr}{dt} = e^{t}\mathbf{i} - 4e^{-t}\mathbf{j}$

$\frac{dt}{du} = 2u$

$\frac{d\mathbf{r}}{du} = \frac{d\mathbf{r}}{dt} \cdot \frac{dt}{du} = (e^{t}\mathbf{i} - 4e^{t}\mathbf{j}).(2u) = 2u\, e^{u2}\mathbf{i} - 8ue^{-u^2}\mathbf{j}$

By expressing **r** in terms of u

$R = e^{u}\mathbf{i} + 4e^{-u^2}\mathbf{j}$

$\frac{d\mathbf{r}}{du} = 2u\, e^{u^2}\mathbf{i} - 8ue^{-u^2}\mathbf{j}$

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Lecture No -30 Exact Differential

If $z = f(x, y)$, then $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$

The result can be extended to functions of more than two independent variables.

If $z = f(x, y, w)$, $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy + \frac{\partial z}{\partial w} dw$

Make a note of these results in differential form as shown.

Exercise

Determine the differential dz for each of the following functions.

1. $z = x^2 + y^2$
2. $z = x^3 \sin 2y$
3. $z = (2x - 1)\, e^{3y}$
4. $z = x^2 + 2y^2 + 3w^2$
5. $z = x^3 y^2 w$.

Finish all five and then check the result.

1. $dz = 2 (x\, dx + y\, dy)$
2. $dz = x^2 (3 \sin 2y\, dx + 2x \cos 2y\, dy)$
3. $dz = e^{3y} \{2dx + (6x - 3)\, dy\}$
4. $dz = 2 (xdx + 2ydy + 3wdw)$
5. $dz = x^2 y (3ywdx + 2xwdy + xydw)$

Exact Differential

We have just established that if $z=f(x, y)$

$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$

We now work in reverse.

Any expression $dz = Pdx + Qdy$, where P and Q are functions of x and y, is an exact differential if it can be integrated to determine z.

$\therefore\ P = \frac{\partial z}{\partial x}$ and $Q = \frac{\partial z}{\partial y}$

Now $\frac{\partial P}{\partial y} = \frac{\partial^2 z}{\partial y \partial x}$ and $\frac{\partial Q}{\partial x} = \frac{\partial^2 z}{\partial x \partial y}$ and we know that $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$

Therefore, for dz to be an exact differential $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ and this is the test we apply.

Example

$dz = (3x^2 + 4y^2)\, dx + 8xy\, dy$.

If we compare the right-hand side with $Pdx + Qdy$, then

$P = 3x^2 + 4y^2 \ \therefore\ \frac{\partial P}{\partial y} = 8y$

$Q = 8xy \qquad \therefore\ \frac{\partial Q}{\partial x} = 8y$

$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \qquad \therefore$ dz is an exact differential

Similarly, we can test this one.

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Example

$dz = (1 + 8xy)\,dx + 5x^2\,dy$.

From this we find dz is not an exact differential

for $dz = (1 + 8xy)\,dx + 5x^2\,dy$

$\therefore\ P = 1 + 8xy \quad \therefore\ \frac{\partial P}{\partial y} = 8x$

$Q = 5x^2 \qquad \therefore\ \frac{\partial Q}{\partial x} = 10x$

$\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$ $\therefore$ dz is not an exact differential

Exercise

Determine whether each of the following is an exact differential.

1. $dz = 4x^3y^3dx + 3x^4y^2\,dy$
2. $dz = (4x^3y+2xy^3)\,dx+(x^4+3x_2y^2)\,dy$
3. $dz=(15y^2e^{3x}+2xy^2)dx+(10ye^{3x}+x^2y)dy$
4. $dz=(3x^2e^{2y}-2y^2e^{3x})dx+(2x^3e^{2y}-2ye^{3x})dy$
5. $dz=(4y^3\cos 4x+3x^2\cos 2y)dx+(3y^2\sin 4x-2x^3\sin 2y)\,dy$.

1. Yes 2. Yes 3. No 4. No 5. Yes

We have just tested whether certain expressions are, in fact, exact differentials–and we said previously that, by definition, an exact differential can be integrated. But how exactly do we go about it? The following examples will show.

Integration Of Exact Differentials

$dz = Pdx+Qdy$ where $P=\frac{\partial z}{\partial x}$ and $Q=\frac{\partial z}{\partial y}$

$\therefore\ z = \int P\,dx$ and also $z = \int Q\,dy$

Example

$dz = (2xy + 6x)\,dx + (x2 + 2y3)\,dy$.

$P = \frac{\partial z}{\partial x} = 2xy + 6x \quad \therefore\ z=\int (2xy+6x)dx$

$\therefore\ z = x^2y + 3x^2 + f(y)$ where f(y) is an arbitrary function of y only, and is akin to the constant of integration in a normal integral.

Also

$Q = \frac{\partial z}{\partial y} = x^2 + 2y^3 \quad \therefore\ z = \int (x^2+2y^3)\,dy$

$\therefore\ z = x2y + \frac{y^4}{2} + F(x)$ where F(x) is an arbitrary function of x only.

$z = x^2y + 3x^2 + f(y)$ (i)

and $z = x^2y + \frac{y^4}{2} + F(x)$ (ii)

For these two expressions to represent the same function, then

f(y) in (i) must be $\frac{y^4}{2}$ already in (i)

and F(x) in (ii) must be $3x^2$ already in (i)

$\therefore\ z = x^2y + 3x^2 + \frac{y^4}{2}$

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Example

Integrate $dz = (8e^{4x} + 2xy^2)\,dx + (4\cos 4y + 2x^2y)\,dy$.

Argue through the working in just the same way, from which we obtain $z = 2e^{4x} + x^2y^2 + \sin 4y$

Here it is.

$dz = (8e^{4x} + 2xy^2)\,dx + (4\cos 4y + 2x^2y)\,dy$

$$P = \frac{\partial x}{\partial x} = 8e^{4x} + 2xy^2$$

$$\therefore\ z = \int (8e^{4x} + 2xy^2)\,dx$$

$$\therefore\ z = 2e^{4x} + x^2y^2 + f(y) \qquad \text{(i)}$$

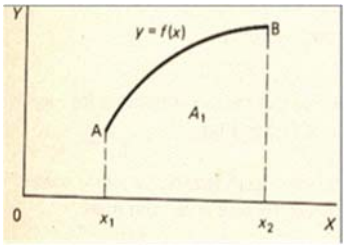
$$Q = \frac{\partial z}{\partial y} = 4\cos 4y + 2x^2y$$

$$\therefore\ z = \int (4\cos 4y + 2x^2y)\,dy$$

$$\therefore\ z = \sin 4y + x^2y^2 + F(x) \qquad \text{(ii)}$$

For (i) and (ii) to agree, $f(y) = \sin 4y$ and $F(x) = 2e^{4x}$

$$\therefore\ z = 2e^{4x} + x^2y^2 + \sin 4y$$



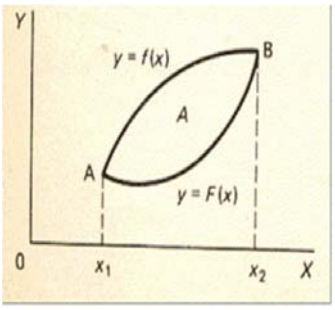
Area enclosed by the closed curve

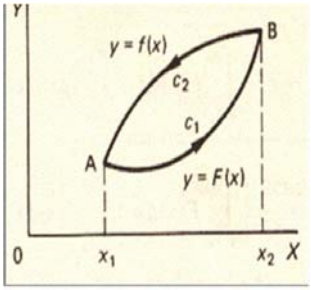
One of the earliest application of integration is finding the area of a plane figure bounded by the x-axis, the curve $y = f(x)$ and ordinates at $x=x_1$ and $x=x_2$.

$$A_1 = \int_{x_1}^{x_2} y\,dx = \int_{x_1}^{x_2} f(x)\,dx$$

If points A and B are joined by another curve, $y = F(x)$

$$A_2 = \int_{x_1}^{x_2} f(x)\,dx$$





Combining the two figures, we have

$$A = A_1 - A_2 \qquad \therefore\ A = \int_{x_1}^{x_2} f(x)\,dx - \int_{x_1}^{x_2} f(x)\,dx$$

The final result above can be written in the form

$$A = -\oint y\,dx$$

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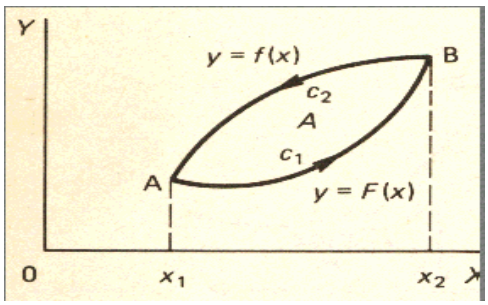
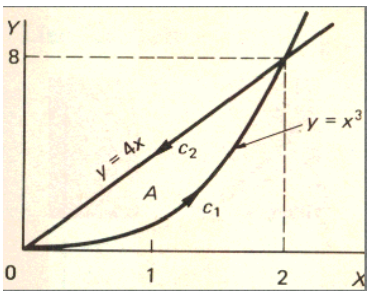
Where the symbol $\oint$ indicates that the integral is to be evaluated round the closed boundary in the positive



Example

Determine the area enclosed by the graph of $y = x^3$ and $y = 4x$ for $x \geq 0$.



First we need to know the points of intersection. These are $x = 0$ and $x = 2$

We integrate in a an anticlockwise manner

c_1: $y = x^3$, limits $x = 0$ to $x = 2$

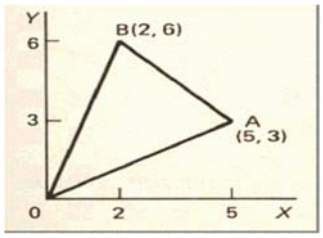
c_2: $y = 4x$, limits $x = 2$ to $x = 0$.

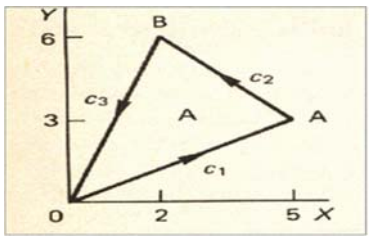
$A = -\oint y\,dx = A = 4$ square units

For $A = -\oint ydx = -\left\{\int_1^2 x^3dx + \int_2^0 4xdx\right\} = -\left\{\left(\frac{x^4}{4}\right)_0^2 + \left[2x^2\right]_0^2\right\} = 4$

Example

Find the area of the triangle with vertices (0, 0), (5, 3) and (2, 6).





The equation of OA is $y = \frac{3}{5}x$, BA is $y = 8 - x$, OB is $y = 3x$

Then $A = -\oint y\,dx$

Write down the component integrals with appropriate limits.

$$A = -\oint ydx = -\left\{\int_0^5 \frac{3}{5}xdx + \int_5^2 (8-x)dx + \int_2^0 3xdx\right\}$$

The limits chosen must progress the integration round the boundary of the figure in an anticlockwise manner. Finishing off the integration, we have

$A = 12$ square units

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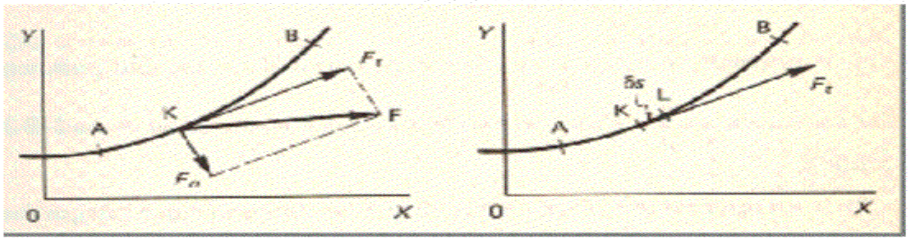
The actual integration is easy enough. The work we have just done leads us on to consider line integrals, so let us make a fresh start in the next frame.

Line Integrals

If a field exists in the xy-plane, producing a force F on a particle at K, then F can be resolved into two components.F_1 along the tangent to the curve AB at K. Fa along the normal to the curve AB at K.

Line Integrals



The work done in moving the particle through a small distance δs from K to L along the curve is then approximately F_1 δs. So the total work done in moving a particle along the curve from A to B is given by

$\lim_{\delta \to 0} \sum F_t \, \delta s = \int F_t \, ds$ from A to B

This is normally written $\int_{AB} F_t \, ds$ where A and B are the end points of the curve,

or as $\int_{C} F_t \, ds$ where the curve c connecting A and B is defined.

Such an integral thus formed, is called a line integral since integration is carried out along the path of the particular curve c joining A and B.

$\therefore \; I = \int_{AB} F_t \, dx = \int_{C} F_t \, ds$

where c is the curve y = f(x) between A(x_1, y_2) and B (x_2, y_2).

There is in fact an alternative form of the integral which is often useful, so let us also consider that.

Alternative form of a line integral

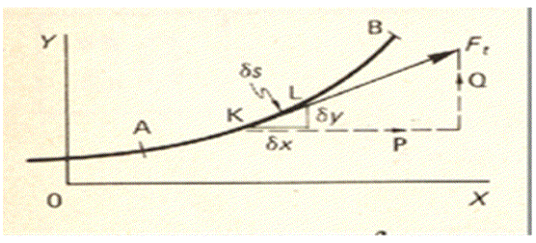
It is often more convenient to integrate with respect to x or y than to take arc length as the variable.

If F_t has a component

P in the x-direction

Q in the y-direction

then the work done from K to L can be stated as Pδx + Qδy



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$\therefore \int_{AB} F_t \, ds = \int_{AB} (P\,dx + Q dy)$

where P and Q are functions of x and y.

In general then, the line integral can be expressed as

$$I = \int_C F_t \, ds = \int_C (P\,dx + Q dy)$$

where c is the prescribed curve and F, or P and Q, are functions of x and y.

Make a note of these results –then we will apply them to one or two examples.

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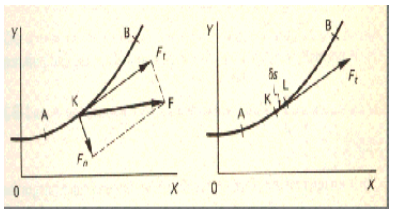
Lecture No -31 Line Integral

The work done in moving the particle through a small distance δs from K to L along the curve is then approximately F_1 δs. So the total work done in moving a particle along the curve from A to B is given by



$\lim_{\delta\to 0} \Sigma F_t\, \delta s = \int F_t\, ds$ from A to B

This is normally written $\int_{AB} F_t\, ds$ where A and B are the end points of the curve, or as $\int_C F_t\, ds$ where the curve c connecting A and B is defined.Such an integral thus formed, is called a line integral since integration is carried out along the path of the particular curve c joining A and B.

$\therefore\ I = \int_{AB} F_t\, dx = \int_C F_t\, ds$

where c is the curve y = f(x) between A(x_1, y_2) and B (x_2, y_2).

There is in fact an alternative form of the integral which is often useful, so let us also consider that.

Alternative form of a line integral

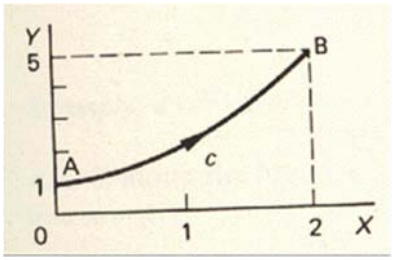
It is often more convenient to integrate with respect to x or y than to take arc length as the variable.

If F_t has a component ,P in the x-direction ,Q in the y-direction

then the work done from K to L can be stated as Pδx + Qδy

Example 1

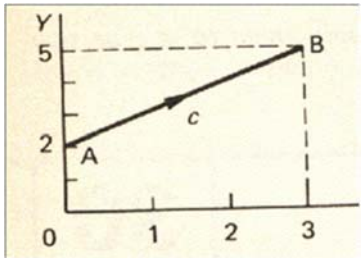
Evaluate $\int_C (x + 3y)\, dx$ from A (0, 1) to B (2, 5) along the curve $y = 1 + x^2$.



The line integral is of the form

$\int_C (P\, dx + Q dy)$ where, in this case, Q = 0 and c is the curve $y = 1 + x^2$.

It can be converted at once into an ordinary integral by substituting for y and applying the appropriate limits of x.



$$I = \int_C (Pdx+Qdy) = \int_C (x+3y)dx = \int_C^2 (x+3+3x^2)dx$$

$$= \left[\frac{x^2}{2}+3x+x^3\right]_0^2 = 16$$

Now for another, so turn on.

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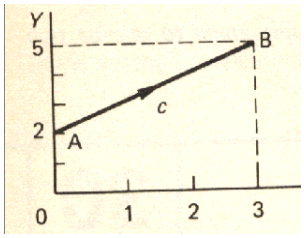
Example 2

Evaluate $I = \int_C (x^2 + y)\,dx + (x - y^2)dy$ from A (0, 2) to B (2, 5) along the curve $y = 2 + x$.

$I = \int_C (Pdx + Qdy)$



$P = x^2 + y = x^2 + 2 + x = x^2 + x + 2$

$Q = x - y^2 = x - (4 + 4x + x^2) = -(x^2 + 3x + 4)$

Also $y = 2 + x$

$\therefore$ $dy = dx$ and the limits are $x=0$ to $x=3$

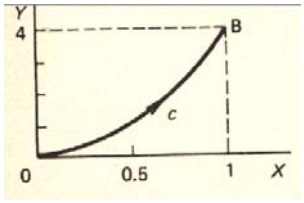
$\therefore$ $I = -15$

for $I = \int_0^2 \{(x^2+x+2)\,dx - (x^2+3x+4)\,dx\}$ or $\int_0^2 -(2x+2)\,dx = \left[-x^2 - 2x\right]_0^2 = -4 - 4 = -8$

Example 3

Evaluate $I = \int_C \{(x^2+2y)dx + xydy\}$ from O(0, 0) to B(1, 4) along the curve $y=4x^2$.

In this case, c is the curve y = 4x2.



$\therefore$ $dy = 8x\,dx$

Substitute for y in the integral and apply the limits.

Then $I = 9.4$

for $I = \int_C \{(x^2+2y)\,dx + xydy\}$

$y = 4x^2$ $\therefore$ $dy = 8xdx$

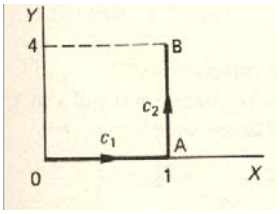
also $x^2 + 2y = x^2 + 8x^2 = 9x^2$; $xy = 4x^3$

$\therefore$ $I = \int_0^1 \{9x^2dx + 32x^4dx\} = \int_0^1 (9x^2 + 32x^4)\,dx = 9.4$

They are all done in very much the same way.

Example 4

Evaluate $I = \int_C \{(x^2 + 2y)\,dx + xydy\}$ from O(0, 0) to A (1, 0) along the line y = 0 and then from A (1, 0) to B (1, 4) along the line x = 1.



(i) OA : c1 is the line $y = 0$ $\therefore$ $dy = 0$.

Substituting $y = 0$ and $dy = 0$ in the given integral gives.

$I_{OA} = \int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$

(ii) AB: Here c_2 is the line $x = 1$ $\therefore$ $dx=0$

$\therefore$ $I_{AB} = 8$

For $I_{AB} = \int_0^4 \{(1 + 2y)(0) + ydy\} = \int_0^4 ydy = \left[\frac{y^2}{2}\right]_0^4 = 8$

Then $I = I_{OA} + I_{AB} = \frac{1}{3} + 8 = 8\frac{1}{3}$ $\therefore$ $I = 8\frac{1}{3}$

If we now look back to Example 3 and 4 just completed, we find that we have evaluated the same integral between the same two end points, but along different paths of integration.If we combine the two diagrams, we have

where c is the curve $y = 4x^2$ and c1 + c^2 are the lines

y = 0 and x = 1. The result obtained were

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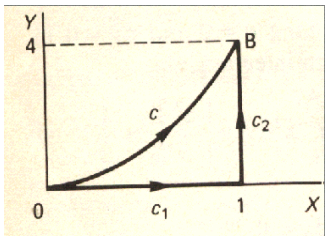
$I_c = 9\frac{2}{3}$ and $I_{c_1+c_2} = 8\frac{1}{3}$



Notice therefore that integration along two distinct paths joining the same two end points does not necessarily give the same results.

Properties of line integrals

1. $\int_C Fds = \int_C \{Pdx + Qdy\}$

2. $\int_{AB} Fds = -\int_{BA} Fds$ and $\int_{AB} \{Pdx+Qdy\} = \int_{BA} \{Pdx+Qdy\}$

i.e. the sign of a line integral is reversed when the direction of the integration along the path is reversed.

3. (a) For a path of integration parallel to the y-axis, i.e. x = k, dx = 0

$\therefore \int_C Pdx = 0 \quad \therefore I_C = \int_C Qdy.$

(b) For a path of integration parallel to the x-axis, i.e. y = k, dy = 0.

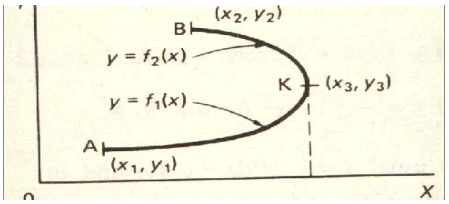
$\therefore \int_C Qdy=0 \therefore I_C=\int_C Pdx.$

4. If the path of integration c joining A to B is divided into two parts AK and KB, then

$I_c = I_{AB} = I_{AK} + I_{KB}$.

5 .If the path of integration c is not single valued for part of its extent, the path is divided into two sections.

$y = f_1(x)$ from A to K, $y = f_2(x)$ from K to B.



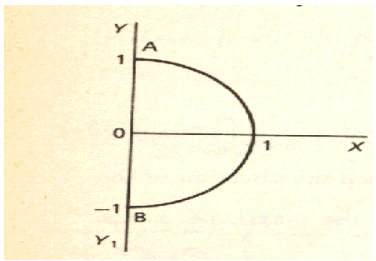
6. In all cases, the actual path of integration involved must be continuous and single-valued.

Example

Evaluate $I = \int_C (x + y)\,dx$ from A(0, 1) to B (0, – 1) along the semi-circle $x^2+y^2=1$ for $x \geq 0$.The first thing we notice is that the path of integration c is not single-valued For any value of x, $y = \pm\sqrt{1 - x^2}$. Therefore, we divided c into two parts

(i) $y = \sqrt{1 - x^2}$ from A to K

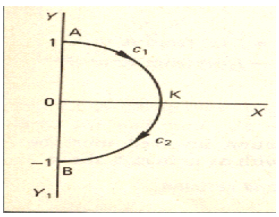
(ii) $y = -\sqrt{1 - x^2}$ from K to B



As usual, $I = \int_C (Pdx + Qdy)$ and in this particular case, Q = 0

$$\therefore I = \int_C Pdx = \int_0^1 (x+\sqrt{1 - x^2})\,dx + \int_1^0 (x\sqrt{1 - x^2})\,dx$$

$$= \int_0^1 (x + \sqrt{1 - x^2} - x+\sqrt{1 - x^2}\,dx = 2\int_0^1 \sqrt{1 - x^2}\,dx$$



Now substitute $x = \sin\theta$ and finish it off. $I = \frac{\pi}{2}$

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for $I = 2\int_0^1 \sqrt{1-x^2}dx$ $\quad x = \sin\theta$

$\therefore \quad dx = \cos\theta\, d\theta \quad \sqrt{1-x^2} = \cos\theta$

Limits : $x = 0, \quad \theta = 0; \quad x = 1, \quad \theta = \frac{\pi}{2}$

$$\therefore \quad I = 2\int_0^{\pi/2} \cos 2\theta d\theta = \int_0^{\pi/2} (1+\cos 2\theta)d\theta = \left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\pi/2} = \frac{\pi}{2}$$

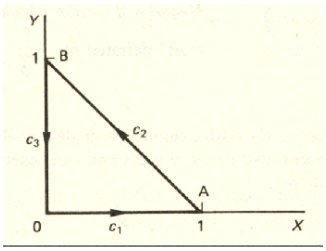
Now let us extend this line of development a stage further.

Example

Evaluate the line integral

$I = \oint (x^2dx - 2xy\, dy)$ where c comparises the three sides of the triangle joining O(0, 0), A (1, 0) and B (0, 1).

First draw the diagram and mark in c_1, c_2 and c_3, the proposed directions of integration. Do just that.The three sections of the path of integration must be arranged in an anticlockwise manner round the figure.



Now we deal with each pat separately.

(a) OA : c_1 is the line $y = 0$

Therefore, $dy = 0$.

Then $I = \oint (x^2dx - 2xy\, dy)$ for this part becomes

$$I_1 = \int_0^1 x^2dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3} \text{ therefore } I_1 = \frac{1}{3}$$

(b) AB : for c_2 is the line $y = 1 - x$

$\therefore dy = -dx.$

$$I_2 = \int_0^1 \{x^2dx + 2x(1-x)dx\} = \int_0^1 (x^2+2x-2x^2)dx = \int_0^1 (2x-x^2)\, dx = \left[x^2 - \frac{x^3}{3}\right]_0^1 = -\frac{2}{3}$$

$$\therefore \qquad I_2 = -\frac{2}{3}$$

Note that anticlockwise progression is obtained by arranging the limits in the appropriate order.

Now we have to determine I_3 for BO.

(c) BO: c_3 is the line $x = 0$

$\therefore \quad dx = 0 \qquad \therefore I_3 = \int 0dy = 0 \quad \therefore \quad I_3 = 0$

Finally, $I = I_1 + I_2 + I_3 = \frac{1}{3} - \frac{2}{3} + 0 = -\frac{1}{3} \qquad \therefore \quad I = -\frac{1}{3}$

Example

Evaluate $\oint_c y\, dx$ when c is the circle $x^2+y^2 = 4$.

$x^2 + y^2 = 4 \qquad \therefore \qquad y = \pm\sqrt{4-x^2}$

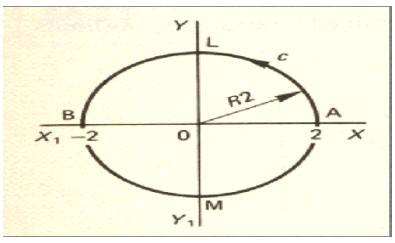
y is thus not single-valued. Therefore use

$y = \sqrt{4-x^2}$ for ALB between

$x = 2$ and $x = -2$ and

$y = -\sqrt{4-x^2}$ for BMA between

$x = -2$ and $x = 2$.



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$$\therefore\ I = \int_{2}^{-2} \sqrt{4 - x^2}\, dx + \int_{-2}^{2} \{-\sqrt{4 - x^2}\}\, dx$$

$$= 2\int_{2}^{-2} \sqrt{4 - x^2}\, dx = -2\int_{-2}^{2} \sqrt{4 - x^2}\, dx \qquad = -4\int_{0}^{2} \sqrt{4 - x^2}\, dx.$$

To evaluate this integral, substitute $x = 2 \sin \theta$ and finish it off. $I = -4\pi$

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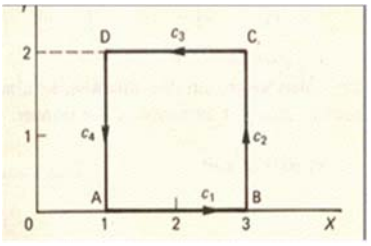
Lecture No -32 Examples

Example

Evaluate $I = \oint \{xydx+(1+y^2)dy\}$ where c is the boundary of the rectangle joining A(1,0), B (3, 0), C(3, 2), D (1, 2).

First draw the diagram and insert c_1,c_2,c_3,c_4.



That give

Now evaluate I_1 for AB; I_2 for BC; I_3 for CD; I_4 for DA; and finally I.

$I_1 = 0$; $I_2 = 4\frac{2}{3}$; $I_3 = -8$ $I_4 = -4\frac{2}{3}$; $I = -8$

Here is the complete working.

$I = \oint \{xydx + (1 + y^2)\, dy\}$

(a) AB: c_1 is $y = 0$ $\therefore$ $dy = 0$ $\therefore$ $I_1 = 0$

(b) BC: c_2 is $x = 3$ $\therefore$ $dx = 0$

$$\therefore I_2 = \int_0^2 (1+y^2)dy = \left[y + \frac{y^3}{3}\right]_0^2 = 4\frac{2}{3} \quad \therefore I_2 = 4\frac{2}{3}$$

(c) CD : c_3 is $y = 2$ $\therefore$ $dy = 0$

$$\therefore I_3 = \int_3^1 2xdx = \left[x^2\right]_3^1 = -8 \quad \therefore I_3 = -8$$

(d) DA : c_4 is $x = 1$ $\therefore$ $dx = 0$

$$\therefore I_4 = \int_2^0 (1+y^2)\, dy = \left[y + \frac{y^3}{3}\right]_2^0 = -4\frac{2}{3}$$

Finally $I = I_1 + I_2 + I_3 + I_4 = 0 + 4\frac{2}{3} - 8 - 4\frac{2}{3} = -8$ $\therefore I = -8$

Remember that, unless we are directed otherwise, we always proceed round the closed boundary in an anticlockwise manner.

Line integral with respect to arc length

We have already established that

$$I = \int_{AB} F_t ds = \int_{AB} \{Pdx+Qdy\}$$

where F_t denoted the tangential force along the curve c at the sample point K(x,y).

The same kind of integral can, of course, relate to any function f(x,y) which is a function of the position of a point on the stated curve, so that

$$I = \int_C f(x, y)\, ds.$$

This can readily be converted into an integral in terms of x:

$$I = \int_C f(x,y)dx = \int_C f(x,y) \frac{ds}{dx} dx$$

where $\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

$$\therefore \int_C f(x,y)\, dx = \int_{x_1}^{x_2} f(x,y)\sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx \text{---------- (1)}$$

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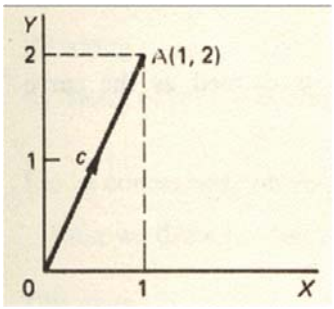
Example

Evalaute $I = \int_C (4x+3xy)ds$ where c is the straight line joining O(0,0) to A (1,2).

c is the line y = 2x $\therefore \frac{dy}{dx} = 2$



$$\therefore \frac{ds}{dx} = \sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{5}$$

$$\therefore I = \int_{x=0}^{x=1} (4x+3xy)ds = \int_0^1 (4x+3xy)(\sqrt{5})\,dx. \text{ But } y = 2x$$

$$\text{for} \quad I = \int_0^1 (4x+6x^2)(\sqrt{5})\,dx = 2\sqrt{5}\int_0^1 (2x+3x^2)\,dx = 4\sqrt{5}$$

Parametric Equations

When x and y are expressed in parametric form, e.g. x = y (t), y = g(t), then

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2} \qquad \therefore ds = \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$$

$$I = \int_C f(x,y)ds = \int_{t_1}^{t_2} f(x,y)\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2} \text{ ----------(2)}$$

Example

Evaluate $I = \oint 4xyds$ where c is defined as the curve x = sin t, y = cos t between t=0 and $t=\frac{\pi}{4}$.

We have $x = \sin t \quad \therefore \frac{dx}{dt} = \cos t$, $y = \cos t \quad \therefore \frac{dy}{dt} = -\sin t$

$$\therefore \quad \frac{ds}{dt} = I$$

$$\text{for } \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2} = \sqrt{\cos^2 t+\sin^2 t} = 1$$

$$\therefore I = \int_{t_1}^{t_2} f(x,y)\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$$

$$= \int_0^{\pi/4} 4\sin t\cos t\,dt = 2\int_0^{\pi/4} \sin 2t\,dt$$

$$= -2\left[\frac{\cos 2t}{2}\right]_0^{\pi/4} = 1$$

Dependence of the line integral on the path of integration

We know that integration along two separate paths joining the same two end points does not necessarily give identical results.With this in mind, let us investigate the following problem.

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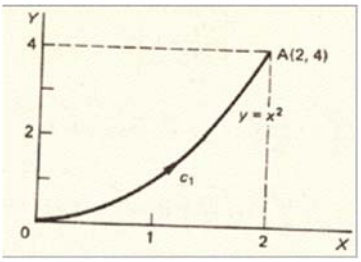
EXAMPLE

Evaluate $I = \oint_C \{3x^2 y^2\, dx + 2x^3 y\, dy\}$ between O (0, 0) and A (2, 4)

(a) along c_1 i.e. $y = x^2$
(b) along c_2 i.e. $y = 2x$
(c) along c_3 i.e. x = 0 from (0,0) to (0,4) and y = 4 from (0,4) to (2,4).

(a).First we draw the figure and insert relevant information.

$$I = \int_C \{3x^2y^2dx + 2x^3ydy\}$$



The path c_1 is $y = x^2$ $\therefore$ $dy = 2x\,dx$

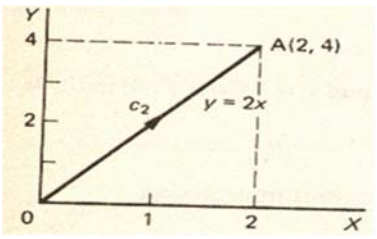
$$\therefore \quad I_1 = \int_0^2 \{3x^2x^4dx + 2x^3x^2 2xdx\} = \int_0^2 (3x^6 + 4x^6)\,dx$$

$$\therefore \quad = \left[x^7\right]_0^2 = 128 \quad \therefore \quad I_1 = 128$$

(b) In (b), the path of integration changes to c_2, i.e. $y = 2x$
So, in this case,for with c_2, $y = 2x$ $\therefore$ $dy = 2dx$

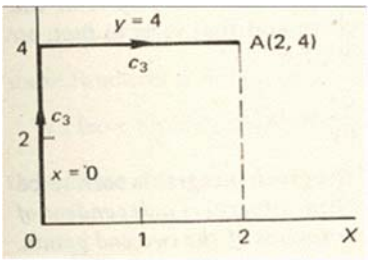
$$\therefore I_2 = \int_0^2 (3x^2\, 4x^2\, dx + 2x^3\, 2x^2\, dx\}$$

$$= \int_0^2 20\, x^4\, dx = 4\left[x^5\right]_0^2 = 128 \qquad \therefore \quad I_2 = 128$$

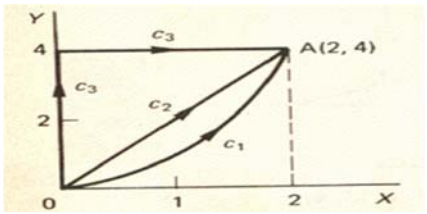


(c) In the third case, the path c_3 is split
x = 0 from (0,0) to (0, 4), y = 4 from (0, 4) to (2, 4)
Sketch the diagram and determine I_3.
from (0,0) to (0,4) x=0 $\therefore$ dx=0 $\therefore$ I_{3a}=0
from (0,4) to (2,4) y=4 $\therefore$ dy=0 $\therefore$ I_{3b}=48

$$\int_0^2 48x^2\, dx = 128 \qquad \therefore \quad I_3 = 128$$



In the example we have just worked through, we took three different paths and in each case, the line integral produced the same result. It appears, therefore, that in this case, the value of the integral is independent of the path of integration taken.



We have been dealing with $I = \int_C \{3x^2y^2dx + 2x^3ydy\}$

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On reflection, we see that the integrand $3x^2 y^2 dx + 2x^3 y dy$ is of the form Pdx+Qdy which we have met before and that it is, in fact, an exact differential of the function

$z = x^3y^2$, for $\frac{\partial z}{\partial x} = 3x^2 y^2$ and $\frac{\partial z}{\partial y} = 2x^3 y$

This always happens. If the integrand of the given integral is seen to be an exact differential, then the value o the line integral is independent of the path taken and depends only on the coordinates of the two end points.

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Lecture No -33 Examples

Example

Evaluate $I = \int_C \{3ydx + (3x+2y)dy\}$ **from A(1, 2) to B (3, 5).**

No path is given, so the integrand is doubtless an exact differential of some function z = f (x,y). In fact $\frac{\partial P}{\partial y} = 3 = \frac{\partial Q}{\partial x}$. We have already dealt with the integration of exact differentials, so there is no difficulty. Compare with

$I = \int_C \{P\,dx + Q\,dy\}$.

$P = \frac{\partial z}{\partial x} = 3y \qquad \therefore\ z = \int 3ydz = 3xy + f(y)$ -------- (i)

$Q = \frac{\partial z}{\partial y} = 3x + 2y \qquad \therefore\ z = \int (3x+2y)\,dy = 3xy + y^2 + F(x)$ (ii)

For (i) and (ii) to agree $f(y) = y^2$; $F(x) = 0$

Hence z = 3xy + y2

$\therefore I = \int_C \{3ydx + (3x+2y)dy\} = \int_{(1,2)}^{(3,5)} d(3xy+y^2) = \left[3xy+y^2\right]_{(1,2)}^{(3,5)} = (45+25) - (6+4) = 60$

Example

Evaluate $I = \int_C \{(x^2+ye^x)dx + (e^x+y)dy\}$ **between A (0, 1) and B (1, 2).**

As before, compare with $\int_C \{Pdx + Q\,dy\}$.

$P = \frac{\partial z}{\partial x} = x^2 + ye^x \qquad \therefore\ z = \frac{x^3}{3} + ye^x + f(y)$

$Q = \frac{\partial z}{\partial y} = e^x + y \qquad \therefore\ z = ye^x + \frac{y^2}{2} + F(x)$

For these expressions to agree,

$f(y) = \frac{y^2}{2}$; $F(x) = \frac{x^3}{3}$ Then $I = \left[\frac{x^3}{3} + ye^x + \frac{y^2}{2}\right]_{(0,1)}^{(1,2)} = \frac{5}{6} + 2e$

So the main points are that, if (Pdx+Qdy) is an exact differential

(a) $I = \int_C (Pdx + Qdy)$ is independent of the path of integration

(b) $I = \oint_C (P\,dx + Q\,dy)$ is zero.

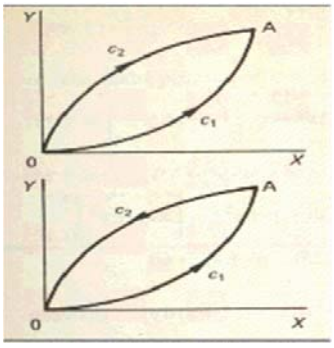
If $I = \int_C \{P\,dx + Q\,dy\}$ and (Pdx + Qdy) is an exact differential,

Then $I_{c_1} = I_{c_2}$

$I_{c_1} + I_{c_2} = 0$

Hence, the integration taken round a closed curve is zero, provided (Pdx+Q dy) is an exact differential.

$\therefore$ If (P dx + Q dy) is an exact differential, $\oint (P\,dx + Q\,dy) = 0$



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Exact differentials in three independent variables

A line integral in space naturally involves three independent variables, but the method is very much like that for two independent variables.

dz = Pdx + Q dy + R dw is an exact differential of z = f(x, y, w)

$$\text{if } \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x};\ \frac{\partial P}{\partial w} = \frac{\partial R}{\partial x};\ \frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$$

If the test is successful, then

(a) $\int_C$ (P dx + Q dy + R dw) is independent of the path of integration.

(b) $\oint_C$ (P dx + Q dy + R dw) is zero.

Example

Verify that $dz=(3x^2yw+6x)dx+(x^3w-8y)dy+(x^3y+1)\,dw$ is inexact differential and hence evaluate $\int_C dz$ from A (1, 2, 4) to B (2, 1 3).

First check that dz is an exact differential by finding the partial derivatives above, when

$P = 3x^2yw + 6x$; $Q = x^3w - 8y$; and $R = x^3y + 1$

$$\frac{\partial P}{\partial y} = 3x^2w;\ \frac{\partial Q}{\partial x} = 3x^2w \ \therefore\ \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

$$\frac{\partial P}{\partial w} = 3x^2y;\ \frac{\partial R}{\partial x} = 3x^2y \ \therefore\ \frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}$$

$$\frac{\partial R}{\partial y} = x^3;\ \frac{\partial Q}{\partial w} = x^3 \ \therefore\ \frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$$

$\therefore$ dz is an exact differential

Now to find z. $P = \frac{\partial z}{\partial x}$; $Q = \frac{\partial z}{\partial y}$; $R = \frac{\partial z}{\partial w}$

$$\therefore \frac{\partial z}{\partial x} = 3x^2yw+6x \ \therefore\ z = \int (3x^2yw+6x)dx = x^3yw+3x^2+f(y)+F(w)$$

$$\therefore \frac{\partial z}{\partial y} = x^3w-8x \ \therefore\ z = \int (x^3w-8y)dy = x^3yw-4y^2+g(x)+F(w)$$

$$\frac{\partial z}{\partial w} = x^3y+1 \quad \therefore\ z = \int (x^3y+1)dw = y^3yw+w+f(y)+g(x)$$

For these three expressions for z to agree

$f(y) = -4y^2$; $F(w) = w$; $g(x) = 3x^2$

$\therefore \quad z = x^3yw + 3x^2 - 4y^2 + w$

$$\therefore \quad I = \left[x^3yw + 3x^2 - 4y^2 + w\right]_{(1,2,4)}^{(2,1,3)}$$

$$\text{for} \quad I = \left[x^3yw + 3x^2 - 4y^2 + w\right]_{(1,2,4)}^{(2,1,3)} = (24+12-4+3)-(8+3-16+4)=36$$

The extension to line integrals in space is thus quite straightforward.

Finally, we have a theorem that can be very helpful on occasions and which links up with the work we have been doing. It is important, so let us start a new section.

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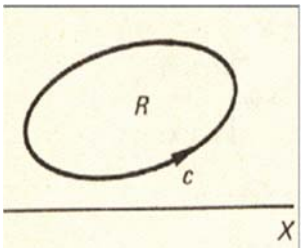
Green's Thorem

Let P and Q be two function of x and y that are finite and continuous inside and the boundary c of a region R in the xy-plane.If the first partial derivatives are continuous within the region and on the boundary, then Green's theorem states that.



$$\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\, dy = -\oint_C (P\, dx + Q\, dy)$$

That is, a double integral over the plane region R can be transformed into a line integral over the boundary c of the region – and the action is reversible.

Let us see how it works.

EXAMPLE

Evaluate $I = \oint_C \{(2x - y)dx + (2y+x)dy\}$ around the boundary c . the ellipse $x^2 + 9y^2 = 16$.

The integral is of the form

$I = \oint_C \{P\, dx + Q\, dy)$ where $P = 2x - y \;\therefore\; \frac{\partial P}{\partial y} = -1$ and $Q = 2y + x \;\therefore\; \frac{\partial Q}{\partial x} = 1.$

$$\therefore I = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx dy = -\iint_R (-1-1) dx\, dy = 2\iint_R dx\, dy$$

But $\iint_R dx\, dy$ over any closed region give the area of the figure.

In this case, then, I = 24 where A is the area of the ellipse

$x^2+9y^2 = 16$ i.e. $\frac{x^2}{16} + \frac{9y^2}{16} = 1$

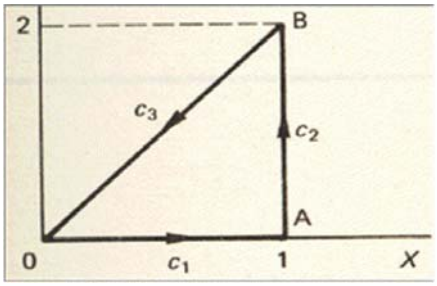
$\therefore\; a = 4;\; b = \frac{4}{3} \qquad \therefore\; A = \frac{16\pi}{3} \qquad \therefore\; I = 2A = \frac{32\pi}{3}$

To demonstrate the advantage of Green's theorem, let us work through the next example (a) by the previous method, and (b) by applying Green's theorem.

Example

Evaluate $I = \oint_C \{(2x+y)\, dx + (3x-2y)\, dy\}$ taken in anticlockwise manner round the triangle with vertices at O (0,0) A (1, 0) B (1, 2).

$I = \oint_C \{(2x + y)\, dx + (3x - 2y)\, dy\}$



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(a) By the previous method

There are clearly three stages with c_1, c_2, c_3. Work through the complete evaluation to determine the value of I. It will be good revision. When you have finished, check the result with the solution in the next frame. $I = 2$

(a) (i) c_1 is $y = 0$ $\quad \therefore \; dy = 0$

$$\therefore \; I_1 = \int_0^1 2x\, dx = \left[x^2\right]_0^1 = 1 \quad \therefore \; I_1 = 1$$

(ii) c_2 is $x = 1$ $\quad \therefore \; dx = 0$

$$\therefore \; I_2 = \int_0^2 (3-2y)\, dy = \left[3y - y^2\right]_1^0 = 2 \; \therefore \; I_2 = 2$$

(iii) c_3 is $y = 2x$ $\quad \therefore \; dy = 2\, dx$

$$\therefore \; I_3 = \int_1^0 \{4x\, dx + (3x - 4x)\, 2\, dx\}$$

$$= \int_1^0 2x\, dx = \left[x^2\right]_1^0 = -1 \quad \therefore \; I_3 = -1$$

$$I = I_1 + I_2 + I_3 = 1 + 2 + (-1) = 2 \quad \therefore \; I = 2$$

Now we will do the same problem by applying Green's theorem, so more

(b) By Green's theorem

$$I = \oint_C \{(2x + y)\, dx + (3x - 2y)\, dy\}$$

$P = 2x + y \quad \therefore \quad \dfrac{\partial P}{\partial y} = 1;$

$Q = 3x - 2y \quad \therefore \quad \dfrac{\partial Q}{\partial x} = 3$

$$I = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\, dy$$

Finish it off. $I = 2$

For $I = -\iint_R (1-3)\, dx\, dy = 2\iint_R dx\, dy = 2A$

$= 2 \times$ the area of the triangle $= 2 \times 1 = 2$

$\therefore \quad I = 2$

Application of Green's theorem is not always the quickest method. It is useful, however, to have both methods available.

If you have not already done so, make a note of Green's theorem.

$$\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\, dy = -\oint_C (P\, dx + Q\, dy)$$

Note: Green's theorem can, in fact, be applied to a region that is not simply connected by arranging a link between outer and inner boundaries, provided the path of integration is such that the region is kept on the left-hand side.

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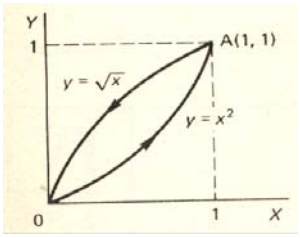
Lecture -34……Examples

Example

Evaluate the line integral $I = \oint_C \{xy\,dx + (2x - y)\,dy\}$ round the region bounded by the curves $y = x^2$ and $x = y^2$ by the use of Green's theorem.

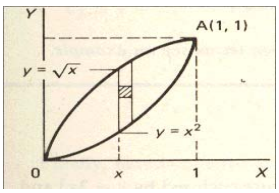


Points of intersection are O(0, 0) and A(1, 1).

$$I = \oint_C \{xy\,dx + (2x - y)\,dy\}$$

$$\oint_C \{P dx + Q dy\} = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy$$

$$P = xy \;\therefore\; \frac{\partial P}{\partial y} = x;\; Q = 2x - y \;\therefore\; \frac{\partial Q}{\partial x} = 2$$



$$I = -\iint_R (x - 2)\,dx\,dy = -\int_0^1 \int_{y=x^2}^{y=\sqrt{x}} (x - 2)\,dy\,dx$$

$$= -\int_0^1 (x - 2)\left[\,y\,\right]_{x^2}^{\sqrt{x}} dx$$

$$\therefore\; I = -\int_0^1 (x - 2)\left(\sqrt{x} - x^2\right) dx = -\int_0^1 (x^{3/2} - x^3 - 2x^{1/2} + 2x^2)\,dx$$

$$= -\left[\frac{2}{5}x^{5/2} - \frac{1}{4}x^4 - \frac{4}{3}x^{3/2} + \frac{2}{3}x^3\right]_0^1 = \frac{31}{60}$$

In this special case when P=y and Q= – x so $\frac{\partial P}{\partial y} = 1$ and $\frac{\partial Q}{\partial x} = -1$

Green's theorem then states $\iint_R \{1 - (-1)\}\,dx\,dy = -\oint_C (P\,dx + Q\,dy)$

i.e. $2\iint_R dx\,dy = -\oint_C (y\,dx - x\,dy) = \oint_C (x\,dy - y\,dx)$

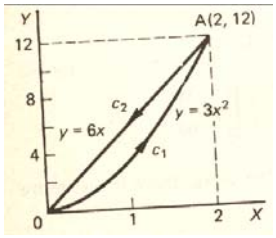
Therefore, the area of the closed region $A = \iint_R dx\,dy = \frac{1}{2}\oint_C (x\,dy - y\,dx)$

Example

Determine the area of the figure enclosed by $y = 3x^2$ and $y = 6x$.

Points of intersection : $3x^2 = 6x$ $\;\therefore\;$ x = 0 or 2

Area $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$



We evaluate the integral in two parts, i.e.
OA along c_1 and AO along c_2

$$2A = \int_{c_1\,(\text{along OA})} (x\,dy - y\,dx) + \int_{c_2\,(\text{along OA})} (x\,dy - y\,dx) = I_1 + I_2$$

I_1: c_1 is $y = 3x^2$ $\;\therefore\;$ $dy = 6x\,dx$

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$\therefore \quad I_1 = \int_0^2 (6x^2dx - 3x^2dx) = \int_0^2 3x^2\,dx = \left[x^3 \right]_0^2 = 8 \;\therefore\; I_1 = 8$

Similarly, for c_2 is $y = 6x \quad \therefore \; dy = 6\,dx$

$\therefore \quad I_2 = \int_2^0 (6x\,dx - 6x\,dx) = 0$

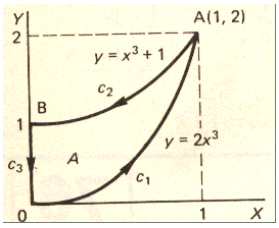
$\therefore \quad I_2 = 0$

$\therefore \quad I = I_1 + I_2 = 8 + 0 = 8$

$\therefore \; A = 4$ square units

Example

Determine the area bounded by the curves $y = 2x^3$, $y = x^3 + 1$ and the axis $x = 0$ for $x \geq 0$.



Here it is $y = 2x^3$; $y = x^3 + 1$; $x = 0$

Point of intersection $2x^3 = x^3 + 1 \quad \therefore \; x^3 = 1 \quad \therefore \; x = 1$

Area $A = \frac{1}{2}\oint_C (x\,dy - y\,dx) \quad \therefore \; 2A = \oint_C (x\,dy - y\,dx)$

(a) OA : c_1 is $y = 2x^3 \quad \therefore \; dy = 6x^2\,dx$

$\therefore I_1 = \int_{c_1} (xdy - ydx) = \int_0^1 (6x^3dx - 2x^3\,dx) \quad = \int_0^1 4x3\,dx = \left[x^4 \right]_0^1 = 1$

$\therefore \; I_1 = 1$

(b) AB: c_2 is $y = x^3 + 1 \quad \therefore \; dy = 3x^2\,dx$

$\therefore \; I_2 = \int_1^0 \{3x^3\,dx - (x^3 + 1)\,dx\} = \int_1^0 (2x^3 - 1)\,dx = \left[\frac{x^4}{2} - x\right]_1^0 = -\left(\frac{1}{2} - 1\right) = \frac{1}{2}$

$\therefore \; I_2 = \frac{1}{2}$

(c) BO: c_3 is $x = 0 \quad \therefore \; dx = 0$

$I_3 = \int_{y=1}^{y=0} (xdy - ydx) = 0 \quad \therefore \; I_3 = 0$

$\therefore \; 2A = I = I_1 + I_2 + I_3 = 1 + \frac{1}{2} + 0 = 1\frac{1}{2} \quad \therefore \; A = \frac{3}{4}$ square units

Revision Summary

Properties of line integrals

- Sign of line integral is reversed when the direction of integration along the path is reversed.
- Path of integration parallel to y-axis, $dx = 0 \quad \therefore \; I_c = \int_c Q\,dy$.
- Path of integration parallel to x-axis, $dy = 0 \quad \therefore \; I_c = \int_c P\,dx$.
- Path of integration must be continuous and single-valued.
- Dependence of line integral on path of integration.
- In general, the value of the line integral depends on the particular path of integration.

- Exact differential
  If $P\,dx + Q\,dy$ is an exact differential

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(a) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

(b) $I = \int_c (P\,dx + Q\,dy)$ is independent of the path of integration

(c) $I = \oint_C (P\,dx + Q\,dy)$ is zero.

- Exact differential in three variables.
  If P dx + Q dy + R dw is an exact differential
  (a) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}\ ;\ \frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}\ ;\ \frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$

  (b) $\int_c (P\,dx + Q\,dy + R\,dw)$ is independent of the path of integration.

  (c) $\oint_C (P\,dx + Q\,dy + R\,dw)$ is zero.
- Green's theorem
  $\oint_C (P\,dx+Q\,dy) = -\iint_R \left\{\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right\} dx\,dy$ and, for a simple closed curve,
  $\oint_C (x\,dy - y\,dx) = 2\iint_R dx\,dy = 2A$
  where A is the area of the enclosed figure.

Gradient of a scalar function

Del operator is given by $\nabla = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right)$

$$\nabla\phi = \text{grad}\,\phi = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right)\phi = \mathbf{i}\frac{\partial \phi}{\partial x} + \mathbf{j}\frac{\partial \phi}{\partial y} + \mathbf{k}\frac{\partial \phi}{\partial z}$$

$$\text{grad}\phi = \nabla\phi = \frac{\partial \phi}{\partial x}\mathbf{i} + \frac{\partial \phi}{\partial y}\mathbf{j} + \frac{\partial \phi}{\partial z}\mathbf{k}$$

Div (Divergence of a vector function)

If $\mathbf{A} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

then $\text{div}\,A = \nabla.A = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right).\left(a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}\right)$

$$\therefore \text{div}\,\mathbf{A} = \nabla.\mathbf{A} = \frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z}$$

Note that

(a) the grad operator ∇ acts on a scalar and gives a vector

(b) the div operator $\nabla.$ acts on a vector and gives a scalar.

Example

If $A = x^2y\mathbf{i} - xyz\mathbf{j} + yz^2\mathbf{k}$ then

$$\text{Div}\,\mathbf{A} = \nabla.\mathbf{A} = \frac{\partial}{\partial x}(x^2y) - \frac{\partial}{\partial y}(xyz) + \frac{\partial}{\partial z}(yz^4) = 2xy - xz + 2yz$$

Example

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If $A = 2x^2y\mathbf{i} - 2(xy^2+y^3)\mathbf{j}+3y^2z^2\mathbf{k}$ determine $\nabla\mathbf{.A}$ i.e. div **A**.

$\mathbf{A}= 2x^2y\mathbf{i} - 2(xy^2 + y^3z)\mathbf{j} + 3y^2z^2\mathbf{k}$

$$\nabla\mathbf{.A}=\frac{\partial a_x}{\partial x}+\frac{\partial a_y}{\partial y}+\frac{\partial a_z}{\partial z}= 4xy-2(2xy+3y^2z)+6y^2z= 4xy - 4xy - 6y^2z+6y^2z = 0$$

Such a vector A for which $\nabla\mathbf{.A} = 0$ at all points, i.e. for all values of x, y, z, is called a solenoid vector. It is rather a special case.

Curl (curl of a vector function)

The curl operator denoted by $\nabla\times$, acts on a vector and gives another vector as a result.

If $\mathbf{A}= a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ then curl $\mathbf{A}=\nabla\times\mathbf{A}$.

$$\text{i.e. curl } \mathbf{A}=\nabla\times\mathbf{A}=\left(\mathbf{i}\frac{\partial}{\partial x}+\mathbf{j}\frac{\partial}{\partial y}+\mathbf{k}\frac{\partial}{\partial z}\right)\times\left(a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}\right)$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$$

$$\therefore \nabla\times\mathbf{A}= \mathbf{i}\left(\frac{\partial a_3}{\partial y}-\frac{\partial a_2}{\partial z}\right)+\mathbf{j}\left(\frac{\partial a_1}{\partial z}-\frac{\partial a_3}{\partial x}\right)+\mathbf{k}\left(\frac{\partial a_2}{\partial x}-\frac{\partial a_1}{\partial y}\right)$$

Curl A is thus a vector function.

Example

If $\mathbf{A}=(y^4-x^2z^2)\mathbf{i}+(x^2+y^2)\mathbf{j}-x^2yzk$,determine curl A at the point (1,3, –2).

$$\text{Curl } \mathbf{A} = \nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y4-x^2z^2 & x^2+y^2 & -x^2yz \end{vmatrix}$$

Now we expand the determinant

$$\nabla \times \mathbf{A}= \mathbf{i}\left\{\frac{\partial}{\partial y}(-x^2yz)-\frac{\partial}{\partial z}(x^2+y^2)\right\}-\mathbf{j}\left\{\frac{\partial}{\partial x}(-x^2yz)-\frac{\partial}{\partial z}(y^4-x^2z^2)\right\}$$
$$+\mathbf{k}\left\{\frac{\partial}{\partial x}(x^2+y^2)-\frac{\partial}{\partial y}(y^4-x^2z^2)\right\}$$

$\nabla\times\mathbf{A}=\mathbf{i}\{-x^2z\}-\mathbf{j}\{-2xyz+2x^2z\}+\mathbf{k}(2x-4y^3\}$. $\therefore$ At (1, 3, – 2),

$\nabla\times\mathbf{A} = \mathbf{i}\,(2)-\mathbf{j}\,(12-4)+\mathbf{k}\,(2-108) \quad = 2\mathbf{i} - 8\mathbf{j} - 106\mathbf{k}$

Example

Determine curl F at the point (2,0,3) given that $\mathbf{F}=ze^{2xy}\mathbf{i}+2xycosy\mathbf{j}+(x+2y)\mathbf{k}$.

$$\text{In determine form, curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ ze^{2xy} & 2xzcosy & x+2y \end{vmatrix}$$

Now expand the determinant and substitute the values for x, y and z, finally obtaining curl

$\nabla \times \mathbf{F} = \mathbf{i}\{2 - 2x\cos y\} - \mathbf{j}\{1 - e^{2xy}\} + \mathbf{k}\,(\{2z\cos y - 2xze^{2xy}\}$

$\therefore$At(2,0,3) $\quad \nabla\times\mathbf{F}= \mathbf{i}(2-4)-\mathbf{j}(1-1)+\mathbf{k}(6-12) = -2\mathbf{i}-6\mathbf{k} = -2\,(\mathbf{i} + 3k)$

Summary of grad, div and curl

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(a) Grad operator ∇ acts on a scalar field to give a vector field.
(b) Div operator $\nabla\cdot$ Acts o a vector field to give a scalar field.
(c) Curl operator $\nabla\times$ acts on a vector field to give a vector field.
(d) With a scalar function $\phi(x,y,z)$

$$\text{Grad } \phi = \nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i}+\frac{\partial\phi}{\partial y}\mathbf{j}+\frac{\partial\phi}{\partial z}\mathbf{k}$$

(e) With a vector function $A=a_x i+a_y \mathbf{j}+a_z \mathbf{k}$

(i) $$\text{div } \mathbf{A} = \nabla.\mathbf{A} = \frac{\partial a_x}{\partial x}+\frac{\partial a_y}{\partial y}+\frac{\partial a_z}{\partial z}$$

(ii) $$\text{Curl } \mathbf{A} = \nabla\times\mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_x & a_y & a_z \end{vmatrix}$$

Multiple Operations

We can combine the operators grad, div and curl in multiple operations, as in the examples that follow.

EXAMPLE

If $A = x^2y\mathbf{i} + yz^3\mathbf{j} - zx^3\mathbf{k}$

Then $$\text{div } \mathbf{A}=\nabla.\mathbf{A} = \left(\mathbf{i}\frac{\partial}{\partial x}+\mathbf{j}\frac{\partial}{\partial y}+\mathbf{k}\frac{\partial}{\partial z}\right).(x^2y\mathbf{i} + yz^3\mathbf{j} - zx^3\mathbf{k})$$

$$= 2xy+z^3 + x^3 = \phi \text{ (say)}$$

Then $$\text{grad (div } \mathbf{A}) = \nabla(\nabla.\mathbf{A}) = \frac{\partial\phi}{\partial x}\mathbf{i}+\frac{\partial\phi}{\partial y}\mathbf{j}+\frac{\partial\phi}{\partial z}\mathbf{k} = (2y+3x^2)\mathbf{i}+(2x)\mathbf{j}+(3z^2)\mathbf{k}$$

i.e., $$\text{grad div } A = \nabla(\nabla.\mathbf{A})= (2y+3x^2)\mathbf{i}+2x\mathbf{j}+3z^2\mathbf{k}$$

Example

If $\phi = xyz - 2y^2z + x^2z^2$ determine div grad ϕ at the point (2, 4, 1).

First find grad ϕ and then the div of the result.

div grad $\phi = \nabla.(\nabla\phi)$

We have $\phi = xyz - 2y^2z + x^2z^2$

$$\text{grad } \phi = \nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i}+\frac{\partial\phi}{\partial y}\mathbf{j}+\frac{\partial\phi}{\partial z}\mathbf{k} = (yz+2xz^2)\mathbf{i}+(xz-4yz)\mathbf{j}+(xy-2y^2+2x^2z)\mathbf{k}$$

$\therefore$ div grad $\phi=\nabla.(\nabla\phi) = 2z^2 - 4z+2x^2$

$\therefore$ At (2,4,1), div grad $\phi = \nabla.(\nabla\phi) = 2 - 4 + 8 = 6$

$$\text{grad } \phi = \frac{\partial\phi}{\partial x}\mathbf{i}+\frac{\partial\phi}{\partial y}\mathbf{j}+\frac{\partial\phi}{\partial z}\mathbf{k}$$

Then $$\text{div grad } \phi = \nabla.(\nabla\phi)= \left(\mathbf{i}\frac{\partial}{\partial x}+\mathbf{j}\frac{\partial}{\partial y}+\mathbf{k}\frac{\partial}{\partial z}\right).\left(\frac{\partial\phi}{\partial x}\mathbf{i}+\frac{\partial\phi}{\partial y}\mathbf{j}+\frac{\partial\phi}{\partial z}\mathbf{k}\right)= \frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial z^2}+\frac{\partial^2\phi}{\partial z^2}$$

$$\therefore \text{div grad } \phi=\nabla.(\nabla\phi) = \frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2}$$

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Example

If $F = x^2yz\mathbf{i} + xyz^2\mathbf{j} + y^2z\mathbf{k}$ determine curl **F** at the point (2, 1, 1).Determine an expression for curl F in the usual way, which will be a vector, and then the curl of the result. Finally substitute values.

Curl curl $\mathbf{F}=\nabla\times(\nabla\times\mathbf{F}) = \mathbf{i}+2\mathbf{j}+6\mathbf{k}$

For curl $\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2yz & xyz^2 & y^2z \end{vmatrix} = (2yz-2xyz)\mathbf{i}+x^2y\mathbf{j}+(yz^2-x^2z)\mathbf{k}$

Then Curl Curl $\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2yz-2xyz & x^2y & yz^2-x^2z \end{vmatrix} = z^2i-(-2xz-2y+2xy)\mathbf{j}+(2xy-2z+2xz)\mathbf{k}$

$\therefore$ At (2, 1, 1), curl cul $\mathbf{F}=\nabla\times(\nabla\times\mathbf{F}) = \mathbf{i}+2\mathbf{j}+6\mathbf{k}$

Two interesting general results

(a) Curl grad ϕ where ϕ is a scalar

$$\text{grad }\phi = \frac{\partial\phi}{\partial x}\mathbf{i}+\frac{\partial\phi}{\partial y}\mathbf{j}+\frac{\partial\phi}{\partial z}\mathbf{k}$$

$$\therefore \text{ curl grad }\phi = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial\phi}{\partial x} & \frac{\partial\phi}{\partial y} & \frac{\partial\phi}{\partial z} \end{vmatrix}$$

$$= \mathbf{i}\left\{\frac{\partial^2\phi}{\partial y\partial z}-\frac{\partial^2\phi}{\partial z\partial x}\right\}-\mathbf{j}\left\{\frac{\partial^2\phi}{\partial z\partial x}-\frac{\partial^2\phi}{\partial x\partial z}\right\}+\mathbf{k}\left\{\frac{\partial^2\phi}{\partial x\partial y}-\frac{\partial^2\phi}{\partial y\partial x}\right\}=0$$

$\therefore$ curl grad $\phi = \nabla\times(\nabla\phi) = 0$

(b) Div curl A where A is a vector.

$$\mathbf{A} = a_x\mathbf{i}+a_y\mathbf{j}+a_z\mathbf{k}$$

$$\text{curl }\mathbf{A} = \nabla\times\mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_x & a_y & a_z \end{vmatrix} = \mathbf{i}\left(\frac{\partial a_z}{\partial y}-\frac{\partial a_y}{\partial z}\right)-\mathbf{j}\left(\frac{\partial a_z}{\partial x}-\frac{\partial a_x}{\partial z}\right)+\mathbf{k}\left(\frac{\partial a_y}{\partial x}-\frac{\partial a_x}{\partial y}\right)$$

$$\text{Then div curl }\mathbf{A} = \nabla.(\nabla\times\mathbf{A}) = \left(\mathbf{i}\frac{\partial}{\partial x}+\mathbf{j}\frac{\partial}{\partial y}+\mathbf{k}\frac{\partial}{\partial z}\right).(\nabla\times\mathbf{A})$$

$$= \frac{\partial^2 a_z}{\partial x\partial y}-\frac{\partial^2 a_y}{\partial z\partial x}-\frac{\partial^2 a_z}{\partial x\partial y}+\frac{\partial^2 a_x}{\partial y\partial z}-\frac{\partial^2 a_y}{\partial z\partial x}-\frac{\partial^2 a_x}{\partial y\partial z}=0$$

$\therefore$ div curl $\mathbf{A} = \nabla\cdot(\nabla\times\mathbf{A}) = 0$

(c) Div grad ϕ where ϕ is a scalar.

$$\text{grad }\phi = \frac{\partial\phi}{\partial x}\mathbf{i}+\frac{\partial\phi}{\partial y}\mathbf{j}+\frac{\partial\phi}{\partial z}\mathbf{k}$$

$$\text{Then div grad }\phi = \nabla.(\nabla\phi) = \left(\mathbf{i}\frac{\partial}{\partial x}+\mathbf{j}\frac{\partial}{\partial y}+\mathbf{k}\frac{\partial}{\partial z}\right).\left(\frac{\partial\phi}{\partial x}\mathbf{i}+\frac{\partial\phi}{\partial y}\mathbf{j}+\frac{\partial\phi}{\partial z}\mathbf{k}\right) = \frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial z^2}+\frac{\partial^2\phi}{\partial z^2}$$

$$\therefore \text{div grad }\phi = \nabla.(\nabla\phi) = \frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2}$$

This result is sometimes denoted by $\nabla^2\phi$.

So these general results are

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(a) curl grad $\phi = \nabla\times(\nabla\phi) = 0$

(b) div curl $\mathbf{A} = \nabla.(\nabla\times\mathbf{A}) = 0$

(c) div grad $\phi=\nabla.(\nabla\phi)=\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}+\frac{\partial^2\phi}{\partial z^2}$

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Lecture No -35 Definite Integrals

Definite integral for $\sin^n x$ and $\cos^n x$, $0 \le x \le \pi/2$

$$\int_0^{\frac{\pi}{2}} \sin^2 x dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} (1-\cos 2x)dx = \frac{1}{2}\left| x - \frac{\sin 2x}{2}\right|_0^{\frac{\pi}{2}} = \frac{1}{2}\left|\frac{\pi}{2} - \frac{\sin \pi}{2}\right| = \frac{\pi}{4}$$

$$\int_0^{\frac{\pi}{2}} \sin^2 x dx = \frac{1}{2}\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \cos^2 x dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} (1+\cos 2x)dx = \frac{1}{2}\left| x + \frac{\sin 2x}{2}\right|_0^{\frac{\pi}{2}} = \frac{1}{2}\left|\frac{\pi}{2} + \frac{\sin \pi}{2}\right| = \frac{\pi}{4}$$

$$\int_0^{\frac{\pi}{2}} \cos^2 x dx = \frac{1}{2}\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \sin^3 x dx = \int_0^{\frac{\pi}{2}} \sin^2 x \sin x dx = \int_0^{\frac{\pi}{2}} (1-\cos^2 x)\sin x dx = \int_0^{\frac{\pi}{2}} \sin x dx + \int_0^{\frac{\pi}{2}} \cos^2 x(-\sin x)dx$$

$$= \left|-\cos x\right|_0^{\frac{\pi}{2}} + \left|\frac{\cos^3 x}{3}\right|_0^{\frac{\pi}{2}} = -\cos\frac{\pi}{2} + \cos 0 + \frac{1}{3}\left[\cos^3\frac{\pi}{2} - \cos^3 0\right] = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \cos^3 x dx = \int_0^{\frac{\pi}{2}} \cos^2 x \cos x dx = \int_0^{\frac{\pi}{2}} (1-\sin^2 x)\cos x dx = \int_0^{\frac{\pi}{2}} \cos x dx - \int_0^{\frac{\pi}{2}} \sin^2 x(\cos x)dx$$

$$= \left|\sin x\right|_0^{\frac{\pi}{2}} - \left|\frac{\sin^3 x}{3}\right|_0^{\frac{\pi}{2}} = \sin\frac{\pi}{2} - \sin 0 - \frac{1}{3}\left[\sin^3\frac{\pi}{2} - \sin^3 0\right] = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \sin^4 x dx = \int_0^{\frac{\pi}{2}} (\sin^2 x)^2 dx = \int_0^{\frac{\pi}{2}} \left[\frac{1-\cos 2x}{2}\right]^2 dx = \frac{1}{4}\int_0^{\frac{\pi}{2}} (1-2\cos 2x + \cos^2 2x)dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} (1-2\cos 2x + \frac{1+\cos 4x}{2})dx = \frac{1}{4}\int_0^{\frac{\pi}{2}} (\frac{3}{2} - 2\cos 2x + \frac{c\cos 4x}{2})dx$$

$$= \frac{1}{4}\left|\frac{3}{2}x - \sin 2x + \frac{\sin 4x}{8}\right|_0^{\frac{\pi}{2}} = \frac{1}{4}\left[\frac{3}{2}\frac{\pi}{2} - \sin\pi + \frac{\sin 2\pi}{8}\right]$$

$$\int_0^{\frac{\pi}{2}} \sin^4 x dx = \frac{1}{4}\left[\frac{3}{2}\frac{\pi}{2}\right] \qquad so \qquad \int_0^{\frac{\pi}{2}} \sin^4 x dx = \frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

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$$\int_0^{\frac{\pi}{2}} \cos^4 x dx = \int_0^{\frac{\pi}{2}} (\cos^2 x)^2 dx = \int_0^{\frac{\pi}{2}} \left[\frac{1+\cos 2x}{2}\right]^2 dx = \frac{1}{4}\int_0^{\frac{\pi}{2}} (1+2\cos 2x+\cos^2 2x)dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} (1+2\cos 2x+\frac{1+\cos 4x}{2})dx = \frac{1}{4}\int_0^{\frac{\pi}{2}} (\frac{3}{2}+2\cos 2x+\frac{c\cos 4x}{2})dx$$

$$= \frac{1}{4}\left|\frac{3}{2}x+\sin 2x+\frac{\sin 4x}{8}\right|_0^{\frac{\pi}{2}} = \frac{1}{4}\left[\frac{3}{2}\frac{\pi}{2}+\sin\pi+\frac{\sin 2\pi}{8}\right]$$

$$\int_0^{\frac{\pi}{2}} \cos^4 x dx = \frac{1}{4}\left[\frac{3}{2}\frac{\pi}{2}\right] \qquad \text{So} \qquad \int_0^{\frac{\pi}{2}} \cos^4 x dx = \frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \sin^5 x dx = \frac{4}{5}\frac{2}{3} \qquad \text{and} \qquad \int_0^{\frac{\pi}{2}} \cos^5 x dx = \frac{4}{5}\frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \sin^6 x dx = \frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2} \qquad \text{and} \qquad \int_0^{\frac{\pi}{2}} \cos^6 x dx = \frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \sin^7 x dx = \frac{6}{7}\frac{4}{5}\frac{2}{3} \qquad \text{and} \qquad \int_0^{\frac{\pi}{2}} \cos^7 x dx = \frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \sin^8 x dx = \frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2} \qquad \text{and} \qquad \int_0^{\frac{\pi}{2}} \cos^8 x dx = \frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \sin^9 x dx = \frac{8}{9}\frac{6}{7}\frac{4}{5}\frac{2}{3} \qquad \text{and} \qquad \int_0^{\frac{\pi}{2}} \cos^9 x dx = \frac{8}{9}\frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \sin^{10} x dx = \frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2} \qquad \text{and} \qquad \int_0^{\frac{\pi}{2}} \cos^8 x dx = \frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

Wallis Sine Formula

When n is even $\int_0^{\frac{\pi}{2}} \sin^n x dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdot\text{---------}\cdot\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$

When n is odd $\int_0^{\frac{\pi}{2}} \cos^n x dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdot\text{---------}\cdot\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}\cdot$

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$$\int_0^{\frac{\pi}{2}} \sin^{11} x dx = \frac{10.8.6.4.2}{11.9.7.5.3} \quad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^{11} x dx = \frac{10.8.6.4.2}{11.9.7.5.3}$$

$$\int_0^{\frac{\pi}{2}} \sin^{12} x dx = \frac{11.9.7.5.3.1}{10.8.6.4.2}\frac{\pi}{2} \quad \text{and} \quad \int_0^{\frac{\pi}{2}} \cos^{12} x dx = \frac{11.9.7.5.3.1}{10.8.6.4.2}\frac{\pi}{2}$$

Integration By Parts

$$\int UVdx = U\int V\,dx - \int\left[\int Vdx.\frac{dU}{dx}\right]dx$$

Example *Evaluate* $\int$ x lnx dx

$\int$ x lnx dx= lnx $\int$ xdx – $\int$ [$\int$ x dx.$\frac{d}{dx}$(ln x)] dx (We are integrating by parts)

$$= \ln x(\frac{x^2}{2}) - \int (\frac{x^2}{2})(\frac{1}{x})dx = (\frac{x^2}{2})\ln x - \int (\frac{x}{2})d x = (\frac{x^2}{2})\ln x - \frac{1}{2}(\frac{x^2}{2})$$

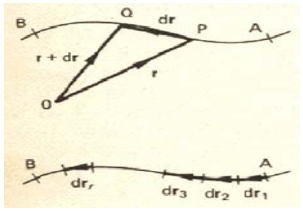
Example *Evaluate* $\int$ x sinx dx

$\int$ **x** sinx dx =x$\int$ sin xdx – $\int$ [$\int$ sinx dx.$\frac{d}{dx}$(x)] dx (We are integrating by parts)

$$= x(-\cos x) - \int (-\cos x)(1)dx = -x(\cos x) + \int \cos x\, d x = -x(\cos x) + \sin x$$

Line Integrals

Let a point p on the curve c joining A and B be denoted by the position vector **r** with respect to origin O. If q is a neighboring point on the curve with position vector **r+dr** , then $\overline{PQ}$ = **r**

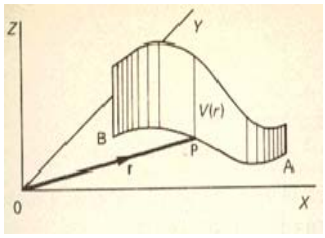


The curve c can be divided up into many n such small arcs , approximating to d**r**$_1$, d**r**$_2$, d**r**$_3$,……. d**r**$_p$,……

so that $\overline{AB}\sum_{p=1}^{n} dr_p$ where d**r**$_p$ is a vector representing the element of the arc in both magnitude and direction. If dr $\to$ 0 , then the length of the curve AB= $\int_c dr$.

Scalar Field

If a scalar field V(r) exists for all points on the curve , the $\sum_{p=1}^{n} V(r)dr_p$ with dr $\to$ 0 , defines the line integral of V i.e line integral = $\int_c V(r)dr$.



We can illustrate this integral by erecting a continuous Ordinate to V(r) at each point of the curve $\int_c V(r)dr$ is then represented by the area of the curved surface between the ends A and B the curve c. To evaluate a line integral , the integrand is expressed in terms of x , y ,z with dr =dx + dy j + dz k

In practice , x , y and are often expressed in terms of parametric equation of a fourth variable (say u), i.e. x = x(u) ; y = y(u) ; z = z(u) . From these , dx, dy and dz can be written in terms of u and the integral evaluate in terms of this parameter u.

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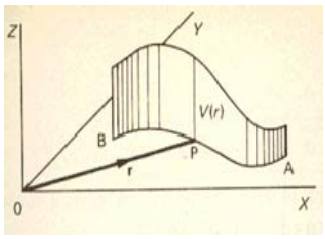
Lecture No -36 Scalar Field

Scalar Field

If a scalar field V(r) exists for all points on the curve ,

the $\sum_{p=1}^{n} V(r)dr_p$ with dr $\to 0$, defines the line integral

of V i.e line integral = $\int_c V(r)dr$.



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In practice , x , y and are often expressed in terms of parametric equation of a fourth variable (say u), i.e. x = x(u) ; y = y(u) ; z = z(u) . From these , dx, dy and dz can be written in terms of u and the integral evaluate in terms of this parameter u.

Example

If V=xy^{2}z , evaluate $\int_c V(r)dr$ along the curve c having parametric equations

x = 3u; y=2u^2 ;z=u^3 between A(0,0,0) and B(3,2,1)

V= xy^{2}z = (3u)(4u^4)(u^3)=12u^8

d**r**= dx**i**+ dy **j**+ dz **k** $\Rightarrow$ **dr= 3**du i +4udu j +**3u^2**du k

for x = 3u ; $\therefore$ dx = 3du ; y = 2u^2 $\therefore$ dy = 4u du ; z = u^3 $\therefore$ dz =**3u^{2}dz**

Limiting : A(0,0,0) corresponds to B(3,2,1) corresponds to u

A(0,0,0) $\equiv$ u=0 ; B(3,2,1) $\equiv$ u = 1

$$\int_c V(r)dr = \int_0^1 12u^8(3\,i + 4u\,j + 3u^2\,k)du = \left| 36\frac{u^9}{9}i + 48\frac{u^{10}}{10}j + 36\frac{u^{11}}{11} \right|_0^1 = 4i + \frac{24}{5}j + \frac{36}{11}k$$

Example

If V = xy + y^{2}z Evaluate $\int_c V(r)dr$ along the curve c defined by x= t^2; y = 2t ; z= t+5

between A(0,0,5) and B(4,4,7) . As before , expressing V and d**r** in term of the parameter t .

since V=xy+y^{2}z

= (t^2)(2t)+(4t^2)(t+5)

= 6t^3 + 20t^2.

x = t^2 dx = 2t dt

y = 2t dy = 2 dt }

z = t+5 dz = dt

$\therefore$ dr = dx**i** + dy**j** + dz**k**

= 2t dt **i** + 2 dt **j** + dt**k**

$\therefore \int_C$ Vdr = $\int_C$ (6t^3+20t^2)(2t **i** +2**j**+**k**) dt

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Limits: A (0, 0, 5) $\equiv$ t = 0;
B (4, 4, 7) $\equiv$ t = 2

$$\therefore \int_C \mathbf{V}\,d\mathbf{r} = \int_0^2 (6t^3+20t^2)(2t\,\mathbf{i}+2\,\mathbf{j}+\mathbf{k})\,dt$$

$$\int_C \mathbf{V}\,d\mathbf{r} = 2\int_0^2 \{6t^4+20t^3)\mathbf{i}+(6t^3+20t^2)\mathbf{j} + (3t^3+10t^2)\mathbf{k}\}\,dt.$$

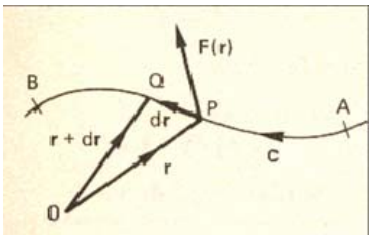
$$= \frac{8}{15}(444\mathbf{i} + 290\mathbf{j} + 145\mathbf{k})$$

Vector Field

If a vector field F(r) exist for all points of the curve c, then for each element of arc we can form the scalar product F.dr. Summing these products for all elements of arc, we have

$$\sum_{p=1}^{n} F.dr_p$$

The line integral of F(r) fr om A to B along the stated curve = $\int_C$ F.dr.



In this case, since **F**.d**r** is a scalar product, then the line integral is a scalar. To evaluate the line integral, F and d **r** are expressed in terms of x,y,z, and the curve in parametric form. We have

$\mathbf{F} = F_1\,\mathbf{i} + F_2\,\mathbf{j} + F_3\,\mathbf{k}$
And $d\mathbf{r} = dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k}$
Then $\mathbf{F}.d\mathbf{r} = (F_1\,\mathbf{i} + F_2\,\mathbf{j} + F_3\,\mathbf{k}).(dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k}) = \int_c (F_1dx + F_2dy + F_3dz)$

Now for an example to show it in operation.

Example

If $\mathbf{F}(r) = x^2y\,\mathbf{i} + xz\,\mathbf{j} + 2yz\,\mathbf{k}$, Evaluate $\int_c F.dr$ between A(0,0,0) and B(4,2,1) along the curve c having parametric equations $x=4t$; $y-2t^2$; $z = t^3$
Expressing everthing in terms of the parameter t, we have
$dx = 4\,dt$; $dy = 4tdt$; $dz = 3t^2\,dt$
$x^2y = (16t^2)(2t^2) = 32\,t^4$
$x = 4t$ $\quad\therefore\quad dx = 4\,dt$
$xz = (4t)(t^3) = 4\,t^4$
$y = 2\,t^2$ $\quad dy = 4t\,dt$
$2yz = (4\,t^2)(\,t^3) = 4\,t^5$
$z = t^3$ $\quad\therefore\quad dz = 3t^2\,dt$
$\mathbf{F} = 32\,t^4\,\mathbf{i} + 4\,t^4\,\mathbf{j} - 4\,t^5\,\mathbf{k}$
$d\mathbf{r} = 4dt\,\mathbf{i} + 4t\,dt\,\mathbf{j} + 3t^2\,\mathbf{k}$

Then $\int \mathbf{F}.d\mathbf{r} = \int (32t^4\mathbf{i}+4t^4\mathbf{j}-4t^5\mathbf{k}).(4dt\,\mathbf{i} + 4t\,dt\,\mathbf{j} + 3t^2\,dt\,\mathbf{k})$

$$= \int (128t^4+16t^5+12t^7)\,dt$$

Limits: A(0,0,0) $\equiv$ t = 0;
B (4, 2, 1) $\equiv$ t = 1

$$\int_C \mathbf{F}.d\mathbf{r} = (128t^4 + 16t^5 + 12t^7)dt = \frac{128}{5}\,t^5 + \frac{16}{6}\,t^6 + \frac{12}{8}\,t^8 = \frac{128}{5} + \frac{8}{3} + \frac{3}{2} = 29.76$$

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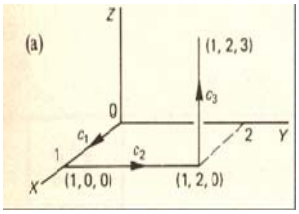
Example

If $\mathbf{F}(\mathbf{r}) = x^2y\mathbf{i} + 2yz\mathbf{j} + 3z^2x\mathbf{k}$

Evaluate $\int_C \mathbf{F}.d\mathbf{r}$ between A(0,0,0) and B(1,2,3)



B (1, 2, 3)

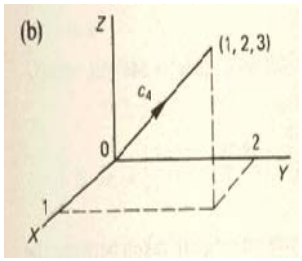
(a) along the straight line

c_1 from (0, 0, 0) to (1, 0, 0)

then c_2 from (1, 0, 0) to (1, 2, 0)

and c_3 from (1, 2, 0) to (1, 2, 3)

(b) along the straight line c_4 joining (0, 0, 0) to (1, 2, 3).



We first obtain an expression for $\mathbf{F}.d\mathbf{r}$ which is

$\mathbf{F}.d\mathbf{r} = (x^2y\mathbf{i} + 2yz\mathbf{j} + 3z^2x\mathbf{k}).$

$(dx\,\mathbf{i} + dy\mathbf{j} + dz\,\mathbf{k})$

$\mathbf{F}.d\mathbf{r} = x^2y\,dx + 2yz\,dy + 3z^2x\,dz$

$\int \mathbf{F}.d\mathbf{r} = \int x^2y dx + \int 2yz dy + \int 3z^2x dz$

Here the integration is made in three sections, along c_1, c_2 and c_3.

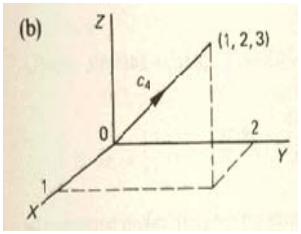
(i) c_1: $y = 0, z = 0, dy = 0, dz = 0$

$\therefore \int_{C_1} \mathbf{F}.d\mathbf{r} = 0+0+0 = 0$

(ii) c_2: The conditions along c_2 are

c_2: $x = 1, z = 0, dx = 0, dz = 0$

$\therefore \int_{C_2} F.dr = 0 + 0 + 0 = 0$



(iii) c_3: $x = 1, y = 2, dx = 0, dy = 0$

$\int_{C_3} F.dr = 0 + 0 + \int_0^3 3z^2dz = 27$

Summing the three partial results

$\int_{(0,0,0)}^{(1,2,3)} \mathbf{F}.d\mathbf{r} = 0 + 0 + 27 = 27$

$\therefore \int_{c_1+c_2+c_3} \mathbf{F}.d\mathbf{r} = 27$

If t taken as the parameter, the parametric equation of c are $x = t; y = 2t; z = 3t$

$(0, 0, 0) \Rightarrow t = 0, (1, 2, 3) \Rightarrow t = 1$ and the limits of t are $t = 0$ and $t = 1$

$\mathbf{F} = 2t^3\mathbf{i} + 12t^2\mathbf{j} + 27t^3\mathbf{k}$

$d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + kdz = dt\,\mathbf{i} + 2dt\,\mathbf{j} + 3dt\,\mathbf{k}$

$$\int_{C_4} \mathbf{F}.d\mathbf{r} = \int_0^1 (2t^3\mathbf{i}+12t^3\mathbf{j}+27t^3\mathbf{k}).(\mathbf{i}+2\mathbf{j}+3\mathbf{k})\,dt = \int_0^1 (2t^3+24t^2+81t^3)dt$$

$$= \int_0^1 (83t^3 + 24t^2)\,dt = \left[83\frac{t^4}{4} + 8t^3\right]_0^1 = \frac{115}{4} = 28.75$$

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So the value of the line integral depends on the path taken between the two end points A and B

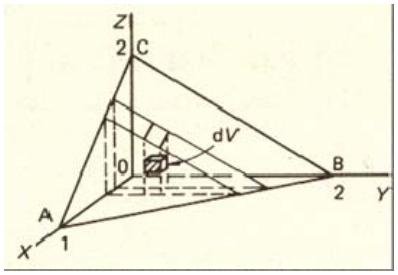
(a) $\int \mathbf{F}.d\mathbf{r}$ via c_1, c_2 and $c_3 = 27$

(b) $\int \mathbf{F}.d\mathbf{r}$ via $c_4 = 28.75$

Example

Evaluate $\int_v F\,dv$ where V is the region bounded by the planes x = 0 ,y = 0 , z = 0 and $2x + y = 2$, and $F = 2z\ \mathbf{i} + y\ \mathbf{k}$. To sketch the surface 2x + y + z = 2, note that

when z = 0, 2x+y=2 i.e. y = 2 – 2x
when y = 0, 2x+z=2 i.e. z = 2 – 2x
when x = 0, y+z=2 i.e. z = 2 – y
Inserting these in the planes
x = 0, y = 0, z = 0 will help.
The diagram is therefore.



So 2x + y + z = 2 cuts the axes at
A(1,0,0); B (0, 2, 0); C (0, 0, 2).
Also $F = 2z\mathbf{i} + y\mathbf{k}$;
$z = 2 - 2x - y = 2(1 - x) - y$

$$\therefore \int_v F dV = \int_0^1 \int_0^{2(1-x)} \int_0^{2(1-x)-y} (2x\mathbf{i}+y\mathbf{k})\,dz\,dy\,dx$$

$$= \int_0^1 \int_0^{2(1-x)} \left[z^2\mathbf{i}+yz\mathbf{k}\right]_{z=0}^{z=2(1-x)-y} dy\,dx$$

$$= \int_0^1 \int_0^{2(1-x)} \{[4(1-x)^2-4(1-x)y+y^2]\mathbf{i}+[2(1-x)y-y^2]k\}\,dy\,dx$$

$$\int_v F dV = \frac{1}{3}(2\mathbf{i}+\mathbf{k})$$

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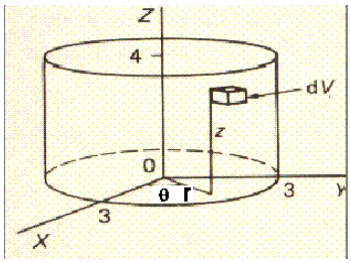
Lecture No -37 Examples

Example

Evaluate $\int_v \mathbf{F}dV$ where $\mathbf{F}=2\mathbf{i}+2z\mathbf{j}+y\mathbf{k}$ and V is the region bounded by the planes z = 0, z = 4 and the surface
$x^2+y^2 = 9$.



It will be convenient to use cylindrical polar coordinates (r, θ, z) so the relevant transformations are
$x = r\cos\theta; \quad y = r\sin\theta \quad z = z; \quad dV = rdrd\theta\, dz$

Then $\int_v FdV = \iiint_v (2i+2zj+yk)dxdydz$

Changing into cylindrical polar coordinates with appropriate change of limits this becomes

$$\int_v FdV = \int_{\theta=0}^{2\pi}\int_{r=0}^{3}\int_{z=0}^{4}(2i+2zj+r\sin\theta k)rdzdrd\theta = \int_{\theta=0}^{2\pi}\int_{r=0}^{3}\left[2z\mathbf{i}+z^2\mathbf{j}+r\sin\theta z\mathbf{k}\right]_{z=0}^{4} rdrd\theta$$

$$= \int_0^{2\pi}\int_0^{3}(8\mathbf{i}+16\mathbf{j}+4r\sin\theta\mathbf{k})rdrd\theta = 4\int_0^{2\pi}\int_0^{3}(2r\mathbf{i}+4r\mathbf{j}+r^2\sin\theta\mathbf{k})\,drd\theta = 4\int_0^{2\pi}\left[r^2\mathbf{i}+2r^2\mathbf{j}+\frac{r^3}{3}\sin\theta\,\mathbf{k}\right]_0^{3}d\theta$$

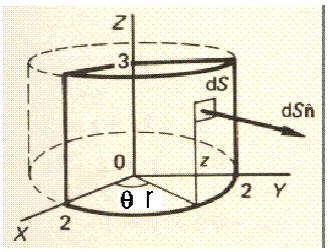
$$= 4\int_0^{2\pi}(9\mathbf{i}+18\mathbf{j}+9\sin\theta\mathbf{k})\,d\theta = 36\int_0^{2\pi}(\mathbf{i}+2\mathbf{j}+\sin\theta\mathbf{k})d\theta = 36\left[\theta\mathbf{i}+2\theta\mathbf{j}-\cos\theta\mathbf{k}\right]_0^{2\pi}$$

$$= 36\{(2\pi\mathbf{i}+4\pi\mathbf{j}-\mathbf{k})-(-\mathbf{k})\} = 72\pi(\mathbf{i}+2\mathbf{j})$$

Scalar Fields

A scalar field F = xyz exists over the curved surface S defined by $x^2+y^2 = 4$ between the planes z = 0 and z = 3 in the first octant.

Evaluate $\int_S F\,dS$ over this surface.



We have F = xyz S: $x^2+y^2-4 = 0$, z = 0 to z = 3

$d\mathbf{S} = \hat{\mathbf{n}}\,dS$ where $\hat{\mathbf{n}} = \dfrac{\nabla S}{|\nabla S|}$

Now $\nabla S = \dfrac{\partial S}{\partial x}\mathbf{i}+\dfrac{\partial S}{\partial y}\mathbf{j}+\dfrac{\partial S}{\partial z}k = 2x\mathbf{i}+2y\mathbf{j}$

$|\nabla S| = \sqrt{4x^2+4y^2} = 2\sqrt{x^2+y^2} = 2\sqrt{4} = 4$

$\therefore \hat{\mathbf{n}} = \dfrac{\nabla S}{|\nabla S|} = \dfrac{x\mathbf{i}+y\mathbf{j}}{2}$

$\therefore d\mathbf{S} = \hat{\mathbf{n}}dS = \dfrac{x\mathbf{i}+y\mathbf{j}}{2}dS$

$$\therefore \int_S FdS = \int_S F\hat{\mathbf{n}}dS = \frac{1}{2}\int_S xyz(x\mathbf{i}+y\mathbf{j})dS = \frac{1}{2}\int_S (x^2yz\mathbf{i}+xy^2z\mathbf{j})dS \quad (1)$$

We have to evaluate this integral over the prescribed surface.
Changing to cylindrical coordinates with r = 2
$x = 2\cos\theta; \quad y = 2\sin\theta z = z; \quad dS = 2d\theta dz$
$\therefore x^2yz = (4\cos^2\theta)(2\sin\theta)(z) = 8\cos^2\theta\sin\theta z$
$xy^2z = (2\cos\theta)(4\sin^2\theta)(z) = 8\cos\theta\sin^2\theta z$

Then result $\int_S FdS = \dfrac{1}{2}\int_S (x^2yz\mathbf{i}+xy^2z\mathbf{j})dS$ becomes.

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$$\int_S F dS = \frac{8}{2}\int_0^{\pi/2}\int_0^3 (\cos^2\theta\sin\theta\mathbf{i}+\cos\theta\sin 2\theta\mathbf{j})2z dz d\theta = 4\int_0^{\pi/2}\int_0^3 (\cos^2\theta\sin\theta\mathbf{i}+\cos\theta\sin^2\theta\mathbf{j})2z dz\, d\theta$$

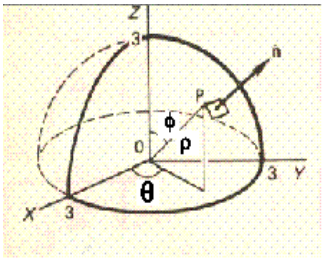
$$= 4\int_0^{\pi/2} (\cos^2\theta\sin\theta\mathbf{i}+\cos\theta\sin^2\theta\mathbf{j})z^2\Big|_0^3 d\theta = 4\int_0^{\pi/2} (\cos^2\theta\sin\theta\mathbf{i}+\cos\theta\sin^2\theta\mathbf{j})9 d\theta$$

$$= 36\left[-\frac{\cos^3\theta}{3}\mathbf{i} + \frac{\sin^3\theta}{3}\mathbf{j}\right]_0^{\pi/2} = 12\,(\mathbf{i}+\mathbf{j})$$

Vector Field

A vector field $\mathbf{F}=y\mathbf{i}+2\mathbf{j}+\mathbf{k}$ exists over a surface S defined by $x^2+y^2+z^2 = 9$ bounded by $x = 0, y = 0, z = 0$ in the first octant,

Evaluate $\int_S \mathbf{F}.dS$ over the surface indicated.



$$dS = \hat{\mathbf{n}}dS ; \qquad \hat{\mathbf{n}} = \frac{\nabla S}{|\nabla S|}$$

$$S : x^2 + y^2 + z^2 - 9 = 0$$

$$\nabla S = \frac{\partial S}{\partial x}\mathbf{i} + \frac{\partial S}{\partial y}\mathbf{j} + \frac{\partial S}{\partial z}\mathbf{k} = 2x\mathbf{i}+2y\mathbf{j}+2z\mathbf{k}$$

$$\therefore |\nabla S| = \sqrt{4x^2+4y^2+4z^2} = 2\sqrt{x^2+y^2+z^2} = 2\sqrt{9} = 6$$

$$\therefore \quad \hat{\mathbf{n}} = \frac{1}{6}(2x\mathbf{i}+2y\mathbf{j}+2z\mathbf{k}) = \frac{1}{3}(x\mathbf{i}+y\mathbf{j}+z\mathbf{k})$$

$$\int_S \mathbf{F}.dS = \int_S \mathbf{F}.\hat{\mathbf{n}}dS = \int_S (y\mathbf{i}+2\mathbf{j}+\mathbf{k}).\frac{1}{3}(x\mathbf{i}+y\mathbf{j}+z\mathbf{k})dS = \frac{1}{3}\int_S (xy + 2y + z)\, dS$$

Before integrating over the surface, we convert to spherical polar coordinates.

$x = 3\sin\phi\cos\theta$; $y = 3\sin\phi\sin\theta$ $z = 3\cos\phi$; $dS = 9\sin\phi d\phi d\theta$

Limits of ϕ and θ are $\phi = 0$ to $\frac{\pi}{2}$; $\theta = 0$ to $\frac{\pi}{2}$

$$\int_S \mathbf{F}.dS = \frac{1}{3}\int_S (xy + 2y + z)\, dS$$

$xy = 3\sin\phi\cos\theta \cdot 3\sin\phi\sin\theta = 9\sin^2\theta\cos\theta$

$2y = 2 \cdot 3\sin\phi\sin\theta = 6\sin\phi\sin\theta$

$z = 3\cos\phi$

$dS = 9\sin\phi\, d\phi\, d\theta$

Putting these values we get

$$\therefore \int_S \mathbf{F}.dS = \frac{1}{3}\int_0^{\pi/2}\int_0^{\pi/2} (9\sin^2\phi\sin\theta\cos\theta+6\sin\phi\sin\theta+3\cos\phi)9\sin\phi d\phi d\theta$$

$$= 9\int_0^{\pi/2}\int_0^{\pi/2} (3\sin^3\phi\sin\theta\cos\theta+2\sin^2\phi\sin\theta+\sin\phi\cos\phi)d\phi d\theta$$

As we know that

$$\int_0^{\pi/2} \sin^3\phi\, d\phi = \frac{2}{3} \qquad \text{by Wallis Formula}$$

$$\text{also} \int_0^{\pi/2} \sin^2\phi\, d\phi = \frac{1}{2}\frac{\pi}{2}$$

So we get

$$\int_S \mathbf{F}.d\mathbf{S} = 9\int_0^{\pi/2}\left(2\sin\theta\cos\theta+\frac{\pi}{2}\sin\theta+\frac{1}{2}\right)d\theta = 9\left[\sin^2\theta-\frac{\pi}{2}\cos\theta+\frac{\theta}{2}\right]_0^{\pi/2} = 9\left[(1-0+\frac{\pi}{4})-(0-\frac{\pi}{2}+0)\right]$$

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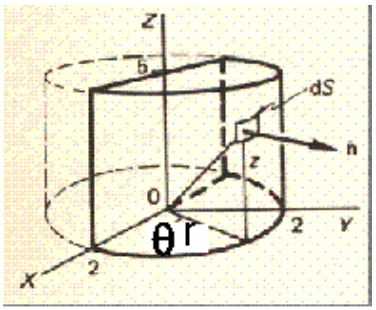
$$= 9\left[1+\frac{\pi}{4}+\frac{\pi}{2}\right] = 9\left(1+\frac{3\pi}{4}\right)$$

Example

Evaluate $\int_S \mathbf{F}.d\mathbf{S}$ where $\mathbf{F} = 2y\mathbf{j} + z\mathbf{k}$ and S is the surface $x^2 + y^2 = 4$ in the first two octants bounded by the planes $z = 0$, $z = 5$ and $y = 0$.



$S : x^2 + y^2 - 4 = 0$

$$\hat{\mathbf{n}} = \frac{\nabla S}{|\nabla S|}$$

$$\nabla S = \frac{\partial \mathbf{S}}{\partial x}\mathbf{i} + \frac{\partial \mathbf{S}}{\partial y}\mathbf{j} + \frac{\partial \mathbf{S}}{\partial z}\mathbf{k} = 2x\mathbf{i} + 2y\mathbf{i}$$

$$\therefore |\nabla \mathbf{S}| = \sqrt{4x^2 + 4y^2} = 2\sqrt{x^2 + y^2} = 2\sqrt{4} = 4$$

$$\therefore \hat{\mathbf{n}} = \frac{\nabla S}{|\nabla S|} = \frac{2x\mathbf{i} + 2y\mathbf{j}}{4} = \frac{1}{2}(x\mathbf{i} + y\mathbf{j})$$

$$\therefore \int_S \mathbf{F}.d\mathbf{S} = \int_S \mathbf{F}.\hat{\mathbf{n}}\, dS$$

$$\int_S \mathbf{F}.\hat{\mathbf{n}}\, d\mathbf{S} = \frac{1}{2}\int_S (2y^2)\, dS = \int_S y^2\, dS$$

This is clearly a case for using cylindrical polar coordinates.

$x = 2\cos\theta; \quad y = 2\sin\theta \quad z = z; \quad dS = 2d\theta\, dz$

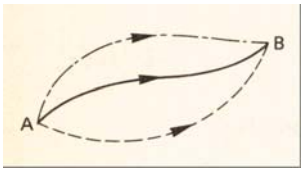
$$\therefore \int_S \mathbf{F}.d\mathbf{S} = \int_S y^2 dS = \iint_S 4\sin^2\theta 2 d\theta dz = 8\iint_S \sin^2\theta\, d\theta\, dz$$

Limits: $\theta = 0$ to $\theta = \pi$; $z = 0$ to $z = 5$

$$\int_S \mathbf{F}.d\mathbf{S} = 4\int_{z=0}^{5}\int_{\theta=0}^{\pi}(1-\cos 2\theta)\, d\theta dz = 4\int_0^5 \left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi} dz = 4\int_0^5 \pi\, dz = 4\pi\left[z\right]_0^5 = 20\pi$$

Conservative Vector Fields

In general, the value of the integral $\int_C \mathbf{F}.d\mathbf{r}$ between two stated points A and B depends on the particular path of integration followed. If, however, the line integral between A and B is independent of the path of integration between the two end points, then the vector field F is said to be conservative.



It follows that, for a closed path in a conservative

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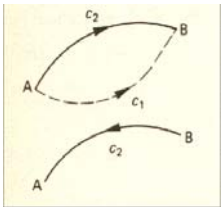
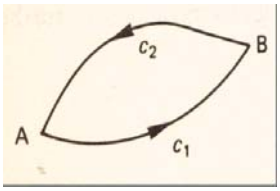
field, $\oint_C \mathbf{F}.d\mathbf{r} = 0$.



For, if the field is conservative,

$\int_{C_1(AB)} \mathbf{F}.d\mathbf{r} = \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$

But $\int_{c_2(BA)} \mathbf{F}.d\mathbf{r} - \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$



Hence, for the closed path $AB_{c_1} + BA_{c_2}$, $\oint \mathbf{F}.d\mathbf{r}$

$= \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} + \int_{c_2(AB)} \mathbf{F}.d\mathbf{r} = \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} - \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$

$= \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} - \int_{c_1(AB)} \mathbf{F}.d\mathbf{r} = 0$

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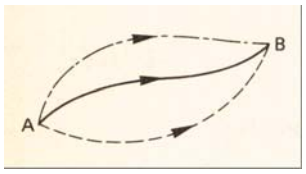
Lecture No -38 Vector Field

Conservative Vector Fields

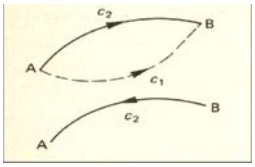
In general, the value of the integral $\int_C \mathbf{F}.d\mathbf{r}$ between two stated points A and B depends on the particular path of integration followed. If, however, the line integral between A and B is independent of the path of integration between the two end points, then the vector field F is said to be conservative.



It follows that, for a closed path in a conservative field, $\oint_C \mathbf{F}.d\mathbf{r} = 0$.

For, if the field is conservative,

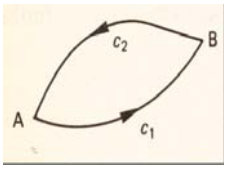
$\int_{C_1(AB)} \mathbf{F}.d\mathbf{r} = \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$ But $\int_{c_2(BA)} \mathbf{F}.d\mathbf{r} - \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$



Hence, for the closed path $AB_{c_1} + BA_{c_2}$, $\oint \mathbf{F}.d\mathbf{r}$

$= \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} + \int_{c_2(AB)} \mathbf{F}.d\mathbf{r}$

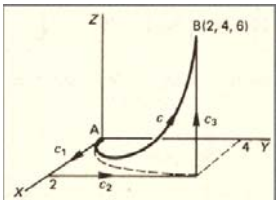
$= \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} - \int_{c_2(AB)} \mathbf{F}.d\mathbf{r} = \int_{C_1(AB)} \mathbf{F}.d\mathbf{r} - \int_{c_1(AB)} \mathbf{F}.d\mathbf{r} = 0$



Note that this result holds good only for a closed curve and when the vector field is a conservative field.
Now for some examples

Example

If $F = 2xyz\mathbf{i} + x^2z\mathbf{j} + x^2y\mathbf{k}$, evaluate the line integral $\int \mathbf{F}.dr$ between A(0,0,0) and B(2,4,6)
(a) along the curve c whose parametric equations are $x = u$, $y = u^2$, $z = 3u$
(b) along the three straight lines c_1:(0,0,0) to (2, 0, 0); c_2 : (2, 0, 0) to (2, 4, 0); c_3: (2, 4, 0) to (2, 4, 6).
Hence determine whether or not F is a conservative field.
First draw the diagram.



(a) $F = 2xyz\mathbf{i} + x^2z\mathbf{j} + x^2y\mathbf{k}$
$x = u$; $y = 2u$; $z = 3u$
$\therefore$ $dx = du$; $dy = 2udu$; $dz = 3du$.
$\mathbf{F}.d\mathbf{r} = (2xyz\mathbf{i}+x^2z\mathbf{j}+x^2y\mathbf{k}).(dx\mathbf{i}+dy\mathbf{j}+dz\mathbf{k})$
$= 2xyz\,dx + x^2z\,dy + x^2y\,dz$
Using the transformation shown above, we can now express $\mathbf{F}.dr$ in terms of u.
for $2xyzdx = (2u)(u^2)(3u)du = 6u^4du$
$x^2zdy = (u2)(3u)(2u)du = 6u^4du$
$x^2ydz = (u^2)(u^2)3du = 3u^4du$
$\therefore$ $\mathbf{F}.dr = 15u^4du$
The limits of integration in u are u = 0 to u = 2

$\therefore \int_C F.dr = \int_0^2 15u^4du = \left[3u^5\right]_0^2 = 96$

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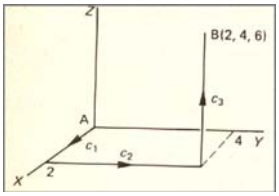
(b) The diagram for (b) is as shown. We consider each straigh



$\int \mathbf{F}.d\mathbf{r} = \int(2xyz\,dx + x^2z dy + x^2y dz)$

c_1: (0,0,0) to (2,0,0); y = 0, z = 0, dy = 0,

$\therefore \quad \int_{C_1} \mathbf{F}.d\mathbf{r} = 0 + 0 + 0 = 0$

In the same way, we evaluate the line integral along c_2 and c_3.

$$\int_{C_1} \mathbf{F}.d\mathbf{r} = 0;$$

$$\int \mathbf{F}.d\mathbf{r} = \int(2xyzdx + x^2zdy + x^2ydz)$$

c_2: (2,0,0) to (2,4,0); x = 2, z = 0, dx=0, dz = 0

$\therefore \quad \int_{C_2} \mathbf{F}.d\mathbf{r} = 0+0+0 = 0 \quad \int_{C_2} \mathbf{F}.d\mathbf{r} = 0$

c_3: (2,4,0) to (2,4,6); x = 2, y = 4, dx=0, dy = 0

$\therefore \quad \int_{C_3} \mathbf{F}.d\mathbf{r} = 0+0+\int_0^6 16dz = \left[16z\right]_0^6 = 96 \quad \therefore \quad \int_{C_3} \mathbf{F}.d\mathbf{r} = 96$

Collecting the three results together

$$\int_{c_1+c_2+c_3} \mathbf{F}.d\mathbf{r} = 0 + 0 + 96 \quad \therefore \quad \int_{c_1+c_2+c_3} \mathbf{F}.d\mathbf{r} = 96$$

In this particular example, the value of the line integral is independent of the two paths we have used joining the same two end points and indicates that F is a conservative field. It follows that

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz & x^2z & x^2y \end{vmatrix}$$

$$= (x^2 - x^2)\mathbf{i} - (2xy - 2xy)\mathbf{j} + (2xz - 2xz)\mathbf{k} = 0$$

$\therefore$ curl $\mathbf{F} = 0$

So three tests can be applied to determine whether or not a vector field is conservative. They are

(a) $\oint \mathbf{F}.d\mathbf{r} = 0$

(b) curl $\mathbf{F} = 0$

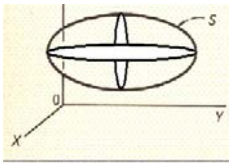
(c) $\mathbf{F}$ = grad V

Any one of these conditions can be applied as is convenient.

Divergence Theorem (Gauss' theorem)

For a closed surface S, enclosing a region V in a vector field F

$$\int_V \text{div } \mathbf{F}\, dV = \int_S \mathbf{F}.d\mathbf{S}$$



In general, this means that the volume integral (tripple integral) on the left-hand side can be expressed as a surface integral (double integral) on the right-hand side.

Example

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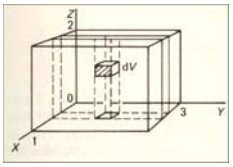
Verify the divergence theorem for the vector field
$\mathbf{F} = x^2\mathbf{i} + z\mathbf{j} + y\mathbf{k}$ taken over the region bonded by the planes $z = 0, z = 2, x = 0, x = 1, y = 0, y = 3$.



$dV = dx\,dy\,dz$

we have to show that $\int_V \text{div } \mathbf{F}\, dV = \int_S \mathbf{F}.d\mathbf{S}$

(a) To find $\int_V \text{div } \mathbf{F}\, dV$

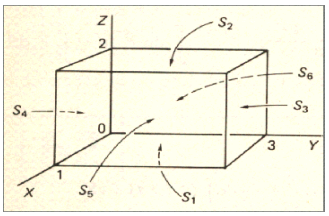
$$\text{div } F = \nabla.F = \left(\frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}\right).(x^2\mathbf{i} + z\mathbf{j} + y\mathbf{k})$$

$$= \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(z) + \frac{\partial}{\partial z}(y) = 2x + 0 + 0 = 2x$$

$$\therefore \int_V \text{div } F\, dV = \int_V 2x\, dv = \iiint_V 2x\,dz\,dy\,dx$$

$$\int_V \text{div } \mathbf{F}\, dV = \int_0^1\int_0^3\int_0^2 2x\,dz\,dy\,dx = \int_0^1\int_0^3 \Big[2xz\Big]_0^2 dy\,dx = \int_0^1 \Big[2xz\Big]_0^3 dx = \int_0^1 12x\,dx = \Big[6x^2\Big]_0^1 = 6$$

(b) to find $\int_S F.dS$ i.e. $\int_S \mathbf{F}.\hat{\mathbf{n}}\, dS$



The enclosing surface S consists of six separate plane faces denoted as
$S_1, S_2, \ldots S_6$ as shown. We consider each face in turn.
$\mathbf{F} = x^2\mathbf{i} + z\mathbf{j} + y\mathbf{k}$

(i) S_1 (base): $z = 0$; $\hat{\mathbf{n}} = -\mathbf{k}$ (outwards and downwards)

$\therefore$ $\mathbf{F} = x^2\mathbf{i} + y\mathbf{k}$ $\quad dS_1 = dx\,dy$

$$\therefore \int_{S_1} \mathbf{F}.\hat{\mathbf{n}}dS = \iint_{S_1} (x2\mathbf{i}+y\mathbf{k}).(-\mathbf{k})dydx = \int_0^1\int_0^3 (-y)\,dydx = \int_0^1 \left[-\frac{y^2}{2}\right]_0^3 dx = -\frac{9}{2}$$

(ii) S_2 (top): $z = 2$: $\hat{\mathbf{n}}=\mathbf{k}$ $\quad dS_2=dx\,dy$

$$\int_{S_2} \mathbf{F}.\hat{\mathbf{n}}dS = \iint_{S_2} (x^2\mathbf{i}+z\mathbf{j}+y\mathbf{k}).(\mathbf{k})dydx = \int_0^1\int_0^3 y\,dy\,dx = \frac{9}{2}$$

(iii) S_3 (right-hand end): $y = 3$;

$\hat{\mathbf{n}}=\mathbf{j}$ $dS_3 = dxdz\mathbf{j}$

$\mathbf{F} = x^2\mathbf{i} + z\mathbf{j} + y\mathbf{k}$

$$\therefore \int_{S_3} \mathbf{F}.\hat{\mathbf{n}}dS = \iint_{S_3} (x^2\mathbf{i}+z\mathbf{j}+3\mathbf{k}).(\mathbf{j})dzdx = \int_0^1\int_0^2 z\,dz\,dx = \int_0^1 \left[\frac{y^2}{2}\right]_0^2 dx = \int_0^1 2\,dx = 2$$

(iv) S_4 (left-hand end): $y = 0$, $\hat{\mathbf{n}} = -\mathbf{j}$, $\quad dS_4 = dx\,dz$

$$\therefore \int_{S_4} \mathbf{F}.\hat{\mathbf{n}}dS = -2$$

$$\text{for } \int_{S_4} \mathbf{F}.\hat{\mathbf{n}}dS = \iint_{S_4} (x^2\mathbf{i}+z\mathbf{j}+y\mathbf{k}).(-\mathbf{j})dzdx = \int_0^1\int_0^2 (-z)\,dz\,dx = \int_0^1 \left[-\frac{z^2}{2}\right]_0^2 dx$$

$$= \int_0^1 (-2)\,dx = -2$$

Now for the remaining two sides S_5 and S_6. Evaluate these in the same manner, obtaining

$$\int_{S_5} \mathbf{F}.\hat{\mathbf{n}}\, dS = 6; \qquad \int_{S_5} \mathbf{F}.\hat{\mathbf{n}}\, dS = 0$$

Check:

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(v) S_5 (front): x = 1; $\hat{\mathbf{n}} = \mathbf{i}$ $dS_5 = dy\,dz$

$\therefore \int_{S_5} F.\hat{\mathbf{n}}dS = \iint_{S_5} (\mathbf{i}+z\mathbf{j}+y\mathbf{k}).(\mathbf{i})dy\,dz = \iint_{S_5} 1\,dy\,dz = 6$

(vi) S_6 (back): x = 0; $\hat{\mathbf{n}} = -\mathbf{i}$ $dS_6 = dy\,dz$

$\therefore \int_{S_6} F.\hat{\mathbf{n}}dS = \iint_{S_6} (z\mathbf{j}+y\mathbf{k}).(-\mathbf{i})dy\,dz = \iint_{S_6} 0\,dy\,dz = 0$

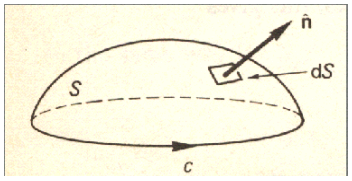
For the whole surface S we therefore have $\int_S \mathbf{F}.dS = -\frac{9}{2}+\frac{9}{2}+2-2+6+0 = 6$

and from our previous work in section (a) $\int_V \text{div } \mathbf{F}\, dV = 6$

We have therefore verified as required that, in this example $\int_V \text{div } \mathbf{F}\, dV = \int_S \mathbf{F}.d\mathbf{S}$

Stokes Theorem

If F is a vector field existing over an open surface S and around its boundary closed curve C, then $\int_S \text{curl } \mathbf{F}.dS = \oint_C \mathbf{F}.d\mathbf{r}$

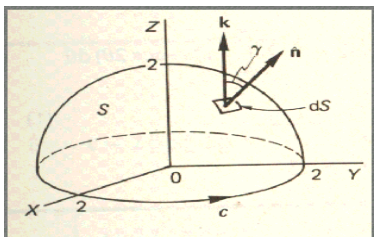


This means that we can express a surface integral in terms of a line integral round the boundary curve. The proof of this theorem is rather lengthy and is to be found in the Appendix. Let us demonstrate its application in the following examples.

Example

A hemisphere S is defined by $x^2 + y^2 + z^2 = 4$ $(z \geq 0)$. A vector field $\mathbf{F} = 2y\mathbf{i} - x\mathbf{j}+xz\mathbf{k}$ exists over the surface and around its boundary c. Verify Stoke's theorem that

$\int_S \text{curl } \mathbf{F}.dS = \oint_C \mathbf{F}.d\mathbf{r}$.



$S : x^2 + y^2 + z^2 - 4 = 0$

$\mathbf{F} = 2y\mathbf{i} - x\mathbf{j} + xz\mathbf{k}$ c is the circle $x^2 + y^2 = 4$.

(a) $\oint_C \mathbf{F}.d\mathbf{r} = \int_C (2y\mathbf{i} - x\mathbf{j} + xz\mathbf{k}) \cdot (\mathbf{i}dx + \mathbf{j}dy + \mathbf{k}dz)$

$= \int_C (2ydx - x\,dy + xz\,dz)$

Converting to polar coordintes. $x = 2\cos\theta$; $y = 2\sin\theta$; $z = 0$

$dx = -2\sin\theta\,d\theta$; $dy = 2\cos\theta\,d\theta$; Limits $\theta = 0$ to 2π

$$\oint_C \mathbf{F}.d\mathbf{r} = \int_0^{2/\pi} (4\sin\theta[(-2\sin\theta d\theta] - 2\cos\theta 2\cos\theta d\theta = -4\int_0^{2/\pi} (2\sin^2\theta + \cos^2\theta)\,d\theta$$

$$= -4\int_0^{2/\pi} (1 + \sin^2\theta)\,d\theta = -2\int_0^{2/\pi} (3 - \cos 2\theta)\,d\theta = -2\left[3\theta - \frac{\sin 2\theta}{2}\right]_0^{2\pi} = -12\,\pi \quad (1)$$

(b) Now we determine $\int_S \text{curl } \mathbf{F}.d\mathbf{S}$

$\int \text{curl } \mathbf{F}.d\mathbf{S} = \int \text{curl } \mathbf{F}.\,\hat{\mathbf{n}}dS$

$\mathbf{F} = 2y\mathbf{i} - x\mathbf{j} + xz\mathbf{k}$

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$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2y & -x & xz \end{vmatrix} = \mathbf{i}(0-0)-\mathbf{j}(z-0)+\mathbf{k}(-1-2) = -z\mathbf{j} - 3\mathbf{k}$$

Now $\hat{\mathbf{n}} = \frac{\nabla S}{|\nabla S|} = \frac{2x\mathbf{i}+2y\mathbf{j}+2z\mathbf{k}}{\sqrt{4x^2+4y^2+4z^2}} = \frac{x\mathbf{i}+y\mathbf{j}+z\mathbf{k}}{2}$

Then $\int_S \text{curl } \mathbf{F}.\hat{\mathbf{n}}dS = \int_S (-z\mathbf{k}-3\mathbf{k}).\left(\frac{x\mathbf{i}+y\mathbf{j}+z\mathbf{k}}{2}\right) dS = \frac{1}{2}\int_S (-yz-3z)\, dS$

$x = 2\sin\phi\cos\theta$; $y = 2\sin\phi\sin\theta$; $z = 2\cos\phi$, $dS = 4\sin\phi\, d\phi\, d\theta$

$\therefore \int_S \text{curl}\mathbf{F}.\hat{\mathbf{n}}dS = \frac{1}{2}\iint_S (-2\sin\phi\sin\theta\, 2\cos\phi - 6\cos\phi)\, 4\sin\phi\, d\phi\, d\theta$

$$= -\int_0^{2\pi}\int_0^{\pi/2} (2\sin^2\phi\cos\phi\sin\theta + 3\sin\phi\cos\phi)\, d\phi d\theta = -4\int_0^{2\pi}\left[\frac{2\sin^3\phi\sin\theta}{3} + \frac{3\sin^2\phi}{2}\right]_0^{\pi/2} d\theta$$

$$= -4\int_0^{2\pi}\left(\frac{2}{3}\sin\theta + \frac{3}{2}\right) d\theta = -12\pi \quad (2)$$

So we have from our two results (1) and (2)

$$\int_S \text{curl } \mathbf{F}.dS = \oint_C \mathbf{F}.d\mathbf{r}$$

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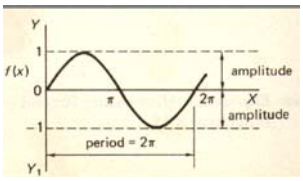
Lecture No -39 Periodic Functions

Periodic functions

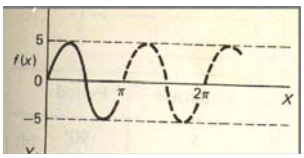
A function f(x) is said to be periodic if its function values repeat at regular intervals of the independent variable. The regular interval between repetitions is the period of the oscillations. $f(x + p) = f(x)$

Graphs of y = A sin nx



(a) y = sin x The obvious example of a periodic function is y = sin x, which goes through its complete range of values while x increases from 0° to 360°.

The period is therefore 360° or 2π radians and the amplitude, the maximum displacement from the position of rest.



y = 5sin 2x

The amplitude is 5. The period is 180° and there are thus 2 complete cycles in 360°.

Example

Functions	Amplitude	Period
y = 3 sin 5x	3	72°
y = 2 cos 3x	2	120°
$y = \sin \frac{x}{2}$	1	720°
y = 4 sin 2x	4	180°

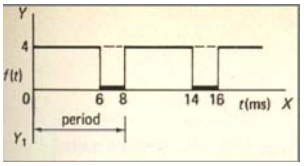
y = A sin nx

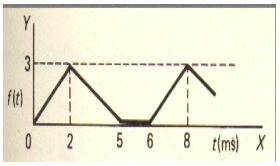
Thanking along the same lines, the function y = A sin nx

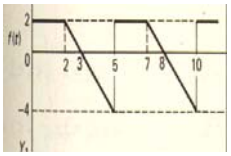
has amplitude = A; period = $\frac{360°}{n} = \frac{2\pi}{n}$; n cycles in 360°.

Graphs of y = A cos nx have the same characteristics

Example







period = 8 ms period = 6 ms period = 5 cm

Analytical description of a periodic function

A periodic function can be defined analytically in many cases.

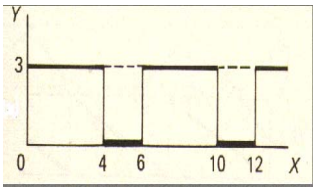
Example

(a) Between x = 0 and x = 4, y = 3,
i.e. $f(x) = 3 \quad 0 < x < 4$

(b) Between x = 4 and x = 6, y = 0,
i.e. $f(x) = 0 \quad 4 < x < 6$.

So we could define the function by

$f(x) = 3 \qquad 0 < x < 4$



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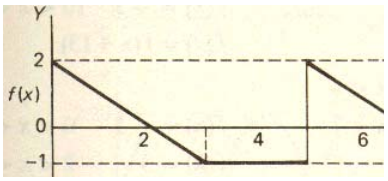
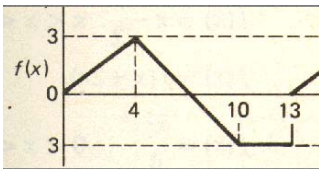
$f(x) = 0 \qquad 4 < x < 6$
$f(x) = f(x + 6)$
the last line indicating that the function is periodic with period 6 units



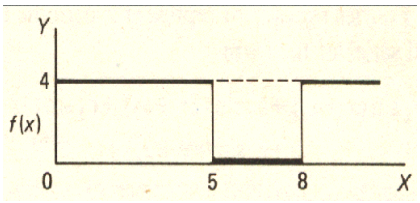
$f(x) = 2 - x \qquad 0 < x < 3$
$f(x) = -1 \qquad 3 < x < 5$
$f(x) = f(x+5)$
$f(x) = \frac{3x}{4} \qquad 0 < x < 4$
$f(x) = 7 - x \qquad 4 < x < 10$
$f(x) = -3 \qquad 10 < x < 13$
$f(x) = f(x + 13)$



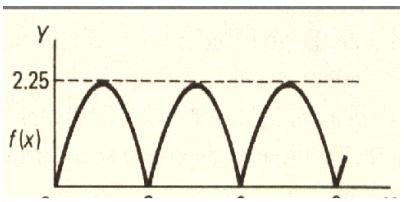
Example

Sketch the graphs of the following inserting relevant values.

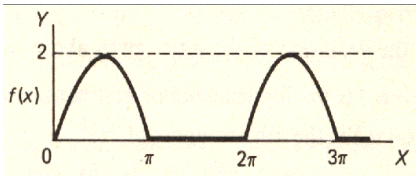
1. $f(x) = 4 \qquad 0 < x < 5$
   $f(x) = 0 \qquad 5 < x < 8$
   $f(x) = f(x + 8)$

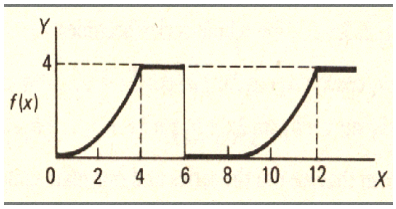


   $f(x) = 3x - x^2 \qquad 0 < x < 3$
   $f(x) = f(x + 3)$



   $f(x) = 2 \sin x \qquad 0 < x < \pi$
   $f(x) = 0 \qquad \pi < x < 2\pi$
   $f(x) = f(x + 2\pi).$
   $f(x) = \frac{x^2}{4} \qquad 0 < x < 4$
   $f(x) = 4 \qquad 4 < x < 6$
   $f(x) = 0 \qquad 6 < x < 8$
   $f(x) = f(x + 8).$





Useful integrals

The following integrals appear frequently in our work on Fourier series, so it will help if we obtain the result in readiness. In each case, m and n are integers other than zero.

(a) $\int_{-\pi}^{\pi} \sin nx\, dx = \left[\frac{-\cos nx}{n}\right]_{-\pi}^{\pi} = \frac{1}{n}\{-\cos n\pi + \cos n\pi\} = 0$

(b) $\int_{-\pi}^{\pi} \cos nx\, dx = \left[\frac{\sin nx}{n}\right]_{-\pi}^{\pi} = \frac{1}{n}\{\sin n\pi + \sin n\pi\} = 0$

(c) $\int_{-\pi}^{\pi} \sin^2 nx\, dx = \frac{1}{2}\int_{-\pi}^{\pi}(1 - \cos 2nx)\, dx = \frac{1}{2}\left[x - \frac{\sin 2nx}{2n}\right]_{-\pi}^{\pi} = \pi \ (n \neq 0)$

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(d) $\int_{-\pi}^{\pi} \cos^2 nx\, dx = \frac{1}{2}\int_{-\pi}^{\pi} (1 + \cos 2nx)\, dx = \frac{1}{2}\left[x + \frac{\sin 2nx}{2n}\right]_{-\pi}^{\pi} = \pi \quad (n \neq 0)$

(e) $\int_{-\pi}^{\pi} \sin nx \cos mx\, dx = \frac{1}{2}\int_{-\pi}^{\pi} \{\sin(n+m)x + \sin(n-m)x\}\, dx = \frac{1}{2}\{0 + 0\} = 0$

from result (a) with $n \neq m$

(f) $\int_{-\pi}^{\pi} \cos nx \cos mx\, dx = \frac{1}{2}\int_{-\pi}^{\pi} \{\cos(n+m)x + \cos(n-m)x\}\, dx = \frac{1}{2}\{0 + 0\} = 0$

from result (b) with $n \neq m$

Note:

If n = m, then

$\int_{-\pi}^{\pi} \cos nx \cos mx\, dx$ becomes $\int_{-\pi}^{\pi} \cos^2 nx\, dx = \pi$ from (d) above.

(g) $\int_{-\pi}^{\pi} \sin nx \sin mx\, dx = -\frac{1}{2}\int_{-\pi}^{\pi} (-2) \sin nx \sin mx\, dx = -\frac{1}{2}\int_{-\pi}^{\pi} \cos(n+m)x - \cos(n-m)x\, dx$

$= \frac{1}{2}\int_{-\pi}^{\pi} \{0 - 0) = 0$ from result (b) with $n \neq m$

Note:

If n = m, then

$\int_{-\pi}^{\pi} \sin nx \sin mx\, dx$ becomes $\int_{-\pi}^{\pi} \sin^2 nx\, dx = \pi$ from (c) above

Summary of integrals

(a) $\int_{-\pi}^{\pi} \sin nx\, dx = 0$, (b) $\int_{-\pi}^{\pi} \cos nx\, dx = 0$, (c) $\int_{-\pi}^{\pi} \sin^2 nx\, dx = \pi \ (\neq 0)$

(d) $\int_{-\pi}^{\pi} \cos^2 nx\, dx = \pi \ (n \neq 0)$ (e) $\int_{-\pi}^{\pi} \sin nx \cos mx\, dx = 0$

(f) $\int_{-\pi}^{\pi} \cos nx \cos mx\, dx = 0 \ (n \neq m)$ (g) $\int_{-\pi}^{\pi} \sin nx \sin mx\, dx = 0 \ (n \neq m)$

$\int_{-\pi}^{\pi} \cos nx \cos mx\, dx = \pi$ $(n = m)$ $\int_{-\pi}^{\pi} \sin nx \sin mx\, dx = \pi$ $(n = m)$

Note

We have evaluated the integrals between $-\pi$ and π, but, provided integration is carried out over a complete periodic interval of 2π, the results are the same. Thus, the limits could just as well be $-\pi$ to π, 0 to 2π, $-\pi/2$ to $3\pi/2$, etc. We can therefore choose the limits to suit the particular problems.

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Fourier series

Periodic functions of period 2π

The basis of a Fourier series is to represent a periodic function by a trigonometrical series of the form.

$f(x) = A_0+c_1 \sin (x+\alpha_1) + c_2 \sin (2x+\alpha_2) +c_3 \sin (3x+\alpha_3)+\ldots+c_n \sin (nx+\alpha_n)+ \ldots$

where A_0 is a constant term

$c_1, c_2, c_3 \ldots c_n$ denote the amplitudes of the compound sine terms $\alpha_1, \alpha_2, \alpha_3 \ldots..$ are constant auxiliary angles.

Each sine term, $c_n \sin (nx + \alpha_n)$ can be expanded thus:

$c_n \sin (nx + \alpha_n) = c_n\{\sin nx \cos\alpha_n+\cos n x \sin\alpha_n\}= (c_n\sin\alpha_n) \cos nx + (c_n\cos\alpha_n) \sin nx$

$= a_n\cos nx + b_n\sin nx$

the whole series becomes.

$$\mathbf{f(x) = A_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}}$$

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Lecture No- 40 Fourier Series

As we know that $A_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$;

which can be written as in the expanded from

$A_0+(a_1\cos x+b_1\sin x)+(a_2\cos 2x+b_2\sin 2x)+ \ldots + (a_n\cos nx+b_n\sin nx)+\ldots\ldots$

$A_0+a_1\cos x+a_2\cos 2x+.+a_n\cos nx+.. .. +b_1\sin x+b_2\sin 2x+... b_n\sin nx+\ldots\ldots$

$f(x) = A_0+a_1\cos x+a_2\cos 2x+.+a_n\cos nx+.. \ldots+b_1\sin x+b_2\sin 2x+... b_n\sin nx+...$

Fourier coefficients

We have defined Fourier series in the form

$f(x) = A_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$; n a positive integer

(a) To find A_0, we integrate f(x) with respect to x from $-\pi$ to π.

$$\int_{-\pi}^{\pi} f(x)dx = \int_{-\pi}^{\pi} A_0\, dx + \sum_{n=1}^{\infty} \left\{ \int_{-\pi}^{\pi} a_n \cos nx\, dx + \int_{-\pi}^{\pi} b_n \sin nx\, dx \right\} = [A_0 x]_{-\pi}^{\pi} + \sum_{n=1}^{\infty} \{ 0 + 0 \} = 2A_0\pi$$

$$2A_0\,\pi = \int_{-\pi}^{\pi} f(x)\, dx$$

$$A_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx = 1/2\, a_0 \; ; \qquad \text{Where } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\, dx$$

(b) To find a_n we multiply f(x) by cos mx and integrate from $-\pi$ to π.

$$\int_{-\pi}^{\pi} f(x)\cos mx\, dx = \int_{-\pi}^{\pi} A_0 \cos mx\, dx + \sum_{n=1}^{\infty} \left\{ \int_{-\pi}^{\pi} a_n \cos nx \cos mx\, dx + \int_{-\pi}^{\pi} b_n \sin nx \cos mx\, dx \right\}$$

$$\int_{-\pi}^{\pi} f(x)\cos mx\, dx = A_0 \{0\} + \sum_{n=1}^{\infty} \{a_n (0) + b_n (0)\} = 0 \qquad \text{for } n \neq m$$

$$= 0 + a_n\pi + 0 = a_n\pi \qquad \text{for } n = m$$

$$\therefore \; \mathbf{a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\, dx}$$

(c) To find b_n we multiply f(x) by sinmx and integrate from $-\pi$ to π.

$$\int_{-\pi}^{\pi} f(x)\sin mx\, dx = \int_{-\pi}^{\pi} A_0 \sin mx\, dx + \sum_{n=1}^{\infty} \left\{ \int_{-\pi}^{\pi} a_n \cos nx \sin mx\, dx + \int_{-\pi}^{\pi} b_n \sin nx \sin mx\, dx \right\}$$

$$\int_{-\pi}^{\pi} f(x)\sin mx\, dx = A_0 \{0\} + \sum_{n=1}^{\infty} \{a_n (0) + b_n (0)\} = 0 \qquad \text{for } n \neq m$$

$$= 0 + 0 + b_n\pi = b_n\pi \qquad \text{for } n = m$$

$$\therefore \; \mathbf{b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\, dx}$$

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Result For Fourier Series

$$\mathbf{f(x)=\frac{1}{2}a_0+\sum_{n=1}^{\infty}\{a_n cosnx+b_n sinnx\};}$$

(a) $a_0=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$ = 2×mean value of f(x) over a period

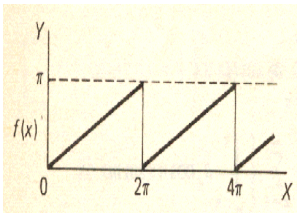
(b) $a_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$=2×mean value of f(x) cosnx over a period.

(c) $b_n=\frac{1}{\pi}\int_{-\pi} f(x)\sin nx\,dx$ = 2 × mean value of f(x) sin nx over a period.

In each case, n = 1, 2, 3, ……

Example

Determine the Fourier series to represent the periodic function shown.

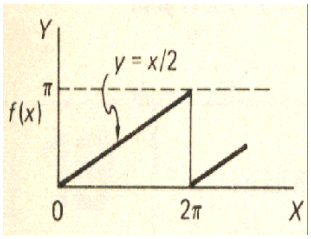


It is more convenient here to take the limits as 0 to 2π.

The function can be defined as

$f(x)=\frac{x}{2}$ $\qquad 0<x<2\pi$

$f(x)=f(x+2\pi)$ $\qquad$ period = 2π.



Now to find the coefficients

(a) $a_0=\frac{1}{\pi}\int_0^{2\pi} f(x)\,dx=\frac{1}{\pi}\int_0^{2\pi}\left(\frac{x}{2}\right)dx=\frac{1}{4\pi}\left[x^2\right]_0^{2\pi}=\pi$

$a_0=\pi$

(b) $a_n=\frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx$

$=\frac{1}{\pi}\int_0^{2\pi}\left(\frac{x}{2}\right)\cos nx\,dx$

$$a_n=\frac{1}{2\pi}\int_0^{2\pi} x\cos nx\,dx=\frac{1}{2\pi}\left\{\left[\frac{x\sin nx}{n}\right]_0^{2\pi}-\frac{1}{n}\int_0^{2\pi}\sin nx\,dx\right\}=\frac{1}{2\pi}\left\{(0-0)-\frac{1}{n}(0)\right\}$$

$\therefore a_n=0$

(a) $b_n=\frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx$ So we now have

$$b_n=\frac{1}{\pi}\int_0^{2\pi}\frac{x}{2}\sin nx\,dx=\frac{1}{2\pi}\left\{\left[-\frac{x\cos nx}{n}\right]_0^{2\pi}+\frac{1}{n}\int_0^{2\pi}\cos nx\,dx\right\}=-\frac{1}{2\pi n}\left[2\pi-0\right]=-\frac{1}{n}$$

$$\mathbf{a_0=\pi;\qquad a_n=0;\qquad b_n=-\frac{1}{n}}$$

Now the general expression for a Fourier series is

$f(x)=\frac{1}{2}a_0+\sum_{n=1}^{\infty}\{a_n\cos nx+b_n\sin nx\}$ Therefore in this case

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$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \{b_n \sin nx\} = \frac{\pi}{2} + \{-\frac{1}{1}\sin x - \frac{1}{2}\sin 2x - \frac{1}{3}\sin 3x - ...\} \quad \text{since } a_n = 0$$

$$f(x) = \frac{\pi}{2} - \{\sin x + \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x + ...\}$$

Dirichlet Conditions

If the Fourier series is to represent a function f(x), then putting $x = x_1$ will give an infinite series in x_1 and the value of this should converge to the value of $f(x_1)$ as more and more terms of the series are evaluated. For this to happen, the following conditions must be fulfilled.

(a) The function *f*(x) must be defined and single-valued.

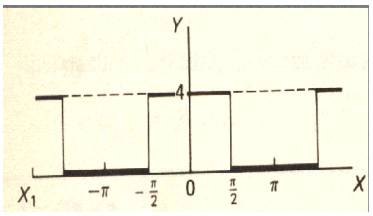
(b) *f*(x) must be continuous or have a finite number of finite discontinuities within a periodic interval.

(c) *f*(x) and $f'(x)$ must be piecewise continuous in the periodic interval.

If these Dirichlet conditions are satisfied, the Fourier series converges to $f(x_1)$, if $x = x_1$ is a point of continuity

Example

Find the Fourier series for the function shown.



Consider one cycle between x=0 and x=π.

The function can be defined by

$$f(x) = 0 \qquad -\pi < x < -\frac{\pi}{2}$$

$$f(x) = 4 \qquad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

$$f(x) = 0 \qquad \frac{\pi}{2} < x < \pi$$

$$f(x) = f(x + 2\pi)$$

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$$

The expression for a_0 is $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$ This gives

$$a_0 = \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2} 0\,dx + \int_{-\pi/2}^{\pi/2} 4dx + \int_{\pi/2}^{\pi} 0\,dx\right\} = \frac{1}{\pi}[4x]_{-\pi/2}^{\pi/2} = 4$$

$$a_0 = 4$$

$$(b)\ a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2} (0)\cos nx\,dx + \int_{-\pi/2}^{\pi/2} 4\cos nx\,dx + \int_{\pi/2}^{\pi} (0)\cos nx\,dx\right\} = \frac{1}{\pi}\left\{\int_{-\pi/2}^{\pi/2} 4\cos nx\,dx\right\}$$

$$= \frac{4}{\pi n}\sin nx\Big|_{-\pi/2}^{\pi/2} = \frac{8}{\pi n}\sin\frac{n\pi}{2}$$

$$a_n = \frac{8}{\pi n}\sin\frac{n\pi}{2}$$

Then considering different integer values of n, we have

If n is even $\quad a_n = 0$

If n = 1.5, 9…. $\quad a_n = \frac{8}{n\pi}$

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If n = 3, 7, 11, …. $a_n = -\frac{8}{n\pi}$

(c) To find b_n

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2}(0)\sin nx\,dx + \int_{-\pi/2}^{\pi/2} 4\sin nx\,dx + \int_{\pi/2}^{\pi}(0)\sin nx\,dx\right\}$$

$$b_n = \frac{4}{\pi}\int_{-\pi/2}^{\pi/2}\sin nx\,dx = \frac{4}{\pi}\left[\frac{-\cos nx}{n}\right]_{-\pi/2}^{\pi/2} = -\frac{4}{n\pi}\left\{\cos\frac{n\pi}{2} - \cos\left(\frac{-n\pi}{2}\right)\right\} = 0$$

$b_n = 0$

So with $a_0 = 4$: an as stated above; $b_n = 0$;

The Fourier series is

$f(x) = 2 + \frac{8}{\pi}\{\cos x - \frac{1}{3}\cos 3x + \frac{1}{5}\cos 5x - \frac{1}{7}\cos 7x + \ldots..\}$

In this particular example, there are, in fact, no sine terms.

Effect Of Harmonics

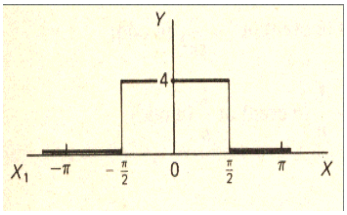
It is interesting to see just how accurately the Fourier series represents the function with which it is associated. The complete representation requires an infinite number of terms, but we can, at least, see the effect of including the first few terms of the series.
Let us consider the waveform shown. We established earlier that the function

$f(x) = 0 \qquad -\pi < x < -\frac{\pi}{2}$

$f(x) = 4 \qquad -\frac{\pi}{2} < x < \frac{\pi}{2}$

$f(x) = 0 \qquad \frac{\pi}{2} < x < \pi$

$f(x) = f(x + 2\pi)$

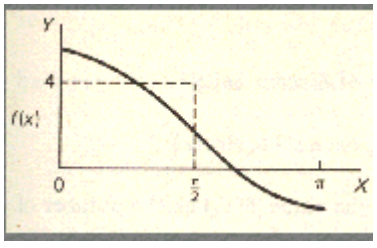


gives the Fourier series

$f(x) = 2 + \frac{8}{\pi}\{\cos x - \frac{1}{3}\cos 3x + \frac{1}{3}\cos 5x - \frac{1}{7}\cos 7x + \ldots.\}$

If we start with just one cosine term, we can then see the effect o including subsequent harmonics. Let us restrict our attention to just the right-hand half of the symmetrical waveform. Detailed plotting of points gives the following development.

(1) $f(x) = 2 + \frac{8}{\pi}\cos x$



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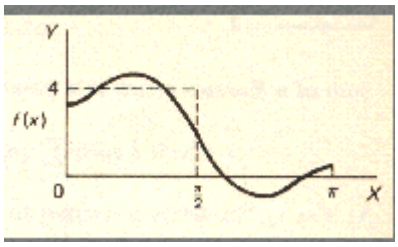
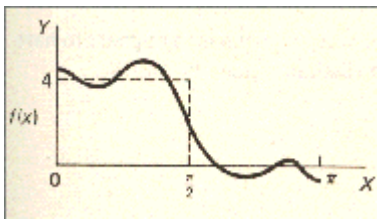
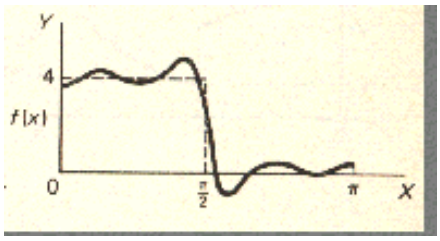
(2) $f(x) = 2 + \frac{8}{\pi} \{\cos x - \frac{1}{3}\cos 3x\}$



(3) $f(x)=2+\frac{8}{\pi} \{\cos x - \frac{1}{3}\cos 3x + \frac{1}{3}\cos 5x)$



(4) $f(x) = 2 + \frac{8}{\pi} \{\cos x - \frac{1}{3}\cos 3x + \frac{1}{3}\cos 5x - \frac{1}{7}\cos 7x\}$



As the number of terms is increased, the graph gradually approaches the shape of the original square waveform. The ripples increase in number and decrease in amplitude, but a perfectly square waveform is unattainable in practice. For practical purpose, the first few terms normally suffice to give an accuracy of acceptable level.

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Lecture No -41 Examples

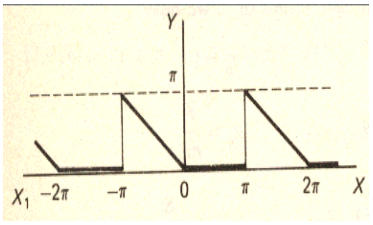
Example

Find the Fourier series for the function defined by

$f(x) = -x \qquad -\pi < x < 0$

$f(x) = 0 \qquad 0 < x < \pi$

$f(x) = f(x + 2\pi)$



The general expressions for a_0, a_n, b_n are

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx, \quad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\int_{-\pi}^{0}(-x)\,dx + \frac{1}{\pi}\int_{0}^{\pi} 0\,dx = \frac{1}{\pi}\int_{-\pi}^{0}(-x)\,dx = \frac{1}{\pi}\left[-\frac{x^2}{2}\right]_{-\pi}^{0} = \frac{\pi}{2}$$

(b) To find a_n

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_{-\pi}^{0}(-x)\cos nx\,dx + \frac{1}{\pi}\int_{0}^{\pi} 0\,dx = -\frac{1}{\pi}\int_{-\pi}^{\pi} x\cos nx\,dx$$

$$= -\frac{1}{\pi}\left\{\left[x\frac{\sin nx}{n}\right]_{-\pi}^{0} - \frac{1}{n}\int_{-\pi}^{0}\sin nx\,dx\right\} = -\frac{1}{\pi}\left\{(0-0) - \frac{1}{n}\left[\frac{-\cos nx}{n}\right]_{-\pi}^{0}\right\}$$

$$= -\frac{1}{\pi}\left\{\frac{1}{n}\left[\frac{\cos nx}{n}\right]_{-\pi}^{0}\right\} = -\frac{1}{\pi n^2}\left\{\left[\frac{\cos nx}{n}\right]_{-\pi}^{0}\right\} = -\frac{1}{\pi n^2}\left[\cos 0 - \cos n\pi\right]$$

$$= -\frac{1}{\pi n^2}\left\{1 - \cos n\pi\right\}$$

But $\cos n\pi = 1$ (n even) and $\cos n\pi = -1$ (n odd)

$a_n = -\frac{2}{\pi n^2}$ (n odd) and $a_n = 0$ (n even)

(c) Now to find b_n

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_{-\pi}^{0}(-x)\sin nx\,dx = -\frac{1}{\pi}\int_{-\pi}^{0} x\sin nx\,dx$$

$$= -\frac{1}{\pi}\left\{\left[x\left(\frac{-\cos nx}{n}\right)\right]_{-\pi}^{0} + \frac{1}{n}\int_{-\pi}^{0}\cos nx\,dx\right\} = -\frac{1}{\pi}\left\{\frac{\pi\cos n\pi}{n} + \frac{1}{n}\left[\frac{\sin nx}{n}\right]_{-\pi}^{0}\right\} = -\frac{\cos n\pi}{n}$$

$b_n = -\frac{1}{n}$ (n even) and $b_n = \frac{1}{n}$ (n odd)

So we have

$a_0 = \frac{\pi}{2}$; $a_n = 0$ (n even) and $a_n = -\frac{2}{\pi n^2}$ (n odd)

$b_n = -\frac{1}{n}$ (n even) and $b_n = \frac{1}{n}$ (n odd)

$$f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left(\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \ldots\right) + \left(\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \frac{1}{4}\sin 4x + \ldots.\right)$$

It is just a case of substituting n = 1, 2, 3, etc.

In this particular example, we have a constant term and both sine and cosine terms.

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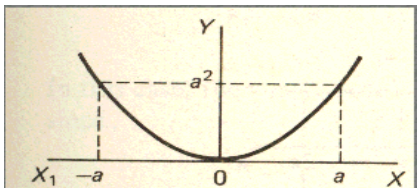
Odd And Even Functions

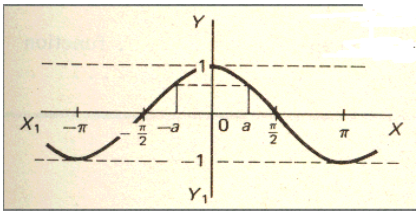
(a) Even functions

A function f(x) is said to be even if $f(-x) = f(x)$ i.e. the function value for a particular negative value of x is the same as that for the corresponding positive value of x. The graph of an even function is therefore symmetrical about the y-axis.

$y = f(x) = x^2$ is an even function
since
$f(-2) = 4 = f(2)$
$f(-3) = 9 = f(3)$ etc.



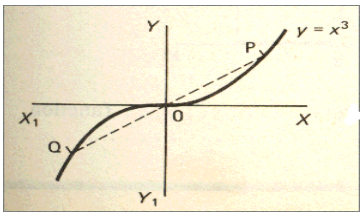
$y = f(x) = \cos x$ is an even function
since $\cos(-x) = \cos x$
$f(-a) = \cos a = f(a)$



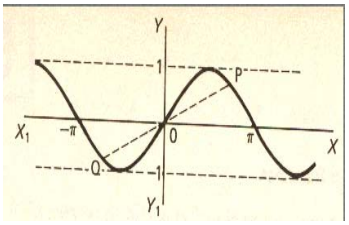
(b) Odd functions

A function f(x) is said to be odd if $f(-x) = -f(x)$
i.e. the function value for a particular negative value of xis numerically equal to that for the corresponding positive value of x but opposite in sign. The graph of an odd function is thus symmetrical about the origin.

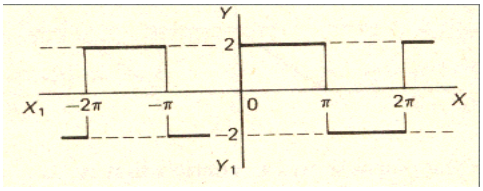
$y = f(x) = x^3$ is an odd function since
$f(-2) = -8 = -f(2)$
$f(-5) = -125 = -f(5)$ etc.



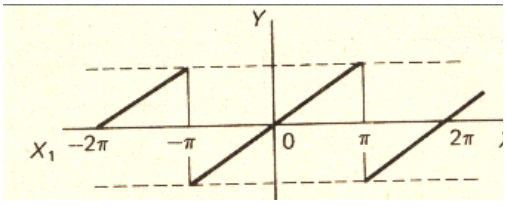
$y = f(x) = \sin x$ is an odd function
Since $\sin(-x) = -\sin x$
$f(-a) = -f(a)$.



So, for an even function $f(-x) = f(x)$ symmetrical about the y-axis
for an odd function $f(-x) = -f(x)$ symmetrical about the origin.



odd.



Odd

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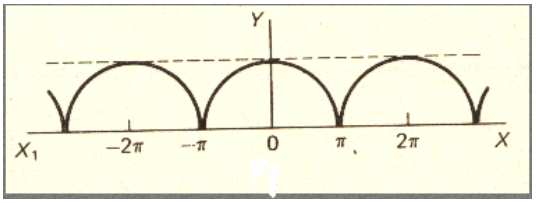
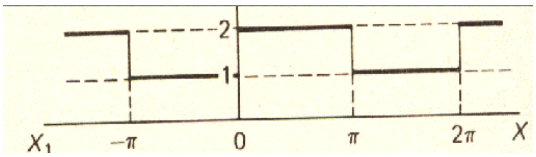
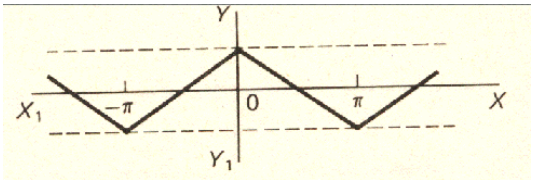


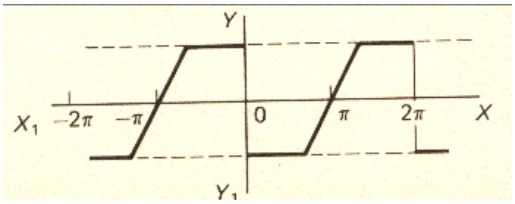
Even



neither



Even



Odd

Products Of Odd And Even Functions

The rules closely resemble the elementary rules of signs.

(even)×(even)=(even) like (+)×(+)=(+) ; (odd)× (odd) = (even) (–)×(–)=(+) ;

(odd)× (even) = (odd) (–)×(+)=(–)

The results can easily be proved.

(a) Two even functions

Let F(x) = f(x) g(x) where f(x) and g(x) are even functions.

Then F(– x) = f(– x) g (– x) = f(x) g(x) since f(x) and g(x) are even.

∴ F(– x) = F(x)

F(x) is even

(b) Two odd functions

Let F(x) = f(x) g(x) where f(x) and g(x) are odd functions.

Then F(– x) = f(– x) g(– x) = {– f (x)} {– g(x)}

since f(x) and g(x) are odd.

= f(x) g(x) = F (x)

∴ F(– x) = F(x)

F (x) is even

Finally

(c) One odd and one even function

Let F(x) = f(x) g(x) where f(x) is odd and g(x) even.

Then F (– x) = f (– x) g(– x) = – f(x) g(x) = – F (x)

∴ F (– x) = – F (x)

F (x) is odd

So if f(x) and g(x) are both even, then f(x) g(x) is even and if f(x) and g(x) are both odd, then f(x) g(x) is even but if either f(x) or g(x) is even and the other odd. Then f(x) g(x) is odd.

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Example

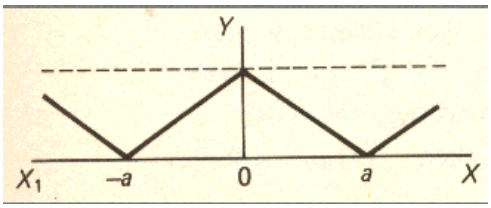
State whether each of the following products is odd, even, or neither.

1. $x^2 \sin 2x$ odd (E) (O) = (O)
2. $x^3 \cos x$ odd (O) (E) = (O)
3. $\cos 2x \cos 3x$ even (E) (E) = (E)
4. $x \sin nx$ even (O) (O) = (E)
5. $3 \sin x \cos 4x$ odd (O) (E) = (O)
6. $(2x + 3) \sin 4x$ neither (N) (O) = (N)
7. $\sin^2 x \cos 3x$ even (E) (E) = (E)
8. $x^3 e^x$ neither (O) (N) = (N)
9. $(x^4 + 4) \sin 2x$ odd (E) (O) = (O)

Two useful facts emerge from odd and even functions.

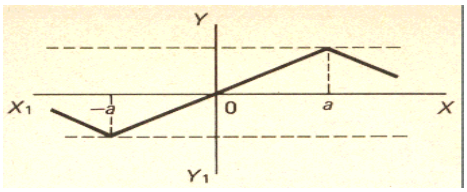
(a) For an even function

$$\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)\,dx$$



(b) For an odd function

$$\int_{-a}^{a} f(x)\,dx = 0$$



THEOREM 1

If f(x) is defined over the interval $-\pi < x < \pi$ and f(x) is even, then the Fourier series for f(x) contains cosine terms only. Included in this is a_0 which maybe regarded as $a_n \cos nx$ with n = 0.

Proof:

(a) $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)dx \quad \therefore \quad a_0 = \frac{2}{\pi}\int_{0}^{\pi} f(x)\,dx$

(b) $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx.$

But f(x) cos nx is the product of two even functions and therefore itself even.

$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx. \quad \therefore a_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx$

(c) $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$

Since f(x) sin nx is the product of an even function and an odd function, it is itself odd.

$\therefore \; b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = 0.$

$\therefore \; b_n = 0$

Therefore, there are no sine terms in the Fourier series for f(x).

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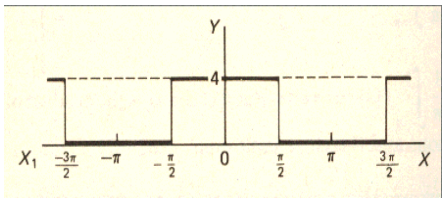
Example

The waveform shown is symmetrical about the y-axis. The function is therefore even and there will be no sine terms in the series.



$$\therefore \quad f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

(a) $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)\,dx. = \frac{2}{\pi}\int_{0}^{\pi/2} 4\,dx$

$= \frac{2}{\pi}[4x]_0^{\pi/2} = 4$

(b) $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_{0}^{\pi/2} 4\cos nx\,dx = \frac{8}{\pi}\left[\frac{\sin nx}{n}\right]_0^{\pi/2}$

$= \frac{8}{\pi n}\sin\frac{n\pi}{2}$

But $\sin\frac{n\pi}{2} = 0$ for n even

$= 1$ for n 1, 5, 9, …..

$= -1$ for n = 3, 7, 11, ….

$a_n = 0$ (n even);

$a_n = \frac{8}{\pi n}$ (n = 1, 5, 9, ….);

$a_n = -\frac{8}{\pi n}$ (n = 3, 7, 11 ……)

(c) $b_n = 0$, since f(x) is an even function.Therefore, the required series is

$$f(x) = 2 + \frac{\pi}{8}\{\cos x - \frac{1}{3}\cos 3x + \frac{1}{3}\cos 5x - \frac{1}{7}\cos 7x + \ldots.\}$$

Theorem 2:

If f(x) is an odd function defined over the interval $-\pi < x < \pi$, then the Fourier series for f(x) contains sine terms only.

Proof:

Since f(x) is an odd function

$$\int_{-\pi}^{0} \mathbf{f(x)\,dx} = -\int_{0}^{\pi} \mathbf{f(x)\,dx.}$$

(a) $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx.$

But f(x) is odd

$\therefore \quad a_0 = 0$

(b) $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$

Remembering that f(x) is odd and cosnx is even, the product f(x) cosnx is odd.

$\therefore\ a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi}$ (odd function) dx

$\therefore\ a_n = 0$

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(b) $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx \, dx$

and since f(x) and sin nx are each odd, the product f(x) sin nx is even.

Then $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi}$ (even function) dx = $\frac{2}{\pi}\int_{0}^{\pi} f(x) \sin nx \, dx$

$\therefore \; b_n = \frac{2}{\pi}\int_{0}^{\pi} f(x) \sin nx \, dx$

So,

If f(x) is odd function then $a_0 = 0$; $a_n = 0$;

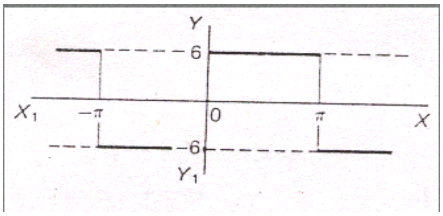
$b_n = \frac{2}{\pi}\int_{0}^{\pi} f(x) \sin nx \, dx$ i.e. the Fourier series contains sine terms only.

Example

f(x) = – 6 $\quad -\pi < x < 0$

f(x) = 6 $\quad 0 < x < \pi$

f(x) = f(x + 2π)



We can see that this is an odd function;

$a_0 = 0$ and $a_n = 0$

$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx \, dx.$

f(x) sin nx is a product of two odd functions and is therefore even.

$\therefore \; b_n = \frac{2}{\pi}\int_{0}^{\pi} f(x) \sin nx \, dx$

$b_n = \frac{2}{\pi}\int_{0}^{\pi} 6 \sin nx \, dx = \frac{12}{\pi}\left[\frac{-\cos nx}{n}\right]_0^{\pi} = \frac{12}{\pi}\left[\frac{\cos nx}{n}\right]_{\pi}^{0} = \frac{12}{\pi}\,[\cos 0 - \cos n\pi]$

$= \frac{12}{\pi n}\,(1 - \cos n\pi).$

$b_n = 0$ $\quad$ (n even) $\quad b_n = \frac{24}{\pi n}$ $\quad$ (n odd)

So the series is $f(x) = \frac{24}{\pi}\{\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \ldots.\}$

Of course,

if f(x) is neither an odd nor an even function, then we must obtain expressions for a_0, a_n and b_n in full.

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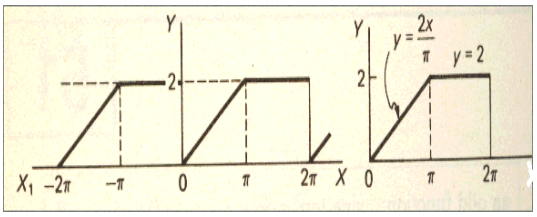
Lecture No -42 Examples

Example

Determine the Fourier series for the function shown.



This is neither odd nor even.

Therefore we must find a_0, a_n and b_n.

$$f(x) = \frac{1}{2}\, a_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$$

$$f(x) = \frac{2}{\pi} x \qquad 0 < x < \pi$$

$$= 2\,, \qquad \pi < x < 2\pi$$

$$(a)\ a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)dx = \frac{1}{\pi}\left\{\int_0^{\pi} \frac{2}{\pi} x\, dx + \int_{\pi}^{2\pi} 2\, dx\right\} = \frac{1}{\pi}\left\{\left[\frac{x^2}{\pi}\right]_0^{\pi} + \left[2x\right]_{\pi}^{2\pi}\right\} = \frac{1}{\pi}\ \{\pi + 4\pi - 2\pi\}$$

$$\therefore\ a_0 = 3$$

$$(b)\ a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\, dx = \frac{1}{\pi}\left\{\int_0^{\pi}\left(\frac{2}{\pi}x\right)\cos nx\, dx + \int_{\pi}^{2\pi} 2\cos nx\, dx\right\}$$

$$= \frac{1}{\pi}\left\{\frac{2}{\pi}\int_0^{\pi} x\cos nx\, dx + 2\int_{\pi}^{2\pi} \cos nx dx\right\} = \frac{2}{\pi}\left\{\frac{1}{\pi}\int_0^{\pi} x\cos nx\, dx + 2\int_{\pi}^{2\pi} \cos nx dx\right\}$$

$$= \frac{2}{\pi}\left[\left\{\frac{1}{\pi}\left[\frac{x\sin nx}{n}\right]_0^{\pi} - \frac{1}{\pi n}\int_0^{2\pi} \sin nx dx\right\} + \int_{\pi}^{2\pi} \cos nx dx\right] = \frac{2}{\pi}\left\{0 - \frac{1}{n\pi}\left[-\frac{\cos nx}{n}\right]_0^{\pi} + \left[\frac{\sin nx}{n}\right]_{\pi}^{2\pi}\right\}$$

$$= \frac{2}{\pi}\left\{\frac{1}{n\pi}\left[\frac{\cos nx}{n}\right]_0^{\pi}\right\} = \frac{2}{\pi^2 n^2}\ \{\cos n\pi - \cos 0\} = \frac{2}{\pi^2 n^2}\ \{\cos n\pi - 1\}$$

$a_n = 0$ (n even); $a_n = \dfrac{-4}{\pi^2 n^2}$ (n odd)

(c) To find b_n, we proceed in the same general manner

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\, dx = \frac{1}{\pi}\left\{\int_0^{\pi}\left(\frac{2}{\pi}x\right)\sin nx\, dx + \int_{\pi}^{2\pi} 2\sin nx\, dx\right\}$$

$$= \frac{2}{\pi}\left\{\frac{1}{\pi}\left[\frac{-x\cos nx}{n}\right]_0^{\pi} + \frac{1}{\pi n}\int_0^{\pi}\cos nx dx + \int_{\pi}^{2\pi} \sin nx dx\right\} = \frac{2}{\pi}\left\{\frac{1}{\pi n}(-\pi\cos n\pi) + \frac{1}{\pi n}\left[\frac{\sin nx}{n}\right]_0^{\pi} + \left[\frac{-\cos nx}{n}\right]_{\pi}^{2\pi}\right\}$$

$$= \frac{2}{\pi}\left\{-\frac{1}{n}\cos n\pi + (0-0) - \frac{1}{n}(\cos 2\pi n - \cos n\pi)\right\} = \frac{2}{\pi}\left\{-\frac{1}{n}\cos n\pi + (0-0) - \frac{1}{n}\cos 2\pi n + \frac{1}{n}\cos n\pi\right\}$$

$$= \frac{2}{\pi}\left\{-\frac{1}{n}\cos 2n\pi\right\} = -\frac{2}{\pi n}\ \cos 2n\pi$$

But $\cos 2n\pi = 1$.

$$\therefore\ b_n = -\frac{2}{\pi n}$$

So the required series is

$$f(x) = \frac{3}{2} - \frac{4}{\pi^2}\left\{\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \ldots\right\} - \frac{2}{\pi}\left\{\sin x + \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x + \frac{1}{4}\sin 4x \ldots\right\}$$

Sum of a Fourier series at a point of discontinuity

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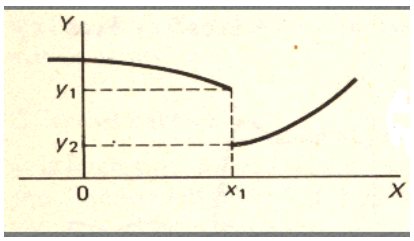
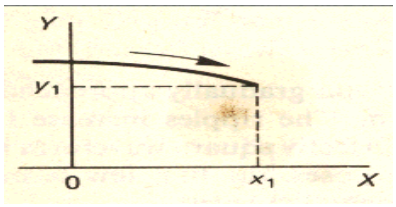
$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$$



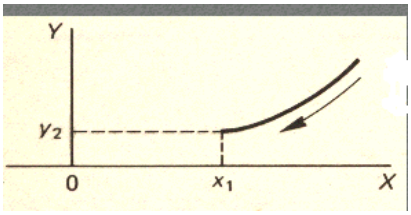
At $x = x_1$, the series converges to the value $f(x_1)$ as the number of terms including increases to infinity.

A particular point of interest occurs at a point of finite discontinuity or `jump' of the function $y = f(x)$. At $x = x_1$, the function appears to have two distinct values, y_1 and y_2.



If we approach $x = x_1$ from below that value, the limiting value of $f(x)$ is y_1.

If we approach $x = x_1$ from above that value, the limiting value of $f(x)$ is y_2.



To distinguish between these two values we write
$y_1 = f(x_1 - 0)$ denoting immediately before $x = x_1$
$y_2 = f(x_1 + 0)$
denoting immediately after $x = x_1$.
In fact, if we substitute $x = x_1$ in the Fourier series for $f(x)$, it can be shown that the series converges to the value

$\frac{1}{2}\{f(x_1 - 0) + f(x_1 + 0)\} = \frac{1}{2}(y_1 + y_2)$, the average of y_1 and y_2.

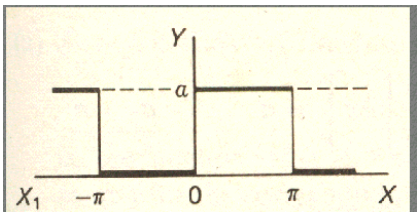
Example

Consider the function

$f(x) = 0 \qquad -\pi < x < 0$

$f(x) = a \qquad 0 < x < \pi$

$f(x) = f(x + 2\pi)$



$$(a)\ a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\int_{0}^{\pi} a\,dx$$

$$= \frac{1}{\pi}[ax]_0^{\pi} = a$$

$$\therefore \quad a_0 = a$$

$$(b)\ a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_{0}^{\pi} a\cos nx\,dx = \frac{a}{\pi}\left[\frac{\sin nx}{n}\right]_0^{\pi} = 0$$

$$\therefore\ a_n = 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_{0}^{\pi} a\sin nx\,dx 0 = \frac{a}{\pi}\left[\frac{-\cos nx}{n}\right]_0^{\pi} = \frac{a}{n\pi}(1 - \cos n\pi)$$

But $\cos n\pi = 1$ (n even) and $\cos n\pi = -1$ (n odd)

$$b_n = 0 \ \text{(n even)} \quad \text{and} \quad b_n = \frac{2a}{n\pi} \ \text{(n odd)}$$

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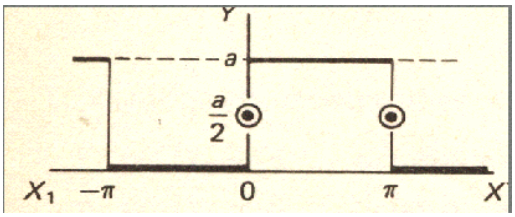
$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} b_n \sin nx$$

$$f(x) = \frac{a}{2} + \frac{2a}{\pi} \{\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + ...\}$$

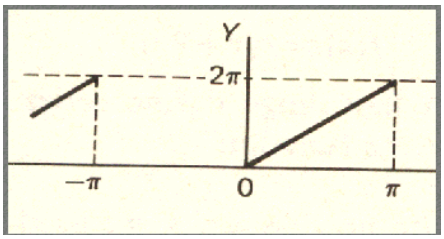
A finite discontinuity, or `jump', occurs at x = 0. If we substitute x = 0 in the series obtained, all the sine terms vanish and we get f(x) = a/2, which is, in fact, the average of the two function values at x = 0.



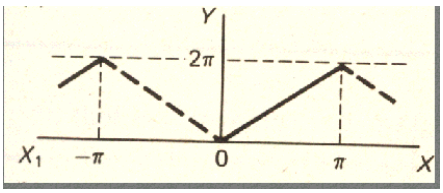
Note also that at x = π, another finite discontinuity occurs and substituting x = π in the series gives the same result. Because of this ambiguity, the function is said to be `undefined' at x = 0, x = π, etc.

Half-Range Series

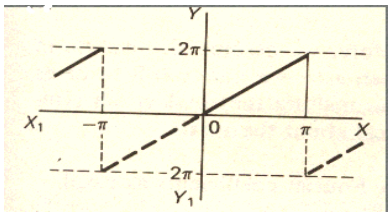
Sometime a function of period 2π is defined over the range 0 to π, instead of the normal – π to π, or 0 to 2π. We then have a choice of how to proceed. For example, if we are told between x = 0 and x = π, f(x) = 2x, then, since the period is 2π, we have no evidence of how the function behave between x = – π and x = 0.



If the waveform were as shown in (a), the function would be an even function, symmetrical about the y-axis and the series would have only cosine terms (including possibly a_0).



On the other hand, if the waveform were as shown in (b), the function would be odd, being symmetrical about the origin and the series would have only sine terms.



Example

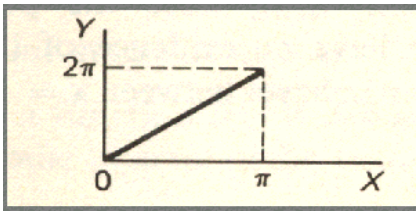
A function f(x) is defined by

f(x) = 2x $0 < x < \pi$

f(x) = f(x + 2π)

Obtain a half-range cosine series to represents the function.

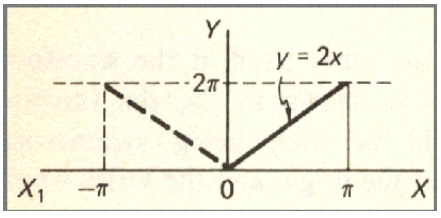
To obtain a cosine series, i.e. a series with no sine terms, we need an even function. Therefore, we assume the waveform between x = – π and x = 0 to be as shown, making the total graph symmetrical about the y-axis.



Now we can find expressions for the Fourier coefficients as usual.

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx = \frac{2}{\pi}\int_0^{\pi} 2x dx = \frac{2}{\pi}\left[x^2\right]_0^{\pi} = 2\pi$$

$\therefore\ a_0 = 2\pi$



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$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} 2x\cos nx\,dx = \frac{4}{\pi}\int_0^{\pi} x\cos nx\,dx = \frac{4}{\pi}\left\{\left[\frac{x\sin nx}{n}\right]_0^{\pi} - \frac{1}{n}\int_0^{\pi}\sin nx\,dx\right\}$$

$$= \frac{4}{\pi}\left\{(0-0) - \frac{1}{n}\left[\frac{-\cos nx}{n}\right]_0^{\pi}\right\} = \left(\frac{4}{\pi n^2}\right)(\cos n\pi - 1)$$

$\cos n\pi = 1$ (n even) and $\cos n\pi = -1$ (n odd)

$\therefore\ a_n = 0$ (n even) and $a_n = -\frac{8}{\pi n^2}$ (n odd)

All that now remains is b_n which is zero, since f(x) is an even function, i.e. $b_n = 0$

So $a_0 = 2\pi$, $a_n = 0$ (n even) and $a_n = -\frac{8}{\pi n^2}$ (n odd),

$b_n = 0$. Therefore

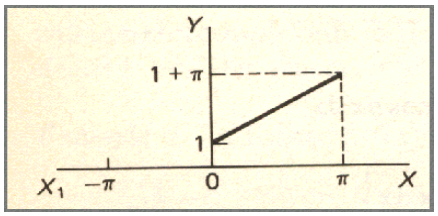
$$f(x) = \pi - \frac{8}{\pi}\left\{\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \ldots\right\}$$

Example

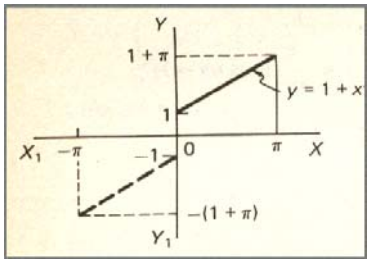
Determine a half-range sine series to represent the function f(x) defined by

$f(x) = 1 + x \quad 0 < x < \pi$

$f(x) = f(x + 2\pi)$



We choose the waveform between $x = -\pi$ and $x = 0$ so that the graph is symmetrical about the origin. The function is then an odd function and the series will contain only sine terms.



$\therefore\ a_0 = 0$ and $a_n = 0$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$b_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)\sin nx\,dx$$

$$b_n = \frac{2}{\pi}\int_0^{\pi}(1+x)\sin nx\,dx = \frac{2}{\pi}\left\{\left[(1+x)\frac{-\cos nx}{n}\right]_0^{\pi} + \frac{1}{n}\int_0^{\pi}\cos nx\,dx\right\}$$

$$= \frac{2}{\pi}\left\{-\frac{1+\pi}{n}\cos n\pi + \frac{1}{n} + \frac{1}{n}\left[\frac{\sin nx}{n}\right]_0^{\pi}\right\} = \frac{2}{\pi}\left\{\frac{1}{n} - \frac{1+\pi}{n}\cos n\pi\right\} = \frac{2}{\pi n}\{1 - (1+\pi)\cos n\pi\}$$

$\cos n\pi = 1$ (n even) and $\cos n\pi = -1$ (n odd)

$\therefore\ b_n = -\frac{2}{n}$ (n even)

$\therefore\ = \frac{4+2\pi}{\pi n}$ (n odd)

Substituting in the general expression $f(x) = \sum_{x=1}^{\infty} b_n \sin nx$ we have

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$$f(x)=\frac{4+2\pi}{\pi}\{\sin x+\frac{1}{3}\sin 3x+\frac{1}{5}\sin 5x+...\}-2\{\frac{1}{2}\sin 2x+\frac{1}{4}\sin 4x+\frac{1}{6}\sin 6x+....)$$

and the required series obtained

$$f(x)=\left(\frac{4}{\pi}+2\right)\{\sin x+\frac{1}{3}\sin 3x+\frac{1}{5}\sin 5x+..\}-2\{\frac{1}{2}\sin 2x+\frac{1}{4}\sin 4x+\frac{1}{6}\sin 6x+.....\}$$

So knowledge of odd and even functions and of half-range series saves a deal of unnecessary work on occasions.

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Lecture No.-43 Functions With Periods Other Than 2π

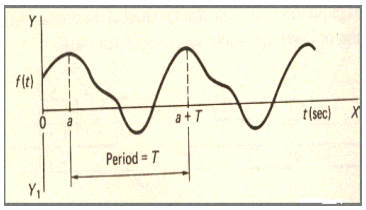
So far, we have considered functions f(x) with period 2π. In practice, we often encounter functions defined over periodic intervals other than 2π, e.g. from 0 to T, $-\frac{T}{2}$ to $\frac{T}{2}$ etc.

Functions With Period T

If y = f(x) is defined in the range $-\frac{T}{2}$ to $\frac{T}{2}$, i.e. has a period T, we can convert this to an interval of 2π by changing the units of the independent variable.

In many practical cases involving physical oscillations, the independent variable is time (t) and the periodic interval is normally denoted by T, i.e.

$$f(t) = f(t + T)$$



Each cycle is therefore completed in T seconds and the frequency f hertz (oscillations per second) of the periodic function is therefore given by $f = \frac{1}{T}$. If the angular velocity, ω radians per seconds, is defined by ω = 2πf, then

$$\omega = \frac{2\pi}{T} \text{ and } T = \frac{2\pi}{\omega}$$

The angle, x radians, at any time t is therefore x = ωt and the Fourier series to represent the function can be expressed **as**

$$\mathbf{f(t) = \frac{1}{2} a_0 + \sum_{x=1}^{\infty} \{a_n Cos\ n\omega t + b_n Sin\ n\omega t\}}$$

which can also be written in the form

$$\mathbf{f(t) = \frac{1}{2} A_0 + \sum_{x=1}^{\infty} B_n \sin(n\omega t + \phi_n)} \qquad n = 1, 2, 3, ...$$

Fourier Coefficients

With the new variable, the Fourier coefficients become:

$$f(t) = \frac{1}{2} a_0 + \sum_{x=1}^{\infty} \{a_n \cos n\omega t + b_n \sin n\omega t\}$$

$$a_0 = \frac{2}{T}\int_0^T f(t)\,dt = \frac{\omega}{\pi}\int_0^{2\pi/\omega} f(t)\,dt$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega t\,dt = \frac{\omega}{\pi}\int_0^{2\pi/\omega} f(t)\cos n\omega t\,dt$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega t\,dt = \frac{\omega}{\pi}\int_0^{2\pi/\omega} f(t)\sin n\omega t\,dt$$

We can see that there is very little difference between these expressions and those that have gone before. The limits can, of course, be 0 to T, $-\frac{T}{2}$ to $\frac{T}{2}$, $-\frac{\pi}{\omega}$ to $\frac{\pi}{\omega}$, 0 to $\frac{2\pi}{\omega}$ etc. as is convenient, so long as they cover a complete period.

Example

Determine the Fourier series for a periodic function defined by

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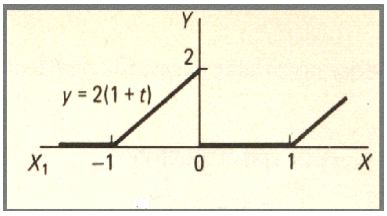
$f(t) = 2(1 + t) \qquad -1 < t < 0$

$f(t) = 0 \qquad 0 < t < 1$

$f(t) = f(t + 2)$

The first step is to sketch the wave which is.



$$f(t) = \frac{1}{2}\, a_0 + \sum_{x=1}^{\infty} \{a_n \cos n\omega t + b_n \sin n\omega t\})$$

$T = 2$

$$a_0 = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\, dt = \frac{2}{2}\int_{-1}^{1} f(t)\, dt = \int_{-1}^{0} 2(1 + t)\, dt + \int_{0}^{1} (0)\, dt = \left[2t + t^2\right]_{1}^{0} = 1$$

$$\therefore \quad a_0 = 1$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cos n\omega t\, dt = \frac{2}{2}\int_{-1}^{1} f(t)\cos n\omega t\, dt = \int_{-1}^{0} 2(1 + t)\cos n\omega t\, dt$$

$$a_n = 2\left\{\left[(1+t)\frac{\sin n\omega t}{n\omega}\right]_{-1}^{0} - \frac{1}{n\omega}\int_{-1}^{0} \sin n\omega t\, dt\right\} = 2\left\{(0-0) - \frac{1}{n\omega}\left[-\frac{\cos n\omega t}{n\omega}\right]_{-1}^{0}\right\}$$

$$= \frac{2}{n^2\omega^2}\,(1 - \cos n\omega)$$

Now $T = \frac{2\pi}{\omega}$

$$\therefore \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi \quad \therefore a_n = \frac{2}{n^2\omega^2}\,(1 - \cos n\pi)$$

$$\therefore \quad a_n = 0 \text{ (n even)}$$

$$= \frac{4}{n^2\omega^2} \text{ (n odd)}$$

Now for b_n

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t) \sin n\omega t\, dt$$

$$b_n = \frac{2}{2}\int_{-1}^{0} 2(1+t) \sin n\omega t\, dt = 2\left\{\left[(1+t)\frac{-\cos n\omega t}{n\omega}\right]_{-1}^{0} + \frac{1}{n\omega}\int_{-1}^{0} \cos n\omega t\, dt\right\}$$

$$= 2\left\{-\frac{1}{n\omega} + \frac{1}{n\omega}\left[\frac{\sin n\omega t}{n\omega}\right]_{-1}^{0}\right\} = 2\left\{-\frac{1}{n\omega} + \frac{1}{n^2\omega^2}(\sin n\omega)\right\}$$

As before $\omega = \pi$

$$\therefore \quad b_n = -\frac{2}{n\omega}$$

So the first few terms of the series give

$$f(t) = \frac{1}{2} + \frac{4}{\omega^2}\left\{\cos \omega t + \frac{1}{9}\cos 3\omega t + \frac{1}{25}\cos 5\omega t + ..\right\} - \frac{2}{\omega}\left\{\sin \omega t + \frac{1}{2}\sin 2\omega t + \frac{1}{3}\sin 3\omega t + \frac{1}{4}\sin 4\omega t..\right\}$$

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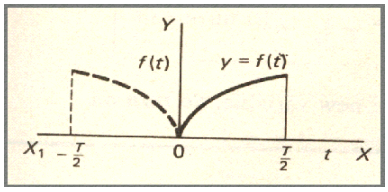
Half-Range Series

The theory behind the half-range sine and cosine series still applies with the new variable.

(a) Even function

Half-range cosine series



$y = f(t) \qquad 0 < t < \frac{T}{2}$

$f(t) = f(t + T)$

symmetrical about the y-axis.

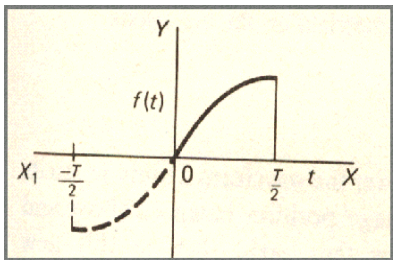
With an even function, we know that $b_n = 0$

$$\therefore \quad f(t) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t$$

where $a_0 = \frac{4}{T}\int_0^{T/2} f(t)\, dt$ and $a_n = \frac{4}{T}\int_0^{T/2} f(t) \cos n\omega t\, dt$

(b) Odd function

Half-range sine series



$y = f(t) \qquad 0 < t < \frac{T}{2}$

$f(t) = f(t + T)$

symmetrical about the origin.

$\therefore a_0 = 0$ and $a_n = 0$

$$f(t) = \sum_{x=1}^{\infty} b_n \sin n\omega t;$$

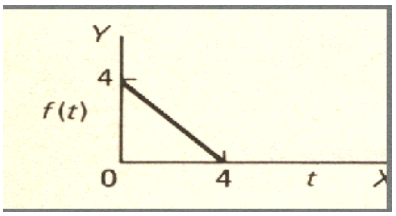
$$b_n = \frac{4}{T}\int_0^{T/2} f(t) \sin n\omega t\, dt$$

Example

A function f(t) is defined by

$f(t) = 4 - t, \qquad 0 < t < 4.$

We have to form a half-range cosine series to represent the function in this interval.

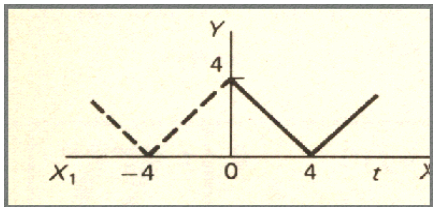


First we form an even function, i.e. symmetrical about the y-axis.

$$a_0 = \frac{4}{T}\int_0^{T/2} f_1(t)\, dt = \frac{4}{8}\int_0^{4} (4 - t)\, dt = \frac{1}{2}\int_0^{4} (4 - t)\, dt$$

$$= \frac{1}{2}\left|4t - \frac{t^2}{2}\right|_0^4 = \frac{1}{2}\left[4(4) - \frac{(4)^2}{2}\right] = \frac{1}{2}[16 - 8] = \frac{1}{2}(8) = 4$$

$$a_n = \frac{4}{T}\int_0^{T/2} f_1(t) \cos n\omega t\, dt = \frac{4}{8}\int_0^{4} (4 - t) \cos n\omega t\, dt$$



Simple integration by parts gives

$$a_n = \frac{1}{2}\left\{-\frac{2 \sin 4n\omega}{n\omega} - \frac{1}{n^2\omega^2}(\cos 4n\omega - 1)\right\}$$

But $\omega = \frac{2\pi}{T} = \frac{2\pi}{8} = \frac{\pi}{4}$

$$\therefore a_n = \frac{1}{2}\left\{-\frac{2 \sin n\pi}{n\omega} - \frac{1}{n^2\omega^2}(\cos n\pi - 1)\right\}$$

$n = 1, 2, 3, \ldots\ldots..$

$\sin n\pi = 0;$

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$\cos n\pi = 1$ (n even); $\cos n\pi = -1$ (n odd)

$\therefore\ a_n = 0$ (n even)

$a_n = \frac{1}{n^2\omega^2}$ (n odd)

$\therefore\ f(t)=2+\frac{1}{\omega^2}\left\{\cos \omega t+\frac{1}{9}\cos 3\omega t+\frac{1}{25}\cos 5\omega t+\ldots\right\}$ where $\omega = \frac{\pi}{4}$.

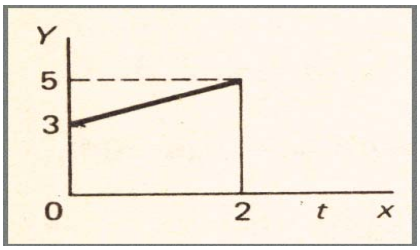
Example

A function f(t) is defined by

$f(t) = 3 + t \qquad 0 < t < 2$

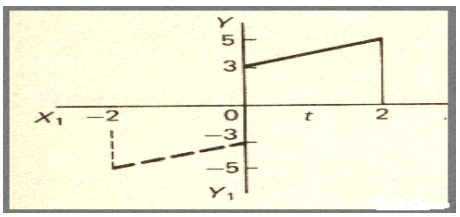
$f(t) = f(t + 4)$

Obtain the half-range sine series for the function in this range.



Sine series required. Therefore, we form an odd function, symmetrical about the origin.

$a_0 = 0; \qquad a_n = 0; \qquad T = 4$



$f(t) = \sum_{n=1}^{\infty} b_n \sin n\omega t$

$\therefore\ b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin n\omega t\, dt = \int_0^2 (3 + t)\sin n\omega t\, dt$

$\int_0^2 (3 + t)\sin n\omega t\, dt = \left|\frac{(3+t)\cos n\omega t}{-n\omega}\right|_0^2 - \int_0^2 \frac{\cos n\omega t}{-n\omega}\, dt$

$= \frac{(3 + 2)\cos n\omega 2}{-n\omega} - \frac{3}{-n\omega} + \frac{1}{n\omega}\left|\frac{\sin n\omega t}{n\omega}\right|_0^2 = \frac{3}{n\omega} - \frac{5}{n\omega}\cos 2n\omega + \frac{1}{n^2\omega^2}\left[\frac{\sin 2n\omega}{n\omega} - 0\right]$

But $T = \frac{2\pi}{\omega} \qquad \therefore \qquad \omega = \frac{2\pi}{T} = \frac{\pi}{2}$

$= \frac{3}{n\omega} - \frac{5}{n\omega}\cos n\pi + \frac{1}{n^2\omega^2}\left[\frac{\sin n\pi}{n\omega}\right]$

$b_n = \frac{1}{n\omega}(3 - 5\cos 2n\omega) + \frac{1}{n^2\omega^2}(\sin 2n\omega)$

$\therefore\ b_n = \frac{1}{n\omega}(3 - 5\cos n\pi) + \frac{1}{n^2\omega^2}(\sin n\pi) = -\frac{2}{n\omega}$ (n even)

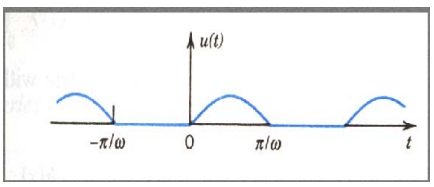
$= \frac{8}{n\omega}$ (n odd)

$\therefore\ f(t)=\frac{2}{\omega}\left\{4\sin\omega t - \frac{1}{2}\sin 2\omega t+\frac{4}{3}\sin 3\omega t - \frac{1}{4}\sin 4\omega t..\right\}$

Half-Wave Rectifier

A sinusoidal voltage E sin ωt, where t is time, is passed through a half-wave rectifier that clips the negative portion of the wave

Find the Fourier series of the resulting periodic functions.



$u(t) = 0 \qquad$ if $\qquad -T/2 < t < 0$

$= E \sin \omega t \qquad 0 < t < T/2$

here $T = \frac{2\pi}{\omega}$

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$$a_0 = \frac{2}{T}\int_{-T/2}^{T/2} u(t)\,dt = \frac{2}{T}\int_{-T/2}^{0} 0\,dt + \frac{2}{T}\int_{0}^{T/2} E\sin\omega t\,dt = \frac{2}{T}\int_{0}^{T/2} E\sin\omega t\,dt = \frac{\omega}{\pi}\int_{0}^{\pi/\omega} E\sin\omega t\,dt$$

$$= \frac{\omega}{\pi} E\left|\frac{-\cos\omega t}{\omega}\right|_0^{\pi/\omega} = \frac{2E}{\pi}$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} u(t)\cos n\omega t\,dt = \frac{2}{T}\int_{0}^{T/2} E\sin\omega t\cos n\omega t\,dt = \frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} 2\sin\omega t\cos n\omega t\,dt$$

$$= \frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} [\sin(1+n)\omega t + \sin(1-n)\omega t]\,dt$$

If n = 1 then integral on the right is zero
and if n = 2, 3, …… then we obtain.

$$a_n = \frac{\omega E}{2\pi}\left[-\frac{\cos(1+n)\omega t}{(1+n)\omega} - \frac{\cos(1-n)\omega t}{(1-n)\omega}\right]_0^{\pi/\omega} = \frac{\omega E}{2\pi}\left[\frac{-\cos(1+n)\pi + 1}{(1+n)\omega} + \frac{-\cos(1-n)\pi + 1}{(1-n)\omega}\right]$$

$$= \frac{\omega E}{2\pi\omega}\left[\frac{-\cos(1+n)\pi + 1}{(1+n)} + \frac{-\cos(1-n)\pi + 1}{(1-n)}\right]$$

if n is odd then $a_n = 0$

if n is even then $a_n = \frac{E}{2\pi}\left(\frac{2}{1+n} + \frac{2}{1-n}\right) = \frac{E}{2\pi}\left[\frac{2 - 2n + 2 + 2n}{(1+n)(1-n)}\right] = \frac{2E}{(1-n)(1+n)\pi}$

$$= \frac{2E}{(1-n^2)\pi}$$

For b_n we have

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} u(t)\sin n\omega t\,dt = \frac{2}{T}\int_{0}^{T/2} E\sin\omega t\sin n\omega t\,dt = -\frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} -2\sin\omega t\sin n\omega t\,dt$$

$$= -\frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} [\cos(1+n)\omega t - \cos(1-n)\omega t]\,dt$$

If n = 1

$$b_n = -\frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} [\cos 2\omega t - 1]\,dt = -\frac{\omega E}{2\pi}\left|\frac{\sin 2\omega t}{2\omega} - t\right|_0^{\pi/\omega} = -\frac{\omega E}{2\pi}(-\pi/\omega) = E/2$$

if $n \neq 1$

$$b_n = -\frac{\omega E}{2\pi}\int_{0}^{\pi/\omega} [\cos(1+n)\omega t - \cos(1-n)\omega t]\,dt = -\frac{\omega E}{2\pi}\left[\frac{\sin(1+n)\omega t}{(1+n)\omega} - \frac{\sin(1-n)\omega t}{(1-n)\omega}\right]_0^{\pi/\omega}$$

$$= -\frac{\omega E}{2\pi}\left[\frac{\sin(1+n)\pi}{(1+n)\omega} - \frac{\sin(1-n)\pi}{(1-n)\omega}\right] = 0 \qquad \text{for } n = 2, 3, 4, \ldots\ldots$$

$$u(t) = \frac{1}{2}a_0 + \sum_{n=2}^{\infty} a_n \cos n\omega t$$

$$u(t) = \frac{E}{\pi} + \frac{E}{2}\sin\omega t - \frac{2E}{\pi}\left(\frac{1}{1.3}\cos 2\omega t + \frac{1}{3.5}\cos 4\omega t + \ldots\right)$$

Lecture No -44 Laplace Transforms

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$\mathcal{L}\{e^{3t}\} = \frac{1}{s-3}$; $\mathcal{L}\{t^3\} = \frac{3!}{s^4}$; $\mathcal{L}\{2\sin 3t + \cos 3t\} = \frac{s+6}{s^2+9}$

And $L^{-1}\{\frac{1}{s-3}\} = e^{3t}$; $L^{-1}\{\frac{3!}{s^4}\} = t^3$; $L^{-1}\{\frac{s+6}{s^2+9}\} = 2\sin 3t + \cos 3t$

We show that $L\{t^3\} = \frac{3!}{s^4}$ For this consider the integral

$$L(t^3) = \int_0^\infty t^3 e^{-st} dt = [t^3 \frac{e^{-st}}{-s}]_0^\infty - \int_0^\infty 3t^2 e^{-st} \frac{1}{-s} dt = \frac{-t^3}{se^{st}} \Big|_0^\infty + \frac{3}{s}\int_0^\infty t^2 e^{-st} dt$$

$$= 0 + \frac{3}{s}\{t^2 \frac{e^{-st}}{-s} \Big|_0^\infty + \frac{2}{s}\int_0^\infty te^{-st} dt\} = \frac{3}{s}\frac{2}{s}\int_0^s te^{-st} dt = \frac{3.2}{s^2}\{t \frac{e^{-st}}{-s} \Big|_0^\infty + \frac{1}{s}\int_0^\infty e^{-st} dt\}$$

$$= \frac{3.2}{s^3}\int_0^\infty e^{-st} dt = \frac{3.2}{s^3}\frac{-1}{s}\int_0^\infty e^{-st}(-s) dt = \frac{-3.2.1}{s^4} e^{-st} \Big|_0^\infty = \frac{-3.2.1}{s^4}(0-1) = \frac{3.2.1}{s^4} = \frac{3!}{s^4}$$

So that $L\{t^3\} = \frac{3!}{s^4}$

Laplace Transform

The Laplace Transform of a function F(t) is denoted by L{F(t)} and is defined as the integral of F(t) e^{-st} between the limits t=0 and t = ∞

$$L\{F(t)\} = \int_0^\infty e^{-st} F(t) dt.$$

In all cases, the constant parameter s is assumed to be positive and large enough to ensure that the product F(t) e^{-st} converges to zero as t $\to \infty$, what ever the function F(t).
In determining the transform of any function , you will appreciate that the limits are substituted for t, so that the result will be a function of s.
Laplace Transform of F(t) = a (constant).
That is

$$L(a) = \int_0^\infty ae^{-st} dt = a\int_0^\infty e^{-st} dt$$

$$= a\frac{e^{-st}}{-s} \Big|_0^\infty = \frac{-a}{s}\{0-1\} = \frac{a}{s}$$

Example

Find the laplace transform of the form e^{at} that F(t)= e^{at} where a is a constant.

$$L(e^{at}) = \int_0^\infty e^{at} e^{-st} dt = \int_0^\infty e^{-(s-a)t} dt = [\frac{e^{-(s-a)t}}{-(s-a)}]_0^\infty$$

$$= -\frac{1}{s-a}\left[\frac{1}{e^{(s-a)t}}\right], \quad s > a$$

$$= -\frac{1}{s-a}[0 - 1] = \frac{1}{s-a}$$

$$L\{e^{at}\} = \frac{1}{s-a}$$

So we already have two standard transforms

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$L\{a\}=\frac{a}{s}$; $L\{e^{at}\}=\frac{1}{s-a}$;

$L\{4\}=\frac{4}{s}$; $L\{e^{4t}\}=\frac{1}{s-4}$;

$L\{-5\}=\frac{-5}{s}$; $L\{e^{-2t}\}=\frac{1}{s+2}$;

Laplace transform is always a function of s.

Complex Numbers Power of i

Every time a factor i^4 occurs , it can be replace by the factor 1, so that the power of i is reduced to one of the four results above.

$i^9 = (i^4)^2 i = (1)^2 i = 1.\ i = i$
$i^{20} = (i^4)^5 = (1)^5 = 1$
$i^{30} = (i^4)^7\ i^2 = (1)^7 (-1) = 1(-1) = -1$
$i^{15} = (i^4)^3\ i^3 = 1(-i) = -i$

Complex Numbers

z = 3+ 5 i , is called a complex number where 3 is real part and 5 is imaginary part of the complex number.

In general z = a+ b i , is called a complex number where a is real part and b is imaginary part of the complex number. So,

Complex Number = (Real Part)+ *i* (Imaginary Part)

Conjugate complex numbers

For a complex number a + *i* b , the complex number a - *i* b is called the conjugate of a + *i* b. Conjugate complex numbers are identical except the signs in the middle for the brackets.

- (4 + 5 i) and (4 - 5 i) are conjugate complex numbers
- (6 + 2 i) and (2 + 6 i) are not conjugate complex numbers
- (5 - 3 i) and (-5 +3 i) are not conjugate complex numbers

Remember

The product of complex number by its conjugate is always entirely real.

(3 + 4 *i*) (3 - 4 *i*) = 9 + 16 = 25

(a+ b *i*) (a - b*i*) = $a^2 + b^2$

Euler Formaula

As we know that the series expansion of e^x ,cos x and sin x are given as

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots\ldots\ldots\ldots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots\ldots\ldots\ldots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots\ldots\ldots\ldots$$

Replace x by (*i*t) , we get

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$$e^{(it)} = 1 + (it) + \frac{(it)^2}{2!} + \frac{(it)^3}{3!} + \frac{(it)^4}{4!} + \ldots\ldots\ldots\ldots$$

$$e^{(it)} = 1 + (it) - \frac{(t)^2}{2!} - \frac{i(t)^3}{3!} + \frac{(t)^4}{4!} + \frac{i(t)^5}{5!} - \frac{(t)^6}{6!} - \frac{i(t)^7}{7!} + \ldots\ldots\ldots\ldots$$

$$e^{(it)} = [1 - \frac{(t)^2}{2!} + \frac{(t)^4}{4!} - \frac{(t)^6}{6!} + \ldots\ldots\ldots\ldots] + i[t - \frac{(t)^3}{3!} + \frac{(t)^5}{5!} - \frac{(t)^7}{7!} + \ldots\ldots\ldots\ldots]$$

where

$$\cos t = 1 - \frac{(t)^2}{2!} + \frac{(t)^4}{4!} - \frac{(t)^6}{6!} + \ldots\ldots\ldots\ldots$$

$$\sin t = t - \frac{(t)^3}{3!} + \frac{(t)^5}{5!} - \frac{(t)^7}{7!} + \ldots\ldots\ldots\ldots$$

$$e^{it} = \cos t + i \sin t$$

R(e^{it})= cost and I(e^{it}) = sin t

The Laplace transform of F(t) = sin at

$$L(\text{Sin at}) = L(I(e^{iat})) = I\int_0^\infty e^{iat}e^{-st}dt = I\int_0^\infty e^{-(s-ia)t}dt = I\left\{\left[\frac{e^{-(s-ia)t}}{-(s-ia)}\right]_0^\infty\right\}$$

$$= I\left\{\frac{1}{-(s-ia)}(0-1)\right\} = I\left\{\frac{1}{(s-ia)}\right\}$$

$$L\{\sin at\} = I\{\frac{s+ia}{s^2+a^2}\} = I[\frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2}]$$

$$\therefore L\{\sin at\} = \frac{a}{s^2+a^2}$$

We can use the same method to determined to determine L{cos at }.
Since cos at is the real part of e^{iat}, written as R(e^{iat})

$$L\{\cos at\} = R\{\frac{s+ia}{s^2+a^2}\} = R[\frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2}]$$

$$L\{\cos at\} = \frac{s}{s^2+a^2}$$

The Transform of F(t) = t^n where n is a positive integer.

By the definition $L(t^n) = \int_0^\infty t^n e^{-st}dt$ integrating by parts

$$L(t^n) = \int_0^\infty t^n e^{-st}dt = \left[t^n\frac{e^{-st}}{-s}\right]_0^\infty + \frac{n}{s}\int_0^\infty t^{n-1}e^{-st}dt = -\frac{1}{s}\left[t^n e^{-st}\right]_0^\infty + \frac{n}{s}\int_0^\infty t^{n-1}e^{-st}dt$$

$$\therefore \left[t^n\frac{e^{-st}}{-s}\right]_0^\infty = 0 - 0 = 0 \qquad \therefore L\{t^n\} = \frac{n}{s}\int_0^\infty t^{n-1}e^{-st}dt \text{ ----------(1)}$$

you will notice that $\int_0^\infty t^{n-1}e^{-st}dt$ is identical to $\int_0^\infty t^n e^{-st}dt$ except that n is replaced by (n-1)

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If $if\ I_n = \int_0^\infty t^n e^{-st} dt, then\ I_{n-1} = \int_0^\infty t^{n-1} e^{-st} dt$ and the result (1) becomes

$$I_n = \frac{n}{s} I_{n-1}$$

This is reduction formula and , if we now replace n by (n-1) we get

$$I_{n-1} = \frac{n-1}{s} I_{n-2}$$

If we replace n by (n-1) again in this last result , we have

$$I_{n-2} = \frac{n-2}{s} I_{n-3}$$

$$So\ I_n = \int_0^\infty t^n e^{-st} dt = \frac{n}{s} I_{n-1} = \frac{n}{s}.\frac{n-1}{s} I_{n-2} = \frac{n}{s}.\frac{n-1}{s}.\frac{n-2}{s} I_{n-3} = \frac{n}{s}.\frac{n-1}{s}.\frac{n-2}{s}.\frac{n-3}{s} I_{n-4}$$

So finally, we have

$$I_n = \frac{n}{s}.\frac{n-1}{s}.\frac{n-2}{s}.\frac{n-3}{s}.\frac{n-4}{s}.....\frac{2}{s}.\frac{1}{s} I_0$$

But

$$I_0 = L\{t^0\} = L\{1\} = \frac{1}{s}$$

$$I_n = \frac{n}{s}\frac{n-1}{s}\frac{n-2}{s}\frac{n-3}{s}\frac{n-4}{s}.....\frac{2}{s}\frac{1}{s} I_0 = \frac{n!}{s^{n+1}}$$

$$L\{t^n\} = \frac{n!}{s^{n+1}}$$

$$L\{t\} = \frac{1}{s^2}; L\{t^2\} = \frac{2!}{s^3}; L\{t^3\} = \frac{3!}{s^4} = \frac{6}{s^4}$$

Laplace Transform of F(t) = sinh at and F(t) = cosh at.

Starting from the exponential definitions of sinh at and cosh at

$$i.e.\quad \sinh at = \frac{1}{2}(e^{at} - e^{-at})\ and\ \cosh at = \frac{1}{2}(e^{at} + e^{-at})$$

We proceed as follow

a) $F(t) = \sinh at$

$$\text{Sinh at} = \int_0^\infty \sinh at\ e^{-st} dt = \frac{1}{2}\int_0^\infty (e^{at} - e^{-at}) e^{-st} dt = \frac{1}{2}\int_0^\infty (e^{at}e^{-st} - e^{-at}e^{-st}) dt = \frac{1}{2}\int_0^\infty (e^{-(s-a)t} - e^{-(s+a)t}) dt$$

$$= \frac{1}{2}\left\{\left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^\infty - \left[\frac{e^{-(s+a)t}}{-(s+a)}\right]_0^\infty\right\} = \frac{1}{2}\left\{\frac{1}{-(s-a)}(0-1) - \frac{1}{-(s+a)}(0-1)\right\} = \frac{1}{2}\left\{\frac{1}{(s-a)} - \frac{1}{(s+a)}\right\}$$

$$= \frac{1}{2}\left\{\frac{s+a-s+a}{(s-a)(s+a)}\right\} = \frac{1}{2}\left\{\frac{2a}{s^2-a^2}\right\} = \left\{\frac{a}{s^2-a^2}\right\}$$

b) $F(t) = \cosh at$

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$$L(\cosh at) = L(\frac{1}{2}(e^{at} + e^{-at})) = \frac{1}{2}\int_0^\infty (e^{at} + e^{at})e^{-st}dt = \frac{1}{2}\left[\int_0^\infty e^{-(s-a)t}dt + \int_0^\infty e^{-(s+a)t}dt\right]$$

$$= \frac{1}{2}\left\{\left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^\infty + \left[\frac{e^{-(s+a)t}}{-(s+a)}\right]_0^\infty\right\} = \frac{1}{2}\left\{\frac{1}{-(s-a)}(0-1) + \frac{1}{-(s+a)}(0-1)\right\} = \frac{1}{2}\left\{\frac{1}{(s-a)} + \frac{1}{(s+a)}\right\}$$

$$= \frac{1}{2}\left\{\frac{s+a+s-a}{(s-a)(s+a)}\right\} = \frac{1}{2}\left\{\frac{2s}{s^2-a^2}\right\} = \left\{\frac{s}{s^2-a^2}\right\}$$

$$L(\cosh at) = \left\{\frac{s}{s^2-a^2}\right\}$$

Several Standard Results

$$L\{a\} = \frac{a}{s}; \qquad L\{e^{at}\} = \frac{1}{s-a}; \qquad L\{t^n\} = \frac{n!}{s^{n+1}}$$

$$L\{\sin at\} = \frac{a}{s^2+a^2}; \qquad L\{\cos at\} = \frac{s}{s^2+a^2}$$

$$L\{\sinh at\} = \frac{a}{s^2-a^2}; \qquad L\{\cosh at\} = \frac{s}{s^2-a^2}$$

We can , of course, combine these transforms by adding or subtracting as necessary , but they must not be multiplied together to form the transform of a product.

Example

$$a)\ L\{2\sin 3t + \cos 3t\} = 2L(\sin 3t) + L(\cos 3t) = 2.\frac{3}{s^2+9} + \frac{s}{s^2+9} = \frac{s+6}{s^2+9}$$

$$b)\ L\{4e^{2t} + 3\cosh 4t\} = 4L(e^{2t}) + L(3\cosh 4t) = 4.\frac{1}{s-2} + 3.\frac{s}{s^2-16} = \frac{4}{s-2} + \frac{3s}{s^2-16}$$

$$= \frac{7s^2 - 6s - 64}{(s-2)(s^2-16)}$$

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Lecture No -45 Theorems

Theorem 1
The First Shift Theorem

The first Shift theorem states that if **L{F(t)}= f(s)** then **$L\{e^{(-at)} F(t)\}= f(s+a)$**
The transform **$L\{e^{(-at)} F(t)\}$** is thus the same as **L{F(t)}** with s everywhere in the result replaced by (s+a)

Example

$$L\{\sin 2t\}= \frac{2}{s^2+4} \quad \textbf{then} \quad L\{e^{-3t} \sin 2t\}= \frac{2}{(s+3)^2+4} = \frac{2}{s^2+6s+13}$$

Example

$L\{t^2\} = \frac{2}{s^3}$; $L\{t^2 e^{4t}\}$ is the same with s replaced by (s-4)

So $L\{t^2 e^{4t}\} = \frac{2}{(s-4)^3}$

Theorem 2
Multiplying by t

If $L\{F(t)\}= f(s)$ **then** $L\{t(F(t)\} = -\frac{d}{ds}\{f(s)\}$

Example $L\{\sin 2t\}= \frac{2}{s^2+4}$

And $L\{t \sin 2t\} = -\frac{d}{ds}\left[\frac{2}{(s^2+4)}\right] = \frac{4s}{(s^2+4)^2}$

Example $L\{t\cos 3t\} = -\frac{d}{ds}\left(\frac{s}{s^2-9}\right) = -\frac{(s^2-9)-s(2s)}{(s^2-9)^2} = -\frac{s^2-9-2s^2}{(s^2-9)^2} = \frac{s^2+9}{(s^2-9)^2}$

We could, if necessary, take this a stage further and find

$$L\{t\cos 3t\} = -\frac{d}{ds}\left(\frac{s^2+9}{(s^2-9)^2}\right) = \frac{2s(s^2+27)}{(s^2-9)^3}$$

Theorem Obviously extends the range of function that we can deal with. So, in general

If $L\{F(t)\}= f(s)$ **then** $L\{t^n(F(t)\} = (-1)^n \frac{d^n}{ds^n}\{f(s)\}$

Theorem
Dividing by t

If $L\{F(t)\}= f(s)$ **then** $L\{\frac{F(t)}{t}\} = \int_s^\infty f(s)ds$

Example Determine $L\{\frac{\sin at}{t}\}$

$As\ L\{\sin at\} = \frac{a}{s^2+a^2}$

$$\therefore L\{\frac{\sin at}{t}\} = \int_s^\infty \frac{a}{s^2+a^2}ds = \left[\tan^{-1}(\frac{s}{a})\right]_s^\infty = \frac{\pi}{2} - \tan^{-1}(\frac{s}{a}) = \tan^{-1}(\frac{a}{s})$$

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Example Determine $L\{\frac{1-\cos 2t}{t}\}$

As $L\{1-\cos 2t\}=\frac{1}{s}-\frac{s}{s^2+4}$

Then by Theorem 3,

$$L\{\frac{1-\cos 2t}{t}\}=\int_s^\infty(\frac{1}{s}-\frac{s}{s^2+4})ds=\left[\ln s-\frac{1}{2}\ln(s^2+4)\right]_s^\infty=\left[\frac{1}{2}.2\ln s-\frac{1}{2}\ln(s^2+4)\right]_s^\infty$$

$$=\left[\frac{1}{2}\ln s^2-\frac{1}{2}\ln(s^2+4)\right]_s^\infty=\frac{1}{2}\left[\ln s-\ln(s^2+4)\right]_s^\infty=\frac{1}{2}\left[\ln\frac{s^2}{(s^2+4)}\right]_s^\infty$$

When $s\to\infty$ *then* $\ln\frac{s^2}{(s^2+4)}\to\ln 1=0$

$$L\{\frac{1-\cos 2t}{t}\}=\int_s^\infty(\frac{1}{s}-\frac{s}{s^2+4})ds=-\frac{1}{2}\left[\ln\frac{s^2}{(s^2+4)}\right]=\ln\left[\frac{s^2}{(s^2+4)}\right]^{-\frac{1}{2}}=\ln\sqrt{\frac{s^2+4}{s^2}}$$

Standard Forms

F(t)	$\mathcal{L}$ {F(t)} = f(s)
a	$\frac{a}{s}$
e^{at}	$\frac{1}{s-a}$
sin at	$\frac{a}{s^2+a^2}$
cos at	$\frac{a}{s^2+a^2}$
sinh at	$\frac{a}{s^2-a^2}$
cosh at	$\frac{s}{s^2-a^2}$
t^n	$\frac{n!}{s^{n+1}}$

(n a positive integer)

Theorem 1 The First Shift Theorem

If L{F(t)}= f(s) then **$L\{e^{(-at)}F(t)\}$= f(s+a)**

Theorem 2 Multiplying by t

If $L\{F(t)\}=f(s)$ **then** $L\{t(F(t)\}=-\frac{d}{ds}\{f(s)\}$

Theorem 3 Dividing by t

If $L\{F(t)\}=f(s)$ **then** $L\{\frac{F(t)}{t}\}=\int_s^\infty f(s)ds$

Provided $\underset{t\to 0}{Lim}\left\{\frac{F(t)}{t}\right\}$ *exists.*

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Inverse Transforms

Here we have the reverse process i.e. given a Laplace transform, we have to find the function of t to which it belongs. For example, we know that $\frac{a}{s^2+a^2}$ is the Laplace Transform of **sin at** , so we can now write $L^{-1}\left[\frac{a}{s^2+a^2}\right]=\sin at$, the symbol L^{-1} indicating the inverse transform and not a reciprocal.

$(a)\qquad L^{-1}\left\{\frac{1}{s-2}\right\}=e^{-2t}$

$(b)\qquad L^{-1}\left\{\frac{s}{s^2+25}\right\}=\cos 5t$

$(c)\qquad L^{-1}\left\{\frac{4}{s}\right\}=4$

$(d)\qquad L^{-1}\left\{\frac{12}{s^2-9}\right\}=4\sinh 3t$

But what about $L^{-1}\left\{\frac{3s}{s^2-s-6}\right\}$, it happens that we can write $\frac{3s}{s^2-s-6}$ as the sum of two simpler functions $\frac{1}{s+2}+\frac{1}{s-3}$ which , of course , makes all the difference , since we can now proceed. $L^{-1}\left\{\frac{3s}{s^2-s-6}\right\}=L^{-1}\left\{\frac{1}{s+2}\right\}+L^{-1}\left\{\frac{1}{s-3}\right\}=e^{-2t}+2e^{3t}$

Rules of Partial Fractions

1. The numerator must be of lower degree than denominator. If it is not , then we first divide out.
2. Factorise the denominator into its prime factors. These determine the shapes of the partial fraction.
3. A linear factor (s+a) gives a partial fraction $\frac{A}{s+a}$ is a constant to be determined.
4. A repeated factor $(s+a)^2$ gives $\frac{A}{s+a}+\frac{B}{(s+a)^2}$
5. Similarly $(s+a)^3$ gives $\frac{A}{s+a}+\frac{B}{(s+a)^2}+\frac{C}{(s+a)^3}$
6. A quadratic Factor (s^2+ps+q) gives $\frac{Ps+Q}{s^2+ps+q}$
7. Repeated quadratic Factor $(s^2+ps+q)^2$ gives $\frac{Ps+Q}{s^2+ps+q}+\frac{Rs+T}{(s^2+ps+q)^2}$

So $\frac{s-19}{(s-2)(s+5)}$ has partial fraction of the form $\frac{A}{(s-2)}+\frac{B}{(s+5)}$ and $\frac{3s^2-4s+11}{(s+3)(s-2)^2}$ has partial fraction $\frac{A}{(s+3)}+\frac{B}{(s-2)}+\frac{C}{(s-2)^2}$

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Example

To determine $L^{-1}\left\{\frac{5s+1}{s^2-s-12}\right\}$

a) First we check that the numerator is of lower degree than the denominator. In fact this is so.

b) Factories the denominator

$$\frac{5s+1}{s^2-s-12}=\frac{5s+1}{(s-4)(s+3)}=\frac{A}{(s-4)}+\frac{B}{(s+3)}$$

We therefore have an identity

$$\frac{5s+1}{s^2-s-12}=\frac{A}{(s-4)}+\frac{B}{(s+3)}$$

which is true for any value of s we care to substitute

If we multiply through by the denominator (s^2-s-12) we have

$5s + 1 \equiv A(s + 3) + B(s - 4)$

We now substitute covenant values for s

i) Let (s -4) =0 that is s = 4 therefore 21 = A(7) + B(0) $\Rightarrow$ A= 3

ii) Let (s + 3) =0 that is s = -3 therefore B = 2

So $\frac{5s+1}{s^2-s-12}=\frac{3}{(s-4)}+\frac{2}{(s+3)}$

$$L^{-1}\left\{\frac{5s+1}{s^2-s-12}\right\}=3e^{4t}+2e^{-3t}$$

Example Determined $L^{-1}\left\{\frac{9s-8}{s^2-2s}\right\}$

$$L\{F(t)\}=\frac{9s-8}{s^2-2s}$$

a) Numerator of first degree ; denominator of second degree.

b) $\frac{9s-8}{s^2-2s}=\frac{A}{s}+\frac{B}{s-2}$

c) Multiply by s(s – 2)

$\therefore 9s - 8 \equiv A(s - 2) + B(s)$

d) Put s = 0

$\therefore -8 \equiv A(-2) + B(0) \quad \therefore A = 4$

e) Put s – 2 , i.e. s = 2

$\therefore 10 = A(0) + B(0) \therefore B = 5$

$\therefore F(t) = L^{-1}\left\{\frac{4}{s}+\frac{5}{s-2}\right\} = 4 + 5e^{2t}$

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Table of inverse transforms

Standard transforms

f(s)	F(t)
$\frac{a}{s}$	a
$\frac{1}{s+a}$	e^{-at}
$\frac{n!}{s^{n+1}}$	t^n (n a positive integer)
$\frac{1}{s^n}$	$\frac{t^{n-1}}{(n-1)!}$ (n a positive integer)
$\frac{a}{s^2+a^2}$	sin at
$\frac{s}{s^2+a^2}$	Cos at
$\frac{a}{s^2-a^2}$	Sinh at
$\frac{s}{s^2-a^2}$	Cos ax

Transforms Of Derivatives

Let $F^{/}(t)$ denote the first derivative of F(t) with respect to t , $F^{//}(t)$ denote the second derivative of F(t) with respect to t, etc.

Then $L\{F^{/}(t)\}=\int_0^\infty e^{-st}F^{/}(t)dt$ by definition ,

Integrating By Parts

$$L\{F^{/}(t)\}=\left[e^{-st}F(t)\right]_0^\infty-\int_0^\infty F(t)(-se^{-st})dt$$

when $t\to 0, e^{-st}F(t)\to 0$

$$L\{F^{/}(t)\}=-F(0)+s\int_0^\infty e^{-st}F(t)dt$$

$$L\{F^{/}(t)\}=-F(0)+sL\{F(t)\}$$

$$L\{F^{//}(t)\}=-F^{/}(0)+sL\{F^{/}(t)\}=-F^{/}(0)+s\left[-F(0)+sL\{F(t)\}\right]$$

$$L\{F^{//}(t)\}=s^2L\{F(t)\}-sF(0)+F^{/}(0)$$

$$L\{F^{///}(t)\}=s^3L\{F(t)\}-s^2F(0)-sF^{/}(0)-F^{//}(0)$$

$$L\{F^{iv}(t)\}=s^4L\{F(t)\}-s^3F(0)-s^2F^{/}(0)-sF^{//}(0)-F^{///}(0)$$

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Differential Equation And Its Solution

$\frac{dx}{dt} - 2x = 4$ ----------(1)

Its Solution is $x = -2 + 3e^{2t}$, To verify it we find $\frac{dx}{dt}$

$\frac{dx}{dt} = 6e^{2t}$ then

$$\frac{dx}{dt} - 2x = 6e^{2t} - 2(-2 + 3e^{2t})$$
$$= 6e^{2t} + 4 - 6e^{2t} = 4$$

So equation (1) is satisfied. Hence $x = -2 + 3e^{2t}$ is solution of $\frac{dx}{dt} - 2x = 4$

Example Solve the differential equation $\frac{dx}{dt} - 2x = 4$ given that at t = 0 , x = 1

Taking Laplace transform as

$$L\left[\frac{d}{dt}(x(t)) - 2L(x(t))\right] = L(4) \Rightarrow L\left[\frac{d}{dt}(x(t))\right] - 2L(x(t)) = L(4) \Rightarrow sL(x(t)) - x(0) - 2L(x(t)) = \frac{4}{s}$$

$$(s-2)L(x(t)) - x(0) = \frac{4}{s} \Rightarrow (s-2)L(x(t)) - 1 = \frac{4}{s} \Rightarrow (s-2)L(x(t)) = \frac{4}{s} + 1 \Rightarrow (s-2)L(x(t)) = \frac{4+s}{s}$$

$$\Rightarrow L(x(t)) = \frac{4+s}{s(s-2)} \Rightarrow x(t) = L^{-1}\left[\frac{4+s}{s(s-2)}\right]$$ ----------(1)

First we do the partial fraction of $\frac{4+s}{s(s-2)}$

$$\frac{4+s}{s(s-2)} = \frac{A}{s} + \frac{B}{(s-2)}$$

$\Rightarrow 4 + s = A(s-2) + B(s)$ ----------(2)

Put s = 0 in equation (2) ; 4 = -2A ; A = -2

Put s = 2 in equation (2) ; 6 = B(2) ; B = 3

$So \ \frac{4+s}{s(s-2)} = \frac{-2}{s} + \frac{3}{s-2}$

Equation # (1) becomes

$$x(t) = L^{-1}\left[\frac{4+s}{s(s-2)}\right] = L^{-1}\left[\frac{-2}{s}\right] + L^{-1}\left[\frac{3}{s-2}\right] = -2 + 3e^{2t}$$

Solution of differential equation by laplace transforms

To solve a differential equation by Laplace transforms, we go through Laplace transforms, we go through four distinct stages.

(a) Re- write the equation in term of Laplace transforms.
(b) Insert the given initial conditions.
(c) Rearrange the equation algebraically to give the transform of the solution.

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(d) Determine the inverse transform to obtain the particular solution.

Solve the equation

$\frac{d^2x}{dt^2} - 3\frac{d\ x}{dt} + 2x = 2e^{3t}$ given that at t = 0 , x = 5 and $\frac{dx}{dt} = 7$

$x''(t) - 3x'(t) + 2x(t) = 2e^{3t}$

Given x (0) = 5 , x′ (0) = 7

$L\left(x''(t)\right) - 3L(x'(t)) + 2L(x(t)) = 2L(e^{3t})$

$s^2L\{x(t)\} - s\,x(0) - x'(0) - 3\{s\ L(x(t)) - x(0)\} + 2L\,x(t) = \frac{2}{s-3}$

We rewrite the equation in term of its transforms.

$L\left[\frac{d^2x}{dt^2}\right] - 3L\left[\frac{d\ x}{dt}\right] + 2L[x] = 2L\left[e^{3t}\right]$

$\left[[s^2L(x(t)) - s\,x(0) - x'(0)] - 3[s\,L(x(t)) - x(0)] + 2L(e^{3t})\right]$

At t = 0 , x = 5 , $\frac{dx}{dt} = 7$

So x (0) = 5 , x ′(0) = 7

$s^2L(x(t)) - s(5) - 7 - 3\{sL(x(t)) + 3(5)\} + 2L(x(t)) = \frac{2}{s-3}$

$s^2L(x(t)) - 3sL(x(t)) + 2L(x(t)) = \frac{2}{s-3} - 8 + 5s$

$(s^2 - 3s + 2)L(x(t)) = \frac{2 - 8s + 24 + 5s^2 - 15s}{s-3}$

$L(x(t)) = \frac{2 - 8s + 24 + 5s^2 - 15s}{(s-1)(s-2)(s-3)} = \frac{5s^2 - 23s + 24}{(s-1)(s-2)(s-3)}$

Making Partial fraction of R.H.S , We have

$L(x(t)) = \frac{A}{(s-1)} + \frac{B}{(s-2)} + \frac{C}{(s-3)}$

After solving these we get A= 3 , B = 2 and C = 0

So $L(x(t)) = \frac{3}{(s-1)} + \frac{2}{(s-2)} + \frac{0}{(s-3)}$

$L(x(t)) = \frac{3}{(s-1)} + \frac{2}{(s-2)}$

$x(t) = L^{-1}\{\frac{3}{(s-1)}\} + L^{-1}\{\frac{2}{(s-2)}\} = 3e^t + 2e^{2t}$

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Practice questions first 22 lectures

calculus II (Virtual University of Pakistan)

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Practice Questions of Lectures 1-22

Lecture 1

Q1: What is the difference between quadrant and octant?

Q2: Describe two applications of Differential Calculus and two applications of Integral Calculus.

Q3: In two dimensions, in which quadrant, are the following points located?

a) $(2, 5)$
b) $(-3, 7)$
c) $(9, -2)$
d) $(-8, -1)$

Q4: In three dimensions, in which octant, are the following points located?

a) $(1, 2, 3)$
b) $(-5, 7, 1)$
c) $(9, -2, 6)$
d) $(-9, -3, 9)$
e) $(-4, -2, -6)$
f) $(-1, 2, -3)$
g) $(2, 2, -2)$

Q5: Write down the equations of the following:

a) x – axis
b) y – axis
c) z – axis
d) xy – plane
e) yz – plane
f) xz – plane

Q6: Identify the functions of single variable, two and three variables.

a) $y = 2x + 1$
b) $z = 5x + 6y + 10$
c) $w = 3xyz + 5$
d) $y = x^2 + 5x + 6$

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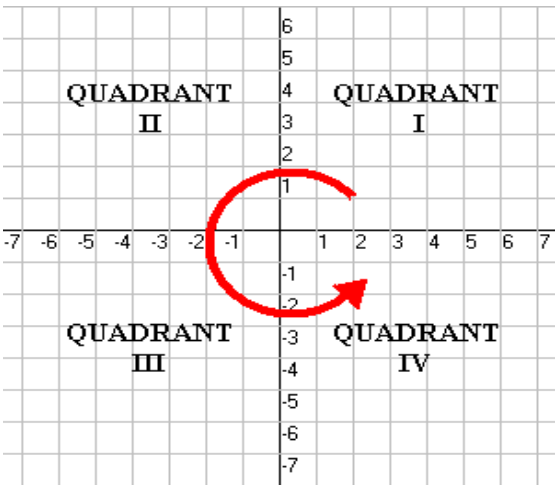
e) $A = \pi r^2$

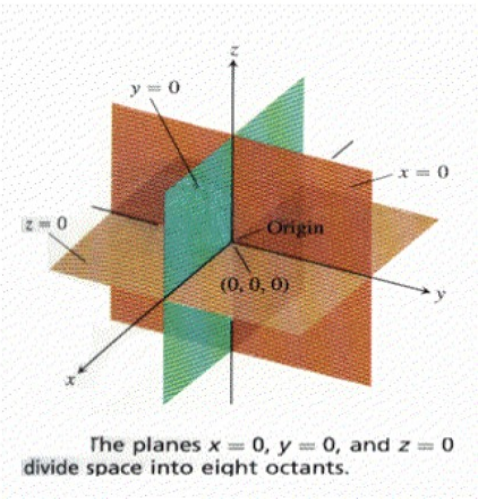
f) $P = 2(x + y)$

Answers

Q1: In two dimension plane, say xy- plane, we divide the plane into four parts. Each part is called quadrant.

In three dimension space, we divide the whole space in eight parts. Each part is called Octant.





The planes $x = 0$, $y = 0$, and $z = 0$ divide space into eight octants.

Q2: See handouts lecture No. 1. The answer is there.

Q3: (a) First quadrant (b) Second quadrant (c) 4th quadrant (d) 3rd quadrant

Q4: (a) First octant (b) 2nd octant (c) 4th octant

(d) 3rd octant (e) 7th octant (f) 6th octant

(g) 5th octant

Q5: See the page 14 Lecture No. 3

Q6: (a) single variable (b) two variables (c) three variables

(d) single variable (e) single variable (f) two variables

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Lecture 2,3

Q#1: Let $f(x,y) = x^2 + 9x^3y - 2y$. Is the function defined at the point (4 , 2)? If yes, what is its value and if not, give the reason.

Sol:

$$f(x,y) = x^2 + 9x^3y - 2y$$
$$f(4,2) = (4)^2 + 9(4)^3(2) - 2(2)$$
$$= 16 + 1152 - 4 = 1164$$

Q#2: Let $f(x, y, z) = x^2 y e^{xyz}$. Is the function defined at point $(1, 3, 6)$? If yes, what is its value and if not, give the reason.

Sol:

$$f(x, y, z) = x^2 y e^{xyz}$$
$$f(1,3,6) = (1)^2(3)\mathrm{e}^{(1)(3)(6)} = 196979907.4$$

Q#3: Let $f(x, y, z)$ such that $f(\mathrm{x}, \mathrm{y}, \mathrm{z}) = \ln\sqrt{x^2 + y^2 + z^2}$. Is the function defined at origin? If yes, what is its value and if not, give the reason.

Sol: f is not defined at origin.

Q#4: Find the points where the parabola, represented by the equation $y = x^2 - 4x$, meets x-axis.

Sol: See solution of Q#5.

Q#5: Find at least two points that lie on the parabola $y = x^2 - 6x + 8$.

Sol: put $y = 0$; $x^2 - 6x + 8 = 0$ (after factorizing $x = 2$ and $x = 4$)

Q#6: Find the mid point between (6,2,4) and (2, 5, 8).

Sol: $\left(\frac{6+2}{2}, \frac{2+5}{2}, \frac{4+8}{2}\right) = \left(\frac{8}{2}, \frac{7}{2}, \frac{12}{2}\right) = \left(4, \frac{7}{2}, 6\right)$

Q#7: Find the distance between the points $A(4, 5, 9)$ and $B(1, 3, 7)$.

Solution: $\sqrt{(4-1)^2 + (5-3)^2 + (9-7)^2} = \sqrt{17}$

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Q#8: For a line joining the points $A(3, 2, 6)$ and $B(0, 3, 4)$, find

a) Direction ratios of the line AB

0-3 , 3-2 , 4-6 = -3 , 1 , -2

b) Direction cosines of the line AB

$\frac{-3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}$

Q#9: Find the vertex point of the parabola $y = 2x^2 + 6x + 3$

Given equation of parabola:

$$y = 2x^2 + 6x + 3$$

x- coordinate of vertex = $x_0 = \frac{-b}{2a} = \frac{-6}{2(2)} = \frac{-6}{4} = \frac{-3}{2}$

y- coordinate of vertex = $y_0 = 2\left(\frac{-3}{2}\right)^2 + 6\left(\frac{-3}{2}\right) + 3$

$$= \frac{18}{4} - \frac{18}{2} + 3$$

$$= \frac{-6}{4} = \frac{-3}{2}$$

So the vertex point is $\left(\frac{-3}{2}, \frac{-3}{2}\right)$

Q#10: Find the points where the parabola $y = x^2 - 5x$ meets x-axis.

Given equation of the parabola is

$$y = x^2 - 5x \quad ----(1)$$

At x-axis, $y = 0$

So eq (1) becomes

$$0 = x^2 - 5x$$

$$0 = x(x - 5)$$

$$0 = x \quad or \quad x - 5 = 0$$

$$0 = x \quad or \quad x = 5$$

So the parabola meets x-axis at (0, 0) and (5, 0).

Q#11: Find at least two points that lie on the parabola $y = x^2 - 7x + 12$.

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put

$y = 0$

$\Rightarrow x^2 - 7x + 12 = 0$

$(x - 4)(x - 3) = 0$

$x = 4$ or x=3

so the two points on parabola be (4,0) and (3,0).

Lecture 4

Q1: Convert the Cartesian coordinates $(2, 3)$ into polar coordinates (r, θ)

Q2: Convert the Rectangular coordinates $(1, 2, 3)$ into Cylindrical coordinates (r, θ, z).

Q3: Convert the spherical coordinates $(10, \frac{\pi}{3}, \pi)$ into

(a) cylindrical coordinates (r, θ, z)

(b) rectangular coordinates (x, y, z)

Answers

Q1: $r = \sqrt{13}, \quad \theta = \tan^{-1}\left(\frac{3}{2}\right)$

Q2: $r = \sqrt{5}, \quad \theta = \tan^{-1}(2), \; z = 3$

Q3: (a) $r = 0, \quad \theta = \frac{\pi}{3}, \; z = -10$

(b) $x = 0, \quad y = 0, \quad z = -10$

Lecture No. 4 (with Solutions)

Q1: Convert the Cartesian coordinates $(3, 4)$ into polar coordinates (r, θ)

Sol: since $x = 3, y = 4.$

Here, we can find the polar coordinates (r, θ)

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$$r = \sqrt{x^2 + y^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{4}{3}\right).$$

Q2: Convert the Rectangular coordinates $(-2, 2\sqrt{3}, 4)$ into Spherical polar coordinates (ρ, θ, ϕ).

Sol Q2(a): since $x = -2, y = 2\sqrt{3}, and\ z = 4$

Here, we can find the spherical coordinates (ρ, θ, ϕ)

Since $\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{4 + 12 + 16} = \sqrt{32} = 4\sqrt{2}.$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{2\sqrt{3}}{-2}\right) = \tan^{-1}(-\sqrt{3}) = \frac{2\pi}{3}$$

$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{4}{4\sqrt{2}}\right) = \frac{\pi}{4}$$

Thus, the point is $(4\sqrt{2}, \frac{2\pi}{3}, \frac{\pi}{4})$ in spherical coordinates.

Question 2 (b): Convert the Rectangular coordinates $(-\sqrt{3}, 1, -2)$ into Spherical polar coordinates (ρ, θ, ϕ).

Solution: Since $x = -\sqrt{3}, y = 1, and\ z = -2$

Here, we can find the spherical coordinates (ρ, θ, ϕ)

Since $\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{3 + 1 + 4} = \sqrt{8} = 2\sqrt{2}.$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{1}{\sqrt{-3}}\right) = \frac{5\pi}{6}$$

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$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{-2}{2\sqrt{2}}\right) = \frac{3\pi}{4}$$

Thus, the point is $(2\sqrt{2}, \frac{5\pi}{6}, \frac{3\pi}{4})$ in spherical coordinates.

Q3: Express the given equation in cylindrical and spherical coordinates:

$$(x+y)^2 - z^2 + 4 = 0$$

Solution:

$$(x+y)^2 - z^2 + 4 = 0$$

$$x^2 + y^2 + 2xy - z^2 + 4 = 0 \quad \text{..... (1)}$$

For cylindrical coordinates, $x = r\cos\theta, y = r\sin\theta, and\ z = z$, so (1) becomes

$$(r\cos\theta)^2 + (r\sin\theta)^2 + 2(r\cos\theta)(r\sin\theta) - z^2 + 4 = 0$$

$$r^2\cos^2\theta + r^2\sin^2\theta + 2r^2\cos\theta\sin\theta - z^2 + 4 = 0$$

$$r^2(\cos^2\theta + \sin^2\theta) + 2r^2\cos\theta\sin\theta - z^2 + 4 = 0$$

$$r^2(1) + 2r^2\cos\theta\sin\theta - z^2 + 4 = 0$$

$$r^2(1 + 2\cos\theta\sin\theta) - z^2 + 4 = 0$$

$$r^2(1 + \sin 2\theta) - z^2 + 4 = 0$$

For spherical coordinates, $x = p\sin\phi\cos\theta, y = p\sin\phi\sin\theta, and\ z = p\cos\phi$, so (1) becomes

$$(p\sin\phi\cos\theta)^2 + (p\sin\phi\sin\theta)^2 + 2(p\sin\phi\cos\theta)(p\sin\phi\sin\theta) - (p\cos\phi)^2 + 4 = 0$$

$$p^2\sin^2\phi(\cos^2 + \sin^2\theta) + 2p^2\sin^2\phi\cos\theta\sin\theta - p^2\cos^2\phi + 4 = 0$$

$$p^2\sin^2\phi(1) + 2p^2\sin^2\phi\cos\theta\sin\theta - p^2\cos^2\phi + 4 = 0$$

$$2p^2\sin^2\phi\cos\theta\sin\theta + 4 = p^2\cos^2\phi - p^2\sin^2\phi$$

$$2p^2\sin^2\phi\cos\theta\sin\theta + 4 = p^2(\cos^2\phi - \sin^2\phi)$$

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$$p^2 \cos 2\phi - 2p^2 \sin^2 \phi \cos\theta \sin\theta = 4$$

$$p^2(\cos 2\phi - 2\sin^2 \phi \cos\theta \sin\theta) = 4$$

$$p^2(\cos 2\phi - \sin^2 \phi \sin 2\theta) = 4$$

$$2p^2 \sin^2 \phi \cos\theta \sin\theta + 4 = p^2(\cos^2 \phi - \sin^2 \phi)$$

Lecture 5

Q.1 Find the limit of the following function:

$$\lim_{(x,y)\to(1,3)} \frac{x^2 - 3xy + y^2}{x^2 y + 4x + 2y^2}$$

Q.2: By considering different path approach, find whether

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y^2}{x + y^3}$$ exists or not

Q.3 By considering different path approach show that the function

$$f(x,y) = \frac{x^3 - y}{y}$$ has not limit as $(x,y) \to (0,0)$

Answers

Q1: $\frac{1}{25}$

Q2: Limit exists.

Lecture No. 5 (Solutions)

Q1: Find the domains of the following functions and explain it through graph.

$$f(x,y) = \ln(9 - x^2 - 9y^2)$$

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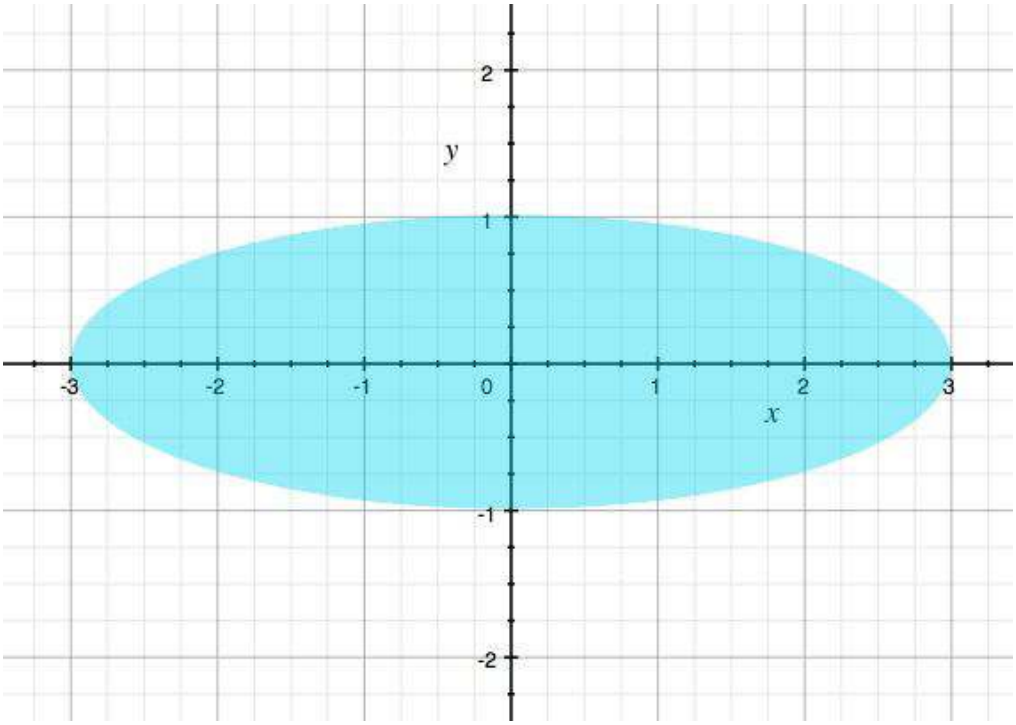
We see that $9 - x^2 - 9y^2$ must be positive.

We have $9 - (x^2 + 9y^2) > 0 \Rightarrow x^2 + 9y^2 < 9 \Rightarrow \frac{x^2}{9} + y^2 < 1$

The domain is the set of all points interior to the ellipse centered at the origin with a=3, b=1.



Q2: Find the limit of the following function.

$$\lim_{(x,y,z)\to(1,1,1)} x^2 \ln \frac{(xyz)}{e^{xz}}$$

$$\lim_{(x,y,z)\to(1,1,1)} \frac{x^2 In(xyz)}{e^{xz}}$$

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By applying limit, we get

$$=\frac{(1)^2 In(1)}{e^1}=\frac{1\times 0}{e}=0.$$

Q3: By using different path approach, find whether $\lim_{(x,y)\to(0,0)} \frac{x^4-y^4}{x^2+y^2}$ exist or not.

$$\lim_{(x,y)\to(0,0)} \frac{x^4 - y^4}{x^2 + y^2} = \lim_{(x,y)\to(0,0)} \frac{(x^2 - y^2)(x^2 + y^2)}{x^2 + y^2} = \lim_{(x,y)\to(0,0)} (x^2 - y^2)$$

Along x-axis $y=0.$

$$\lim_{(x,y)\to(0,0)} x^2 = 0$$

Along y-axis $x=0$.

$$\lim_{(x,y)\to(0,0)} - y^2 = 0$$

Along the line $y=mx$.

$$\lim_{(x,y)\to(0,0)} x^2 - (mx)^2 = \lim_{(x,y)\to(0,0)} x^2(1 - m^2) = 0.$$

Along the parabola $y=x^2$.

$$\lim_{(x,y)\to(0,0)} x^2 - (x^2)^2 = \lim_{(x,y)\to(0,0)} 1 - x^2 = 1$$

Thus the limit does not exist

Q4: Find the limit, if (x, y) $\to(0,0)$ along the line $y=x^3$.

$$\lim_{(x,y)\to(0,0)} \frac{x^3 y}{x^2+y^2}$$

If $(x,y)\to(0,0)$ along the line $y=x^3$, then

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$$\lim_{(x,y)\to(0,0)} \frac{x^3x^3}{x^2+(x^3)^2} = \lim_{(x,y)\to(0,0)} \frac{x^6}{x^2+x^6} = \lim_{(x,y)\to(0,0)} \frac{x^6}{x^2(1+x^4)} = \lim_{(x,y)\to(0,0)} \frac{x^4}{(1+x^4)} = \frac{0}{1} = 0.$$

Lecture 6

Q.1: Check whether the following function is continuous or not.

$$f(x,y) = \begin{cases} \frac{x^2y}{2x^2 - y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

Q.2: Find the region where the following function is continuous.

$$f(x,y) = 2x\sin x - \frac{2y}{x-1}$$

Q.3 Given that $f(t) = \frac{2t}{t+1}$ and $g(x,y) = x+y$. Find $f \circ g(x,y)$

Answers

Q1: Continuous

Q2: on entire *xy*-plane where $x \neq 1$

Q3: $\frac{2(x+y)}{x+y+1}$

Lecture No. 6 (Solutions)

Q1: : Evaluate the given limits.

$$\lim_{(x,y)\to(1,-1)} e^{-xy}\cos(x+y)$$

Solution: $\lim_{(x,y)\to(1,-1)} e^{-xy}\cos(x+y) = e^{-1(-1)}\cos(1+(-1)) = e^1\cos(0) = e$

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Q2: Find the region where the function $f(x,y)=\frac{x^2}{x^3-y^2}$ is continuous.

Solution: $f(x,y)$ is continuous in the entire two dimensional space except at the point

where $y=\sqrt{x^3}$

Q3: Examine the continuity of f at the point (0, 0).

$$f(x,y)=\begin{cases}\frac{2xy^2}{x^3+y^3} & if\,(x,y)\neq(0,0)\\ 0 & if\,(x,y)=(0,0)\end{cases}$$

Solution:

Let $(x,y)\to(0,0)$ along the line $y=mx$ so that $y\to 0\,as\,x\to 0$, then

$$f(x,y)=\frac{2x\ (m^2x^2)}{x^3+m^3x^3}$$

$$\lim_{(x,y)\to(0,0)} f(x,y)=\lim_{x\to 0}\frac{2m^2x^3}{x^3(1+m^3)}$$

$$=\lim_{x\to 0}\frac{2m^2}{(1+m^3)}$$, since $x\neq 0$

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$$=\frac{2m^2}{1+m^3}$$

So that limit value will vary as m changes. Therefore $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist

so f is discontinuous at (0, 0).

Q4: Let $f(x,y)=e^{x-y}$ be a surface. Find the partial derivatives of f with respect to *x* and *y* .

Solution: $f_x = e^{x-y}, f_y = -e^{x-y}$

Q5: Find the first order partial derivatives of the given function.

$$f(x,y)=\tan^{-1}\left(\frac{y}{x}\right)$$

Solution:

$$f_x = \frac{1}{1+\frac{y^2}{x^2}}\frac{\partial}{\partial x}\left(\frac{y}{x}\right) = \frac{x^2}{x^2+y^2}.\frac{-y}{x^2} = \frac{-y}{x^2+y^2}$$

$$f_y = \frac{1}{1+\frac{y^2}{x^2}}\frac{\partial}{\partial y}\left(\frac{y}{x}\right) = \frac{x^2}{x^2+y^2}.\frac{1}{x} = \frac{x}{x^2+y^2}$$

Lecture#7

Q1. If the function $f(x,\ y) = e^{2x}\cos y$, then find $\frac{\partial f}{\partial x}$ at point $(\ln 3,\ \pi)$.

Answer (-18)

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Q2. If $f(x, y) = \sin(x^4 y^3)$, then find $\frac{\partial f}{\partial x}$.

Answer ($4x^3 y^3 \cos(x^4 y^3)$)

Q3. If $f(x, y) = x \ln(1 + xy^{-1/4})$, find $\frac{\partial f}{\partial y}$.

Answer ($-\frac{x^2}{4(y^{5/4} + xy)}$)

Q4. Given that $h(u,v) = \ln uv^2 + \cos v$. Find $\frac{\partial^2 h}{\partial v^2}$.

Answer ($-\frac{2}{v^2} - \cos v$)

Q5. If $g(x, y) = (x+3)\ln(xy)$, then find both first order partial derivatives.

Answer ($(x+3)\frac{x}{xy} = \frac{x+3}{y}$)

Q.6: Verify Euler's theorem if $f(x,y) = e^{xy} \sin y$

Q7 : Let $g = \ln(3x - 3y) + \sin(x+y)$

a) Find $\frac{\partial^2 g}{\partial x^2}$

b) Find $\frac{\partial^2 g}{\partial y^2}$

c) Check whether $\frac{\partial^2 g}{\partial x^2} = \frac{\partial^2 g}{\partial y^2}$

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Answer $\frac{\partial^2 g}{\partial x^2} = \frac{\partial^2 g}{\partial y^2}$ holds true.

Lecture#8

Q1. Suppose $f(x, y) = \frac{x}{y}$ $where$ $x = e^u$ and $y = \ln u$. Use chain rule to find $\frac{df}{du}$ and write the answer in variable *u* only.

Answer $\frac{[u(\ln u) - 1]e^u}{u(\ln u)^2}$

Q2.

If $h = g(u, v, w)$

$u = u(t)$

$v = v(t)$

$w = w(t)$, then state the chain rule for derivative of h with respect to t.

Answer $\frac{dh}{dt} = \frac{\partial h}{\partial u}\frac{du}{dt} + \frac{\partial h}{\partial v}\frac{dv}{dt} + \frac{\partial h}{\partial w}\frac{dw}{dt}$

Q3. Let $u = f(x, y) = xy - \cos y$

$x(t) = e^t$

$y(t) = \sqrt{t^2 + 1}$

Use chain rule to find $\frac{du}{dt}$ and show it as a function of t.

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Answer ($\dfrac{e^t(t^2+1)+t(e^t+\sin\sqrt{t^2+1})}{\sqrt{t^2+1}}$)

Q4. $Let \quad w=g(u,v)=\ln(u+v)$
$u(t)=e^{2t}$
$v(t)=t^2+t$

Use chain rule to find $\dfrac{dw}{dt}$ and express it as a function of t.

Answer ($\dfrac{dw}{dt}=\dfrac{2e^{2t}+2t+1}{e^{2t}+t^2+t}$)

Q5: Find the partial derivatives with respect to *x* and *y* at point $(1,\ 2).$

$$g(x,y)=x^3y^2\ln(xy)$$

Answer $g_x(1,2)=12\ln 2+4, \quad g_y(1,2)=4\ln 2+2$

Lecture#9

Q1. Let $f(x,\ y)=\dfrac{\cos x}{y}, \quad x=u^2-v \text{ and } \quad y=v+e^u.$ Find $\dfrac{\partial f}{\partial u}$ and simplify it as a function of u and v.

Answer ($-\dfrac{2u\sin(u^2-v)e^u}{(v+e^u)}-\dfrac{e^{2u}\cos(u^2-v)}{(v+e^u)^2}$)

Q2. Let $z=f(x,\ y)=e^{xy}, \ x(s,t)=3s\cos t \text{ and } y(s,t)=4t^2s$. Use chain rule to find $\dfrac{\partial z}{\partial t}$ and simplify it as a function of s and t.

Answer ($12s^2te^{12s^2t^2\cos t}(-t\sin t+2\cos t)$)

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Q3.

If $h = g(x, y, z)$

$u = u(r, s, t)$

$v = v(r, s, t)$

$w = w(r, s, t)$, then state the chain rule for derivative of h with respect to r.

Answer

$$\frac{\partial g}{\partial r} = \frac{\partial g}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial g}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial g}{\partial z}\frac{\partial z}{\partial r}$$

Lecture 10

Q.1: Find a vector that is perpendicular to both of the vectors $\hat{i} - 2\hat{j} + 3\hat{k}$ and

$$2\hat{i} + 4\hat{j} - \hat{k}$$

Q.2: Find the scalar product of the vectors $3\hat{i} + 2\hat{j} + \hat{k}$ and $4\hat{i} - \hat{j} + 2\hat{k}$.

Q.3: Find the value of '$\bar{a}$' if $a\hat{i} + \hat{j} + 5\hat{k}$ and $\hat{i} - 3\hat{j} + \hat{k}$ are orthogonal to each other.

Answers

Q1: $-10\hat{i} + 7\hat{j} + 8\hat{k}$

Q2: 12

Q3: -2

Lecture 11

Q1: Let $\bar{a} = 2\hat{i} + 5\hat{j} + 6\hat{k},\ \bar{b} = -\hat{i} + 3\hat{j} - 7\hat{k},\ \bar{c} = 3\hat{i} + 5\hat{j} + 3\hat{k}.$ Find $\bar{a} \times \bar{b} . \bar{c}$

Q2: Find the gradient of $f(x, y) = ye^x + \cos(xy)$

Q3: Find the gradient of $g(x, y) = 5x^3 y^2$

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Q4: Find the directional derivative of the function $f(x,y) = xy + \sin(xy)$ at $(2,0)$ in the direction of $\bar{a} = 4\hat{i} + 3\hat{j}$.

Answers

Q1: -86

Q2: $\nabla f = \left(ye^x - y\sin(xy)\right)\hat{i} + \left(e^x - x\sin(xy)\right)\hat{j}$

Q3: $\nabla g = 15x^2y^2\,\hat{i} + 10x^3y\,\hat{j}$

Q4: $\nabla f(2,0).\hat{a} = \frac{12}{5}$

Lecture No. 12

Q.1: Find an equation of the tangent plane to the surface $z = 3y^2 - 2x^2 + x$ at the point $(2,-1,-3)$.

Q.2 : Find an equation of the tangent plane to the surface $z = x\sin(x+y)$ at the point $(-1,1,0)$.

Q.3: Find an equation of the plane passing through point $(2,1,3)$ and perpendicular to the vector $n = \hat{i} + \hat{j} + 2\hat{k}$.

Answers

Q1: $z = -7x - 6y + 5$

Q2: $x + y + z = 0$

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Q3: $x + y + 2z = 9$

Lecture#13

Q1.

Find equation of normal line to the given surface $f(x,y,z) = x^2 - 2y^2 - z^2 - 4$ at the point $P(-6, 2, \sqrt{24})$

Solution

$f(x,y,z) = x^2 - 2y^2 - z^2 - 4$

$f_x = 2x$, $f_y = -4y$, $f_z = -2z$

$f_x(-6, 2, \sqrt{24}) = -12$, $f_y(-6, 2, \sqrt{24}) = -8$, $f_z(-6, 2, \sqrt{24}) = -2\sqrt{24}$

Equation of normal line of the surface is

$$\frac{x+6}{-12} = \frac{y-2}{-8} = \frac{z-\sqrt{24}}{-2\sqrt{24}}$$

$$\frac{x+6}{3} = \frac{y-2}{2} = \frac{z-\sqrt{24}}{\sqrt{6}}$$

Q: Find the Equation of the tangent plane to the surface $f(x,y,z) = xz - 4$ at the point

P (-2, 2,-2)

Solution

$f(x,y,z) = xz - 4$

$f_x = z$, $f_y = 0$, $f_z = x$

$f_x(-2, 2, -2) = -2$, $f_y(-2, 2, -2) = 0$, $f_z(-2, 2, -2) = -2$

Equation of tangent plane to the surface is

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$-2(x+2)-2(z+2)=0$

$2(x+2)+2(z+2)=0$

$2x+2z+8=0$

$2(x+z+4)=0$

Q3 Show that the following two surfaces intersect orthogonally.

$x^2+y^2+2z^2-16=0$

$x^2+y^2-4z=0$

Solution

let

$f(x,\ y,\ z)=x^2+y^2+2z^2-16=0$ - - - - - - - - - - - - - - - (1)

$g(x,\ y,\ z)=x^2+y^2-4z=0$ - - - - - - - - - - - - - - - - - (2)

$f_x(x,\ y,\ z)=2x$

$f_y(x,\ y,\ z)=2y$

$f_z(x,\ y,\ z)=4z$

$g_x(x,\ y,\ z)=2x$

$g_y(x,\ y,\ z)=2y$

$g_z(x,\ y,\ z)=-4$ 1+1+1

$$f_xg_x+f_yg_y+f_zg_z=(2x)(2x)+(2y)(2y)+(4z)(-4)$$
$$=4x^2+4y^2-16z$$
$$=4(x^2+y^2-4z)$$

Since from equation (2) , $x^2+y^2-4z=0$ *So*

$f_xg_x+f_yg_y+f_zg_z=4(x^2+y^2-4z)=0$

Thus, the surfaces $x^2+y^2+2z^2-16=0$ and $x^2+y^2-4z=0$ are orthogonal at every point of intersection.

Lecture#14

Q1.

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Find the critical points for the function

$f(x,y) = 2x^2 - 4x + xy^2 - 1$

Solution

$f(x,y) = 2x^2 - 4x + xy^2 - 1$

$f_x = 4x - 4 + y^2$

$f_y = 2xy$

The critical points of f satisfy the equations

$4x - 4 + y^2 = 0 - - - - - - - - - - - - (1)$

$2xy = 0 - - - - - - - - - (2)$

$2xy = 0 \Rightarrow xy = 0 \Rightarrow x = 0 \text{ or } y = 0$

If $x = 0$ *then from* $(1), y = \pm 2.$

If $y = 0$ *then from* $(1), x = 1.$

Thus the critical point are $(1, 0), (0, 2), (0, -2)$

Q2.

Find the critical points of the function.

$f(x,y) = \sin x + \sin y$

Solution

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$f(x,y) = \sin x + \sin y$

$f_x = \cos x$

$f_y = \cos y$ 1+1

The critical points of f satisfy the equations

$\cos x = 0$ - - - - - - - - - - - - - - (1)

$\cos y = 0$ - - - - - - - - - - - - - - (2)

From (1)

$x = \cos^{-1}(0) \Rightarrow \quad x = m\pi + \frac{\pi}{2}$

From (2)

$y = \cos^{-1}(0) \Rightarrow \quad y = n\pi + \frac{\pi}{2}$

Thus the critical points of a given function are $(m\pi + \frac{\pi}{2}, n\pi + \frac{\pi}{2})$

Q3

Find the extrema for the following function.

$f(x,y) = x^2 - xy + y^2 + 6x$

Solution

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$f(x, y) = x^2 - xy + y^2 + 6x$

$f_x = 2x - y + 6$

$f_y = -x + 2y$

The critical points of f satisfy the equations

$f_x = 2x - y + 6 = 0$ - - - - - - - - - - - - (1)

$f_y = -x + 2y = 0$ - - - - - - - - - - - - (2)

Now put x=2y in (1), we have

$2(2y) - y + 6 = 0 \Rightarrow$ 4y-y+6=0 $\Rightarrow$ 3y=-6 $\Rightarrow$ y=-2

Put y=-2 in (2)

$-x + 2(-2) = 0 \Rightarrow x$=-4

Thus the only critical points is $(-4, -2)$

$f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = -1$

$D = f_{xx} f_{yy} - f_{xy}^2$

For $(-4, -2)$, $D = 3 > 0$ Re$lative$ min$imum$

Q4.

Find the critical points of the function.

$f(x, y) = 3x^2 y + y^3 - 3x^2 - 3y^2 + 2$

Solution

$f(x, y) = 3x^2 y + y^3 - 3x^2 - 3y^2 + 2$

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$f_x = 6xy - 6x$

$f_y = 3x^2 + 3y^2 - 6y$

The critical points of f satisfy the equations

$6xy - 6x = 0$ - - - - - - - - - - - - (1)

$3x^2 + 3y^2 - 6y = 0$ - - - - - - - - - (2)

From (1)

$6x(y - 1) = 0 \Rightarrow x = 0$ *and* $y = 1$

For $x = 0$, *we have*

$3y(y-2) = 0 \Rightarrow y = 0, y = 2$

For y=1

$3x^2 + 3(1)^2 = 6(1) \Rightarrow 3x^2 = 3 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$

Thus the critical poi nt *are* $(0,0), (0,2), (1,1), (-1,1)$

Lecture#15

Q1. Find the critical points of the function $f(x, y) = x^2 + 4xy + 2y^2 + 4x - 8y + 7$ and then classify them into relative maxima, relative minima and saddle points.

Solution

$f(x, y) = x^2 + 4xy + 2y^2 + 4x - 8y + 7$

$f_x = 2x + 4y + 4$

$f_y = 4x + 4y - 8$

The critical points of f satisfy the equations

$f_x = 2x + 4y + 4 = 0$ - - - - - - - - - - - - (1)

$f_y = 4x + 4y - 8 = 0$ - - - - - - - - - - - - (2)

Subtracting (2) from (1), we have

$-2x+12=0 \Rightarrow 2x=12 \Rightarrow x=6$

Now put x=2 in (1), we have

$4y+16=0 \Rightarrow 4y=-16 \Rightarrow y=-4$

Thus the only critical points is $(6, -4)$

$f_{xx} = 2$, $f_{yy} = 4$, $f_{xy} = 4$

$D = f_{xx}\ f_{yy} - f_{xy}^{\ 2} = 8 - 16 = -8$

For $(6, -4)$, $D = -8 < 0$ saddle point

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Q2. Consider a function $f(x,y) = x^3 + 6xy + 3y^2 - 9x$ .One of its critical point is (1, -1). Find, whether (1, -1) is relative maxima, relative minima or saddle point of $f(x,y)$

Solution

$f(x,y) = x^3 + 6xy + 3y^2 - 9x$

$f_x = 3x^2 + 6y - 9$

$f_y = 6x + 6y$

$f_{xx} = 6x$,

$f_{yy} = 6,$

$f_{xy} = 6$

$f_{xx}(1,-1) = 6$,

$f_{yy}(1,-1) = 6$,

$f_{xy}(1,-1) = 6$

$D = 36 - 36 = 0$

since $D > 0,\ f_{xx} > 0,$ therefore, $f(x,y)$ *has Saddle* point *at* $(1,-1)$.

Q3. Find and classify the critical points of $f(x,y) = 2xye^{-x^2-y^2}$

Solution

$f(x,y) = 2xye^{-x^2-y^2}$

$f_x = 2ye^{-x^2-y^2} - 4x^2ye^{-x^2-y^2}$

$= 2ye^{-x^2-y^2}(1-2x^2)$

$f_y = 2xe^{-x^2-y^2} - 4y^2xe^{-x^2-y^2}$

$= 2xe^{-x^2-y^2}(1-2y^2)$

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The critical points of f satisfy the equations

$f_x = 2ye^{-x^2-y^2}(1-2x^2) = 0 - - - - - - - - - - - - (1)$

$f_y = 2xe^{-x^2-y^2}(1-2y^2) = 0 - - - - - - - - - - - - (2)$

$(1) \Rightarrow ye^{-x^2-y^2}(1-2x^2) = 0 \Rightarrow y = 0 \; or \; 1 = 2x^2 \Rightarrow x = \pm\frac{1}{\sqrt{2}}$

$(2) \Rightarrow xe^{-x^2-y^2}(1-2y^2) = 0 \Rightarrow x = 0 \; or \; 1 = 2y^2 \Rightarrow y = \pm\frac{1}{\sqrt{2}}$

critical points are $(0,0), (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}), (\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}), (\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}), (\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

$$fxx = 2\left(-4xye^{-x^2-y^2} - 2x(1-2x^2)ye^{-x^2-y^2}\right)$$

$$= -4xy(3-x^2)e^{-x^2-y^2}$$

$$fxy = 2(1-2x^2)(1-2y^2)e^{-x^2-y^2}$$

$$fyy = -4xy(3-y^2)e^{-x^2-y^2},$$

$D = f_{xx} f_{yy} - f_{xy}^2$

$For \; (0,0), \; D = -4 < 0 \; Saddle \; point$

$For \; (1/\sqrt{2}, 1/\sqrt{2}) = (-1/\sqrt{2}, -1/\sqrt{2}), \; D = 49 > 0$

$fxx(1/\sqrt{2}, 1/\sqrt{2}) = fxx(-1/\sqrt{2}, -1/\sqrt{2}) < 0, \; Relative \; maximum.$

$For \; (-1/\sqrt{2}, 1/\sqrt{2}) = (1/\sqrt{2}, -1/\sqrt{2}), \; D = 49 > 0$

$fxx(1/\sqrt{2}, 1/\sqrt{2}) = fxx(-1/\sqrt{2}, -1/\sqrt{2}) > 0, \; Relative \; minimum$

Q4. For a function $f(x,y) = x^2 + (y+1)^2$, $f_x = 2x, \; f_y = 2(y+1)$. One of the critical points is $(0,-1)$. Check whether it is relative maxima, relative minima or a saddle point.

Solution

$f_{xx} = 2, \; f_{yy} = 2, \; f_{xy} = 0$

$f_{xx}(0,-1) = 2 > 0, \; f_{yy}(0,-1) = 2$

$D = f_{xx}(0,-1) f_{yy}(0,-1) - f_{xy}^2(0,-1)$

$= (2)(2) - 0 = 4 > 0$

since $D > 0, \; f_{xx} > 0,$ therefore, $f(x,y)$ *has* $Relative \; minimum \; at \; (0,-1)$.

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Lecture 16

Q1. If $f(x,y) = 2x - 3y^2 + xy$ and R is the triangular region with vertices O (0, 0), A (3, 0) and B (1, 4), then find the critical points along the line segment OB at which the absolute extreme values of $f(x,y)$ can occur.

Solution

As the equation of the line for two points is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

So the equation of line segment OB from $(0,0)$ to $(1,4)$ is $y = 4x$ as compared to general equation of line. By putting $y=4x$ in $f(x,y) = 2x - 3y^2 + xy$, we have

$$\begin{aligned} f(x,x) &= 2x - 3(4x)^2 + x(4x) \\ &= 2x - 48x^2 + 4x^2 \\ &= 2x - 44x^2 \end{aligned}$$

$$f_x(x,x) = 2 - 88x$$

Equating with zero, we get

$$2 - 88x = 0 \Rightarrow x = \frac{1}{44}$$

$$y = 4\left(\frac{1}{44}\right) = \frac{1}{11}$$

Thus, the critical point at which the function may assume the absolute extrema is $\left(\frac{1}{44}, \frac{1}{11}\right)$.

Q 2. If $f(x,y) = xy + x + 1$ and R is the triangular region with vertices O (0, 0), A (6, 0) and B (0, 5), then find the critical points along the line segment AB at which the absolute extrema of $f(x,y)$ can occur.

Solution

Equation of the line segment from $(6,0)$ to $(0,5)$ is $y = -\frac{5}{6}x + 5$ as compared to general equation of straight line. By putting $y = -\frac{5}{6}x + 5$ in $f(x,y) = xy + x + 1$, we have

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$$f(x,-\frac{5}{6}x+5)=x(-\frac{5}{6}x+5)+x+1$$

$$=-\frac{5}{6}x^2+6x+1$$

$$f_x(x,-\frac{5}{6}x+5)=-\frac{5}{3}x+6$$

Equating with zero, we get

$$-\frac{5}{3}x+6=0\Rightarrow x=\frac{18}{5}$$

$$y=-\frac{5}{6}(\frac{18}{5})+5=2$$

Thus, the critical point at which $f(x,y)$ may assume the absolute extrema is $\left(\frac{18}{5},2\right)$.

Q3. Find the critical point for the given function $f(x,y)=x^2-2xy+y^2$ along the line $y=2x-3$ at which the absolute extrema of the function can occur.

Solution

As $f(x,y)=x^2-2xy+y^2$, simplifies it to single variable x

$$f(x,2x-3)=x^2-2x(2x-3)+(2x-3)^2$$

$$=x^2-6x+9$$

As critical point exist when $f'(x,2x-3)=0$

So here $2x-6=0$

$\Rightarrow x=3$

$y=2(3)-3=3$

Thus the critical point exists at (3, 3).

Q4. Let $f(x,y)=xy-x-y$ and R is the triangular region with vertices O (0, 0), A (0, 4) and B (5, 0). Find the critical points along the line segment OA at which the absolute extrema of $f(x,y)$ can occur.

Solution

Since along the line segment OA $x=0$, so the given function becomes

$f(0,y)=-y$

$f_y(0,y)=-1$

As the critical point exist where $f_y(0,y)=0$.

Since $-1\neq 0$, therefore, there does not exist any critical point along the line OA at which the function $f(x,y)$ may assume the extreme values.

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Lecture#17

Q1. Consider three positive integers whose sum is 35 and the product of two integers with the cube of third integer will be maximum. Write down the function to be maximized.

Solution

The function to be maximized is
$f(x,y)=xy(35-x-y)^3$.

Q2. Show that $f(x,y)=xy(11-x-y)$ is maximum at $x=5, y=5$, by using second partial derivative test.

Solution

As $f(x,y)=11xy-x^2y-xy^2$

$f_x=11y-2xy-y^2,\ f_y=11x-x^2-2xy$

$f_{xx}=-2y,\ f_{yy}=-2x,\ f_{xy}=11-2x-2y$

at $x=5,\ y=5$

$f_{xx}=-10<0,\ f_{yy}=-10,\ f_{xy}=-9$

$D=(f_{xx})(f_{yy})-(f_{xy})^2$

$=(-10)(-10)-(-9)^2=19>0$

Hence, given function is maximum at the given points.

Q3. Using second partial derivative test show that the function $f(x,y)=2xy(5-x-y)$ is maximum at $x=1, y=1$.

Solution

As $f(x,y)=10xy-2x^2y-2xy^2$

$f_x=10y-4xy-2y^2,\ f_y=10x-2x^2-4xy$

$f_{xx}=-4y,\ f_{yy}=-4x,\ f_{xy}=10-4x-4y$

at $x=1, y=1$

$f_{xx}=-4<0,\ f_{yy}=-4,\ f_{xy}=2$

$D=(f_{xx})(f_{yy})-(f_{xy})^2$

$=(-4)(-4)-4=12>0$

Hence, given function is maximum at given points.

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Lecture#18

Q1. Evaluate the following double integral.

$$\iint \left(x^3 y - 6x^2 \right) \, dy \, dx$$

Solution

$$\iint \left(x^3 y - 6x^2 \right) \, dy \, dx = \int \left(\frac{x^3 y^2}{2} - 6x^2 y \right) \, dx$$

$$= \frac{x^4 y^2}{8} - 2x^3 y$$

Q2. Evaluate the double integral

$$\int_2^6 \int_1^3 \left(7xy^2 + 3x^3 y \right) \, dx \, dy.$$

Solution

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$$\int_2^6 \int_1^3 (7xy^2 + 3x^3y)\, dx\, dy = \int_2^6 \left| \frac{7x^2y^2}{2} + \frac{3x^4y}{4} \right|_{-1}^{3} dy$$

$$= \int_2^6 \left(\frac{7}{2} y^2 \left(3^2 - (-1)^2\right) + \frac{3}{4} y \left(3^4 - (-1)^4\right) \right) dy$$

$$= \int_2^6 \left(\frac{7}{2} y^2 (9-1) + \frac{3}{4} y (81-1) \right) dy$$

$$= \int_2^6 \left(\frac{7}{2} y^2 (8) + \frac{3}{4} y (80) \right) dy$$

$$= \int_2^6 (28y^2 + 60y)\, dy$$

$$= \left| \frac{28y^3}{3} + 30y^2 \right|_2^6$$

$$= \left| \frac{28}{3}(6^3 - 2^3) + 30(6^2 - 2^2) \right|_2^6$$

$$= \frac{28}{3}(216 - 8) + 30(36 - 4)$$

$$= \frac{8704}{3}$$

Q3. Evaluate $\int_1^3 \int^4 (3x^3y + x^2) dx\, dy$.

Solution

$$\int_1^3 \int^4 (3x^3y + x^2) dx\, dy = \int_1^3 \left| \frac{3x^4y}{4} + \frac{x^3}{3} \right|_1^4 dy$$

$$= \int_1^3 \left(\frac{765y}{4} + 21\right) dy$$

$$= \left| \frac{765y^2}{8} + 21y \right|_{-1}^{3}$$

$$= \frac{765}{8}[(3)^2 - (-1)^2] + 21[(3) - (-1)]$$

$$= 765 + 84 = 849$$

Q4. Evaluate each of the following integrals over the given region *D*.

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$$\iint_D e^{\frac{x}{y}} dA, D = \left\{ \{x, y\} \mid 1 \le y \le 2, y \le x \le y^3 \right\}$$

Solution

$$\iint_D \mathbf{e}^{\frac{x}{y}} dA = \int_1^2 \int_y^{y^3} \mathbf{e}^{\frac{x}{y}} dx\, dy = \int_1^2 y\, \mathbf{e}^{\frac{x}{y}} \Big|_y^{y^3} dy$$

$$= \int_1^2 y\, \mathbf{e}^{y^2} - y\mathbf{e}^1 dy$$

$$= \left(\frac{1}{2} \mathbf{e}^{y^2} - \frac{1}{2} y^2 \mathbf{e}^1 \right) \Big|_1^2 = \frac{1}{2} \mathbf{e}^4 - 2\mathbf{e}^1$$

Answer ($\frac{1}{2}e^4 - 2e^1$)

Practice Questions Lectures 19-22

Q#1: Evaluate $\int_0^3 \int_{x^2}^{3x} (3x+1)\, dy\, dx$

Q#2: If the order of integration for the integral $\int_0^6 \int_{y/3}^2 e^{x^2} dx dy$ is changed. Find the change in the limits of new integral.

Q#3: Evaluate the integral $\int_0^3 \int_0^{2x} e^{x^2} dy dx$.

Q#4: Evaluate the integral $\int_0^3 \int_0^{y^2} \cos y^3 dx dy.$

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Solution of Practice Question Lecture 19-22

Q#1: Evaluate $\int_0^3 \int_{x^2}^{3x} (3x+1)\, dy\, dx$

Solution:

$$\int_0^3 \int_{x^2}^{3x} (3x+1)\, dy\, dx$$

$$= \int_0^3 [3x(y)_{x^2}^{3x} + (y)_{x^2}^{3x}]\, dx$$

$$= \int_0^3 [3x(3x - x^2) + (3x - x^2)]\, dx$$

$$= = \int_0^3 [9x^2 - -3x^3 + 3x - x^2]\, dx = \frac{99}{4}$$

If the order of integration for the integral $\int_0^6 \int_{y/3}^{2} e^{x^2} dx dy$ is changed. Find the change in the limits of new integral.

Solution:

$$\frac{y}{3} \le x \le 2 \Rightarrow y \le 3x \le 6$$

$$0 \le y \le 3x \le 6$$

$$0 \le 3x \le 6$$

$$0 \le x \le 2$$

Q#3: Evaluate the integral $\int_0^3 \int_0^{2x} e^{x^2} dy dx$.

Solution:

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$$\int_0^3 \int_0^{2x} e^{x^2} dy dx$$

$$= \int_0^3 e^{x^2} (y)_0^{2x} dx$$

$$= \int_0^3 e^{x^2} 2x dx = (e^{x^2})_0^3 = e^{3^2} - 1$$

$$= e^9 - 1$$

Q#4: Evaluate the integral $\int_0^3 \int_0^{y^2} \cos y^3 dx dy.$

Solution:

$$\int_0^3 \int_0^{y^2} \cos y^3 dx dy$$

$$= \int_0^3 (x)_0^{y^2} . \cos y^3 \ dy$$

$$= \int_0^3 y^2 . \cos y^3 \ dy$$

$$= \frac{1}{3} \int_0^3 3y^2 . \cos y^3 \ dy$$

$$= \frac{1}{3} \left(\sin y^3 \right)_0^3 = = \frac{1}{3} \sin 27$$

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